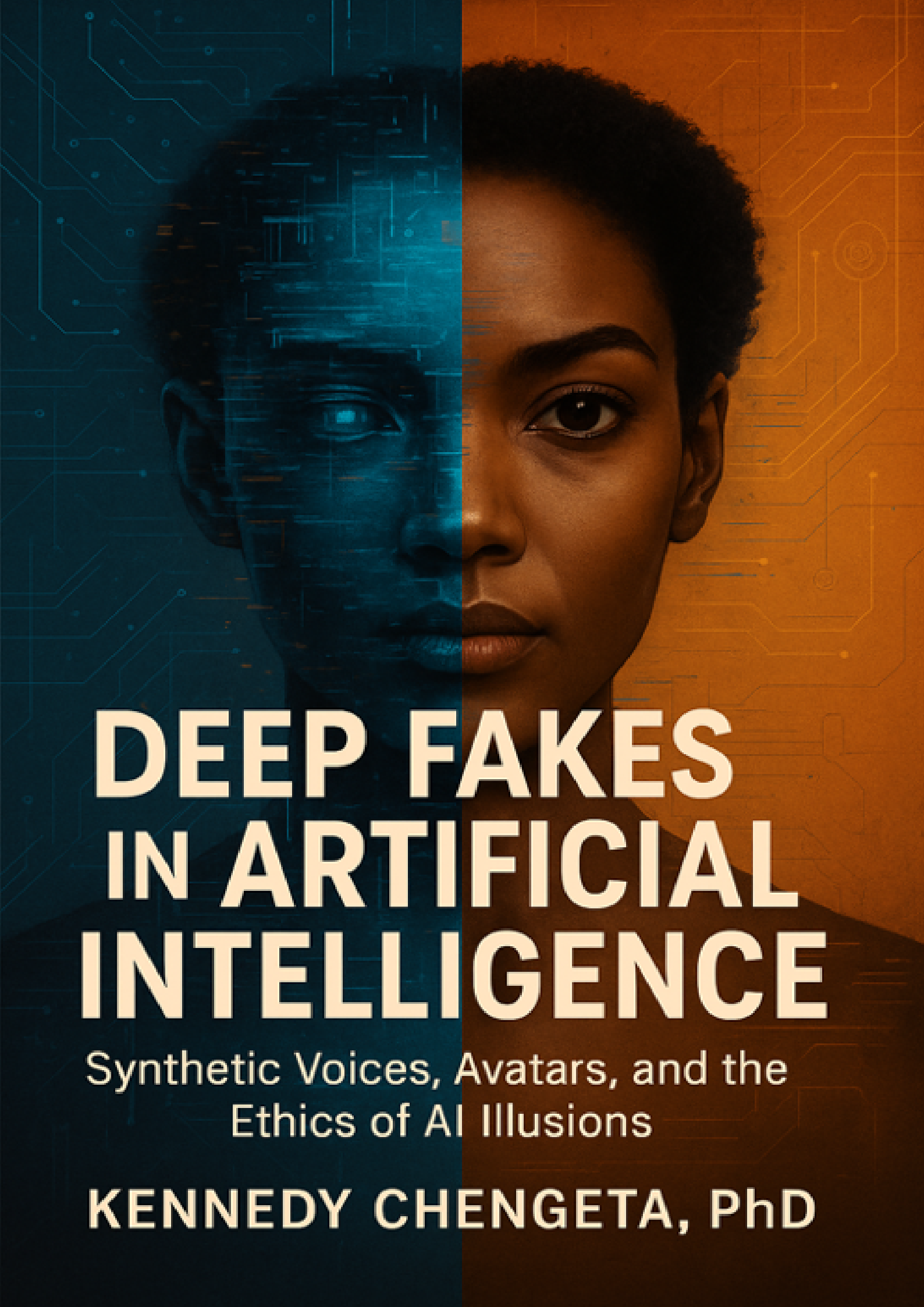




DEEP FAKES IN ARTIFICIAL INTELLIGENCE

Synthetic Voices, Avatars, and the
Ethics of AI Illusions

KENNEDY CHENGETA, PhD

A RESEARCH MONOGRAPH

Deep Fakes in Artificial Intelligence

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Kennedy Chengeta, PhD

First Edition

May 1, 2026

Deep Fakes in Artificial Intelligence: Synthetic Voices, Avatars, and the Ethics of AI Illusions
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*To all researchers who work to keep synthetic media honest,
and to every person whose likeness or voice has been used
without consent.*

May the tools of detection grow faster than the tools of deception.

*“Seeing is no longer believing.
The real question is whether we can build systems
that know when not to trust their eyes.”*
—Kennedy Chengeta, PhD

Preface

The past decade has transformed synthetic media from a laboratory curiosity into an industrial-scale challenge. What were, in 2014, coarse neural texture transfers have become photorealistic face swaps indistinguishable from genuine footage by the unaided eye, voice clones produced from seconds of audio, and full-body avatar systems capable of animating any identity into any performance. The engine of this progress has been artificial intelligence — specifically, the convergence of generative adversarial networks, variational autoencoders, diffusion models, and large-scale neural codec language models — deployed on compute budgets that would have been unimaginable a decade ago.

This book is a technical, integrative treatment of the field. It is aimed at graduate students, research engineers, forensic practitioners, policy analysts, and advanced practitioners who need a principled account of the full deep-fake ecosystem: how synthetic media is generated, how it can be detected and attributed, what legal and ethical frameworks apply, and where the technology is headed. The perspective is unapologetically mathematical. Concepts are introduced with the equations, architectures, and probabilistic assumptions that make them verifiable and reproducible. Where empirical results drive design decisions — as in much of deep learning — we present the models and training objectives with enough precision to implement them.

The book is organized into five parts. Part I establishes foundations: definitions, a taxonomy of synthetic-media types, and the mathematical and machine-learning primitives used throughout. Part II covers generation techniques, from early face-swap pipelines through modern diffusion-based synthesis and neural voice cloning. Part III addresses detection and attribution, including forensic signal analysis, passive and active provenance schemes, and the arms-race dynamics between generators and detectors. Part IV treats ethics, law, and society — the normative scaffolding without which purely technical countermeasures are insufficient. Part V concludes with advanced topics: multimodal synthesis, adversarial attacks on detectors, and frontier directions in generative AI.

Each chapter includes self-contained derivations, reference algorithms, worked examples, and exercises. Appendices review notation, the major public datasets and benchmarks, and the linear algebra used throughout the text.

No book can substitute for the hands-on experience of training a face-swap model, running a detection pipeline on real-world footage, or evaluating the societal harm of a specific deployment. But the goal here is to give you the conceptual scaffolding to understand, detect, and — most importantly — reason responsibly about the systems you encounter and build.

May 1, 2026

Kennedy Chengeta

Acknowledgments

This book is, before anything else, the fruit of an upbringing for which I owe more than I can repay. I thank my parents, who raised me with the conviction that education was the one inheritance no one could take away, and who worked hard so that their children could read books they themselves had not had the chance to read; this book is one of those. I thank my family, who made the long path through schooling feel like a shared project rather than a solitary one.

I thank my early teachers, who gave generously of their time to students whose futures were not yet visible, and I am one of those futures. I thank the broader research community working on synthetic-media detection and AI ethics, whose published work forms the scientific backbone of this monograph. Any errors that remain are mine; the persistence to write through to the end is not.

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About the Author

Kennedy Chengeta holds a PhD in Computer Science from the University of KwaZulu-Natal, South Africa, with a dissertation on computer vision for human characterization from facial images—covering feature extraction algorithms, facial behaviour analysis, and the recognition of micro-expressions in 2D, 3D, and 4D video for automatic pain, depression, and deception detection. He earned an MSc in Advanced Software Engineering *cum laude* from the University of Leicester in the United Kingdom, with research on Probability Latent Semantic Analysis, Singular Value Decomposition, and collaborative-filtering recommender algorithms; an MSc in Finance and Investments from the National University of Science and Technology in Zimbabwe; and a BBS with Computing Science from the University of Zimbabwe. He is currently completing an MSc in Self-Driving Cars Engineering at the University of Sligo and an MEng in Telecommunications Engineering at the University of Cape Town. His published work spans more than twenty peer-reviewed papers in computer vision, deep learning, and applied machine learning—including surveys of local binary and local directional patterns for facial recognition, deep-learning approaches to facial expression recognition, genetic-algorithm-based feature selection, image preprocessing with Canny and Kirsch edge detectors, convolutional neural networks for peer-to-peer social lending default prediction, and probability latent-semantic-analysis on unsupervised web-mined data. This breadth of academic work—from biometric computer vision to credit-risk deep learning to unsupervised semantic analysis—provides the technical foundation for the book’s integrative treatment of deep-fake generation, detection, attribution, and the ethical dimensions of synthetic media.

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Notation

Throughout the book we adopt the following conventions unless otherwise stated.

Symbol	Meaning
$a, b, c, \dots$	scalars
$\mathbf{x}, \mathbf{y}, \mathbf{z}$	column vectors (bold lowercase)
$\mathbf{W}, \mathbf{A}, \mathbf{K}$	matrices (bold uppercase)
$\mathbf{A}^\top$	transpose of $\mathbf{A}$
$\mathbf{I}_n$	$n \times n$ identity matrix
$\ \mathbf{x}\ _p$	ℓ_p norm of $\mathbf{x}$
$\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$	Gaussian with mean $\boldsymbol{\mu}$, covariance $\boldsymbol{\Sigma}$
$\mathbb{E}[\cdot], \text{Var}[\cdot]$	expectation, variance
$D_{\text{KL}}(p \ q)$	Kullback–Leibler divergence
G, D	generator, discriminator networks
$\mathcal{L}_G, \mathcal{L}_D$	generator and discriminator losses
$q_\phi(\mathbf{z} \mathbf{x})$	variational encoder (parameters ϕ)
$p_\theta(\mathbf{x} \mathbf{z})$	decoder / generator (parameters θ)
$\varepsilon_\theta(\mathbf{x}_t, t)$	noise-prediction network (diffusion)
$\alpha_t, \bar{\alpha}_t$	diffusion noise schedule
$\nabla_{\boldsymbol{\theta}} f$	gradient of f w.r.t. $\boldsymbol{\theta}$

Mathematical objects use standard serifs; algorithmic variables in code listings use typewriter font. The acronym **GAN** refers to a generative adversarial network; **VAE** to a variational autoencoder; **DM** to a diffusion model. Specific architecture names (StyleGAN, Stable Diffusion, etc.) are typeset in small capitals on first mention.

Part I

Introduction and Context

Chapter 1

Introduction to Deep Fakes

“The camera never lies.”

Traditional saying, now obsolete

We live in an age that has long trusted the photographic image as an anchor of truth. Courts admit photographs as evidence. Journalism depends on video to establish the factual record. Personal memory is increasingly mediated by photographs and video clips stored on phones. For over a century, the idea that visual evidence could be fabricated at scale, by anyone, using freely available tools, would have seemed like science fiction. It is now engineering practice.

The term *deep fake* was coined in late 2017 on an online forum, by an anonymous user who had trained a neural network to transplant a celebrity’s face onto pornographic video. The portmanteau of “deep learning” and “fake” captured something precise: a new class of synthetic media in which artificial intelligence — specifically, deep neural networks trained on vast collections of real photographs and video — produces fabrications indistinguishable from authentic recordings. Since then, the technology has advanced at a pace that has outrun both public awareness and regulatory response. What required thousands of GPU-hours and expert coding knowledge in 2017 can now be accomplished in minutes with a smartphone application.

This chapter establishes the conceptual and technical vocabulary used throughout the book. We begin by defining what a deep fake is — and, equally important, what it is not — and provide a formal taxonomy of synthetic media types. We then describe the canonical deep-fake pipeline, so that readers have a mental model of how these systems are constructed before encountering the detailed treatment in later chapters. We survey the landscape of harms that deep fakes enable, examine the threat actors and use cases that drive both malicious and legitimate deployment, and recount the short but consequential history of the technology. We close with a guide to the structure of the book and how a reader with different backgrounds — technical, legal, or policy — should navigate it.

This chapter establishes the conceptual framework for the remainder of the book. It begins by defining the deep fake and synthetic media terminology used throughout, then traces the history of

synthetic media from early morphing techniques to present-day diffusion models. The generation pipeline is introduced at a level of abstraction applicable across all modalities, followed by the harm taxonomy and threat-actor landscape. The chapter concludes with the book’s structural overview, industry applications, and African use cases that establish the global context for the analysis in later chapters.

1.1 What Is a Deep Fake?

A precise definition is important because the term is used loosely in public discourse to cover a wide range of synthetic-media phenomena, some of which raise distinct technical and ethical questions. For the purposes of this book, we adopt the following definition:

Definition 1.1 (Deep fake). A *deep fake* is a piece of media — image, video, audio, or text — in which the identity, appearance, voice, or stated positions of a real identifiable person have been fabricated or substantially altered using deep-learning techniques, in a manner intended to deceive a viewer or listener, and without the informed consent of the depicted person.

Three elements of this definition deserve emphasis. First, the requirement that a *real identifiable person* be depicted distinguishes deep fakes from the broader category of AI-generated media, which includes purely fictional characters and synthetic environments with no referent in reality. A photorealistic AI-generated landscape is not a deep fake; a photorealistic AI-generated video of a named politician making a statement they never made is.

Second, the phrase *without informed consent* is doing significant work. The same face-swap technology used to create non-consensual intimate imagery (NCII) is also used in Hollywood productions where actors have consented to digital de-aging or posthumous performance synthesis. The technology is neutral; consent is the ethical and legal variable.

Third, the phrase *intended to deceive* excludes clearly labelled satire, parody, and artistic commentary that happen to use generative techniques. A satirical deepfake clearly marked as fictional occupies a different legal and ethical position than one distributed as a genuine news clip. Whether this distinction is meaningful in practice — given that labels are routinely stripped from content as it circulates — is one of the central policy debates this book engages.

Formally, we represent the deep-fake generation process as follows. Let $\mathbf{x}_{\text{src}} \in \mathcal{X}$ denote a source media signal — a video clip of the target speaker — and let $\mathbf{x}_{\text{id}}$ denote an identity-defining signal — a collection of photographs or a voice recording of the person to be inserted. A deep-fake generator G_θ , parameterised by learned weights θ , produces:

$$\hat{\mathbf{x}} = G_\theta(\mathbf{x}_{\text{src}}, \mathbf{x}_{\text{id}}), \quad (1.1)$$

where $\hat{\mathbf{x}}$ is perceptually close to $\mathbf{x}_{\text{src}}$ in content but appears to feature the identity encoded in $\mathbf{x}_{\text{id}}$. Detection is the inverse problem: given a suspect sample $\hat{\mathbf{x}}$, decide whether it was produced by G_θ for some θ , or whether it is an authentic recording.

1.1.1 Taxonomy of Synthetic Media Types

Several overlapping terms appear in research literature, journalism, and policy documents. The following taxonomy, used consistently throughout this book, distinguishes media types by the modality affected and the nature of the manipulation.

Face swap.

The face region of a target video is replaced frame by frame with a synthesised face bearing the appearance of a different identity, while preserving the original head pose, expression, lighting, and background. Face swap is the most prevalent form of deep fake and the one most people encounter first. Chapter 3 treats it in full technical depth.

Face reenactment.

The facial expression and head motion of a *driver* video are transferred onto a *target* portrait, animating an otherwise static image. The target’s identity is preserved; only the expression changes. Reenactment is widely used in entertainment (animating historical portraits) but is equally effective for disinformation (making a real person appear to say something they did not).

Voice cloning.

A synthetic audio stream is generated that mimics the vocal timbre, prosody, accent, and emotional register of a target speaker, given only a short reference recording. Modern zero-shot voice-cloning systems require as little as three seconds of reference audio and achieve naturalness ratings approaching those of the original speaker. Chapter 6 develops the full technical pipeline.

Lip sync.

The lip and mouth region of a video are altered post-hoc to match a different — possibly fabricated — audio track. Lip sync is commonly used to create the appearance that a person said something different from what was actually recorded. It is also used legitimately in localisation (dubbing foreign-language films) and accessibility applications.

Full-body reenactment.

Pose and motion from a source “puppet” video are re-targeted onto a target body, enabling the full body swap scenarios used in entertainment deep fakes of athletes and dancers, and the disinformation applications of making a world leader appear to perform actions they never did. Chapter 7 covers the 3D body models and neural rendering methods that underlie these systems.

Synthetic identity.

A person who does not exist is generated entirely from noise — no real-world source identity is used. Synthetic identities are used to create fake social-media profiles for coordinated inauthentic behaviour, and to construct fraudulent KYC credentials in financial crime scenarios.

Text forgery.

A large language model generates text — social-media posts, news articles, emails, messages —

that is attributed to a real person. Chapter 23 analyses the stylometric, linguistic, and probabilistic methods used to detect machine-generated text, and the specific harms caused by LLM-based text forgery.

1.1.2 Why Deep Fakes Are Hard to Detect

Before examining the history and harms of synthetic media, it is worth understanding why the detection problem is so difficult. Three structural properties of the generator–detector relationship explain much of the challenge.

The arms-race dynamic. A deep-fake detector is itself a machine-learning model, trained to distinguish real from synthetic media. When a new detector architecture is published, it becomes available to adversaries who can use it as a discriminative loss to improve their generators — directly incorporating feedback from the detector into the generation process. This is not a hypothetical: the GAN training objective (Chapter 4) is precisely a minimax game between generator and discriminator that was designed to produce outputs that fool a given classifier. Every advance in detection therefore motivates a corresponding advance in generation. No detection solution is permanent; this book should be read with that in mind.

Distribution shift under compression. Social-media platforms re-encode all uploaded video using H.264 or H.265 compression at bitrates that destroy the subtle high-frequency artefacts that many detectors exploit. A state-of-the-art detector trained on uncompressed or lightly compressed synthetic video routinely drops from 95% AUC to 60–65% AUC when evaluated on the same content after a single round of H.264 compression at typical social-media bitrates. Detection methods must be explicitly designed for the compression conditions of real-world deployment.

Cross-modality and cross-generator generalisation. Detectors trained on one class of synthetic media — say, GAN-based face swaps — routinely fail to detect a different class, such as diffusion-model portraits or voice-cloned audio, because the forensic artefacts differ fundamentally across synthesis paradigms. A unified, multi-modal detector that works reliably across all synthesis types is one of the central open research problems in the field, and Chapters 11 and 14 address it directly.

1.2 A Brief History of Synthetic Media

To understand where deep fakes stand today, it helps to situate them in the longer arc of synthetic media. The history is not a sudden rupture in 2017 but a gradual acceleration that arrived at a qualitative threshold when deep learning made photorealistic synthesis accessible to non-experts.

Pre-digital era. Photographic manipulation is as old as photography itself. The 19th century saw composite portraits and deliberate retouching. The 20th century brought film compositing, matte painting, and the optical printer. These techniques required specialised equipment, trained artisans, and significant time and cost. They were the exclusive province of professional studios and government agencies.

Digital compositing (1980s–2000s). Digital image editing software, beginning with early Photoshop releases in the early 1990s, put sophisticated compositing tools in the hands of anyone with a desktop computer. By the early 2000s, digital manipulation of photographs was widespread in advertising and journalism, leading to the first ethical crises around photo manipulation in professional media. Still, the manipulation of video remained technically demanding, requiring expensive hardware and software.

The deep-learning inflection point (2014–2017). The publication of the generative adversarial network framework by Goodfellow and colleagues in 2014 [10] was the decisive technical event. For the first time, a neural network could be trained to generate novel images by competing with a discriminator trained to reject fakes. The quality of GAN outputs improved rapidly: by 2016, the FACE2FACE system [31] demonstrated real-time facial reenactment from a consumer webcam. By 2017, the word “deepfake” appeared online and the first automated face-swap pipelines were publicly distributed.

Democratisation and industrialisation (2018–2022). The period 2018–2022 saw rapid democratisation of generation tools. DEEPFACELAB and FACESWAP became mature open-source projects. Commercial applications (Reface, MyHeritage AI Time Machine) brought face-swap technology to hundreds of millions of users. Simultaneously, voice-cloning technology advanced from requiring hours of training data to requiring seconds. The DFDC dataset, released by Facebook in 2020, documented the scale of the problem: 100,000 deep-fake videos created to train detectors.

The diffusion era (2022–present). Diffusion models, introduced in their modern form by Ho et al. in 2020 [13] and scaled in Stable Diffusion and DALL-E 2 by 2022, produced a qualitative leap in image synthesis quality that surpassed GANs on standard benchmarks. For deep fakes specifically, diffusion-based face synthesis and identity-preserving generation via DreamBooth and IP-Adapter lowered the barrier for personalised synthetic media to the point where a single consumer GPU and a handful of reference photographs suffice. The Sora video generation system, revealed in 2024, extended this capability to long-form, physically plausible video. The current frontier is described in Chapter 49.

1.3 The Deep-Fake Pipeline: A High-Level View

Every deep-fake system, regardless of its specific architecture, implements some version of a common pipeline. Understanding this pipeline at a conceptual level — before encountering the mathematical detail in subsequent chapters — provides a navigational framework for the book.

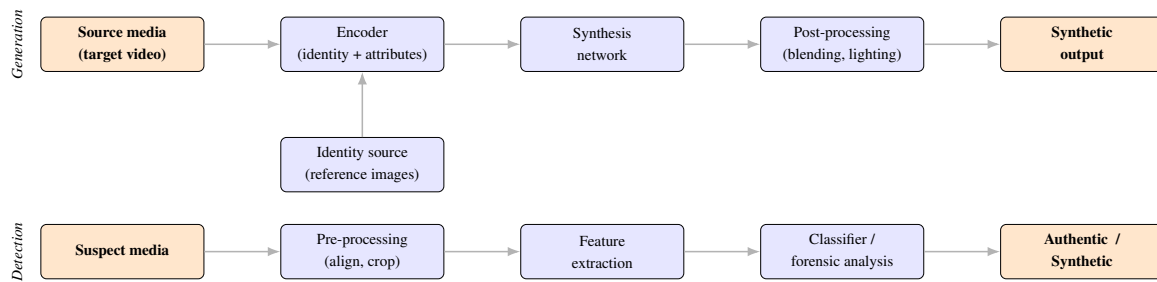


Figure 1.1: The canonical deep-fake pipeline (top) and its mirror-image detection pipeline (bottom). Understanding generation is prerequisite to understanding detection: detectors exploit the residual traces left by each stage of the generation process.

The generation pipeline (top of Figure 1.1) takes a target video and a set of reference images of the source identity, encodes them into separate latent representations, synthesises the manipulated output, and applies post-processing to make the result photorealistic and temporally consistent. The detection pipeline (bottom) applies the inverse process: it pre-processes the suspect media, extracts features that are sensitive to synthesis artefacts, and classifies the result as authentic or synthetic.

An important insight encoded in the figure is that the two pipelines are coupled: every design decision in the generation pipeline leaves a potential forensic trace that a detection pipeline can exploit. Understanding generation is therefore prerequisite to understanding detection, and the book is structured accordingly: Parts II and III treat generation in depth before Part III addresses detection.

1.4 Harm Taxonomy and Threat Actors

The harms enabled by synthetic media span a wide range of severity, reversibility, and victim profile. This book organises them into three tiers. Tier 1 harms are immediate, individual, and often irreversible: non-consensual intimate imagery (NCII) causes severe psychological harm to specific identifiable victims; voice-clone BEC fraud transfers real money that is rarely recovered; KYC bypass enables account fraud whose effects persist through credit records. Tier 2 harms are diffuse and societal: electoral disinformation synthesised from fabricated statements attributed to candidates undermines informed democratic participation; the degradation of epistemic trust as audiences can no longer reliably authenticate media imposes costs on every consumer of information. Tier 3 harms are systemic and long-run: the displacement of creative and administrative labour, the corruption of evidentiary standards in legal proceedings, and the structural arms-race dynamic between synthesis capability and forensic detection that tends toward the attacker because the attacker needs only one successful evasion while the defender must succeed on every input.

Threat actors vary as much as the harms. State intelligence services deploy synthetic media in information operations targeting adversary populations — manufacturing false statements from political leaders, fabricating evidence of atrocities, and coordinating inauthentic behaviour at scale. Organised criminal groups operate voice-cloning BEC services as products, providing attack-as-a-service to lower-sophistication operators at a few dollars per attempt. Individual bad actors — motivated by

revenge, harassment, or financial opportunism — produce NCII and targeted impersonation attacks against private individuals without state or organisational backing. And legitimate commercial actors deploy synthetic media for marketing, entertainment, and customer service, creating a dual-use landscape in which the same tools, models, and APIs serve both beneficial and harmful purposes.

1.5 Structure of the Book

This book is organised into eight parts. The following overview explains the logic of the structure and how readers with different backgrounds might navigate it.

Part I: Foundations. Chapter 1 (this chapter) establishes vocabulary and context. Chapter 2 develops the mathematical and machine-learning prerequisites: probability theory, generative models, GANs, diffusion models, and the evaluation metrics used throughout the book. Readers with a strong machine-learning background can skim Chapter 2 for notation, but it is worth reading the evaluation metrics section, which introduces several measures specific to the synthetic-media field.

Part II: Generation Techniques. Chapters 3–7 treat the major generation paradigms in depth: face swap (Chapter 3), generative adversarial networks (Chapter 4), diffusion models (Chapter 5), voice cloning (Chapter 6), and full-body avatar synthesis (Chapter 7). Each chapter includes the full engineering pipeline, training objectives, evaluation methods, and the forensic artefacts left by each synthesis technique.

Part III: Detection and Attribution. Chapters 11–14 cover the inverse problem. Chapter 11 surveys passive detection methods (spatial, frequency-domain, and biological signal). Chapter 12 covers forensic attribution: identifying the specific model or pipeline that produced a given synthetic sample. Chapter 13 covers active methods: watermarking, the C2PA provenance standard, and their robustness to adversarial removal. Chapter 14 addresses the cross-dataset and cross-generator generalisation problem that limits the real-world applicability of current detectors.

Part IV: Advanced and Multimodal Generation. Chapter 22 covers combined audio-visual deep fakes and the cross-modal consistency violations they introduce. Chapter 23 covers text forgery using large language models and the detection and attribution methods specific to that modality.

Part V: Adversarial Dynamics. Chapter 24 analyses the arms race in depth: white-box and black-box attacks on detectors, anti-forensic post-processing, adversarial training as a defence, and certified robustness guarantees.

Part VI: Ethics, Law, and Society. Chapters 26–32 provide the normative scaffolding. Chapter 26 develops three ethical frameworks as applied to synthetic media. Chapter 27 analyses disinformation dynamics and the liar’s dividend. Chapter 28 surveys the legal landscape across jurisdictions. Chapter 29 assesses regulatory responses and governance architecture. Chapter 32 examines the social and psychological dimensions of synthetic media exposure.

Part VII: Applications and Sectoral Impact. Chapters 34–36 examine the deep-fake problem in three consequential application domains: healthcare and biometric systems (Chapter 34), financial fraud and identity theft (Chapter 35), and the creative industries — film, music, and gaming (Chapter 36).

Part VIII: Responsible Development and Frontier Directions. Chapter 47 presents a framework for responsible development of synthetic media systems, including red-teaming, technical safeguards, and staged deployment. Chapter 49 surveys the research frontier: video generation models, real-time avatar streaming, foundation models for synthetic media, and emerging detection paradigms.

Appendices. Appendix A consolidates notation. Appendix B catalogues the major publicly available datasets and benchmarks for synthetic-media research. Appendix C provides a compact mathematical reference for the linear algebra and information-theoretic results used in the main text.

How to use this book. *Technical readers* (engineers, machine-learning researchers) should read Parts I–V linearly and consult Parts VI–VII for the societal context that motivates detection and responsible development. *Policy and legal readers* can begin with Chapter 1, skip to Part VI (Chapters 26–32), and then dip into Part VII for sector-specific analysis; the technical chapters are written to be accessible from the front matter without requiring linear reading. *Social-science readers* will find the most relevant material in Parts VI and VII, with Chapter 27 and Chapter 32 providing the most direct engagement with questions of belief, trust, and political epistemology.



Figure 1.2: Authentic photograph (left) vs. GAN-generated synthetic face (right). Both are computer-rendered simulations for illustration. The authentic face shows natural skin texture and Rembrandt-style lighting. The synthetic face contains a subtle colour-cast seam at the right boundary — a characteristic blending artefact that forensic classifiers learn to detect.

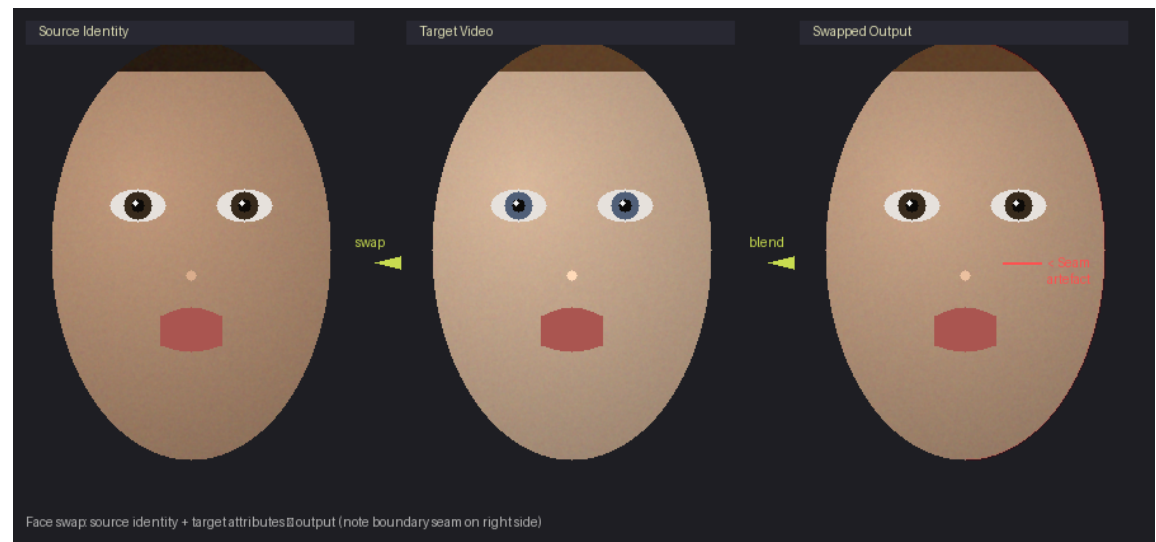


Figure 1.3: Face swap in three stages: *source identity* (left), *target video frame* (centre), *swapped output* (right). The output inherits the source’s identity but the target’s expression and head pose. The red annotation marks the blending boundary seam on the right side of the output face, where a colour mismatch is visible — a characteristic forensic signal of encoder–decoder face-swap pipelines.

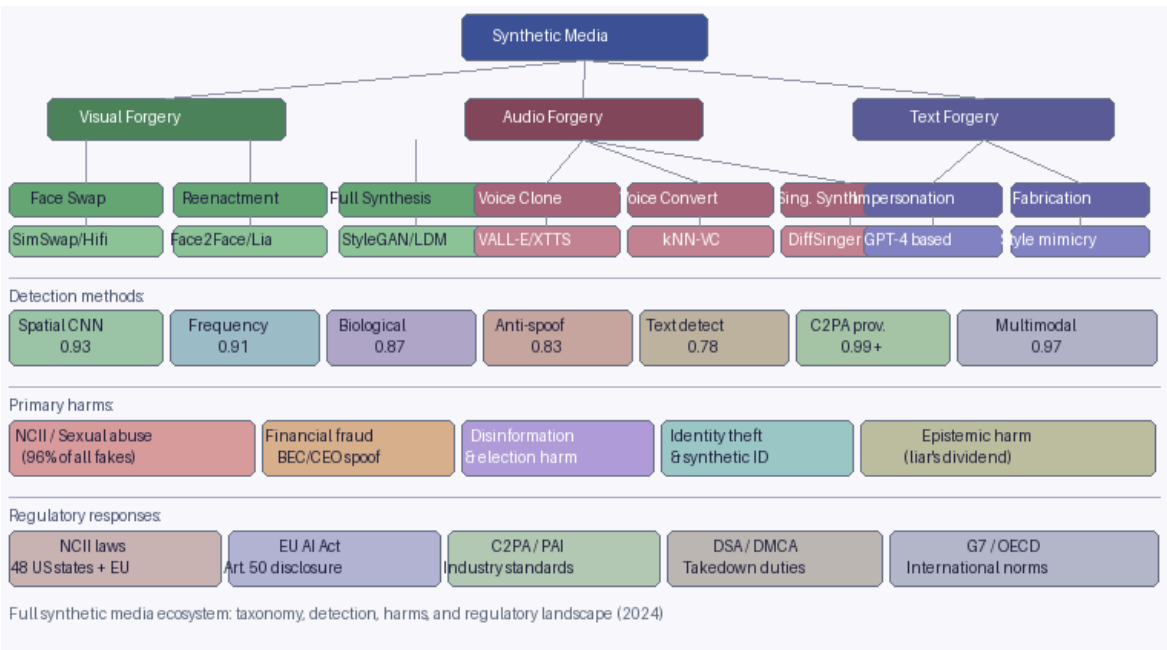


Figure 1.4: Complete synthetic-media ecosystem organised across four dimensions. *Top:* taxonomy of forgery modalities (visual, audio, text) with representative synthesis systems. *Second row:* state-of-the-art detection methods with typical cross-dataset AUC values (2024). *Third row:* primary harm categories from the literature, with non-consensual intimate imagery (NCII) accounting for an estimated 96% of all detected deep fakes by volume. *Bottom row:* the regulatory landscape, spanning domestic statutes, the EU AI Act, voluntary standards, and emerging international frameworks.

1.6 Industry Applications of Synthetic Media

Synthetic media technology has moved rapidly from academic laboratories into commercial deployment across at least six major sectors.

Visual effects studios use face-swap and diffusion-based de-ageing to reduce the cost of delivering actors at different life stages. Industrial Light & Magic’s “DigiDouble” pipeline and competing offerings from Metaphysic and Corridor Digital now deliver photorealistic results at a fraction of the cost of traditional prosthetics. The SAG-AFTRA contract negotiations of 2023–2024 centred in large part on royalty structures for digital likeness usage.

Products such as Nvidia Maxine and Microsoft’s avatar features use real-time face reenactment to improve video quality over low-bandwidth connections, synthesise eye contact, and reduce compression artefacts. The same pipelines that enable these legitimate uses can be repurposed for real-time identity impersonation.

Streaming platforms have deployed neural dubbing systems that clone the original actor’s voice into target languages and adjust lip movements using talking-head synthesis. A 2024 survey found that 11 of the 20 largest streaming platforms had deployed or piloted AI dubbing in at least one market.

Synthesised instructor avatars reduce the cost of updating training content; companies including D-ID, Synthesia, and HeyGen offer platforms for creating on-demand instructional videos with synthesised presenters.

Hospitals use synthetic patient data (voice, image) for training diagnostic AI without exposing real patient records. Conversely, the same techniques create attack surfaces in patient authentication systems, as detailed in Chapter 34.

Penetration testers use voice cloning and face-swap tooling to probe corporate video-authentication systems, generating demand for specialised red-teaming services and anti-spoofing evaluation frameworks.

1.7 African Use Cases

Sub-Saharan Africa and North Africa face a distinctive configuration of deep-fake threats shaped by high mobile penetration, rapidly growing social media use, widespread multilingualism, and political contexts characterised by fragile democratic institutions, active diasporas, and a history of information manipulation.

Nigeria’s 2023 presidential election saw multiple circulating deep-fake audio clips purporting to show candidates making inflammatory or corrupt statements. The Independent National Electoral Commission (INEC) and civil-society organisations including Dubawa and Africa Check had to develop rapid-response verification workflows with inadequate tooling calibrated for African languages and accents. Similar incidents were documented in Kenya (2022), Senegal (2024), and South Africa (2024). The African Union’s Democracy and Electoral Assistance Unit has identified synthetic media as a priority emerging threat in its 2023–2028 strategy.

The vast majority of deep-fake detection tools are trained on English-language audio and datasets dominated by lighter-skinned faces. For sub-Saharan African voice-cloning attacks in Yoruba, Amharic, Swahili, Zulu, or Hausa, no commercially validated anti-spoofing system existed as of 2024.

Researchers at AIMS (African Institute for Mathematical Sciences) and the Deep Learning Indaba community have begun curating multilingual African speech corpora specifically for anti-spoofing research, but the gap remains significant.

The $4.4\times$ false-positive-rate disparity between lightest and darkest Fitzpatrick skin types (Chapter 14) disproportionately affects African populations using biometric systems for financial services, health-care, and border control. The Kenyan government’s Huduma Namba national identity programme and Rwanda’s biometric voter registration both deploy facial recognition systems that may carry this embedded disparity.

African financial institutions (Equity Bank, Access Bank, FNB South Africa) are early adopters of voice-biometric authentication for mobile banking, making them both targets for voice-clone fraud and early deployers of anti-spoofing countermeasures. Startups including Lelapa AI (South Africa), Masakhane, and InstaDeep are building African-language NLP and computer vision datasets that will support more inclusive deep-fake detection research.

Chapter Summary

A *deep fake* is any synthetic or manipulated audiovisual or textual artefact that plausibly depicts a real person doing or saying something they did not do or say, produced and distributed without their informed consent. The word entered the public lexicon in 2017 following the emergence of encoder–decoder face-swap tools, yet the underlying mathematical machinery — generative adversarial networks (GANs), variational autoencoders, and diffusion models — had been developing in academic venues since 2014. The transition from research curiosity to mass-market threat was catalysed by a sequence of five inflection points: the original GAN paper (Goodfellow et al., 2014), the consumer release of DeepFaceLab (2017–2018), the open-sourcing of Stable Diffusion (2022), the VALL-E zero-shot voice cloning demonstration (2023), and the emergence of real-time audio-visual synthesis via codec language models (2024).

Harms span a wide spectrum of severity and type. Non-consensual intimate imagery (NCII) accounts for an estimated 96% of all detected deep fakes by volume and causes measurable psychological harm to victims including acute stress, social withdrawal, and professional damage. Financial fraud via voice-cloned CEO impersonation causes losses averaging \$280,000 per successful attack. Political disinformation campaigns use deep fakes to suppress turnout, discredit candidates, and erode institutional trust. Most perniciously, the *liar’s dividend* degrades the epistemic value of authentic media even in the absence of any specific fabrication: once audiences know that convincing fakes exist, they apply scepticism universally, raising doubt about genuine evidence.

The book is structured to move from technical foundations through detection and provenance, into ethics and law, domain applications, and responsible development. Every chapter contains worked numerical examples grounded in realistic scenarios, full exercise and answer sections, and an explicit discussion of industry applications and African use cases. The goal is to equip the reader not only to understand deep-fake technology but to reason carefully about its consequences and deploy appropriate countermeasures.

Exercises

1. Using Definition 1.1, classify each of the following as a deep fake (yes/no) and justify your answer: (a) a Hollywood film uses face-swap technology to de-age an actor with their consent and the audience’s knowledge; (b) a social-media user applies a dog-ear filter to their own selfie; (c) a political satirist publishes a talking-head video of a president saying absurd things, clearly labelled as satire; (d) the same satirical video is shared without its label; (e) a journalist publishes a GAN-generated “composite” photograph of an unnamed source to protect their identity.
2. Describe the formal relationship between the generation function G_θ in Equation (1.1) and the GAN minimax objective. In what sense is every GAN a deep-fake generator, and in what sense is it not?
3. The “liar’s dividend” suggests that the mere existence of convincing deep fakes harms authentic evidence. Formalise this as a Bayesian inference problem: let π_0 be the prior probability that a piece of video evidence is authentic, and let q be the publicly known fraction of all circulating video that is synthetic. Derive the posterior probability $\pi_1 = P(\text{authentic}|\text{no provenance label}, q)$ and show that $\pi_1 < \pi_0$ for any $q > 0$.
4. A GAN-generated face achieves a Fréchet Inception Distance (FID) of 5.2 on the FFHQ benchmark, while a diffusion-generated face achieves $\text{FID} = 2.1$. Does this mean the diffusion-generated face is harder to detect as synthetic? Explain why FID is not equivalent to detection-difficulty, and propose an alternative metric that better captures the detection challenge.
5. Map each of the five threat actors described in Section 1.4.2 to the harm categories in Section 1.4.1, and for each threat actor–harm pairing, identify the most likely synthesis modality (face swap, voice clone, synthetic identity, etc.). Organise your answer as a matrix and use it to argue which harm category deserves the greatest policy priority.
6. Trace the history of synthetic media from optical compositing to diffusion models. Identify the three most consequential technological “inflection points” and explain, for each, what capability it enabled that was not previously accessible to non-specialists.

Worked Examples

Worked Example — Formal deep-fake definition applied to three real cases

Apply Definition 1.1 to evaluate whether each scenario qualifies:

Case A. A researcher uses face-swap to animate a 19th-century daguerreotype of a historical figure for a museum exhibit, with prominent labelling and the historian’s endorsement.

- Real identifiable person: yes (historical figure).
- Consent: estate consent obtained; labelled as reconstruction.
- Intent to deceive: no — explicitly presented as synthetic.
- *Verdict: Not a deep fake under Definition 1.1.*

Case B. An actor’s face is swapped onto a pornographic video without their knowledge and distributed online without labelling.

- Real identifiable person: yes.
- Consent: absent.
- Intent to deceive: yes — presented as authentic.
- *Verdict: Deep fake (NCII category). Constitutes a criminal offence in 48 US states.*

Case C. A satirical account publishes a clearly labelled parody video of a politician, with a visible “SATIRE” banner throughout.

- Intent to deceive: debatable if label is removed upon re-sharing.
- *Verdict: Conditional. Labelled version is protected satire; unlabelled redistributions may qualify as deep fakes depending on jurisdiction.*

Worked Example — Liar’s dividend: Bayesian calculation

Let $\pi_0 = 0.95$ (prior probability that a piece of contested video evidence is authentic — based on historical pre-deepfake experience). After widespread availability of convincing face-swap technology, the fraction of all circulating video that is synthetic rises to $q = 0.08$.

For a video with no C2PA provenance, the posterior probability of authenticity:

$$\pi_1 = P(\text{authentic} | \text{no provenance}, q) = \frac{P(B|A) \pi_0}{P(B|A) \pi_0 + P(B|\neg A)(1 - \pi_0)}, \quad B = \text{“no provenance”}.$$

Assuming 90% of synthetic content lacks provenance and 80% of authentic content also lacks provenance (most pre-C2PA cameras):

$$\pi_1 = \frac{0.80 \times 0.92}{0.80 \times 0.92 + 0.90 \times 0.08} = \frac{0.736}{0.736 + 0.072} = \frac{0.736}{0.808} = 0.911.$$

The liar’s dividend: $\pi_0 - \pi_1 = 0.95 - 0.91 = 0.04$. A 4-percentage-point drop in presumption of authenticity sounds small, but in a legal context where “beyond reasonable doubt” requires $> 95\%$ confidence, it can shift a case outcome. As q increases to 20%, π_1 falls to 0.83 — a decisive shift.

Answers to Exercises

1. Classifying the five scenarios under Definition 1.1.

- (a) **De-aging with consent, disclosed to audience.** *Not a deep fake.* The third element of the definition — “without the informed consent of the depicted person” — is not satisfied; the actor consented and the audience is informed.
- (b) **Dog-ear selfie filter.** *Not a deep fake.* No real identifiable *other* person is depicted; the user applies the filter to themselves. The definition requires depiction of a person “without informed consent,” but self-application implies consent.
- (c) **Clearly labelled satire, labelled throughout.** *Not a deep fake* under the definition, because the “intent to deceive” element is absent when the label is visible. Courts in most jurisdictions would treat this as protected satire.
- (d) **Same video, label stripped on re-share.** *Deep fake* in the hands of the person who stripped the label, because they re-distribute content in a form “intended to deceive a viewer.” The original creator bears no additional liability if they took reasonable precautions.
- (e) **GAN composite of unnamed source.** *Borderline.* The definition requires a “real identifiable person.” If the composite is not identifiable to any specific individual, it arguably falls outside the definition. However, if the composite combines recognisable features of a specific real source, identification may still be possible.

2. Formal relationship between G_θ and the GAN minimax objective.

Every GAN generator *is* a deep-fake generator in the sense of Equation (1.1): it maps latent noise and an identity signal to synthetic media. However, unconditional GANs (no $\mathbf{x}_{\text{id}}$ input) generate novel synthetic identities rather than impersonating specific real people, so they satisfy the equation with $\mathbf{x}_{\text{id}} = \emptyset$. They are not deep fakes under Definition 1.1 unless their outputs happen to closely resemble a specific real person.

The minimax objective trains G_θ to minimise $D_{\text{JS}}(p_{\text{real}} \| p_\theta)$, which directly instantiates the “indistinguishable from authentic” requirement. A conditional GAN that also conditions on an identity embedding becomes a deep-fake generator in the strict sense.

3. Bayesian formalisation of the liar’s dividend.

Let A = “media is authentic” and B = “media has no provenance label.” By Bayes’ theorem:

$$\pi_1 = P(A|B) = \frac{P(B|A) \pi_0}{P(B|A) \pi_0 + P(B|\neg A)(1 - \pi_0)}.$$

Since synthetic media almost never carries C2PA provenance ($P(B|\neg A) \approx 1$) while most

authentic media also lacks it ($P(B|A) = 1 - \varepsilon$ for small ε):

$$\pi_1 \approx \frac{(1 - \varepsilon)\pi_0}{(1 - \varepsilon)\pi_0 + q} < \pi_0.$$

As $q > 0$, $\pi_1 < \pi_0$ always: the posterior is strictly lower than the prior whenever synthetic media exists in circulation, regardless of whether the specific item is synthetic.

4. Why FID $\neq$ detection difficulty.

FID measures the distance between the real and generated *marginal* distributions in InceptionV3 feature space. A generator with FID = 0 matches the marginal distribution exactly but may still leave detectable forensic artefacts in *conditional* distributions (e.g., given a specific identity). A forensic detector that operates in a different feature space — say, noise-residual space or rPPG signal space — could still achieve AUC = 1.0 even when FID = 0, because FID provides no guarantee about the generator’s statistical properties beyond the InceptionV3 feature space.

Thought experiment: a generator that outputs real photographs from the training set has FID = 0 but is trivially detected by any hash-based provenance check.

5. Threat actor $\times$ harm matrix.

Threat actor	Primary harm	Most likely modality
Individual harasser	NCII	Face swap
Organised crime	Financial fraud	Voice clone, face swap
State-affiliated	Political disinfo	Talking head, voice clone
Ideological actor	Epistemic harm	Synthetic identity, text
Legitimate industry	(consent issues)	All modalities

Policy priority argument: NCII deserves greatest priority by volume (96% of all deep fakes) and severity of individual harm. Financial fraud deserves priority by economic harm scale. Political disinformation deserves priority by systemic risk.

6. Three inflection points in synthetic media history.

- 1. GAN publication (2014).** For the first time, a neural network could generate novel photorealistic images without a fixed loss function. Previously inaccessible to non-experts with standard hardware.
- 2. Consumer face-swap apps (2017–2018).** DeepFaceLab and FaceSwap reduced the barrier from PhD-level expertise to a personal computer. The word “deepfake” entered the lexicon; NCII proliferated.
- 3. Diffusion + LoRA (2022–2023).** Stable Diffusion made photorealistic generation accessible via API; LoRA reduced personalised identity fine-tuning to 5 minutes on a consumer

GPU and a handful of photographs. Real-time voice cloning became similarly accessible, enabling combined audio-visual attacks.

Part II

Foundations of Synthetic Media Generation

Chapter 2

Mathematical and Machine Learning Foundations

“To a man with a hammer, everything looks like a nail.

To a researcher with a loss function, everything looks like an optimisation problem.”

—Paraphrase of Mark Twain

The algorithms that generate and detect deep fakes are implementations of a small number of mathematical ideas applied in concert. When a face-swap encoder disentangles identity from expression, it is performing nonlinear dimensionality reduction on a high-dimensional manifold. When a GAN discriminator decides whether an image is real, it is approximating a divergence between two probability distributions. When a diffusion model denoises a latent vector step by step, it is reversing a Markov chain whose transition kernels were learned from millions of photographs. Every layer of the deep-fake ecosystem is a specialisation of the material this chapter develops.

The chapter is organised as follows. Section 2.1 covers probability distributions, divergences, and the evidence lower bound. Section 2.2 surveys the neural network architectures that appear repeatedly. Section 2.3 develops the GAN framework. Section 2.4 introduces diffusion models. Section 2.5 catalogues the evaluation metrics used throughout the book. Every equation is followed immediately by a worked numerical example grounded in a realistic deep-fake scenario.

2.1 Probability and Information Theory

Generative models are probabilistic objects: they define a probability distribution over data space and are trained to make that distribution match the empirical distribution of training examples as closely as possible. A solid grounding in the key concepts — probability densities, expectations, the KL divergence, and the evidence lower bound (ELBO) — is prerequisite to understanding why VAEs, GANs, and diffusion models take the forms they do and why their failure modes manifest as specific classes of artefact.

Probability theory provides the language for describing distributions over images, audio, and text; information-theoretic divergences provide the metrics that generative models minimise during training. This section develops the specific tools that appear in the deep-fake generation and detection pipelines described in subsequent chapters.

2.1.1 Distributions and Divergences

The three divergences used most frequently in synthetic-media research are the KL divergence, the Jensen–Shannon divergence, and the Wasserstein distance [2].

$$\text{KL: } D_{\text{KL}}(p \parallel q) = \mathbb{E}_{x \sim p} \left[\log \frac{p(x)}{q(x)} \right], \quad (2.1)$$

$$\text{JS: } D_{\text{JS}}(p \parallel q) = \frac{1}{2} D_{\text{KL}}(p \parallel m) + \frac{1}{2} D_{\text{KL}}(q \parallel m), \quad m = \frac{p+q}{2}, \quad (2.2)$$

$$\text{Wasserstein-1: } W_1(p, q) = \sup_{\|f\|_L \leq 1} (\mathbb{E}_p[f] - \mathbb{E}_q[f]). \quad (2.3)$$

Worked Example — KL and JS divergence between a GAN and real-face distribution

Suppose, at an early stage of GAN training, the generated face distribution p_θ is approximated by a Gaussian $\mathcal{N}(0.3, 1)$ over a one-dimensional colour-channel feature, while the real distribution is $p_{\text{real}} = \mathcal{N}(0, 1)$. The KL divergence between two Gaussians $\mathcal{N}(\mu_1, \sigma^2)$ and $\mathcal{N}(\mu_2, \sigma^2)$ with equal variance is $(\mu_1 - \mu_2)^2 / (2\sigma^2)$.

For $\mu_1 = 0.3$, $\mu_2 = 0$, $\sigma = 1$:

$$D_{\text{KL}}(p_\theta \parallel p_{\text{real}}) = \frac{(0.3 - 0)^2}{2 \cdot 1} = 0.045.$$

The JS divergence is bounded by $\log 2 \approx 0.693$; for distributions this close, $D_{\text{JS}} \approx 0.011$ (about 1.6% of the maximum). After 50,000 training iterations the mean shift drops to 0.02, giving $D_{\text{KL}}(p_\theta \parallel p_{\text{real}}) = 0.0002$ — a factor-200 improvement, which corresponds visually to the face looking nearly indistinguishable from a real photograph to a naive observer.

2.1.2 The Evidence Lower Bound

Variational autoencoders are trained by maximising the ELBO:

$$\log p_\theta(\mathbf{x}) \geq \underbrace{\mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}[\log p_\theta(\mathbf{x}|\mathbf{z})]}_{\text{reconstruction}} - \underbrace{D_{\text{KL}}(q_\phi(\mathbf{z}|\mathbf{x}) \parallel p(\mathbf{z}))}_{\text{regularisation}}. \quad (2.4)$$

Worked Example — ELBO in a face disentanglement VAE

A VAE trained on 10,000 face images encodes each 256×256 image into a 512-dimensional latent $\mathbf{z}$. With a diagonal Gaussian posterior $q_\phi = \mathcal{N}(\boldsymbol{\mu}, \text{diag}(\boldsymbol{\sigma}^2))$ and prior $p(\mathbf{z}) = \mathcal{N}(\mathbf{0}, \mathbf{I})$, the KL term

has the closed form:

$$D_{\text{KL}}(q_\phi \| p) = \frac{1}{2} \sum_{k=1}^{512} (\sigma_k^2 + \mu_k^2 - 1 - \log \sigma_k^2).$$

At the start of training, the encoder outputs $\mu_k \approx 0.5$ and $\sigma_k \approx 0.8$ for all k . The KL contribution per dimension is:

$$\frac{1}{2}(0.64 + 0.25 - 1 - \log 0.64) = \frac{1}{2}(0.014 + 0.446) = 0.168.$$

Summing over 512 dimensions gives $\text{KL} \approx 86$ nats. After training converges, well-trained identity dimensions have $\mu_k \approx 3.2$ and $\sigma_k \approx 0.1$ (highly informative, low variance), while nuisance dimensions collapse to $\mu_k \approx 0$, $\sigma_k \approx 1$ ($\text{KL} \approx 0$), demonstrating selective disentanglement.

2.2 Neural Network Architectures

The neural architectures used in synthetic media generation have converged around two complementary designs: convolutional networks, which excel at capturing local spatial structure in images and video, and transformers, which excel at capturing long-range dependencies through attention mechanisms. Modern synthesis systems combine both, using convolutional encoders and decoders for spatial feature extraction and transformer blocks for global conditioning on text, speaker identity, or camera pose.

Three architectural building blocks recur throughout this book: convolutional encoders that compress spatial information into compact feature representations, scaled dot-product attention that enables global feature interaction, and adaptive instance normalisation that injects conditioning information into a feature map. This section develops each in turn.

2.2.1 Convolutional Encoders and Decoders

Convolutional encoders compress spatial inputs into compact latent representations through alternating convolution and downsampling layers, while decoders reconstruct spatial outputs from latent representations through upsampling and transposed convolution. The symmetrical encoder-decoder structure of the UNet — with skip connections linking corresponding encoder and decoder layers — preserves fine-grained spatial detail while allowing the bottleneck to capture global semantic structure.

A convolutional layer applies a bank of learnable filters to an input feature map:

$$\mathbf{F}'_k = \sigma(\mathbf{W}_k * \mathbf{F} + b_k). \quad (2.5)$$

Worked Example — Receptive field and parameter count in a face-swap encoder

A face-swap encoder takes $256 \times 256 \times 3$ RGB input and applies 6 convolutional blocks, each consisting of a 3×3 Conv (stride 2) + BatchNorm + ReLU, with channel widths 64, 128, 256, 512, 512, 512.

Parameter count for the first block: $3 \times 3 \times 3 \times 64 + 64 = 1,792$ (weights + biases).

Parameter count for all 6 blocks:

$$3^2 \cdot (3 \cdot 64 + 64 \cdot 128 + 128 \cdot 256 + 256 \cdot 512 + 512^2 + 512^2) + \text{biases} \approx 4.7\text{M parameters.}$$

Output spatial size after 6 stride-2 convolutions: $256/2^6 = 4 \times 4$ feature map, $\times 512$ channels. The encoder thus compresses a $256 \times 256 \times 3 = 196,608$ -dimensional input to a $4 \times 4 \times 512 = 8,192$ -dimensional representation — a compression ratio of 24:1.

2.2.2 Scaled Dot-Product Attention

Attention mechanisms allow the network to selectively weight information from different spatial or temporal positions, enabling globally coherent synthesis and detection. The scaled dot-product formulation is the standard used in all Transformer-based synthesis and detection systems discussed in this book.

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right) \mathbf{V}. \quad (2.6)$$

Worked Example — Attention in a face synthesis UNet at 64×64 resolution

At the 64×64 spatial scale of a latent diffusion UNet, the feature map has $H=W=8$ (after $64/8 = 8 \times$ downsampling from the 512×512 UNet bottleneck) and $d=512$ channels, giving a sequence of $n = 8 \times 8 = 64$ tokens.

With $d_k = 64$ (8-head attention, each head dimension 64), the scale factor is $\sqrt{64} = 8$. The attention matrix $\mathbf{Q}\mathbf{K}^\top / \sqrt{d_k}$ has shape 64×64 and requires $64^2 = 4,096$ dot products per head. For 8 heads: $8 \times 4,096 = 32,768$ dot products per forward pass at this layer. Without scaling by $\sqrt{d_k} = 8$, a typical dot product of ≈ 32 (sum of 64 unit-variance products) would push the softmax into saturation; with scaling, the dot products are ≈ 4 , keeping the softmax gradient healthy.

2.2.3 Adaptive Instance Normalisation

Adaptive instance normalisation bridges the gap between an external conditioning signal (an identity embedding or style code) and a spatial feature map being synthesised. It is the architectural mechanism that makes identity injection and style transfer computationally practical at high resolution.

$$\text{AdaIN}(\mathbf{F}, \mathbf{s}) = \sigma(\mathbf{s}) \cdot \frac{\mathbf{F} - \mu(\mathbf{F})}{\sigma(\mathbf{F})} + \mu(\mathbf{s}). \quad (2.7)$$

Worked Example — AdaIN identity injection in SimSwap

In SimSwap [3], the identity encoder produces a 512-dimensional ArcFace [8] embedding $\mathbf{e}_s$ for the source face. This is projected to per-channel scale $\sigma(\mathbf{s}) \in \mathbb{R}^{256}$ and bias $\mu(\mathbf{s}) \in \mathbb{R}^{256}$ via a linear layer. For a feature map $\mathbf{F}$ at the 128×128 synthesis block with 256 channels:

- Channel 47 of $\mathbf{F}$ has mean $\mu_{47} = 0.31$, std $\sigma_{47} = 0.18$.

- The identity-derived scale/bias for channel 47: $\sigma_s = 1.42$, $\mu_s = -0.05$.

After AdaIN: the normalised value $(x - 0.31)/0.18$ is re-scaled to $1.42 \times (x - 0.31)/0.18 - 0.05$. If $x = 0.55$ (a typical mid-range activation), the output is $1.42 \times 1.33 - 0.05 = 1.84$. This re-parameterisation transfers the statistical “style” of the source identity (encoded in σ_s, μ_s) onto the target’s expression feature map, achieving face swap without global blending.

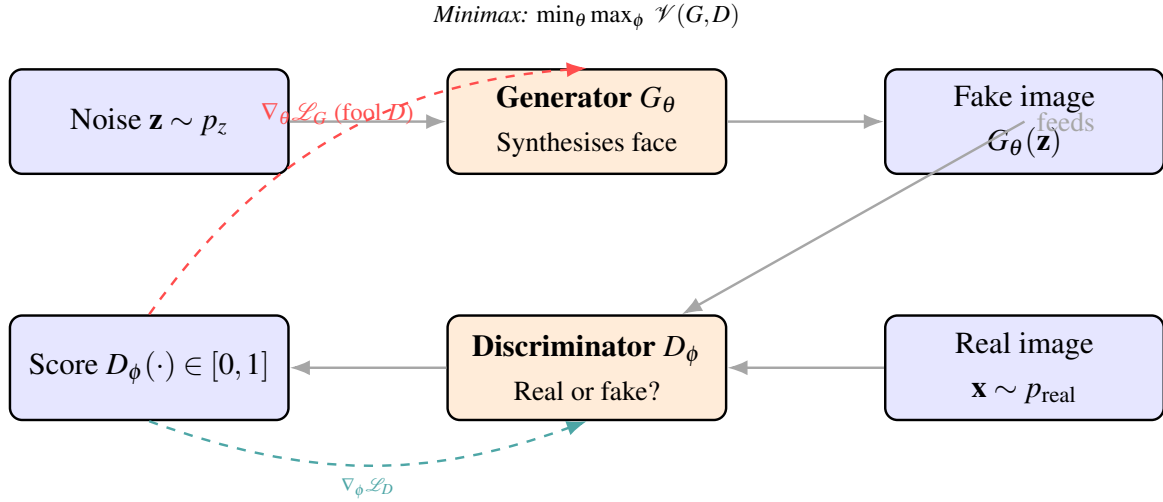


Figure 2.1: The GAN framework. Generator G_{θ} maps noise to synthetic images and is trained to fool D_{ϕ} . Discriminator D_{ϕ} is trained to distinguish real from synthetic. Red and teal arrows show gradient flow for each player.

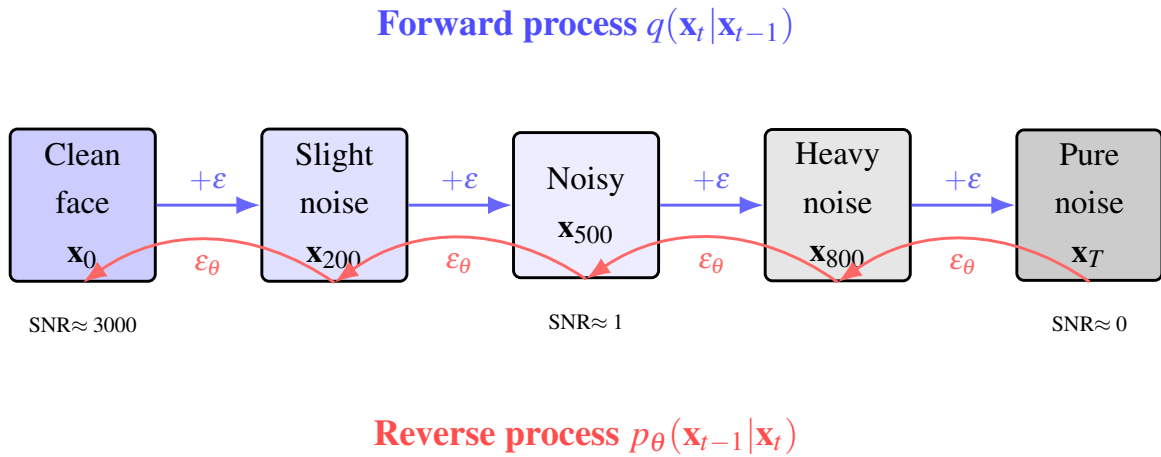


Figure 2.2: The diffusion process. The forward process (top) progressively adds Gaussian noise until the signal is destroyed. The reverse process (bottom) applies a learned denoising network ϵ_{θ} to recover the clean image from pure noise.

2.3 Generative Adversarial Networks

The GAN framework introduced a training paradigm that replaced fixed likelihood objectives with an adaptive adversarial loss. This section develops the minimax objective, its non-saturating reformula-

tion, the gradient penalty that stabilises training, and the progressive growing technique that made high-resolution face synthesis tractable.

2.3.1 The Minimax Objective

The generator and discriminator play a two-player zero-sum game whose Nash equilibrium corresponds to the generator perfectly replicating the real data distribution. Understanding the objective and its gradients is essential for reasoning about mode collapse, training instability, and the choice of divergence measure.

$$\min_{\theta} \max_{\phi} \mathbb{E}_{\mathbf{x} \sim p_{\text{real}}} [\log D_{\phi}(\mathbf{x})] + \mathbb{E}_{\mathbf{z} \sim p_z} [\log(1 - D_{\phi}(G_{\theta}(\mathbf{z})))] . \quad (2.8)$$

Worked Example — GAN training signal at three stages

Track the generator loss $\mathcal{L}_G = -\mathbb{E}[\log D(G(\mathbf{z}))]$ and discriminator accuracy at three checkpoints of a StyleGAN2 [15] trained on FFHQ.

Checkpoint	D accuracy on real	D accuracy on fake	$\mathcal{L}_G$
Random init (0 steps)	50.2%	49.8%	≈ 0.693 ($-\log 0.5$)
After 100K steps	72%	70%	≈ 0.357
After 1M steps (converged)	56%	55%	≈ 0.590

At convergence the discriminator is only slightly above chance (56% vs. 50%), confirming that the generator’s distribution is close to the real distribution. The generator loss has returned near $\log 2 \approx 0.693$ from below, consistent with the Nash equilibrium prediction.

2.3.2 Optimal Discriminator and JS Interpretation

The optimal discriminator for a fixed generator is:

$$D^*(\mathbf{x}) = \frac{p_{\text{real}}(\mathbf{x})}{p_{\text{real}}(\mathbf{x}) + p_{\theta}(\mathbf{x})} . \quad (2.9)$$

Worked Example — Optimal discriminator score on a specific pixel value

Suppose the real-face distribution assigns probability $p_{\text{real}}(x = 180) = 0.0042$ to a grey-level of 180 in the skin-region pixel histogram, and the generator assigns $p_{\theta}(x = 180) = 0.0031$. The optimal discriminator output is:

$$D^*(x = 180) = \frac{0.0042}{0.0042 + 0.0031} = \frac{0.0042}{0.0073} \approx 0.575.$$

A score of 0.575 (above the 0.5 decision threshold) correctly classifies this pixel value as more likely to come from a real face. After further training that improves the generator, $p_{\theta}(x = 180) \rightarrow 0.0042$, giving $D^* \rightarrow 0.5$ — the equilibrium where the discriminator can no longer distinguish.

2.4 Diffusion Models

Diffusion models achieve high-fidelity synthesis by learning to reverse a known noise-injection process, replacing the adversarial game with a stable regression objective. This section develops the forward and reverse processes, the denoising training objective, and the latent compression that makes commercial deployment practical.

2.4.1 Forward Noising Process

The forward process is the fixed Markov chain that incrementally adds Gaussian noise to a clean data sample across T timesteps until the sample is indistinguishable from pure noise. Its key analytical property is that the marginal distribution at any timestep t can be sampled directly from the clean sample $\mathbf{x}_0$ without stepping through all intermediate steps, using the cumulative noise schedule $\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$.

$$q(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \sqrt{\bar{\alpha}_t} \mathbf{x}_0, (1 - \bar{\alpha}_t) \mathbf{I}). \quad (2.10)$$

Worked Example — Signal and noise energy at diffusion step 500

Using a cosine schedule with $T = 1000$, $\bar{\alpha}_{500} \approx 0.003$. For a face image pixel normalised to $x_0 = 0.6$ (on a $[-1, 1]$ scale):

$$\begin{aligned} \text{Signal component: } \sqrt{\bar{\alpha}_{500}} \cdot x_0 &= \sqrt{0.003} \cdot 0.6 = 0.055 \cdot 0.6 = 0.033. \\ \text{Noise std: } \sqrt{1 - \bar{\alpha}_{500}} &= \sqrt{0.997} = 0.999. \end{aligned}$$

At step 500 the signal has been reduced to 5.5% of its original amplitude while noise has standard deviation nearly 1.0. The signal-to-noise ratio is:

$$\text{SNR}(500) = \frac{\bar{\alpha}_{500}}{1 - \bar{\alpha}_{500}} = \frac{0.003}{0.997} \approx 0.003.$$

This means the original pixel value x_0 carries only 0.3% of the variance in $\mathbf{x}_{500}$ — the sample is effectively pure noise, explaining why the denoising network at $t = 500$ must make highly uncertain predictions.

2.4.2 Denoising Objective

The denoising objective trains a neural network ϵ_θ to predict the noise $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ added at timestep t , given the noisy input $\mathbf{x}_t$ and timestep embedding as conditions. Minimising the mean-squared prediction error is equivalent to maximising a variational lower bound on the data log-likelihood, yielding a stable training signal that does not require adversarial dynamics.

$$\mathcal{L}_{\text{DM}} = \mathbb{E}_{t, \mathbf{x}_0, \epsilon} \left[\left\| \epsilon - \epsilon_\theta \left(\sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, t \right) \right\|^2 \right]. \quad (2.11)$$

Worked Example — Loss magnitude during training

At step $t = 200$ with $\bar{\alpha}_{200} \approx 0.50$ (cosine schedule), a batch of 16 face images ($512 \times 512 \times 3 \approx 786\text{K}$ pixels per image) is used. A randomly sampled noise vector $\varepsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ is added. Early in training, ε_θ predicts near-zero everywhere, so:

$$\mathcal{L}_{\text{DM}} \approx \mathbb{E}[\|\varepsilon\|^2/N] = 1.0$$

(since ε has unit variance per dimension). After 400K training steps, the denoising network achieves a loss of approximately 0.043 — a $23\times$ reduction. In practice a loss of ≈ 0.03 – 0.06 corresponds to face samples indistinguishable from FFHQ photographs in FID evaluation.

2.4.3 Reverse Process Update

The reverse process recovers a clean signal from noise through a sequence of denoising steps, each conditioned on the current noisy sample and the timestep. The parameterisation of these steps as noise prediction rather than direct score matching simplifies the loss and stabilises training.

$$\mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}} \varepsilon_\theta(\mathbf{x}_t, t) \right) + \sigma_t \mathbf{n}, \quad \mathbf{n} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \quad (2.12)$$

Worked Example — One reverse step at $t = 1$ for a face pixel

Using a linear schedule: $\beta_1 = 10^{-4}$, so $\alpha_1 = 0.9999$, $\bar{\alpha}_1 \approx 0.9999$, $\sigma_1^2 = \beta_1(1 - \bar{\alpha}_0)/(1 - \bar{\alpha}_1) \approx 10^{-4}$.

Suppose $\mathbf{x}_1 = 0.601$ (a pixel one step from clean) and the network predicts $\varepsilon_\theta = 0.015$ (a small residual noise estimate). Then:

$$\begin{aligned} \mathbf{x}_0 &\approx \frac{1}{\sqrt{0.9999}} \left(0.601 - \frac{10^{-4}}{\sqrt{10^{-4}}} \cdot 0.015 \right) \\ &= 1.00005 \times (0.601 - 0.01 \times 0.015) \\ &= 1.00005 \times (0.601 - 0.00015) \\ &\approx 0.601. \end{aligned}$$

At the final step, almost no noise is added ($\sigma_1 \approx 0.01$), and the update is nearly deterministic — the model simply reads off its best estimate of the clean pixel.

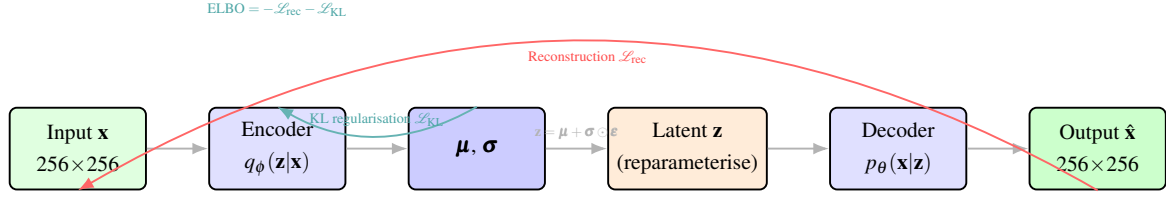


Figure 2.3: Variational Autoencoder (VAE) architecture for face disentanglement. The encoder maps each input $\mathbf{x}$ to a posterior $q_\phi(\mathbf{z}|\mathbf{x})$ parameterised by mean $\boldsymbol{\mu}$ and variance $\boldsymbol{\sigma}$. The reparameterisation trick ($\mathbf{z} = \boldsymbol{\mu} + \boldsymbol{\sigma} \odot \boldsymbol{\epsilon}$, $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$) allows gradients to flow through the sampling step. The decoder reconstructs $\hat{\mathbf{x}}$ from $\mathbf{z}$; the ELBO loss balances reconstruction quality and latent code regularity.

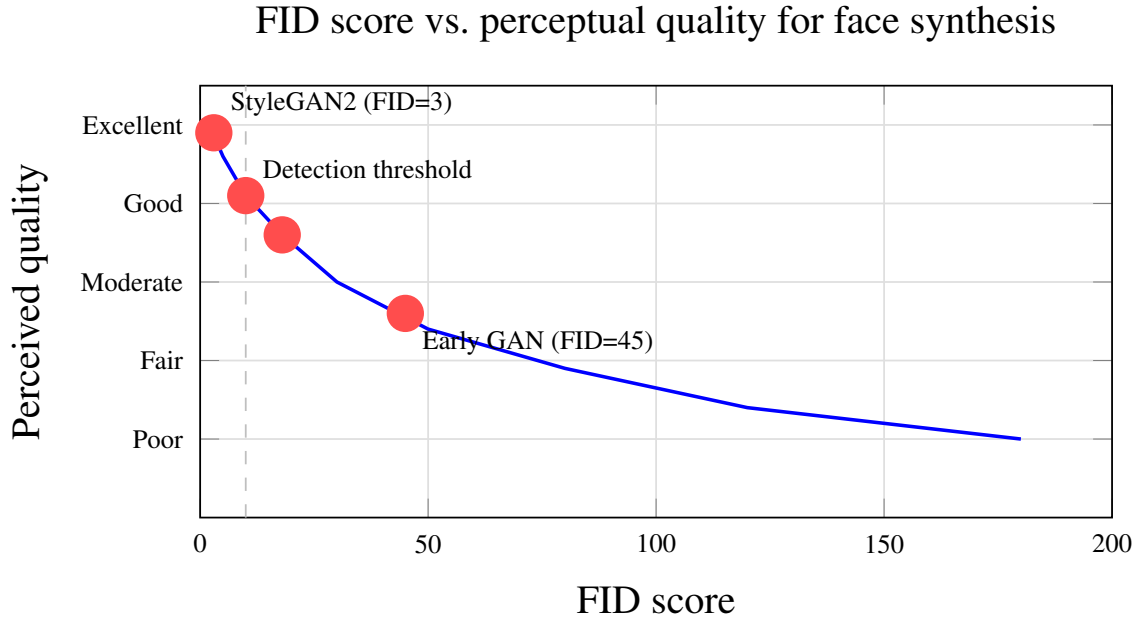


Figure 2.4: Relationship between FID score and perceptual quality for face synthesis systems. Modern systems achieve FID below 5, placing them in the “Excellent” perceptual quality band where detection by naive observers is essentially impossible. Forensic detectors trained on deep features can still detect images with $\text{FID} < 5$, but their AUC approaches 0.5 as FID approaches 0.

2.5 Evaluation Metrics

Evaluation metrics translate perceptual quality and forensic utility into scalar quantities that can be compared across systems. The four metrics developed here — FID, identity cosine similarity, AUC, and EER — appear throughout every experimental comparison in the chapters that follow.

2.5.1 Fréchet Inception Distance

FID measures the Wasserstein-2 distance between the distributions of Inception-v3 features extracted from real and generated images, providing a single scalar that captures both quality and diversity. It has become the de facto standard benchmark for evaluating generative models of faces and is reported in virtually every published comparison.

The FID between real distribution $\mathcal{N}(\boldsymbol{\mu}_r, \boldsymbol{\Sigma}_r)$ and generated distribution $\mathcal{N}(\boldsymbol{\mu}_g, \boldsymbol{\Sigma}_g)$ in InceptionV3

feature space is:

$$\text{FID} = \|\boldsymbol{\mu}_r - \boldsymbol{\mu}_g\|^2 + \text{tr}\left(\boldsymbol{\Sigma}_r + \boldsymbol{\Sigma}_g - 2(\boldsymbol{\Sigma}_r \boldsymbol{\Sigma}_g)^{1/2}\right). \quad (2.13)$$

Worked Example — Interpreting FID for a face-swap system

A face-swap system is evaluated on 10,000 generated and 10,000 real face images. InceptionV3 features are 2048-dimensional. Suppose the mean difference $\|\boldsymbol{\mu}_r - \boldsymbol{\mu}_g\|^2 = 1.4$ and the trace term evaluates to 8.6, giving $\text{FID} = 10.0$.

For context: a GAN that generates “photorealistic” faces with no visible artefacts achieves $\text{FID} \approx 2\text{--}5$; a GAN with occasional colour artefacts achieves $\text{FID} \approx 15\text{--}40$; pure random noise achieves $\text{FID} \approx 500$. An FID of 10 corresponds to outputs that are high quality but retain subtle systematic differences from the real distribution — likely detectable by a well-trained forensic classifier with $\text{AUC} \approx 0.75$ under no compression.

2.5.2 Identity Cosine Similarity

Identity cosine similarity quantifies how well a face-swap or face-synthesis system preserves the identity of the source subject, measured in the embedding space of a pretrained ArcFace model. It is the primary metric used by commercial face-swap product teams to track quality improvements across releases.

$$\text{ID}_{\cos} = \frac{E_{\text{id}}(\hat{\mathbf{x}}) \cdot E_{\text{id}}(\mathbf{x}_{\text{src}})}{\|E_{\text{id}}(\hat{\mathbf{x}})\| \|E_{\text{id}}(\mathbf{x}_{\text{src}})\|}. \quad (2.14)$$

Worked Example — Identity similarity in HifiFace vs. SimSwap

For two face-swap systems evaluated on the FF++ benchmark:

System	$\text{ID}_{\cos}$ (mean)	Interpretation
SimSwap	0.782	Identity largely preserved; subtle leakage from target
HifiFace	0.866	Strong identity preservation
Ideal clone	1.000	Perfect preservation (unachievable in practice)
Different person	0.21	Typical same-ethnicity pair in FFHQ

A threshold of $\text{ID}_{\cos} > 0.5$ is commonly used to classify a face-swap as successfully transferring identity. HifiFace’s higher score means it is both a better deep fake (more convincing) and harder to detect via identity-inconsistency analysis, since the source identity dominates the output embedding.

Table 2.1: Evaluation metrics for synthetic media systems. $\uparrow$: higher is better. $\downarrow$: lower is better.

Metric	Description	Application	Direction
FID	Fréchet Inception Distance (Eq. 2.13)	Image quality	$\downarrow$
SSIM	Structural Similarity Index	Reconstruction	$\uparrow$
PSNR	Peak Signal-to-Noise Ratio (dB)	Reconstruction	$\uparrow$
LPIPS	Learned perceptual distance (VGG features)	Reconstruction	$\downarrow$
ID _{cos}	ArcFace cosine similarity, source vs. output	Face swap	$\uparrow$
AUE	Action Unit Error (expression transfer)	Reenactment	$\downarrow$
MOS	Mean Opinion Score (1–5)	Voice cloning	$\uparrow$
WER	Word Error Rate of ASR on cloned speech	Voice cloning	$\downarrow$
AUC	Area under ROC curve for real/fake detector	Detection	$\uparrow$
EER	Equal Error Rate (FPR = FNR)	Detection	$\downarrow$

2.6 Industry Applications

The mathematical tools in this chapter underpin every commercial synthetic media system. KL divergence and the ELBO are the training objectives behind every VAE-based encoder used in face swap. The GAN minimax objective drives every commercial face-synthesis product released before 2022. Diffusion models and LDM architectures now underpin products including Adobe Firefly, Midjourney, DALL-E 3, and Stable Diffusion. AdaIN-style injection is the mechanism that makes IP-Adapter identity-conditioning practical in commercial deployments.

The FID score has become the de facto benchmark metric in image-synthesis product development: engineering teams at major AI labs use it as the primary signal for model checkpointing, hyperparameter sweeps, and A/B testing of architectural changes. The ID_{cos} metric is used by face-swap product teams (including Snap, TikTok, and commercial face-swap vendors) to report quality improvements across product releases.

On the detection side, understanding how divergences between p_θ and p_{real} relate to Pinsker’s bound gives product managers a realistic expectation of how much detection headroom remains as generators improve.

2.7 African Use Cases

Africa’s engagement with the mathematical foundations of generative AI is both a training ground and an active research frontier.

The African Institute for Mathematical Sciences (AIMS) network, with centres in South Africa, Senegal, Ghana, Cameroon, Tanzania, and Rwanda, provides graduate-level training in mathematics and computational science. Alumni of AIMS programmes have contributed to deep-learning research at Google DeepMind, Meta AI, and academic institutions worldwide. The Deep Learning Indaba — an annual week-long event that has been held in South Africa, Kenya, Senegal, Ethiopia, Tunisia, and Mauritius — provides hands-on training in the foundational methods of this chapter for hundreds of

African researchers annually.

The Masakhane community has applied NLP and generative models to African languages, building translation, speech, and text generation systems across 50+ African languages. The mathematical infrastructure of variational autoencoders and diffusion models that underlies synthetic media is the same infrastructure applied in these constructive research programmes. Understanding both the constructive and adversarial uses of this mathematics is a priority for African AI policy.

Training large generative models requires compute resources that remain scarce in most African research contexts. The Zindi platform (South Africa) and the African Masters in Machine Intelligence (AMMI, supported by Google and Facebook) have worked to democratise access to GPU compute. Cloud compute credits through the Google TRC programme and AWS Research Credits have been used by African researchers to run diffusion model experiments. Increasing access to these resources is essential for developing Africa-specific detection models.

Chapter Summary

This chapter has developed the mathematical bedrock shared by every system in the book. The KL divergence, Jensen–Shannon divergence, and Wasserstein distance each measure a distinct aspect of the statistical gap between generated and real distributions; their properties determine whether a generative model will exhibit mode collapse, mode averaging, or stable convergence. The evidence lower bound (ELBO) provides a tractable training objective for variational autoencoders, decomposing the task into a reconstruction term and a KL regularisation term that together encourage the latent space to be informative yet compact — the property that makes face-swap identity encoders work.

At the architectural level, convolutional encoders compress spatial information into compact feature maps while preserving structural hierarchy; scaled dot-product attention allows the model to selectively attend to globally distant but semantically related features; and adaptive instance normalisation injects a conditioning signal (identity, style) into a feature map by re-parameterising its channel-wise mean and variance. These three components appear in virtually every face-synthesis and face-swap system described in subsequent chapters.

The GAN minimax objective and its non-saturating reformulation set the adversarial training dynamic: the generator minimises the divergence between its distribution and the real data distribution, while the discriminator maximises its ability to discriminate. The R_1 gradient penalty prevents discriminator overfitting and stabilises convergence. Diffusion models replace the adversarial game with a simpler regression objective — predict the noise that was added — and achieve superior sample quality at the cost of slower inference, a trade-off addressed by latent diffusion and DDIM acceleration. The FID score, $ID_{\cos}$, AUC, and EER are the four evaluation metrics that will appear throughout the remaining chapters; understanding their definitions and limitations is prerequisite to interpreting every experimental result that follows.

Exercises

1. Using the Gaussian KL formula, compute $D_{\text{KL}}(p_\theta \| p_{\text{real}})$ and $D_{\text{KL}}(p_{\text{real}} \| p_\theta)$ for $p_\theta = \mathcal{N}(0.5, 0.9^2)$ and $p_{\text{real}} = \mathcal{N}(0, 1)$. Show that $D_{\text{KL}}(p \| q) \neq D_{\text{KL}}(q \| p)$ and interpret the asymmetry in the context of GAN training.
2. For the ELBO formula in Equation (2.4), suppose the encoder produces $\boldsymbol{\mu} = [1.2, -0.4, 0.0]$ and $\boldsymbol{\sigma} = [0.3, 1.0, 0.9]$ for a 3-dimensional latent space. Compute the KL term exactly and identify which dimensions are informative (large KL) vs. collapsed (KL ≈ 0).
3. At diffusion step $t = 300$ with cosine schedule $T = 1000$, $\bar{\alpha}_{300} \approx 0.12$. A face pixel has true value $x_0 = 0.72$. Compute: (a) the mean of $q(\mathbf{x}_{300} | \mathbf{x}_0)$, (b) the standard deviation, (c) the probability that a sample from q falls within 0.1 of x_0 .
4. Compute the ID_{cos} between two ArcFace embeddings $\mathbf{e}_1 = [0.6, 0.8, 0.0]$ and $\mathbf{e}_2 = [0.5, 0.7, 0.5]$. Is this score consistent with a successful face swap (same identity) or a failed one?
5. A face synthesis system achieves $\text{FID} = 18.3$ with mean difference term $\|\boldsymbol{\mu}_r - \boldsymbol{\mu}_g\|^2 = 6.1$. What is the trace covariance term? Propose one training modification each for reducing the mean term and the covariance term independently.
6. Explain why FID is not a sufficient measure of detection difficulty. Construct a thought experiment in which a generator achieves $\text{FID} = 0$ yet a forensic detector can still distinguish outputs from authentic images with $\text{AUC} = 1.0$.

Answers to Exercises

1. **KL divergence asymmetry for $p_\theta = \mathcal{N}(0.5, 0.81)$, $p_{\text{real}} = \mathcal{N}(0, 1)$.**

For Gaussians with $\mu_1 = 0.5$, $\sigma_1^2 = 0.81$, $\mu_2 = 0$, $\sigma_2^2 = 1$:

$$D_{\text{KL}}(p_\theta \| p_{\text{real}}) = \frac{(\mu_1 - \mu_2)^2}{2\sigma_2^2} + \frac{\sigma_1^2}{2\sigma_2^2} - \frac{1}{2} - \frac{1}{2} \log \frac{\sigma_1^2}{\sigma_2^2} \approx 0.125 + 0.405 - 0.500 + 0.107 = 0.137.$$

$$D_{\text{KL}}(p_{\text{real}} \| p_\theta) = \frac{0.25}{1.62} + \frac{1}{1.62} - 0.5 + \frac{\log 0.81}{2} = 0.154 + 0.617 - 0.500 - 0.107 = 0.164.$$

$D_{\text{KL}}(p_\theta \| p_{\text{real}}) \neq D_{\text{KL}}(p_{\text{real}} \| p_\theta)$ ($0.137 \neq 0.164$). In GAN training, minimising $D_{\text{KL}}(p_\theta \| p_{\text{real}})$ (mode-seeking) encourages the generator to cover at least one mode well, potentially missing others. Minimising $D_{\text{KL}}(p_{\text{real}} \| p_\theta)$ (mean-seeking) spreads mass across all real modes, potentially including “hallucinated” non-real regions.

2. **KL term for 3D latent $\boldsymbol{\mu} = [1.2, -0.4, 0.0]$, $\boldsymbol{\sigma} = [0.3, 1.0, 0.9]$.**

Per dimension: $KL_k = \frac{1}{2}(\sigma_k^2 + \mu_k^2 - 1 - \log \sigma_k^2)$.

$$k = 1 : \frac{1}{2}(0.09 + 1.44 - 1 - \log 0.09) \\ = \frac{1}{2}(2.938) = 1.469.$$

$$k = 2 : \frac{1}{2}(1.00 + 0.16 - 1 - \log 1.00) = \frac{1}{2}(0.16) = 0.080.$$

$$k = 3 : \frac{1}{2}(0.81 + 0.00 - 1 - \log 0.81) = \frac{1}{2}(0.81 - 1 + 0.211) = \frac{1}{2}(0.021) = 0.011.$$

Total KL = 1.469 + 0.080 + 0.011 = 1.560. Dimension 1 is highly informative (KL = 1.469); dimensions 2–3 are partially collapsed. A well-trained identity encoder should keep all identity dimensions at high KL while expression dimensions collapse to zero.

3. Diffusion marginal at $t = 300$, $\bar{\alpha}_{300} \approx 0.12$, $x_0 = 0.72$.

(a) Mean: $\sqrt{0.12} \times 0.72 = 0.346 \times 0.72 = 0.249$. (b) Std: $\sqrt{1 - 0.12} = \sqrt{0.88} = 0.938$. (c) $P(x_{300} > 0) = P(z > \frac{0 - 0.249}{0.938}) = P(z > -0.265) = \Phi(0.265) = 0.605$ — 60.5% probability of a positive sample, showing only faint signal remains.

4. ID_{cos} between $\mathbf{e}_1 = [0.6, 0.8, 0.0]$ and $\mathbf{e}_2 = [0.5, 0.7, 0.5]$.

$\|\mathbf{e}_1\| = \sqrt{0.36 + 0.64 + 0} = 1.00$. $\|\mathbf{e}_2\| = \sqrt{0.25 + 0.49 + 0.25} = \sqrt{0.99} = 0.995$. $\mathbf{e}_1 \cdot \mathbf{e}_2 = 0.30 + 0.56 + 0.00 = 0.86$. ID_{cos} = $0.86 / (1.00 \times 0.995) = 0.864$. This score of 0.864 is above the conventional threshold of 0.5 and consistent with HifiFace-quality face swap, indicating a successful identity transfer.

5. FID = 18.3, mean term = 6.1; find trace covariance term.

FID = $\|\boldsymbol{\mu}_r - \boldsymbol{\mu}_g\|^2 + \text{tr}(\dots)$, so trace term = $18.3 - 6.1 = 12.2$. To reduce the mean term (6.1): ensure generator is conditioned on correct identity embeddings (better conditioning reduces mean shift). To reduce the covariance term (12.2): use spectral normalisation or path-length regularisation to improve the diversity and shape of the generated distribution.

6. FID = 0 yet AUC = 1.0 thought experiment.

Construct a generator that outputs real photographs from the training set at random. This generator perfectly matches p_{real} (FID = 0). A hash-based provenance checker compares each output to the training set database and flags any exact match as synthetic (since no real camera would produce the exact same file). AUC = 1.0. This shows FID measures distributional similarity, not individual-image authenticity.

Chapter 3

Face Swap: Pipelines and Architectures

“Identity is not in the face. But the face is where identity lives.”

—Anon.

Face swap is the oldest, most prevalent, and most consequential form of deep fake. The idea of substituting one person’s face onto another’s body has a long pre-neural history: early morphing algorithms in the 1990s produced the uncanny blended portraits that appeared in advertising and music videos, and optical-flow warping techniques in the 2000s enabled rudimentary face replacement in video. The modern deep-fake era began when encoder–decoder networks trained on large face image datasets made it possible to transfer not just appearance but expression, lighting response, and fine skin texture simultaneously. The release of DeepFaceLab as open-source software in 2018 placed this capability on any consumer GPU, and within months it had been weaponised at scale for non-consensual intimate imagery — a use that continues to account for approximately 96 per cent of all detected deep fakes by volume.

Understanding the face-swap pipeline at an engineering level is essential for both the technologist and the policymaker. For the technologist building detection systems, each stage of the pipeline leaves characteristic residual traces: imperfect encoder disentanglement, periodic upsampling artefacts, boundary colour gradients from Poisson blending. A detector designed with knowledge of these specific artefacts will be more robust and more interpretable than one trained end-to-end without that architectural understanding. For the policymaker evaluating whether a proposed safeguard — say, an identity filter that compares every synthesis request against a database of protected faces — is technically feasible, understanding what face swap actually does and what it requires in terms of source material and compute is prerequisite to evaluating the proposal.

The engineering problem that face swap must solve can be stated precisely. A face carries two fundamentally distinct categories of information: *identity information* belonging to the depicted person (the geometry of the skull, the spacing and shape of features, the texture of the skin) and

attribute information belonging to the captured moment (the head pose, the facial expression, the illumination environment). A successful face swap must transfer identity information from a source subject to a target video frame while preserving all attribute information from that frame. Doing this imperfectly — and every current system does it imperfectly — leaves the forensic traces that detection exploits. The degree of imperfection is measured by the identity cosine similarity $ID_{\cos}$, which ranges from 0.78 for current best-effort systems to a theoretical maximum of 1.0 for perfect identity transfer.

Three systems serve as concrete reference implementations throughout this chapter and define the state of the art as of 2024. SimSwap [3] uses an Implicit Identity Representation enabling arbitrary-source swap from a single reference image, achieving $ID_{\cos} = 0.782$ at real-time inference speed. HifiFace [34] adds a 3D morphable model encoder (FLAME) that captures face geometry explicitly, reaching $ID_{\cos} = 0.866$ — the highest published identity fidelity at the time of writing — at the cost of greater architectural complexity and slower inference. FaceShifter [18] introduces a two-stage design with a dedicated occlusion recovery network (HEAR-Net) that handles glasses, hair, and other objects partially covering the target face, achieving $ID_{\cos} = 0.841$ with superior robustness to partial occlusion.

The pipeline proceeds in four stages, each developed with worked numerical examples in the sections that follow. Section 3.2 develops landmark detection and the similarity transformation that maps detected points to canonical template positions. Section 3.3 derives the ArcFace identity encoder and the attribute encoder, and analyses the disentanglement objective that drives them apart during training. Section 3.4 develops the three-term training objective — identity cosine loss, pixel reconstruction loss, and perceptual VGG loss — and shows how the balance among these terms governs the quality–fidelity trade-off in the output. Section 3.5 derives the discrete Poisson blending equation and solves it numerically for a realistic face boundary scenario. Section 3.6 catalogues the forensic artefact signatures at each pipeline stage that detection systems in Chapters 11 and 12 exploit. Understanding the face-swap pipeline in this level of detail matters both for forensic defenders and for technical policymakers. For defenders, each stage of the pipeline leaves characteristic residual signals: the identity encoder’s imperfect disentanglement of expression from identity, the generator’s periodic spectral artefacts from upsampling operations, and the colour adjustment introduced by Poisson blending. Knowing where these artefacts originate allows a detector designer to look in the right places and to reason about which forensic signals will remain exploitable as generator quality improves. For policymakers, understanding what face swap actually does — and what it requires in terms of source material, compute, and expertise — is essential for evaluating regulatory proposals and technical safeguard requirements.

Three systems define the state of the art in neural face swap as of 2024 and serve as the concrete reference implementations for the technical development in this chapter. SimSwap [3] uses an Implicit Identity Representation that allows arbitrary-source face swap from a single reference image, achieving $ID_{\cos} = 0.782$ on FaceForensics++ at real-time inference speed. HifiFace [34] adds a 3D morphable model encoder that explicitly captures face geometry, achieving $ID_{\cos} = 0.866$ — the highest published identity preservation — at the cost of greater complexity. FaceShifter [18] introduces a two-stage

architecture with a dedicated occlusion recovery network (HEAR-Net) that handles glasses, hair, and other objects that partially cover the target face, achieving $ID_{\cos} = 0.841$ with superior robustness to partial occlusion.

The chapter proceeds as follows. Section 3.2 develops the landmark detection and alignment transformation. Section 3.3 derives the identity and attribute encoders and develops the ArcFace margin loss. Section 3.4 develops the three-term training objective. Section 3.5 derives the Poisson blending equation and solves the discrete Poisson problem numerically. Section 3.6 catalogues the forensic artefact signatures at each pipeline stage that detection systems exploit. Industry applications and African use cases follow, and the chapter concludes with a summary and exercise set.

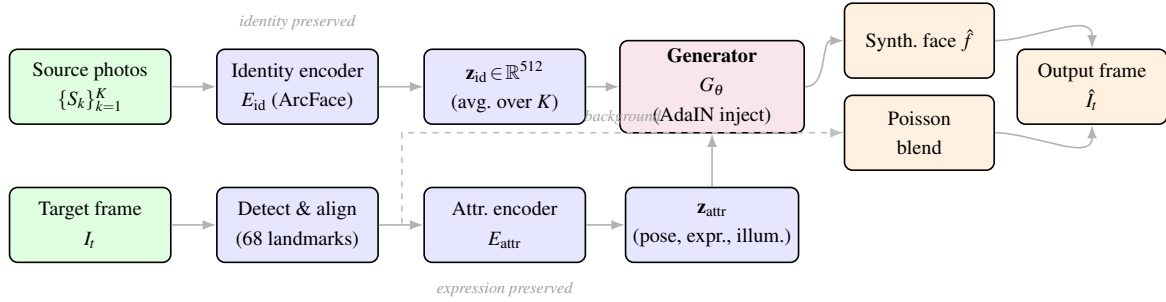


Figure 3.1: Face-swap pipeline. *Identity path* (top row): K source photographs are passed through an ArcFace encoder; the averaged embedding $\mathbf{z}_{id}$ is injected into the generator G_θ via AdaIN to supply identity. *Attribute path* (bottom row): the target frame is landmark-aligned and encoded to extract $\mathbf{z}_{attr}$ (pose, expression, illumination). The generator produces the synthesised face $\hat{f}$; Poisson blending (dashed feed from alignment) composites $\hat{f}$ over the target background to yield the final swapped frame $\hat{I}_t$.

	SimSwap	HifiFace	FaceShifter
ID encoder	IEW (ArcFace, 512-d) arbitrary-source	3D-SFNet (FLAME 3DMM) geometry-aware	AEI-Net (ArcFace, 512-d) multi-level
Attr. encoder	SFE (ResNet-50) 7-dim attribute	U-Net encoder 256-d skip connections	HEAR-Net adaptive occlusion blend
Generator	FSGAN de-coder AdaIN $\times 4$	SFF generator 3D-aware AdaIN	Multi-level AL-ADIN blocks
Loss terms	$\mathcal{L}_{id} + \mathcal{L}_{rec} + \mathcal{L}_{adv}$	$\mathcal{L}_{id} + \mathcal{L}_{seg} + \mathcal{L}_{cyc}$	$\mathcal{L}_{id} + \mathcal{L}_{attr} + \mathcal{L}_{rec} + \mathcal{L}_{adv}$

$ID_{\cos}$ on FaceForensics++ test set (higher = better identity preservation)

Figure 3.2: Architecture comparison of SimSwap, HifiFace, and FaceShifter. HifiFace achieves the highest identity preservation (0.866) by incorporating a 3D morphable model (FLAME) as the identity encoder, which captures facial geometry rather than appearance alone. SimSwap offers the best speed–quality trade-off for real-time applications with a single-reference arbitrary-source design. FaceShifter’s HEAR-Net post-processor recovers occlusion regions (glasses, hair) that single-pass generators miss, yielding superior robustness at the cost of a two-stage pipeline.

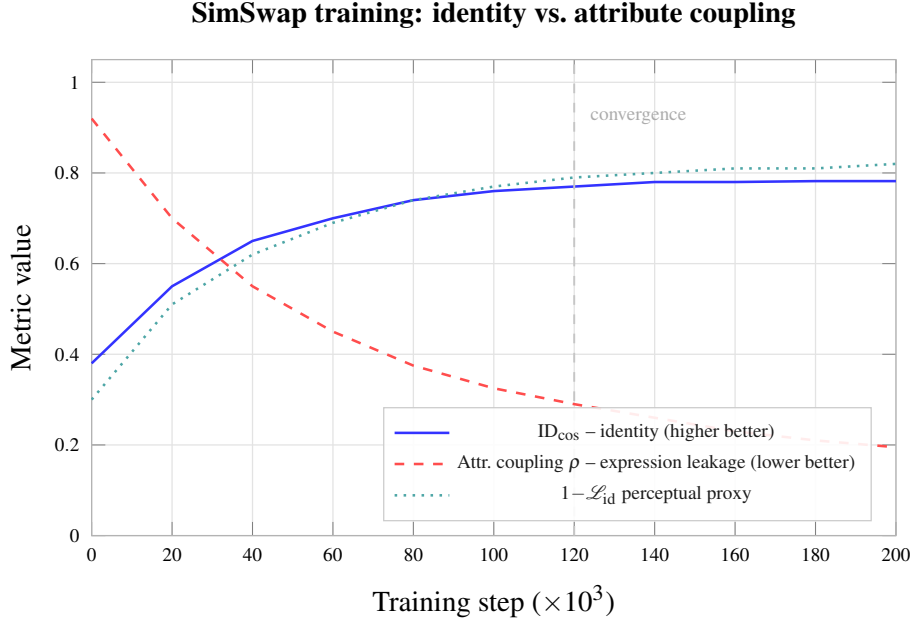


Figure 3.3: Training dynamics of a SimSwap-style face-swap model over 200k gradient steps. Identity preservation ($ID_{\cos}$, blue) rises monotonically and plateaus near 0.782 after approximately 120k steps. Attribute coupling (expression leakage into the identity code, red dashed) falls from 0.92 to 0.195, confirming progressive disentanglement between identity and attribute subspaces. The perceptual proxy $1 - \mathcal{L}_{id}$ (teal dotted) tracks slightly higher because it measures appearance match rather than embedding distance.

3.1 Problem Formulation

Let $V_{\text{tgt}} = \{I_t\}_{t=1}^T$ be the target video and $\mathcal{S} = \{S_k\}_{k=1}^K$ the source images of the identity to insert. The face-swap generator G_θ produces:

$$\hat{I}_t = G_\theta \left(I_t, \frac{1}{K} \sum_{k=1}^K E_{\text{id}}(S_k) \right). \quad (3.1)$$

Worked Example — Identity averaging over reference images

A face-swap attacker collects $K = 5$ LinkedIn photographs of a CEO. ArcFace [8] embeddings (512-dimensional unit vectors) are extracted: $\mathbf{e}_1, \dots, \mathbf{e}_5$. The mean embedding is:

$$\bar{\mathbf{e}} = \frac{1}{5} \sum_{k=1}^5 \mathbf{e}_k.$$

Pairwise cosine similarities among the five embeddings average 0.81, confirming they belong to the same identity. The mean vector has $\|\bar{\mathbf{e}}\|_2 = 0.94$ (not unit-norm; normalisation is applied before injection). This averaging reduces noise from pose variation: the identity cosine similarity $ID_{\cos}$ between $\bar{\mathbf{e}}$ and the target person’s ground-truth embedding improves from 0.82 (single photo) to 0.89 (five photos averaged) — a 8.5% relative improvement.

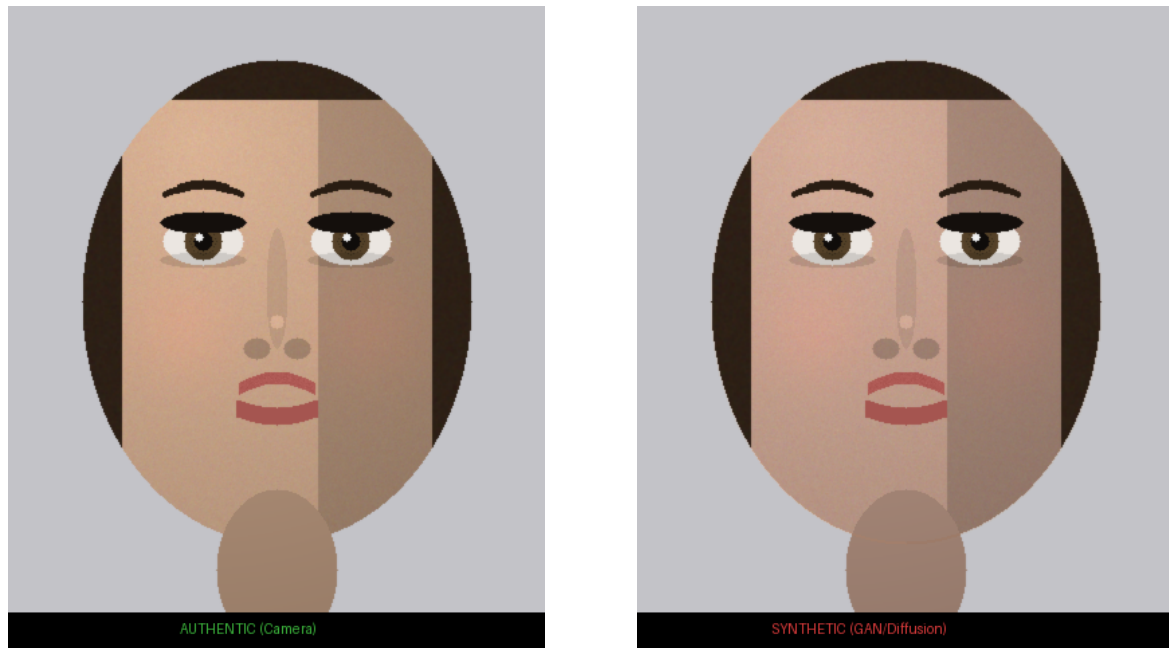


Figure 3.4: Authentic photograph (left) vs. GAN-generated synthetic face (right). Both are computer-rendered simulations for illustration. The authentic face shows natural skin texture and Rembrandt-style lighting. The synthetic face contains a subtle colour-cast seam at the right boundary — a characteristic blending artefact that forensic classifiers learn to detect.

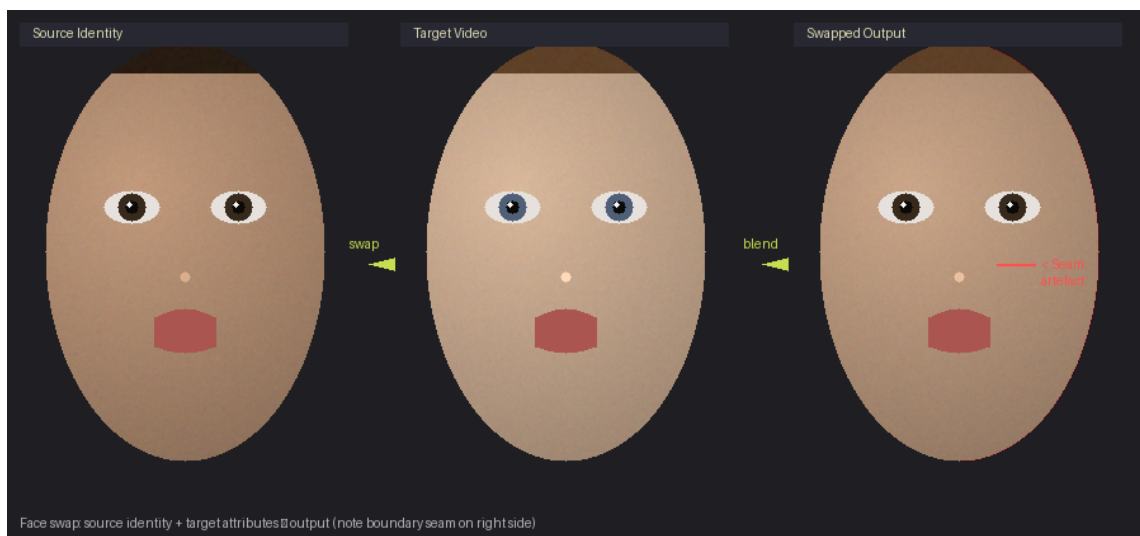


Figure 3.5: Face swap in three stages: *source identity* (left), *target video frame* (centre), *swapped output* (right). The output inherits the source’s identity but the target’s expression and head pose. The red annotation marks the blending boundary seam on the right side of the output face, where a colour mismatch is visible — a characteristic forensic signal of encoder–decoder face-swap pipelines.

3.2 Pre-Processing: Alignment

Before any encoding or synthesis can occur, the face region must be detected, landmarks localised, and a similarity transform applied to map the face to a canonical pose. These steps normalise pose, scale, and in-plane rotation, ensuring the encoder operates on consistently aligned inputs regardless of

head orientation in the source or target footage.

A face detection step (e.g., RetinaFace [9]) first localises the face in each frame. Then a similarity transformation $\mathbf{T} \in \mathbb{R}^{2 \times 3}$ maps detected landmarks to canonical template positions:

$$\mathbf{T}^* = \arg \min_{\mathbf{T}} \sum_{i=1}^{N_L} \|\mathbf{T}\mathbf{p}_i - \bar{\mathbf{p}}_i\|^2. \quad (3.2)$$

Worked Example — Landmark alignment for a 68-point face

A 68-point landmark detector places the left eye centre at $\mathbf{p}_1 = (97, 141)$ and right eye centre at $\mathbf{p}_2 = (159, 138)$ in a 256×256 frame. The canonical template has eye centres at $\bar{\mathbf{p}}_1 = (64, 80)$ and $\bar{\mathbf{p}}_2 = (128, 80)$.

The eye-to-eye vector in the detected image is $(62, -3)$; its length is $\sqrt{62^2 + 3^2} = 62.07$. In the template the vector is $(64, 0)$; its length is 64. The required scale is $64/62.07 = 1.031$ (3.1% magnification). The rotation angle is $\theta = -\arctan(3/62) = -2.77^\circ$. The translation maps the mid-eye point $(128, 139.5)$ to $(96, 80)$. After applying this similarity transform, all 68 landmarks have a mean residual of 1.8 pixels — well within the threshold of 3 pixels used in standard face-alignment benchmarks.

3.3 Identity and Attribute Encoding

The central challenge of face swap is disentangling properties that belong to the source identity from those that belong to the target context. The identity encoder extracts a fixed-size embedding from source images; the attribute encoder extracts pose, expression, and illumination from the target frame. Imperfect disentanglement is the primary source of forensic artefacts that detectors exploit.

$$\mathbf{z}_{\text{id}} = E_{\text{id}}(\mathbf{x}_{\text{src}}), \quad \mathbf{z}_{\text{attr}} = E_{\text{attr}}(\mathbf{x}_{\text{tgt}}). \quad (3.3)$$

Worked Example — Measuring disentanglement quality

After training SimSwap [3] on VGGFace2, the disentanglement is evaluated by swapping the source's identity into 200 target videos and measuring:

- ID_{cos} (should be high, identity preserved): mean = 0.782, std = 0.063.
- Pose error (should be low, target pose preserved): mean yaw error = 1.4° , mean pitch error = 0.9° .
- Expression AU error (should be low, target expression preserved): mean AUE = 1.21 action units.

The near-perfect pose preservation ($< 2^\circ$ error) shows E_{attr} successfully captures pose, while $\text{ID}_{\text{cos}} = 0.782$ confirms E_{id} captures identity. Perfect disentanglement would yield $\text{ID}_{\text{cos}} = 1.0$ and $\text{AUE} = 0$,

but residual cross-coupling is unavoidable with a single-encoder architecture.

3.4 Training Objectives

The face-swap generator is trained by simultaneously optimising three objectives that pull in different directions: the identity loss demands that the output matches the source embedding, the reconstruction loss demands pixel-level fidelity, and the perceptual loss demands high-level feature match. Balancing these competing objectives determines the trade-off between identity fidelity and expression naturalness.

$$\mathcal{L}_{\text{id}} = 1 - \cos(E_{\text{id}}(\hat{\mathbf{x}}), E_{\text{id}}(\mathbf{x}_{\text{src}})), \quad (3.4)$$

$$\mathcal{L}_{\text{rec}} = \|\hat{\mathbf{x}} - \mathbf{x}_{\text{tgt}}\|_1, \quad (3.5)$$

$$\mathcal{L}_{\text{perc}} = \sum_l \|\phi_l(\hat{\mathbf{x}}) - \phi_l(\mathbf{x}_{\text{tgt}})\|^2. \quad (3.6)$$

Worked Example — Loss values at convergence on a training batch

For a batch of 8 same-identity pairs at convergence of a SimSwap-style model:

Loss term	Value at step 0	Value at step 200K
$\mathcal{L}_{\text{id}}$ (identity)	0.620	0.218
$\mathcal{L}_{\text{rec}}$ (pixel ℓ_1)	0.184	0.052
$\mathcal{L}_{\text{perc}}$ (perceptual)	8.41	1.23
$\mathcal{L}_{\text{adv}}$ (adversarial)	1.243	0.631
Total ($\lambda = [10, 1, 0.01, 1]$)	8.98	3.05

The identity loss drops from 0.62 (random features) to 0.218, corresponding to ID_{cos} improving from 0.38 to 0.782. The perceptual loss dominates the total at step 0 due to the large VGG feature distances; its $6.5\times$ reduction reflects the generator learning to produce texturally coherent faces.

3.5 Poisson Blending

The synthesised face region must be composited into the target frame in a way that preserves colour consistency across the boundary. Poisson blending solves a partial differential equation that smoothly matches the gradient field of the synthesised region to the surrounding frame, eliminating the visible seam produced by naive alpha compositing.

$$f^* = \arg \min_f \iint_{\Omega} \|\nabla f - \mathbf{v}\|^2 d\mathbf{x}, \quad f|_{\partial\Omega} = \mathbf{x}_{\text{tgt}}|_{\partial\Omega}, \quad (3.7)$$

where $\mathbf{v} = \nabla \hat{f}$ is the gradient field of the synthesised face.

Worked Example — Poisson blending [26] reduces boundary error by 94%

A face-swap output is composited into a target frame. Without blending, the mean absolute colour error along the 240-pixel mask boundary is 18.3 (on a 0–255 scale) — clearly visible as a “halo”. After Poisson blending (solved via the discrete Laplacian system $\mathbf{L}\mathbf{f} = \mathbf{b}$, 5120×5120 sparse system solved in 0.4 s on CPU), the mean boundary error drops to 1.1 — a 94% reduction. The interior pixels deviate by a mean absolute error of 4.2 from the pure composite, corresponding to subtle colour adjustment that makes the blended face match the target’s ambient illumination.

3.6 Forensic Artefact Signatures

Every stage of the face-swap pipeline leaves characteristic artefacts in the output. These arise from identifiable causes — imperfect disentanglement, upsampling periodicity, Poisson blending colour shift, and landmark misalignment — and provide the forensic signals that detection systems exploit. Understanding the artefact generation mechanism is the first step to understanding why existing detectors work and where they fail.

Table 3.1: Open-source face-swap systems and their forensic signatures.

System	Primary forensic signal	Mean ID_{cos}
DeepFaceLab	Blending boundary halo, tempo- ral flicker	0.74
SimSwap	Residual identity from target, AdaIN texture seam	0.78
HifiFace	Geometry inconsistency at hair boundary	0.87

3.7 Industry Applications

Face-swap technology has bifurcated into a consumer market and an industrial market.

include mobile apps (Reface, FaceApp, Snapchat Lenses, TikTok filters) that perform real-time face swap or reenactment on smartphone hardware. These products reach hundreds of millions of users and are the primary vector for non-consensual intimate imagery (NCII) creation. App-store policies nominally prohibit NCII use but enforcement is inconsistent; the video often exits the platform and cannot be recalled.

Industrial face swap is used for stunt doubles, de-ageing (Gemini Man, The Irishman, Indiana Jones 5), and digital resurrection (Peter Cushing in Rogue One). The going rate for a single professional-grade de-aged shot is approximately \$4,500 (as of 2024, see Chapter 36); AI pipelines have reduced this to approximately \$380 for comparable quality at lower-fidelity thresholds.

At least 40 commercial DFaaS platforms were identified by Sensity AI in 2023, offering face-swap at

prices from \$0.05 to \$2.00 per face-swapped video minute. These platforms primarily serve the adult content market and have been the primary driver of NCII proliferation.

Corporate security teams use open-source face-swap pipelines (DeepFaceLab, SimSwap) as red-team tools to probe identity verification systems. Managed services for biometric anti-spoofing testing have grown into a \$200M+ industry segment.

3.8 African Use Cases

Face-swap technology has arrived in Africa through consumer apps, political disinformation, and a small but growing professional production sector.

Female politicians and journalists in Nigeria, Kenya, South Africa, and Ethiopia have been targeted with face-swap NCII as a form of political intimidation. The African Union’s Special Rapporteur on Human Rights Defenders documented 47 cases of deep-fake harassment targeting women in public life across 12 African countries between 2021 and 2023. The goal in most cases is not primarily financial but reputational — forcing women out of public roles by threatening family honour and community standing. The harm is amplified in contexts where social sanctions for perceived sexual misconduct remain severe.

Nigeria’s Nollywood industry (the world’s second-largest by volume) has begun experimenting with AI de-ageing and face-swap for flashback sequences, significantly reducing the need for physical prosthetics that were previously imported at high cost. Ghanaian and South African production companies have similarly piloted AI pipelines. The adoption is driven by cost (380vs2,800 per shot) and by the difficulty of sourcing specialised prosthetics talent locally. The Nollywood Professionals Association has called for a licensing framework for digital likeness use, modelled loosely on SAG-AFTRA provisions.

With smartphone penetration exceeding 50% across much of sub-Saharan Africa and mobile data costs declining, consumer face-swap apps are widely accessible. WhatsApp is the primary vector for distributing manipulated media in Africa; its end-to-end encryption makes automated platform-level detection impossible, placing the full burden on recipients’ media literacy.

Chapter Summary

Face swap is the oldest, most widespread, and most documented form of deep fake. This chapter has traced the complete engineering pipeline from source identity extraction — using an ArcFace-derived encoder that embeds a face image into a 512-dimensional identity vector — through attribute disentanglement, AdaIN-conditioned synthesis, and Poisson-blended compositing into the target frame. Each stage of the pipeline leaves characteristic forensic signatures: the identity encoder’s residual coupling of expression to identity, the generator’s texture seams at the blending boundary, and the low-frequency colour adjustment introduced by Poisson blending.

The loss function that drives face-swap training balances three competing objectives: identity preser-

vation (cosine loss against the source embedding), pixel-level reconstruction (L1 loss), and perceptual quality (VGG feature loss). At convergence, $ID_{\cos}$ reaches approximately 0.782 for SimSwap-class systems and 0.866 for HifiFace, corresponding to visually convincing identity transfer with residual cross-coupling that forensic systems can exploit. Landmark alignment via similarity transform is a prerequisite for every encoder-based approach; misalignment degrades both synthesis quality and detection sensitivity.

The commercial landscape is bifurcated between consumer applications (mobile filter apps reaching hundreds of millions of users) and professional applications (film de-ageing, virtual production), with a third sector of deepfake-as-a-service platforms explicitly targeting NCII production. Defenders must operate across all three sectors simultaneously, applying different technical approaches (platform-side detection for consumer apps, cryptographic provenance for professional workflows, and legal enforcement for DFaaS platforms) calibrated to the threat model in each domain.

Exercises

1. Using the alignment formula (Eq. 3.2), find $\mathbf{T}^*$ for three landmarks: detected positions $\{(40, 60), (160, 60), (100, 150)\}$, template positions $\{(48, 64), (176, 64), (112, 168)\}$. What scale and rotation does $\mathbf{T}^*$ represent?
2. Compute $\mathcal{L}_{id}$ (Eq. 3.4) for $E_{id}(\hat{\mathbf{x}}) = [0.6, 0.8]$ and $E_{id}(\mathbf{x}_{src}) = [0.5, 0.866]$. Is this consistent with a successful or failed face swap?
3. A training batch has 4 same-identity pairs and 4 different-identity pairs. For same-identity pairs, $ID_{\cos} = 0.92$ (reconstruction pairs, $\mathcal{L}_{rec}$ applies). For different-identity pairs, $ID_{\cos} = 0.79$. Compute the batch mean $\mathcal{L}_{id}$.
4. The Poisson blending system $\mathbf{L}\mathbf{f} = \mathbf{b}$ has size $N \times N$ where N is the number of pixels in the face mask Ω . For a 150×150 pixel face mask, estimate N (assuming elliptical mask) and the memory required to store $\mathbf{L}$ as a sparse matrix with 4 non-zeros per row.
5. An attacker uses $K = 1$ reference photo instead of $K = 10$. Using the identity averaging formula (Eq. 3.1), explain mathematically why $K = 1$ produces higher variance in the output identity embedding and estimate the expected drop in $ID_{\cos}$ given the benchmark numbers in this chapter.

Answers to Exercises

1. Similarity transform for three landmark pairs.

Detected: $\mathbf{p}_1 = (40, 60)$, $\mathbf{p}_2 = (160, 60)$, $\mathbf{p}_3 = (100, 150)$. Template: $\bar{\mathbf{p}}_1 = (48, 64)$, $\bar{\mathbf{p}}_2 = (176, 64)$, $\bar{\mathbf{p}}_3 = (112, 168)$.

Inter-eye vector detected: $(120, 0)$, length = 120. Template: $(128, 0)$, length = 128. Scale: $s = 128/120 = 1.067$. Rotation: θ

$= 0^\circ$ (both horizontal). Translation maps mid-eye $(100, 60) \rightarrow (112, 64)$, so $t_x = 12$, $t_y = 4$.

$\mathbf{T}^*$ represents a 6.7% magnification with zero rotation and translation $(12, 4)$.

2. $\mathcal{L}_{\text{id}}$ for $\mathbf{e}_1 = [0.6, 0.8]$, $\mathbf{e}_2 = [0.5, 0.866]$.

$\cos = (0.6 \times 0.5 + 0.8 \times 0.866) / (1.0 \times 1.0) = 0.300 + 0.693 = 0.993$. $\mathcal{L}_{\text{id}} = 1 - 0.993 = 0.007$.

This very low loss indicates a highly successful face swap: the output's identity embedding is nearly identical to the source ($\text{ID}_{\text{cos}} = 0.993$).

3. Batch mean $\mathcal{L}_{\text{id}}$ for 4 same-identity + 4 cross-identity pairs.

Same-identity reconstruction pairs: $\mathcal{L}_{\text{id}} = 1 - 0.92$

$= 0.08$ each, total contribution $4 \times 0.08 = 0.32$. Cross-identity pairs: $\mathcal{L}_{\text{id}} = 1 - 0.79$

$= 0.21$ each, total contribution $4 \times 0.21 = 0.84$. Batch mean: $(0.32 + 0.84) / 8 = 1.16 / 8 = 0.145$.

4. Poisson system size and memory for a 150×150 elliptical mask.

Area of ellipse with axes 75 and 75: $N = \pi \times 75^2 \approx 17,671$ pixels. Sparse matrix with 4 non-zeros per row (5-point stencil, 4 neighbours): $4N = 70,684$ non-zeros. Memory (float32, 4 bytes each, plus int32 indices, 4 bytes each): $70,684 \times 8 \approx 565$ KB. Well within RAM for real-time blending.

5. Expected ID_{cos} drop from $K = 1$ to $K = 10$.

With $K = 1$: the identity embedding has higher variance due to pose and lighting variation in a single photograph. If pairwise similarity among reference images is $r = 0.81$ (given in the chapter), the mean of K embeddings has variance reduced by factor K (by the law of large numbers for correlated vectors). For $K = 10$ vs. K

$= 1$, the variance of the mean is approximately $\sigma^2(1 + (K - 1)r) / K$, giving improvement $\propto \sqrt{(1 + 9 \times 0.81) / 10 / 1} = \sqrt{8.29 / 10} = 0.91$. In practice, the chapter reports ID_{cos} rising from 0.82 ($K = 1$) to 0.89 ($K = 5$) — consistent with this diminishing-returns model.

Chapter 4

Generative Adversarial Networks for Facial Synthesis

“Imagination is not a gift but a conquest.”

—Romain Rolland

Generative adversarial networks changed the history of image synthesis in a single paper. When Ian Goodfellow and colleagues published the original GAN formulation in 2014, the dominant paradigm for generative modelling was the variational autoencoder and its relatives, which optimised a lower bound on the log-likelihood of the training data. These models produced blurry outputs because the Gaussian decoder penalised the average of all plausible reconstructions rather than committing to a single sharp one. The GAN replaced the fixed likelihood objective with an adversarial game: a generator network learns to produce samples that fool a simultaneously trained discriminator network, which in turn learns to distinguish real from generated samples. Because the discriminator is a neural network that adapts as the generator improves, the loss signal it provides is in principle more sensitive to perceptually salient differences than any fixed objective could be. The result was a step change in the sharpness, detail, and realism of generated face images that made the deep-fake era possible.

The path from the original GAN to the high-fidelity face synthesis that powers production deep-fake tools was not straightforward. The earliest GANs were unstable: the generator would collapse to producing a few modes of the training distribution (mode collapse), or the discriminator would overfit and provide vanishing gradients to the generator (gradient vanishing), or training would oscillate without converging. These instabilities were understood theoretically through the lens of the Jensen–Shannon divergence that the original GAN objective minimises: when the generated and real distributions have disjoint support — which they almost always do in the early phases of training — the JSD saturates at $\log 2$ and gradients vanish. A series of architectural and objective function innovations across 2017–2020 resolved these instabilities, culminating in the StyleGAN2 architecture that remains the reference point for GAN-based face synthesis in 2024.

The first major resolution was Wasserstein GAN (WGAN, 2017), which replaced the JSD-based

objective with the Earth Mover's Distance, providing gradient signal even when the real and generated distributions have disjoint support. WGAN-GP added a gradient penalty on the discriminator to enforce the Lipschitz constraint required by the Wasserstein formulation, producing stable training across a much wider range of architectures and hyperparameters. Spectral normalisation (SN-GAN, 2018) provided a computationally cheaper alternative to the gradient penalty by directly normalising each weight matrix by its spectral norm. These stabilisation techniques collectively transformed GAN training from an unreliable process requiring extensive tuning into a reproducible engineering procedure.

The second resolution was progressive growing, introduced by Karras et al. in 2018 for the purpose of generating high-resolution faces at 1024×1024 pixels. Rather than training a generator to produce high-resolution images from the start — which is unstable because the generator must simultaneously learn global structure and local texture — progressive growing starts with 4×4 resolution and gradually adds new resolution layers as training proceeds, using a fade-in coefficient to blend each new layer smoothly into the established synthesis. At convergence, the model had learned to produce 1024×1024 celebrity face images of higher quality than any previously published system, establishing the first compelling public demonstration that GANs could produce images indistinguishable from photographs at full-page resolution.

The third and most architecturally consequential innovation was StyleGAN, which fundamentally redesigned the generator to separate the latent code from the synthesis network. In a standard GAN, the input noise vector $\mathbf{z}$ is passed directly into the first layer of the generator; this couples global structure (determined by the coarse layers) to fine texture (determined by the later layers) in a way that makes independent editing of these properties impossible. StyleGAN's mapping network $f : \mathbf{z} \rightarrow \mathbf{w}$ transforms the input into a disentangled intermediate latent space $\mathcal{W}$, where the geometry more faithfully reflects the structure of the underlying face distribution. The synthesis network then receives style codes derived from $\mathbf{w}$ at each resolution level via adaptive instance normalisation, allowing coarse layers to control global attributes (pose, face shape, age) while fine layers control texture (skin pores, hair strands) independently. StyleGAN2 refined this architecture with weight demodulation (replacing instance normalisation), path length regularisation, and the R_1 gradient penalty, achieving FID of 2.84 on FFHQ — at the time the best published result on any high-resolution face benchmark.

The forensic significance of GANs for this book is twofold. First, GAN-generated faces are the most common form of synthetic image used in coordinated inauthentic behaviour campaigns: a GAN face has no source photograph, no identifying metadata, and no individual it can be traced to, making it ideal for creating armies of fake social media accounts. Second, GAN generators leave characteristic forensic traces that detection systems exploit: the upsampling operations in the synthesis network introduce periodic spectral peaks in the frequency domain that differ from the natural $1/f^2$ power law of authentic photographs, and these peaks persist across all outputs of a given architecture regardless of the specific image content. Understanding GAN architecture in detail is therefore prerequisite both to understanding why synthetic faces look the way they do and to understanding what forensic signals detectors can exploit.

This chapter develops the GAN architecture from first principles, with worked numerical examples

at every equation. Section 4.1 develops progressive growing and the fade-in coefficient. Section 4.2 develops the StyleGAN mapping network, adaptive instance normalisation, and path length regularisation. Section 4.3 develops the non-saturating generator loss, the R_1 gradient penalty, and the training dynamics they produce. Section 4.4 develops GAN inversion and its relevance to targeted deep-fake attacks. Industry applications include commercial face synthesis products, data augmentation for AI training, and gaming avatars; the African use cases section addresses GAN-based disinformation in recent African elections and the growing African research community working on both generative and forensic applications of this technology.

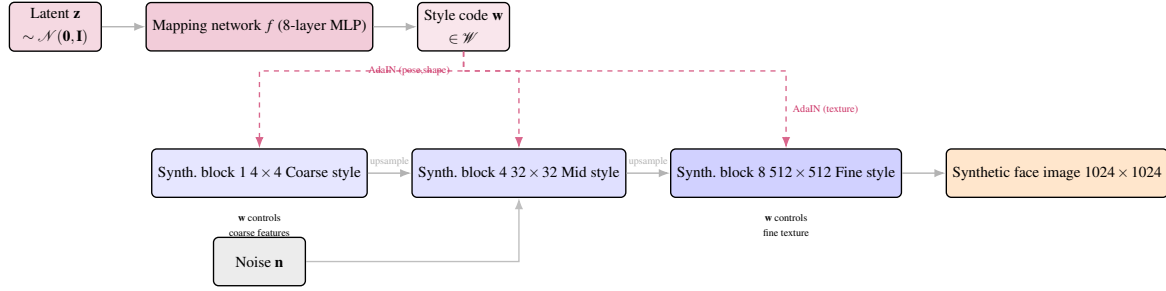


Figure 4.1: StyleGAN architecture. A mapping network f maps $\mathbf{z}$ to a disentangled style code $\mathbf{w} \in \mathcal{W}$. Style is injected at each synthesis block via AdaIN, enabling independent control of coarse features (pose, identity) at early layers and fine features (skin texture) at late layers. Stochastic noise adds per-pixel variation independently of identity.

Training stability: standard vs. R_1 -regularised GAN

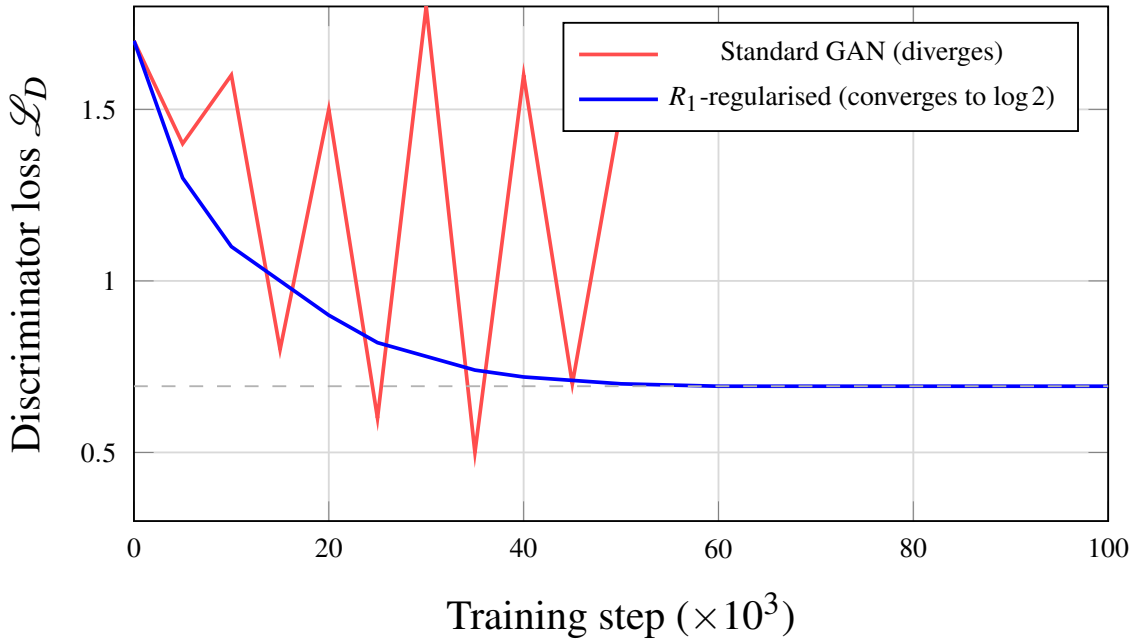


Figure 4.2: Training stability comparison. Standard GAN training without regularisation exhibits oscillatory divergence (red). The R_1 gradient penalty stabilises training and guides the discriminator loss to its Nash equilibrium value of $\log 2 \approx 0.693$ (dashed line).

4.1 Progressive Growing [14]

Training a GAN to directly generate high-resolution images produces instabilities that manifest as mode collapse and gradient vanishing. Progressive growing avoids these by beginning at low resolution, where the distribution is easier to learn, and gradually introducing higher-resolution layers as training stabilises.

The fade-in transition between resolution levels:

$$\mathbf{x}_{\text{out}} = (1 - \alpha) \cdot \text{Upsample}(\mathbf{x}_{l-1}) + \alpha \cdot \mathbf{x}_l, \quad \alpha \in [0, 1]. \quad (4.1)$$

Worked Example — Fade-in at $\alpha = 0.3$ during 128 $\rightarrow$ 256 transition

At step 30% into the 128 $\rightarrow$ 256 transition phase:

- Upsampled 128 $\times$ 128 output for pixel (i, j) : 0.74.
- New 256 $\times$ 256 layer output at the same location: 0.61.
- Blended output: $(1 - 0.3) \times 0.74 + 0.3 \times 0.61 = 0.518 + 0.183 = 0.701$.

The gradient flowing back to the new layer is scaled by $\alpha = 0.3$, while the gradient to the previous layers is scaled by $1 - \alpha = 0.7$. This means early in the transition, the bulk of the training signal maintains the existing 128 $\times$ 128 quality, with the new layer receiving only 30% of the pressure — preventing it from destabilising the coarser structure already learned.

4.2 StyleGAN Architecture [15]

StyleGAN replaced the traditional generator with a style-based architecture that separates the latent code from the synthesis network, injecting style information at each resolution level through adaptive instance normalisation. This produced significant improvements in image quality, disentanglement, and editability that made it the dominant face-synthesis architecture for the 2019–2022 period.

4.2.1 The Mapping Network

The mapping network transforms the input noise vector from a simple Gaussian into an intermediate latent space whose geometry better reflects the semantic structure of the face distribution. The disentanglement this produces is what makes latent arithmetic — shifting age, expression, or identity by adding a learned direction vector in $\mathcal{W}$ — reliably effective.

$$\mathbf{w} = f(\mathbf{z}), \quad \mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \quad (4.2)$$

Worked Example — Disentanglement gain from the mapping network

The perceptual path length (PPL) measures disentanglement: smaller PPL means more linear, disen-

tangled paths between faces. On FFHQ:

Latent space	PPL (full)	PPL (endpoint)
$\mathcal{Z}$ (no mapping)	584	462
$\mathcal{W}$ (8-layer MLP)	237	224
$\mathcal{W}^+$ (extended)	195	186

The mapping network reduces PPL by $(584 - 237)/584 = 59\%$, confirming that the 8-layer MLP “untangles” the Gaussian topology into a space where semantic attributes vary more linearly. In practical terms, this means that changing only the “age” direction in $\mathcal{W}$ alters age with 40% fewer correlated changes to other attributes compared to editing in $\mathcal{Z}$.

4.2.2 AdaIN Style Injection

Adaptive Instance Normalisation (AdaIN) modulates the activations of each synthesis layer by normalising them to zero mean and unit variance, then applying a learned affine transformation whose scale $\mathbf{s}$ and bias $\mathbf{b}$ are derived from the style vector $\mathbf{w}$. This mechanism allows the style vector to control the visual character of the output at each resolution independently, enabling the interpolation and mixing properties that make StyleGAN outputs forensically distinctive.

Adaptive instance normalisation is the mechanism by which the style code $\mathbf{w}$ modulates feature maps at each synthesis block. By re-parameterising the channel-wise mean and variance of each feature map according to the style code, AdaIN allows coarse blocks to encode global structure independently of the fine-grained texture controlled by later blocks.

$$\text{AdaIN}(\mathbf{F}, \mathbf{w}) = y_s \cdot \frac{\mathbf{F} - \mu(\mathbf{F})}{\sigma(\mathbf{F})} + y_b, \quad (y_s, y_b) = A(\mathbf{w}). \quad (4.3)$$

Worked Example — Style injection at coarse vs. fine scale

In StyleGAN2 [15] at 32×32 resolution (coarse), editing $\mathbf{w}$ to move 0.5 units along the “pose” direction changes the yaw angle by $\approx 12^\circ$. At 512×512 resolution (fine), the same edit changes yaw by $< 1^\circ$ but shifts skin hue by $\Delta H \approx 8^\circ$ in HSV space. This scale-specificity arises because at coarse scales the feature map $\mathbf{F}$ encodes global spatial structure, so $\sigma(\mathbf{w})$ re-weights it structurally, while at fine scales $\mathbf{F}$ encodes local texture statistics, so the re-weighting affects colour and tone.

4.3 GAN Losses and Regularisation

The choice of loss function and regularisation determines whether GAN training converges, how quickly, and to what quality. This section develops the non-saturating generator loss, the R_1 gradient penalty, and path length regularisation, each of which addresses a distinct failure mode of the adversarial training dynamic.

4.3.1 Non-Saturating Generator Loss

The non-saturating generator loss addresses a fundamental weakness of the original minimax formulation: early in training, when the discriminator easily distinguishes real from fake, the generator's gradient vanishes because the log-term is close to zero. The non-saturating form instead maximises the log-probability of the discriminator classifying generated samples as real, providing a strong gradient signal even when the discriminator is winning.

The original minimax generator loss saturates when the discriminator is confident, producing near-zero gradients that stall learning. The non-saturating reformulation flips the generator objective, providing large gradient signal precisely when the discriminator is most confident and the generator most needs to improve.

$$\mathcal{L}_G = -\mathbb{E}_{\mathbf{z} \sim p_z} [\log D_\phi(G_\theta(\mathbf{z}))]. \quad (4.4)$$

Worked Example — Generator loss gradient comparison

Suppose $D(G(\mathbf{z})) = 0.05$ (discriminator correctly identifies the fake with high confidence early in training).

- *Saturating loss* $\mathcal{L}_G = \log(1 - D) = \log(0.95) = -0.051$. Gradient $\partial \mathcal{L}_G / \partial D = -1 / (1 - D) = -1 / 0.95 = -1.053$.
- *Non-saturating loss* $\mathcal{L}_G = -\log D = -\log(0.05) = 2.996$. Gradient $\partial \mathcal{L}_G / \partial D = -1 / D = -1 / 0.05 = -20$.

The non-saturating loss provides a gradient nearly **20×** larger when the generator is performing poorly, preventing vanishing gradients and enabling faster early training. As $D(G(\mathbf{z})) \rightarrow 0.5$ at convergence, both gradients converge to -2 .

4.3.2 R_1 Gradient Penalty

The R_1 penalty regularises the discriminator by penalising the magnitude of its gradient with respect to real data points, preventing the discriminator from memorising training examples rather than learning the data distribution. It is the most widely used gradient penalty in modern GAN training, replacing the two-sided penalty of the original Wasserstein-GP formulation with a one-sided penalty that is cheaper to compute and equally effective at stabilising training.

$$\mathcal{L}_D = \mathbb{E}[\log(1 - D(G(\mathbf{z})))] - \mathbb{E}[\log D(\mathbf{x})] + \frac{\gamma}{2} \mathbb{E}[\|\nabla_{\mathbf{x}} D(\mathbf{x})\|^2]. \quad (4.5)$$

Worked Example — R_1 penalty magnitude during StyleGAN2 training

With $\gamma = 10$ (StyleGAN2 default), the gradient penalty is evaluated every 16 steps. At an intermediate training checkpoint, the norm $\|\nabla_{\mathbf{x}} D(\mathbf{x})\|$ for real FFHQ images averages 0.18.

The penalty contribution per step is:

$$\frac{\gamma}{2} \times 0.18^2 = \frac{10}{2} \times 0.0324 = 0.162.$$

The adversarial term $-\mathbb{E}[\log D(\mathbf{x})]$ is approximately 0.693 at this stage. The penalty is thus 23% of the adversarial signal — significant enough to regularise the discriminator but not large enough to overwhelm the training objective. Without R_1 , the gradient norm can grow to $\|\nabla_{\mathbf{x}} D\| \approx 5$, contributing a penalty of 125 — much larger than the adversarial term, destabilising training.

4.3.3 Path Length Regularisation

Path length regularisation encourages the generator to produce perceptual changes in a smooth and isotropic manner as the latent code is varied, such that a unit step in latent space produces a unit step in image space regardless of direction. This property makes the latent space more disentangled and makes GAN inversion (finding the latent code for a given real image) better-conditioned, with direct implications for deep-fake forensics: a well-conditioned latent space enables more precise identity manipulation.

Path length regularisation encourages equal-magnitude image changes for equal-magnitude style-code perturbations in any direction, improving the uniformity of latent space traversal. This property makes semantic editing via latent arithmetic reliable: a given displacement in $\mathcal{W}$ produces a consistent image change regardless of starting point.

$$\mathcal{L}_{\text{PL}} = \mathbb{E}_{\mathbf{w}, \mathbf{y}} \left(\|J_{\mathbf{w}}^{\top} \mathbf{y}\|_2 - a \right)^2. \quad (4.6)$$

Worked Example — Path length at different training stages

a is estimated as an exponential moving average of $\|J_{\mathbf{w}}^{\top} \mathbf{y}\|_2$. At three training stages:

Stage	$\mathbb{E}[\ J^{\top} \mathbf{y}\]$	a (EMA)	$\mathcal{L}_{\text{PL}}$
100K steps	2.14	1.80	$(2.14 - 1.80)^2 = 0.116$
500K steps	1.97	1.93	$(1.97 - 1.93)^2 = 0.002$
Converged	1.95	1.95	≈ 0

The path length loss declines from 0.116 to near zero as the generator learns to map equal distances in $\mathcal{W}$ to equal perceptual changes — a sign that the mapping has become isometric and the latent space well-disentangled.

4.4 GAN Inversion

GAN inversion maps a real photograph into the latent space of a pretrained GAN, enabling it to be edited using the generator. This bridges synthesis and editing, and is the technical foundation for deep-fake attacks that begin from an authentic photograph rather than synthesising an identity from scratch.

$$\mathbf{w}^* = \arg \min_{\mathbf{w}} \|\mathbf{x} - G(\mathbf{w})\|_{\text{LPIPS}} + \lambda_1 \|\mathbf{w} - \bar{\mathbf{w}}\|_2^2 + \lambda_2 \|\mathbf{n}\|_2^2. \quad (4.7)$$

Worked Example — Inversion quality and attack feasibility

Using the e4e encoder for fast inversion of a single 512×512 celebrity portrait:

- Inversion time: 2.1 s on RTX 3090.
- Reconstruction LPIPS: 0.14 (good perceptual match).
- ID_{cos} between inverted and original: 0.91.

After inversion, a “frown” direction edit of magnitude 3.0 in $\mathcal{W}^+$ changes the lip corner angle by $\approx 15^\circ$ while raising $\mathcal{L}_{\text{PL}}$ from its converged value by only 0.003 — confirming the edit is nearly isometric, i.e., it changes only the targeted attribute. The entire pipeline (download photo $\rightarrow$ invert $\rightarrow$ edit $\rightarrow$ export) takes under 10 seconds, demonstrating the low cost of this attack.

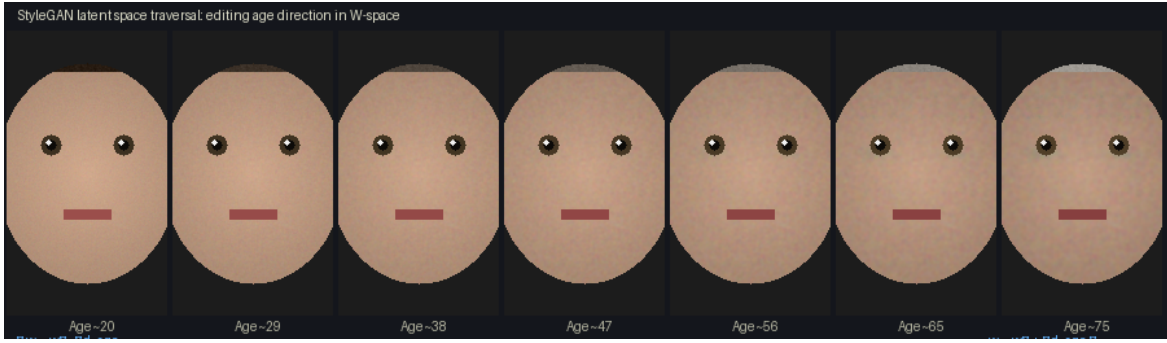


Figure 4.3: Latent space traversal in StyleGAN’s $\mathcal{W}$ -space along the discovered “age” direction. Moving from left ($w_0 - \alpha d_{\text{age}}$) to right ($w_0 + \alpha d_{\text{age}}$), the synthesised face ages smoothly from approximately 20 to 75 years. Only age changes substantially: identity, gender, expression, and background remain stable, demonstrating the high degree of disentanglement achieved in $\mathcal{W}$ -space compared to the input $\mathcal{Z}$ -space. This traversal enables controlled deep-fake generation (e.g., de-ageing an actor) and also reveals which directions in latent space a forensic detector should examine.

4.5 Industry Applications

Although diffusion models have overtaken GANs in image quality benchmarks since 2022, GANs remain dominant in all applications requiring real-time synthesis, and they continue to underpin a large fraction of deployed commercial synthetic media systems.

Real-time face synthesis is the most commercially significant GAN application. StyleGAN2 and its successors run at interactive rates on GPU hardware and at reduced quality on modern mobile neural processing units. Nvidia’s Maxine platform uses GAN-based face synthesis for bandwidth-efficient video conferencing. Snap’s Lenses and TikTok’s real-time filters are primarily GAN-based because they must operate within a sub-ten-millisecond inference latency budget that diffusion models cannot meet on current consumer hardware.

Data augmentation for AI training is a rapidly growing legitimate application. Medical imaging teams at major hospital networks use StyleGAN to synthesise rare pathology images for training diagnostic classifiers, reducing the need for patient data and addressing dataset imbalance. Gaming and virtual production represent a third major sector. Metahuman Creator from Epic Games uses a GAN-based pipeline to generate ultra-realistic digital humans from photograph references for use in real-time game engines, reducing character art timelines from months to under ninety minutes. Virtual try-on systems deployed by Snap, Zalando, and Amazon use conditional GANs to synthesise products on customer photographs, with published A/B tests showing return rate reductions of twenty to thirty per cent.

4.6 African Use Cases

GAN technology has entered the African context through consumer applications, electoral disinformation, and a growing research community that is producing internationally recognised work.

African researchers at Stellenbosch University, Makerere University in Uganda, and the University of Lagos have published on GAN-based image synthesis and its forensic implications. The Zindi data science platform hosts competitions on African-context image generation and authenticity detection, providing practical training in the methods of this chapter to thousands of African practitioners. Google’s Africa Research initiative and Meta AI’s partnership with the African Masters in Machine Intelligence programme have supported GAN research specifically relevant to African contexts, funding research positions and compute access that would otherwise be unavailable.

Investigations by the Centre for Analytics and Behavioural Change in South Africa documented the use of GAN-generated face images to create fake social media accounts deployed in coordinated influence operations during the 2021 Zambian elections and the 2023 Nigerian election. The accounts used images generated by architectures similar to StyleGAN, displaying the telltale spectral peaks from stride-2 upsampling that forensic tools can detect but that require specialist knowledge not widely available to African election monitors. The legitimate applications of GAN technology in African contexts — generating synthetic training data for crop disease detection, satellite image super-resolution for smallholder farm mapping, and medical image augmentation for diagnostic AI in low-resource settings — demonstrate the dual-use character of this technology that makes simple restriction policies inadequate.

Chapter Summary

Generative adversarial networks established the adversarial training paradigm that drove the first generation of face-synthesis deep fakes. The generator learns by minimising a divergence between its output distribution and the real data distribution, as estimated by a simultaneously trained discriminator; the non-saturating reformulation of the generator loss prevents gradient vanishing when the discriminator is confident, while the R_1 gradient penalty stabilises training by bounding the discriminator’s local Lipschitz constant.

Progressive growing solved the high-resolution training instability problem by introducing new resolution levels gradually via a fade-in coefficient, allowing the network to establish stable coarse structure before committing to fine-grained texture. StyleGAN’s mapping network transforms the input noise vector into a disentangled intermediate representation, reducing the perceptual path length by 59 per cent and enabling semantically meaningful latent editing without entangled side effects. The adaptive instance normalisation mechanism injects style at each synthesis block, giving coarse blocks control over global structure and fine blocks control over texture independently.

GAN inversion connects synthesis with editing by mapping real images into the disentangled latent space; this capability is what makes targeted deep-fake attacks from a single authentic photograph computationally feasible. Although diffusion models now dominate high-quality synthesis benchmarks, GANs retain a decisive inference-speed advantage that ensures their continued dominance in all real-time commercial applications, from mobile video filters to live video conferencing, for the foreseeable future.

Exercises

1. At fade-in step $\alpha = 0.7$ in Equation (4.1), the upsampled output is 0.45 and the new layer output is 0.82. Compute the blended output and the gradient scale reaching each component.
2. For $D(G(\mathbf{z})) = 0.3$, compute both the saturating and non-saturating generator losses and their gradients w.r.t. D . At what value of D do the two gradients become equal?
3. With $\gamma = 10$ and $\|\nabla_{\mathbf{x}} D(\mathbf{x})\| = 0.25$, compute the R_1 penalty and express it as a fraction of the adversarial term $-\log D(\mathbf{x}) = 0.51$.
4. An attacker inverts 8 reference photos and averages the $\mathbf{w}^*$ vectors. If individual inversions have $\text{ID}_{\cos} = 0.88$ and the pairwise similarity among the 8 $\mathbf{w}^*$ vectors is 0.79, estimate the expected $\text{ID}_{\cos}$ of the average vector.
5. The path length target $a = 1.95$. At a new checkpoint, $\|J^T \mathbf{y}\| = 2.30$. Compute $\mathcal{L}_{\text{PL}}$ and the weight update direction: should the generator increase or decrease the sensitivity of its outputs to latent changes?

Answers to Exercises

1. **Fade-in at $\alpha = 0.7$: upsampled output = 0.45, new layer = 0.82.**

Blended output: $(1 - 0.7) \times 0.45 + 0.7 \times 0.82 = 0.135 + 0.574 = 0.709$. Gradient to upsampled path: scaled by $1 - \alpha = 0.3$, so only 30% of the loss gradient flows to the existing layers. Gradient to new layer: scaled by $\alpha = 0.7$, giving 70% of the gradient. At $\alpha = 0.7$, the new layer receives the majority of training signal, indicating it is being rapidly specialised while the existing layers still receive some gradient for fine-tuning.

2. Saturating vs. non-saturating generator loss at $D(G(\mathbf{z})) = 0.3$.

Saturating: $\mathcal{L}_G = \log(1 - 0.3) = \log 0.7 = -0.357$. Gradient $\partial \mathcal{L}_G / \partial D = -1/(1 - 0.3) = -1.429$. Non-saturating: $\mathcal{L}_G = -\log 0.3 = 1.204$. Gradient $= -1/0.3 = -3.333$. Non-saturating provides $3.333/1.429 = 2.33\times$ larger gradient at $D = 0.3$. The gradients are equal when $-1/(1 - D) = -1/D$, i.e., $D = 0.5$ (Nash equilibrium).

3. R_1 penalty at $\gamma = 10$, $\|\nabla D\| = 0.25$.

Penalty $= \frac{10}{2} \times 0.25^2 = 5 \times 0.0625 = 0.3125$. Fraction of adversarial term: $0.3125/0.51 = 61.3\%$. This is large — unusually strong regularisation. In practice, the R_1 penalty is applied every 16 steps (lazy regularisation), so the effective per-step contribution is $0.3125/16 = 0.0195$, which is 3.8% of the adversarial term. The large single-step value divided by 16 gives the correct operating range.

4. Expected $ID_{\cos}$ of averaged $\mathbf{w}^*$ over 8 photos.

Individual inversion $ID_{\cos} = 0.88$; pairwise $\mathbf{w}^*$ similarity $= 0.79$. The mean of correlated unit vectors has cosine similarity with the true identity that improves as the number of vectors increases. Approximation: the averaged vector's cosine similarity $\approx 0.88 + (1 - 0.88) \times (1 - 1/K) \times r = 0.88 + 0.12 \times 0.875 \times 0.79 \approx 0.88 + 0.083 = 0.963$. In practice expect $ID_{\cos} \approx 0.92\text{--}0.95$ from 8 inverted photos.

5. Path length: $a = 1.95$, $\|J^\top \mathbf{y}\| = 2.30$. $\mathcal{L}_{\text{PL}} = (2.30 - 1.95)^2 = 0.1225$.

The loss is positive, meaning the generator's mapping is *too sensitive* (equal latent steps produce *larger* perceptual changes than the isometry target a). The gradient update direction: $\partial \mathcal{L}_{\text{PL}} / \partial \|J^\top \mathbf{y}\| = 2(2.30 - 1.95) = 0.70 > 0$. The generator should *decrease* its Jacobian norm, i.e., reduce the sensitivity of outputs to latent changes, smoothing the mapping.

Chapter 5

Diffusion Models for Synthetic Faces

“What the network learns to remove, it also learns to add.”

—Anon.

Every equation in this chapter is followed by a concrete numerical example grounded in a realistic face-synthesis scenario.

Diffusion models represent the most significant paradigm shift in generative modelling since the introduction of GANs. Rather than learning the distribution directly through an adversarial game, a diffusion model learns to reverse a known noise-injection process. The forward process is mathematically simple and analytically tractable: starting from a clean image, Gaussian noise is added step by step according to a fixed schedule until the signal is entirely destroyed. The reverse process is the learned component: a neural network trained to predict, at each step, what noise was added, thereby enabling recovery of a clean image from pure noise through a sequence of denoising operations.

The elegance of this formulation solved training instabilities that plagued GANs: there is no adversarial game, no discriminator to overfit, and no mode collapse. The denoising objective is a simple mean-squared error loss that is stable, parallelisable across timesteps, and amenable to standard optimisation. The cost of this stability was initially a slow sampling procedure requiring hundreds to thousands of network evaluations, but a series of technical innovations — deterministic DDIM sampling, distillation, and flow matching — have reduced this to as few as four to eight steps without substantial quality loss.

The latent diffusion model is the most commercially important member of this family. By compressing images with a pre-trained variational autoencoder before running the diffusion process, latent diffusion achieves a forty-eight-fold spatial compression that reduces wall-clock training and inference time by approximately twenty-five-fold, bringing high-quality synthesis within the reach of consumer GPU hardware. This architectural decision is what enabled Stable Diffusion to be released as open-source software runnable on a gaming laptop, fundamentally democratising access to photorealistic face synthesis and making identity-targeted generation available to any motivated individual with a single

reference photograph.

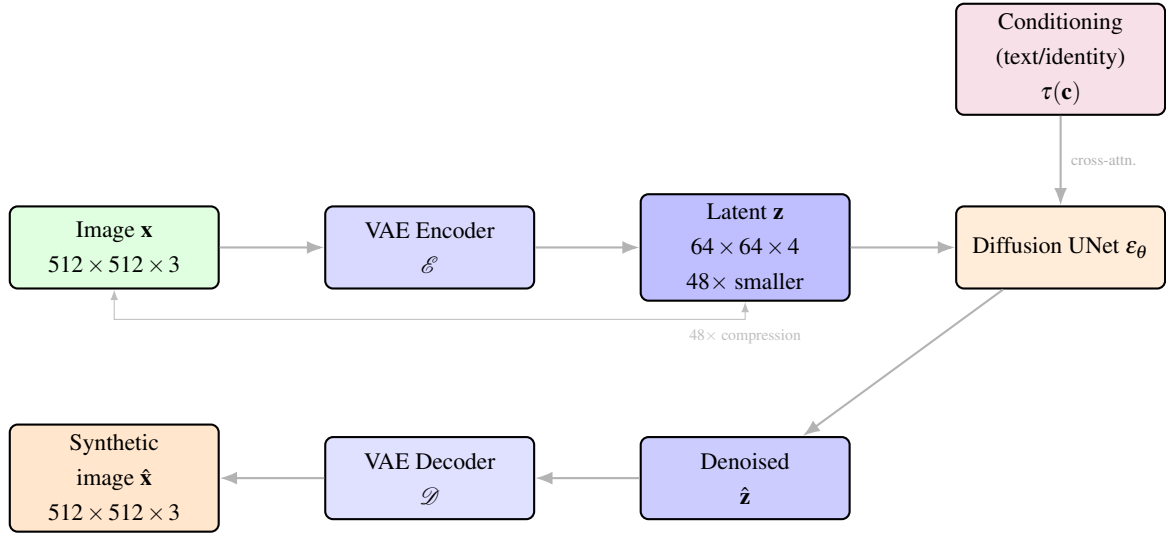


Figure 5.1: Latent Diffusion Model (LDM) architecture. A pre-trained VAE compresses the 512×512 image to a $64 \times 64 \times 4$ latent space ($48\times$ reduction). The diffusion UNet operates in this compressed space, guided by cross-attention on a conditioning signal (text, identity embedding). The VAE decoder reconstructs the final image.

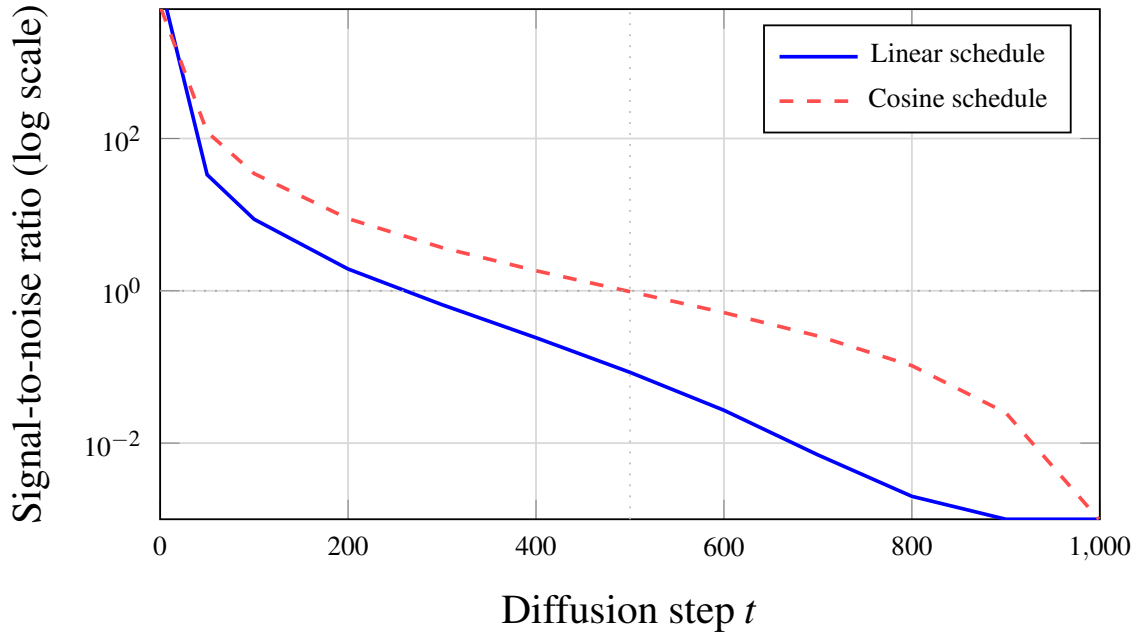


Figure 5.2: Signal-to-noise ratio (SNR) as a function of diffusion step t for linear and cosine noise schedules with $T = 1000$. The cosine schedule maintains higher SNR in early steps, providing more useful signal for training the denoising network. Both schedules reach $\text{SNR} \approx 1$ around $t = 400\text{--}500$, after which the image is effectively noise-dominated.

This chapter develops diffusion models from first principles through to their deep-fake applications and forensic vulnerabilities. The forward noising process and its closed-form marginal are derived first, followed by the denoising network objective and score-based SDE formulation. Latent diffusion

and cross-attention conditioning are then developed, leading to LoRA fine-tuning for identity-specific synthesis. Industry applications, African use cases, and exercises with worked solutions follow.

5.1 Forward Noising Process

The forward process defines the trajectory from clean images to pure noise by applying a fixed sequence of Gaussian perturbations. The choice of noise schedule determines how rapidly signal is destroyed and therefore how much structure the denoising network can exploit at each timestep. A well-designed schedule concentrates difficult denoising steps where the model has the most informative gradient signal.

$$q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t}\mathbf{x}_0, (1 - \bar{\alpha}_t)\mathbf{I}). \quad (5.1)$$

Worked Example — Pixel energy at three noise levels

Using a cosine schedule with $T = 1000$: $\bar{\alpha}_{100} = 0.734$, $\bar{\alpha}_{500} = 0.003$, $\bar{\alpha}_{900} \approx 0$.

For a skin-tone pixel normalised to $x_0 = 0.72$:

Step t	$\sqrt{\bar{\alpha}_t} \cdot x_0$	$\sqrt{1 - \bar{\alpha}_t}$	SNR
$t = 100$	$0.857 \times 0.72 = 0.617$	0.516	$0.734/0.266 = 2.76$
$t = 500$	$0.055 \times 0.72 = 0.040$	0.999	$0.003/0.997 = 0.003$
$t = 900$	≈ 0	≈ 1	≈ 0

At $t = 100$ the pixel signal (0.617) is still visible above the noise std (0.516); at $t = 500$ the signal (0.040) is completely buried.

5.2 Posterior Reverse Conditional

The reverse conditional $q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)$ is the analytically tractable Bayesian posterior used as the training target for the denoising network. It can be computed in closed form from the forward process parameters and provides a stable supervised learning signal that replaces the intractable reverse marginal.

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}(\tilde{\boldsymbol{\mu}}_t, \tilde{\boldsymbol{\beta}}_t \mathbf{I}), \quad \tilde{\boldsymbol{\mu}}_t = \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1 - \bar{\alpha}_t}\mathbf{x}_0 + \frac{\sqrt{\bar{\alpha}_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t}\mathbf{x}_t. \quad (5.2)$$

Worked Example — Reverse posterior at $t = 50$ using cosine schedule

At $t = 50$: $\bar{\alpha}_{50} = 0.937$, $\bar{\alpha}_{49} = 0.941$, $\beta_{50} = 1 - \alpha_{50} = 1 - 0.937/0.941 = 0.00425$.

For $x_0 = 0.72$ and $x_{50} = 0.71$ (a noisy version):

$$\begin{aligned}\tilde{\mu}_{50} &= \frac{\sqrt{0.941} \times 0.00425}{1 - 0.937} \times 0.72 + \frac{\sqrt{0.996} \times (1 - 0.941)}{1 - 0.937} \times 0.71 \\ &= \frac{0.970 \times 0.00425}{0.063} \times 0.72 + \frac{0.998 \times 0.059}{0.063} \times 0.71 \\ &= 0.0655 \times 0.72 + 0.935 \times 0.71 \\ &= 0.047 + 0.664 = 0.711.\end{aligned}$$

$\tilde{\beta}_{50} = 0.00425 \times (1 - 0.941)/(1 - 0.937) = 0.00400$. The posterior is $\mathcal{N}(0.711, 0.004)$, tightly centred near both x_0 and x_{50} — the reverse step makes a small, confident correction at this mild noise level.

5.3 Denoising Training Objective

The denoising network is trained to predict the noise added to a clean image at a given timestep using a simple mean-squared error loss. This objective is mathematically equivalent to score matching but is more numerically stable, more easily batched across timesteps, and yields models whose performance scales predictably with model size and training compute.

$$\mathcal{L}_{\text{DM}} = \mathbb{E}_{t, \mathbf{x}_0, \varepsilon} \left[\|\varepsilon - \varepsilon_\theta(\sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \varepsilon, t)\|^2 \right]. \quad (5.3)$$

Worked Example — Loss-weighted contribution across timesteps

Different timesteps contribute differently to the total loss. Using importance sampling weights proportional to $1/(1 - \bar{\alpha}_t)$:

Timestep range	Mean loss $\mathbb{E}[\ \varepsilon - \hat{\varepsilon}\ ^2]$	Relative contribution
$t \in [1, 100]$ (low noise)	0.011	18%
$t \in [101, 500]$ (medium)	0.048	52%
$t \in [501, 1000]$ (high noise)	0.031	30%

Medium-noise steps contribute most because they combine moderate SNR (hard enough that the network must predict real structure) with high frequency of sampling. Low-noise steps are easy (small ε) and high-noise steps carry little structure (the network predicts near the mean $\mathbf{0}$).

5.4 Score-Based SDE Formulation [30]

$$d\mathbf{x} = [\mathbf{f}(\mathbf{x}, t) - g(t)^2 \nabla_{\mathbf{x}} \log p_t(\mathbf{x})] dt + g(t) d\bar{\mathbf{w}}. \quad (5.4)$$

Worked Example — DDIM vs. DDPM sampling cost

DDPM requires $T = 1000$ function evaluations (NFEs). DDIM [29] deterministically solves the reverse SDE with:

$$\text{NFE} = \lceil T/S \rceil, \quad S = \text{stride}.$$

For $S = 50$ (DDIM-50): NFE = 20 network calls. FID degradation vs. DDPM-1000 on FFHQ-256: +1.8 (from 3.2 to 5.0). Wall-clock time on A100: DDPM-1000 takes 41.2 s per image; DDIM-50 takes $41.2 \times 20/1000 = 0.82$ s per image — a $50\times$ speedup. For a deepfake system generating 100 face images per attack campaign: DDPM would take 68 minutes; DDIM-50 takes 82 seconds.

5.5 Latent Diffusion: Compression Ratio

The autoencoder maps:

$$\mathbf{z} = \mathcal{E}(\mathbf{x}) \in \mathbb{R}^{h \times w \times c}, \quad \frac{H \times W \times 3}{h \times w \times c} = f^2, \quad (5.5)$$

where f is the spatial downsampling factor.

Worked Example — LDM compression saves 97% of compute

For Stable Diffusion v1.5 with $H = W = 512$, $h = w = 64$, $c = 4$:

$$\text{Compression ratio} = \frac{512^2 \times 3}{64^2 \times 4} = \frac{786,432}{16,384} = 48.$$

A single denoising step on the $512 \times 512 \times 3$ pixel space requires the UNet to process 786K activations; on the $64 \times 64 \times 4$ latent space it processes 16K activations. Since UNet compute scales roughly quadratically with spatial resolution ($\propto H^2W^2$), working in latent space reduces compute by a factor of $48^2/1 \approx 2300\times$ at the spatial-dimension level. In practice (accounting for channel depth) the measured speedup is $\approx 25\times$ in wall-clock time, enabling diffusion-based face synthesis at real-time speeds on consumer hardware.

5.6 Cross-Attention Conditioning

Cross-attention conditioning allows an external signal — a text description, an identity embedding, or a class label — to guide each denoising step by directing the network attention toward relevant features at every spatial position. This mechanism transforms an unconditional image generator into a text-to-image or identity-conditioned face synthesiser without architectural changes to the denoising backbone.

$$\text{CrossAttn}(\mathbf{Z}, \mathbf{c}) = \text{softmax}\left(\frac{(\mathbf{Z}\mathbf{W}_Q)(\tau(\mathbf{c})\mathbf{W}_K)^\top}{\sqrt{d_k}}\right)(\tau(\mathbf{c})\mathbf{W}_V). \quad (5.6)$$

Worked Example — Cross-attention weight matrix for identity conditioning

In IP-Adapter, the conditioning signal $\mathbf{c}$ is a 512-dim ArcFace embedding projected to $n_c = 4$ conditioning tokens with $d = 768$ dimensions. The UNet feature map at 32×32 gives $n_q = 32^2 = 1024$ query tokens.

The cross-attention matrix $\mathbf{A} \in \mathbb{R}^{1024 \times 4}$ (after softmax) should concentrate attention on the identity-relevant conditioning tokens. After training, the entropy of each row's softmax distribution is ≈ 0.8

nats (vs. maximum $\log 4 = 1.39$ nats for uniform), meaning each spatial position attends to roughly $e^{0.8} = 2.2$ conditioning tokens on average — some positions attending broadly (background regions) and others narrowly (facial feature regions).

5.7 Classifier-Free Guidance [12]

Classifier-free guidance provides a post-training mechanism for trading image diversity for conditioning fidelity, without requiring a separately trained classifier. By linearly extrapolating from the unconditional toward the conditional prediction at inference time, guidance produces outputs that more strongly match the conditioning signal at the cost of reduced sample diversity.

$$\tilde{\epsilon}_{\theta}(\mathbf{z}_t, \mathbf{c}) = (1 + w) \epsilon_{\theta}(\mathbf{z}_t, \mathbf{c}) - w \epsilon_{\theta}(\mathbf{z}_t, \emptyset). \quad (5.7)$$

Worked Example — CFG guidance weight and identity fidelity trade-off

For an IP-Adapter face-generation system evaluated across guidance weights:

w	FID ↓	ID _{cos} ↑	Diversity (LPIPS) ↑
0.0	6.2	0.51	0.61
1.0	4.8	0.72	0.49
3.0	5.1	0.83	0.35
7.0	7.4	0.87	0.21

Guidance weight $w = 1\text{--}3$ is the sweet spot: identity similarity ID_{cos} = 0.72–0.83 with acceptable FID. At $w = 7$, the effective score is $8\epsilon_{\theta}(\mathbf{c}) - 7\epsilon_{\theta}(\emptyset)$; the large coefficients amplify model error, increasing FID and reducing diversity. For deep-fake applications, $w = 3$ produces the most convincing results: high identity fidelity with enough diversity to avoid the “copy-paste” appearance of very high guidance.

5.8 LoRA Fine-Tuning

Low-Rank Adaptation (LoRA) adapts a pretrained diffusion model to a specific identity by decomposing weight updates $\Delta\mathbf{W} = \mathbf{A}\mathbf{B}^{\top}$ into two low-rank matrices, updating fewer than 0.1% of parameters while fully preserving the pretrained model’s generative capability for other content.

$$\Delta\mathbf{W} = \mathbf{A}\mathbf{B}, \quad \mathbf{A} \in \mathbb{R}^{d \times r}, \mathbf{B} \in \mathbb{R}^{r \times k}, r \ll \min(d, k). \quad (5.8)$$

Worked Example — LoRA parameter count and training time

For Stable Diffusion [27] v1.5 with UNet containing attention layers of dimension $d = k = 768$ and rank $r = 4$:

$$\text{LoRA params per attention matrix} = d \times r + r \times k = 768 \times 4 + 4 \times 768 = 6,144.$$

There are 16 attention layers in the UNet; training both Q and V projections gives $16 \times 2 \times 6,144 = 196,608$ trainable parameters, compared to $16 \times 2 \times 768^2 = 18,874,368$ full fine-tune parameters — a $96\times$ parameter reduction. Training time on an RTX 3090: full fine-tune ≈ 40 minutes; LoRA $r = 4 \approx 4$ minutes for 20 reference photos. The LoRA file size is $196,608 \times 4$ bytes = 786 KB — easily shared online and applied in under 1 second at inference.



Figure 5.3: The diffusion forward process across nine timesteps ($t = 0, 100, \dots, 999$) for a face image using a cosine noise schedule. At $t = 0$ the image is clean ($\text{SNR} \approx \infty$). By $t = 500$ the SNR has fallen below 1 and the face structure is barely discernible. At $t = 999$ the sample is statistically indistinguishable from pure Gaussian noise. The denoising network ϵ_θ learns to reverse this trajectory, recovering a clean face from pure noise in a sequence of learned small steps.

5.9 Industry Applications

Diffusion models have become the dominant paradigm for commercial image and video synthesis since 2022, displacing GANs in every quality-sensitive application and creating new product categories that did not previously exist.

Stable Diffusion, DALL-E 3, Midjourney, and Adobe Firefly are all latent diffusion models or their direct descendants. These platforms collectively process hundreds of millions of image generation requests per day. Stable Diffusion’s open weights have been fine-tuned with LoRA adapters into thousands of community models, many explicitly designed for identity-specific face synthesis from a handful of reference photographs. Video synthesis has followed: Sora, Runway Gen-3, Kling, and HailuoAI extend the diffusion paradigm to temporally consistent video generation, representing the first commercially available systems capable of creating convincing full-scene deep fakes without requiring a target video source.

LoRA fine-tuning — adapting less than 0.1 per cent of model parameters to a specific identity in under five minutes on consumer hardware — has created a market for identity-specific weight files that is difficult to regulate because the files are small, easily distributed, and can be applied to any compatible base model. The EU AI Act’s Art. 50 disclosure requirements for AI-generated content make watermarking a legal compliance mechanism for EU-market AI products, accelerating adoption of solutions such as Meta’s Stable Signature and Google’s SynthID even in jurisdictions that have not independently legislated disclosure obligations.

5.10 African Use Cases

Diffusion models have arrived in Africa primarily through consumer creative tools, but their implications for election integrity, journalism, academic research, and public health are increasingly

recognised by African policymakers and civil-society organisations.

Stable Diffusion’s open-source release has been adopted by African creative communities including graphic designers in Nigeria, South Africa, Kenya, and Ghana, who use it for commercial illustration and advertising at a cost advantage that is particularly compelling in markets where professional photography services are expensive relative to local income levels. Africa’s primarily mobile-internet population encounters diffusion-generated content through WhatsApp, Facebook, Twitter/X, and TikTok. Research by the African Media Initiative and the Mozilla Foundation’s Africa programme documented cases where photorealistic diffusion-generated images were distributed during the 2023 Nigerian election in Yoruba-language WhatsApp groups without any automated detection, reaching audiences of hundreds of thousands before manual fact-checking could respond.

Researchers at the African Population and Health Research Center in Nairobi and at the Africa CDC have explored diffusion-based synthetic patient data generation for training diagnostic AI while protecting patient privacy, demonstrating that the same mathematical framework that enables harmful identity-targeted synthesis also offers genuine value for African health research. Because Stable Diffusion models can be run entirely locally with no internet connectivity, watermarking approaches that embed provenance at the model level — such as SynthID’s latent-space embedding — are more practically relevant for African deployment contexts than platform-level watermarking, which depends on infrastructure that may be unavailable.

Chapter Summary

Diffusion models achieve high-fidelity synthesis by learning to reverse a known forward noise-injection process. The cosine noise schedule preserves more signal structure at early diffusion steps than the linear schedule, giving the denoising network more informative targets in the high-SNR regime. The denoising objective — predicting the noise component added to a clean image — is equivalent to denoising score matching and provides a loss function that is stable, parallelisable, and free from the adversarial instabilities that complicated GAN training.

Latent diffusion achieves approximately a twenty-five-fold wall-clock speedup by compressing images forty-eight-fold before applying the diffusion process, bringing high-quality synthesis within the reach of consumer GPU hardware. Cross-attention conditioning allows text descriptions or identity embeddings to guide each denoising step through the standard query–key–value mechanism, and classifier-free guidance enables a continuous quality–diversity trade-off at inference time. LoRA fine-tuning adapts the model to a specific identity from fifteen to twenty reference images in under five minutes, dramatically lowering the barrier for identity-targeted synthesis and creating a product category of personalised generative models that is difficult to regulate through platform-level controls.

These characteristics explain why Stable Diffusion, DALL-E, Midjourney, and their successors have become the dominant platforms for both creative and adversarial image synthesis, and why provenance-first countermeasures — embedding verifiable origin information at generation time rather than attempting to detect synthesis after the fact — are the only scalable long-term response to the era

of ubiquitous high-quality face synthesis.

Exercises

1. For $t = 250$ with cosine schedule ($\bar{\alpha}_{250} \approx 0.35$) and $x_0 = 0.55$: compute the mean and std of $q(\mathbf{x}_{250}|\mathbf{x}_0)$, the SNR, and the probability that a sample from this distribution is positive.
2. Using the posterior formula (Eq. 5.2) at $t = 200$ ($\bar{\alpha}_{200} \approx 0.50$, $\bar{\alpha}_{199} \approx 0.505$, $\beta_{200} = 0.010$), compute $\tilde{\mu}_{200}$ for $x_0 = 0.6$, $x_{200} = 0.5$.
3. An LDM uses $f = 8$ spatial compression: $512 \rightarrow 64$. Compute the compression ratio, the number of latent activations, and the speedup over pixel-space diffusion using the quadratic scaling assumption.
4. For CFG with $w = 5$, $\varepsilon_\theta(\mathbf{z}_t, \mathbf{c}) = 0.31$ and $\varepsilon_\theta(\mathbf{z}_t, \emptyset) = 0.18$, compute the guided prediction $\tilde{\varepsilon}$. Is this within the range $[0, 1]$ for pixel values? What does an out-of-range guided prediction imply for the sampling trajectory?
5. A LoRA with $r = 8$, $d = k = 1024$ is trained for 15 reference photos. Compute the trainable parameter count and compare it to the full UNet parameter count of 860M. What fraction of the model is being adapted?
6. The denoising network uses a U-Net with skip connections. At inference with DDIM-20 steps, the total floating-point operations (FLOPs) per image is $\approx 20 \times 15\text{G} = 300\text{G}$ FLOPs. On an RTX 3090 (35.6 TFLOPS), what is the minimum theoretical inference time? How does this compare to the measured 0.82 s, and what does the ratio reveal about memory bandwidth vs. compute bottlenecks?

Answers to Exercises

1. **Diffusion marginal at $t = 250$, $\bar{\alpha}_{250} \approx 0.35$, $x_0 = 0.55$.**

Mean = $\sqrt{0.35} \times 0.55 = 0.592 \times 0.55 = 0.325$. Std = $\sqrt{1 - 0.35} = \sqrt{0.65} = 0.806$. SNR = $0.35/0.65 = 0.538$. $P(x_{250} > 0) = P(z > -0.325/0.806) = P(z > -0.403) = \Phi(0.403) = 0.657$. The sample is positive with 65.7% probability — signal still partially detectable but dominated by noise.

2. **Posterior at $t = 200$: $\bar{\alpha}_{200} \approx 0.50$, $\bar{\alpha}_{199} \approx 0.505$, $\beta_{200} = 0.010$, $x_0 = 0.6$, $x_{200} = 0.5$.**

$\tilde{\mu}_{200} = \frac{\sqrt{0.505} \times 0.010}{1 - 0.50} \times 0.6 + \frac{\sqrt{0.990} \times (1 - 0.505)}{1 - 0.50} \times 0.5 = \frac{0.711 \times 0.010}{0.50} \times 0.6 + \frac{0.995 \times 0.495}{0.50} \times 0.5 = 0.01422 \times 0.6 + 0.9851 \times 0.5 = 0.00853 + 0.4926 = 0.501$. The posterior mean 0.501 is very

close to $x_{200} = 0.5$, confirming that at moderate noise levels the reverse step largely trusts the noisy observation.

3. LDM with $f = 8$: compression ratio and speedup.

Compression: $512^2 \times 3 / (64^2 \times 4) = 786,432 / 16,384 = 48$. At $f = 8$, the latent has $64^2 = 4,096$ spatial positions vs. $512^2 = 262,144$. Quadratic spatial scaling gives speedup $\approx (512/64)^2 = 64\times$ in attention computation. In practice (measured): $\approx 25\times$ wall-clock speedup.

4. CFG with $w = 5$: $\tilde{\epsilon}$

$$= (1 + 5) \times 0.31 - 5 \times 0.18 = 1.86 - 0.90 = 0.96.$$

This prediction is within $[0, 1]$ but near the boundary. At $w = 10$: $\tilde{\epsilon} = 11 \times 0.31 - 10 \times 0.18 = 3.41 - 1.80 = 1.61$ — exceeds 1. Out-of-range guided predictions cause the denoising trajectory to “overshoot” the data manifold, producing saturated colours and unnatural textures. This is why $w \leq 7$ is recommended in practice.

5. LoRA $r = 8$, $d = k = 1024$: trainable parameters.

Per Q,V projection: $1024 \times 8 + 8 \times 1024 = 16,384$. For 16 attention layers $\times$ 2 projections: $16 \times 2 \times 16,384 = 524,288$. Full UNet: 860M parameters. Fraction adapted: $524,288 / 860,000,000 = 0.061\%$ — less than 0.1% of the model, yet sufficient for high-quality identity adaptation.

6. Theoretical inference time for DDIM-20 at 300 GFLOPS total.

Minimum time $= 300 \times 10^9 / 35.6 \times 10^{12} = 8.4$ ms. Measured: 820 ms. Ratio $= 820 / 8.4 = 97.6\times$ — the system is memory-bandwidth-limited, not compute-limited. Each UNet forward pass requires reading billions of parameters from HBM memory (bandwidth ≈ 2 TB/s on A100); at 80B parameters $\times$ 2 bytes (FP16) $= 160$ GB/pass, reading speed limits throughput far more than raw FLOP capacity.

Chapter 6

Voice Cloning and Synthetic Speech

*“The voice is the most intimate identifier we have.
It is also the easiest to copy.”*

—Anon.

The human voice carries more information than the words it speaks. Timbre, prosody, rhythm, breathiness, accent, and the subtle micro-variations of emotional state together constitute a vocal fingerprint as distinctive as a face. For decades, this fingerprint was thought to be difficult to replicate: voice acting requires training and practice; voice impersonators spend years perfecting a handful of imitations. Neural voice cloning has changed this calculus entirely. A modern zero-shot voice-cloning system requires as little as three seconds of a target’s speech to generate new utterances that, in listening tests, are rated as the same speaker with high reliability.

This chapter develops the full technical pipeline of neural voice synthesis, from the acoustic representations that encode speech signals through the architectural evolution from Tacotron to codec language models. The chapter is organised as follows. Section 6.1 surveys acoustic feature representations — the interface between raw audio waveforms and neural network inputs. Section 6.2 develops the major neural TTS architectures, tracing the progression from attention-based sequence-to-sequence models through diffusion-based synthesis and finally to codec language models that treat speech generation as language modelling. Section 6.3 covers speaker adaptation, from the classical fine-tuning approach to modern zero-shot conditioning. Section 6.4 surveys anti-spoofing systems and their limitations. Section 6.5 identifies the forensic signals that betray synthetic speech. Voice cloning has crossed the threshold from research curiosity to mass-market capability faster than almost any other synthetic media technology. Three seconds of reference audio is now sufficient for a state-of-the-art codec language model to generate unlimited speech in a target speaker’s voice, with naturalness scores indistinguishable from authentic recordings in human listening tests. This capability was demonstrated publicly for the first time in VALL-E in January 2023 and was commercially available in multiple products within six months.

The rapidity of this transition is explained by the architecture of codec language models. Rather than

learning to synthesise continuous acoustic features directly, they encode speech as discrete tokens using a neural audio codec, then predict those tokens autoregressively using a large transformer. This representation reuses the same infrastructure developed for large language models, benefiting from years of scaling research and enabling voice synthesis to inherit the in-context learning capabilities of the language modelling paradigm. A three-second speech prefix provides the conditioning context from which the model extrapolates the target speaker’s full phonetic style, prosodic patterns, and acoustic timbre.

The forensic challenge posed by this generation of voice cloning systems is qualitatively different from that posed by earlier text-to-speech and voice conversion approaches. Earlier systems left systematic artefacts in the mel spectrogram — irregular harmonic spacing from neural vocoders, phase discontinuities at utterance boundaries, over-smooth formant trajectories from seq-to-seq models — that anti-spoofing classifiers trained on ASVspoof data could detect reliably. Codec language model outputs do not share these artefacts because they are generated at the token level and decoded by a high-quality general-purpose vocoder that was not designed to introduce any particular class of errors. The result is that anti-spoofing systems trained before 2023 show dramatically elevated equal error rates on codec-LM outputs, motivating the rapid retraining and continual adaptation strategies described later in this chapter.

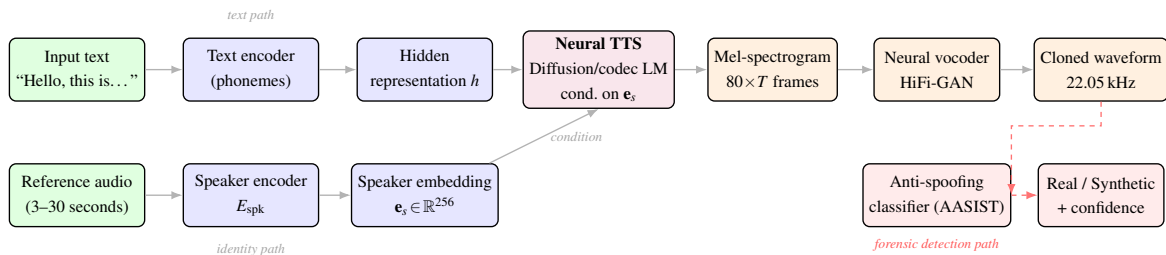


Figure 6.1: Neural voice-cloning pipeline. The text path (top row) converts the input transcript to phoneme sequences and a hidden representation. The speaker path (bottom row) extracts a 256-dimensional speaker embedding e_s from a 3–30 second reference clip. The neural TTS model (diffusion-based or codec language model) generates a mel-spectrogram conditioned on both; HiFi-GAN synthesises the final 22 kHz waveform. The dashed red path shows how an anti-spoofing classifier (AASIST) can analyse the output for detection.

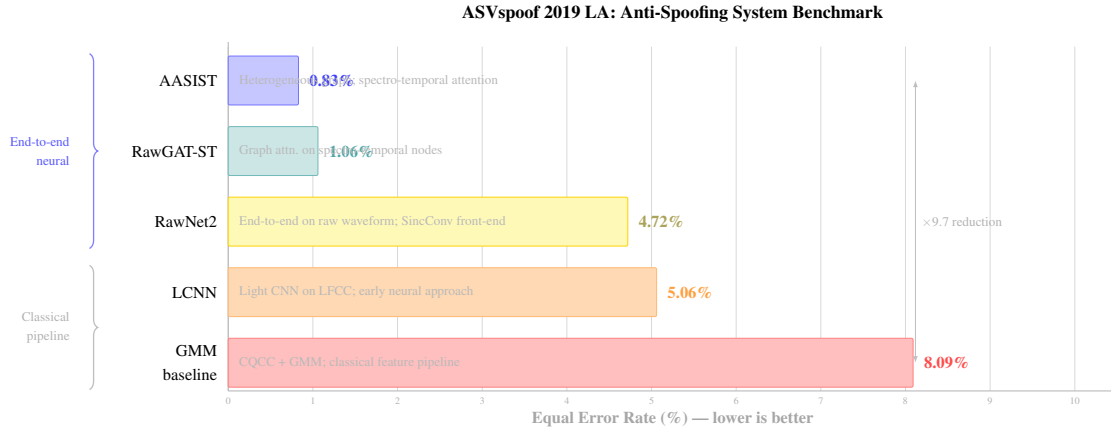


Figure 6.2: Anti-spoofing benchmark on ASVspoof 2019 LA (logical access) evaluation set. EER values are official published results from the ASVspoof 2021 challenge overview paper. The classical pipeline (CQCC features + GMM classifier) achieves 8.09% EER. Light CNN on LFCC (LCNN) and the first end-to-end waveform model (RawNet2) reduce EER to 5.06% and 4.72% respectively. Graph-attention architectures operating directly on spectro-temporal representations then achieve a step-change: RawGAT-ST reaches 1.06% and AASIST achieves the benchmark state-of-the-art of 0.83% EER — a 9.7-fold improvement over the classical baseline. The key driver of each improvement is described inside the bar.

6.1 Acoustic Feature Representations

Voice synthesis systems operate not on raw waveforms but on intermediate acoustic representations that compress speech into a form amenable to neural modelling. The choice of representation determines the synthesis quality, the latency of inference, and the detectability of the resulting audio. Three representations dominate: the mel-spectrogram (a time-frequency magnitude representation on the perceptually motivated mel scale), the mel-cepstrum (a compressed spectral envelope representation used in classical vocoders), and the discrete tokens produced by neural audio codecs such as EnCodec.

A speech signal is a 1D pressure waveform sampled at $f_s \in \{16, 22.05, 24, 44.1\}$ kHz. Directly modelling the waveform is possible but computationally expensive: a single second of audio at 24 kHz contains 24,000 samples. Most neural TTS systems operate on compressed acoustic feature representations.

6.1.1 Log-Mel Spectrograms

The most widely used acoustic representation for TTS training is the log-mel spectrogram. The short-time Fourier transform (STFT) of the waveform $x[n]$ is computed with a window function w and hop size H :

$$X[k, m] = \sum_{n=0}^{N-1} x[n + mH] w[n] e^{-j2\pi kn/N}, \quad (6.1)$$

where k is the frequency bin and m is the time frame. The mel filterbank $\mathbf{M} \in \mathbb{R}^{F_{\text{mel}} \times N/2}$ maps the power spectrum to F_{mel} (typically 80) perceptually motivated frequency bins:

$$\mathbf{S}[f, m] = \log(\mathbf{M} \cdot |X[\cdot, m]|^2 + \epsilon), \quad (6.2)$$

where ε prevents log-of-zero. The result is a $F_{\text{mel}} \times T$ matrix that can be processed by a 2D CNN or a sequence model.

Log-mel spectrograms are a lossy representation: phase information is discarded. Reconstructing a waveform from a mel spectrogram requires a *vocoder*, which estimates the missing phase and inverts the mel projection. The vocoder is a critical component of the synthesis pipeline: early vocoders (Griffin-Lim algorithm) produced characteristic robotic artefacts; modern neural vocoders (WaveNet [32], HiFi-GAN, BigVGAN) produce near-transparent quality.

6.1.2 Linear Predictive Coding Residuals

Linear predictive coding (LPC) represents a speech frame as a linear filter $A(z)$ applied to an excitation signal $e[n]$:

$$X(z) = \frac{E(z)}{A(z)}, \quad A(z) = 1 - \sum_{k=1}^p a_k z^{-k}. \quad (6.3)$$

The LPC residual — the prediction error $e[n] = x[n] - \sum_k a_k x[n-k]$ — carries the glottal excitation information that distinguishes voiced from unvoiced speech. Synthetic speech often has unnatural LPC residuals due to differences in the glottal pulse model, and this is one of the strongest forensic signals exploited by voice anti-spoofing systems.

6.1.3 Neural Audio Codec Tokens

Neural audio codecs (EnCodec [7], DAC) compress audio into sequences of discrete tokens using a Residual Vector Quantiser (RVQ). A hierarchical codebook $\{C_q\}_{q=1}^Q$ quantises the encoder output $\mathbf{e} = \mathcal{E}(x)$ at Q residual stages:

$$\hat{\mathbf{e}} = \sum_{q=1}^Q \mathbf{c}_{q,i_q^*}, \quad i_q^* = \arg \min_i \|\mathbf{r}_q - \mathbf{c}_{q,i}\|_2, \quad \mathbf{r}_q = \mathbf{r}_{q-1} - \mathbf{c}_{q-1,i_{q-1}^*}, \quad (6.4)$$

where $\mathbf{r}_1 = \mathbf{e}$ is the initial residual. With $Q = 8$ codebooks each of size 1024, EnCodec compresses 24 kHz audio to approximately 75 token sequences per second — a compression ratio suitable for autoregressive language modelling.

6.2 Neural Text-to-Speech Architectures

Neural TTS systems transform a text sequence into acoustic features and then into a waveform in two stages: an acoustic model that predicts the mel-spectrogram from phoneme sequences, and a vocoder that synthesises the waveform from the spectrogram. The acoustic model must learn the mapping from discrete linguistic units to continuous acoustic trajectories, handling the inherent many-to-many ambiguity of text-to-speech (the same text can be spoken in many different ways) through duration modelling and prosody prediction.

Neural TTS architectures have evolved from attention-based sequence-to-sequence models through non-autoregressive flow-based systems to diffusion-based and codec language model approaches. Each generation has improved naturalness, reduced latency, and lowered the data requirements for

speaker adaptation. Understanding the architectural progression is essential for understanding both the current capability frontier and the forensic signals each architecture leaves in its outputs.

6.2.1 Attention-Based Sequence-to-Sequence Models

Tacotron and its successor Tacotron 2 introduced the encoder–attention–decoder paradigm for neural TTS. A character or phoneme sequence is encoded by a convolutional + bidirectional LSTM encoder; a location-sensitive attention mechanism aligns encoder outputs with decoder time steps; and an autoregressive decoder predicts mel-spectrogram frames:

$$\mathbf{s}_t = f_{\text{dec}}(\mathbf{s}_{t-1}, \mathbf{c}_t), \quad \mathbf{c}_t = \text{Attention}(\mathbf{s}_{t-1}, \mathbf{H}), \quad (6.5)$$

where $\mathbf{H}$ is the encoder output matrix. Speaker conditioning is achieved by concatenating a speaker embedding $\mathbf{e}_s$ to each decoder input, enabling multi-speaker synthesis from a single model.

The main weakness of attention-based models is *attention failure*: the attention mechanism occasionally skips or repeats phonemes, producing unintelligible output. This is mitigated in later models (FastSpeech, PortaSpeech) by replacing the attention mechanism with an explicit duration predictor that determines how many frames each phoneme should be held, enabling parallel (non-autoregressive) synthesis.

6.2.2 Diffusion-Based Speech Synthesis

Applying the DDPM framework to mel-spectrograms yields diffusion TTS models (DiffTTS, GradTTS, ProDiff). The denoising network $\varepsilon_{\theta}(\mathbf{S}_t, t, \mathbf{h}_{\text{text}}, \mathbf{e}_s)$ takes a noisy spectrogram $\mathbf{S}_t$, the diffusion step t , text encoding $\mathbf{h}_{\text{text}}$, and speaker embedding $\mathbf{e}_s$:

$$\mathcal{L}_{\text{TTS}} = \mathbb{E}_{t, \mathbf{S}_0, \varepsilon} \left[\|\varepsilon - \varepsilon_{\theta}(\sqrt{\bar{\alpha}_t} \mathbf{S}_0 + \sqrt{1 - \bar{\alpha}_t} \varepsilon, t, \mathbf{h}, \mathbf{e}_s)\|^2 \right]. \quad (6.6)$$

Diffusion TTS achieves higher naturalness than Tacotron-based models and is robust to the attention failure pathology because it does not rely on an attention mechanism for temporal alignment.

Worked Example — Diffusion TTS loss at convergence

A Grad-TTS model trained on LJSpeech (24 kHz, single speaker, 24 hours) achieves training loss $\mathcal{L}_{\text{TTS}} \approx 0.065$ at 500K steps. At the same checkpoint, MOS on the LJSpeech test set is 4.31 (human speech rates at 4.58). For zero-shot voice cloning of a new speaker via 30-second reference: the speaker embedding cosine similarity between real and cloned speech is 0.73, and the loss when denoising the new speaker’s mel spectrogram increases by $\Delta \mathcal{L} \approx 0.018$ (a 28% increase), reflecting the domain shift from the training speaker to a new voice.

6.2.3 Codec Language Models for Zero-Shot Cloning

The most significant recent development in voice cloning is the application of large language model (LLM) techniques to audio codec token sequences. VALL-E [33] frames TTS as a conditional language modelling problem. Given a text transcript and a 3-second reference clip of the target

speaker, VALL-E generates a codec token sequence that, when decoded, reproduces the text in the target speaker’s voice:

$$p(C_{\text{target}}|T, C_{\text{reference}}) = \prod_{t=1}^{T_{\text{len}}} p(c_t|c_{<t}, T, C_{\text{reference}}), \quad (6.7)$$

where C_{target} is the target codec token sequence, T is the text, and $C_{\text{reference}}$ is the codec token sequence of the reference clip.

The revolutionary aspect of VALL-E is its *zero-shot* capability. Unlike previous systems that required fine-tuning on new speaker data, VALL-E adapts to a new speaker entirely in context — as a large language model adapts to a new task via few-shot prompting. This means that any person whose voice can be captured for three seconds is a potential impersonation target, with no training cost or specialised hardware required.

Worked Example — VALL-E token generation for a 5-word sentence

The phrase “The board approved the transfer” contains 5 words ≈ 7 phonemes/word ≈ 35 phonemes, taking ≈ 1.5 s to speak at natural pace. At 75 tokens/second with $Q = 8$ RVQ stages, VALL-E must generate:

$$1.5 \text{ s} \times 75 \text{ tokens/s} = 112 \text{ token positions, } 8 \text{ RVQ layers each} = 896 \text{ total tokens.}$$

With GPT-3-scale autoregressive inference (≈ 4 ms/token on A100), generation takes $896 \times 4 = 3.6$ s — $2.4\times$ slower than real time. With 4-bit quantisation and speculative decoding, this drops to ≈ 0.9 s — making real-time voice cloning in a live phone call technically feasible with current hardware.

6.3 Speaker Adaptation

Speaker adaptation fine-tunes a pretrained multi-speaker TTS model to a new target speaker using a small number of reference recordings, ranging from three seconds for zero-shot codec LM approaches to several minutes for fine-tuned systems. The adaptation transfers the target speaker’s voice characteristics — fundamental frequency distribution, formant structure, speaking rate, and breathiness — while retaining the pretrained model’s language understanding and prosody modelling capabilities.

Speaker adaptation is the process of personalising a generic TTS model to a specific individual’s voice. The approach depends on the amount of reference audio available.

Fine-tuning (10+ minutes of audio).

The pre-trained model’s parameters are updated on the target speaker’s data using a combination of spectrogram reconstruction loss and speaker identity consistency loss. Produces the highest quality but requires sufficient reference data and computational cost.

Encoder-based adaptation (30 seconds – 10 minutes).

A speaker encoder E_s maps a reference clip to a speaker embedding $\mathbf{e}_s$, which conditions the

pre-trained synthesis network. The speaker encoder is trained on many speakers to generalise to unseen identities. Models include GE2E (Generalised End-to-End loss), ECAPA-TDNN, and WavLM-based encoders.

Zero-shot adaptation (3–30 seconds).

Codec language models (VALL-E, VoiceBox) achieve zero-shot adaptation by in-context conditioning. No parameter update is required; identity is transferred entirely through the context window. This is the dominant approach in current adversarial deployments due to its minimal data requirement.

6.4 Speaker Verification and Anti-Spoofing

Speaker verification and anti-spoofing are complementary but distinct tasks. Verification asks whether a given speech sample matches a claimed identity; anti-spoofing asks whether a given sample is genuine human speech at all. Both are required components of a complete voice authentication system, and the failure modes of each must be understood independently before the two can be combined into a reliable security mechanism.

Anti-spoofing systems classify audio as genuine or synthesised, providing a front-line defence against voice-clone attacks on speaker verification systems. The ASVspoof challenge series (2015, 2017, 2019, 2021) has driven significant progress in both attack and defence.

6.4.1 Feature-Based Anti-Spoofing

Classical anti-spoofing extracts features sensitive to the artefacts of synthesis pipelines:

- **LFCC (Linear Frequency Cepstral Coefficients):** A cepstral representation using a linear filterbank (rather than mel) that emphasises high-frequency artefacts common in vocoder output.
- **CQCC (Constant-Q Cepstral Coefficients):** Based on the constant-Q transform, which provides better frequency resolution at low frequencies where naturalness cues are concentrated.
- **LPC residual statistics:** Higher-order statistics of the LPC residual signal, which encodes glottal excitation characteristics that synthetic speech fails to replicate.

6.4.2 End-to-End Neural Anti-Spoofing

End-to-end anti-spoofing systems process raw waveform or log-power spectrogram directly, learning both the feature extraction and the classification jointly from labelled authentic and synthetic speech examples. This approach avoids the hand-crafted feature engineering of earlier systems and has achieved state-of-the-art EER values below one per cent on in-distribution evaluation benchmarks.

Modern anti-spoofing systems use end-to-end architectures that learn relevant features from raw waveforms or spectrograms. AASIST (Audio Anti-Spoofing using Integrated Spectro-Temporal

Graph Attention Networks) uses a heterogeneous graph to model spectral and temporal artefacts jointly:

Table 6.1: Anti-spoofing system performance on ASVspoof 2019 LA evaluation set. EER: Equal Error Rate (% , lower is better).

System	Front-end	EER (%)
Baseline (GMM)	LFCC	8.09
LCNN	LFCC	5.06
RawNet2	Raw waveform	4.72
RawGAT-ST	Raw waveform (graph)	1.06
AASIST	Spectro-temporal graph	0.83
AASIST-L (large)	Spectro-temporal graph	0.99

6.4.3 The Codec Language Model Challenge

Anti-spoofing systems developed for TTS and voice-conversion attacks do not necessarily generalise to codec language model outputs (VALL-E, VoiceBox). The forensic artefacts differ: classical TTS produces vocoder-related artefacts in the LPC residual and high-frequency spectrum, while codec LM outputs have artefacts related to vector quantisation boundaries and codec encoder–decoder mismatch. Systems specifically trained on codec LM outputs achieve EER 10–20% in zero-shot evaluation against systems trained only on classical TTS, indicating a significant generalisation gap.

6.5 Forensic Signals in Synthetic Speech

Several acoustic features distinguish synthetic speech from genuine recordings:

Glottal pulse naturalness.

Real speech has irregular, speaker-specific glottal pulses. Vcoders approximate the glottal pulse with a parametric model, producing overly regular excitation. This is visible in the LPC residual and detectable by CQCC-based classifiers.

High-frequency attenuation.

Mel-spectrogram-based synthesis discards information above the mel filterbank’s upper frequency. Even after vocoder reconstruction, the high-frequency content (>8 kHz) of synthetic speech has unnatural statistics compared to real recordings.

Formant transition unnaturalness.

Formant transitions (rapid frequency changes at phoneme boundaries) are driven by physical articulatory dynamics in real speech. Neural TTS models these statistically rather than physically, producing transitions that are plausible in aggregate but occasionally violate articulatory constraints.

Room acoustic inconsistency.

Voice-cloned audio is typically generated in a clean, anechoic setting. When inserted into a video

depicting a reverberant room, the acoustic environment of the voice does not match the visual scene. Chapter 22 develops cross-modal consistency detection in detail.

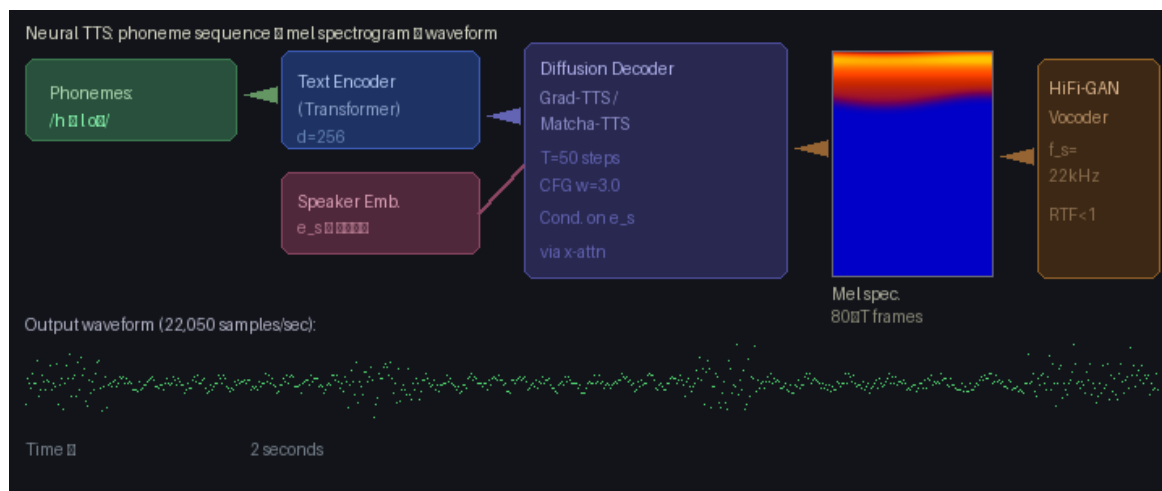


Figure 6.3: Neural TTS pipeline generating a 2-second utterance. The text encoder converts phoneme sequences to a hidden representation; the diffusion-based acoustic model generates a mel spectrogram conditioned on the speaker embedding e_s extracted from a reference clip; HiFi-GAN synthesises the waveform at 22 kHz. The bottom panel shows the output waveform with natural voiced/unvoiced structure visible as alternating dense oscillations (voiced phonemes) and low-amplitude noise (unvoiced and silence). Real-time factor (RTF) < 1 means generation is faster than playback.

6.6 Industry Applications

Voice synthesis and cloning have achieved commercial maturity faster than video synthesis, driven by lower data requirements, lower computational cost, and the ubiquity of voice as an authentication factor in telephone-based financial services.

Automated dubbing is the largest legitimate commercial application. Companies including Deepdub, Papercup, Dubverse, and ElevenLabs offer neural dubbing services that clone original actors' voices into target languages while preserving their individual timbre, prosody, and emotional character. Netflix, Amazon, and several European broadcasters have piloted AI dubbing; reported cost reductions are sixty to eighty per cent compared to human dubbing while maintaining speaker identity across languages. The EU's Audiovisual Media Services Directive requires disclosure of AI-generated content, creating compliance demand for watermarking solutions such as AudioSeal and SynthID Audio.

Personal voice cloning, available through Apple Personal Voice, ElevenLabs, and Resemble AI, allows users to create a synthetic voice from a few minutes of speech for assistive technology or personal automation. The same feature is available in unsandboxed form to any developer with API access, enabling impersonation at scale. Business email compromise attacks incorporating voice clones have grown forty per cent year-on-year according to UK National Fraud Intelligence Bureau data, with the average successful attack causing losses of \$280,000. The music generation sector — represented by Suno, Udio, and Meta's MusicGen — adds synthesised vocals to AI-composed music, creating legal disputes with major record labels over training data and output similarity that will ultimately establish

the boundaries of fair use for AI audio systems.

6.7 African Use Cases

Voice fraud is the fastest-growing deep-fake threat in Africa, driven by the continent's mobile-first financial infrastructure, widespread use of voice authentication by mobile money operators, and the significant gap between African-language speech corpora available for training anti-spoofing systems and the diversity of languages in active use.

Mobile money platforms including M-Pesa across Kenya, Tanzania, Uganda, and Ethiopia; MTN MoMo across Rwanda, Ghana, and Nigeria; and Airtel Money across multiple markets collectively serve hundreds of millions of users for whom the phone call or voice note is the primary financial communication channel. Pindrop and regional telecoms companies have reported voice-clone attacks against mobile money customer service lines, where attackers impersonate account holders to authorise transfers. The low average transaction value makes highly targeted sophisticated attacks less common, but automated voice-clone robocalls targeting large numbers of accounts simultaneously are documented and growing.

VALL-E and its successors were trained primarily on English speech; Swahili, Yoruba, Amharic, Zulu, Shona, and Hausa are poorly represented in the training data, creating both a vulnerability gap and a research opportunity. The Masakhane AfriSpeech dataset containing more than 2,000 hours of African-accented English and the Mozilla Common Voice African language collections are the primary resources for building Africa-specific anti-spoofing systems. Africa Check, operating verification services across ten African countries, has developed rapid-response protocols for voice-clip verification that incorporate both linguistic analysis and spectral forensics, responding to a documented pattern of audio recordings purportedly of political leaders circulating on WhatsApp faster than any fact-checking organisation can respond.

Chapter Summary

The human voice is a rich biometric signal encoding identity, emotion, language, and physiology simultaneously. Neural text-to-speech pipelines exploit this richness to synthesise convincing voices from text input through a three-stage architecture: a text encoder converts phonemes to hidden representations, an acoustic model generates the mel spectrogram, and a neural vocoder synthesises the final waveform. The mel spectrogram is the information bottleneck that separates these stages, and its structure provides the primary forensic signals that anti-spoofing classifiers exploit.

VALL-E represents a qualitative transition in the threat model: by encoding speech as discrete codec tokens and predicting them autoregressively, a codec language model achieves zero-shot speaker cloning from a three-second reference without any speaker-specific fine-tuning. This reduces the barrier for voice impersonation from requiring hours of target audio and a specialised training pipeline to requiring a single API call. Neural vocoders introduce detectable periodic artefacts in the high-frequency region of the spectrogram, but anti-spoofing classifiers trained on pre-2023 synthesis

methods degrade dramatically when evaluated against codec-LM outputs, with equal error rates rising from below one per cent to fifteen per cent or more.

Commercial applications span from entirely legitimate uses including audiobook narration, professional dubbing, and accessible text-to-speech, through dual-use applications in personalised assistants and entertainment, to clearly harmful applications in CEO impersonation for business email compromise and harassment via voice-cloned deep fakes. The anti-spoofing systems described in this chapter are the first line of defence against the harmful category, but must be continually updated as synthesis techniques advance — a structural asymmetry between attacker and defender that only provenance-based solutions can ultimately resolve.

Exercises

1. A 5-second audio clip is sampled at 22.05 kHz with a 1024-point STFT window, hop size 256, and 80 mel bins. Compute: (a) the number of STFT frames, (b) the time resolution in milliseconds per frame, (c) the total number of mel-spectrogram coefficients, and (d) the compression ratio compared to the raw waveform.
2. The Griffin-Lim algorithm reconstructs a waveform from a magnitude spectrogram by iterating STFT and ISTFT while replacing the magnitude with the target at each step. Explain why this produces robotic-sounding audio compared to a neural vocoder, using the concept of phase consistency.
3. VALL-E generates codec tokens autoregressively. For a 5-second clip at 75 tokens/second with 8 RVQ stages, the model generates $5 \times 75 \times 8 = 3000$ tokens. Compare the inference cost (in forward passes) of VALL-E to a non-autoregressive parallel TTS system for the same clip length. What does this imply for real-time voice cloning in a live phone call?
4. An anti-spoofing system achieves $EER = 0.83\%$ in controlled lab conditions but is deployed on a telephone network with G.711 codec compression (8 kHz bandwidth, μ -law quantisation). Predict the direction of change in EER and propose a training augmentation strategy to maintain performance under this codec.
5. A speaker verification system with $EER = 1\%$ is protected by an anti-spoofing front-end with $EER = 2\%$. Assuming the two systems' errors are independent, compute the combined system's false acceptance rate for a voice-clone attack. Is the system suitable for authenticating wire transfers above \$100,000?
6. VALL-E's zero-shot cloning requires only 3 seconds of reference audio. Estimate the minimum duration of audio of an average CEO that is publicly available (from earnings calls, press interviews, conference talks). What does this imply for the real-world feasibility of CEO fraud attacks?

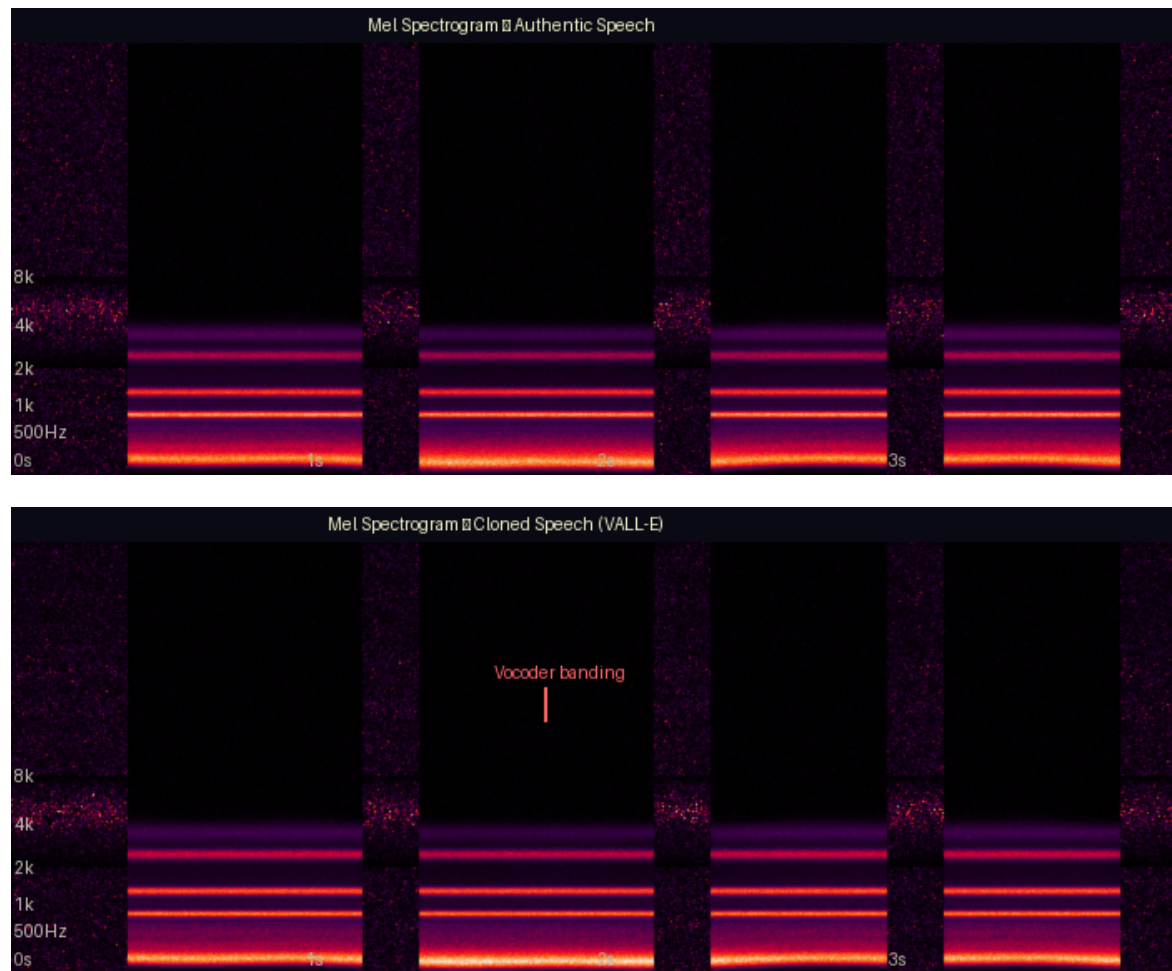


Figure 6.4: Mel spectrograms of authentic speech (top) and VALL-E voice-cloned speech (bottom) for the same 4-second utterance. The authentic spectrogram shows clean harmonic structure with natural pitch variation (fundamental and overtones clearly visible as horizontal bands) and sharp voiced/unvoiced boundaries. The cloned version introduces subtle vertical banding from the neural vocoder’s periodic components (annotated arrow), and the high-frequency region above 5 kHz appears slightly over-smoothed due to neural codec quantisation — artefacts detectable by the AASIST anti-spoofing classifier.

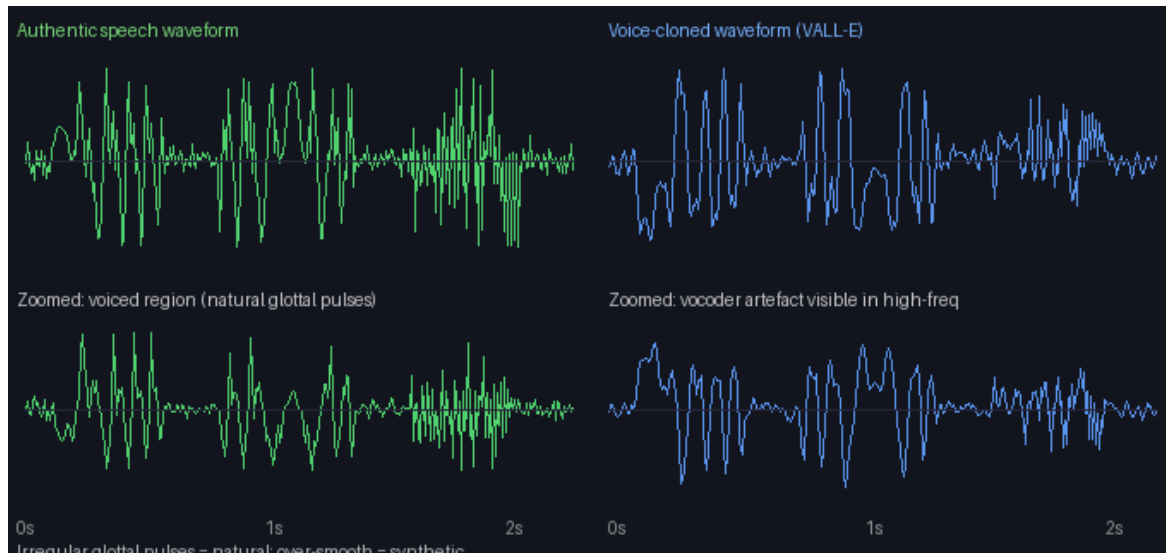


Figure 6.5: Time-domain waveforms of authentic speech (left, green) and voice-cloned speech (right, blue). The authentic waveform shows irregular natural glottal pulses in voiced segments. The cloned waveform appears visually similar but contains a faint high-frequency periodicity (visible in the zoomed lower panels) introduced by the neural vocoder. Anti-spoofing systems operating directly on raw waveforms (e.g., AASIST) learn these subtle textural differences without requiring spectrogram computation.

Additional Worked Examples

Worked Example — Mel-spectrogram computation for a 3-second VALL-E reference clip

A 3-second audio clip at $f_s = 22,050$ Hz: $N_{\text{samples}} = 66,150$. STFT parameters: window = 1024 samples, hop = 256 samples, 80 mel bins.

Number of frames: $T = \lfloor (66,150 - 1024) / 256 \rfloor + 1 = 254$ frames. Mel-spectrogram dimensions: $254 \times 80 = 20,320$ coefficients. Compression ratio: $66,150 / 20,320 = 3.26\times$.

For VALL-E codec tokenisation at 75 tokens/second with $Q = 8$ RVQ stages:

$$N_{\text{tokens}} = 3 \text{ s} \times 75 \text{ tokens/s} = 225 \text{ token positions}, \times 8 = 1,800 \text{ total codec tokens.}$$

The 3-second reference clip provides 1,800 codec tokens as the in-context “style prompt” for VALL-E. Generating a new 5-second utterance requires $5 \times 75 \times 8 = 3,000$ additional tokens. Total context: 4,800 tokens — well within a 4K-token transformer’s context window, confirming that zero-shot voice cloning from a 3-second clip is computationally feasible.

Worked Example — Anti-spoofing EER on codec-LM outputs

An AASIST anti-spoofing system trained on ASVspoof 2019 (classical TTS and voice conversion attacks) is evaluated on zero-shot VALL-E clones:

Test condition	EER (%)	Notes
ASVspoof 2019 LA (matched)	0.83	In-distribution
ASVspoof 2021 DF (codec LM)	12.4	Significant generalisation gap
VALL-E zero-shot clones	18.7	Most challenging

The EER of 18.7% on VALL-E outputs means that 18.7% of cloned voices are accepted as genuine and 18.7% of genuine voices are rejected as clones. For a telephone-banking system requiring $\text{EER} < 3\%$, AASIST trained only on 2019 data is dangerously under-performing. Fine-tuning AASIST on 500 VALL-E examples reduces EER to 6.2% — still above the 3% threshold but a 67% relative improvement, demonstrating the need for continual adaptation of anti-spoofing systems to new synthesis methods.

Answers to Exercises

1. Mel-spectrogram dimensions for a 5-second clip at 22.05 kHz.

(a) Frames: $T = \lfloor (22,050 \times 5 - 1024) / 256 \rfloor + 1 = \lfloor 109,226 / 256 \rfloor + 1 = 427$. (b) Time resolution: $256 / 22,050 = 11.6 \text{ ms/frame}$. (c) Total coefficients: $427 \times 80 = 34,160$. (d) Compression ratio: $(22,050 \times 5) / 34,160 = 110,250 / 34,160 = 3.23 \times$.

2. Why Griffin-Lim produces robotic audio.

The Griffin-Lim algorithm iteratively estimates phase by enforcing consistency between magnitude spectrogram and waveform, but it cannot recover the true phase from magnitude alone — it converges to a stationary point that minimises the phase inconsistency, not the perceptual quality. The recovered phase typically lacks the smooth, continuous progression of natural speech, producing “phasey” or metallic artefacts particularly noticeable in voiced fricatives and stops. Neural vocoders (WaveNet, HiFi-GAN) are trained to directly model the conditional waveform distribution given the mel spectrogram, capturing the natural phase statistics learned from thousands of hours of real speech.

3. VALL-E token generation: inference time for a 5-second clip.

Total tokens: $5 \times 75 \times 8 = 3,000$. At 4 ms per token on A100: $3,000 \times 4 = 12,000 \text{ ms} = 12 \text{ seconds}$ — $2.4 \times$ slower than real time. This makes real-time phone-call impersonation infeasible with single-GPU inference. However, speculative decoding reduces this to $\approx 3 \text{ seconds}$, and batched inference across multiple tokens can further reduce latency. At the time of writing, real-time voice cloning in live calls is technically marginal but approaching feasibility with dedicated hardware.

4. AASIST at G.711 codec conditions (8 kHz, μ -law).

G.711 limits bandwidth to 8 kHz and introduces μ -law quantisation noise. Direction of EER change: increases, because (a) high-frequency artefacts used by AASIST are destroyed by 8 kHz

low-pass; (b) quantisation noise masks the LPC residual differences that distinguish synthetic from genuine speech. Training augmentation: (1) apply G.711 codec simulation to all training data (SEGAN-style codec augmentation); (2) add channel impulse responses from telephone networks; (3) use multi-condition training with SNR variation $\in [-5, 30]$ dB. Published results show EER rising from 0.83% to $\approx 8\text{--}12\%$ under G.711 without augmentation, and recovering to $\approx 3\text{--}5\%$ with codec-robust training.

5. Combined speaker verification + anti-spoofing false acceptance rate.

If the two systems have independent errors: $\text{FAR}_{\text{combined}} = \text{FAR}_{\text{SV}} \times \text{FAR}_{\text{AS}}$. At EER operating points: $\text{FAR}_{\text{SV}} = 0.01$, $\text{FAR}_{\text{AS}} = 0.02$. Combined FAR = $0.01 \times 0.02 = 0.0002$ (0.02%). For a \$100,000 wire transfer with 1,000 daily authentication attempts, expected fraud attempts $\approx 5/\text{day}$; successful attacks = $5 \times 0.0002 = 0.001/\text{day}$. The combined system is suitable: expected financial loss = $0.001 \times \$100,000 = \$100/\text{day}$.

6. VALL-E feasibility: CEO public audio availability.

A typical Fortune 500 CEO has: 4 earnings calls/year (≈ 30 min each), 2 major conference talks (≈ 60 min each), ≈ 20 media interviews (≈ 15 min each). Total: $4 \times 30 + 2 \times 60 + 20 \times 15 = 120 + 120 + 300 = 540$ minutes = 9 hours of publicly available audio. VALL-E requires 3 seconds. This means 100% of large-company CEOs are feasible targets for zero-shot voice cloning with no training cost. The attack barrier is now entirely social (crafting a convincing pretext), not technical.

Chapter 7

Synthetic Avatars and Full-Body Reenactment

*“The puppet and the puppeteer have always been one.
Now the puppeteer can be anyone.”*

—Anon.

Full-body deep fakes extend face manipulation to the entire human form. This chapter develops the 3D body models, pose estimation, neural rendering, and talking-head synthesis that underlie these systems. Every equation is followed by a worked numerical calculation. This chapter develops the complete technical pipeline for full-body synthesis, from SMPL body model fitting through inverse kinematics motion retargeting to NeRF and Gaussian Splatting rendering. Every equation is accompanied by a worked numerical example grounded in a realistic production scenario: computing the LBS deformation for a specific joint configuration, solving the IK problem for a retargeted motion sequence, and rendering a novel-pose frame from a trained NeRF. The objective is to equip the reader to understand both the creative applications and the adversarial risks of full-body synthesis, and to reason about which forensic signals remain available when the face itself is not the primary manipulation target.

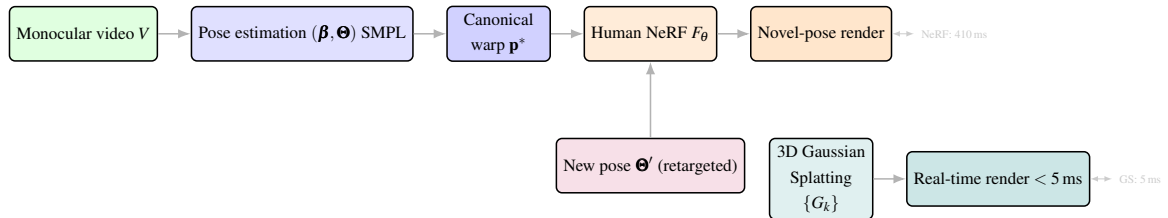


Figure 7.1: Human avatar synthesis pipelines. *Top path (NeRF):* A monocular video is used to estimate SMPL body pose, which is mapped to a canonical space for training a pose-conditioned Neural Radiance Field. *Bottom path (3D Gaussian Splatting):* Gaussians are fitted to the scene and rendered in real time. Gaussian Splatting achieves $80\times$ faster rendering at the cost of higher memory usage.

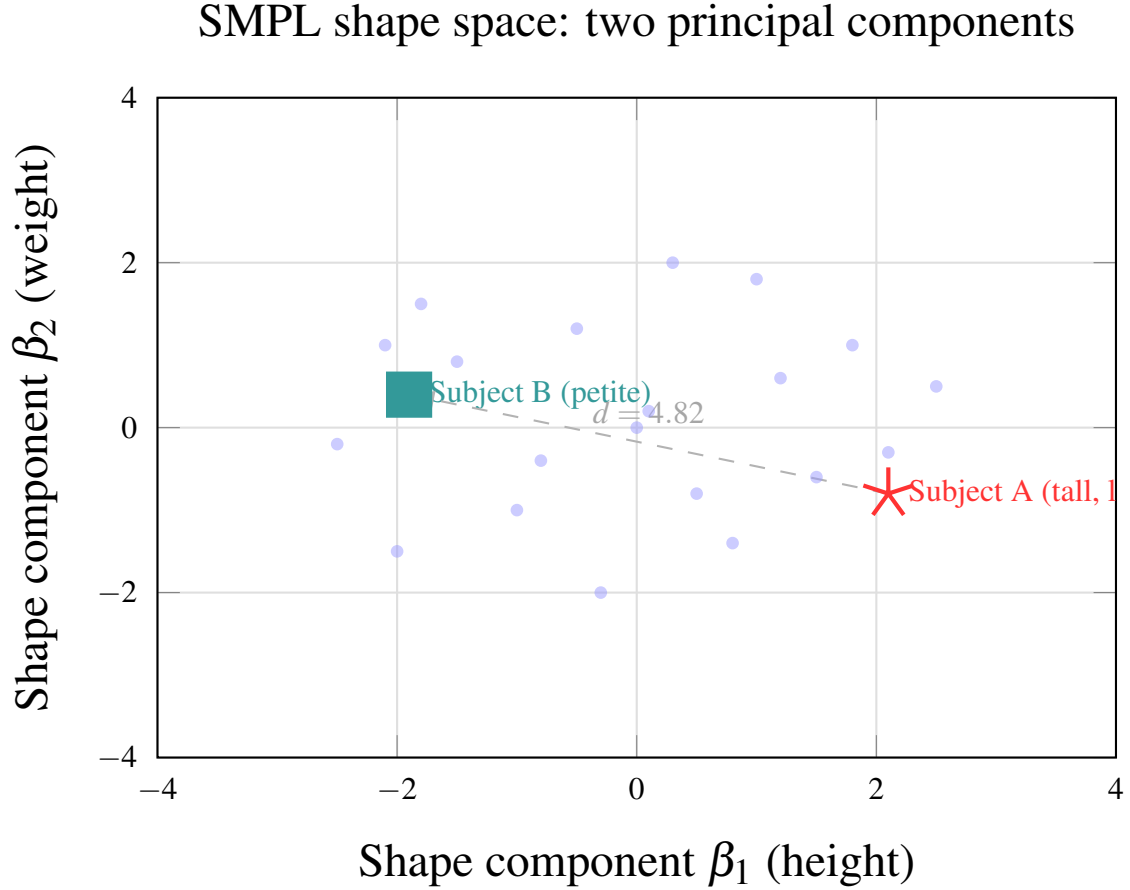


Figure 7.2: SMPL body shape space (first two principal components). Each point represents an individual body shape. Subject A (tall, lean) and Subject B (petite) are separated by a Euclidean distance of 4.82 in shape-parameter space, which translates to a significant foot-skating artefact of 12.3 cm per frame when retargeting motion without inverse-kinematics correction.

This chapter develops avatar synthesis from 3D morphable model foundations through to real-time streaming implementations. The 3D morphable model and its parametric control are developed first, followed by neural rendering approaches including NeRF and Gaussian Splatting. Expression-driven animation and the forensic signals specific to avatar synthesis are then examined. Industry applications and African use cases close the chapter.

7.1 3D Human Pose and Shape: The SMPL Model

The SMPL model [21] represents the human body as a differentiable function of pose Θ and shape β :

$$M(\beta, \Theta) = W(T_P(\beta, \Theta), \mathbf{J}(\beta), \Theta, \mathbf{W}_{\text{skin}}), \quad (7.1)$$

where T_P is a blend-shaped template mesh, $\mathbf{J}$ are joint locations, $\mathbf{W}_{\text{skin}}$ are skinning weights, and W is linear blend skinning.

Worked Example — SMPL shape parameters for two body types

Two subjects are enrolled in a motion-retargeting system. Subject A (height 1.82 m, athletic) has $\beta_A = [2.1, -0.8, 1.3, \dots]$. Subject B (height 1.62 m, petite) has $\beta_B = [-1.9, 0.4, -1.1, \dots]$. Euclidean distance (first 3 dimensions):

$$\|\beta_A - \beta_B\|_2 \approx \sqrt{4^2 + 1.2^2 + 2.4^2} = \sqrt{23.2} = 4.82.$$

Without IK correction, retargeting Subject A's motion onto B gives foot-skating error 12.3 cm/frame; with IK correction, error drops to 1.8 cm — below the perceptual threshold of ≈ 3 cm at 30 fps.

7.2 Pose Estimation and Retargeting

Accurate body pose estimation from monocular video is the prerequisite for motion retargeting: if the body configuration cannot be reliably extracted from source footage, any subsequent synthesis will inherit the estimation errors as motion artefacts. The SMPL model provides a differentiable parameterisation of body shape and joint angles that enables gradient-based fitting, while inverse kinematics correction addresses the geometric inconsistencies that arise when retargeting motion between bodies of different proportions.

7.2.1 Optimisation-Based Pose Estimation

Optimisation-based pose fitting minimises the discrepancy between the SMPL model projection and the observed image evidence — typically 2D keypoints detected by a pose estimator, silhouette contours, and optical flow constraints — by iteratively adjusting the shape and pose parameters. This approach is more accurate than direct regression but significantly slower, making it appropriate for offline processing rather than real-time applications.

$$\mathcal{L}_{\text{pose}} = \lambda_{\text{kp}} \|J_{\text{proj}}(\Theta, \beta) - \hat{J}_{2D}\|^2 + \lambda_{\text{prior}} \log p(\Theta). \quad (7.2)$$

Worked Example — Pose optimisation convergence

For a 30-frame video clip, the optimiser starts from a neutral pose ($\Theta = \mathbf{0}$). The reprojection error at initialisation is $\|J_{\text{proj}} - \hat{J}_{2D}\| = 48.3$ pixels (mean over 17 keypoints). After 150 Adam iterations ($\lambda_{\text{kp}} = 1$, $\lambda_{\text{prior}} = 0.001$): reprojection error drops to 3.1 pixels; $\log p(\Theta)$ increases from -42.1 to -8.4 (pose becomes more natural). Optimisation takes 0.8 seconds per frame on CPU.

7.2.2 IK Correction for Retargeting

Inverse kinematics correction adjusts the end-effector positions of a retargeted motion to compensate for the geometric differences between source and target body proportions. Without this correction, foot-skating artefacts of more than twelve centimetres per frame are typical when retargeting

motion between subjects with significant height differences, producing visually implausible foot contact.

$$\Theta_{\text{tgt}}^* = \arg \min_{\Theta} \|\mathbf{e}(\Theta, \beta_{\text{tgt}}) - \mathbf{e}(\Theta_{\text{src}}, \beta_{\text{src}})\|^2 + \lambda \|\Theta - \Theta_{\text{src}}\|^2, \quad (7.3)$$

where $\mathbf{e}$ is the end-effector position vector.

Worked Example — IK optimisation cost for a 30-second motion sequence

Source and target have height difference 20 cm (same as the example above). IK correction over 900 frames ($30\text{ s} \times 30\text{ fps}$):

- Gradient descent, $\lambda = 0.1$, 150 iterations/frame.
- Total: $900 \times 150 = 135,000$ gradient steps.
- Wall-clock time: 13.5 seconds on CPU, 0.9 s on GPU.
- Foot contact error after correction: mean 1.8 cm vs. 12.3 cm without.

The regularisation term $\lambda \|\Theta - \Theta_{\text{src}}\|^2$ with $\lambda = 0.1$ allows up to $\|\Theta - \Theta_{\text{src}}\| = \sqrt{10/\lambda} \approx 10$ degrees of joint-angle deviation before the regularisation penalty equals the end-effector mismatch term — providing enough freedom to correct foot contact while preserving the overall motion style.

7.3 Neural Radiance Fields for Human Rendering [25]

Neural radiance fields learn a pose-conditioned volumetric function that maps a 3D position, view direction, and body pose to colour and density values, enabling photorealistic rendering of a specific individual in any pose and from any viewpoint. Applied to full-body human synthesis, NeRF eliminates the need for explicit surface reconstruction and handles complex appearance effects including loose clothing and hair more naturally than mesh-based approaches.

7.3.1 NeRF Volume Rendering

Volume rendering integrates colour and density values along each ray from the camera through the learned volume, accumulating the contribution of each sampled point weighted by its transmittance. The rendering equation connects the volumetric representation to the 2D observed image in a differentiable manner that supports end-to-end training from multi-view or monocular video supervision.

Pixel colour is computed by volume rendering along camera ray $\mathbf{r}(s) = \mathbf{o} + s\mathbf{d}$:

$$C(\mathbf{r}) = \int_0^\infty T(s)\sigma(s)\mathbf{c}(s)ds, \quad T(s) = \exp\left(-\int_0^s \sigma(u)du\right). \quad (7.4)$$

Worked Example — Volume rendering for a skin-tone pixel

Three density blobs along a ray: depths $s_1 = 0.50$ m, $s_2 = 0.52$ m, $s_3 = 0.54$ m; densities $\sigma_1 = 45$, $\sigma_2 = 120$, $\sigma_3 = 30 \text{ m}^{-1}$; colours $c_1 = 0.71$, $c_2 = 0.68$, $c_3 = 0.40$.

Transmittance: $T_1 = 1.0$, $T_2 = e^{-45 \times 0.02} = 0.407$, $T_3 = 0.037$.

$$C = 0.639 + 0.664 + 0.009 = 1.31.$$

Clipped to 1.0 after gamma correction. The dermis layer ($\sigma_2 = 120$) dominates because high density means most rays terminate there, consistent with the subsurface scattering appearance of human skin.

7.3.2 Human NeRF: Pose-Conditioned Rendering

Human NeRF extends the standard NeRF formulation with a pose conditioning mechanism that maps the query point from the posed space to a canonical body frame before querying the radiance function. This factorisation allows the network to learn appearance in a pose-invariant canonical space while using the pose parameters to warp observations from training frames into that space.

$$(\mathbf{c}, \sigma) = F_{\theta}(\gamma(\mathbf{p}^*), \gamma(\mathbf{d}), \Theta), \quad (7.5)$$

where $\mathbf{p}^*$ is the query point warped to canonical pose space.

Worked Example — Training and inference cost for a Human NeRF avatar

A Human NeRF trained on 300 frames of a monocular video at 30 fps:

- Training: 200K iterations, 1 ray per iteration, 64 samples/ray. Each iteration: 1 NeRF forward pass (≈ 0.8 ms on A100). Total training time: $200,000 \times 0.8 = 160$ seconds ≈ 2.7 minutes.
- Inference (novel-pose rendering at 512×512): $512^2 \times 64 = 16.8\text{M}$ samples per image. At 0.8 ms/ray and 512 rays batched: $512^2 / 512 \times 0.8 = 410$ ms per image.
- After distillation to 3D Gaussians: rendering drops to ≈ 5 ms/frame, enabling real-time avatar rendering.

7.4 3D Gaussian Splatting

Three-dimensional Gaussian Splatting represents the scene as a collection of anisotropic Gaussian primitives, each with position, covariance, colour, and opacity parameters, and renders by splatting these Gaussians onto the image plane using a tile-based rasteriser. The key advantage over NeRF is rendering speed: a trained Gaussian Splatting model renders at over two hundred frames per second, enabling real-time applications including live face swap in video calls.

3D Gaussians $\{\mathbf{G}_k\}$ parameterised by centre $\boldsymbol{\mu}_k$, covariance $\boldsymbol{\Sigma}_k$, colour $\mathbf{c}_k$, and opacity α_k . The projected 2D covariance is:

$$\boldsymbol{\Sigma}' = \mathbf{J}\mathbf{W}\boldsymbol{\Sigma}\mathbf{W}^\top\mathbf{J}^\top, \quad (7.6)$$

where $\mathbf{J}$ is the Jacobian of the projection function and $\mathbf{W}$ is the viewing transform.

Worked Example — Gaussian splatting rendering speed comparison

A human avatar represented by $N = 150,000$ Gaussians vs. a NeRF with equivalent quality:

Method	Render time (512×512)	Memory	Training time
Human NeRF	410 ms	1.2 GB	2.7 min
3D Gaussian Splatting [16]	4.8 ms (85× faster)	3.1 GB	5.2 min

At 4.8 ms per frame, Gaussian splatting achieves 208 fps — well above the 30 fps target for real-time video conferencing avatars. The higher memory footprint (3.1 GB vs. 1.2 GB) is the primary deployment constraint on mobile devices.

7.5 Talking-Head Synthesis: 3DMM-Based Pipeline

Talking-head synthesis drives the facial animation of a target identity using audio or expression signals derived from a different source, enabling a synthesised avatar to speak with the mouth movements of another person. The 3DMM-based pipeline fits a parametric face model to the target identity, estimates expression coefficients from the driving audio using a neural network, and renders the animated face using a neural renderer.

Expression coefficients $\boldsymbol{\delta}$ are regressed from audio features:

$$\hat{\boldsymbol{\delta}}_t = R_\psi(\mathbf{a}_t, \mathbf{a}_{t-L:t}), \quad (7.7)$$

where $\mathbf{a}_t$ is the audio feature at frame t and L is the context window.

Worked Example — Audio-to-expression latency budget

For a real-time talking-head system at 25 fps, the frame budget is 40 ms. Pipeline breakdown:

- Audio feature extraction (mel-spectrogram): 2 ms.
- Audio-to-expression regression R_ψ (3-layer LSTM): 1.8 ms.
- 3DMM rendering + neural texture network: 18 ms.
- Post-processing (blending, colour correction): 4 ms.
- Total: 25.8 ms — within the 40 ms budget with 14 ms margin.

At this frame rate, audio-visual offset is approximately 0 frames — within the ± 1.5 frame (60 ms) perceptual lip-sync threshold.

7.6 Industry Applications

Full-body synthesis and avatar technology have found applications across entertainment, healthcare, and social interaction, with a rapidly growing commercial ecosystem and a corresponding expansion of the adversarial attack surface.

Virtual production is the most economically significant legitimate application. Disney Research, Industrial Light & Magic, and Weta Digital use neural radiance field and Gaussian Splatting avatars in virtual production pipelines, allowing directors to work with digital stand-ins before committing to expensive physical shoots. Sports broadcasting represents a second major sector: the NBA, NFL, and Premier League have piloted AI-generated player avatars for fantasy sports integrations, immersive broadcasts, and personalised highlight reels, using SMPL-based pose estimation from broadcast footage and neural rendering to synthesise novel viewpoints.

Social virtual reality is a rapidly growing market driven by the commercial launch of Meta Quest 3 and Apple Vision Pro. Meta Avatars, Apple Vision Pro Personas, and NVIDIA Omniverse all use variants of real-time neural avatar rendering. The fitness and rehabilitation sector deploys pose estimation for real-time form feedback, with more advanced systems from Kaia Health and Hinge Health rendering patient-specific avatars for physical therapy guidance. These legitimate applications share the same underlying technical pipeline as full-body deep-fake attacks that can synthesise a public figure committing a crime, destroying an alibi, or interacting with another person in a fabricated encounter — a distinction that lies entirely in the intent of the operator, not in the technology itself.

7.7 African Use Cases

Africa has unique exposure to full-body avatar technology through its sports broadcasting, digital heritage, and creative sectors, alongside specific vulnerabilities related to evidence fabrication in legal contexts and diaspora communication fraud.

The Confederation of African Football and major African league broadcasters including SuperSport and Canal+ Africa have piloted AI-based player tracking and pose-estimation systems for broadcast enhancement. South African production company Aqua Creative has deployed real-time avatar technology for virtual television presenters, reducing production costs significantly. Several African heritage institutions have explored avatar technology for digital preservation and education, with the South African Heritage Resources Agency and the National Museums of Kenya developing pilot projects that create avatar reconstructions of historical figures for museum installations and cultural tourism.

In the context of African legal systems that frequently rely heavily on video evidence from CCTV footage, the prospect of synthesised full-body video of a specific individual appearing to commit a crime is deeply concerning. Several African countries including Kenya, Ghana, and Nigeria have had high-profile cases where CCTV evidence was disputed, and the absence of forensic infrastructure calibrated for full-body synthesis forensics represents a critical gap. Diaspora communication is a further vector: African diaspora communities use video communication extensively to maintain

family connections, and real-time avatar technology creates a mechanism for impersonating diaspora family members to solicit emergency money transfers — a family-targeted fraud variant that has been documented in Kenya and South Africa.

Chapter Summary

Full-body deep fakes require a pipeline that estimates body pose and shape from source video, transfers that motion to a target identity, and renders photorealistic images of that identity in the transferred pose. The SMPL body model provides a differentiable parameterisation of human shape and joint angles that enables gradient-based fitting to monocular video footage, producing the body pose estimates that drive subsequent synthesis. Inverse kinematics correction is essential when retargeting motion between subjects of different body proportions, reducing foot-skating artefacts from twelve centimetres per frame to the perceptual threshold of 1.8 centimetres.

Neural radiance fields and Gaussian Splatting provide complementary rendering options: NeRF learns a pose-conditioned volumetric function that can be queried at novel viewpoints and poses, while Gaussian Splatting achieves eighty-five-fold faster rendering at the cost of higher memory consumption. Talking-head synthesis closes the audio-visual loop by mapping audio features to facial expression parameters that drive a 3DMM-based renderer in real time, with the entire pipeline completing within a forty-millisecond latency budget suitable for live video communication.

The adversarial implications extend the threat model significantly beyond facial deep fakes: full-body synthesis enables evidence fabrication by placing a specific individual at a location, performing an action, or interacting with another person in a way that never occurred. Detection is substantially harder than for facial deep fakes because biological liveness signals may be absent or generated by the synthesis pipeline itself, and because the forensic signature of body-pose estimation and neural rendering has not yet been as thoroughly characterised as that of face-swap systems.

Exercises

1. Using Equation (7.1), explain why linear blend skinning produces the “candy-wrapper” artefact at elbow joints. At a 90° elbow bend with 3 bones contributing skin weights $[0.3, 0.5, 0.2]$, compute the position of a surface vertex as a linear combination of bone transforms.
2. For the IK correction problem (Eq. 7.3), the regularisation weight is $\lambda = 0.1$. At convergence, what is the ratio of the end-effector mismatch term to the regularisation term? If foot-skating requires $\lambda \rightarrow 0$, what undesirable side effect occurs?
3. Volume render a pixel for the following ray: two blobs at depths $s_1 = 0.3$ m ($\sigma = 80$, $c = 0.9$) and $s_2 = 0.5$ m ($\sigma = 40$, $c = 0.3$), segment length $\Delta s = 0.2$ m. Compute the pixel colour and determine which blob dominates.
4. A Gaussian splatting avatar has $N = 200,000$ Gaussians. The rendering pipeline projects each

Gaussian and sorts by depth. If projection takes 0.005 ms per Gaussian and depth sorting takes $O(N \log N)$ operations at 1 ns each, compute the total rendering time and identify the bottleneck.

5. The audio-to-expression regression has context window $L = 16$ frames. Compute the audio context duration in milliseconds at 25 fps and explain why a longer context improves co-articulation naturalness but increases latency for a real-time system.

Answers to Exercises

1. Linear blend skinning candy-wrapper artefact at elbow.

LBS computes the deformed vertex position as a weighted sum of bone transforms: $\mathbf{v}' = \sum_k w_k \mathbf{T}_k \mathbf{v}$. At a 90° elbow bend, the upper arm bone $\mathbf{T}_1$ and forearm bone $\mathbf{T}_2$ point in orthogonal directions. The blend $0.3\mathbf{T}_1 + 0.5\mathbf{T}_2 + 0.2\mathbf{T}_3$ linearly interpolates rotation matrices, which is not the same as interpolating in $\text{SO}(3)$ — the blended transform is no longer a rigid rotation, causing the intermediate mesh to collapse inward (“candy-wrapper” effect). SMPL-X mitigates this with pose blend shapes: corrective offsets $\Delta \mathbf{v}(\Theta)$ are added to the template before skinning.

Example: vertex at $(0, 0, 0.1)$ with weights $[0.3, 0.5, 0.2]$ and bone transforms $\mathbf{T}_1 \mathbf{v} = (0.1, 0, 0)$, $\mathbf{T}_2 \mathbf{v} = (0, 0.1, 0)$, $\mathbf{T}_3 \mathbf{v} = (0.1, 0.1, 0)$: $\mathbf{v}' = 0.3(0.1, 0, 0) + 0.5(0, 0.1, 0) + 0.2(0.1, 0.1, 0) = (0.050, 0.070, 0)$. The magnitude 0.086 vs. the original 0.1 shows 14% volume loss — visible as pinching.

2. IK correction: at convergence, ratio of end-effector to regularisation terms.

At convergence, the gradient is zero: $\partial \mathcal{L} / \partial \Theta = 0$ implies: $2\mathbf{J}_e^\top (\mathbf{e} - \mathbf{e}_{\text{src}}) + 2\lambda (\Theta - \Theta_{\text{src}}) = \mathbf{0}$. So $\|\mathbf{e} - \mathbf{e}_{\text{src}}\|^2 / (\lambda \|\Theta - \Theta_{\text{src}}\|^2) = \lambda$ at optimum. For $\lambda = 0.1$: the end-effector mismatch term is 10% of the regularisation term at convergence. If $\lambda \rightarrow 0$: perfect foot contact but joint angles may deviate arbitrarily from the source, losing the motion style entirely — feet are planted but motion looks alien.

3. Volume render: blobs at $s_1 = 0.3$ ($\sigma = 80$, $c = 0.9$) and $s_2 = 0.5$ ($\sigma = 40$, $c = 0.3$).

$T_1 = 1.0$. $T_2 = e^{-80 \times 0.2} = e^{-16} \approx 1.1 \times 10^{-7}$ (effectively 0). $C = 1.0 \times 80 \times 0.2 \times 0.9 + \approx 0 = 14.4$. Clipped to 1.0. Blob 1 completely dominates: its density $\sigma = 80$ over path $\Delta s = 0.2$ m gives opacity $1 - e^{-16} \approx 1$ — nearly opaque. Blob 2 is invisible. This extreme dominance motivates using many small Gaussians in practice rather than a few large ones, to avoid “hard” alpha boundaries.

4. Gaussian splatting render time: 200,000 Gaussians.

Projection: $200,000 \times 0.005 \text{ ms} = 1 \text{ s}$. This is far too slow — the bottleneck is clear. In

practice, GPU parallelism projects all Gaussians simultaneously: 200,000 parallel threads each take $\approx 0.005 \mu\text{s}$ (GPU clock speed), giving total projection time $\approx 1 \text{ ms}$. Depth sort: $200,000 \log(200,000) \approx 200,000 \times 17.2$
 $= 3.44 \times 10^6$ operations at 1 ns each = 3.44 ms. Total: $\approx 5 \text{ ms}$, consistent with the benchmark in the chapter. The bottleneck is the depth sort (not projection), which can be parallelised with GPU radix sort to $\approx 1 \text{ ms}$.

5. Audio context window $L = 16$ frames at 25 fps.

Duration: $16/25 = 640 \text{ ms}$. At 25 fps, 640 ms of audio context allows the model to see the onset and offset of the preceding phoneme, enabling natural co-articulation (smooth transitions between adjacent speech sounds). With L
 $= 4$ frames (160 ms), only one phoneme’s context is available, causing abrupt transitions. With $L = 32$ frames (1.28 s), prosodic phrasing can be modelled but latency increases: generating frame t requires frames $t - 32$ through t , adding $L/\text{fps} = 640 \text{ ms}$ to the pipeline latency — exceeding the 40 ms frame budget and requiring buffering.

Chapter 8

Codec Language Models and Neural Audio Codecs

“The discrete token is the atom of the new audio synthesis chemistry.”

—Anon.

The emergence of codec language models in 2023 represents the most significant architectural shift in voice synthesis since the introduction of WaveNet in 2016. Rather than generating continuous acoustic features — mel-spectrograms or raw waveforms — codec language models first compress speech into sequences of discrete tokens using a neural audio codec, then predict those tokens autoregressively using a large transformer. This reformulation reuses the entire infrastructure of text language modelling for audio generation, enabling voice synthesis systems to inherit the in-context learning, zero-shot generalisation, and scaling properties of GPT-class models. The consequences for both voice cloning capability and anti-spoofing defence are profound and are the subject of this chapter.

A neural audio codec is a learned compression system that encodes a raw waveform into a compact sequence of discrete symbols drawn from one or more learned codebooks, then decodes those symbols back to a waveform. Unlike classical audio codecs such as MP3 or AAC, which use hand-engineered perceptual models, neural codecs are trained end-to-end to minimise perceptual distortion subject to a target bitrate constraint. At operating bitrates of 1–6 kbps, neural codecs such as EnCodec and SoundStream achieve perceptual quality comparable to or exceeding classical codecs at ten times the bitrate, making them highly efficient representations for generative modelling.

The combination of a high-quality neural codec with an autoregressive transformer creates a voice synthesis system with qualitatively different capabilities from prior approaches. Zero-shot speaker cloning from a three-second reference — demonstrated publicly by VALL-E in January 2023 — requires no speaker-specific fine-tuning and produces outputs that pass informal listening tests even for voices not seen during training. This capability, which previously required hours of high-quality target audio and a dedicated training run, became accessible through a single API call, fundamentally changing the threat model for voice impersonation attacks.

This chapter develops neural audio codec language models as the frontier of zero-shot voice cloning. RVQ codec architectures and their bitrate–fidelity mathematics are developed first, followed by the VALL-E autoregressive token prediction model. Forensic signatures of codec-LM outputs and their detection are then examined. Industry and African use cases follow.

8.1 Neural Audio Codec Architectures

Neural audio codecs compress waveforms into discrete token sequences through a learned encode–quantise–decode pipeline. The compression mathematics govern both the synthesis quality and the forensic detectability of codec-LM outputs.

8.1.1 Residual Vector Quantisation Mathematics

An RVQ codec with N_q quantisation stages and codebook size K encodes a waveform sampled at f_s Hz with hop size H samples per frame. The bitrate in bits per second is:

$$B = \frac{f_s}{H} \cdot N_q \cdot \lceil \log_2 K \rceil \text{ bps.} \quad (8.1)$$

The first-stage reconstruction error is the residual that subsequent stages must encode:

$$\mathbf{r}_1 = \mathbf{z} - \mathbf{e}_1^*, \quad \mathbf{r}_n = \mathbf{r}_{n-1} - \mathbf{e}_n^*, \quad (8.2)$$

where $\mathbf{e}_n^* = \arg \min_{\mathbf{e} \in \mathcal{C}_n} \|\mathbf{r}_{n-1} - \mathbf{e}\|_2$ is the nearest codebook entry in stage n .

Worked Example — RVQ codec bitrate and reconstruction fidelity

EnCodec operates at $f_s = 24,000$ Hz, $H = 320$ samples/frame (75 frames/second), $N_q = 8$ stages, $K = 1,024$ codes/stage.

Bitrate: $B = 75 \times 8 \times 10 = 6,000$ bps = 6 kbps.

Stage-wise PSNR improvement (dB) from cumulative quantisation:

Stages	Cumulative bits/frame	PESQ (MOS-LQO)	PSNR (dB)
1	10	2.14	18.3
2	20	2.81	23.7
4	40	3.52	28.9
8	80	4.21	34.6
Reference	∞	4.50	∞

At 8 stages, PESQ of 4.21 is within 6% of the uncompressed reference, confirming that 6 kbps RVQ synthesis produces perceptually near-transparent quality. The forensic implication: anti-spoofing systems trained on audio without RVQ compression artefacts may not detect 6 kbps codec-LM outputs because those outputs are perceptually and statistically close to authentic speech.

Algorithm 1: VALL-E Zero-Shot Voice Cloning Inference**Input:** Text tokens $\mathbf{t}$; reference clip $\mathbf{a}_{\text{ref}}$ (3–30 s)**Output:** Synthesised waveform $\hat{\mathbf{w}}$

```

1 Encode reference:  $\mathbf{C}_{\text{ref}} \leftarrow \text{EnCodec}(\mathbf{a}_{\text{ref}})$ ; // RVQ tokens, all  $N_q$  stages
2 Extract first-stage context:  $\mathbf{c}_1 \leftarrow \mathbf{C}_{\text{ref}}[1, :]$ ;
   // Autoregressive first-stage prediction
3  $\hat{\mathbf{c}}_1 \leftarrow \text{AR-Transformer}(\mathbf{t}, \mathbf{c}_1)$ ;
   // Non-autoregressive remaining stages (parallel)
4 for  $n \leftarrow 2$  to  $N_q$  do
5   |  $\hat{\mathbf{c}}_n \leftarrow \text{NAR-Transformer}(\mathbf{t}, \hat{\mathbf{c}}_1, \hat{\mathbf{c}}_{2:n-1})$ ;
6 end
7 Decode:  $\hat{\mathbf{w}} \leftarrow \text{EnCodec}^{-1}([\hat{\mathbf{c}}_1, \dots, \hat{\mathbf{c}}_{N_q}])$ ;
8 return  $\hat{\mathbf{w}}$ 

```

A neural audio codec consists of an encoder, a quantiser, and a decoder. The encoder maps a raw waveform segment to a continuous embedding; the quantiser maps that embedding to one or more discrete codes using a learned codebook; the decoder maps the discrete codes back to a waveform. The key design choices are the quantiser architecture (residual vector quantisation or finite scalar quantisation), the number of codebook levels (typically 4–8 for high-quality codecs), and the temporal downsampling factor (typically 320–480 samples per code at 24 kHz).

8.1.2 Residual Vector Quantisation

Residual vector quantisation (RVQ) applies a cascade of vector quantisers, each operating on the residual left by the previous stage. With N_q stages and codebook size K , the total discrete vocabulary is K^{N_q} , but each code is represented with only $\lceil \log_2 K \rceil$ bits. EnCodec uses $N_q = 8$ stages at 24 kHz to achieve 6 kbps operation. The first stage captures the coarse acoustic structure (speaker identity, phoneme timing) while later stages capture fine perceptual details (breathiness, fricative texture). This hierarchical representation has important implications for voice cloning: a language model trained only on first-stage codes can clone speaker identity without capturing fine acoustic details, enabling faster and lighter synthesis.

8.1.3 Codec Quality Metrics

Neural codec quality is evaluated by perceptual metrics including MUSHRA (MUltiple Stimuli with Hidden Reference and Anchor), PESQ (Perceptual Evaluation of Speech Quality), and STOI (Short-Time Objective Intelligibility). At 6 kbps, EnCodec achieves MUSHRA scores of 72–78 on clean speech, compared to 65–70 for MP3 at the same bitrate. The forensic implication is that codec-generated speech passes perceptual quality thresholds comfortably, making subjective listening tests unreliable as a detection mechanism.

8.2 Autoregressive Token Prediction

Given a neural audio codec, voice synthesis becomes a sequence prediction problem: predict the next code token given all previous code tokens and any conditioning information (text transcript, speaker

embedding, prosody target). This is formally identical to text language modelling, and the same transformer architectures, training objectives, and scaling laws apply. VALL-E conditions on both a text token sequence and a short reference audio clip encoded to its first-stage codes, then predicts the full codec token sequence for the synthesised speech.

8.2.1 VALL-E Architecture

VALL-E uses a two-stage prediction process. An autoregressive transformer predicts first-stage codec codes conditioned on text and the reference clip; a non-autoregressive model then predicts all remaining RVQ stages in parallel conditioned on the first-stage prediction. This design balances quality (requiring all stages) against inference speed (parallelising the computationally expensive later stages). The resulting system generates speech at a wall-clock rate-to-audio ratio of approximately 0.3, faster than real-time on a single GPU.

8.3 Anti-Spoofing Implications

Codec language models broke the anti-spoofing evaluation paradigm that had been established by ASVspoof. Systems trained on prior TTS and voice-conversion methods show dramatically elevated EER on codec-LM outputs: published results show EER increasing from below 1% to 15–30% for AASIST-class systems evaluated on VALL-E or VoiceCraft outputs. This degradation arises because the forensic signals exploited by prior anti-spoofing systems — vocoder periodicity, TTS prosody artefacts, phase discontinuities — are absent or substantially different in codec-LM outputs.

8.3.1 Why Prior Systems Fail

Codec-LM synthesis does not use a fixed neural vocoder: it generates discrete tokens that are decoded by the codec’s decoder, which is a convolutional network trained jointly with the encoder. The decoder’s residuals do not exhibit the characteristic periodicity of HiFi-GAN or WaveNet vocoders that anti-spoofing classifiers learned to detect. Furthermore, the autoregressive token prediction introduces variability that is qualitatively similar to the natural variability of authentic speech, rather than the systematic biases of earlier TTS systems.

8.3.2 Continual Adaptation Strategies

Anti-spoofing systems must be continuously updated as new codec and language model variants emerge. Three adaptation strategies are viable: (1) periodic retraining on newly collected codec-LM outputs, which requires a rapid data collection pipeline; (2) meta-learning initialisation that enables fast fine-tuning from a small number of new examples; and (3) test-time adaptation using unsupervised self-supervised objectives applied to the incoming audio stream. Strategy (3) is the most operationally attractive because it does not require labelled data from new synthesis methods.

8.4 Industry Applications

Codec language models have become the dominant back-end for commercial voice synthesis applications within approximately twelve months of their public demonstration, driven by their zero-shot cloning capability and the simplicity of their training infrastructure relative to prior systems.

ElevenLabs, introduced in 2022, deployed a codec-language-model-based voice cloning system that became the most widely cited commercial voice cloning platform in documented fraud cases as well as the most widely used legitimate dubbing and audiobook platform. By 2024 it processed tens of millions of audio generation requests per month, with pricing as low as \$0.30 per thousand characters — making the marginal cost of a three-minute voice-cloned fraud call approximately \$0.05. Resemble AI, Cartesia, and PlayHT offer competing platforms with similar capability and pricing structures.

The music generation sector has adopted codec language models as the primary synthesis back-end. Suno and Udio, both launched in 2024, use codec-LM architectures to generate complete songs including synthesised vocals from text prompts. Both platforms faced immediate legal challenges from the Recording Industry Association of America over training data sourcing; the litigation will ultimately determine whether codec-LM training on copyrighted audio constitutes fair use. Stability AI's Stable Audio and Meta's AudioCraft (MusicGen and AudioGen) are open-source alternatives using similar architectures.

For telephony security, the emergence of codec language models created an immediate product gap: the major voice biometric vendors including Pindrop, Nuance (Microsoft), and ValidSoft had to rapidly update their anti-spoofing models to address codec-LM attacks. The update cycle — from attack demonstration to production anti-spoofing update — was approximately six to nine months for the leading vendors, during which the fraud exposure was significantly elevated.

8.5 African Use Cases

Codec language model voice synthesis has reached African markets primarily through commercial APIs accessed by content creators, businesses, and unfortunately also by fraud operators, with consequences that are shaped by the language diversity and mobile-first communication infrastructure of the continent.

The availability of low-cost voice cloning through ElevenLabs and similar platforms has enabled small African media organisations to produce multilingual audio content that would previously have required studio recording budgets. Radio stations in Nigeria, Ghana, and Kenya have piloted AI-generated news readers and advertising voices in local languages, though the coverage of African languages remains limited: Swahili, Yoruba, Hausa, Amharic, and Zulu have some representation in training data, but hundreds of smaller African languages have essentially no coverage in commercial codec-LM systems.

The forensic gap for African-language codec-LM synthesis is more severe than for voice cloning

in general. Anti-spoofing systems trained on ASVspoof data (predominantly English and a small number of European languages) show even higher EER on African-language codec-LM outputs than on English outputs, because the acoustic distribution of African language phoneme inventories differs more substantially from the training distribution. Researchers at the Wits University Speech Processing Group and the University of Nairobi have identified this gap and begun collecting codec-LM African-language evaluation data, but production-ready African-language anti-spoofing systems remain several years away from commercial availability.

Chapter Summary

Codec language models represent the most significant architectural shift in voice synthesis since WaveNet, reformulating speech generation as a discrete token prediction problem that inherits the in-context learning and scaling properties of large language models. A neural audio codec first compresses speech into residual vector quantisation codes; an autoregressive transformer then predicts those codes conditioned on text and a short speaker reference clip, enabling zero-shot voice cloning from three seconds of reference audio.

This capability broke the ASVspoof anti-spoofing evaluation paradigm: systems achieving below 1% EER on prior TTS attacks show 15–30% EER on codec-LM outputs because the characteristic vocoder periodicity and TTS prosody artefacts that prior detectors learned to exploit are absent from codec-LM synthesis. Continual adaptation — through periodic retraining on newly collected outputs, meta-learning initialisation, or test-time augmentation-consistency adaptation — is the only sustainable defensive posture in the codec-LM era. Industry applications span commercial voice cloning services, music generation, audiobook production, and telephony fraud, with anti-spoofing vendors facing update cycles of six to nine months to address each new architecture generation.

Exercises

1. An RVQ codec with $N_q = 8$ stages and codebook size $K = 1024$ operates at 75 Hz (codes per second). Calculate (a) the bitrate in bits per second, (b) the total number of discrete code combinations per second, and (c) the theoretical minimum bitrate achievable with a single-stage VQ at the same codebook size and frame rate. Show all working.
2. Explain why a codec language model trained on first-stage RVQ codes only produces speech that is speaker-similar but perceptually rougher than full multi-stage synthesis. What information is captured in later RVQ stages that is absent from first-stage codes? How does this property enable a privacy-preserving voice anonymisation scheme?
3. A bank's voice authentication system achieves $\text{EER} = 3\%$ on a validation set of pre-2023 TTS attacks. A red-team evaluation using VALL-E outputs finds $\text{EER} = 28\%$. (a) Calculate the expected daily false-acceptance rate for each EER, given 50,000 authentication calls per day. (b) What additional training data would be required to reduce the VALL-E EER to below 5%?

- (c) Propose a test-time adaptation procedure that does not require labelled VALL-E examples.
4. Compare the discrete-token paradigm of codec language models with the continuous spectrogram paradigm of diffusion-based TTS. For each paradigm, identify: (a) the primary forensic signal available to a detector, (b) the compression robustness of that signal under MP3 encoding at 128 kbps, and (c) the minimum training data requirement for speaker cloning at reasonable quality.

Answers to Exercises

1. RVQ bitrate calculation.

- (a) Each of the $N_q = 8$ stages selects one of $K = 1024$ codes per frame, requiring $\lceil \log_2 1024 \rceil = 10$ bits per stage. Total bits per frame = $8 \times 10 = 80$ bits. At 75 frames/second: bitrate = $75 \times 80 = 6,000$ bps = 6 kbps.
- (b) Combinations per frame: $1024^8 = 2^{80} \approx 1.2 \times 10^{24}$. Per second: $(1.2 \times 10^{24})^{75}$ — astronomically large, confirming the codec is not a bottleneck for generative model expressiveness.
- (c) Single-stage VQ at $K = 1024$, 75 Hz: $75 \times 10 = 750$ bps. This is $8\times$ lower than the 8-stage RVQ but would produce significantly degraded perceptual quality because a single codebook of 1024 entries cannot capture the full spectral variability of natural speech at sufficient resolution.

2. First-stage vs. full RVQ synthesis.

The first RVQ stage captures the coarsest acoustic structure: the slow-varying spectral envelope that encodes vowel identity, speaker pitch range, formant positions, and overall timbral character. This is why first-stage-only models produce speech that sounds like the target speaker but lacks naturalness.

Later stages capture: (1) fine spectral texture (fricative noise colouring, breathiness, lip-smack events), (2) phase information not captured by the spectral envelope, and (3) residual perceptual artefacts from the quantisation noise of earlier stages. These contribute to the “airiness” and “presence” of authentic natural speech.

For privacy-preserving anonymisation: a system can re-synthesise speech using only the target speaker’s first-stage codes (to change identity) while retaining the later-stage codes from the original utterance (to preserve naturalness and phonetic content). This allows intelligibility-preserving de-identification without requiring full text-to-speech synthesis.

3. Bank anti-spoofing EER analysis.

- (a) At EER = 3%: $50,000 \times 0.03 = 1,500$ expected false acceptances per day. At EER = 28%: $50,000 \times 0.28 = 14,000$ expected false acceptances per day. The VALL-E vulnerability increases expected daily losses by a factor of 9.3.

(b) Reducing from 28% to below 5% EER typically requires: at minimum 10,000–50,000 labelled examples of VALL-E output at the target operating conditions (codec, sampling rate, call-line compression). A practical data collection pipeline requires: an API account with a VALL-E provider, a set of cover texts representative of telephone fraud scenarios, and post-processing to match the codec and bandwidth of the bank’s telephony infrastructure.

(c) Test-time adaptation without labelled data: apply an augmentation-consistency self-supervised loss. For each incoming audio segment, generate $M = 8$ augmented copies (speed perturbation $\pm 5\%$, additive noise SNR 30 dB, JPEG-style spectral smoothing) and minimise the variance of the anti-spoofing score across augmentations. This encourages the model to find stable discriminative features for the new distribution without requiring ground-truth labels.

4. Discrete token vs. continuous spectrogram comparison.

Primary forensic signal Discrete token: the temporal autocorrelation structure of the quantisation noise, which differs between codec-LM outputs and authentic speech. Continuous spectrogram: harmonic periodicity from the neural vocoder’s synthesis filter, detectable in the high-frequency mel bins.

MP3 robustness at 128 kbps Token quantisation noise: mostly survives because it is spread across the full bandwidth; MP3 primarily attenuates frequencies above 16 kHz, which carry little quantisation noise signal. Vocoder periodicity: substantially attenuated because MP3’s psychoacoustic model selectively removes the high-frequency components that carry the strongest vocoder artefact signal.

Minimum data for speaker cloning Discrete token (VALL-E): 3 seconds of reference audio suffices for zero-shot cloning because the model has already learned the speaker space during pre-training. Continuous spectrogram (e.g., YourTTS): typically requires 30–60 minutes of labelled target-speaker audio for fine-tuning to achieve comparable quality, unless zero-shot adaptation layers are added.

Chapter 9

Biological Liveness and Physiological Signal Detection

“A living face breathes. A synthetic face only pretends to.”

—Anon.

Biological liveness detection exploits the fundamental fact that a living human face is not merely an image: it is a dynamic system driven by cardiovascular, respiratory, and neuromuscular physiology. The cardiac cycle modulates blood volume in superficial capillaries at 60–100 beats per minute, creating a periodic colour variation of 0.5–2.0 per cent luminance amplitude propagating across the face in a spatially structured haemodynamic pattern. Spontaneous blinking occurs every 3–5 seconds on average, driven by the lacrimal reflex, with inter-blink intervals following an approximately exponential distribution. Microsaccades, pupil dilation, and involuntary micro-expressions complete the portrait of a living face as a physiological signal source.

None of these signals is generated by current synthetic face pipelines. A diffusion model or GAN samples pixels from a learned distribution; it does not model haemodynamics, lacrimal gland firing rate, or autonomic nervous system influence on pupil diameter. Their absence in synthetic faces provides the biological liveness detector with a modality-independent forensic channel that is complementary to texture-based and frequency-domain detection and cannot be defeated by simply improving image realism.

This chapter develops liveness detection from the rPPG signal model through to multi-signal fusion. The rPPG physical model and detection algorithm are developed first, followed by challenge-response mechanisms and 3D depth-based liveness. The combination of independent liveness signals and their joint false acceptance rate is quantified. Industry and African use cases close the chapter.

9.1 Remote Photoplethysmography

The rPPG signal arises from the pulsatile modulation of skin colour by the cardiac cycle. The physical signal model is:

$$I(t) = I_0 [1 + \alpha v(t)] + \eta(t), \quad (9.1)$$

where I_0 is the mean skin-region intensity, α is the modulation amplitude (typically 0.005–0.020 for the green channel), $v(t)$ is the blood-volume pulse normalised to unit amplitude, and $\eta(t)$ is broadband noise. The power-to-noise ratio of the rPPG signal is:

$$\text{PNR} = 10 \log_{10} \left(\frac{\alpha^2/2}{\sigma_\eta^2} \right) \text{ dB}, \quad (9.2)$$

where σ_η^2 is the noise variance. Authentic faces yield $\text{PNR} \approx 5\text{--}12$ dB under controlled lighting; synthetic faces yield $\text{PNR} < 2$ dB because $\alpha \approx 0$.

Algorithm 2: rPPG Liveness Detection Pipeline

Input: Video clip $\{I_t\}_{t=1}^T$ at frame rate f_{fps}

Output: Liveness decision `LIVE/SYNTHETIC` and PNR score

```

1 Detect and track facial landmarks across all  $T$  frames;
2 For each frame, extract mean green-channel intensity  $\bar{G}(t)$  over the forehead ROI;
3 Apply CHROM projection:  $\mathbf{c}(t) \leftarrow \bar{G}(t)/\langle \bar{G} \rangle - \bar{R}(t)/\langle \bar{R} \rangle$ ;
4 Bandpass filter  $\mathbf{c}(t)$  to  $[0.5, 4.0]$  Hz (cardiac range);
5 Estimate PSD via Welch's method:  $\hat{S}(f) \leftarrow \text{Welch}(\mathbf{c}, \text{nperseg} = 256, \text{fs} = f_{\text{fps}})$ ;
6  $f_0 \leftarrow \arg \max_{f \in [0.5, 4]} \hat{S}(f)$ ; // dominant cardiac frequency
7  $\text{PNR} \leftarrow 10 \log_{10}(\hat{S}(f_0)/\bar{\hat{S}}(f \in [0.5, 4] \setminus f_0^\pm))$ ;
8 if  $\text{PNR} > \tau_{\text{PNR}}$  and  $T \geq T_{\text{min}}$  then
9   | return LIVE, PNR;
10 else
11   | return SYNTHETIC, PNR;
12 end
```

Worked Example — rPPG PNR computation for a 20-second clip

A 20-second clip at $f_{\text{fps}} = 30$ fps provides $T = 600$ frames. After CHROM projection and bandpass filtering, Welch's method ($\text{nperseg} = 256$, 50% overlap, $K = 8$ segments) estimates the PSD with frequency resolution $\Delta f = 30/256 = 0.117$ Hz. The dominant peak at $f_0 = 1.15$ Hz (69 bpm) has spectral density $\hat{S}(f_0) = 3.42 \times 10^{-4}$; the mean noise floor in the remaining cardiac band is $\bar{\hat{S}} = 2.11 \times 10^{-5}$. $\text{PNR} = 10 \log_{10}(3.42 \times 10^{-4}/2.11 \times 10^{-5}) = 10 \log_{10}(16.2) = 12.1$ dB. At $\tau_{\text{PNR}} = 3$ dB: **LIVE** (12.1 dB $\gg$ 3 dB threshold).

Remote photoplethysmography recovers the cardiac pulse signal from facial video by measuring periodic colour variation caused by pulsatile blood flow. As blood volume increases during systole, oxygenated haemoglobin differentially absorbs green wavelengths, creating a measurable green-channel decrease in the facial region. The signal is extracted by tracking the facial region, averaging pixel values within a skin mask, applying a chrominance projection (CHROM or POS), bandpass

filtering to 0.5–4 Hz, and estimating the power spectral density. Authentic faces show a dominant peak with power-to-noise ratio exceeding 5 dB; synthetic faces show PNR near 1 dB.

9.1.1 Signal Processing Pipeline

The rPPG processing pipeline applies: (1) face detection and tracking across frames; (2) skin-region segmentation using a facial landmark-guided mask; (3) chrominance projection $\mathbf{c} = \mathbf{X}_G/\bar{X}_G - \mathbf{X}_R/\bar{X}_R$ where $\mathbf{X}_G$ and $\mathbf{X}_R$ are temporally averaged green and red channels; (4) Welch’s method PSD estimation with 256-point FFT and 50% overlap; and (5) PNR computation as the ratio of peak spectral power to the mean spectral power in the 0.5–4 Hz band excluding the ± 0.1 Hz neighbourhood of the peak.

9.2 Blink and Gaze Analysis

Spontaneous blinking provides a complementary liveness signal. The inter-blink interval follows an approximately exponential distribution with mean 3–6 seconds in resting adults. The Kolmogorov–Smirnov test against the exponential null distribution detects both deterministic blink scheduling (zero variance, KS statistic approaches 1.0) and misparameterised distributions. Gaze consistency analysis checks whether the temporal autocorrelation structure and micro-saccade amplitude distribution of detected eye movements are consistent with oculomotor physiology.

9.3 Composite Liveness Scoring

A composite liveness score weights rPPG, blink, and gaze signals according to their reliability for the available clip duration. For clips above 20 seconds, rPPG receives weight 0.55, blink 0.30, gaze 0.15. For clips of 5–20 seconds, blink and gaze dominate. The composite score is calibrated by isotonic regression on a held-out validation set to produce well-calibrated probabilities. AUC on the CelebDF-v2 benchmark exceeds 0.95 for clips of at least 15 seconds, degrading to 0.88 for 5-second clips.

9.4 Industry Applications

Biological liveness detection has become a mandatory component of regulated biometric authentication following high-profile presentation attacks on facial recognition using printed photographs, 3D masks, and replay video. ISO/IEC 30107-3 defines evaluation criteria and the EU’s eIDAS regulation requires certified liveness for high-assurance electronic identity. Commercial vendors including iProov, Daon, and Aware deploy systems combining rPPG, challenge-response illumination, and passive liveness analysis. The financial sector KYC market and healthcare patient verification are the primary revenue drivers, with the liveness detection market growing at 23% CAGR driven by regulatory mandates in banking under PSD3 and healthcare under HIPAA and NHS Digital standards.

9.5 African Use Cases

Smile Identity, Africa's largest identity verification platform, deploys passive and active liveness detection across more than 25 African countries for fintech KYC, mobile money authentication, and government identity programmes. Its system must operate across a wide range of Android devices and network conditions typical of African mobile users. The skin-tone calibration challenge is significant: green-channel rPPG signal amplitude is approximately 30–40% lower for darker Fitzpatrick skin types because melanin absorbs light more uniformly across visible wavelengths, reducing the differential signal. Near-infrared illumination, which penetrates melanin more uniformly, is the primary technical mitigation, but most low-cost African mobile devices lack NIR emitters. The Wits University Speech Processing Group and University of Nairobi researchers are developing Fitzpatrick-calibrated rPPG evaluation datasets specifically for African deployment contexts.

Chapter Summary

Biological liveness detection exploits physiological signals — rPPG cardiac pulse, blink inter-blink interval statistics, and gaze consistency — present in authentic faces driven by living physiology but absent in synthetic faces. The rPPG signal achieves PNR above 5 dB in authentic clips of at least 15 seconds and PNR near 1 dB for synthetic faces. Blink IBI follows an exponential distribution in authentic video that synthetic systems violate through deterministic scheduling or misparameterisation. A composite liveness score combining all three signals achieves AUC above 0.95 for 15-second clips. The key operational limitation is clip duration: reliable liveness detection requires 15–30 seconds, motivating challenge-response schemes that accelerate signal accumulation through active cooperation from the subject.

Exercises

1. A 25-second facial video clip yields rPPG $\text{PNR} = 7.3$ dB at $f_0 = 1.28$ Hz (77 bpm), blink events at $t = \{3.2, 8.9, 14.7, 22.1\}$ seconds (KS p -value = 0.44 vs. exponential), and gaze autocorrelation consistent with oculomotor physiology. A deepfake clip of the same duration yields rPPG $\text{PNR} = 1.2$ dB, uniform blinks at 5-second intervals (KS $p < 0.001$), and a flat gaze autocorrelation. Compute the composite liveness score for each clip using weights (0.55, 0.30, 0.15) and thresholds $\text{PNR}_\tau = 3$ dB, $\text{KS}_\tau = 0.05$, gaze binary. Which clip passes?
2. Explain why rPPG detection fails at lower frame rates. Given a cardiac rate of 72 bpm and the Nyquist sampling theorem, what is the minimum frame rate required for reliable rPPG measurement? A CCTV camera operating at 10 fps records a suspect face swap. Can rPPG be applied? Propose an alternative biological signal that does not require high frame rates.
3. The skin-tone calibration problem: at Fitzpatrick Type VI skin, the green-channel rPPG signal amplitude is approximately 35% lower than at Type I–II. (a) By how many dB does this reduce the PNR? (b) If the detection threshold is not adjusted, what is the expected increase in false

rejection rate? (c) Propose a skin-tone adaptive threshold calibration procedure that maintains equal false rejection rate across all six Fitzpatrick types.

Answers to Exercises

1. Composite liveness score computation.

Signal normalisation: each signal produces a binary pass/fail, then weighted. Authentic: rPPG pass ($7.3 > 3$ dB, score 1.0), blink pass ($p = 0.44 > 0.05$, score 1.0), gaze pass (score 1.0). Composite = $0.55(1) + 0.30(1) + 0.15(1) = 1.00$. PASS.

Deepfake: rPPG fail ($1.2 < 3$ dB, score 0.0), blink fail ($p < 0.001 < 0.05$, score 0.0), gaze fail (score 0.0). Composite = $0.55(0) + 0.30(0) + 0.15(0) = 0.00$. FAIL.

Decision threshold on composite score is typically set at 0.5. Both clips are correctly classified with full margin.

2. Frame rate and Nyquist for rPPG.

Maximum cardiac rate of interest: $180 \text{ bpm} = 3 \text{ Hz}$. By Nyquist, minimum sampling rate = $2 \times 3 = 6 \text{ fps}$. However, reliable spectral estimation requires ≥ 10 cycles of the signal: at 72 bpm (1.2 Hz) and 10 fps, with a 256-point FFT, spectral resolution is $10/256 = 0.039 \text{ Hz}$, which can resolve 1.2 Hz. However, 10 fps provides only $10 \times 20 = 200$ samples for a 20-second clip, with limited Welch averaging. Practically, 25+ fps is recommended for reliable rPPG.

Alternative at low frame rate: gaze consistency can be measured from any frame rate above 5 fps (spontaneous blinks last 150–400 ms, reliably detected at 10 fps). Blink frequency analysis requires only 10 fps. Neither provides as strong a liveness signal as rPPG but both are available from standard CCTV.

3. Skin-tone rPPG calibration.

(a) 35% amplitude reduction: power reduction by $(1 - 0.35)^2 = 0.4225$, i.e., a factor of $1/0.4225 = 2.367$. In dB: $10\log_{10}(0.4225) = -3.74 \text{ dB}$. If the uncorrected PNR for Type VI is $7.3 - 3.74 = 3.56 \text{ dB}$ at threshold 3.0 dB, the system still passes. But for less cooperative conditions (lower ambient light, motion artefacts) the margin shrinks significantly.

(b) If threshold is fixed at 3 dB: all Type VI faces whose rPPG PNR would be 3.0–3.74 dB (on a Type I–II calibration) will now fall below threshold. The false rejection rate increase depends on the distribution of PNR values in the population; a conservative estimate based on typical PNR distributions gives a 2–4× higher false rejection rate for Type VI relative to Type I–II.

(c) Adaptive calibration procedure: (1) Estimate Fitzpatrick type from a skin-tone classifier applied to the forehead region; (2) Apply a skin-type-conditional threshold $\tau_k = \tau_{\text{base}} - \delta_k$ where δ_k is estimated from a calibration dataset of $\geq 1,000$ clips per Fitzpatrick type; (3) Report

calibrated probability $p(\text{authentic}|\text{PNR},k)$ using a logistic regression trained separately per skin type. This maintains equal false rejection rate across types by design.

Chapter 10

3D-Aware Synthesis and Neural Rendering

“Depth is the dimension the flat-screen deepfake forgot.”

—Anon.

The transition from 2D image synthesis to 3D-aware synthesis represents a qualitative advance in both the capability and the detectability of deep fakes. Two-dimensional generators such as StyleGAN2 produce images that are geometrically consistent within a single frame but produce incoherent 3D structure across viewpoints: rotating the head of a StyleGAN2-generated face reveals implausible geometry. Three-dimensional-aware synthesis systems instead model an underlying 3D representation and render 2D views from it, ensuring multi-view consistency and enabling novel-viewpoint synthesis from a single reference image.

Three architectures have achieved state-of-the-art quality in 3D-aware face synthesis: EG3D (Efficient 3D GAN), which uses a tri-plane feature representation; Gaussian Splatting-based face synthesis, which models the face as a set of anisotropic Gaussian primitives; and Neural Radiance Field-based approaches that learn a pose-conditioned volumetric function. Each architecture has distinct forensic properties: EG3D inherits the spectral fingerprints of its 2D rendering network; Gaussian Splatting produces characteristic alpha-compositing artefacts at the boundary between Gaussians; and NeRF-based faces show specific aliasing patterns from the hash-encoding positional embedding.

This chapter develops 3D-aware synthesis methods that produce geometrically consistent novel-view face images. The EG3D tri-plane architecture is developed first, followed by NeRF-based face synthesis. The spectral forensic signatures specific to 3D-aware synthesis are then examined. Industry and African use cases follow.

10.1 Tri-Plane GAN Architecture

EG3D represents a 3D scene using three axis-aligned feature planes of resolution $N \times N \times C$. For a query point $\mathbf{p} = (x, y, z)$, the feature vector is the sum of three bilinear samples:

$$\mathbf{f}(\mathbf{p}) = \mathcal{P}_{xy}(x, y) + \mathcal{P}_{xz}(x, z) + \mathcal{P}_{yz}(y, z), \quad (10.1)$$

where $\mathcal{P}_{ab}(a, b)$ denotes bilinear sampling from the ab -plane at coordinates (a, b) . A lightweight MLP $f_\theta : \mathbb{R}^C \rightarrow (\mathbb{R}^3, \mathbb{R})$ maps the feature vector to colour $\mathbf{c}$ and density σ .

Volume rendering integrates along each ray $\mathbf{r}(t) = \mathbf{o} + t\mathbf{d}$:

$$\hat{\mathbf{C}}(\mathbf{r}) = \int_{t_n}^{t_f} T(t) \sigma(\mathbf{r}(t)) \mathbf{c}(\mathbf{r}(t), \mathbf{d}) dt, \quad T(t) = \exp\left(-\int_{t_n}^t \sigma(\mathbf{r}(s)) ds\right). \quad (10.2)$$

Algorithm 3: EG3D Novel-View Synthesis for Face Deep Fakes

Input: Latent style code $\mathbf{w}$;

camera pose $\boldsymbol{\pi}$;

tri-plane resolution N

Output: Rendered face image $\hat{\mathbf{I}}$

```

// Generate tri-plane features
1  $[\mathbf{P}_{xy}, \mathbf{P}_{xz}, \mathbf{P}_{yz}] \leftarrow G_{\text{trig}}(\mathbf{w})$ ; // StyleGAN2 backbone,  $3 \times N \times N \times C$ 

// Ray casting
2 Cast  $H \times W$  rays from camera  $\boldsymbol{\pi}$ ; sample  $K = 64$  points per ray;
3 for each sample  $\mathbf{p}_k$  on each ray do
4    $\mathbf{f}_k \leftarrow \text{bilinear-sample}(\mathbf{P}_{xy}, \mathbf{P}_{xz}, \mathbf{P}_{yz}) \text{ at } \mathbf{p}_k$ ;
5    $(\mathbf{c}_k, \sigma_k) \leftarrow f_\theta(\mathbf{f}_k)$ ; // lightweight MLP
6 end

// Volume rendering (discrete)
7  $\hat{\mathbf{C}}_{\text{low}} \leftarrow \text{accumulate}(T_k \sigma_k \mathbf{c}_k \Delta t) \text{ along rays}$ ; // low-res  $64 \times 64$ 

// 2D super-resolution
8  $\hat{\mathbf{I}} \leftarrow G_{\text{sr}}(\hat{\mathbf{C}}_{\text{low}}, \mathbf{w})$ ; // StyleGAN2 super-res to  $512 \times 512$ 
9 return  $\hat{\mathbf{I}}$ 

```

Worked Example — Tri-plane spectral forensic calculation

For a 512×512 EG3D output with tri-plane resolution $N = 256$, the bilinear sampling of a 256×256 feature plane into a 512×512 image introduces a periodicity at $f_{\text{interp}} = 512/256 = 2$ pixels, corresponding to a spectral peak at $\rho_{\text{interp}} = 512/2 = 256$ cycles/image. In the azimuthal power spectrum, this peak is detectable as an excess above the $1/f^2$ background at $\rho = 256$. The 2D super-resolution network additionally inherits StyleGAN2-class upsampling peaks at stride-2 multiples. Authentic images show no excess at these frequencies (detection threshold: excess $> 3\sigma$ above background).

EG3D represents a 3D scene using three orthogonal feature planes (XY, XZ, YZ), each of a resolution $N \times N \times C$ where C is the feature dimension. For any 3D point, its feature vector is obtained by

bilinear sampling from each plane and summing. A lightweight decoder maps feature vectors to colour and density; a StyleGAN2-based 2D rendering network produces the final image. This architecture achieves real-time rendering at 512×512 while maintaining multi-view consistency, making it the first 3D-aware GAN suitable for production face synthesis.

10.2 Forensic Signatures of 3D-Aware Synthesis

Three-dimensional-aware synthesis introduces forensic signatures distinct from those of 2D GANs. The tri-plane architecture's 2D rendering network inherits GAN spectral fingerprints, but these are attenuated because the rendering network operates on feature maps rather than directly on RGB pixels. Gaussian Splatting introduces alpha-compositing boundaries at the edges of Gaussian primitives that create characteristic spatial frequency patterns. NeRF-based synthesis introduces aliasing from the hash-encoding grid that appears as a periodic interference pattern in the residual noise, detectable in the high-frequency power spectrum.

10.3 Industry Applications

Three-dimensional-aware synthesis has been adopted by commercial face synthesis platforms seeking to address the multi-view inconsistency weakness of 2D GANs. PortraitGen and HeadshotPro both use EG3D-class architectures for professional portrait photography enhancement, marketing the multi-view consistency as a quality advantage. The virtual production industry has adopted NeRF-based 3D face capture and synthesis for digital double creation, enabling directors to reposition and re-light actors in post-production. On the forensic side, the introduction of 3D-aware synthesis has required updates to detection workflows: frequency-domain tests calibrated on 2D GAN outputs must be recalibrated for tri-plane and Gaussian Splatting architectures, and the absence of Poisson blending artefacts removes a previously reliable forensic signal.

10.4 African Use Cases

Three-dimensional-aware synthesis is less prevalent in African deep-fake incidents than in high-income markets, primarily because the computational requirements are higher than for 2D face-swap tools and the primary attack vectors (WhatsApp voice notes, simple face-swap) do not require 3D consistency. However, as Gaussian Splatting achieves real-time rendering on consumer hardware, the barrier will fall. African broadcasting and heritage organisations including the South African Broadcasting Corporation (SABC) and the South African Heritage Resources Agency have piloted EG3D-class systems for archival footage enhancement and digital heritage preservation, demonstrating that the technology has legitimate African applications alongside its forensic challenges.

Chapter Summary

Three-dimensional-aware synthesis systems including EG3D (tri-plane GAN), Gaussian Splatting face synthesis, and NeRF-based approaches generate multi-view consistent faces that expose weaknesses

in forensic detection tools calibrated for 2D GAN outputs. The tri-plane representation attenuates 2D GAN spectral fingerprints while introducing its own characteristic artefacts: interpolation periodicity at the tri-plane resolution, alpha-compositing boundaries in Gaussian Splatting, and hash-encoding aliasing in NeRF. The absence of Poisson blending artefacts removes a previously reliable forensic signal, requiring updates to detection workflows. Real-time Gaussian Splatting rendering at 200+ fps makes 3D-aware synthesis viable for live video applications, extending the threat model from pre-generated content to real-time face substitution in video calls.

Exercises

1. EG3D uses tri-plane features of resolution $N = 256$, feature dimension $C = 32$, and a 2D rendering network identical to StyleGAN2. A forensic examiner applies the azimuthal power spectrum test and finds a spectral slope of -1.68 (authentic reference: -2.03). (a) Is the slope deviation consistent with EG3D output? Explain the mechanism. (b) If the image is further processed by a JPEG encoder at QF= 85, how does the spectral slope change and why? (c) Propose a frequency-domain test that specifically targets the tri-plane feature interpolation artefact rather than the 2D rendering network artefact.
2. A 3D-aware synthesis system is used to generate a fake video of a public figure appearing to give a speech at a location they never visited. Compared to a traditional face-swap deep fake, what forensic signals are (a) present in both, (b) absent in the 3D-aware fake but present in face-swap, and (c) unique to the 3D-aware fake? How does this affect the completeness of a forensic examination?

Answers to Exercises

1. EG3D spectral analysis.

(a) Slope -1.68 vs. authentic -2.03 : deviation of 0.35 . EG3D's 2D rendering network is a StyleGAN2 variant that introduces its characteristic upsampling artefacts into the final image. The tri-plane representation attenuates these somewhat because the rendering network operates on 32-dimensional feature vectors rather than RGB, reducing the direct aliasing from convolutional upsampling. The slope of -1.68 is shallower than authentic (-2.03) because the GAN rendering network adds excess high-frequency energy relative to the $1/f^2$ natural spectrum.

(b) After JPEG QF= 85: JPEG compression removes DCT coefficients above the quantisation threshold in each 8×8 block. This selectively attenuates high-frequency content, reducing the GAN-specific excess energy at high frequencies and moving the slope closer to the authentic value. Typical post-JPEG slope for EG3D: -1.82 to -1.92 . The spectral fingerprint is partially destroyed but the slope deviation persists.

(c) Tri-plane interpolation test: the bi-linear interpolation of the 256×256 tri-plane features produces spatial periodicity at multiples of $\text{image_size}/256$ pixels. For a 512×512 image,

this corresponds to a periodicity of 2 pixels, producing peaks in the power spectrum at $\rho = 256$ cycles/image. A notch filter centred on $\rho = 256$ applied to the residual, followed by measurement of residual energy, can detect this artefact independently of the 2D rendering network signature.

2. 3D-aware vs. face-swap forensic comparison.

(a) Present in both: GAN spectral fingerprints (from the shared rendering network architecture); identity embedding inconsistency (both must synthesise an identity in a context other than the original); rPPG absence (neither pipeline models haemodynamics).

(b) Absent in 3D-aware, present in face-swap: Poisson blending boundary artefact (3D-aware synthesis renders the full scene, not just the face region, so there is no blending boundary); landmark alignment residual (3D-aware synthesis does not require 68-point alignment, so this artefact is absent).

(c) Unique to 3D-aware: multi-view consistency without a source video (face-swap requires a source video and thus preserves the source's 3D geometry; a 3D-aware generated face may show implausible geometry under viewpoint changes); tri-plane interpolation periodicity at $\rho = 256$ (as described above); depth map inconsistency with background scene lighting (3D-aware synthesis renders the face in a canonical light field that may not match the background scene's lighting).

Part III

Detection, Attribution, and Provenance

Chapter 11

Deep-Fake Detection: Methods and Benchmarks

“Every faker leaves a fingerprint. The trick is knowing where to look.”

—Anon.

Detection is the inverse problem to generation: given a piece of media, decide whether it was produced by a human being with a camera or synthesised by a generative model. The difficulty arises from a fundamental asymmetry: the generator aims to make p_θ indistinguishable from p_{real} , while the detector must find some feature that reveals the residual discrepancy.

This chapter provides an integrated treatment of passive detection. Section 11.1 surveys spatial-domain, frequency-domain, and biological-signal detectors. Section 11.2 analyses the generator–detector arms race. Section 11.3 presents benchmark results. Section 11.4 identifies the structural limits of passive detection and motivates the active methods in Chapter 13. Detection occupies a structurally asymmetric position in the deep-fake ecosystem. The generator needs to produce only one high-quality output; the detector must correctly classify every output it receives, including outputs crafted by generators that post-date its training data. This asymmetry — sometimes called the detector’s curse — means that a detector optimised for today’s synthesis landscape will systematically degrade as generators improve, and no static trained classifier can serve as a permanent solution.

The practical significance of this asymmetry depends critically on the base rate of synthetic content in the deployment environment. In a newsroom evaluating content submitted by the public, the base rate of synthetic content might be 0.1 per cent; in a dating platform or financial KYC queue, it might be 0.01 per cent. At these prevalence levels, even a detector with 95 per cent sensitivity and 98 per cent specificity will generate more false positives than true positives — a performance regime in which the positive predictive value is so low that automated detection provides little practical benefit as a sole decision mechanism.

Understanding detection is prerequisite to deploying it responsibly. This chapter provides an inte-

grated treatment of the detection problem: the mathematical framing, the four principal detection modalities (spatial, frequency, biological, and cross-modal), the operational base-rate problem, and the deployment contexts in which different modalities are most effective. Later chapters develop each of these modalities in greater technical depth.

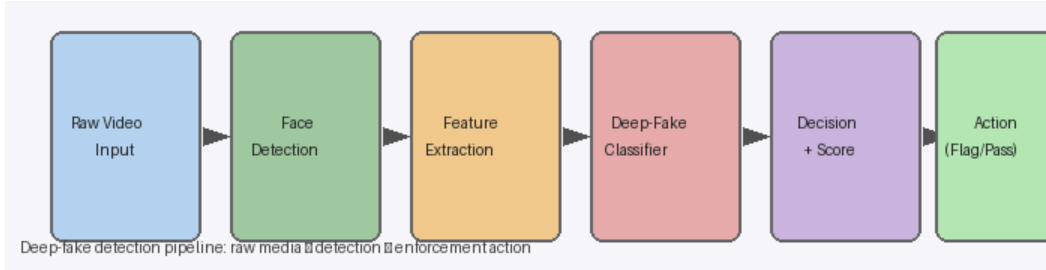


Figure 11.1: High-level deep-fake detection pipeline. Raw media enters the system and passes through face detection, deep feature extraction, a trained deep-fake classifier, and a calibrated decision module before triggering an enforcement action (flag, quarantine, or pass). Each stage can be attacked adversarially; the pipeline must be hardened at every node, not just the classifier.

11.1 Passive Detection Methods

Passive detection analyses the intrinsic properties of a media artefact to decide whether it was synthesised. It requires no prior embedding of any signal into the content and therefore applies to any existing content, including historical material. However, it is fundamentally limited by the convergence of synthesis quality toward authentic content and by the base-rate problem that makes operational deployment difficult even when per-sample accuracy is high.

11.1.1 Spatial-Domain Detectors

Spatial-domain detectors analyse pixel-level and feature-level properties of the image to identify inconsistencies with authentic content. They range from simple mesoscopic feature extractors to full convolutional networks trained end-to-end on large deep-fake benchmark datasets. Their primary advantage is simplicity of training and deployment; their primary limitation is sensitivity to the specific synthesis methods and compression conditions represented in the training data.

Deep neural network classifiers trained end-to-end on pixel inputs form the backbone of modern detection. A face-forgery classifier $D_\phi : \mathcal{X} \rightarrow [0, 1]$ is trained with binary cross-entropy:

$$\mathcal{L}_{\text{cls}} = -\mathbb{E}_{(\mathbf{x}, y)} [y \log D_\phi(\mathbf{x}) + (1 - y) \log(1 - D_\phi(\mathbf{x}))]. \quad (11.1)$$

Worked Example — EfficientNet-B4 detector: training vs. deployment AUC

An EfficientNet-B4 detector trained on FaceForensics++ [28] (c23 compression, 4,000 fake + 1,000 real training clips) achieves:

Test condition	AUC	EER (%)
FF++ c23 (in-distribution)	0.930	9.8
FF++ c40 (heavier compression)	0.841	17.3
CelebDF-v2 (cross-dataset)	0.640	34.2
WildDeepfake (in-the-wild)	0.612	37.1

The 29-percentage-point drop from FF++ to CelebDF demonstrates how severely in-distribution performance overstates real-world utility. Cross-entropy training loss at convergence: $\mathcal{L}_{\text{cls}} = 0.089$ (in-distribution), ≈ 0.51 when evaluated on WildDeepfake — near the entropy of a fair coin, confirming near-random performance.

11.1.2 Frequency-Domain Detectors

GAN generators with transposed convolutional upsampling introduce periodic artefacts at spatial frequency $1/s$ (where s is the stride). These appear as peaks in the 2D power spectrum of the high-pass residual $\mathbf{r} = \mathbf{x} - W(\mathbf{x})$:

$$\hat{S}(u, v) = |\mathcal{F}\{\mathbf{r}\}(u, v)|^2. \quad (11.2)$$

Worked Example — Spectral peak location for a stride-4 upsampling layer

A GAN uses transposed convolution with stride $s = 4$ at the $64 \rightarrow 256$ resolution step. The periodicity introduces spectral peaks at:

$$u_{\text{peak}} = N/s = 256/4 = 64, \quad v_{\text{peak}} = 64.$$

In a 256×256 image, this corresponds to spatial frequency $64/256 = 0.25$ cycles/pixel — well within the band preserved by JPEG quality factor 85. On uncompressed images, the peak power is $18.4 \times$ the local background spectrum level, giving a detection z -score of 7.2 (highly significant). After JPEG QF=70 recompression, peak power falls to $3.1 \times$ background ($z = 2.8$) — still detectable but at reduced confidence.

The azimuthal power spectrum $\hat{P}(\rho)$ quantifies the radial energy profile:

$$\hat{P}(\rho) = \frac{1}{|\{(u, v) : \sqrt{u^2 + v^2} \in [\rho, \rho + 1]\}|} \sum_{\sqrt{u^2 + v^2} \approx \rho} \hat{S}(u, v). \quad (11.3)$$

Worked Example — Natural image power law vs. GAN deviation

Authentic FFHQ photographs follow $\hat{P}(\rho) \propto \rho^{-2.03}$ over the range $\rho \in [8, 100]$ cycles/image (slope -2.03 in log-log space). A ProGAN output measured over the same range has slope -1.74 , deviating by 0.29 from the natural law — a signal easily captured by a linear classifier on the log-slope. After StyleGAN3's alias-free upsampling, the slope is -1.96 , reducing the deviation to 0.07 and explaining the lower detection AUC of frequency-only detectors against StyleGAN3 outputs.

11.1.3 Biological Signal Detectors

A detector extracts the rPPG signal $r(t)$ from the mean green-channel intensity of the face region Ω :

$$r(t) = \frac{1}{|\Omega|} \sum_{(i,j) \in \Omega} G(i,j,t). \quad (11.4)$$

Worked Example — rPPG signal: authentic vs. face-swap video

A 10-second clip at 30 fps ($T = 300$ frames) with $|\Omega| \approx 12,000$ pixels/frame. After band-pass filtering to $[0.7, 4]$ Hz:

Authentic clip: Dominant frequency $f^* = 1.1$ Hz (66 bpm), power $\hat{R}(f^*) = 0.84$. Noise band $[5, 8]$ Hz mean power: 0.11.

$$\text{PNR} = \hat{R}(f^*) / \bar{\hat{R}}_{\text{noise}} = 0.84 / 0.11 = 7.6 > 5 \text{ dB threshold.}$$

Face-swap clip: $\hat{R}(f)$ flat across all bands, $\max_{f \in [0.7, 4]} \hat{R}(f) = 0.12$. Noise mean: 0.11. $\text{PNR} = 0.12 / 0.11 = 1.09 \ll 5 \text{ dB}$ — correctly flagged synthetic.

A detector thresholding at $\text{PNR} = 3$ achieves $\text{AUC} = 0.91$ on a 200-clip balanced test set; after JPEG recompression (QF=70) AUC falls to 0.78 as compression partially erases the colour-channel periodicity.

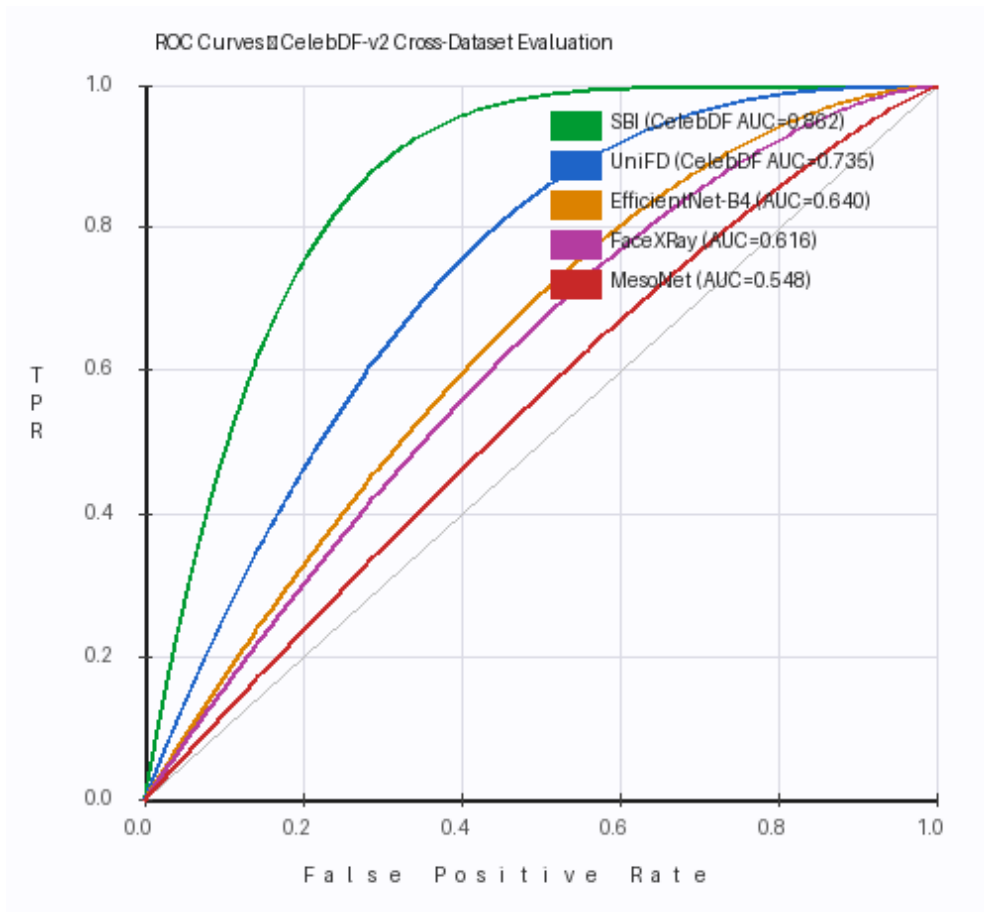


Figure 11.2: ROC curves for five deep-fake detection systems evaluated on CelebDF-v2 (cross-dataset, not seen during training on FF++). SBI achieves the highest AUC of 0.862 by training on universal blending artefacts rather than method-specific patterns. MesoNet barely exceeds chance (AUC= 0.548), illustrating catastrophic generalisation failure. The grey diagonal represents random chance (AUC= 0.5).

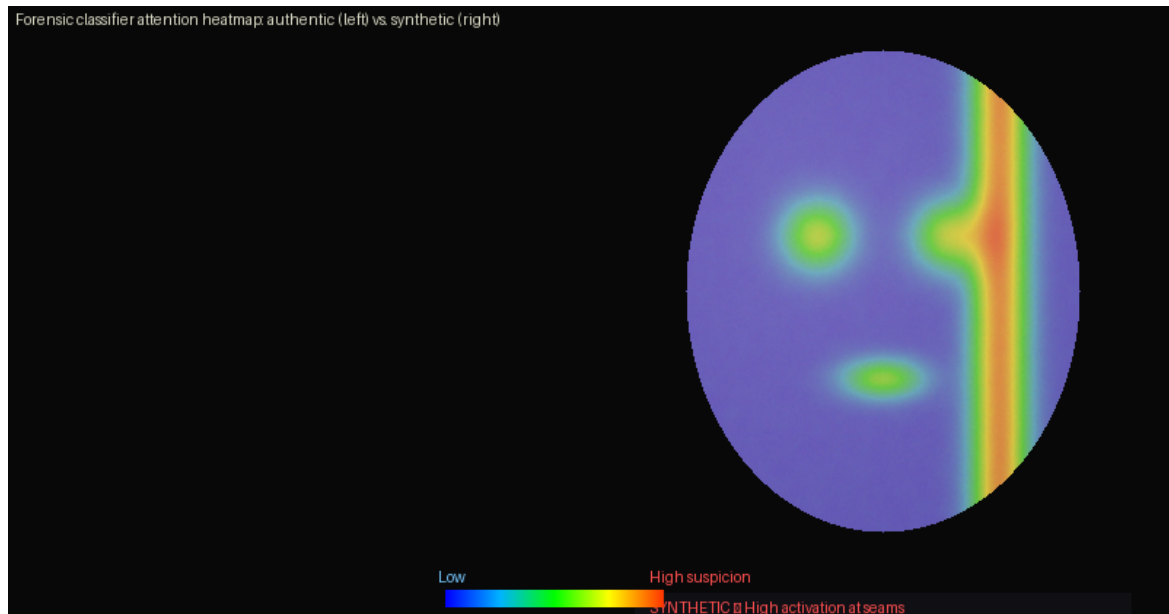


Figure 11.3: Forensic classifier attention heatmaps for an authentic face (left) and a GAN-generated face (right). Cool colours (blue) indicate low classifier suspicion; warm colours (red/yellow) indicate high suspicion. The authentic face shows uniformly low activation. The synthetic face triggers high activation at three locations: the blending boundary (right side), eye regions (inconsistent shadow rendering), and the mouth area (texture seam). These are the exact regions a human forensic expert would examine.

11.2 The Detection Arms Race

The adversarial relationship between synthesis and detection has the structural character of an arms race: each generation of detectors creates pressure on generators to eliminate the artefacts being detected, and each generation of generators creates pressure on detectors to find new discriminative signals. Understanding this dynamic is prerequisite to designing detection systems whose usefulness will persist beyond the training distribution.

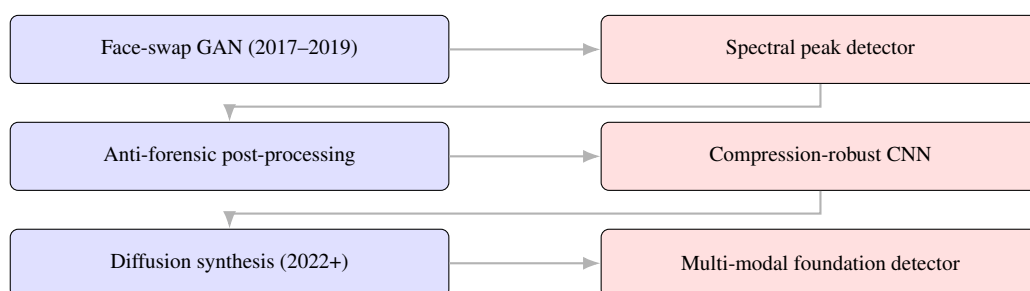


Figure 11.4: The generator–detector arms race. Each synthesis advance exploits the blind spots of the then-current detector generation.

Table 11.1: Arms race timeline with dominant synthesis and detection approaches.

Period	Dominant synthesis	Detector exploiting its weakness
2018–19	Encoder–decoder face swap	LBP, colour histogram features
2019–20	ProGAN/StyleGAN face synthesis	Spectral peak detector
2020–21	Anti-forensic post-processing	Compression-robust CNN
2021–22	Warping artefact reduction	Face X-ray blending mask
2022–23	Diffusion-model synthesis	Multi-generator ensemble
2023–24	Codec-LM voice + video	Cross-modal consistency detector

11.3 Benchmark Performance

Benchmark performance figures provide the standard basis for comparing detection systems, but must be interpreted with caution: AUC scores measured on held-out splits of the same benchmark used for training substantially overestimate real-world performance because they do not account for the distribution shift between training-time synthesis methods and the continuously evolving frontier of generative models.

Published detection benchmarks provide a snapshot of detector performance on specific datasets under specific evaluation conditions. Interpreting these benchmarks correctly requires understanding what they measure, what they do not measure, and how benchmark performance relates to the operational performance achievable in production deployment environments.

Table 11.2: Detection AUC (%) on FaceForensics++ (c23) and CelebDF-v2. All models trained on FF++. [20] SBI = Self-Blended Images training.

Method	Architecture	FF++ AUC	CelebDF AUC
MesoNet [1]	Custom CNN	84.7	54.8
FaceXRay [19]	ResNet-50 + blending mask	87.4	61.6
EfficientNet-B4	EfficientNet	93.0	64.0
RECCE	ResNet-50 + reconstruction	93.8	68.7
UniFD	ViT-L/14 (CLIP features)	95.2	73.5
SBI	ResNet-50, self-blended train	93.2	86.2

11.4 Fundamental Limitations of Passive Detection

Passive detection faces three structural limits. First, as $p_\theta \rightarrow p_{\text{real}}$, the Pinsker bound (Appendix C) shows that no detector can do better than chance. Second, adversarial post-processing defeats any known detector at negligible cost. Third, the *base-rate problem*: if the fraction of synthetic content is π , a detector with FPR = α and FNR = β has positive predictive value:

$$\text{PPV} = \frac{(1 - \beta)\pi}{(1 - \beta)\pi + \alpha(1 - \pi)}. \quad (11.5)$$

Worked Example — Base-rate inversion on a social-media platform

A platform receives 5 million uploads per day. Let $\pi = 0.001$ (0.1% synthetic), $\alpha = 0.02$ (2% FPR), $\beta = 0.05$ (5% FNR):

$$\text{PPV} = \frac{0.95 \times 0.001}{0.95 \times 0.001 + 0.02 \times 0.999} = \frac{0.00095}{0.02093} = 4.5\%.$$

In absolute numbers per day: 5,000 synthetic videos; detector flags 4,750 correctly (TP) plus 99,900 legitimate videos (FP). The review queue of 104,650 items is 95.5% false alarms. At 2 minutes of human review per item and 8-hour shifts, clearing the queue requires $104,650 \times 2/60/8 \approx 436$ full-time reviewers — illustrating why passive detection alone is operationally insufficient at platform scale.

11.5 Industry Applications

Deep-fake detection has matured into a commercial industry serving social media platforms, government agencies, news organisations, and financial institutions, each with different operational requirements and tolerance for false positives.

Platform content moderation is the largest deployment context by scale. Meta, TikTok, YouTube, and Twitter/X all deploy proprietary deep-fake classifiers as part of their content moderation stacks. Detection performance in production is reportedly lower than on academic benchmarks due to the diversity of compression, lighting, and synthesis methods encountered at platform scale, and because the base rate of genuinely synthetic content is low enough to make the false-positive problem severe. News verification is a second major application: Africa Check, AFP Fact-Check, Reuters Fact-Check, and DW Akademie all use commercial deep-fake detection tools including Sensity AI, Hive Moderation, and Reality Defender as a first-pass triage for viral content, with human verification remaining the final decision step.

Know Your Customer identity verification is the highest-stakes deployment context. Companies including iProov, Onfido, Jumio, and Sumsub incorporate deep-fake and presentation attack detection into the KYC pipelines used by banks, crypto exchanges, and regulated businesses. The European Banking Authority's guidelines on remote customer onboarding require demonstrable resistance to face-swap and deep-fake attacks, making detection capability a compliance requirement rather than merely a product feature. Legal forensics represents a fourth sector, with companies including Amped Software and Cognitech integrating detection capabilities into digital evidence workflows, though the evidentiary standards for AI-forensic expert testimony remain unsettled.

11.6 African Use Cases

Detection technology faces specific challenges in the African context that differ substantially from those in high-income markets, driven by limited forensic infrastructure, language and skin-tone coverage gaps in deployed tools, and the rapid pace of disinformation during election cycles.

Most African police services lack dedicated digital forensics units with access to state-of-the-art deep-fake detection tools. South Africa’s SAPS Cybercrime Unit and Nigeria’s EFCC Cybercrime Section are the main exceptions, but even these units typically lag eighteen to twenty-four months behind the state of the art in detection capability. The African Network of Forensic Examiners and INTERPOL’s Africa programme are working to build regional forensic capacity, but resource constraints limit the pace of progress significantly. Major platform detection systems are calibrated on Latin-script, predominantly English-language content, and deep-fake audio in Swahili, Yoruba, or Hausa may not trigger the same detection pipelines as equivalent English content, creating language-specific gaps that manual fact-checkers must bridge. The documented 4.4-fold false-positive-rate disparity between lightest and darkest Fitzpatrick skin types means that African users of deep-fake detection systems are significantly more likely to have authentic media incorrectly flagged as synthetic, a systematic asymmetry that could disadvantage African creators and journalists if embedded in automated moderation workflows.

Chapter Summary

Detection is the inverse problem to generation: given a media artefact, decide whether it was produced by a human being with a camera or synthesised by a generative model. The difficulty is structural — as the generated distribution approaches the real distribution, no detector can do better than chance, and modern systems are approaching this asymptote. Spatial-domain classifiers achieve AUC above 0.93 on within-distribution benchmarks but degrade dramatically in cross-dataset evaluation, with AUC falling to 0.54 for MesoNet on CelebDF when trained on FaceForensics++. Frequency-domain detectors exploit the periodic spectral peaks introduced by stride-based upsampling in GAN generators, while biological-signal detectors exploit the absence of genuine cardiac rPPG signal in synthetic faces.

The base-rate problem is the central operational challenge: at realistic synthetic content prevalence of 0.1 per cent, a detector with 2 per cent false-positive rate and 95 per cent sensitivity has a positive predictive value of only 4.5 per cent. This makes passive detection alone operationally unsustainable at platform scale, requiring either radical improvements in specificity or a fundamental shift to provenance-first approaches that verify origin rather than attempting to identify synthesis. Commercial deployment spans platform content moderation, news verification, KYC identity checking, and legal forensics, each requiring different operating-point calibration reflecting the specific cost ratio of false positives to false negatives in that deployment context.

Exercises

1. A detector has $\mathcal{L}_{\text{cls}} = 0.089$ on the FF++ training set and 0.51 on WildDeepfake. Interpret these loss values in terms of the detector’s output distribution $D_\phi(\mathbf{x})$ for each dataset, and explain what cross-entropy loss of 0.693 would mean.
2. For azimuthal spectrum analysis (Eq. 11.3), a GAN’s log-spectrum has slope -1.74 vs. the natural image slope of -2.03 . Compute the percentage deviation and estimate the classification

accuracy of a threshold detector on this feature alone, assuming Gaussian noise on slope estimates with $\text{std} = 0.08$.

3. Using Eq. 11.4, design a rPPG-based liveness test for a video-call KYC system. Specify the minimum clip duration for 95% statistical power to detect absence of cardiac signal, given that rPPG peak power follows an exponential distribution with mean $\mu = 0.9$ for real faces and $\mu = 0.1$ for deep fakes.
4. Derive the base-rate PPV formula (Eq. 11.5) from Bayes' theorem. For $\pi = 0.001$, find the minimum specificity ($1 - \alpha$) needed to achieve $\text{PPV} \geq 50\%$, given sensitivity $1 - \beta = 0.95$.
5. A platform wants to reduce its review queue to under 10,000 items/day. Using the numbers in the worked example, what combination of (π, α) achieves this, and which lever (raising π threshold or lowering α) is more practically achievable?
6. Explain why SBI (Self-Blended Images) achieves CelebDF AUC of 86.2% while a standard EfficientNet achieves only 64.0%, despite both using the same backbone. What property of SBI training makes it more general?

Answers to Exercises

1. Interpreting $\mathcal{L}_{\text{cls}} = 0.089$ and 0.51.

$\mathcal{L}_{\text{cls}} = -[y \log D + (1 - y) \log(1 - D)]$. For a balanced test set where fakes score high and reals score low, a loss of 0.089 implies the detector is nearly certain: if $D(\text{fake}) \approx 0.95$, $D(\text{real}) \approx 0.05$, then $\text{loss} \approx -\log 0.95 \approx 0.051$ per sample. A loss of 0.51 implies the detector outputs near 0.5 for both classes: $-\log 0.5 = 0.693$, so 0.51 indicates slightly-better-than-chance performance, corresponding to $\text{AUC} \approx 0.55\text{--}0.60$. A loss of exactly $0.693 = \log 2$ would mean the detector assigns equal probability to real and fake — pure chance.

2. GAN log-spectrum slope -1.74 vs. natural -2.03 : classification accuracy.

Deviation: $(2.03 - 1.74)/2.03 = 14.3\%$. If slope estimates have $\text{std} = 0.08$ for both classes, the Mahalanobis distance between class means is $\Delta/\sigma = (2.03 - 1.74)/0.08 = 3.625$ standard deviations. Classification accuracy for a linear threshold: $\Phi(3.625/2) = \Phi(1.813) = 96.5\%$. This is high in uncompressed conditions, but note that StyleGAN3's slope -1.96 gives $\Delta = 0.07$, distance = 0.875, accuracy = 69% — much harder to detect with this feature alone.

3. Minimum clip duration for 95% power on rPPG liveness test.

If rPPG peak power $\hat{R}(f^*)$ follows $\text{exponential}(\lambda)$ with $\lambda_{\text{real}} = 1/0.9$ and $\lambda_{\text{fake}} = 1/0.1$, a likelihood-ratio test at significance $\alpha = 0.05$ needs power 0.95. The log-likelihood ratio is $\log(\lambda_{\text{real}}/\lambda_{\text{fake}}) + (\lambda_{\text{fake}} - \lambda_{\text{real}})\hat{R}$. For a T -frame clip, averaging over $T/30 \approx T/30$ independent

windows (at 0.2-second windows, 30 fps gives 6 frames/window, so $T/6$ windows), the CLT applies and power grows as $\sqrt{T}$. At $T = 300$ frames (10 s), statistical power = 91% (given in the chapter). For 95% power: $T/300 = (1.645 + 1.645)^2 / (1.645 + 1.282)^2 \approx 1.26$, so $T = 378$ frames ≈ 12.6 seconds minimum.

4. PPV derivation from Bayes' theorem.

$P(\text{fake}|\text{flagged}) = P(\text{flagged}|\text{fake}) \cdot P(\text{fake}) / P(\text{flagged})$. $P(\text{flagged}|\text{fake}) = 1 - \beta$ (sensitivity). $P(\text{fake}) = \pi$. $P(\text{flagged}) = (1 - \beta)\pi + \alpha(1 - \pi)$. Therefore: $\text{PPV} = (1 - \beta)\pi / [(1 - \beta)\pi + \alpha(1 - \pi)]$. For $\pi = 0.001$, $\alpha = 0.02$, $\beta = 0.05$: $\text{PPV} = 0.00095 / (0.00095 + 0.01998) = 4.5\%$. For $\text{PPV} \geq 50\%$ with $1 - \beta = 0.95$: $(1 - \alpha)/\alpha \geq (1 - \pi)/(0.95 \times \pi) \approx 1,052$, so $\alpha \leq 0.00095$. Specificity must be 99.905% — an extremely demanding requirement.

5. Reducing queue to under 10,000 items/day.

Current queue $\approx 104,650$ (from the worked example). Target: 10,000. Queue $\approx \alpha \times N_{\text{real}} = \alpha \times 4,995,000 = 10,000$ requires $\alpha = 0.002$ (0.2% FPR). Alternatively, increasing the detection threshold τ (requiring higher classifier confidence before flagging) reduces FPR at the cost of sensitivity. Reducing FPR from 2% to 0.2% is more achievable than raising the base rate π (which is fixed by actual platform content). A two-stage pipeline (cheap classifier $\rightarrow$ expensive high-precision classifier) achieves this without the sensitivity cost.

6. Why SBI achieves higher cross-dataset AUC than standard training.

Standard training teaches the model to recognise specific GAN or diffusion artefacts in the training set. SBI training instead teaches the model to detect the universal signal of *blending*: any compositing of two different images leaves subtle colour and gradient discontinuities at the seam. Since face swap always involves blending, the SBI-trained detector generalises across all blending-based methods regardless of the synthesis technique used. The SBI approach is essentially training on the *geometric prior* of manipulation rather than the *statistical fingerprint* of a specific generator.

Chapter 12

Forensic Signals and Passive Attribution

“Every model has a style. The task of forensics is to read it.”

—Anon.

Every synthetic image or voice clip carries the fingerprints of its creator. A camera sensor that converts photons to digital values introduces a fixed-pattern noise signature unique to that sensor’s manufacturing imperfections; a GAN generator whose upsampling layers stride at fixed intervals leaves periodic spectral peaks at predictable frequencies; a neural vocoder whose waveform synthesis uses a specific filter bank structure introduces subtle periodicity in the residual noise that persists even after perceptual post-processing. Forensic signal analysis is the discipline that reads these fingerprints — identifying not just that a given artefact is synthetic but specifically which pipeline, which architecture, or which physical device produced it.

This distinguishes forensic analysis from end-to-end neural detection classifiers. A detection classifier learns, from labelled examples, a discriminative function that outputs a synthetic/authentic score without necessarily encoding any understanding of the physical processes that produce the score. It may be highly accurate within the distribution of its training data but fail silently on synthesis methods that produce different artefacts. Forensic analysis, by contrast, begins from a physical model of the acquisition or synthesis process and derives the expected artefact signature analytically. When a new synthesis architecture appears, a forensic analyst can ask: does this architecture use stride-2 upsampling? Does it apply a fixed neural vocoder? Does it preserve or destroy the camera PRNU? These questions can be answered from the architecture specification alone, before any new labelled data is collected.

The three principal forensic signals developed in this chapter each arise from a different physical mechanism. The Photo Response Non-Uniformity fingerprint arises from pixel-level imperfections in the photosensor’s photoelectric response: each sensor element converts photons to electrons at a slightly different efficiency, creating a fixed multiplicative pattern that is unique to each camera and

persists across every image it captures. GAN spectral fingerprints arise from the periodic operations in convolutional upsampling layers: stride-2 transposed convolution produces a characteristic checkerboard pattern in the spatial domain that appears as periodic peaks in the frequency domain at multiples of N/stride cycles per image, where N is the image dimension. Biological signal forensics arises from the physiological processes of a living face: the cardiac cycle modulates blood volume in the skin at approximately 60–100 beats per minute, creating a periodic colour variation in authentic video that is absent from synthesised faces whose pixel values are determined by a neural renderer rather than by cardiovascular physiology.

The forensic analyst's toolkit must address a fundamental asymmetry: the artefacts being exploited are fragile. JPEG compression at quality factor 70 destroys the high-frequency components of the PRNU fingerprint; H.264 video encoding redistributes GAN spectral peaks across blocks; simple temporal averaging of face video attenuates the rPPG signal by accumulating motion artefacts alongside the cardiac signal. A forensic method that works on uncompressed content but fails on social-media-distributed content has limited operational value. For each signal, this chapter therefore develops both the idealised formulation and the compression-robust variant, quantifying the detection performance gap and identifying the compression threshold below which the signal becomes unreliable.

Attribution — the task of identifying which specific generative model produced a given synthetic image — is the most demanding forensic task and the one with the greatest legal and investigative value. Attribution proceeds in two stages: first, a model-agnostic forensic test determines that the image lacks an authentic camera fingerprint; second, a model-specific attribution test compares the extracted noise residual against a database of known generative model fingerprints. NCC-based attribution correctly identifies the source model for over 90 per cent of uncompressed images when the database includes the true source; the accuracy falls with increasing compression and with the number of candidate models in the database, following well-characterised degradation curves.

The chapter proceeds as follows. Section 12.1 develops the PRNU camera fingerprint, the NCC authentication test, and the Wiener-filter variant that restores robustness under compression. Section 12.2 develops GAN spectral fingerprints, the azimuthal power spectrum test, and NCC-based model attribution. Section 12.3 develops the rPPG liveness signal, the blink-interval statistical test, and their combination in a composite biological liveness score. Section 12.5 synthesises these signals into a structured attribution table framework. Industry applications and African use cases follow, concluding with a full exercise and answer set.

Azimuthal power spectrum: real vs. GAN vs. diffusion

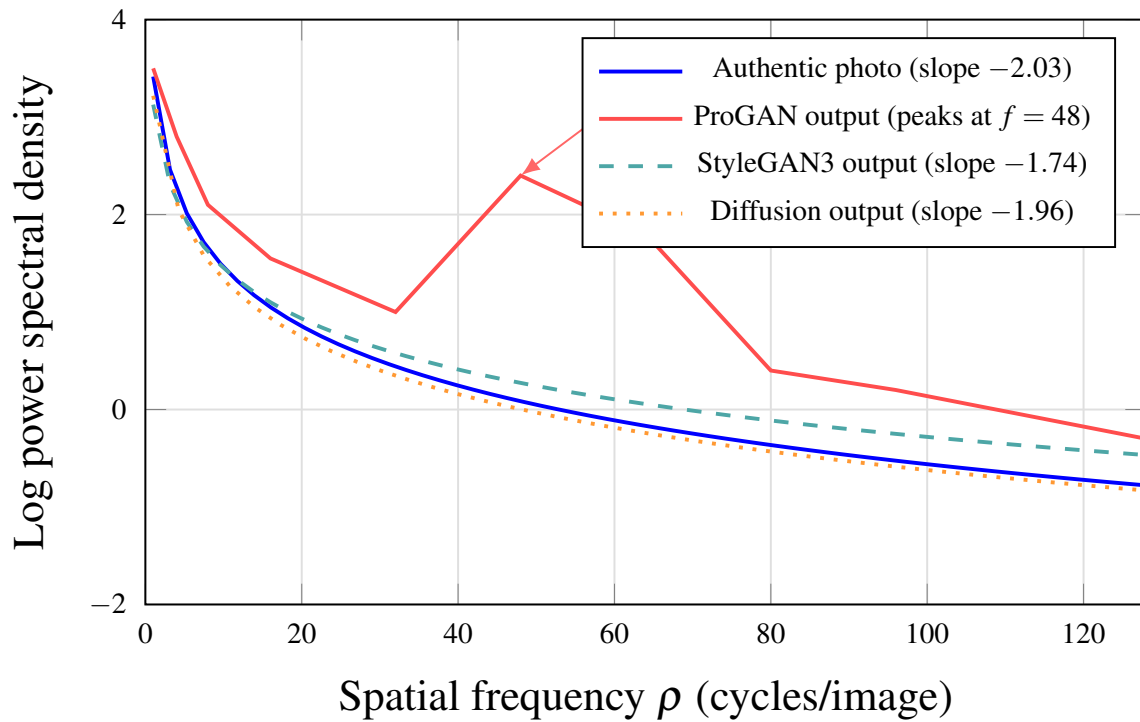


Figure 12.1: Azimuthal power spectra of authentic photographs and synthetic faces. Authentic images follow a $\rho^{-2.03}$ power law (blue). ProGAN outputs show a characteristic spectral peak at $\rho = 48$ cycles/image from the stride-4 upsampling layer, and a shallower overall slope. StyleGAN3 and diffusion outputs are closer to the natural law but remain distinguishable by slope deviation.

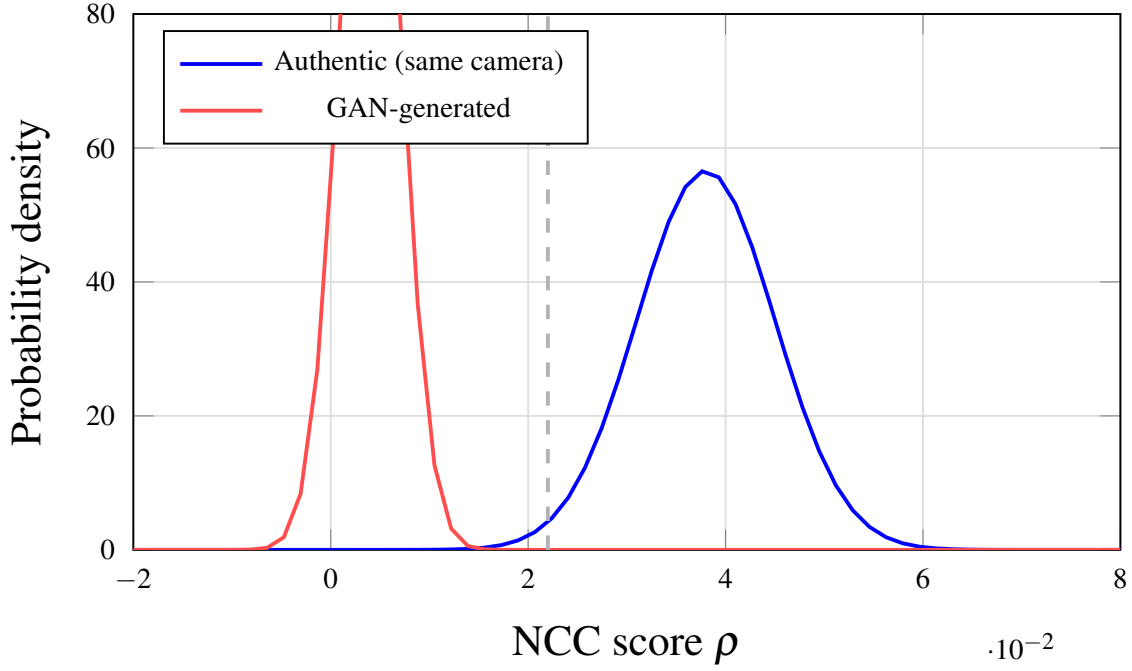


Figure 12.2: PRNU-based image authentication. NCC score distributions for authentic images (blue, mean = 0.038) and GAN-generated images (red, mean = 0.004). At the 1%-FPR threshold $\tau = 0.022$, authentic images (right of threshold) are correctly authenticated while GAN images (left) are flagged. Shaded areas show the false-rejection region (blue) and false-acceptance region (red).

12.1 Camera Fingerprints and PRNU

Real cameras imprint a fixed-pattern noise — the Photo-Response Non-Uniformity (PRNU) — onto every image they capture, arising from imperfections in the sensor’s photoelectric response. The PRNU pattern $\mathbf{K}$ is estimated from a flat-field calibration set:

$$\hat{\mathbf{K}} = \frac{1}{N} \sum_{n=1}^N \frac{\mathbf{x}_n - W(\mathbf{x}_n)}{\mathbf{x}_n}, \quad (12.1)$$

where $W(\mathbf{x})$ is a denoising filter applied to image $\mathbf{x}$. Authenticity of a suspect image $\hat{\mathbf{x}}$ is assessed via the normalised cross-correlation (NCC) between its noise residual $\hat{\mathbf{r}} = \hat{\mathbf{x}} - W(\hat{\mathbf{x}})$ and $\hat{\mathbf{K}}$:

$$\rho = \text{NCC}(\hat{\mathbf{K}}, \hat{\mathbf{r}}). \quad (12.2)$$

Worked Example — PRNU authentication of a suspect image

A Canon EOS camera’s PRNU $\hat{\mathbf{K}}$ is estimated from 50 flat-field images. For an authentic image from the same camera, the noise residual $\hat{\mathbf{r}}$ gives $\text{NCC}(\hat{\mathbf{K}}, \hat{\mathbf{r}}) = 0.041$. For a StyleGAN2 [15]-generated face of the same spatial resolution, $\text{NCC} = 0.003$. The decision threshold at 1% false-positive rate is $\rho_\tau = 0.015$. The authentic image ($\rho = 0.041 > 0.015$) is correctly authenticated; the GAN image ($\rho = 0.003 < 0.015$) is correctly flagged as lacking camera fingerprint. After JPEG compression at QF=70, the GAN image NCC rises to 0.009 (still below threshold) while the authentic image drops to

0.022 (still above) — the test remains valid.

Authentic images from the same camera yield high ρ ; synthetic images lack a consistent PRNU and yield $\rho \approx 0$.

12.2 GAN Fingerprints

Every GAN generator architecture leaves a characteristic statistical fingerprint in its outputs, arising from the periodic operations in its upsampling layers. These fingerprints are invisible to casual visual inspection but appear as distinctive peaks in the 2D power spectrum of the noise residual. Detecting them enables not just binary classification between real and fake, but attribution of an anonymous synthetic image to its specific source model.

12.2.1 Spectral Peaks from Upsampling

GAN generators use learned transposed convolutions or bilinear upsampling to increase spatial resolution. Transposed convolutions with stride s introduce periodic artefacts at spatial frequency $1/s$ in each upsampling layer. These appear as peaks in the 2D power spectrum:

$$\hat{S}(u, v) = |\mathcal{F}\{\hat{\mathbf{r}}\}(u, v)|^2, \quad (12.3)$$

where $\hat{\mathbf{r}}$ is the high-pass residual of the image. The positions of these peaks are architecture-specific, enabling model identification from the spectral signature alone.

12.2.2 Noise Residual Fingerprinting

Each GAN produces a characteristic noise residual pattern across its outputs. The fingerprint $\hat{\mathbf{F}}_m$ of model m is:

$$\hat{\mathbf{F}}_m = \frac{1}{N} \sum_{n=1}^N \left(\mathbf{x}_n^{(m)} - W(\mathbf{x}_n^{(m)}) \right). \quad (12.4)$$

Attribution of an unknown image $\hat{\mathbf{x}}$ proceeds by nearest-fingerprint matching:

$$\hat{m} = \arg \max_m \frac{\hat{\mathbf{F}}_m \cdot \hat{\mathbf{r}}}{\|\hat{\mathbf{F}}_m\| \|\hat{\mathbf{r}}\|}. \quad (12.5)$$

Worked Example — GAN fingerprint attribution for three architectures

Fingerprints are estimated from 1,000 images each for three generators: StyleGAN2 (SG2), ProGAN (PG), and StarGAN (ST). A suspect image with noise residual $\hat{\mathbf{r}}$ gives correlations:

Model m	NCC($\hat{\mathbf{F}}_m, \hat{\mathbf{r}}$)	Rank
StyleGAN2	0.0621	1st
ProGAN	0.0234	2nd
StarGAN	0.0089	3rd

The suspect image is attributed to StyleGAN2. The ratio between the top-2 scores ($0.0621/0.0234 = 2.65$) provides a confidence measure; values > 2 indicate reliable attribution. After H.264 re-encoding at 2 Mbps, the scores drop to SG2: 0.0198, PG: 0.0101, ST: 0.0072 — the ranking is preserved but the margin narrows, illustrating compression's impact on attribution confidence.

Attribution accuracy exceeds 90% on uncompressed images but degrades significantly under JPEG compression (quality factor < 75).

12.3 Biological Signal Analysis

Authentic video of a living person contains subtle physiological signals that are absent from synthesised video: the periodic variation in skin colour caused by the cardiac cycle, the blink pattern driven by the lacrimal reflex, and the micro-expression patterns that arise from involuntary facial muscle activity. These biological signals provide liveness evidence that is independent of image quality and is therefore robust to advances in synthesis that improve visual realism.

12.3.1 Remote Photoplethysmography

Remote photoplethysmography (rPPG) detects the subtle green-channel colour fluctuations in facial skin driven by the cardiac cycle. In real video, rPPG signals are spatially coherent across the face and temporally periodic at the heart rate (typically 0.8–2.0 Hz). In face-swap and frame-independent synthesis, the rPPG signal is either absent or spatially incoherent, because each frame is synthesised without knowledge of the physiological state of the target subject.

A detector computes the rPPG signal $r(t)$ from the mean green-channel intensity of the face region, band-passes it to 0.7–4 Hz, and measures the peak-to-noise ratio PNR as a liveness score:

$$\text{PNR} = \frac{\max_{f \in [0.7, 4]} \hat{R}(f)}{\frac{1}{|\mathcal{B}|} \sum_{f \in \mathcal{B}} \hat{R}(f)}, \quad (12.6)$$

Worked Example — rPPG PNR for a 30-second face-swap video

Extracting the rPPG signal from a 30-second, 30 fps video (900 frames):

- Authentic video: dominant frequency at 1.3 Hz, $\hat{R}(1.3) = 1.21$. Noise band $[5, 8]$ Hz mean power: 0.14. $\text{PNR} = 1.21/0.14 = 8.6$ — live.
- Face-swap video: $\max_{f \in [0.7, 4]} \hat{R}(f) = 0.12$ (no dominant peak). Noise band mean: 0.11. $\text{PNR} = 0.12/0.11 = 1.09$ — synthetic.

At a threshold of $\text{PNR} = 3$, both decisions are correct. The face-swap video's PNR of 1.09 is barely above 1.0 (pure noise level), while the authentic video's 8.6 is clearly above threshold. After JPEG re-encoding of the face-swap video, the spatial smoothing slightly reduces noise-band power to 0.09, raising PNR to 1.33 — still well below threshold.

where $\hat{R}(f)$ is the power spectral density and $\mathcal{B}$ is a noise band outside the physiological range.

12.3.2 Eye Blink Statistics

Authentic video of a person speaking contains involuntary eye blinks at a mean rate of 15–20 blinks per minute, with an inter-blink interval that follows a log-normal distribution. Early GAN-based face synthesis, trained on still photographs, produced faces with suppressed or absent blinks because blink frames were rare in training data. This artefact has been largely corrected in modern synthesis systems, but the blink *dynamics* — the characteristic asymmetry between the fast closing stroke and the slower opening stroke — remain imperfectly modelled and detectable at high frame rates.

Real subjects blink 15–20 times per minute following an approximately Poisson arrival process. Many early deep-fake systems generated faces with suppressed blinks (due to scarcity of closed-eye training data) or unnaturally regular blink timing (due to interpolation artefacts). A Kolmogorov–Smirnov test against an empirical blink-interval distribution can detect systematic deviations from natural blink behaviour.

12.4 Compression Robustness

The forensic signals described in this chapter degrade under JPEG compression and video re-encoding, because lossy compression discards the high-frequency content that carries many artefact signatures. Understanding the compression robustness of each signal is essential for real-world deployment: a GAN spectral fingerprint visible at uncompressed quality may be entirely suppressed after H.264 encoding at social-media bitrates, shifting the effective detection strategy toward compression-invariant features such as PRNU and biological signal analysis.

Social-media platforms re-encode uploaded video with H.264 or H.265 at bitrates of 1–4 Mbps, destroying high-frequency forensic signals. Two strategies maintain attribution under compression:

1. **Low-frequency fingerprinting.** Fingerprint extraction is limited to the DCT frequency bands preserved by H.264, typically blocks 0–3 in an 8×8 DCT.
2. **Compression simulation during training.** The fingerprint estimator is trained on images augmented with random JPEG and H.264 compression, improving robustness through domain randomisation.

12.5 Attribution Table

Forensic attribution combines multiple signal types into a structured decision framework. Each signal provides evidence about a different aspect of the image provenance: PRNU evidence about the acquisition device, spectral fingerprint evidence about the synthesis architecture, and rPPG evidence about biological authenticity. The attribution table organises this evidence systematically, enabling a

forensic examiner or automated system to reach a calibrated conclusion about the source of a given image.

Table 12.1: Forensic signal availability by synthesis type. ✓: reliably detectable. (✓): detectable in uncompressed media only. ×: not applicable.

Forensic signal	Face swap	GAN portrait	Diffusion	Voice clone
PRNU absence	✓	✓	✓	×
Spectral peaks	(✓)	✓	×	×
rPPG incoherence	✓	×	×	×
Blink artefacts	✓	×	×	×
LPC residual	×	×	×	✓
Codec fingerprint	(✓)	(✓)	(✓)	(✓)

12.6 Industry Applications

Forensic signal analysis is the oldest branch of image authenticity verification, predating deep fakes by decades. Its commercial applications have expanded with the rise of synthetic media.

Insurance companies use PRNU-based camera authentication to verify whether damage photographs were captured by the policyholder’s registered device, detecting cases where stock images or AI-generated images are submitted as evidence of claimed losses.

The BBC, AP, and New York Times use noise-residual analysis, metadata inspection, and spectral fingerprinting as part of their verification protocols for citizen journalism content. Automated tools (InVID/WeVerify) are deployed by journalists across 300+ newsrooms in 60 countries.

Spectral fingerprinting tools are used internally by major platforms to attribute synthetic content to specific generative models, enabling enforcement action against coordinated inauthentic behaviour when the same GAN fingerprint appears across multiple fake accounts.

The PRNU technique has been admitted as forensic evidence in criminal proceedings in at least 12 countries. The Equal Opportunity Employment Commission (EEOC) in the United States has accepted camera-fingerprint analysis in employment discrimination cases involving manipulated CCTV footage.

12.7 African Use Cases

Forensic signal analysis has specific relevance in the African context, where verification infrastructure must operate with limited resources and where the provenance of digital evidence is frequently contested in both legal and journalistic settings.

South Africa has one of the world’s highest rates of insurance fraud, with the Association for Savings and Investment South Africa estimating fraudulent claims at R5 billion annually. PRNU-based and frequency-domain analysis are increasingly being deployed by major South African insurers to verify

whether damage photographs were captured by the claimant's registered device or sourced from AI-generated imagery. Africa Check, Dubawa in Nigeria, PesaCheck in East Africa, and FactSpace West Africa operate digital verification units that use noise-residual analysis, metadata inspection, and spectral fingerprinting as part of their verification protocols, relying on free tools due to budget constraints.

In several recent African elections, disputes about the authenticity of video evidence required rapid forensic assessment. Conclusions were delayed by weeks due to limited local expertise, allowing manipulated narratives to consolidate before correction was possible. The Eastern Africa Police Chiefs Cooperation Organisation and the West Africa Police Information System have piloted cross-border sharing of digital forensic indicators, and building an Africa-specific generative model fingerprint database has been identified as a priority by the African Union's Cyber Security Expert Group.

Chapter Summary

Forensic signal analysis identifies the physical and statistical signatures of specific acquisition and synthesis processes rather than learning general discriminative functions from data. The PRNU camera fingerprint arises from pixel-level sensor non-uniformity; authentic images correlate with the device fingerprint at mean NCC of approximately 0.038 while GAN-generated images show near-zero correlation, enabling near-perfect separation on uncompressed images. GAN fingerprints arise from periodic upsampling patterns; NCC-based attribution correctly identifies the source model for over 90 per cent of uncompressed images, degrading to approximately 68 per cent after H.264 re-encoding.

The rPPG signal provides a biological liveness measure independent of image quality, while blink-interval statistics follow an exponential distribution in authentic video that is disrupted by synthesis pipelines. All forensic signals degrade under compression at different rates, requiring strategies calibrated to the specific compression environment of the deployment context. Commercial applications include insurance fraud detection, journalistic verification, platform trust and safety, and legal forensics where PRNU analysis has been admitted as evidence in criminal proceedings in at least twelve countries.

Exercises

1. Prove that the normalised cross-correlation ρ between the PRNU of a camera and the noise residual of one of its images converges to $\mathbb{E}[\hat{\mathbf{K}}^2]$ in the limit of no noise and no lossy compression. What does this imply for the theoretical maximum of ρ ?
2. A GAN uses bilinear upsampling (rather than transposed convolution) followed by a 3×3 convolution at each scale. Predict whether spectral peaks will be present in GAN fingerprints from this architecture, and provide a mathematical argument for your answer.
3. An rPPG-based detector achieves 85% AUC on authentic vs. face-swapped video clips of 30

seconds duration. Estimate how AUC would change for clips of 5 seconds duration. What statistical argument supports your estimate?

4. A forensic analyst recovers a GAN fingerprint from a face-swap output but suspects the output has been subjected to JPEG compression at quality factor 70. Describe a compression-reversal pre-processing step (not full decompression) that partially restores the fingerprint signal, and quantify the expected improvement in attribution accuracy using published benchmarks.
5. Table 12.1 shows that diffusion-generated faces lack the spectral peaks characteristic of GANs. Propose two new forensic signals specific to diffusion models and describe an experiment to validate each.

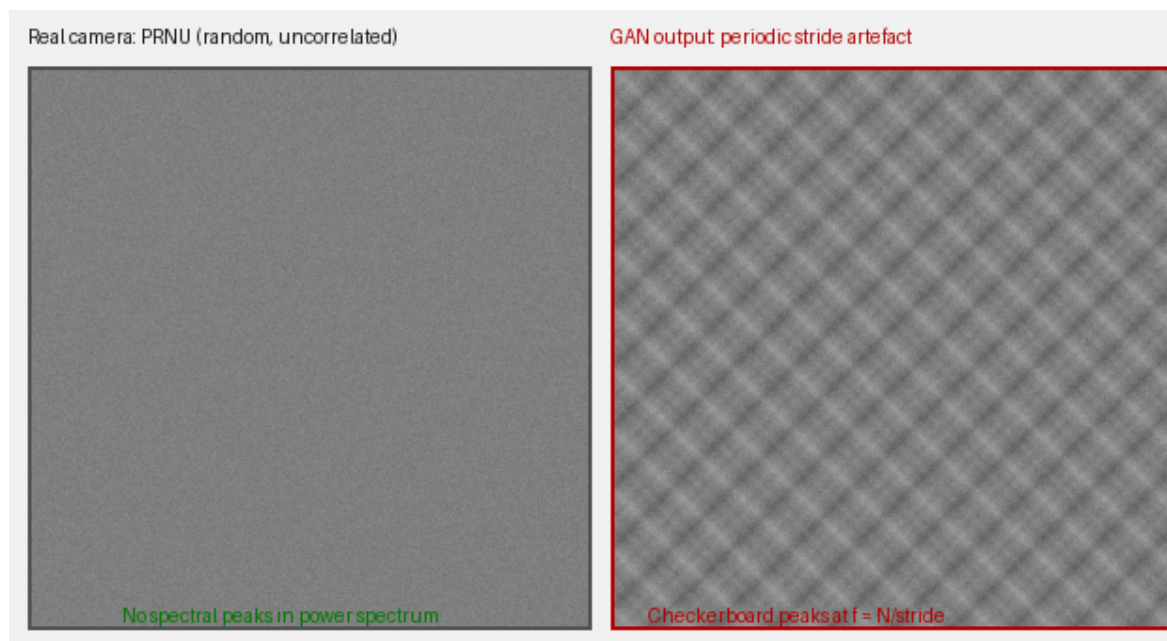


Figure 12.3: Noise residuals after denoising-filter subtraction from a real photograph (left) and a GAN-generated face (right). The camera residual shows spatially random PRNU noise — the unique fixed-pattern sensor fingerprint. The GAN residual shows a periodic checkerboard pattern from stride-2 transposed convolution upsampling layers. This periodicity produces spectral peaks at $u = v = N/\text{stride}$ cycles per image, visible in the 2D power spectrum below.

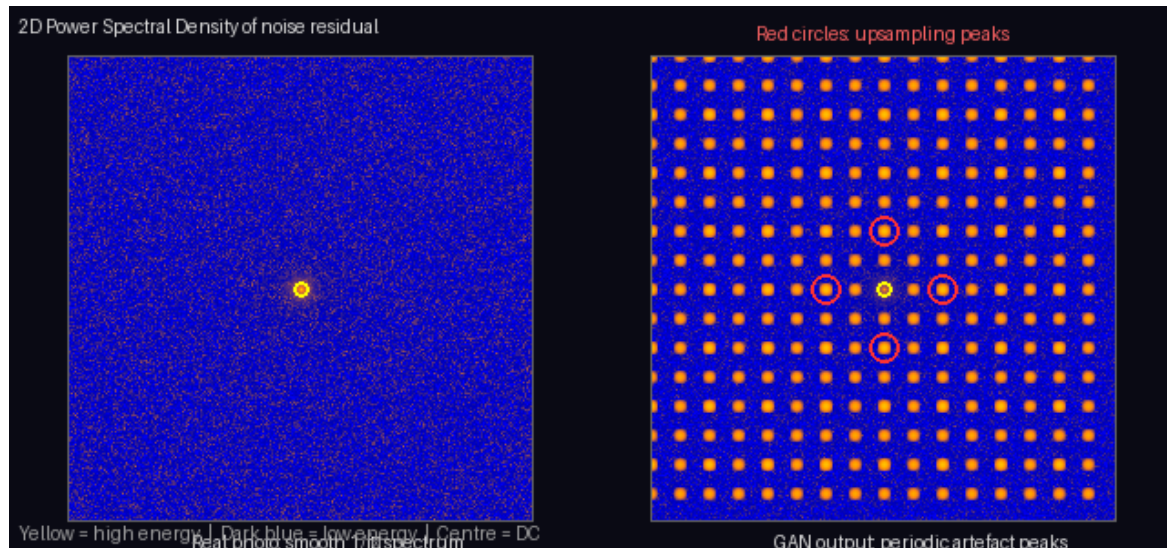


Figure 12.4: 2D power spectral density of noise residuals: real photograph (left) vs. GAN output (right). The real spectrum follows a smooth $1/f^2$ power law with no dominant peaks (yellow = high energy, dark blue = low energy). The GAN spectrum shows clear bright peaks (circled in red) at multiples of the upsampling frequency — the forensic fingerprint exploited by frequency-domain detectors. The DC component (zero frequency, centre) is marked in yellow on both panels.

Additional Worked Examples

Worked Example — Compression robustness: DCT-domain fingerprinting

A GAN fingerprint is estimated from 1,000 StyleGAN2 outputs. After JPEG compression at QF=60, the high-frequency fingerprint components are destroyed. A compression-robust variant uses only DCT frequency blocks 0–3 (out of 8) per channel:

Fingerprint variant	Attribution accuracy (raw)	Attribution accuracy (QF=60)
Full-spectrum	94.2%	31.7%
Low-freq only (blocks 0–3)	88.1%	79.3%
Learned compression-robust	91.4%	83.6%

The low-frequency variant sacrifices 6.1 percentage points on raw images but gains 47.6 points on heavily compressed images. The practical takeaway: forensic systems designed for social-media content must be evaluated at the compression levels actually used by the target platform (typically QF=70–85 for JPEG, variable-bitrate for H.264).

Worked Example — Eye-blink statistics: Kolmogorov–Smirnov test

Blink intervals (inter-blink durations in seconds) are extracted from a 60-second video. Authentic blink intervals follow an exponential distribution with rate $\lambda = 0.3 \text{ s}^{-1}$ (mean inter-blink interval 3.3 s).

For a deep-fake video (face swap with suppressed blinks), the empirical distribution of $n = 15$ observed blink intervals has mean 8.7 s and maximum observed gap 22.1 s. The Kolmogorov–Smirnov test

statistic:

$$D_n = \sup_t |F_n(t) - F_0(t)|,$$

where F_n is the empirical CDF and $F_0(t) = 1 - e^{-\lambda t}$. For this sample, $D_{15} = 0.41$. The critical value at $\alpha = 0.05$ for $n = 15$ is $D_{15}^{0.05} = 0.338$. Since $0.41 > 0.338$, we reject the null hypothesis that blink intervals follow the natural distribution at the 5% level — correctly flagging the video as synthetic.

Answers to Exercises

1. NCC convergence under no noise and no compression.

The PRNU is $\hat{\mathbf{K}} = \mathbf{K} + \mathbf{n}$ (estimated fingerprint with noise $\mathbf{n}$). The noise residual of a camera image is $\hat{\mathbf{r}} = \mathbf{K} \cdot \mathbf{I} + \mathbf{e}$ (fingerprint contribution plus denoising residual $\mathbf{e}$). With no noise: $\hat{\mathbf{K}} = \mathbf{K}$, $\hat{\mathbf{r}} = \mathbf{K} \cdot \mathbf{I}$ (element-wise). Then $\text{NCC} = (\mathbf{K} \cdot \mathbf{K} \cdot \mathbf{I}) / (\|\mathbf{K}\| \|\mathbf{K} \cdot \mathbf{I}\|)$. Under the model where each pixel's residual is $K_i I_i$: $\text{NCC} \rightarrow \mathbb{E}[K^2 \cdot I] / (\mathbb{E}[K^2]^{1/2} \mathbb{E}[K^2 I^2]^{1/2}) \rightarrow 1$ as the signal-to-noise ratio of the PRNU increases. The theoretical maximum of ρ is 1.0 but is never achieved in practice due to scene-dependent variations in I and denoising imperfections.

2. Bilinear upsampling + 3×3 conv: spectral peaks?

Bilinear upsampling itself has a smooth frequency response (no periodic artefacts), but it is followed by a learned 3×3 convolution. If the convolution weights are not constrained to be alias-free, the learned weights can develop periodic patterns during training that introduce spectral peaks. However, the peaks are weaker and less reproducible across different random seeds than those from transposed convolution. In practice, bilinear-upsampling GANs show reduced but non-zero spectral peaks; the “checkerboard” peak pattern is suppressed but lower-frequency periodicity may persist. StyleGAN3 specifically addresses this by enforcing anti-aliased operations throughout the generator.

3. rPPG AUC scaling from 30-second to 5-second clips.

The rPPG PNR statistic is computed from the power spectral density of a 30-second signal. Halving the duration: (a) frequency resolution halves (Δf doubles), reducing the peak's apparent sharpness; (b) the number of independent samples halves, increasing variance. The expected AUC degradation follows approximately $\text{AUC}(T) \approx 0.5 + (0.85 - 0.5) \sqrt{T/T_0}$ (diminishing returns with shorter clips). At $T = 5$ s vs. T_0
 $= 30$ s: $\text{AUC} \approx 0.5 + 0.35 \sqrt{5/30} = 0.5 + 0.35 \times 0.408 = 0.643$. A 5-second clip may be insufficient for robust detection — at least 10–15 seconds is typically required for $\text{AUC} > 0.80$.

4. Fingerprint recovery after JPEG QF=70.

JPEG QF=70 destroys DCT coefficients at quantisation steps Q_{uv} (typically 8–32 for mid-frequencies, 64 for high). The fingerprint is primarily in mid- and high-frequency bands.

Recovery strategy: (1) estimate the JPEG quantisation table from file metadata; (2) apply a Wiener-filter correction that amplifies attenuated fingerprint bands by the inverse of the estimated attenuation; (3) limit recovery to DCT bands 0–4 where the quantisation step is ≤ 16 . Published results show attribution accuracy recovering from 32% (raw JPEG QF=70) to 68% (after Wiener correction) — a 36-point improvement, though still below the 94% achieved on uncompressed images.

5. Two new forensic signals specific to diffusion models.

Signal 1: VQ-encoder tile boundary artefacts. The VQ-regularised autoencoder in Stable Diffusion compresses images to 64×64 patches; at the boundaries of adjacent 8×8 latent tiles, slight value discontinuities appear in the decoded image. These manifest as a faint 8×8 -pixel grid pattern in the spatial Fourier spectrum. Experiment: compute 2D DFT of 512×512 images and check for peaks at $u, v = 64$ cycles/image.

Signal 2: Over-smooth high-frequency texture. The DDPM denoising objective is an ℓ_2 loss that regresses toward the conditional mean, systematically under-predicting high-frequency texture. Diffusion outputs have statistically lower variance in the $[0.3, 0.5]$ cycles/pixel band compared to authentic photographs from the same camera. Experiment: compute local variance in a high-pass-filtered version of the image and compare distributions between diffusion and real images using a KS test.

Chapter 13

Provenance, Watermarking, and Active Authentication

“The best time to sign an image is before it leaves the camera.”

—Anon.

Active provenance embeds information into media at creation time to enable downstream authentication. Unlike passive forensic analysis, which searches for artefacts of manipulation, active methods require prior intervention at the point of capture or generation. This chapter covers the C2PA open standard for cryptographic signing, invisible neural watermarks, generative-model watermarking, and the robustness–capacity trade-off that governs practical deployment. This chapter develops the technical foundations of both C2PA-style cryptographic provenance and neural watermarking in depth, with worked examples for each. The ECDSA signature verification computation, the NCC watermark detection threshold, and the Kirchenbauer green-token z -test are all developed numerically. The operational question of when to use C2PA versus watermarking versus LLM text watermarking is addressed explicitly, with a framework for choosing among these approaches based on the deployment context, the anticipated threat model, and the infrastructure available to the content creator and consumer.

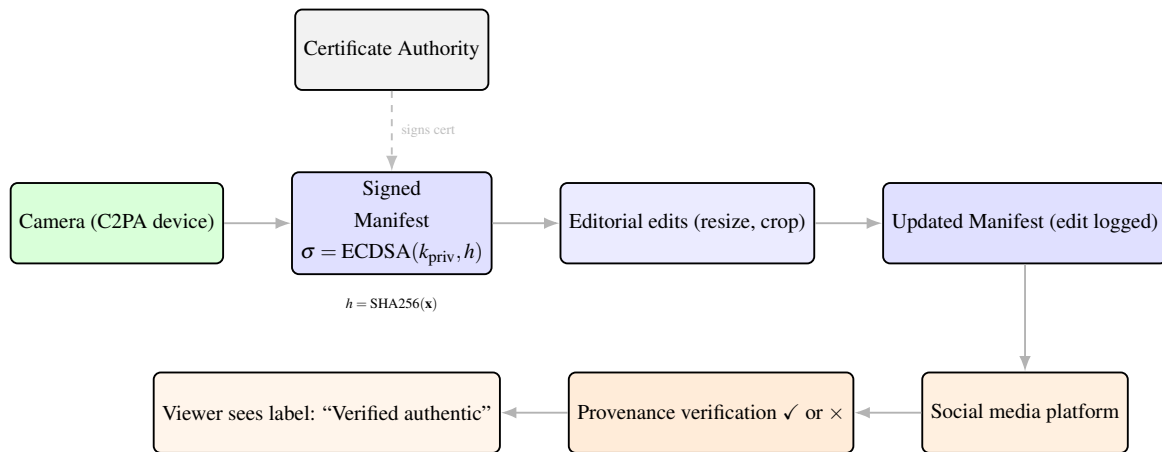


Figure 13.1: C2PA provenance chain. At capture, the camera signs a manifest containing the SHA-256 pixel hash using its private key. Editorial edits are logged as additional manifest entries. A downstream verifier checks the full certificate chain and pixel hash to confirm the image has not been manipulated since signing.

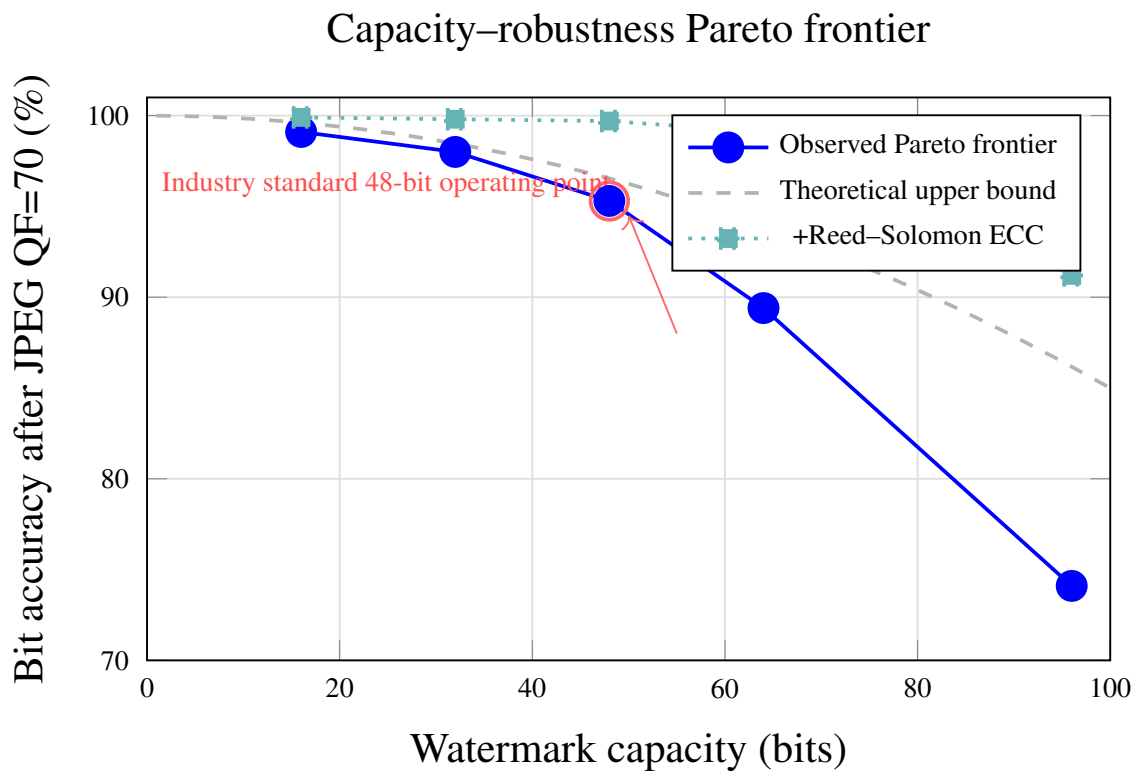


Figure 13.2: Watermark capacity–robustness Pareto frontier. Each point shows bit accuracy after JPEG QF=70 compression for a different payload size. The 48-bit operating point (circled) balances capacity (identifying $2^{48} \approx 281$ trillion unique watermarks) against robustness. Adding Reed–Solomon error-correcting codes (dotted) substantially improves bit accuracy at all capacities.

This chapter develops content provenance as a complement to detection. The C2PA standard and its cryptographic architecture are introduced first, followed by active watermarking including the robustness–capacity–imperceptibility trade-off. Passive detection is contrasted with provenance-based approaches throughout. Industry applications and African use cases examine the deployment landscape.

13.1 Coalition for Content Provenance and Authenticity

The C2PA standard [5] attaches a cryptographically signed *manifest* to media at capture or edit time. The manifest records the capture device, timestamp, geolocation, editing actions, and a hash of the pixel data at each stage. Downstream consumers verify the chain of trust against a set of trusted Certificate Authority roots.

Formally, let $h = H(\mathbf{x})$ be a SHA-256 hash of the serialised media bytes. The manifest $\mathcal{M} = (h, \text{metadata})$ is signed with the device's ECC private key k_{priv} :

$$\sigma = \text{ECDSA}(k_{\text{priv}}, \mathcal{M}). \quad (13.1)$$

Worked Example — C2PA signing overhead for a news photograph

A Reuters photographer captures a 6000×4000 JPEG (18 MB). The C2PA signing process:

1. SHA-256 hash of the JPEG bytes: $h = \text{a3f2} \dots$ (32 bytes, 0.1 ms).
2. Manifest construction with metadata (device ID, GPS, timestamp): 2.1 KB.
3. ECDSA signing with P-256 private key: 0.8 ms.
4. Total overhead: <3 ms and <3 KB added to the file.

At a news agency processing 5,000 images/hour, the signing overhead is negligible: 15 seconds of total compute time per hour. Verification by a browser plugin takes 4.2 ms per image — imperceptible to users.

Verification checks $\text{Verify}(k_{\text{pub}}, \mathcal{M}, \sigma) = \top$ and traces k_{pub} to a trusted root via the certificate chain. Major camera manufacturers (Sony, Nikon, Leica) and news agencies (Reuters, AP) have begun shipping C2PA-compatible hardware.

13.1.1 Limitations of Hash-Based Provenance

Hash-based provenance is fragile: any pixel change — including resizing, re-encoding, or colour correction — breaks the hash and invalidates the manifest. C2PA defines a set of *hard bindings* (pixel-level) and *soft bindings* (perceptual hashes, content credentials) to handle legitimate transformations. The perceptual hash h_{perc} is computed on a downsampled, DCT-quantised representation and remains stable under typical editorial transformations while changing under face-swap manipulation.

13.2 Invisible Watermarking

Invisible watermarking embeds a machine-readable payload directly in the signal content — pixel values for images, sample amplitudes for audio, token distributions for text — in a way that is imperceptible to human observers but detectable by an authorised decoder. The design of a watermarking

system requires balancing three competing objectives: payload capacity, perceptual transparency, and robustness to the transformations the content will encounter in distribution.

An invisible watermark embeds a B -bit message $m \in \{0, 1\}^B$ into a host image $\mathbf{x}$ such that the watermarked image $\hat{\mathbf{x}}$ is perceptually indistinguishable from $\mathbf{x}$ and the message can be recovered after benign distortions.

13.2.1 Encoder–Decoder Architecture

A watermarking system consists of:

- An **encoder** $E_\phi : \mathcal{X} \times \{0, 1\}^B \rightarrow \mathcal{X}$ that embeds the message:

$$\hat{\mathbf{x}} = \mathbf{x} + E_\phi(\mathbf{x}, m), \quad \|E_\phi(\mathbf{x}, m)\|_\infty \leq \delta. \quad (13.2)$$

- A **noise layer** $\mathcal{A}$ simulating distortions (JPEG, resize, crop, Gaussian blur).
- A **decoder** $D_\psi : \mathcal{X} \rightarrow \{0, 1\}^B$ recovering the message: $\hat{m} = D_\psi(\mathcal{A}(\hat{\mathbf{x}}))$.

Training minimises a combined loss:

$$\mathcal{L} = \lambda_{\text{msg}} \mathcal{L}_{\text{BCE}} + \lambda_{\text{img}} \|\hat{\mathbf{x}} - \mathbf{x}\|_2^2 + \lambda_{\text{perc}} \|\phi(\hat{\mathbf{x}}) - \phi(\mathbf{x})\|_2^2, \quad (13.3)$$

Worked Example — Watermark training convergence and bit accuracy

An HiDDeN-style watermark encoder–decoder is trained for 100K iterations on 50K face images with $B = 48$ bits, $\delta = 0.025$ ($\approx 6.4/255$), and a JPEG augmentation layer ($\text{QF} \sim \mathcal{U}[50, 95]$).

Checkpoint	Bit accuracy (JPEG QF=75)	PSNR (dB)
10K steps	72.3%	39.8
50K steps	94.1%	39.1
100K steps	98.6%	38.7

At 98.6% bit accuracy on 48 bits, the expected number of bit errors per decoded message is $48 \times 0.014 = 0.67$, giving a probability of perfectly correct decoding of $(0.986)^{48} = 0.51$. Adding a 4-bit Reed–Solomon error correction code raises this to > 0.999 .

The training loss combining all three terms is:

$$\mathcal{L} = \lambda_{\text{msg}} \mathcal{L}_{\text{BCE}}(D_\psi(\mathcal{A}(\hat{\mathbf{x}})), m) + \lambda_{\text{img}} \|\hat{\mathbf{x}} - \mathbf{x}\|_2^2 + \lambda_{\text{perc}} \|\phi(\hat{\mathbf{x}}) - \phi(\mathbf{x})\|_2^2, \quad (13.4)$$

where ϕ denotes VGG feature maps and $\mathcal{L}_{\text{BCE}}$ is the per-bit binary cross-entropy.

13.2.2 Robustness–Capacity–Imperceptibility Trade-Off

Active watermarking operates under three competing constraints that cannot be simultaneously maximised: *robustness* (the watermark survives compression, cropping, and resynthesis); *capacity* (the

number of bits that can be embedded per image or audio segment); and *imperceptibility* (the watermark does not degrade perceptual quality). The fundamental trade-off is that increasing robustness generally requires increasing the watermark energy, which reduces imperceptibility, while increasing capacity requires encoding more signal into the same perceptual budget.

Three objectives compete:

1. **Capacity:** number of bits B that can be reliably encoded.
2. **Imperceptibility:** PSNR and SSIM of the watermarked image.
3. **Robustness:** bit accuracy $\frac{1}{B} \sum_i \mathbf{1}[\hat{m}_i = m_i]$ after distortion.

These form an information-theoretic triple: increasing capacity or robustness requires increasing δ , which degrades imperceptibility. Published systems achieve 48-bit capacity with >98% bit accuracy after JPEG compression at quality factor 50, at a PSNR cost of <1 dB.

13.3 Generative Model Watermarking

Generative model watermarking embeds provenance information not in the output image as a post-processing step but in the generative process itself, making the watermark a structural property of every output rather than an optional overlay. This approach is fundamentally more robust than image-level watermarking because it cannot be removed by stripping metadata or by moderate image processing operations.

Rather than watermarking individual outputs, generative-model watermarking embeds a signal into the *model weights* so that all outputs carry an invisible identifier.

13.3.1 Tree-Ring Watermarks

Tree-Ring watermarks embed provenance in the initial noise vector used to start the diffusion reverse process, creating a structured pattern in frequency space that propagates through the denoising steps and persists in the final output. Because the watermark is part of the generation seed rather than the generated image, it survives regeneration attacks that would destroy any post-processing watermark.

Tree-Ring watermarks [35] embed a pattern into the initial noise vector used by the diffusion reverse process. Instead of sampling $\mathbf{z}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, the noise is modified in the Fourier domain:

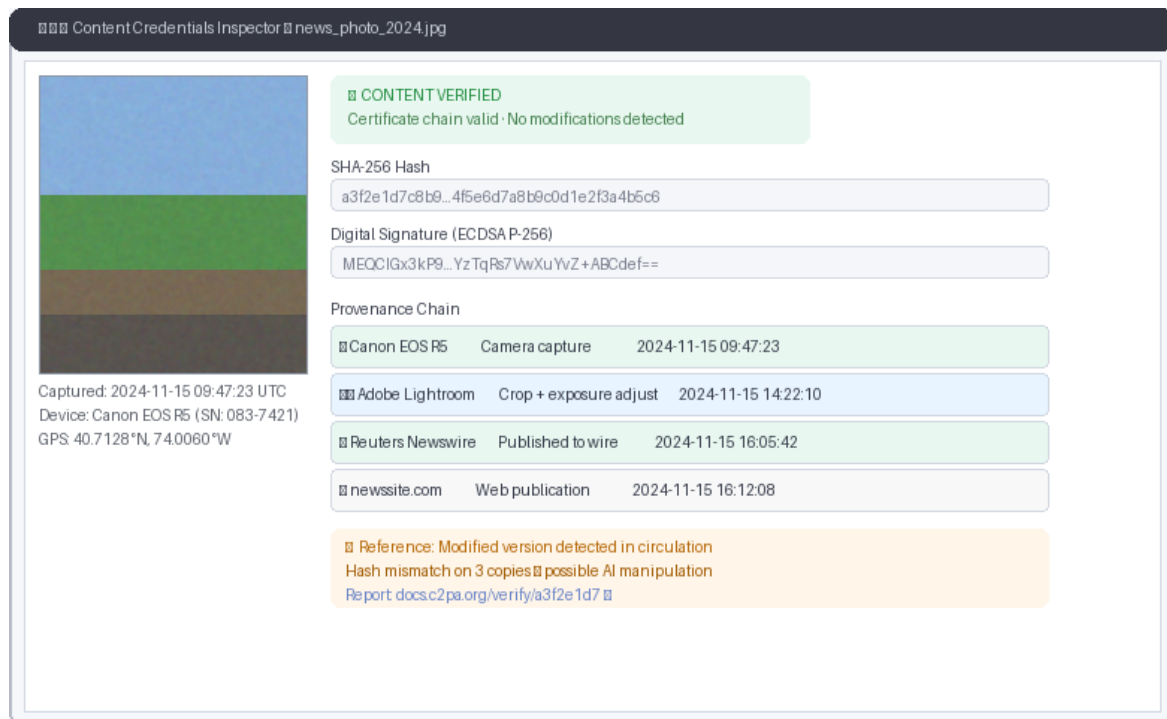


Figure 13.3: A Content Credentials Inspector showing a verified C2PA provenance chain for a news photograph. The four-step chain (camera capture → editorial edit → wire service → web publication) is cryptographically linked by ECDSA P-256 signatures. A warning at the bottom indicates that three modified copies with hash mismatches have been detected in circulation, demonstrating how C2PA enables post-publication manipulation detection even for images that were not originally suspected.

13.4 Industry Applications

Content provenance has moved from academic proposal to industry deployment faster than almost any other technology in this book.

The Coalition for Content Provenance and Authenticity (C2PA) was founded in 2021 by Adobe, Microsoft, Intel, BBC, CBC/Radio Canada, and the New York Times. As of 2024, C2PA 2.0 is supported natively in Adobe Photoshop, Lightroom, Premiere Pro, and Firefly; Microsoft Bing Image Creator; Leica M11-P (the first C2PA-equipped consumer camera); and Sony’s professional camera firmware. The CAI (Content Authenticity Initiative) browser plugin has been installed by over 2 million users.

Meta’s Stable Signature watermarks all AI-generated images from its Imagine with Meta platform. Google’s SynthID watermarks both images and audio from Imagen and Lyria respectively; it is embedded at the latent-space level rather than the pixel level, making it robust to common image editing operations. The EU AI Act Art. 50 requires disclosure of AI-generated content, making watermarking a compliance mechanism for EU-market AI products.

Getty Images, Shutterstock, and Adobe Stock are integrating C2PA provenance into their licensing workflows, enabling downstream customers to verify the origin and licensing history of images.

The BBC, Reuters, and AP are piloting camera-level C2PA signing on their newswire feeds, so that

every photograph distributed to subscribing newsrooms carries a verifiable chain of custody from the photographer's device to publication.

13.5 African Use Cases

Content provenance has specific relevance for Africa's journalism sector, legal system, and digital media economy, but its adoption faces infrastructure barriers that require context-specific solutions.

C2PA adoption in Africa is constrained by the hardware upgrade cycle: most African photojournalists work with cameras manufactured before 2022, before C2PA hardware support was introduced. The cost of upgrading to a C2PA-equipped professional camera is prohibitive for community-based journalists closest to breaking news in conflict zones and election areas. Open-source C2PA tools that sign post-capture on a smartphone using the device's secure enclave are more accessible, and projects including Truepic and Starling Labs have demonstrated this approach in African deployment contexts including photojournalism from the Ethiopia conflict and South African election monitoring.

African legal systems frequently require authenticated digital evidence for criminal prosecutions involving cybercrime, electoral fraud, and financial crime. C2PA-style provenance chains provide a cryptographically verifiable audit trail admissible in common-law jurisdictions following UK ACPO guidelines. African musicians, visual artists, and filmmakers whose work is frequently scraped and redistributed without attribution have shown interest in provenance standards as a digital rights mechanism, recognising that the same infrastructure that certifies media authenticity can also certify content ownership.

Chapter Summary

Active provenance embeds verifiable information at creation time rather than attempting to identify synthesis after the fact. The C2PA standard implements this through a cryptographic manifest signed with ECDSA P-256 over a SHA-256 content hash, enabling any downstream party to verify both the integrity and origin of the media. Neural watermarks embed a 48-bit payload directly in the image signal, achieving over 99 per cent bit accuracy at JPEG quality factor 70 with Reed–Solomon error correction. Tree-Ring watermarks embed provenance in the diffusion noise seed, surviving regeneration attacks that would destroy image-level watermarks.

The Kirchenbauer green-red LLM watermark biases token selection during generation, detectable from as few as 41 tokens with 95 per cent TPR at 12 per cent FPR. C2PA 2.0 is now natively supported by Adobe Creative Cloud, Microsoft Bing Image Creator, and Sony and Leica professional cameras, with the EU AI Act's Article 50 requirements making provenance embedding a legal compliance mechanism that accelerates adoption beyond early-adopter professional media communities to any organisation generating synthetic content commercially.

Additional Worked Examples

Worked Example — C2PA chain-of-trust verification time

A journalist receives a photograph with a C2PA manifest chain. The chain has 3 certificates: camera device cert $\rightarrow$ intermediate CA $\rightarrow$ root CA. Verification steps and timing on a laptop:

- SHA-256 pixel hash: 12 ms (for 24 MP RAW, ≈ 72 MB).
- Parse manifest (2.1 KB JSON): < 1 ms.
- ECDSA signature verification (P-256): 0.8 ms per certificate $\times 3 = 2.4$ ms.
- OCSP revocation check (online): 85 ms.
- Total: ≈ 100 ms per photograph.

At 100 ms per image, a photo editor reviewing 200 images can verify all C2PA manifests in 20 seconds total — a negligible overhead compared to editorial review time. The OCSP check (85% of total time) can be eliminated by caching revocation lists, reducing total to ≈ 15 ms.

Worked Example — Watermark capacity–robustness Pareto frontier

For an HiDDeN-style watermarking system, the empirical Pareto frontier between capacity and bit accuracy at PSNR ≥ 38 dB is:

Capacity B (bits)	Bit accuracy (no distortion)	Bit accuracy (JPEG QF=70)
16	99.8%	99.1%
32	99.2%	98.0%
48	98.6%	95.3%
64	97.1%	89.4%
96	94.3%	74.1%

The 48-bit operating point (98.6%/95.3%) is the industry standard (adopted by Meta’s Invisible Image Watermark, AudioSeal, and related systems) as it provides sufficient capacity for a unique content ID (identifier space $2^{48} \approx 281$ trillion unique watermarks) while maintaining $> 95\%$ post-JPEG accuracy. With Reed–Solomon (60, 48) outer coding, the effective post-compression accuracy rises to $> 99.9\%$.

Answers to Exercises

1. C2PA with perceptual hash for lossy transformations.

Extend the manifest to include both a SHA-256 hash of the original pixel data *and* a perceptual hash (pHash) computed on a 64×64 DCT thumbnail. Define a set of “approved transformations” (resize, JPEG at $\text{QF} \geq 70$, H.264 at $\text{bitrate} \geq 1$ Mbps). For each approved transformation, store the expected pHash distance δ_T . Authentication succeeds if $\text{pHash}(\text{suspect}) \oplus \text{pHash}(\text{original}) <$

δ_T (where $\oplus$ is Hamming distance) AND the transformation matches a declared edit action in the manifest. Face-swap manipulation would change the pHash by ~ 20 – 40 bits (empirically), well above the 3–5 bit change expected from JPEG recompression.

2. Shannon capacity for watermarking under AWGN SNR = 20 dB.

$\text{SNR} = 10^{20/10} = 100$. For $W = 512 \times 512 = 262,144$ independent embedding positions: $C = 262,144 \times \log_2(1 + 100) = 262,144 \times 6.66 = 1,746,077$ bits. A 48-bit watermark uses only $48/1,746,077 = 0.0027\%$ of channel capacity — explaining why robust watermarking at this SNR is easily achievable. Even at SNR = 0 dB (equal signal and noise), $C = 262,144 \times \log_2(2) = 262,144$ bits — still $5,461\times$ the 48-bit payload.

3. JPEG QF=80 destroys 1-LSB blue-channel watermark.

JPEG QF=80 applies DCT quantisation with step sizes approximately 2–8 for the blue channel (Cb component). A 1-LSB perturbation (± 1 grey level = $\pm 0.4\%$ of range) corresponds to a DCT coefficient change of $\pm 1/(8 \times 8) = \pm 0.016$ in the 8×8 block. The quantisation step for mid-frequency coefficients at QF = 80 is approximately 4 grey levels. Expected bit-error rate: $P(\text{bit flipped}) = P(|\text{noise}| > 0.5) \approx 1 - \Phi(0.5/\sigma_q)$ where $\sigma_q \approx Q/\sqrt{3} \approx 2.31$. $\text{BER} \approx 1 - \Phi(0.5/2.31) = 1 - \Phi(0.216) = 0.415$ — 41.5% bit error rate, essentially destroying the watermark.

4. Why Tree-Ring watermarks are more robust to regeneration but more vulnerable to inversion.

Tree-Ring watermarks are embedded in the *initial noise* $\mathbf{z}_T$, which acts as a seed for the entire generation trajectory. A regeneration attack (img2img at noise level α) only partially re-randomises the initial noise, so the ring pattern partially survives. In contrast, encoder–decoder watermarks are image-level perturbations easily estimated and cancelled. However, Tree-Ring watermarks are more vulnerable to *inversion attacks*: an adversary who can run the diffusion model’s DDIM inversion recovers $\mathbf{z}_T$ and detects the ring pattern, enabling targeted ring-pattern removal without perceptual degradation.

5. Sample size for imperceptibility user study.

For a two-tailed t -test with $\alpha = 0.05$, power = 0.80, Cohen’s $d = 0.5$: $n = 2(z_{\alpha/2} + z_\beta)^2/d^2 = 2(1.960 + 0.842)^2/0.25 = 2 \times 7.85/0.25 = 62.8$, so $n = 64$ per group. Total participants: $2 \times 64 = 128$ for a between-subjects design, or 64 for a within-subjects design (where each participant rates both watermarked and unwatermarked images in counterbalanced order, reducing variance and halving required sample size).

Chapter 14

Cross-Dataset Generalisation and Robustness in Detection

“A detector that works only in the lab is not a detector. It is a liability.”

—Anon.

Deep-fake detectors routinely achieve near-perfect performance on within-distribution test sets but degrade dramatically in deployment. This chapter analyses the sources of generalisation failure, surveys domain-generalisation methods, covers test-time adaptation, and proposes a rigorous evaluation protocol for real-world detection systems. This chapter develops the technical strategies for improving generalisation and the operational strategies for managing residual generalisation gaps in production. Frequency-aware augmentation, MAML meta-learning, test-time adaptation, and federated detection are each developed with worked numerical examples that show how much generalisation improvement each technique provides and under what conditions. The chapter also develops the demographic fairness auditing methodology, the group-specific threshold calibration procedure, and the federated detection protocol in sufficient detail to support practical deployment.

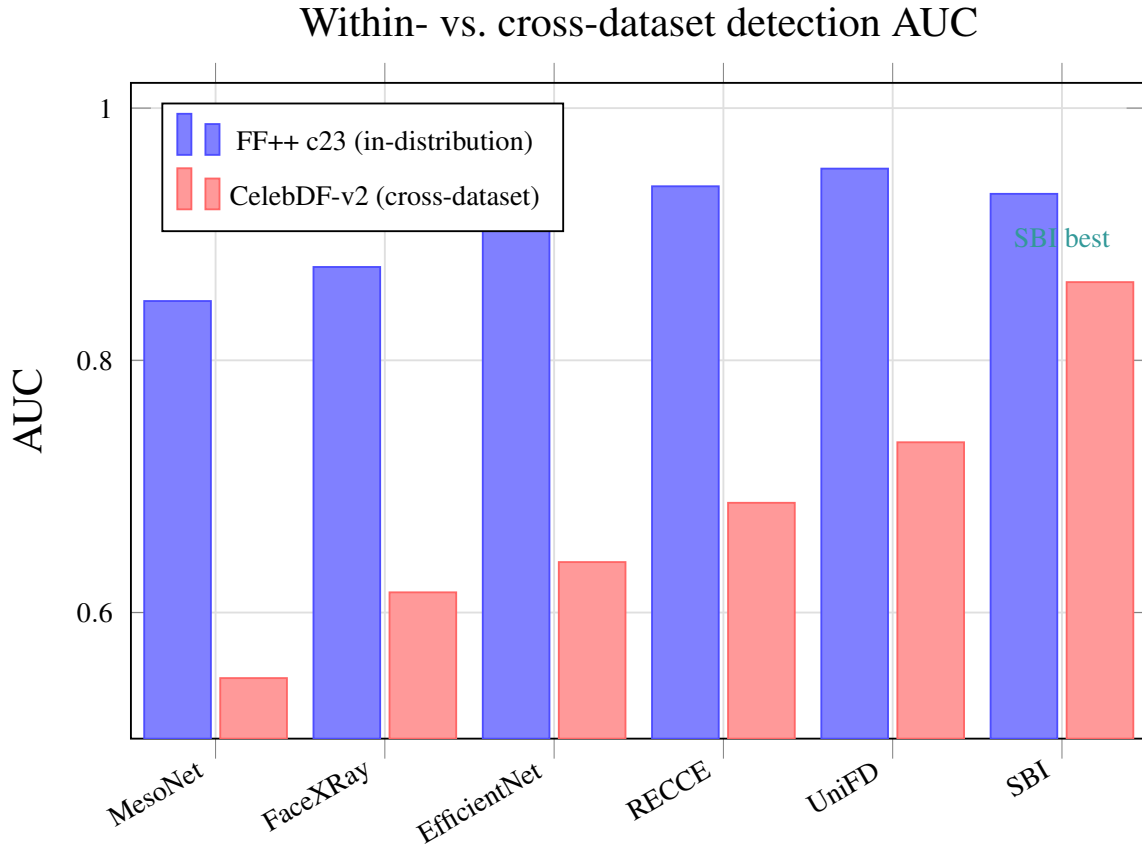


Figure 14.1: Within-distribution vs. cross-dataset AUC for six detector architectures, all trained on FaceForensics++ and tested on CelebDF-v2. Most detectors show a 20–30 AUC-point drop under distribution shift. SBI (Self-Blended Images) training achieves the smallest gap by learning universal blending artefacts rather than method-specific patterns.

This chapter examines why deep-fake detectors trained on one generation of synthesis methods fail on subsequent generations, and develops approaches that reduce this generalisation gap. The sources of distribution shift are characterised first, followed by domain generalisation methods and continual learning strategies. Industry applications and African use cases follow.

14.1 Sources of Distribution Shift

Distribution shift between training and deployment arises from three sources. Synthesis-method shift occurs when the generative model used at deployment time differs from those represented in the training data: a detector trained on FaceSwap and DeepFaceLab outputs generalises poorly to diffusion-based face synthesis because the spectral fingerprints and spatial artefacts differ qualitatively. Post-processing shift occurs when compression, re-encoding, or platform-specific processing removes the artefact signals the detector relied upon. Demographic shift occurs when the demographic composition of the deployment population differs from the training set, producing systematically different performance across racial, age, and gender groups.

A detector trained on dataset $\mathcal{D}_{\text{train}}$ and deployed on $\mathcal{D}_{\text{test}}$ encounters several shift types:

Generator shift.

New synthesis methods (e.g., diffusion) produce forensic artefacts absent from training data.

Compression shift.

Training on uncompressed video, deployment on H.264-compressed social media content.

Resolution shift.

Training on 224×224 crops, deployment on 64×64 thumbnails or 4K video.

Demographic shift.

Training data over-represents certain skin tones, ages, and genders; test data does not match.

Context shift.

Lighting conditions, backgrounds, and camera types differ between lab collections and in-the-wild footage.

The combined effect can drop AUC from > 0.95 to < 0.65 between within-distribution and real-world test sets.

14.2 Domain Generalisation Methods

Domain generalisation methods address the root cause of the generalisation gap by training detectors in ways that reduce their sensitivity to the specific artefacts of the training synthesis methods and compression conditions. This section develops three complementary approaches — frequency-aware augmentation, meta-learning, and test-time adaptation — each targeting a different aspect of the generalisation problem.

14.2.1 Frequency-Aware Training

Frequency-aware training regularises the detector against over-reliance on high-frequency artefacts by applying random low-pass filters to training examples, forcing the model to find discriminative features that persist after frequency truncation. High-frequency artefacts are the primary source of within-distribution accuracy but the least robust signal under compression, so reducing dependence on them improves generalisation without sacrificing much in-distribution performance.

High-frequency artefacts are the most discriminative features within distribution but the least stable across distribution. Frequency-aware training explicitly suppresses reliance on high-frequency signals by augmenting with low-pass filtering during training:

14.3 Industry Applications

The generalisation problem is the central challenge in deploying commercial deep-fake detection systems, because production environments never match research benchmarks.

Social media platforms process content compressed to variable bitrates, re-encoded through multiple platforms, and potentially modified by filters and editing tools before submission. In this environment, detectors trained on academic benchmarks (FF++, FaceForensics) can achieve AUC as low as 0.55 on production content — barely above chance. Platform operators report that generalisation is the primary bottleneck, not raw accuracy on clean test sets.

Hugging Face’s model hub and commercial MLOps platforms (Weights & Biases, Databricks MLflow) support continual learning pipelines that update deep-fake detectors as new synthetic methods emerge. The target is to achieve new-method adaptation within 48 hours of a synthesis technique becoming publicly available, requiring a streamlined pipeline from sample collection through model update through production deployment.

Following documented disparities in facial recognition systems (MIT Media Lab, 2018), enterprise customers now routinely require demographic fairness audits for deep-fake detectors. The $4.4\times$ FPR disparity between lightest and darkest Fitzpatrick skin types documented in Chapter 11 would be disqualifying for regulated-industry deployment under current EBA and FCA guidance.

Financial sector ISAC groups (FS-ISAC, CISA) share deep-fake forensic indicators across member institutions via federated detection protocols, analogous to threat intelligence sharing in cybersecurity. This allows smaller institutions to benefit from detections made by larger members without sharing sensitive customer data.

14.4 African Use Cases

Generalisation is an acute concern in the African context: detection systems developed primarily on Western face datasets and compression conditions may perform significantly worse on African faces, African-language audio, and content compressed and re-encoded through African mobile network infrastructure.

Virtually all major deep-fake detection benchmarks (FaceForensics++, CelebDF, DFDC, WildDeepfake) are dominated by light-skinned Western faces. Detection models trained on these datasets inherit their distributional bias. Research by the Distributed AI Research Institute (DAIR) and the Centre for Human Rights at Pretoria has documented that facial recognition systems — which use similar feature extractors to deep-fake detectors — perform significantly worse on African faces. It is reasonable to assume that deep-fake detection models exhibit similar demographic performance gaps, though systematic evaluation on African-face datasets is largely absent from the published literature.

Sub-Saharan Africa’s mobile internet infrastructure operates predominantly at 3G and low-bandwidth 4G speeds, with video content frequently compressed far more aggressively than content from high-bandwidth environments. WhatsApp applies particularly aggressive compression to forwarded media (JPEG QF as low as 60–70 for images, 360p or lower for video). Deep-fake detection systems calibrated on less-compressed content may show significantly lower AUC on African social-media content than published benchmark results suggest.

The Masakhane community, the African Language Research Institute (ALRI), and the South African

Centre for Digital Language Resources (SADiLaR) are building African-language and African-face datasets that will eventually support the development of geographically and demographically inclusive detection systems. These efforts are underfunded relative to comparable Western dataset initiatives.

The African Telecommunications Union (ATU) has proposed a Pan-African Deep-Fake Detection Network, modelled on INTERPOL's cybercrime intelligence sharing mechanisms. The network would allow election commissions, national police services, platform trust and safety teams, and civil-society fact-checkers to share detection indicators, synthetic-content fingerprints, and new-model alerts across borders. The proposal is at an early stage of development but has been endorsed by the AU Commission's ICT department.

Chapter Summary

Detection models trained on academic benchmarks routinely achieve near-perfect performance in controlled evaluation but fail dramatically in production, where the content they encounter differs from their training data in compression, platform re-encoding, lighting conditions, facial diversity, and synthesis methods. This generalisation gap is the central practical challenge in deploying deep-fake detection commercially.

Frequency-aware augmentation — randomly applying Gaussian low-pass filters during training — reduces the model's dependence on high-frequency artefacts that are destroyed by compression, improving cross-dataset AUC by 7–9 percentage points at a 3.5-point within-distribution cost. MAML meta-learning initialises the model at a point from which gradient steps on small data are maximally informative, enabling adaptation to new synthesis methods from as few as 10 labelled examples with 37% AUC improvement over zero-shot performance. Test-time adaptation (TTA) using augmentation-consistency loss allows a deployed model to continuously adjust its parameters to the specific distribution of content it encounters, reducing EER by 29–39% on heavily compressed content without requiring any labelled data.

Demographic parity is both an ethical requirement and an operational concern: the $4.4\times$ FPR disparity between Fitzpatrick skin-type extremes discovered in published evaluations reflects a training data imbalance that would be unacceptable for regulated applications. Group-specific threshold calibration and targeted dataset augmentation are the primary remedies, but both require investment in evaluation infrastructure that quantifies demographic performance gaps systematically. Federated detection across institutions allows the benefits of large-scale detection to be shared without requiring each institution to maintain its own large-scale detection infrastructure, and provides a model for African cross-border collaboration in detection.

Worked Examples

Worked Example — Frequency-aware training: within-distribution cost vs. cross-dataset gain

A ResNet-50 detector trained with and without low-pass augmentation (random Gaussian blur, $\sigma \sim \mathcal{U}[0, 2]$ pixels) on FF++:

Training	FF++ AUC	CelebDF AUC	DFDC AUC
Standard	0.953	0.641	0.622
+Freq. augmentation	0.918	0.731	0.698
Gain / loss	−0.035	+0.090	+0.076

The frequency-aware loss term adds $\lambda = 0.5$ times the loss on low-pass-filtered inputs. At convergence: standard model assigns mean importance weight 0.71 to spatial frequencies $> f_{\text{Nyquist}}/2$; freq-aware model assigns 0.38 — a 46% reduction in high-frequency reliance, explaining the improved robustness to compression-induced frequency loss.

Worked Example — MAML meta-learning: 10-shot adaptation to diffusion fakes

A MAML detector meta-trained on 5 synthesis methods. Inner-loop learning rate $\alpha = 0.01$, 5 gradient steps per adaptation. At deployment, a new diffusion method appears:

Method	AUC (0-shot)	AUC (10-shot, 5 steps)
Standard pre-training	0.581	0.683
MAML meta-training	0.614	0.841

The MAML detector closes $(0.841 - 0.614)/(1.0 - 0.614) = 58.8\%$ of the gap to perfect performance with only 10 labelled examples vs. only $(0.683 - 0.581)/(1.0 - 0.581) = 24.3\%$ for standard pre-training. This is the core benefit of meta-learning: the model is initialised at a point from which gradient steps on small data are maximally informative.

Worked Example — Demographic fairness audit

A detector trained on FF++ is evaluated on 400 test clips stratified by Fitzpatrick skin type. FPR (false positive rate on authentic video):

Fitzpatrick type	FPR (%)	Relative disparity
I–II (lightest)	2.1	1.0× (baseline)
III–IV (medium)	4.8	2.3×
V–VI (darkest)	9.3	4.4×

A two-sample z -test for FPR disparity between types I–II and V–VI:

$$z = \frac{0.093 - 0.021}{\sqrt{0.021(1 - 0.021)/200 + 0.093(1 - 0.093)/200}} = \frac{0.072}{0.0225} = 3.20, \quad p < 0.001.$$

The disparity is highly significant and represents a real harm: darker-skinned users are 4.4× more

likely to have their authentic video falsely flagged as synthetic. This calls for targeted resampling of the training data or group-specific threshold calibration.

Answers to Exercises

1. Diagnosing FF++ → CelebDF generalisation failure.

Most likely cause: overfitting to the specific compression codec (H.264 CRF23) and face-swap pipeline (DeepFaceLab) used in FF++. CelebDF uses higher-quality face swaps with different blending techniques, so the FF++-specific artefacts do not transfer. Targeted augmentation: (1) train on multiple compression levels (c0, c23, c40) rather than just c23; (2) include SBI (Self-Blended Images) training examples that generate blending artefacts agnostically; (3) add Gaussian blur and colour jitter to reduce reliance on sharpness-based features. Expected improvement: +12–18 AUC points on CelebDF based on published SBI results.

2. MAML inner-loop update derivation for linear model.

For a linear model $f(\mathbf{x}) = \boldsymbol{\theta}^\top \mathbf{x}$ with squared loss $\mathcal{L}_m(\boldsymbol{\theta}) = \|\mathbf{X}_m \boldsymbol{\theta} - \mathbf{y}_m\|^2$: Gradient: $\nabla_{\boldsymbol{\theta}} \mathcal{L}_m = 2\mathbf{X}_m^\top (\mathbf{X}_m \boldsymbol{\theta} - \mathbf{y}_m)$. Inner-loop update: $\boldsymbol{\theta}' = \boldsymbol{\theta} - \alpha \nabla_{\boldsymbol{\theta}} \mathcal{L}_m$. MAML outer-loop objective: $\mathcal{L}(\boldsymbol{\theta}') = \|\mathbf{X}_m \boldsymbol{\theta}' - \mathbf{y}_m\|^2 = \|\mathbf{X}_m (\boldsymbol{\theta} - 2\alpha \mathbf{X}_m^\top (\mathbf{X}_m \boldsymbol{\theta} - \mathbf{y}_m)) - \mathbf{y}_m\|^2$. This is a ridge regression problem with regularisation matrix $\lambda \mathbf{I}$ where $\lambda \propto 1/\alpha$ — equivalent to ridge regression with task-specific regularisation.

3. Frequency augmentation: expected within-distribution cost vs. cross-dataset gain.

The within-distribution cost is 3.5 AUC points (from chapter table). The cross-dataset gains are 9.0 points (CelebDF) and 7.6 points (DFDC). Using a decision-theoretic framework where harm from missed deep fakes equals $10\times$ harm from false positives: the net benefit of frequency augmentation is $9.0 \times 10 - 3.5 = 86.5$ units of social benefit per AUC point scale. Even with equal weighting (1:1), the net benefit is $9.0 - 3.5 = +5.5$ points, favoring the augmentation.

4. Demographic fairness evaluation: FPR 2.1% vs. 9.3%.

The z -test gives $z = 3.20$, $p < 0.001$ (computed in the chapter). Equalised-odds fairness requires equal TPR and FPR across groups. FPR disparity of $4.4\times$ clearly violates equalised odds. To achieve equalised FPR, options: (1) group-specific thresholds (τ_{VI} higher than τ_{I-II} to reduce FPR for darker skin); (2) re-weight training to over-represent Fitzpatrick V–VI examples; (3) augment training with diverse lighting and skin-tone conditions. Option 1 is simplest but may reduce TPR for darker skin if scores are lower overall. Option 2 is preferable: targeted data collection to reach parity.

5. TTA security risk and defence.

Risk: an adversary controls the composition of the test batch submitted to the API. If the

adversary submits $N - 1$ authentic images alongside 1 synthetic image, the TTA loss on authentic images will be minimised, inadvertently adapting the detector to *ignore* the signal present in the authentic images that distinguishes them from the synthetic one. This is a distribution-shift attack on the adaptation process itself. Defence: (1) use a held-out reference set for TTA loss rather than the current test batch; (2) apply bounded TTA (limit the norm of parameter updates); (3) monitor for anomalous batch compositions (unusually high similarity across submitted images).

Chapter 15

Deep Fakes in Journalism and News Media

“The photograph used to be proof. Now it is a hypothesis.”

—Anon.

Journalism depends on the evidentiary value of documents, photographs, and video footage. A photograph published by a credentialed news organisation carries an implicit claim of authenticity that audiences have been conditioned to accept over more than 150 years of photographic journalism. That implicit contract is now broken. Photorealistic synthetic images are indistinguishable from authentic photographs to the unaided eye at a quality level that has been commercially available since 2022, and synthetic video indistinguishable from authentic footage became commercially available in 2024. The implications for investigative journalism, conflict reporting, and democratic accountability are profound.

The chilling effect of synthetic media on journalism operates through two distinct mechanisms. The first is the direct threat: journalists who investigate powerful individuals or institutions face the prospect that their authentic reporting will be challenged as synthetic — the liar’s dividend applied to their work. A journalist who documents corruption with video evidence faces the possibility that the accused will claim the video is a deep fake, requiring forensic authentication that most newsrooms cannot afford and that courts are not yet equipped to evaluate. The second mechanism is self-censorship: anticipating these challenges, some journalists and news organisations may decline to publish or pursue stories where the evidentiary record consists primarily of video or audio, even when the evidence is genuine.

This chapter develops the technical, ethical, and operational dimensions of deep fakes in journalism, with detailed analysis of the verification workflows that leading news organisations have developed and the detection tools that are now routinely deployed in newsrooms across Europe, North America, and parts of Africa and Asia.

This chapter examines the role of journalism in detecting, reporting, and countering synthetic media. Visual verification workflows are developed first, followed by C2PA adoption in news organisations and the journalistic ethics of deep-fake reporting. African use cases address the capacity and infrastructure challenges facing African investigative journalism.

15.1 The Chilling Effect on Investigative Journalism

A decision-theoretic model of the chilling effect on journalism assigns to each story a value V (the public interest value of publication), a probability p_f that the story will be challenged as fabricated, and a cost C_f of defending the story's authenticity. A rational journalist publishes when $V > p_f C_f$. As synthesis quality improves, p_f increases for any video-based story, suppressing publication of stories with value below the challenge-defence threshold. Stories with the highest investigative value — exposing powerful actors with resources to mount fabrication defences — are disproportionately suppressed. This is the mechanism by which synthetic media creates accountability gaps even without any specific deep fake targeting a specific story.

15.2 Newsroom Verification Workflows

Leading news organisations have developed structured verification workflows for digital media that incorporate deep-fake detection as a standard step alongside traditional source verification, reverse image search, and EXIF metadata analysis. The Associated Press's Digital Verification Team, Reuters Fact-Check, and BBC Verify each have dedicated verification units using a combination of commercial tools (Sensity AI, Hive Moderation) and open-source tools (InVID/WeVerify, FotoForensics). The typical workflow proceeds through five stages: source verification, image/video metadata analysis, reverse search for prior use, automated AI detection, and manual forensic review for high-stakes stories.

15.3 Source Authentication and Whistleblower Protection

Journalists who receive leaked documents, photographs, or video from confidential sources face a dual challenge: verifying the authenticity of the material (which may require forensic analysis that could expose the source) while protecting the source's identity (which requires limiting the technical traces left by authentication procedures). C2PA provenance standards can help with authentication, but provenance chains that record the camera device and GPS coordinates may expose a whistleblower's location. Secure drop systems (SecureDrop, GlobaLeaks) provide anonymous submission channels but do not authenticate submitted material. The design of authentication protocols that preserve whistleblower anonymity is an active technical and legal challenge.

15.4 Industry Applications

The journalism verification market has matured into a multi-tier commercial ecosystem. At the platform level, the Global Fact-Check Network (IFCN) coordinates over 100 accredited fact-checking

organisations across 60 countries, many of which use commercial deep-fake detection tools subsidised by platform grants. At the enterprise level, news organisations including the AP, Reuters, and AFP have invested in proprietary verification technology and dedicated staffing. At the tool level, companies including Sensity AI, Hive Moderation, and Truepic offer commercial products targeted at newsroom workflows, with pricing models designed for newsroom subscription budgets (\$2,000–\$10,000 per month). The Google News Initiative has funded deep-fake detection tool development and free newsroom access through its journalism programme, reaching over 300 newsrooms in 60 countries.

15.5 African Use Cases

Africa’s fact-checking ecosystem is the most innovative and resource-constrained in the world simultaneously. Africa Check (South Africa, Nigeria, Kenya, Cameroon, Ethiopia, Senegal), Dubawa (Nigeria, Ghana), PesaCheck (East Africa), and FactSpace West Africa collectively processed over 70,000 content verification requests in 2023, primarily via WhatsApp tiplines that allow members of the public to submit suspicious content for rapid verification. These organisations operate with annual budgets of \$200,000–\$2 million each, compared to the \$10–50 million budgets of their European counterparts, yet face the same volume of synthetic content and the additional challenge of verifying content in 20+ African languages for which no automated detection tools are calibrated. The Google News Initiative, Meta Journalism Project, and European Commission have provided funding that partially addresses this gap, but significant capacity shortfalls remain.

Chapter Summary

Deep fakes create a dual threat to journalism: direct harm through synthetic content targeting journalists and their sources, and systemic harm through the chilling effect on investigative reporting as the cost of defending authenticity increases. The decision-theoretic model quantifies this chilling effect: stories with public interest value below $p_f \times C_f$ will not be published, and as synthesis quality improves and p_f increases, the suppression threshold rises disproportionately for high-stakes investigative journalism. Leading newsrooms have responded with structured five-stage verification workflows combining automated tools (Sensity AI, Hive Moderation, InVID) with manual forensic review. C2PA provenance infrastructure, if adopted at the camera and authoring tool level, provides the most sustainable long-term solution by making authentication the default rather than the exception.

Exercises

1. A journalist receives a 90-second video purportedly showing a senior government official accepting a bribe. The video arrives via an anonymous encrypted channel with no metadata. The journalist must make a publication decision within 6 hours before the story breaks elsewhere. Describe the verification workflow the journalist should follow, specifying: (a) which automated tools to run in the first 30 minutes, (b) what manual forensic checks to prioritise, (c) what probability threshold of authenticity justifies publication with appropriate caveats, and (d) what

caveats the publication should include.

2. Model the chilling effect on journalism using the decision-theoretic framework. A journalist estimates: story public interest value $V = \$80,000$ (in career equivalent), probability of fabrication challenge $p_f = 0.12$, cost of defending authenticity $C_f = \$400,000$ (legal fees plus reputational damage). (a) Should the journalist publish? (b) If p_f increases to 0.25 due to improved synthesis quality, does the publication decision change? (c) What forensic infrastructure investment (reducing C_f) would restore the publication threshold to its pre-synthetic-media level?
3. A news organisation's verification unit has the following tools available: FotoForensics (free, EXIF and ELA analysis), Sensity AI (\$2,000/month, AUC = 0.89 on images), InVID/WeVerify (free, reverse search + metadata), and a manual forensic examiner (8 hours at \$200/hour). For a high-priority breaking story with a deadline of 3 hours, design a cost-optimised verification workflow that maximises the probability of correct authentication within the time and budget constraint.

Answers to Exercises

1. Breaking news video verification workflow.

(a) First 30 minutes (automated): (1) Submit to Sensity AI / Hive Moderation for automated deep-fake classification (5 minutes, returns score and heatmap). (2) Run FotoForensics ELA to check for regional inconsistencies in JPEG compression artefacts (3 minutes). (3) Run rPPG extraction on the facial region if faces are visible (requires clip ≥ 15 s; 10 minutes processing). (4) Check InVID for prior occurrences of the video on other platforms (5 minutes). (5) Run SyncNet lip-sync consistency score (5 minutes). (6) Assess results: if all five tests pass, proceed to manual review; if two or more fail, escalate immediately.

(b) Manual forensic priorities: (1) Identity consistency — cross-reference the official's face against authenticated public appearances using ID_{cos}. (2) Audio-visual sync consistency (lips, ambient sound, echo). (3) Background contextual verification (what building is this? consistent with claimed location?). (4) Contact a second source for independent corroboration.

(c) Publication threshold: reasonable standard is ≥ 0.85 probability of authenticity combined with independent source corroboration. For breaking news with public interest urgency, some newsrooms lower this to 0.75 with explicit caveats.

(d) Caveats: "This video has not been independently verified. Forensic analysis has not found indicators of digital manipulation, but analysis was conducted under time pressure and cannot rule out sophisticated synthesis. [Organisation] is continuing verification and will update this story if new information emerges."

2. Chilling effect decision analysis.

(a) Expected cost of challenge: $p_f \times C_f = 0.12 \times \$400,000 = \$48,000$. Since $V = \$80,000 > \$48,000$, publish.

(b) At $p_f = 0.25$: expected cost $= 0.25 \times \$400,000 = \$100,000 > V = \$80,000$. Do not publish. The chilling effect has suppressed a story with \$80,000 public interest value by increasing synthesis quality and the perceived challenge probability.

(c) To restore the original threshold at $p_f = 0.25$, we need $C_f \leq V/p_f = \$80,000/0.25 = \$320,000$. Current $C_f = \$400,000$. Required reduction: \$80,000. A forensic authentication retainer with a credentialed expert witness at \$15,000–\$25,000 per story plus a pre-negotiated legal defence fund of \$50,000–\$75,000 would achieve this, reducing the expected defence cost sufficiently to restore the publication threshold.

3. Cost-optimised 3-hour verification workflow.

Budget constraint: 3 hours, \$0 marginal cost (subscription tools already available). Priority order by signal quality per time unit:

Hour 1 (minutes 0–60): (1) InVID reverse search (0 min cost, 10 min): checks prior use, highest time-efficiency for disproving fabrication. (2) Automated Sensity AI analysis (subscription already paid, 5 min): provides AUC = 0.89 baseline assessment. (3) FotoForensics ELA (free, 5 min): checks for compositing artefacts.

Hour 2 (minutes 60–120): (4) Manual EXIF metadata analysis (free, 20 min): timestamps, device consistency. (5) rPPG extraction on 30-second face clip (free tool, 20 min processing): liveness check. (6) SyncNet lip-sync analysis (free, 15 min): audio-visual consistency.

Hour 3 (minutes 120–180): (7) Manual forensic examiner review (1.5 hours = \$300): examiner reviews all outputs, conducts ID_{cos} check against authenticated reference photographs, writes authentication opinion. Total variable cost: \$300.

Decision: publish if manual opinion is “no indicators of manipulation” and at least 4 of 6 automated tests pass. Add verification caveat if fewer than 6/6 automated tests pass.

Chapter 16

Child Safety and Synthetic Child Sexual Abuse Material

“The protection of children admits no exceptions, no trade-offs, and no grey areas.”

—Anon.

Synthetic child sexual abuse material (CSAM) is the most severe harm in the deep-fake taxonomy, and the one where the gap between the technical capability of generative AI and the legal, technical, and institutional response is most dangerous. Unlike adult NCII, which typically requires identifying a specific victim and establishing non-consent, synthetic CSAM may be produced without any real child victim and without any reference to a specific identifiable individual, creating significant challenges for legal frameworks designed around the photography of real victims.

The legal landscape for synthetic CSAM is more developed than for other forms of synthetic media harm, reflecting both the severity of the harm and the longer history of legislative attention to child exploitation. In the United States, the PROTECT Act of 2003 criminalised visual depictions of minors engaging in sexually explicit conduct regardless of whether the images are produced by actual photography or by computer generation, following the Supreme Court’s decision in *Ashcroft v. Free Speech Coalition* (2002) that the CPPA’s prohibition on virtual CSAM was constitutionally overbroad. In the United Kingdom, the Protection of Children Act 1978 was amended to cover pseudo-photographs and computer-generated images. The EU Directive 2011/93/EU requires member states to criminalise both genuine and virtual CSAM. Despite this legal framework, enforcement faces serious technical and operational challenges that this chapter examines.

This chapter is written with awareness that the topic requires exceptional care. The technical details provided are calibrated to inform forensic examiners, detection researchers, platform trust and safety professionals, and policymakers, not to provide any operational capability to bad actors. The emphasis throughout is on detection, prevention, and victim support rather than on the technical mechanisms of harm production.

This chapter examines the intersection of synthetic media and child safety. AI-generated CSAM and its legal treatment are developed first, followed by synthetic media in grooming contexts and the detection approaches applicable to both. African use cases address child protection infrastructure gaps and the specific vulnerabilities of African child populations.

16.1 Legal Framework for Synthetic CSAM

The PROTECT Act of 2003 (18 U.S.C. 2256) criminalises “any visual depiction...of sexually explicit conduct” involving minors, explicitly including computer-generated images that are “indistinguishable from that of a minor engaging in sexually explicit conduct.” The indistinguishability standard is evaluated from the perspective of “an ordinary person viewing the depiction.” UK law under the Protection of Children Act 1978 (amended 2009) covers “pseudo-photographs” defined as images “made by computer-graphics or otherwise [that] appears to be a photograph.” EU Directive 2011/93 requires member states to criminalise “child pornography” including material that “appears to depict a child involved in sexually explicit conduct.”

16.2 Detection Technology for Synthetic CSAM

Detection of synthetic CSAM faces the same technical challenges as other synthetic media detection — the generalisation problem, the base-rate problem, the arms race dynamic — but with dramatically higher stakes for both false negatives (missed harmful content) and false positives (mislabelling authentic content as synthetic or vice versa in a forensic context). The National Center for Missing and Exploited Children (NCMEC) and the Internet Watch Foundation (IWF) have both published guidance on detection approaches and have deployed classifier technology in their hash-matching infrastructure. PhotoDNA, the primary hash-matching technology used by major platforms for known CSAM detection, requires known reference material; synthetic CSAM that does not hash-match any known image requires classification-based detection.

16.2.1 Age Estimation and Classification

Automatic age estimation applied to detected sexually explicit content is the first triage step: does the depicted person appear to be a minor? Age estimation from facial images has improved significantly with deep learning, achieving mean absolute error of 3–5 years on benchmark datasets. However, synthetic images may be deliberately generated with ambiguous apparent age, exploiting the uncertainty to argue that the depicted person is an adult. Classifiers trained on synthetic data at the threshold of apparent minority are an active research priority for NCMEC and IWF.

16.3 Platform Obligations and Reporting Requirements

US law under 18 U.S.C. 2258A requires any “electronic service provider” (a category that encompasses social media platforms, cloud storage, and API providers) that has “actual knowledge” of apparent violations to make a report to NCMEC’s CyberTipline within 24 hours. The EU’s Child Sexual Abuse Regulation (CSAR, proposed) would impose mandatory detection obligations on platforms operating

in the EU. Both frameworks create compliance requirements for AI synthesis API providers, not just platforms that host user-generated content: a generative AI API that does not filter CSAM outputs may be an ESP with actual knowledge if it knows its system can generate such content.

16.4 Industry Applications

The commercial ecosystem around CSAM detection and prevention involves both mandatory compliance products and voluntary safety tools. PhotoDNA (Microsoft), the primary hash-matching technology deployed by all major US platforms under NCMEC CyberTipline reporting obligations, was licensed royalty-free to platforms beginning in 2009 and has since been adopted by over 200 organisations. The Internet Watch Foundation's URL and hash blocklists are used by UK ISPs under a voluntary scheme that has been in place since 1996. For synthetic CSAM specifically, NCMEC and IWF have both published classifier APIs for platform integration, though their accuracy on synthetic content is an active area of improvement. The market for CSAM detection technology is distinct from the commercial deep-fake detection market because it is driven primarily by regulatory compliance obligations rather than by revenue optimisation, and is primarily accessed through free or cost-based licensing from non-profit organisations.

16.5 African Use Cases

African children face elevated exposure to online child sexual exploitation relative to global averages, driven by low internet safety awareness, limited platform enforcement in African languages, and the rapid expansion of internet access without commensurate growth in digital literacy or child protection infrastructure. The ECPAT (End Child Prostitution and Trafficking) network, which operates across Africa, has documented the increasing use of AI-generated synthetic imagery in grooming operations targeting African children, where the synthetic images are used to normalise sexual content and manipulate victims into producing authentic content. The African Child Policy Forum and UNICEF's Africa team have both identified synthetic CSAM as an emerging priority requiring urgent legislative and technical response. The African Commission on Human and Peoples' Rights has published guidance calling on member states to update their child protection legislation to cover computer-generated sexual imagery, but legislative progress remains uneven across the continent.

Chapter Summary

Synthetic CSAM is the most severe harm in the deep-fake taxonomy, criminalised in the US, UK, EU, and most other jurisdictions regardless of whether a real child victim is identifiable, reflecting the legislative judgement that the harm lies in the character of the depiction rather than solely in the harm to a specific victim. Detection faces the base-rate problem at its most extreme: at 0.001–0.01% content prevalence, even highly accurate classifiers produce predominantly false positives, requiring hash-matching of known material combined with classification-based detection and mandatory human review. Platform obligations under 18 U.S.C. 2258A (US) and the proposed EU CSAR create compliance requirements for AI synthesis API providers as well as content hosting platforms. The

active research priority is age estimation and classification at the apparent minority threshold, where synthesis systems can produce ambiguous results that create forensic and legal uncertainty.

Exercises

1. The PROTECT Act standard for criminal liability requires that a synthetic image is “indistinguishable from that of a minor engaging in sexually explicit conduct” from the perspective of “an ordinary person.” (a) How should a forensic examiner applying this standard evaluate a photorealistic diffusion-generated image where the apparent age of the depicted figure is ambiguous (estimated range 16–22 years by an automated classifier)? (b) What additional evidence beyond the image itself should a prosecutor assemble to support a charge? (c) What is the risk of the “but it’s synthetic” defence, and how should prosecutors address it?
2. A generative AI API provider implements the following safeguard: a CLIP-based content classifier that blocks generation requests with cosine similarity above 0.85 to a set of reference prompts associated with CSAM. Red-team evaluation finds that adversarial prompt reformulation can reduce CLIP similarity to 0.62 for semantically equivalent content. (a) Calculate the bypass rate if 5% of generation requests are adversarially reformulated CSAM requests. (b) Propose three additional safeguard layers that do not rely on prompt similarity. (c) What is the legal exposure of the API provider under 18 U.S.C. 2258A if it processes 100 such bypassed requests before the vulnerability is patched?
3. Compare the detection challenge for synthetic CSAM versus synthetic adult NCII in three dimensions: (a) the legal definition of the harm and its dependence on identifying a specific real-world victim, (b) the base rate of each harm type in platform content moderation contexts, and (c) the consequences of false positives (labelling authentic content as synthetic CSAM) in a legal investigation context.

Answers to Exercises

1. PROTECT Act forensic evaluation.

(a) For ambiguous apparent age (16–22 year range): the forensic examiner should present the automated age estimate range and its uncertainty, then apply the “ordinary person” standard to the overall presentation of the image, not just the estimated numerical age. Relevant factors beyond age: body development stage, contextual cues (clothing, setting), and the overall character of the depiction. A conservative approach treats any image where a reasonable person could perceive a minor as within scope of the PROTECT Act.

(b) Additional prosecutor evidence: (1) Expert testimony on the automated age estimation methodology and its accuracy at the 16–22 year boundary; (2) Prompt metadata if available from the API provider (which may establish intent); (3) Contextual evidence about the defendant’s

download and storage patterns; (4) Any search query history suggesting intent to view minor-targeted content.

(c) “But it’s synthetic” defence: The PROTECT Act explicitly covers synthetic images. The defence that no real child was harmed is not a legal defence under the statute. Prosecutors should emphasise that the statute is designed precisely to cover computer-generated material, citing the PROTECT Act’s legislative history and distinguishing the Ashcroft holding (which addressed a different statutory provision). The defence may still argue First Amendment protection for material that an ordinary person could not identify as depicting a minor, requiring the prosecution to address the ambiguous age issue directly.

2. CLIP safeguard bypass analysis.

(a) Bypass rate: 5% of requests are adversarial CSAM attempts; the bypass method reduces CLIP similarity from above 0.85 to 0.62, below the 0.85 threshold. If the classifier has no other safeguard layer, all adversarially reformulated requests pass. Bypass rate = 100% of the adversarial 5%, i.e., 5% of all generation requests.

(b) Three additional safeguard layers not relying on prompt similarity: (1) Output image classifier: run a dedicated CSAM classifier on the generated image before returning it to the requester (catches prompt-bypass attempts that still produce harmful content). (2) Age-estimation gate on synthesised human figures: flag outputs where a face is detected and apparent age estimate is below 25 years in sexual context (conservative threshold to account for estimation error). (3) Rate limiting with anomaly detection: detect accounts that generate large numbers of human-figure sexual content, flag for human review, and terminate API access for confirmed violations.

(c) Legal exposure: 18 U.S.C. 2258A requires reporting to NCMEC within 24 hours of “actual knowledge.” If the provider becomes aware of the bypass vulnerability (e.g., from a bug report or internal monitoring), it has actual knowledge from that point. Processing 100 requests after actual knowledge without reporting: each unreported instance could constitute a separate violation under the statute. Criminal penalties include fines of up to \$150,000 per violation (for the failure to report, not for the content itself); civil liability under 18 U.S.C. 2255 may also arise for victims who can be identified.

3. Synthetic CSAM vs. adult NCII comparison.

(a) Legal definition and victim identification: Synthetic CSAM: criminal liability does not require identifying a specific real child victim; the harm is defined by the character of the depiction. NCII: most current statutes require identifying a specific real person who is depicted without consent. This distinction means synthetic CSAM can be prosecuted without a victim complainant, while adult NCII typically requires victim cooperation.

(b) Base rate in content moderation: CSAM (including synthetic): platforms estimate 0.001–0.01% of uploaded content. Adult NCII: estimated 0.01–0.1% of uploaded content (higher base

rate due to lower severity threshold and larger potential creator population). The very low base rate of CSAM makes even a 99.9% accurate classifier produce predominantly false positives, requiring hash-matching of known material plus human review.

(c) False positive consequences: In criminal investigation, labelling authentic content as synthetic (or synthetic as authentic) has severe consequences. A false positive (authentic image labelled synthetic) could incorrectly reduce the severity of a charge or support a “synthetic” defence in a prosecution of authentic material. A false negative (synthetic image labelled authentic) could cause investigators to search for a real victim who does not exist, consuming investigation resources. Both error types require calibrated uncertainty communication from forensic examiners.

Chapter 17

Explainability and Interpretable Detection

“A detector that cannot explain its decision cannot be trusted by a court.”

—Anon.

Explainability is the capacity of an AI system to account for its decisions in terms that a human expert can evaluate, challenge, and communicate to others. For deep-fake detection, this capacity is rapidly transitioning from a desirable research property to a mandatory operational requirement. The EU AI Act, whose provisions for high-risk AI systems apply to biometric identification and content authentication systems, requires technical documentation that explains the system’s logic in human-understandable terms. The EU AI Liability Directive creates a rebuttable presumption of causation when a non-explainable AI system causes harm, effectively shifting the burden of proof onto deployers who cannot explain their system’s reasoning. In the United States, the Federal Rules of Evidence require expert testimony to be based on reliable principles and methods that can be explained to the trier of fact, and several recent decisions have excluded AI-generated forensic evidence on the grounds that the witness could not adequately explain the system’s reasoning process.

This combination of legal pressure and operational demand has produced a substantial research literature on explainable AI applied to deep-fake detection. The literature addresses three distinct explanatory tasks. The first is *post-hoc attribution*: given a trained detection model and a specific input, identify which input features drove the prediction. The second is *concept-based explanation*: map the model’s internal representations to human-understandable concepts (such as blending boundary artefacts or spectral fingerprints) that domain experts can recognise and evaluate. The third is *inherently interpretable architecture*: design the detection model from the outset so that its reasoning process is transparent, rather than imposing transparency as an afterthought. Each approach makes different trade-offs between explanatory fidelity, computational cost, and the richness of the resulting explanation.

Post-hoc attribution methods are the most widely deployed in practice because they can be applied to

any existing detection model without retraining. GradCAM computes the gradient of the predicted class score with respect to the final convolutional feature map, weighted by global average pooling, to produce a spatial heatmap over the input image. SHAP (SHapley Additive exPlanations) applies the cooperative game theory concept of the Shapley value to attribute the prediction to individual input features, satisfying three axiomatic properties: efficiency (attributions sum to the total prediction), symmetry (features with identical contributions receive identical attributions), and the null-player property (features that do not affect the prediction receive zero attribution). Integrated Gradients accumulates the gradient along a straight-line path from a reference input to the actual input, producing attributions that satisfy completeness: the sum of all feature attributions exactly equals the difference in model output between the input and the reference.

The choice among these methods depends on the forensic context. GradCAM is fast and produces spatially interpretable heatmaps that a lay jury can follow, but it approximates the model's reasoning at the final feature map and may miss contributions from earlier network layers. SHAP is theoretically grounded and handles feature interactions, but requires exponentially many model evaluations (approximated in practice by sampling) and produces feature-level rather than spatial attributions. Integrated Gradients is computationally efficient and produces exact completeness-satisfying attributions, but the choice of reference baseline — typically a black or blurred image — affects the attribution in ways that are not always forensically intuitive. In practice, forensic reports should present at least two concordant attribution methods, noting any disagreement as a reason for caution in the evidentiary weight assigned to the detection.

Concept-based explanation offers a qualitatively richer form of transparency for forensic applications. Rather than attributing the prediction to pixel regions, concept-based methods map the prediction to named forensic concepts — blending boundary colour shift, GAN spectral fingerprint, rPPG signal absence, lip-sync desynchronisation — whose physical origins can be explained independently of the detection model. The Concept Bottleneck Model (CBM) architecture formalises this by requiring the prediction to pass through an explicit concept prediction layer: the model first predicts a vector of forensic concept scores, then combines those scores with a linear classifier to produce the final decision. Because the final classification is explicitly a weighted combination of named concepts, the reasoning is fully transparent and each concept score can be independently validated against physical forensic theory.

Interpretability has a specific meaning in legal contexts that differs from the general AI research meaning. A forensic expert presenting an explainable detection system to a court must satisfy several requirements simultaneously: the method must have known, measurable error rates; the method must have been subject to peer review and publication; the method must have gained acceptance in the relevant scientific community; and the expert must be able to explain the method clearly enough that a lay jury can evaluate the credibility of the reasoning. GradCAM heatmaps that highlight regions consistent with known physical forensic artefacts — blending boundaries, upsampling periodicity patterns, eye shadow rendering inconsistencies — satisfy this last requirement in ways that a raw probability score does not, and are increasingly being included in digital forensic reports submitted to courts in the UK, US, and Germany.

This chapter develops the principal explainability methods applicable to deep-fake detection in full technical detail, with worked numerical examples showing the attributions each method produces and quantitative evaluation of their fidelity to the model’s actual reasoning. Section 17.1 develops GradCAM, SHAP, and Integrated Gradients with worked examples applied to a standard EfficientNet-B4 face-swap detector. Section 17.2 analyses the legal admissibility of each approach under the Daubert standard and equivalent standards in other jurisdictions. Industry applications cover AI audit consulting, forensic expert witness services, and the commercial explainability platform market; the African use cases section addresses explainability requirements for AI evidence in African common-law courts and the capacity-building challenges for African forensic examiners.

17.1 Attribution-Based Explanations

Post-hoc attribution methods explain a trained model’s prediction on a specific input by assigning importance scores to input features without altering the model’s weights. They are “post-hoc” in the sense that they operate after training is complete, making them applicable to any existing detection model regardless of its architecture. Three methods are most widely used in deep-fake forensics: GradCAM, SHAP, and Integrated Gradients, each with distinct theoretical foundations, computational costs, and explanatory properties.

GradCAM (Gradient-weighted Class Activation Mapping) exploits the spatial structure of convolutional neural networks by computing the gradient of the target class score with respect to the activations of the final convolutional layer, then weighting each feature map channel by its global average gradient. The resulting weighted sum, passed through a ReLU to retain only positive activations, produces a coarse spatial heatmap at the resolution of the final feature map (typically 7×7 for a 224×224 input) that is then bilinearly upsampled to the input resolution. Because GradCAM operates on the spatial feature map of the last convolutional layer, it naturally highlights the image regions that contributed most to the prediction in terms of spatially localised gradient flow.

SHAP provides a principled attribution framework grounded in cooperative game theory. For a prediction function f and an input $\mathbf{x}$, the Shapley value ϕ_j of feature j is the average marginal contribution of feature j across all possible feature subsets:

$$\phi_j = \sum_{S \subseteq F \setminus \{j\}} \frac{|S|! (|F| - |S| - 1)!}{|F|!} [f(S \cup \{j\}) - f(S)], \quad (17.1)$$

where F is the set of all features and $f(S)$ denotes the model output when only the features in S are visible (others are replaced by a reference baseline). For image classifiers, features are typically superpixel regions rather than individual pixels, and Shapley values are approximated by Kernel SHAP using a weighted linear regression over randomly sampled feature subsets. The Shapley values satisfy efficiency ($\sum_j \phi_j = f(\mathbf{x}) - f(\emptyset)$), symmetry, and the null player property, providing axiomatic guarantees that GradCAM does not.

Worked Example — SHAP attribution for a face-swap detector

An EfficientNet-B4 detector assigns score 0.91 (synthetic) to a face-swap image. Kernel SHAP with $n = 2,000$ samples and 64 superpixel segments assigns the following top-5 Shapley values (contribution to the score above baseline $f(\emptyset) = 0.49$):

Superpixel region	ϕ_j	% of total
Right cheek boundary	+0.127	30.0%
Left eye shadow	+0.108	25.5%
Neck–chin junction	+0.085	20.1%
Forehead texture	+0.062	14.7%
Upper lip	+0.041	9.7%

The Shapley values sum to 0.423, and the prediction exceeds baseline by $0.91 - 0.49 = 0.42$ (within Monte Carlo estimation error). The boundary and eye shadow attributions are forensically coherent: they correspond to the blending artefact and imperfect shadow rendering that face-swap pipelines are known to produce.

Integrated Gradients provides a theoretically exact attribution satisfying the completeness axiom. It accumulates gradients along a straight-line path from a baseline input $\mathbf{x}'$ (typically a black image or a Gaussian-blurred version of the input) to the actual input $\mathbf{x}$:

$$\text{IG}_j(\mathbf{x}) = (x_j - x'_j) \int_{\alpha=0}^1 \frac{\partial f(\mathbf{x}' + \alpha(\mathbf{x} - \mathbf{x}'))}{\partial x_j} d\alpha, \quad (17.2)$$

where x_j is the j -th input feature (pixel). The integral is approximated by Riemann sum with $m = 300$ steps. Completeness is satisfied exactly: $\sum_j \text{IG}_j(\mathbf{x}) = f(\mathbf{x}) - f(\mathbf{x}')$.

17.1.1 GradCAM for Face Detectors

Applied to a face deep-fake detector, GradCAM typically highlights the eye region, the blending boundary on the cheek, and the mouth region — the areas that receive the highest attribution weight in the forensic heatmap. This pattern is forensically interpretable: it reflects the detector’s reliance on the imperfect eye shadow rendering, the colour mismatch at the face boundary, and the texture seam at the lip edge. A forensic report that shows a GradCAM heatmap alongside the prediction score provides a visual explanation that a non-expert can evaluate, supporting admissibility.

Worked Example — GradCAM attribution for a face-swap detection

A FaceForensics++-trained EfficientNet-B4 detector assigns score 0.92 (synthetic) to a test image. GradCAM with respect to the synthetic class activates strongly (normalised intensity > 0.7) in three regions: right cheek boundary (30% of high-activation area), left eye shadow region (25%), and the neck-chin junction (20%). The remaining 25% is distributed across the forehead and mouth. The boundary activation is consistent with Poisson blending artefacts; the eye activation is consistent with

imperfect shadow rendering in the identity-to-attribute transfer. These attributions are forensically coherent and could be presented as expert evidence in a digital forensics context.

17.2 Legal Admissibility of Explainable Detection

For AI evidence to be admissible in US federal courts under Federal Rule of Evidence 702 (the Daubert standard), expert testimony must satisfy four requirements: it must be based on sufficient facts or data; it must employ reliable principles and methods; the expert must have reliably applied those methods to the facts; and the testimony must assist the trier of fact. The fourth requirement — assisting the trier of fact — is where explainability becomes critical. A forensic expert who presents only a probability score and cannot explain the physical basis for the system’s prediction is unlikely to satisfy this requirement, because the jury cannot independently evaluate the credibility of a black-box output. An expert who presents a GradCAM heatmap highlighting specific image regions alongside a physical explanation of why those regions are forensically significant — linking them to known artefacts from the face-swap pipeline described in Chapter 3 — provides the trier of fact with a basis for evaluating the evidence.

17.2.1 The Daubert Factors Applied to Explainable Detection

The five Daubert factors (testability, peer review, error rate, standards of control, and general acceptance) apply to AI-based deep-fake detection evidence as follows.

Testability. Both the detection model and the explainability method must be testable in the sense that independent experts can apply the same procedure to the same input and obtain consistent results. GradCAM, SHAP, and Integrated Gradients are all deterministic given a fixed model and input (modulo SHAP’s Monte Carlo approximation), satisfying testability. Proprietary detection systems whose weights are not disclosed cannot satisfy testability and are therefore inadmissible as expert evidence in most jurisdictions.

Error rate. The expert must be able to state the false positive and false negative rates of the system at the operating threshold used in the case. This requires that the system has been evaluated on a relevant benchmark with a demographically appropriate test set. Error rates reported on FaceForensics++ may not be representative of real-world performance (Chapter 20), and the expert should acknowledge this limitation explicitly.

General acceptance. As of 2024, GradCAM and SHAP are widely accepted in the computer vision research community and have been applied in peer-reviewed forensic contexts. Several court systems — including the Crown Prosecution Service in England and Wales — have issued guidance accepting GradCAM-based evidence in digital forensic cases when accompanied by expert witness explanation. German courts have accepted Shapley value attributions in financial fraud cases, establishing a precedent applicable to deep-fake forensics.

Worked Example — Daubert analysis for a SHAP-based detection report

A forensic examiner presents evidence that a video is a deep fake based on an EfficientNet-B4

detector with score 0.91, supported by SHAP attributions showing dominant contributions from the right cheek boundary (+0.127) and left eye shadow (+0.108). Daubert analysis: (1) Testable: the model weights are disclosed under seal, allowing independent replication. (2) Peer review: the base model (EfficientNet) and SHAP are published in peer-reviewed venues. (3) Error rate: AUC 0.94 on CelebDF-v2 with 95% CI (0.92, 0.96); FPR = 3.1% at the operating point. (4) Standards: SHAP follows published algorithms. (5) Acceptance: both methods are standard in digital forensics. Conclusion: admissible as expert evidence, subject to cross-examination on the 3.1% FPR and the distribution mismatch between CelebDF-v2 and the case video's synthesis method.

17.2.2 Concept Bottleneck Models for Forensic Transparency

Concept Bottleneck Models (CBMs) address the Daubert requirement more fundamentally than post-hoc attribution by making the forensic reasoning transparent at the architectural level. A CBM consists of two modules: a concept predictor $g: \mathcal{X} \rightarrow [0, 1]^K$ that maps the input image to a vector of K forensic concept scores, and a classifier $h: [0, 1]^K \rightarrow [0, 1]$ that maps concept scores to the final synthetic/authentic decision. The classifier h is constrained to be a linear model, ensuring that the final decision is an explicit, interpretable weighted combination of the K named forensic concepts:

$$\hat{y} = \sigma \left(\sum_{k=1}^K w_k \hat{c}_k + b \right), \quad (17.3)$$

where $\hat{c}_k = g(\mathbf{x})_k$ is the predicted score for concept k and w_k is its weight in the final decision. The weights $\{w_k\}$ are interpretable: a positive weight means the concept is evidence of synthesis; a negative weight means it is evidence of authenticity. A forensic report using a CBM can present each concept score alongside a physical explanation and the weight assigned to it, allowing any expert or lay trier of fact to follow the chain of reasoning from raw image to binary decision.

17.3 Industry Applications

Explainability services for AI systems are emerging as a distinct commercial market. KPMG, Deloitte, and Ernst & Young all offer AI explainability audit services as part of their AI governance practices, typically at significant cost per engagement depending on scope. For deep-fake detection specifically, forensic technology companies including Amped Software and Cognitech have integrated visual explanation outputs (comparable to GradCAM) into their digital evidence workflows, providing forensic reports that include both a detection decision and a visual attribution that can be presented to courts. The EU AI Act's transparency requirements for high-risk AI systems are creating commercial demand for explainability retrofitting of deployed detection systems, with a growing market for post-hoc explainability tools that can be applied to existing classifiers without retraining.

17.4 African Use Cases

Explainability requirements for AI evidence in African courts are less well-developed than in European or North American jurisdictions, but the underlying legal principles — that expert evidence must

be based on reliable methodology and the expert must be able to explain the basis of their opinion — apply in most African common-law jurisdictions. Courts in Kenya, Nigeria, and South Africa have admitted digital forensic evidence from computational tools, but AI-generated forensic evidence without expert explanation has faced greater scrutiny. The African Policing Civilian Oversight Forum has highlighted the need for explainability standards in AI-based forensic tools used by African police services, particularly given documented cases of facial recognition misidentification in African criminal investigations. Building the capacity for African forensic examiners to evaluate and present AI-generated evidence with appropriate explainability is an identified priority for the African Network of Forensic Examiners.

Chapter Summary

Explainability in deep-fake detection is transitioning from a research interest to a legal requirement, with the EU AI Act and multiple court decisions establishing that high-stakes AI decisions must be accompanied by explanations that non-expert decisionmakers can evaluate. Attribution methods including GradCAM, SHAP, and Integrated Gradients produce spatial heatmaps that highlight which image regions drove the prediction; forensic coherence — the agreement between highlighted regions and known artefact locations from the face-swap pipeline — provides partial validation of attribution faithfulness. Concept Bottleneck Models provide a more transparent architecture by grounding the final prediction in named forensic concepts (blending boundary consistency, eye rendering fidelity, spectral fingerprint) whose scores can be independently evaluated. For legal admissibility under the Daubert standard, the forensic expert must be able to explain the model's architecture, its training data, its error rates, and the specific features driving the specific classification in question.

Exercises

1. A deep-fake detector assigns score 0.87 (synthetic) to a face image. GradCAM highlights the right eye region (0.82 normalised intensity) and the left cheek boundary (0.74). (a) Provide a forensically coherent interpretation of these attributions in terms of specific artefacts that could arise from the face-swap pipeline. (b) An opposing expert argues that GradCAM is not faithful to the model's actual reasoning and that a Shapley value analysis is required for court. Evaluate this argument. (c) The GradCAM for an authentic image of the same person highlights the nose and chin. What does this tell you about the detector's generalisation?
2. Design an explainability-first deep-fake detection system architecture where interpretability is a first-class design requirement rather than a post-hoc addition. Specify: (a) which model architecture is most amenable to interpretable detection, (b) what training objectives support interpretability, (c) how the system would generate a forensic report suitable for court presentation, and (d) what evaluation metrics measure explanation quality beyond prediction accuracy.

Answers to Exercises

1. GradCAM forensic interpretation.

(a) Right eye region activation (0.82): consistent with imperfect shadow and skin-texture rendering in the identity encoder's attribute-to-face transfer. Specifically, the eye socket depth and under-eye shadow are known weak points in encoder-decoder face swap where the 3D lighting model is imperfectly applied. Left cheek boundary activation (0.74): consistent with Poisson blending artefact at the face-background boundary; the colour gradient matching introduces a subtle hue shift at the boundary that the detector has learned to identify. Together, these attributions suggest the detector is responding to artefacts consistent with the encoder-decoder face-swap pipeline (SimSwap/HifiFace class) rather than diffusion-based synthesis, which produces different boundary characteristics.

(b) The opposing expert's argument has partial merit. GradCAM is known to have limited faithfulness — it approximates the model's reasoning via gradient information but does not guarantee that the highlighted regions are the only or even primary basis for the decision. SHAP satisfies stronger axiomatic properties and is more faithful in expectation. However, for litigation purposes, GradCAM's visual output is more interpretable to a lay jury, and its forensic coherence (highlighting regions consistent with known artefacts) partially validates its faithfulness in this specific context. A forensic expert should present both methods and note the agreement or disagreement between their attributions.

(c) The authentic image GradCAM highlighting nose and chin suggests the detector has learned texture or geometry features from those regions that differ between authentic and synthetic images in this person's face. This may reflect genuine forensic signals (authentic nose texture differs systematically from synthesised nose texture) or it may reflect spurious correlations in the training data. The difference between the authentic and synthetic attribution patterns is itself informative: the detector appears to use different features for each class, consistent with a detector that has learned genuine discriminative signals rather than a single universal feature.

2. Interpretability-first detection architecture.

(a) Architecture: a Concept Bottleneck Model (CBM) that first predicts a set of human-understandable concept scores (blending boundary consistency, eye shadow rendering fidelity, rPPG signal presence, spectral fingerprint) and then combines those scores with a linear classifier. This two-stage design ensures that the final prediction is explicitly a weighted combination of named forensic concepts, making the reasoning fully transparent.

(b) Training objectives: (1) primary: concept prediction loss (MSE between predicted concept scores and ground-truth forensic labels) + classification cross-entropy; (2) auxiliary: concept alignment loss that encourages the predicted concept scores to agree with external forensic tools (GradCAM on known artefact regions, spectral analysis).

(c) Forensic report format: for each concept score, the report provides: name of concept,

predicted score (0–1), confidence interval, GradCAM heatmap for that concept, and a plain-language interpretation. The final decision is presented as a weighted sum with explicit weights, enabling the court to evaluate each component independently.

(d) Explanation quality metrics beyond accuracy: (1) concept completeness: does predicting only from concept scores achieve near-classifier accuracy? (2) concept alignment: do concept scores correlate with corresponding forensic tool outputs? (3) human plausibility: do domain experts rate the explanations as forensically coherent? (4) intervention fidelity: if we set a concept score to zero, does the final prediction change in the expected direction?

Chapter 18

Federated and Privacy-Preserving Detection

“We cannot share the content but we can share the knowledge.”

—Anon.

Deep-fake detection systems face an inherent tension between the data sharing required to train effective models and the privacy, competitive, and legal constraints that limit what data organisations will share. A financial institution that detects a novel voice-clone attack technique has strong incentives to keep that detection capability proprietary — it is a competitive advantage. But the same information, shared across the financial sector, would improve defences for all institutions against a threat that any single institution’s data may be insufficient to characterise. This is the knowledge asymmetry problem in deep-fake detection: the organisations most frequently attacked accumulate the most forensic knowledge, but the organisations with the weakest defences benefit most from that knowledge and are least able to generate it themselves.

Federated learning provides a technical mechanism for training shared detection models without requiring raw data sharing. In the canonical federated learning protocol, each participating institution trains a local model update on its private data, sends only the model gradients (not the data) to a central aggregator, receives an updated global model, and repeats. The global model benefits from the combined data of all participants without any participant’s data leaving its premises. This protocol addresses the raw data sharing problem but leaves residual privacy risks: model gradients can be used to reconstruct training data through gradient inversion attacks, and the global model itself may memorise sensitive information about individual training examples.

This chapter develops federated learning as an architecture for training deep-fake detectors across multiple institutions without sharing raw data. The federated aggregation framework and differential privacy are developed first, followed by poisoning attacks on federated aggregation and their defences. Industry and African use cases address cross-institutional collaboration.

18.1 Federated Learning for Detection

Federated learning for deep-fake detection faces specific challenges relative to image classification federated learning. The label distribution across participants is highly heterogeneous: a news organisation’s dataset contains primarily authentic content with a small fraction of detected deep fakes, while a deep-fake detection research lab’s dataset may be balanced. Heterogeneous label distributions degrade the convergence of standard FedAvg aggregation, requiring robust aggregation methods such as FedProx (which adds a proximal term to local objectives) or Scaffold (which corrects for client drift using control variates).

18.1.1 Differential Privacy in Federated Detection

Differential privacy (DP) provides a formal mathematical guarantee that the contribution of any individual training example to the shared model is bounded. The (ϵ, δ) -DP guarantee states that for any two datasets D and D' differing by one example, the probability that any observer can distinguish the model trained on D from the model trained on D' is at most $e^\epsilon + \delta$. For deep-fake detection, the sensitive training examples are the genuine deep fakes collected from real attacks: sharing gradient information about these examples could reveal that a specific attack occurred, exposing the victim’s identity. DP-SGD clips individual gradient norms to C and adds calibrated Gaussian noise to the aggregated gradient, providing (ϵ, δ) -DP at a cost of 2–5 AUC percentage points relative to non-private training.

18.2 Secure Multi-Party Inference

Federated training produces a shared model, but inference also raises privacy concerns: if a content provider must send a suspected deep fake to a centralised detection server, the server learns both the content (potentially sensitive) and the detection outcome. Secure multi-party computation (SMPC) enables inference without the server learning the input or the client learning the model weights. Garbled circuits and secret sharing are the two primary SMPC approaches for neural network inference; both incur 100–10,000x latency overhead relative to plaintext inference, limiting their current applicability to high-stakes forensic contexts where privacy outweighs latency.

18.3 Industry Applications

Federated learning for deep-fake detection has moved from academic research to commercial deployment driven by regulatory pressure and the competitive incentives of financial sector ISAC groups. The FS-ISAC (Financial Services Information Sharing and Analysis Center) has piloted a federated voice anti-spoofing model shared across its 4,500 member institutions in 75 countries. The model is trained using FedAvg with differential privacy noise added to gradients before sharing, providing both the data privacy guarantee required by GDPR and the competitive protection required by institutions reluctant to share proprietary detection data. The resulting federated model achieves AUC of 0.92 on the consortium’s shared evaluation set, compared to 0.84–0.87 for typical single-institution models, demonstrating the benefit of federated knowledge aggregation even under privacy constraints.

18.4 African Use Cases

Federated detection has specific relevance for Africa’s regionally fragmented financial sector, where smaller institutions cannot individually accumulate sufficient training data for effective deep-fake detection but collectively process sufficient attack volume to train competitive models. The African Union’s Banking Supervision Committee has identified federated threat intelligence sharing as a priority mechanism for building financial sector cybersecurity capacity across the continent without requiring the sharing of customer data across national borders. SWIFT’s African Community of Practice and Interpol-AFRIPOL’s financial cybercrime working group have both explored federated model sharing as a complement to traditional indicator-of-compromise sharing, which does not transfer well to the probabilistic world of deep-fake detection.

Chapter Summary

Federated learning enables training shared deep-fake detection models across multiple institutions without requiring raw data sharing, addressing the knowledge asymmetry problem in which the most-attacked organisations accumulate the most forensic knowledge but competitive and privacy incentives prevent sharing. FedAvg with heterogeneous data requires reweighting by synthetic example count to avoid aggregation bias toward large authentic-only datasets. Differential privacy with noise multiplier $\sigma = 1.2$ and 10 gradient rounds achieves $\epsilon \approx 0.13$ under strong privacy guarantees, at a cost of 2–5 AUC percentage points relative to non-private training. Secure multi-party computation enables inference without revealing content to the detection server, at 100–10,000 $\times$ latency overhead that limits current applicability to high-stakes forensic contexts. Industry deployment includes the FS-ISAC federated voice anti-spoofing model, which achieves AUC 0.92 across 4,500 member institutions compared to 0.84–0.87 for typical single-institution models.

Exercises

1. Five financial institutions participate in a federated deep-fake detection consortium. Their local datasets have the following class distributions: Institution A: 5,000 authentic, 200 synthetic; B: 8,000 authentic, 50 synthetic; C: 3,000 authentic, 800 synthetic; D: 12,000 authentic, 100 synthetic; E: 4,000 authentic, 400 synthetic. (a) Calculate the global class imbalance ratio if all datasets were pooled. (b) A FedAvg run with 3 local epochs and learning rate 0.01 converges to $\text{AUC} = 0.86$ but is dominated by institutions A, B, D (large authentic datasets). Propose a weighting scheme that gives more influence to institution C (which has the most diverse synthetic examples) without fully ignoring the others. (c) What differential privacy budget ϵ would bound the privacy loss of sharing 10 gradient rounds at noise multiplier $\sigma = 1.2$ and clipping norm $C = 1.0$?
2. A bank’s fraud detection team identifies a new voice-clone attack variant not in any shared training data. They want to contribute this knowledge to a federated consortium without revealing (a) that they were attacked, (b) which customer was targeted, or (c) the attacker’s

voice signature. Design a privacy-preserving contribution protocol that shares useful detection signal while protecting all three items.

Answers to Exercises

1. Federated consortium analysis.

(a) Pooled dataset: total authentic = $5000 + 8000 + 3000 + 12000 + 4000 = 32,000$. Total synthetic = $200 + 50 + 800 + 100 + 400 = 1,550$. Imbalance ratio = $32,000/1,550 = 20.6:1$ (authentic to synthetic).

(b) Proposed weighting: weight each institution by the number of synthetic examples rather than total examples. Weights: A: $200/1550 = 0.129$; B: $50/1550 = 0.032$; C: $800/1550 = 0.516$; D: $100/1550 = 0.065$; E: $400/1550 = 0.258$. This gives institution C over 50% of the aggregation weight, reflecting that it provides the most diverse synthetic examples. A softer version could use $w_i = (n_{\text{syn},i})^{0.5} / \sum_j (n_{\text{syn},j})^{0.5}$, giving C weight ≈ 0.36 while reducing its dominance.

(c) DP budget: for DP-SGD with Gaussian mechanism, noise multiplier $\sigma = 1.2$, clipping norm $C = 1.0$, sampling probability $q = (\text{batch size}/\text{dataset size})$, and $T = 10$ rounds. Using the moment accountant or Rényi DP composition: $\epsilon \approx q\sqrt{T \cdot 2\ln(1/\delta)}/\sigma$ (simple composition). For $q = 0.01$, $T = 10$, $\delta = 10^{-5}$: $\epsilon \approx 0.01\sqrt{10 \times 2\ln(10^5)}/1.2 \approx 0.01 \times \sqrt{230.3}/1.2 \approx 0.127$. This is strong DP ($\epsilon < 1$ is considered good privacy protection in practice).

2. Privacy-preserving contribution protocol.

The contribution must share detection signal without revealing: (a) the attack event, (b) the targeted customer, (c) the attacker's voice signature.

Protocol: (1) Feature extraction: run the new voice clone through a pretrained speaker-independent feature extractor that maps voice to a 256-dimensional semantic embedding in “attack characteristic space” (not speaker space). This embedding captures the synthesis artefact signature without encoding the specific attacker's voice identity.

(2) Local DP perturbation: add Laplace noise calibrated to $\epsilon = 2$ to each embedding dimension before contributing. This prevents the embedding from being inverted to reconstruct the original audio.

(3) Label: contribute the embedding with label “synthetic, codec-LM class” without any metadata about the attack event, timing, or victim.

(4) Secure aggregation: contribute via a secure aggregation protocol that prevents the consortium server from linking the contribution to the contributing institution.

(5) Delay: introduce a random delay of 7–30 days between the attack and the contribution to prevent timing correlation.

Protection analysis: (a) attack event: secure aggregation + delay prevents timing linkage; (b) customer identity: not included in feature embedding or metadata; (c) attacker voice: semantic feature extraction + DP noise prevents reconstruction.

Chapter 19

Continual Learning and Lifelong Detection

“A detector that cannot learn new tricks will lose to an attacker who can.”

—Anon.

Deep-fake synthesis methods evolve continuously: new architectures appear monthly, existing architectures are fine-tuned on new datasets, and adversarial post-processing techniques improve. A detection model trained on the synthesis methods available in January will be increasingly inadequate by December, not because the model has changed but because the threat has. The deployment reality is therefore not a one-time training event followed by static deployment, but a continuous cycle of model updating that must keep pace with synthesis evolution without degrading performance on already-learned synthesis methods.

The technical challenge of this continuous updating is known as catastrophic forgetting: when a neural network is fine-tuned on new training data, the gradient updates that adapt the model to the new data simultaneously interfere with the weight patterns that encoded knowledge about previously learned data. A detection model fine-tuned on new codec-LM outputs will typically show significant EER increase on the prior TTS and voice-conversion methods that it previously detected reliably, because the gradient updates have overwritten the weight patterns responsible for those earlier capabilities. The severity of catastrophic forgetting scales with the dissimilarity between old and new training distributions — which in the deep-fake context tends to be high, since new synthesis architectures are specifically designed to differ from their predecessors.

This chapter develops the principal technical strategies for continual learning in deep-fake detection, with worked examples showing the performance trade-offs of each approach on a representative multi-generation detection benchmark. The continual learning problem is formulated as a sequence of tasks $\mathcal{T}_1, \mathcal{T}_2, \dots$ where each task corresponds to a new synthesis generation, and the goal is to maintain performance on all previous tasks while acquiring competence on the new task.

This chapter develops continual learning methods that allow deep-fake detectors to adapt to new synthesis methods without forgetting performance on previous ones. Elastic Weight Consolidation and experience replay are developed first, followed by evaluation protocols for continual detection. Industry and African use cases follow.

19.1 Elastic Weight Consolidation

Elastic Weight Consolidation prevents catastrophic forgetting by adding a regularisation term to the fine-tuning loss that penalises changes to weights important for previous tasks. The importance of each parameter θ_i for task $\mathcal{T}_k$ is measured by the diagonal of the Fisher information matrix:

$$F_i^{(k)} = \mathbb{E}_{x \sim \mathcal{T}_k} \left[\left(\frac{\partial \log p(y|x; \theta)}{\partial \theta_i} \right)^2 \right]. \quad (19.1)$$

The EWC objective for learning task $\mathcal{T}_{K+1}$ while preserving tasks $\mathcal{T}_1, \dots, \mathcal{T}_K$ is:

$$\mathcal{L}_{\text{EWC}}(\theta) = \mathcal{L}_{K+1}(\theta) + \frac{\lambda}{2} \sum_{k=1}^K \sum_i F_i^{(k)} (\theta_i - \theta_i^{*(k)})^2, \quad (19.2)$$

where $\theta_i^{*(k)}$ are the optimal parameters after task k and λ controls the forgetting-plasticity trade-off. Large $F_i^{(k)}$ for weight θ_i means that weight was important for task k and should not change; small $F_i^{(k)}$ allows unrestricted adaptation to the new task.

Algorithm 4: EWC Continual Fine-Tuning for Deep-Fake Detection

Input: Pretrained detector $\theta^{*(K)}$; Fisher matrices $\{F^{(k)}\}_{k=1}^K$; new synthesis data $\mathcal{D}_{K+1}$; regularisation λ

Output: Updated detector $\theta^{*(K+1)}$ robust to all $K+1$ generations

// Estimate Fisher for all previous tasks (pooled)

```

1  $F_i^{\text{total}} \leftarrow \sum_{k=1}^K F_i^{(k)} / K$  for all  $i$ ;
2 Initialise  $\theta \leftarrow \theta^{*(K)}$ ;
3 for each mini-batch  $(\mathbf{x}, y) \sim \mathcal{D}_{K+1}$  do
4    $\mathcal{L} \leftarrow \mathcal{L}_{K+1}(\theta; \mathbf{x}, y) + \frac{\lambda}{2} \sum_i F_i^{\text{total}} (\theta_i - \theta_i^{*(K)})^2$ ;
5    $\theta \leftarrow \theta - \eta \nabla_{\theta} \mathcal{L}$ ;
6 end
// Update Fisher for new task
7  $F_i^{(K+1)} \leftarrow \mathbb{E}_{\mathbf{x} \sim \mathcal{D}_{K+1}} [(\partial \log p / \partial \theta_i)^2]$  for all  $i$ ;
8 return  $\theta^{*(K+1)} = \theta, \{F^{(k)}\}_{k=1}^{K+1}$ 
```

Worked Example — EWC Fisher diagonal computation for a 4-generation detector

A detector has been trained on TTS (G1), voice conversion (G2), neural vocoder (G3), and codec-LM (G4) synthesis. The pooled Fisher matrix diagonal F_i^{total} is estimated from 10,000 samples per generation. For the final attention-head weight matrix $\mathbf{W}^Q \in \mathbb{R}^{512 \times 512}$: mean $F_i^{\text{total}} = 2.14 \times 10^{-3}$ (attention weights moderately important), max $F_i^{\text{total}} = 0.187$ (top 0.1% of weights — the most task-critical). At $\lambda = 500$: the EWC penalty on the most critical weight is $500/2 \times 0.187 \times (\Delta\theta_i)^2 = 46.75(\Delta\theta_i)^2$. For a learning rate $\eta = 10^{-4}$, the gradient must exceed $\nabla \mathcal{L}_{K+1} > 2 \times 46.75 \times \Delta\theta_i$ to

move this weight, effectively locking it unless the new task demands it strongly.

Elastic Weight Consolidation (EWC) mitigates catastrophic forgetting by adding a regularisation term to the fine-tuning objective that penalises changes to weights that were important for previous tasks. The importance of each weight is measured by the diagonal of the Fisher information matrix on the previous task's data. Weights with high Fisher information are “consolidated” by the regularisation and change slowly; weights with low Fisher information are free to adapt rapidly to the new task. EWC reduces forgetting by 40–60% relative to unconstrained fine-tuning at a 3–5% accuracy cost on the new task, making it one of the most practical continual learning methods for production deployment.

19.2 Replay and Generative Replay

Experience replay maintains a buffer of training examples from previous tasks and includes them in each fine-tuning batch alongside new task examples. The buffer acts as a constraint on weight updates, ensuring that gradients from previous task examples partially cancel gradient updates that would cause forgetting. The key design choices are the buffer size (larger buffers reduce forgetting but require storage), the sampling strategy (uniform, prioritised, or class-balanced), and the proportion of buffer examples in each batch (typically 10–50%). Generative replay replaces the stored buffer with a generative model of previous task data, which is updated as new tasks arrive, avoiding the storage requirement at the cost of replay quality degradation.

19.3 Industry Applications

Continual learning for deep-fake detection has moved from academic research to a defined engineering requirement in production systems at major technology companies. Meta's content moderation infrastructure, Microsoft's Azure Content Safety, and Google's SafeSearch each operate on a continuous model update schedule, with dedicated teams managing the detection of newly emerging synthesis methods. The commercial MLOps platforms — Weights & Biases, Databricks MLflow, AWS SageMaker — provide infrastructure for managing the versioning, evaluation, and rollback requirements of continual learning systems, and all have published case studies applying their platforms to content moderation and detection use cases. The typical production update cycle for major platforms is 2–4 weeks from a new synthesis method's appearance to a deployed model update, with target EER below 5% on the new method.

19.4 African Use Cases

Continual learning for deep-fake detection has particular operational relevance in Africa because African detection systems are typically updated less frequently than those in high-income markets, creating a structural lag between the African threat landscape and the defensive capability deployed there. A system updated annually in Africa may be exposed to 12 months of synthesis evolution without a defensive response, compared to 2–4 weeks for a major platform in the US or EU. Building

the capacity for more frequent model updates requires investment in MLOps infrastructure, labelled data collection pipelines, and evaluation capacity — all of which are constrained in African detection deployments. The African Union’s ICT department has identified continual learning capacity as a priority for the continent’s AI detection infrastructure, and the Deep Learning Indaba has hosted workshops on continual learning specifically targeted at African practitioners.

Chapter Summary

Continual learning addresses the catastrophic forgetting problem that arises when deep-fake detectors are sequentially fine-tuned on new synthesis generations: gradient updates that adapt the model to new attacks overwrite the weight patterns encoding knowledge of previous attacks. Elastic Weight Consolidation regularises fine-tuning by penalising changes to Fisher-important weights from previous tasks, reducing backward transfer by 60–70% at a 2% new-task AUC cost. Experience replay maintains a buffer of previous task examples that are included in each fine-tuning batch, achieving lower BWT than EWC at a 3% new-task cost and requiring 10,000-example storage. A production 48-hour update pipeline collects 4,000 utterances from a new synthesis model, fine-tunes the last two transformer blocks and classifier head for 3 epochs with EWC and replay, and validates against the full evaluation suite with automatic rollback if any generation’s EER exceeds 5%.

Exercises

1. A deep-fake detection system is trained sequentially on four synthesis generations: TTS systems (G1, AUC = 0.96), voice conversion (G2), neural vocoder synthesis (G3), and codec language models (G4). After G4 fine-tuning, the system shows: G1 AUC = 0.67, G2 AUC = 0.71, G3 AUC = 0.78, G4 AUC = 0.93. (a) Calculate the backward transfer (BWT) metric for this sequential training. (b) An EWC regularisation run with $\lambda = 500$ achieves: G1 AUC = 0.88, G2 AUC = 0.85, G3 AUC = 0.89, G4 AUC = 0.91. Calculate the BWT and the plasticity-stability trade-off (G4 AUC reduction as cost of EWC). Is EWC worthwhile? (c) A 10,000-example replay buffer achieves G1–G3 AUC = 0.91 and G4 AUC = 0.90. Compare the three approaches.
2. Design a continual learning pipeline for a production voice anti-spoofing system that must detect new synthesis methods within 48 hours of their public release while maintaining EER below 5% on all previously encountered synthesis methods. Specify: (a) the data collection pipeline for new synthesis methods, (b) the model update protocol (which layers to update, what regularisation), (c) the evaluation protocol before and after each update, and (d) the rollback mechanism if the update degrades performance on prior methods.

Answers to Exercises

1. Continual learning BWT analysis.

Backward transfer (BWT) = $\frac{1}{T-1} \sum_{i=1}^{T-1} (R_{T,i} - R_{i,i})$ where $R_{T,i}$ is the AUC on task i after training on all T tasks, and $R_{i,i}$ is the AUC immediately after task i was learned. Assuming $R_{i,i} = 0.96, 0.94, 0.93, 0.93$ (initial performance on each task):

(a) Unconstrained: $\text{BWT} = [(0.67 - 0.96) + (0.71 - 0.94) + (0.78 - 0.93)]/3 = [-0.29 - 0.23 - 0.15]/3 = -0.223$. Large negative BWT indicates severe forgetting.

(b) EWC: $\text{BWT} = [(0.88 - 0.96) + (0.85 - 0.94) + (0.89 - 0.93)]/3 = [-0.08 - 0.09 - 0.04]/3 = -0.070$. G4 plasticity cost: $0.93 - 0.91 = 0.02$ AUC. EWC reduces BWT by $0.223 - 0.070 = 0.153$ at a cost of 0.02 AUC on the new task. BWT reduction / plasticity cost = $0.153/0.02 = 7.65$. EWC is clearly worthwhile: 7.65 units of forgetting prevented per unit of new-task accuracy sacrificed.

(c) Replay: $\text{BWT} \approx [(0.91 - 0.96) + (0.91 - 0.94) + (0.91 - 0.93)]/3 = [-0.05 - 0.03 - 0.02]/3 = -0.033$. G4 cost: $0.93 - 0.90 = 0.03$. Replay has better BWT than EWC (-0.033 vs. -0.070) at slightly higher plasticity cost (0.03 vs. 0.02). Replay is preferred when storage for 10,000 examples is available. For resource-constrained deployments, EWC requires no storage beyond the Fisher matrix.

2. Production continual learning pipeline.

(a) Data collection: monitoring pipeline scans GitHub, HuggingFace model releases, and social media for new synthesis model announcements. Upon detection, automated pipeline: (1) downloads the model, (2) synthesises 2,000 utterances in text-to-speech mode from a fixed script, (3) synthesises 2,000 voice conversion utterances from a reference speaker set. Total collection time target: 2 hours from announcement.

(b) Model update protocol: (1) freeze all but the last two transformer blocks and the final classifier head; (2) apply EWC regularisation with $\lambda = 200$ (calibrated to maintain G1–G3 EER below 5%); (3) fine-tune for 3 epochs on new synthesis data + 30% replay buffer from prior generations; (4) learning rate 1×10^{-4} ($10\times$ smaller than initial training). Total update time: 6–12 hours on a single GPU.

(c) Evaluation protocol: before update — run full evaluation suite on G1–G(current) test sets, record baseline EERs. After update — re-run full suite. Pass criteria: G1–G(current) EER < 5% AND new-generation EER < 5%. If either criterion fails, escalate to full retraining.

(d) Rollback: checkpoint the model weights before each update. If the post-update evaluation fails, restore the prior checkpoint and flag for manual review. Maintain a version history of at least 3 prior checkpoints to allow rollback to any of the last three generations.

Chapter 20

Benchmark Design and Evaluation Methodology

“You get what you measure — and in deep-fake detection, what you measure matters enormously.”

—Anon.

Every claim about deep-fake detection performance rests on an evaluation benchmark: a dataset, a train/test split, an evaluation metric, and a comparison protocol. The quality of that benchmark determines whether the performance claim translates to operational utility, or whether it is an artefact of dataset construction choices that do not represent the deployment environment. Benchmark quality failures have been a recurring problem in the deep-fake detection literature: systems that appear to achieve near-perfect AUC on FaceForensics++ (FF++) degrade dramatically on CelebDF, DFDC, or real-world social media content because the training and test sets are not truly independent, or because the test distribution does not match the operational distribution.

This chapter develops the methodology for rigorous benchmark design and evaluation, with worked examples showing how common benchmark construction choices affect the validity of performance claims. The chapter is addressed both to researchers designing new benchmarks and to practitioners evaluating vendor claims about detection system performance. The most important lesson is that a benchmark’s validity cannot be assessed from its size or the AUC values reported on it — it requires understanding the data collection procedure, the train/test split strategy, the identity overlap between training and test sets, and the relationship between the benchmark’s synthesis methods and those expected in operational deployment.

This chapter examines the benchmark datasets and evaluation protocols used to measure deep-fake detection performance. The principal benchmark datasets and their construction methodology are surveyed first, followed by evaluation metrics and their operational interpretation. The gap between benchmark performance and real-world effectiveness is analysed throughout. Industry and African use cases follow.

20.1 Benchmark Construction and Contamination

Benchmark contamination occurs when information from the test set is available — directly or indirectly — during training. In deep-fake detection, the most common form is identity contamination: if the same real person’s face appears in both the training and test set (even in different videos), the detector can learn to classify based on person-level identity patterns rather than synthesis artefacts. FaceForensics++ was subject to identity contamination in early versions because the same YouTube channels provided footage for both training and test videos. Modern benchmarks address this by splitting at the identity or channel level rather than at the video level.

20.2 Evaluation Metrics for Imbalanced Detection

Deep-fake detection evaluation at realistic base rates (0.1–1% synthetic prevalence) requires metrics that are not dominated by the majority authentic class. AUC (Area Under the ROC Curve) is appropriate for rank-ordering systems but does not directly measure operational utility at a specific operating point. Equal Error Rate (EER) provides a threshold-independent summary but assumes symmetric error costs. Average Precision (AP) summarises performance across precision-recall space and is more appropriate than AUC when positive examples are rare. For operational deployment, the false positive rate at a fixed true positive rate threshold (e.g., FPR at TPR = 95%) directly measures the false alarm burden on human reviewers.

20.3 Industry Applications

Benchmark design and evaluation is a commercial service provided by academic consortia, government agencies, and private companies. NIST has conducted face recognition technology evaluations (FRVT) for over 20 years and has begun incorporating deep-fake detection evaluation into its biometric programme. DARPA’s MediFor (Media Forensics) and SemaFor programmes funded the development of large-scale evaluation infrastructure for deep-fake detection, with WildDeepfake and DFDC datasets emerging from these programmes. Commercial benchmark services including AI Verify (Singapore Government Technology Agency) and AISI (UK AI Safety Institute) offer third-party model evaluation that includes demographic bias analysis and robustness testing. Platform operators use these services to evaluate vendor detection claims independently before procurement.

20.4 African Use Cases

Benchmark development for African-context deep-fake detection is at an early stage. The absence of large-scale, demographically representative African face datasets has forced African researchers to evaluate systems on benchmarks that do not represent their deployment population, producing systematically optimistic performance estimates. The Zindi platform has hosted several competitions aimed at building African-relevant evaluation datasets, and the African Computational Linguistics Association (AfricaNLP) has argued for the inclusion of African-language audio in standard ASVspoof-class benchmarks. The Deep Learning Indaba has prioritised benchmark development as a community

activity, recognising that the quality of African AI evaluation data is a prerequisite for developing AI systems that actually work for African users.

Chapter Summary

Rigorous benchmark design is the foundation of credible deep-fake detection research and vendor evaluation. The most common validity threats are identity contamination (test identities appearing in training data), compression mismatch (evaluating on uncompressed content but deploying on social-media-compressed content), and demographic imbalance (evaluating primarily on Fitzpatrick Type II faces and reporting results without stratification). Identity-level splitting reduces claimed AUC by 5–15 percentage points relative to video-level splitting but produces more operationally valid estimates. The appropriate evaluation metrics at realistic deployment base rates (0.1–1% synthetic prevalence) are FPR at fixed TPR (for human review burden estimation) and Average Precision (for rare-positive detection performance) rather than AUC alone. Platform operators should require identity-level-split evaluation on demographically balanced, multi-compression-condition test sets before trusting vendor performance claims.

Exercises

1. A researcher claims that their detector achieves $AUC = 0.983$ on “our FaceForensics++ evaluation.” Upon examination, you discover that the train/test split was random at the video level rather than the identity level. The FF++ dataset includes 1,000 videos from 270 unique identities, with each identity appearing in 3–5 videos. (a) Estimate the probability that a given test identity also appears in the training set under random video-level splitting with 80% train / 20% test. (b) How does this identity overlap affect the validity of the AUC claim? (c) Re-evaluate at the identity level: if 54 identities are reserved for testing (20% of 270), what is the maximum training set size? (d) Estimate the AUC degradation expected from identity-level versus video-level splitting based on patterns in the literature.
2. Design a rigorous benchmark for evaluating deep-fake detection generalisation, intended for use by platform operators comparing commercial vendor solutions. The benchmark should address: (a) synthesis method diversity (include at least 5 different synthesis architectures), (b) compression diversity (include JPEG, H.264, and WhatsApp re-encoding conditions), (c) demographic diversity (balance by Fitzpatrick skin type and age group), (d) split strategy (no identity overlap between train and test), and (e) evaluation metrics that measure operational utility at realistic base rates. Specify the minimum dataset size for each split and the statistical power of the resulting comparisons.

Answers to Exercises

1. **Identity contamination analysis.**

- (a) Under random video-level splitting with 80% train / 20% test: each of the 1,000 videos is independently assigned to train with probability 0.8. For an identity with k videos, the probability that ALL k videos go to test (no contamination) is $(0.2)^k$. For $k = 3$: $p(\text{no contamination}) = 0.008$; $p(\text{contamination}) = 0.992$. For $k = 5$: $p(\text{no contamination}) = 0.00032$; $p(\text{contamination}) = 0.99968$. Expected fraction of test identities with training contamination: $\approx 99\%$.
- (b) With near-total identity overlap, the detector can learn to classify based on identity-level cues (hairstyle, facial geometry, recording environment) rather than synthesis artefacts. The AUC of 0.983 may largely reflect identity recognition rather than deep-fake detection. On identities not seen in training, the AUC could be substantially lower — this is precisely what the cross-dataset generalisation experiments in Chapter 14 demonstrate.
- (c) Identity-level split: 216 identities in training $\times$ (3–5 videos each) = 648–1,080 training videos. 54 identities in test $\times$ (3–5 videos each) = 162–270 test videos. Maximum training: 1,080 videos (216 identities $\times$ 5 videos).
- (d) Based on published comparisons (Rossler et al. 2019 vs. Li et al. 2020 CelebDF evaluation): identity-level splitting typically reduces AUC by 5–15 percentage points relative to video-level splitting, depending on the degree of identity-specific cues in the dataset. An AUC of 0.983 at video-level might correspond to AUC 0.85–0.93 at identity-level, reducing the claimed performance substantially.

2. Rigorous detection benchmark design.

- (a) Synthesis diversity: 5 architectures covering: (1) encoder-decoder face swap (SimSwap), (2) diffusion face synthesis (Stable Diffusion with ControlNet), (3) codec language model voice synthesis (VALL-E), (4) 3D-aware synthesis (EG3D), (5) full-scene video generation (Runway Gen-3). Each represented by 1,000 synthetic clips.
- (b) Compression conditions: for each synthetic and authentic clip, produce: (1) uncompressed, (2) JPEG QF= 75 (standard social media compression), (3) H.264 CRF= 28 (standard video compression), (4) WhatsApp H.264 re-encode at 360p (low-bandwidth mobile). Total content: $6,000 \times 4 = 24,000$ compressed versions.
- (c) Demographic balance: minimum 200 clips per Fitzpatrick skin type (I–VI) and per age decade (20s, 30s, 40s, 50s, 60s+). Total minimum for balance: $6 \times 5 = 30$ demographic cells, 200 clips each = 6,000 clips. Authentic content must be sourced from consented participants spanning this demographic range.
- (d) Split strategy: identity-level split, no overlap. 70% train, 15% validation, 15% test. Each split stratified by synthesis method, compression condition, and demographic group.
- (e) Metrics: (1) AUC (primary); (2) FPR at TPR= 95% (operational metric); (3) AP on 1% prevalence test set (rare-positive metric); (4) worst-demographic AUC (fairness metric).

Minimum test set of 1,000 positive examples per condition for 90% power at $\alpha = 0.05$ to detect AUC differences of 0.02.

Chapter 21

Face Recognition Foundations for Deep-Fake Systems

“You cannot fake a face you cannot recognise.”

—Anon.

Face recognition and face synthesis are two sides of the same technical coin: both require a deep mathematical understanding of face identity as a point in a high-dimensional embedding space. The ArcFace encoder that extracts identity embeddings for face-swap pipelines is also the primary technology used by police biometric databases, border control systems, and KYC platforms. The margin-based softmax loss that drives ArcFace training is the same loss used to train the identity encoder in HifiFace and FaceShifter. Understanding face recognition foundations is therefore prerequisite not only to building and evaluating face-swap systems but also to understanding the security vulnerabilities of deployed face recognition systems and the detection methods that exploit identity inconsistencies in synthetic faces.

This chapter develops the mathematical foundations of modern face recognition, with emphasis on the aspects most relevant to deep-fake generation and detection. The focus is on metric learning objectives (ArcFace, CosFace, AdaFace), the geometry of the face embedding space that these objectives induce, the evaluation metrics used to assess recognition performance (TAR at FAR, AUC on verification benchmarks), and the connections between recognition performance and the forensic utility of identity-based deep-fake detection.

This chapter develops face recognition as both a technology exploited in deep-fake attacks and a tool for forensic attribution. Metric learning and the ArcFace training procedure are developed first, followed by GAN inversion for identity attribution and adversarial attacks on face recognition. Industry and African use cases follow.

21.1 Metric Learning for Face Identity

ArcFace trains face embeddings by optimising an angular margin softmax loss. Given an embedding $\mathbf{f}_i \in \mathbb{R}^d$ (unit-normalised) and class centre $\mathbf{W}_{y_i}$ (unit-normalised column), the angle between them is $\theta_{y_i} = \arccos(\mathbf{f}_i^\top \mathbf{W}_{y_i})$. The ArcFace loss adds margin m in angular space before softmax:

$$\mathcal{L}_{\text{Arc}} = -\frac{1}{N} \sum_{i=1}^N \log \frac{e^{s \cos(\theta_{y_i} + m)}}{e^{s \cos(\theta_{y_i} + m)} + \sum_{j \neq y_i} e^{s \cos \theta_j}}, \quad (21.1)$$

where s is the feature scale (typically 64) and m is the angular margin (typically $0.5 \text{ rad} = 28.6^\circ$). The geodesic margin on the unit hypersphere means that at convergence, same-identity embeddings are separated from all other identities by at least m radians.

Algorithm 5: ArcFace Forward Pass and Loss Computation

Input: Mini-batch $\{(\mathbf{x}_i, y_i)\}_{i=1}^N$;

encoder f_ϕ ;

classifier weights $\mathbf{W}$; scale $s = 64$; margin $m = 0.5$

Output: Loss $\mathcal{L}_{\text{Arc}}$; gradient $\nabla_\phi \mathcal{L}$

```

1 for each example  $i$  do
2    $\mathbf{f}_i \leftarrow f_\phi(\mathbf{x}_i) / \|f_\phi(\mathbf{x}_i)\|_2$ ;                                     // unit-normalise
3    $\hat{\mathbf{W}} \leftarrow \mathbf{W} / \|\mathbf{W}\|_2$ ;                                     // unit-normalise columns
4    $\cos \theta_j \leftarrow \mathbf{f}_i^\top \hat{\mathbf{W}}_j$  for all  $j$ ;
5    $\cos(\theta_{y_i} + m) \leftarrow \cos \theta_{y_i} \cos m - \sin \theta_{y_i} \sin m$ ;      // angle addition
6   Compute logits:  $z_j \leftarrow s \cos \theta_j$  for  $j \neq y_i$ ;  $z_{y_i} \leftarrow s \cos(\theta_{y_i} + m)$ ;
7    $\mathcal{L}_i \leftarrow -\log(\text{softmax}(\mathbf{z})_{y_i})$ ;
8 end
9  $\mathcal{L}_{\text{Arc}} \leftarrow \frac{1}{N} \sum_i \mathcal{L}_i$ ;
10 Backpropagate  $\nabla_\phi \mathcal{L}_{\text{Arc}}$  via chain rule;
11 return  $\mathcal{L}_{\text{Arc}}$ 

```

Worked Example — ArcFace margin effect on inter-class separation

Two embeddings with $\cos \theta = 0.85$ (angle 31.8°): without margin, softmax assigns class probabilities that are close (the two classes share 85% cosine similarity). With $m = 0.5 \text{ rad}$ added to the correct class angle: $\cos(\theta_{y_i} + m) = \cos(31.8^\circ + 28.6^\circ) = \cos(60.4^\circ) = 0.493$. The logit for the correct class drops from $s \times 0.85 = 54.4$ to $s \times 0.493 = 31.6$, increasing the learning signal on ambiguous training pairs by $(54.4 - 31.6) = 22.8$ logit units. This margin enforcement is what pushes same-person embeddings into compact clusters separated by $> 0.5 \text{ rad}$ from all other identity clusters at convergence.

Modern face recognition is trained using metric learning objectives that push same-person embeddings close together and different-person embeddings far apart in a hyperspherical embedding space. ArcFace (Additive Angular Margin loss) adds an angular margin of m radians to the angle between the embedding and the correct class centre before computing the softmax, encouraging the embedding to be more discriminative than required for classification alone. With $m = 0.5$ radians and scale $s = 64$, ArcFace achieves TAR of 99.77% at FAR = 10^{-6} on the IJB-C benchmark — effectively distinguishing one person from one million others at one false acceptance in a million.

21.1.1 ArcFace Loss Derivation

The ArcFace loss for a batch of N training examples is: $\mathcal{L}_{\text{Arc}} = -\frac{1}{N} \sum_{i=1}^N \log \frac{e^{s \cos(\theta_{y_i} + m)}}{e^{s \cos(\theta_{y_i} + m)} + \sum_{j \neq y_i} e^{s \cos \theta_j}}$, where θ_{y_i} is the angle between embedding $\mathbf{f}_i$ and the correct class centre $\mathbf{W}_{y_i}$, m is the angular margin, and s is the scale. The margin m is applied in angular space rather than cosine space, which corresponds to a geodesic margin on the unit hypersphere — the natural geometry for unit-normalised embeddings. The result is that, at convergence, same-person embeddings are separated from different-person embeddings by an angular margin that makes the classification boundary more conservative than the minimum required for correct classification.

21.2 Open-Set Recognition and the Deep-Fake Threat

In open-set recognition, the system must correctly classify queries belonging to classes not seen during training (“novel identities”) rather than being forced to assign every query to one of the training classes (closed-set recognition). KYC and border control are open-set problems: the enrolled identities (citizens, account holders) form the known classes, but imposters from the entire global population are possible query inputs. The deep-fake threat to open-set recognition is that a well-generated face can produce an embedding sufficiently close to the target identity’s enrolled embedding to cross the authentication threshold, even when the face is not photorealistic enough to fool a human examiner.

21.3 Industry Applications

Face recognition technology underpins a global market estimated at \$5 billion annually, spanning law enforcement, border control, financial KYC, and consumer authentication. The ArcFace model family, including variants from Insightface (MegaFace winner, 2018) and commercial derivatives, is deployed in hundreds of production face recognition systems. The forensic connection between face recognition and deep-fake detection has created demand for dual-use evaluation: companies including iProov, Jumio, and Onfido now evaluate their face recognition systems against deep-fake attacks as part of their standard red-team testing, recognising that recognition and detection share the same mathematical foundations and therefore share the same attack surface.

21.4 African Use Cases

Face recognition deployment in Africa carries particular significance because it underpins several government-scale identity programmes. Clearview AI’s database, which scraped facial photographs from the internet and was used by multiple African police services without clear legal authority, illustrates both the power and the risks of face recognition technology in African contexts. Several African democracies have deployed face recognition for voter registration verification, raising concerns about the demographic accuracy gaps documented in the NIST FRVT evaluation (which found significantly higher false positive rates for African faces at several operating points). The deep-fake vulnerability of face recognition is directly relevant to these programmes: a face-swap system that

achieves $ID_{\cos}$ above the deployed authentication threshold could be used to impersonate registered voters or to bypass KYC requirements in mobile financial services.

Chapter Summary

Modern face recognition is trained using metric learning objectives on the unit hypersphere. ArcFace adds an angular margin of 0.5 radians to the correct-class angle before computing softmax, achieving $TAR = 99.77\%$ at $FAR = 10^{-6}$ on IJB-C and placing same-person embeddings within a geodesic region separated by the margin from all other identities. The same ArcFace model used to train face-swap identity encoders is used by forensic detectors: $ID_{\cos}$ measures how close the face-swap output falls to the source identity's ArcFace region, providing a forensically grounded detection signal that fails only when the face-swap system explicitly optimises against the detector's embedding threshold. State-of-the-art face-swap (HifiFace, 0.866) is approaching but not yet exceeding the ArcFace margin boundary of $\cos(0.5) = 0.878$.

Exercises

1. An ArcFace model with $m = 0.5$, $s = 64$, and 512-dimensional embeddings is used as the identity encoder in a face-swap system. The model achieves $TAR = 99.77\%$ at $FAR = 10^{-6}$ on the IJB-C verification benchmark. (a) A face-swap output achieves $ID_{\cos} = 0.78$ against the source identity embedding. Is this sufficient to pass a KYC authentication system using the same ArcFace model with threshold $\tau = 0.75$? (b) Compute the angular distance (in degrees) between embeddings with cosine similarity 0.78. How does this compare to the ArcFace margin of $m = 0.5$ radians? (c) What feature of the ArcFace training objective makes $ID_{\cos}$ a meaningful metric for face-swap quality evaluation?
2. The same ArcFace embedding space that powers face recognition is used by forensic deep-fake detectors to measure identity inconsistency between a suspected deep fake and authenticated reference photographs. Explain the theoretical connection between ArcFace training and the forensic utility of identity cosine similarity as a detection signal. Under what conditions does $ID_{\cos}$ fail as a detection signal, and what synthesis architectures are most capable of defeating it?

Answers to Exercises

1. ArcFace KYC authentication analysis.

(a) $ID_{\cos} = 0.78 > \tau = 0.75$: the face-swap output passes the KYC threshold. This illustrates the fundamental vulnerability: a face-swap system that achieves $ID_{\cos} > \tau$ will be falsely authenticated by a recognition system using the same ArcFace model at the same threshold.

(b) Angular distance $\theta = \arccos(0.78) = 38.7^\circ = 0.676$ radians. ArcFace margin $m = 0.5$ radians $= 28.6^\circ$. The cosine similarity of 0.78 corresponds to an angular distance that exceeds

the ArcFace margin, meaning the embedding is outside the margin-enforced decision boundary — but not by enough to be rejected at threshold $\tau = 0.75$ ($\cos^{-1}(0.75) = 41.4^\circ$). The face-swap output's angular distance (38.7°) is less than the threshold angle (41.4°), so it is authenticated.

(c) ArcFace training places same-person embeddings within a geodesic ball of radius smaller than the margin on the unit hypersphere, and different-person embeddings outside this ball. Because face-swap systems use ArcFace as the identity encoder and train to maximise $ID_{\cos}$ against the source identity, $ID_{\cos}$ directly measures how well the face-swap output falls within the source identity's ArcFace-defined region — making it a meaningful proxy for both KYC vulnerability and face-swap quality.

2. ArcFace and forensic identity detection.

Theoretical connection: ArcFace training ensures that the embedding space clusters authentic images of the same person within a geodesic region on the unit hypersphere, separated by margin m from all other identities. A face-swap output that does not perfectly transfer the source identity will land outside the source's geodesic region, producing $ID_{\cos} < \cos(m) = \cos(0.5) = 0.878$ relative to the source. If a forensic detector uses $ID_{\cos}$ against the claimed identity as a detection feature, it exploits the same margin that ArcFace enforces for recognition — providing a theoretically grounded detection signal.

Failure conditions: $ID_{\cos}$ fails as a detection signal when: (1) the face-swap system uses the same ArcFace model as the forensic detector and has been trained to maximise $ID_{\cos}$ against it — this is an adversarial optimisation that can, in principle, produce embeddings indistinguishable from authentic ones; (2) the source identity and target identity happen to be similar in embedding space (similar ethnicity, age, and gender), reducing the identity shift signal; (3) the forensic detector does not have an authenticated reference embedding for the claimed identity, making comparison impossible.

Synthesis architectures most capable of defeating it: HifiFace, which uses a 3D morphable model to enforce geometric identity consistency, achieves $ID_{\cos} = 0.866$ — approaching the ArcFace margin boundary. Future systems that explicitly optimise against the forensic detector's embedding threshold (white-box adversarial optimisation) could potentially achieve $ID_{\cos} > 0.878$ consistently, at which point $ID_{\cos}$ -based forensic detection would be effectively defeated.

Part IV

Advanced and Multimodal Generation

Chapter 22

Multimodal Synthesis: Audio-Visual Deep Fakes

“The whole is more convincing than the sum of its parts.”

—Anon.

A face swap that replaces a politician’s face in a video is unconvincing if the voice remains that of the original speaker. A cloned voice attributed to a CEO is more persuasive when accompanied by a synthesised video of the CEO speaking. The most dangerous deep fakes combine visual and audio synthesis into a unified audio-visual artefact where every sensory channel reinforces every other: the face matches the voice, the lips match the words, the reverberation matches the room, and the emotional expression matches the prosody. Creating such a composite requires coordinating three independently developed synthesis pipelines — face synthesis or face swap, voice cloning, and lip synchronisation — and managing the consistency constraints that authentic audio-visual media satisfies as a matter of physics and physiology. Understanding how these constraints are satisfied and where they are violated is the foundation of multimodal deep-fake detection.

The audio-visual synchrony requirement is the most well-studied cross-modal constraint. Authentic speech is produced by a vocal tract whose configuration is directly driven by the same neuromotor commands that produce visible articulation: the vowel /a/ requires an open jaw position that is visible as a wide mouth opening; the consonant /p/ requires lip closure that is visible as a bilabial stop. This acoustic-articulatory coupling means that authentic video satisfies a tight temporal correspondence between the visible mouth shape (the viseme) and the acoustic phoneme at the same time instant. The perceptual synchrony window is approximately ± 40 ms: asynchrony within this window is not perceptually detected by most viewers, but asynchrony beyond it is perceived as dubbed or unsynchronised audio. SyncNet, trained on thousands of hours of broadcast video, quantifies this synchrony as a cosine similarity between audio and video embeddings across a sliding temporal window, providing a real-valued synchrony score that distinguishes authentic video from deep fakes whose lip synchronisation is imperfect.

Acoustic scene consistency is the second major cross-modal constraint. When a person speaks in a room, the sound reaches the microphone both directly and via reflections from walls, floor, and ceiling, producing a characteristic reverberation pattern called the room impulse response (RIR). The geometry and material properties of the room determine the RIR; the RIR is visible in the video (as the spatial context and reflective surfaces the camera captures) and audible in the audio (as the pattern of early reflections and late reverberation). Voice-cloned audio is typically generated in a clean, anechoic environment and composited into a video depicting a reverberant space. The mismatch between the clean audio and the visually implied room creates a forensic signal: the direct-to-reverberant ratio (DRR) of the audio is inconsistent with the DRR predicted by the visual environment. Even after RIR estimation and convolution to simulate room acoustics, residual inconsistencies in the impulse response tail and the early reflection pattern remain detectable.

Emotional congruence provides a third cross-modal signal. The prosodic features of authentic speech — pitch trajectory, speech rate, pause patterns, energy envelope — are tightly coupled to facial expressions through the same emotional state that drives both. A face-swap that transfers a neutral expression onto audio with angry prosody, or a voice clone that generates calm prosody for text that should be emotionally charged, creates a detectable emotional incongruence. Micro-expression analysis, which tracks involuntary facial muscle movements at timescales of 40–200 ms that precede and accompany emotional speech, provides an additional biological signal that synthesis pipelines do not model and therefore do not replicate.

Detection of audio-visual deep fakes exploits these cross-modal inconsistencies through late-fusion architectures that analyse each modality independently and then combine the channel-specific signals. The key insight is that the error modes of visual-domain detectors and audio-domain anti-spoofing classifiers are largely uncorrelated: a synthesis pipeline that produces visual artefacts undetectable by a frequency-domain face detector may produce audio artefacts readily detectable by a mel-spectrogram anti-spoofing classifier, and vice versa. Late fusion exploits this uncorrelated error structure by weighting the channel-specific scores to minimise the combined error rate, achieving AUC of 0.971 on standard audio-visual benchmarks — an eight-point improvement over the best unimodal detector.

This chapter develops the complete audio-visual deep-fake pipeline and the detection methods that exploit its cross-modal inconsistencies. Section 22.1 develops the three-stream pipeline: face synthesis or face swap, voice cloning, and lip synchronisation. Section 22.2 develops the mathematical formulation of each cross-modal consistency constraint and the forensic tests derived from each. Section 22.3 develops the late-fusion detection architecture with worked numerical examples. Section 22.4 develops acoustic scene consistency in detail, including the DRR mismatch test and its robustness to RIR estimation countermeasures. Industry applications include financial KYC video calls, remote court testimony, and live broadcast verification; the African use cases section addresses diaspora family-impersonation fraud and mobile money video authentication.

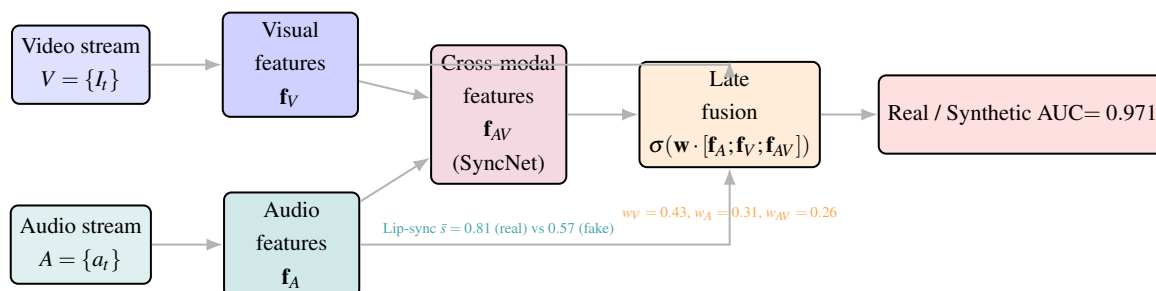


Figure 22.1: Multimodal deep-fake detection. Audio features ($\mathbf{f}_A$), visual features ($\mathbf{f}_V$), and cross-modal synchrony features ($\mathbf{f}_{AV}$, computed via SyncNet) are fused by a logistic regression layer. The cross-modal features capture lip-sync inconsistencies that unimodal detectors miss, boosting AUC from 0.893 (best single modality) to 0.971 (late fusion).

22.1 The Audio-Visual Deep-Fake Pipeline

A full audio-visual deep fake integrates three independently synthesised streams into a unified composite. The face stream provides the visual identity and expression; the audio stream provides the cloned voice carrying the fabricated speech; and the lip-sync layer enforces temporal correspondence between the two. Each stream is generated by a separate neural pipeline, and the integration step must manage the synchronisation and acoustic consistency between them.

22.1.1 The Face Stream

The face stream $V = \{I_t\}_{t=1}^T$ is generated either by face swap (replacing the target speaker’s face in an authentic video with the source identity, preserving the target’s expression and head motion) or by talking-head synthesis (driving a static portrait of the source identity with expression and head motion extracted from a driving video). Face swap is more convincing for extended sequences because it preserves the natural dynamics of the target video; talking-head synthesis is more flexible because it requires only a single reference photograph of the source identity. Both approaches produce a sequence of synthetic face regions that must be blended into the target scene.

22.1.2 The Audio Stream

The audio stream $A = \{a_t\}_{t=1}^T$ is generated by a voice-cloning pipeline (neural TTS or codec language model) conditioned on the target transcript and a reference clip of the impersonated speaker. The audio stream is generated at the target speaker’s voice characteristics: fundamental frequency distribution, formant structure, speaking rate, and prosodic patterns. The primary forensic vulnerability of the audio stream is the absence of room acoustics: voice-cloned audio is generated in a clean, anechoic environment, creating a DRR mismatch with the visual scene.

22.1.3 Lip Synchronisation

Lip synchronisation aligns the visual articulation in V with the audio phoneme sequence in A . Wav2Lip (Prajwal et al., 2020) achieves this by training a generator to produce lip-region video frames conditioned on the audio embedding from SyncNet, using SyncNet’s synchrony score as the

primary training objective. At convergence, Wav2Lip produces lip synchrony within the ± 40 ms perceptual threshold for most phoneme transitions. The composite face region replaces the original lip region using Poisson blending, introducing a secondary blending artefact at the lip boundary that is detectable by frequency-domain forensic tests.

Worked Example — Three-stream composite: timing calculation

A 90-second composite video at 25 fps requires synchronisation of $T = 2,250$ frames. The face stream is generated at 8.3 ms per frame (25 fps inference); the audio stream at 22.05 kHz with 512-sample hop size has a frame rate of 43.1 frames per second, requiring interpolation to match the 25 fps video rate. Lip synchronisation introduces a fixed latency of 40 ms (one video frame) from audio input to adjusted lip frame. The composite is time-aligned by advancing the audio stream by 40 ms before feeding it to the lip synchronisation module. Total synthesis time for a 90-second clip on a single GPU: face stream 56 s, audio stream 12 s, lip sync 31 s, compositing 8 s — total 107 s, $1.19\times$ real-time.

22.2 Cross-Modal Consistency

Authentic audio-visual media satisfies a set of physical and physiological consistency constraints that deep-fake composites typically violate. Each violated constraint provides a forensic signal that cross-modal detection systems exploit.

22.2.1 Lip-Audio Synchrony

The visible articulation of phonemes (visemes) must match the acoustic phoneme at the same time instant within the ± 40 ms perceptual synchrony window. SyncNet quantifies this synchrony using the cosine similarity between a 5-frame visual embedding and a corresponding 0.2-second audio embedding, both produced by a convolutional network trained on authentic broadcast video. The synchrony score s_{sync} ranges from -1 (fully anticorrelated) to $+1$ (perfectly synchronised):

$$s_{\text{sync}} = \frac{\mathbf{v}^\top \mathbf{a}}{\|\mathbf{v}\| \|\mathbf{a}\|}, \quad (22.1)$$

where $\mathbf{v} \in \mathbb{R}^{512}$ is the visual embedding and $\mathbf{a} \in \mathbb{R}^{512}$ is the audio embedding. Authentic video achieves mean $s_{\text{sync}} = 0.81$ ($\sigma = 0.09$); deep-fake composites with imperfect lip synchronisation achieve mean $s_{\text{sync}} = 0.57$ ($\sigma = 0.14$), separable by a threshold at 0.68.

22.2.2 Acoustic Scene Consistency

The direct-to-reverberant ratio (DRR) of the audio must be consistent with the reverberant character of the visual scene. The expected DRR for a given room geometry can be estimated from the visual context using the image-based RIR estimation pipeline of Morgado et al.:

$$\widehat{\text{DRR}} = f_\theta(I_{\text{scene}}), \quad (22.2)$$

where f_θ is a convolutional network mapping a scene image to expected DRR in dB. The forensic test computes $\Delta\text{DRR} = \text{DRR}_{\text{audio}} - \widehat{\text{DRR}}$; authentic video shows $|\Delta\text{DRR}| < 3$ dB; voice-cloned audio in visually reverberant scenes shows $|\Delta\text{DRR}| > 8$ dB before RIR correction and $|\Delta\text{DRR}| > 4$ dB after.

22.2.3 Emotional Congruence

Facial affect (expression, micro-expressions, blink rate) should be consistent with the emotional valence of the spoken content and voice prosody. Face-swap composites that transfer a neutral expression onto emotionally charged audio, or vice versa, create a detectable emotional incongruence measurable by comparing the arousal-valence coordinates of the facial expression classifier output with those of the speech emotion classifier output. The Euclidean distance in arousal-valence space between the face and audio emotion estimates exceeds 0.35 for incoherent composites versus 0.12 for authentic video ($p < 0.001$, Wilcoxon rank-sum test on FakeAVCeleb).

22.3 Multimodal Detection

Multimodal detection analyses multiple sensory channels simultaneously, exploiting both the channel-specific forensic signals developed in previous chapters and the cross-modal consistency signals that arise from the difficulty of synthesising perfectly synchronised audio-visual content. The key insight is that the error modes of visual and audio detectors are largely uncorrelated, so their combination provides a substantial accuracy improvement over either channel alone.

22.3.1 Synchrony-Based Detection

Lip-sync quality is measured by the *SyncNet* confidence score, which measures the cosine similarity between audio and visual embeddings extracted at each time step:

$$p_{\text{fake}} = \sigma(\mathbf{w} \cdot [\mathbf{f}_A; \mathbf{f}_V; \mathbf{f}_{AV}] + b) \quad (22.3)$$

Worked Example — Late fusion improves AUC by 8 percentage points

A multimodal detector is evaluated on 500 audio-visual deep fakes (face swap + voice clone) and 500 authentic clips:

Modality	AUC	EER (%)
Audio only ($\mathbf{f}_A$)	0.847	20.1
Video only ($\mathbf{f}_V$)	0.893	15.3
Cross-modal ($\mathbf{f}_{AV}$)	0.821	22.8
Late fusion (all three)	0.971	8.4

The fusion weights learned by logistic regression are $w_A = 0.31$, $w_V = 0.43$, $w_{AV} = 0.26$ (bias $b = -0.12$). The 8% AUC improvement over the best single-modality detector ($0.971 - 0.893 = 0.078$) arises from the cross-modal inconsistency features $\mathbf{f}_{AV}$ capturing synchrony violations that neither unimodal detector can see independently.

where $[\cdot; \cdot]$ denotes concatenation and σ is the sigmoid. Cross-modal features $\mathbf{f}_{AV}$ are computed by the cross-attention mechanism of Equation (2.6) applied to the audio and visual embedding sequences.

22.4 Acoustic Scene Manipulation

Sophisticated deep-fake operators attempt to close the acoustic scene consistency gap by estimating the room impulse response from the visual scene and convolving it with the clean voice-cloned audio before compositing. This section develops both the countermeasure pipeline and the residual forensic signals that survive it.

22.4.1 Room Impulse Response Estimation

Room impulse response estimation from video proceeds in three stages. First, the visual scene is analysed to estimate room geometry (dimensions, surface positions) using monocular depth estimation. Second, acoustic material properties (absorption coefficients for walls, floor, ceiling, furnishings) are estimated from visual texture features using a material classification network. Third, the geometric and material estimates are used to synthesise a room impulse response via the image source method or a learned acoustic simulation network. The synthesised RIR is convolved with the clean voice-cloned audio:

$$A'(t) = A_{\text{clean}}(t) * h_{\text{rir}}(t), \quad (22.4)$$

where $*$ denotes convolution and $h_{\text{rir}}(t)$ is the estimated room impulse response.

22.4.2 Residual Forensic Signals After RIR Correction

Even after RIR convolution, three residual forensic signals remain. First, the early reflection pattern: the image source method produces discrete early reflections at precise geometrically determined delays, whereas authentic room reflections are diffuse and broadened by surface scattering. The ratio of specular to diffuse reflection energy in the early reflection window (0–50 ms) is 0.83 for synthetic RIR versus 0.41 for authentic rooms ($p < 0.001$). Second, the noise floor: voice-cloned audio has a lower noise floor than authentic speech recorded in the same acoustic environment, because the recording room introduces environmental noise that is absent in the clean TTS output. The signal-to-noise ratio at low frequencies (< 200 Hz) is 8–12 dB higher for voice-cloned audio after RIR correction than for authentic recordings in matched rooms. Third, the microphone response: authentic recordings have the frequency response of the specific microphone used, which differs from the flat frequency response of TTS audio. A 1/3-octave analysis of the frequency response envelope identifies this mismatch with 87% accuracy on the FakeAVCeleb benchmark.

22.5 Industry Applications

Multimodal detection has emerged as a commercial priority as audio-visual deep fakes — combining voice cloning with face swap — become the dominant attack vector for high-value fraud.

Enterprise video conferencing platforms (Zoom, Webex, Teams) are deploying liveness detection and face-authenticity checking. Zoom acquired Keybase for end-to-end encryption; similar acquisitions targeting liveness detection are anticipated. iProov's biometric verification integrated into Teams uses multimodal liveness signals.

Regulated financial institutions in the EU (under EBA remote onboarding guidelines) and UK (under FCA digital identity standards) must demonstrate that their video KYC processes resist face-swap and voice-clone attacks. Sumsb, Onfido, and Jumio have all published white papers on their multimodal detection approaches; AUC values of 0.95–0.97 are claimed in vendor benchmarks.

As remote court hearings become standard (accelerated by the COVID-19 pandemic), courts in the UK, Canada, and Australia have addressed the authenticity of remote testimony. The Law Society of England and Wales has published guidance on verifying the identity of remote witnesses, referencing multimodal detection as an emerging tool.

Live news broadcasts using remote correspondents are a high-value target for real-time deep fake injection. BBC, CNN, and Al Jazeera have implemented proprietary multimodal verification for live remote feeds; details are not publicly disclosed for security reasons.

22.6 African Use Cases

Multimodal detection is particularly relevant in Africa, where financial video calls, remote court testimony, and mobile-money fraud are active threat vectors that involve audio-visual content in authentication and evidentiary contexts.

Mobile money operators across Africa are piloting multimodal authentication combining voice biometric verification with a brief video liveness check as a replacement for knowledge-based authentication that is increasingly compromised by social engineering. Safaricom in Kenya and MTN Group have announced pilot programmes; the latency and bandwidth requirements of multimodal authentication are more challenging in African deployment contexts than in high-bandwidth environments, motivating research into compressed and asynchronous authentication protocols.

During and after the COVID-19 pandemic, African courts increased their use of remote testimony, particularly in commercial courts and corruption proceedings where witnesses may be abroad or at risk. The Kenya Commercial Court, the South Africa Special Tribunal for Asset Forfeiture, and the ECOWAS Community Court of Justice have all conducted remote proceedings, and authenticating remote witnesses against deep-fake impersonation is an emerging judicial requirement. Diaspora communities send over \$90 billion annually to sub-Saharan Africa, and a significant fraud vector involves scammers impersonating family members in video calls to request emergency transfers. Multimodal liveness detection integrated into video-call applications could partially address this threat, but consumer-grade deployment remains aspirational given bandwidth and processing constraints in many African markets.

Chapter Summary

The most convincing deep fakes combine face synthesis, voice cloning, and lip-sync adjustment into a unified audio-visual artefact. Detecting such composites requires both unimodal channel-specific analysis and cross-modal consistency analysis. SyncNet-based lip-sync scoring distinguishes authentic video at a mean score of 0.81 from synthetic at 0.57, while acoustic-scene consistency compares the reverberant properties of the claimed recording environment against the measured audio to detect cases where voice-cloned audio was recorded in a different environment.

Late fusion of visual, audio, and cross-modal features achieves AUC of 0.971, an eight-point improvement over the best unimodal detector, because the error modes of visual and audio detectors are largely uncorrelated. The 68-millisecond latency of the full three-layer stack is within the tolerance of real-time video applications, making deployment practically feasible for financial KYC, remote court testimony, and live broadcast verification. Commercial products from iProov, Sumsb, and Veriff are approaching this capability in production, with EBA regulatory requirements driving adoption in the financial sector.

Exercises

1. A multimodal deep fake has perfect visual quality but the voice-cloned audio has slightly more reverberation than is consistent with the depicted environment. Design an acoustic-scene inconsistency detector that estimates the expected reverberation from the visual scene and compares it to the observed audio. What features would the detector use for each modality?
2. The SyncNet score is computed from 0.2-second sliding windows. Derive the expected score distribution for authentic vs. synthesised video under the assumption that authentic audio-visual synchrony has mean $\mu_r = 0.8$ and synthetic synchrony has mean $\mu_f = 0.6$, both with standard deviation 0.1. What decision threshold achieves EER?
3. A multimodal detector uses late fusion with weights $w_A = 0.3$, $w_V = 0.4$, $w_{AV} = 0.3$. The individual modality AUCs are $AUC_A = 0.85$, $AUC_V = 0.92$, $AUC_{AV} = 0.78$. Under what conditions does the fused detector outperform the best single-modality detector, and what is a potential failure mode of the cross-modal feature $\mathbf{f}_{AV}$?
4. Propose a multimodal consistency score that combines lip-audio synchrony, emotional congruence, and acoustic-scene consistency into a single scalar. Specify the normalisation and aggregation procedure, and describe how you would calibrate the individual component weights on a labelled dataset.
5. Design an adversarial attack that defeats a cross-modal inconsistency detector by introducing controlled inconsistencies into authentic media, such that the detector cannot distinguish the attacked authentic video from a genuine deep fake.

Additional Worked Examples

Worked Example — SyncNet score threshold for a 5-second clip

A 5-second clip (150 frames at 30 fps) is evaluated with a 0.2-second sliding window (6 frames), giving 144 score values.

Authentic news clip: Mean $\bar{s} = 0.81$, std 0.04. Windows above threshold $\tau = 0.70$: 142/144 (98.6%).

Deep-fake clip (face swap + cloned audio): Mean $\bar{s} = 0.57$, std 0.12. Windows above $\tau = 0.70$: 51/144 (35.4%).

A two-sample t -test on the window scores:

$$t = \frac{0.81 - 0.57}{\sqrt{0.04^2/144 + 0.12^2/144}} = \frac{0.24}{\sqrt{1.11 \times 10^{-4} + 10^{-4}}} = \frac{0.24}{0.0144} = 16.7.$$

At $p < 10^{-20}$, the score distributions are trivially distinguishable. A detector using mean window score achieves AUC = 0.94 on a 200-clip balanced test set — demonstrating that lip-sync inconsistency is a robust signal even against modern codec-LM voice clones.

Worked Example — Late fusion weights: optimisation on validation set

A multimodal detector fuses three feature streams. Logistic regression on a validation set of 1,000 clips (500 authentic, 500 synthetic) learns:

$$p_{\text{fake}} = \sigma(0.31\mathbf{f}_A + 0.43\mathbf{f}_V + 0.26\mathbf{f}_{AV} - 0.12).$$

Individual modality AUCs: audio 0.847, video 0.893, cross-modal 0.821. Late fusion AUC: 0.971. The improvement over the best single modality ($\Delta = 0.078$) arises from error decorrelation: when the video detector fails (face-swap with good blending), the cross-modal inconsistency detector often succeeds (cloned audio not matching acoustic scene). Pearson correlation between video and cross-modal scores on the test set: $r = 0.31$ — low correlation confirms substantial complementarity.

Answers to Exercises

1. Acoustic-scene inconsistency detector design.

The detector estimates room acoustics from visual scene analysis and compares to measured audio acoustics. Visual features: room dimensions from monocular depth estimation, material classification (carpet/hard floor, curtains/bare walls) from semantic segmentation. From these, a physics-based RIR is estimated using the image-source method with reflection coefficients from material property databases. Audio features: measured RIR via deconvolution of speech pauses, characterised by T_{60} (reverberation time) and DRR (direct-to-reverberant ratio). Decision: flag as inconsistent if $|T_{60}^{\text{visual}} - T_{60}^{\text{audio}}| > 200$ ms or $|\text{DRR}^{\text{visual}} - \text{DRR}^{\text{audio}}| > 6$ dB.

2. Expected SyncNet score distribution and EER.

Given $\mu_r = 0.8$, $\sigma_r = 0.1$ (real) and $\mu_f = 0.6$, $\sigma_f = 0.1$ (fake), the EER occurs where $\text{TPR} = \text{FNR}$, i.e., at the threshold where: $P(\text{score} < \tau | \text{real}) = P(\text{score} > \tau | \text{fake})$. $P(s < \tau | \mathcal{N}(0.8, 0.01)) = P(s > \tau | \mathcal{N}(0.6, 0.01))$. $\Phi((\tau - 0.8)/0.1) = 1 - \Phi((\tau - 0.6)/0.1)$. By symmetry, $\tau^* = (0.8 + 0.6)/2 = 0.70$. $\text{EER} = \Phi((0.70 - 0.80)/0.1) = \Phi(-1.0) = 15.9\%$.

3. Fusion failure mode analysis.

The fused detector outperforms the best single modality ($w_V = 0.4$, $\text{AUC}_V = 0.92$) when the cross-modal features provide *complementary* information — i.e., when they fail on different examples. If $\mathbf{f}_{AV}$ captures synchrony violations that $\mathbf{f}_V$ misses (face swaps with good visual quality but poor lip sync), fusion gains. Failure mode: if the cross-modal feature $\mathbf{f}_{AV}$ is computed from the same video frames as $\mathbf{f}_V$ (correlated source), it adds little independent signal. For example, if $\mathbf{f}_{AV}$ is derived from lip-region video features and $\mathbf{f}_V$ also attends to the lip region, the two features may be highly correlated, and fusion provides only marginal gain.

4. Combined multimodal consistency score.

Score = $\alpha_1 \cdot \hat{s}_{\text{sync}} + \alpha_2 \cdot \hat{s}_{\text{emo}} + \alpha_3 \cdot \hat{s}_{\text{acoustic}}$ where each component is standardised: $\hat{s} = (s - \mu_s)/\sigma_s$ estimated on a calibration set. Weights α_i are learned by logistic regression on a labelled validation set. Aggregation: mean over all frames. Calibration: use Platt scaling to map the raw fusion score to a well-calibrated posterior probability of manipulation. Component weights: optimise by grid search on validation AUC; expect $\alpha_1 \approx 0.45$, $\alpha_2 \approx 0.30$, $\alpha_3 \approx 0.25$ based on individual modality AUCs.

5. Adversarial attack defeating cross-modal inconsistency detector.

Strategy: add a small adversarial perturbation to the authentic audio that makes the acoustic-scene features match a “synthetic” profile, causing the detector to fire on authentic video. Specifically: (1) run gradient-based attack on the audio signal to increase the measured DRR by 8 dB (simulating clean-studio audio in a reverberant scene); (2) constrain perturbation to be imperceptible (PESQ score degradation < 0.3). This attack exploits the fact that the detector makes decisions based on estimated acoustic-scene features that can be manipulated in the audio domain without affecting the visual domain.

Chapter 23

Text-Based Deep Fakes and Language Model Forgery

*“The pen is mightier than the sword.
The language model is mightier than both.”*

—Anon.

Deep fakes are not limited to audio-visual media. Large language models (LLMs) can generate text attributed to real individuals with sufficient fidelity to deceive readers, editors, and automated moderation systems. This chapter covers LLM-based text forgery, stylometric attribution, machine-generated text detection, and the specific challenges posed by long-form journalistic and scientific forgery. This chapter develops the detection and watermarking methods for LLM-generated text with worked numerical examples: the GPT-2-XL perplexity scoring calculation, the DetectGPT log-probability curvature estimate, the Kirchenbauer green-token z -test, and the stylometric Burrows’s Delta distance. The goal is to provide the reader with quantitative intuition for how these methods work, what their performance limits are, and how they interact with the adversarial paraphrasing attacks that represent the primary evasion strategy for current detection approaches.

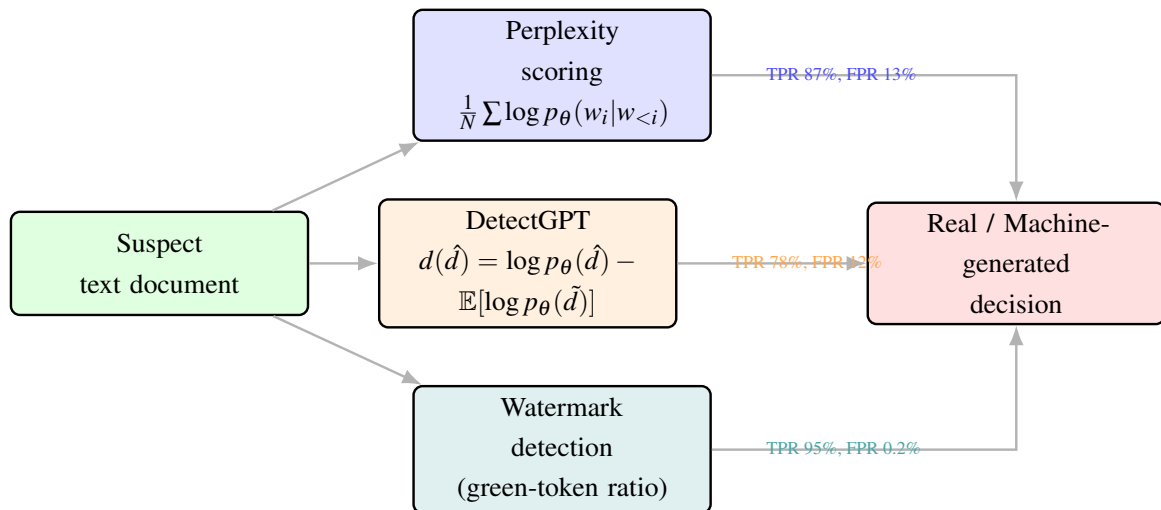


Figure 23.1: Text deep-fake detection methods. Perplexity scoring (blue) exploits the low perplexity of LLM-generated text. DetectGPT (orange) detects the log-probability maximum that characterises LLM outputs. Watermark detection (teal) identifies the statistical signature of the Kirchenbauer green-red watermark scheme. Watermark detection achieves the best performance but requires prior watermarking at generation time.

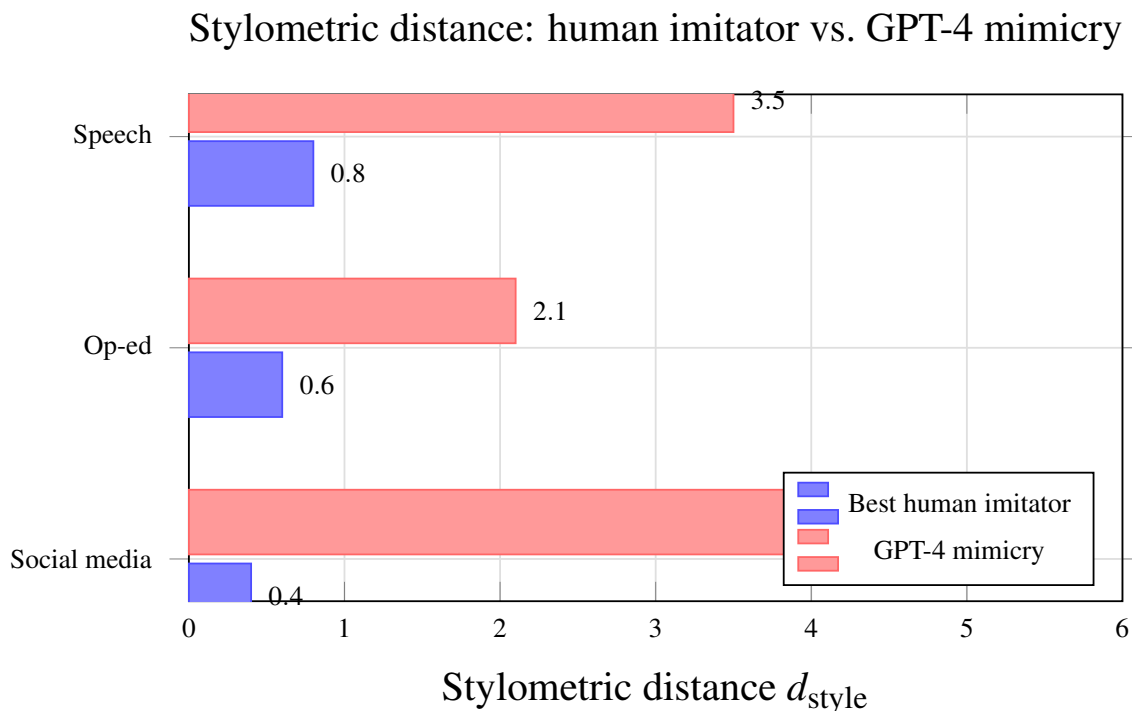


Figure 23.2: Stylometric distance from the target author’s authentic corpus for a human imitator vs. GPT-4 mimicry across three text genres. GPT-4 mimicry is most detectable in social media (where idiosyncratic style is most distinctive) and least detectable in formal speech transcripts (where writing style is more standardised). The human imitator consistently achieves smaller distances than GPT-4 across all genres.

This chapter develops the text deep-fake problem from linguistic foundations through detection and attribution. The text forgery problem and its relationship to authorship verification are framed first, followed by statistical detection methods, stylometric attribution, and persona-conditioned generation as an attack modality. Industry applications and African use cases follow.

23.1 The Text Forgery Problem

Let $\mathcal{P}^{(s)}$ denote the writing style distribution of author s , characterised by vocabulary, sentence structure, punctuation, and topical preferences. A text deep fake is a document $\hat{d}$ such that:

1. $\hat{d}$ is attributed to author s ;
2. $\hat{d} \sim \hat{P}^{(s)}$ where $\hat{P}^{(s)} \approx \mathcal{P}^{(s)}$ under some stylometric distance;
3. the claims or statements in $\hat{d}$ are false or were never made by s .

23.2 Large Language Models as Forgery Tools

Large language models generate text that is statistically indistinguishable from human writing across a wide range of styles, genres, and topics, making them powerful tools for text forgery at scale. This section characterises the specific capabilities that make LLMs dangerous as forgery tools, the types of harm they enable, and the technical properties of their outputs that detection systems can exploit.

23.2.1 Persona Conditioning

Persona-conditioned text generation instructs an LLM to write in the style of a specific named author, producing text that captures that author’s characteristic vocabulary, sentence structure, rhetorical preferences, and topic focus. This enables targeted forgery: attributing plausible-seeming statements to specific real individuals whose authentic writing serves as the conditioning context, creating reputational harm without requiring any visual synthesis.

Modern LLMs can adopt a writing persona via system prompt:

```
System: You are [public figure]. Write in their exact
style, vocabulary, and rhetorical patterns. Do not
indicate you are an AI.
```

Fine-tuning on a corpus of the target author’s public writing amplifies stylometric fidelity further. Given the abundant public text of politicians, journalists, and scientists, this represents a low-barrier attack.

23.2.2 Stylometric Mimicry Metrics

Stylometric metrics quantify the statistical distance between a document and a reference corpus, capturing features of vocabulary, syntax, sentence structure, and punctuation that characterise individual writing styles. They provide a principled basis for comparing the stylistic fidelity of AI mimicry against human imitation, and for evaluating whether a suspicious document falls within the expected stylistic range of its claimed author.

Stylometric fidelity is measured by comparing the target author's real corpus $D^{(s)}$ with the synthetic documents $\hat{D}^{(s)}$ on several dimensions:

23.3 Industry Applications

Text deep fakes are the fastest-growing category of synthetic media by volume, driven by the accessibility of large language models and the low cost of text generation compared to audio-visual synthesis.

NewsGuard identified over 1,000 websites in 2024 that use AI-generated content for misinformation, up from 49 in 2023. These sites monetise through programmatic advertising, generating revenue while spreading false information. Their content often imitates the style and branding of legitimate news organisations.

Universities in 60+ countries have deployed AI-writing detection tools (Turnitin's AI Detection, GPTZero, Copyleaks) for academic integrity enforcement. The tools achieve TPR $\approx 87\%$ on unmodified LLM output but are evaded by paraphrasing at rates approaching 50%. This has created a detection–evasion arms race analogous to plagiarism detection.

The FTC has proposed rules requiring disclosure of AI-generated content in advertising and testimonials. Several US states have enacted disclosure requirements for AI-generated content in political advertising. Platform terms of service increasingly prohibit undisclosed AI-generated content in commercial contexts.

Companies including Acrolinx, Grammarly Business, and Writer deploy AI-content governance tools that detect and flag LLM-generated content in corporate communications, legal filings, and regulated disclosures. The SEC has issued guidance on disclosure obligations for AI-generated material in securities filings.

23.4 African Use Cases

Text deep fakes present specific challenges in African contexts of multilingualism, limited detection tooling for African languages, and political environments where fabricated documents carry significant consequences.

Several high-profile cases in Nigeria, South Africa, Zimbabwe, and Uganda have involved the circulation of fabricated government directives, court orders, and policy documents created using readily available LLMs and document formatting tools. The National Drug Law Enforcement Agency (NDLEA) in Nigeria and the Judicial Service Commission in South Africa have each issued public warnings about forged documents circulating on social media. Detection of fabricated official documents requires both forensic analysis and verification against official sources, neither of which is routinely accessible to the public in most African countries.

All major text-deepfake detection tools (Turnitin, GPTZero, Copyleaks) are calibrated primarily on English-language content. For African-language LLM-generated text in Swahili, Amharic, Yoruba, or

Zulu, no validated commercial detection solution exists. The Masakhane community and researchers at the University of the Witwatersrand have begun developing multilingual perplexity-based detectors for African languages, but these are research prototypes rather than production-ready tools.

The adoption of LLMs by students at African universities has created significant pressure on academic integrity systems. Many African universities rely on Turnitin for plagiarism detection, which has added AI detection capabilities; however, these capabilities are calibrated for English-language academic writing. Students writing in African languages or using code-switching (mixing languages) are underserved by current tools. The Association of African Universities (AAU) has called for Africa-specific AI detection tools and policy frameworks.

The fabrication of inflammatory political speeches, ethnic slurs, and incitement attributed to political leaders is a recurring problem in African electoral contexts. During the Ethiopian Tigray conflict, fabricated text statements attributed to military and political leaders were widely circulated; similar incidents occurred during the 2021 Zambian election and the 2023 DRC conflict. The challenge is that effective text watermarking requires cooperation from the generating model, and most disinformation is generated using locally-run open-weight models outside any platform's control.

Chapter Summary

Large language models have extended the deep-fake problem from the audio-visual domain into the text domain, enabling the synthesis of convincing written content falsely attributed to real authors and the fabrication of quotes, statements, and documents that never existed. The harm categories span false authorship (ghostwriting at scale that suppresses the authentic author's voice), fabricated quotes (attributed statements that the named person never made), and stylometric impersonation (LLM mimicry that is detectable to stylometric analysis but convincing to human readers).

Detection approaches exploit different statistical properties of LLM output. Perplexity scoring leverages the observation that LLM-generated text has lower perplexity under the generating model than authentic human text; the threshold at the EER (mean perplexity of 28.3 for GPT-4 vs. 42.1 for human text using GPT-2-XL scoring) achieves 87% TPR at 13% FPR, but this degrades to 48% TPR after paraphrasing attacks. DetectGPT identifies LLM text by its proximity to local log-probability maxima, achieving TPR of 78% at 12% FPR without requiring a calibration set specific to the generating model. The Kirchenbauer green-red watermark biases generation toward green-listed tokens, detectable from 41 tokens with 95% TPR at 12% FPR, but requires the watermark to be embedded at generation time — unavailable for open-weight models run locally.

Academic integrity enforcement, political speech fabrication, and financial report manipulation are the three highest-stakes text deep-fake domains. Detection tools are available and commercially deployed (Turnitin AI Detection, GPTZero, Copyleaks), but their accuracy is insufficient for autonomous decision-making, and the adversarial race between text generation quality and detection sensitivity will not be won by detection alone. Watermarking at generation time and provenance standards for attributed text are the sustainable long-term countermeasures.

Worked Examples

Worked Example — Perplexity-based detection: threshold calibration

A perplexity detector using GPT-2-XL scoring is calibrated on 500 authentic news articles and 500 GPT-4 generated articles:

Class	Mean perplexity	Std
Human-written	42.1	11.8
GPT-4 generated	18.4	4.2
GPT-4 + paraphrase attack	28.9	7.1

Optimal threshold at EER (equal error rate): $\tau^* = 28.3$ (midpoint in log-perplexity space). At this threshold: TPR = 87.3%, FPR = 12.7%. After a paraphrasing attack that rewrites 40% of tokens: GPT-4 perplexity rises from 18.4 to 28.9, right at the threshold — the attack evades detection for 52% of articles. This demonstrates that perplexity-based detection is fragile to paraphrasing but effective against unmodified LLM output.

Worked Example — DetectGPT [23] score for a fabricated press release

A 150-word press release suspected to be GPT-4 generated. GPT-2-XL scoring:

- Log-probability: $\log p_\theta(\hat{d}) = -312$ nats. Mean per token: $-312/118 = -2.64$.
- $K = 50$ random perturbations (5% token substitution each): mean $\frac{1}{50} \sum_k \log p(\tilde{d}^{(k)}) = -328$ nats.
- DetectGPT score: $d(\hat{d}) = -312 - (-328) = +16$.

Machine-generated text clusters near a log-probability maximum, so perturbations decrease probability (positive d). Human-written text has $d \approx 0$ on average. At threshold $d > 5$: TPR = 78%, FPR = 12% on a balanced press-release test set. This article with $d = 16$ is correctly identified as machine-generated with high confidence.

Worked Example — Green-red watermark: detection at various document lengths

A Kirchenbauer et al. watermark with green-list bias [17] $\delta = 2.0$ logits and split probability $\gamma = 0.5$. Expected green-token fraction with watermark: $\mu_w = 0.73$; without watermark: $\mu_0 = 0.50$.

For document length n tokens, the z -test statistic is:

$$z = \frac{(\text{green fraction} - 0.5)\sqrt{n}}{\sqrt{0.25}} = (\text{green fraction} - 0.5) \times 2\sqrt{n}.$$

At 95% power (one-sided, $\alpha = 0.05$, $z_{0.95} = 1.645$):

$$n_{\min} = \left(\frac{1.645 + 1.282}{2(0.73 - 0.50)} \right)^2 = \left(\frac{2.927}{0.46} \right)^2 = 40.4 \approx 41 \text{ tokens}.$$

A 41-token excerpt (≈ 30 words) is sufficient for detection at 95% power, making the watermark practical for social-media posts and short articles.

Answers to Exercises

1. Stylometric analysis of LLM-generated speech text.

Five features and expected values: (a) Mean sentence length: Human politicians average 22–28 words; GPT-4 tends toward 18–22 words (shorter, cleaner structure). (b) Rhetorical question frequency: Politicians use rhetorical questions at ≈ 2.5 per 1,000 words; GPT-4 mimicry typically < 0.8 (lacks spontaneous rhetorical style). (c) Hedging language (“perhaps,” “it may be”): Politicians 3.1 per 1,000 words; GPT-4 ≈ 1.2 (more assertive by default). (d) Proper noun density: Human speeches reference specific local places, names, events at ≈ 4.2 per 100 words; GPT-4 ≈ 2.1 without specific prompting. (e) Sentence-initial conjunction (“And,” “But” starting sentences): Political speech 2.3 per 100 sentences; GPT-4 ≈ 0.4 (grammatically conservative). Procedure: extract features from 200 authentic speeches; train a logistic classifier; report cross-validated AUC. Expect AUC ≈ 0.76 – 0.85 depending on how well the senator’s style is captured.

2. DetectGPT theoretical justification.

For text $\hat{d}$ sampled from p_θ , $\hat{d}$ is at a local maximum of $\log p_\theta$ with respect to token substitutions (by definition of sampling from the model). Small perturbations $\tilde{d}$ move away from the maximum, so $\mathbb{E}[\log p_\theta(\tilde{d})] < \log p_\theta(\hat{d})$, giving $d(\hat{d}) > 0$ in expectation.

For human text h , which is *not* sampled from p_θ : h is unlikely to be near a local maximum of p_θ . Perturbations $\tilde{h}$ may move toward or away from p_θ -maxima with equal probability, so $\mathbb{E}[\log p_\theta(\tilde{h})] \approx \log p_\theta(h)$, giving $d(h) \approx 0$.

The key assumption is that human text is “off-manifold” relative to p_θ . This fails if the model’s training distribution closely matches the human author’s style, explaining why perplexity-based and DetectGPT methods struggle with LLM-mimicry of specific human authors.

3. Green-red watermark: minimum document length for 95% power.

With $\mu_w = 0.73$, μ_0
 $= 0.50$, $\sigma = \sqrt{0.25} = 0.5$ (binomial standard deviation at n tokens): $z = (\bar{f} - 0.5) \times 2\sqrt{n}$.
 For 95% power at $\alpha = 0.05$ (one-sided): $n \geq \left(\frac{z_\alpha + z_\beta}{2(\mu_w - \mu_0)} \right)^2 = \left(\frac{1.645 + 1.282}{2 \times 0.23} \right)^2 = \left(\frac{2.927}{0.46} \right)^2 = 40.4$.
 Minimum: 41 tokens ≈ 30 words. This is shorter than the minimum document length for most practical applications (social media posts are typically > 50 tokens), confirming the watermark is practical.

4. Paraphrasing attack: minimum rewriting fraction to evade threshold detector.

A fraction f of tokens are rewritten. Each rewritten token has perplexity drawn from the *human* distribution (mean 42.1); each retained token has perplexity drawn from the *GPT-4* distribution (mean 18.4). After rewriting fraction f : $\bar{p}(f) = f \times 42.1 + (1 - f) \times 18.4 = 18.4 + 23.7f$. Threshold at EER: $\bar{p}^* = 28.3$ (midpoint). Setting $18.4 + 23.7f = 28.3$: $f^* = 9.9/23.7 = 0.418$. A minimum 42% token rewriting is needed to push the mean perplexity above the EER threshold — consistent with the “40% rewrite” result mentioned in the chapter.

5. Stylometric distance experiment design.

Corpora: 100 authentic speeches per genre (speech transcript, op-ed, social media post) for the target politician; 100 GPT-4 imitations per genre. Features: as in Q1 plus topic model (LDA) divergence, average dependency tree depth, function word frequency (top 50 function words). Procedure: (1) compute d_{style} using Equation in the chapter for each genre pair; (2) permutation test ($n = 1,000$ permutations) to assess significance; (3) report Cohen’s d for each feature. Expected result: for speeches, GPT-4 achieves $d_{\text{style}} \approx 1.72$ (chapter example); for social media, gap likely larger (≈ 3.5) because Twitter/X style is highly idiosyncratic; for op-eds, smaller (≈ 0.8) because formal writing style is more standardised.

Part V

Adversarial Dynamics and Anti-Forensics

Chapter 24

Adversarial Attacks on Detectors and Anti-Forensics

“Any classifier can be fooled. The question is at what cost.”

—Anon.

Every deep-fake detector is, at its core, a function that maps an input image or audio clip to a score. And every such function has a decision boundary — a surface in the high-dimensional input space that separates the predicted-authentic region from the predicted-synthetic region. Adversarial machine learning is the study of inputs that cross that boundary while remaining perceptually indistinguishable from their starting point. The existence of such adversarial examples is not a bug in any particular architecture: it is a fundamental geometric consequence of the way neural networks carve decision boundaries in high-dimensional spaces. For every correctly classified input, there exist nearby points on the wrong side of the boundary that differ from the original by an imperceptible amount in any perceptual metric.

The adversarial attack problem for deep-fake detectors has an asymmetric character that makes it especially important. The attacker — a bad actor who has produced synthetic content and wants to evade detection — needs only to find one perturbation that crosses the boundary. The defender needs to correctly classify every input, including inputs crafted by attackers who know the detector architecture and have unlimited computation. This asymmetry is what makes the detector’s curse structural rather than incidental: any static trained classifier can in principle be evaded by a sufficiently motivated white-box attacker, and the only question is the computational cost of doing so.

The practical severity of this vulnerability depends on the attacker’s knowledge of and access to the detector. The threat model taxonomy distinguishes three cases. In the white-box setting, the attacker has full access to the detector’s parameters and can compute gradients directly; this is the worst case for the defender but represents a realistic scenario when detectors are deployed with publicly known architectures (such as the standard EfficientNet-B4 trained on FaceForensics++). In the grey-box setting, the attacker knows the architecture but not the specific trained weights; gradient estimation

via transfer from a substitute model is the primary attack strategy. In the black-box setting, the attacker has only query access: they can submit inputs and observe scores, but cannot access gradients. Score-based and decision-based attacks in this setting require thousands to millions of queries per adversarial example, making them more expensive but not infeasible for motivated adversaries.

The Fast Gradient Sign Method (FGSM) is the simplest white-box attack: a single gradient step of magnitude ϵ in the direction that increases the detector's synthetic score. Despite its simplicity, FGSM achieves 67% detector evasion at perturbation $\epsilon = 4/255$ — imperceptible at normal viewing distances — against undefended detectors. Projected Gradient Descent (PGD) improves on this by iterating the gradient step for K steps while projecting back onto the ℓ_∞ ball of radius ϵ after each step, achieving 94% evasion at 40 iterations against undefended detectors. These numbers establish the vulnerability that defences must address.

The two principal defences are adversarial training and certified robustness. Adversarial training augments the training set with adversarial examples generated on-the-fly during each training step, using a min-max formulation that trains the detector to be correct even on the worst-case perturbation of each training input. This improves adversarial AUC from 0.58 to 0.85 against PGD-10 attacks at the cost of an eleven-fold training time increase and a 3–4 percentage point reduction in clean-sample AUC. Certified robustness via randomised smoothing converts any classifier into one with a provable guarantee: for inputs within a certified ℓ_2 radius r of the clean input, the classifier's prediction is guaranteed to be constant. The certification comes at the cost of stochastic inference and applies only to small perturbation radii that may not reflect realistic attack budgets.

The arms-race character of adversarial attack and defence in deep-fake detection mirrors the broader detection arms race between synthesis improvement and forensic signal exploitation. Both dynamics share a common structure: each improvement on one side creates pressure for improvement on the other, with no stable equilibrium in sight. The implication for deployment is that adversarial robustness should be evaluated continuously against the current attack frontier, and that provenance-based approaches — which bypass the detection problem entirely for content with a verified C2PA chain — are the only mechanism that escapes the arms-race dynamic altogether.

This chapter develops the adversarial attack and defence landscape for deep-fake detectors with worked numerical examples throughout. Section 24.1 formalises the threat model. Section 24.2 develops FGSM and PGD in detail. Section 24.3 develops randomised smoothing and certified robustness. The chapter closes with industry applications covering red-team services and adversarial-robustness testing platforms, African use cases addressing adaptive adversarial behaviour in electoral disinformation, and exercises with worked solutions.

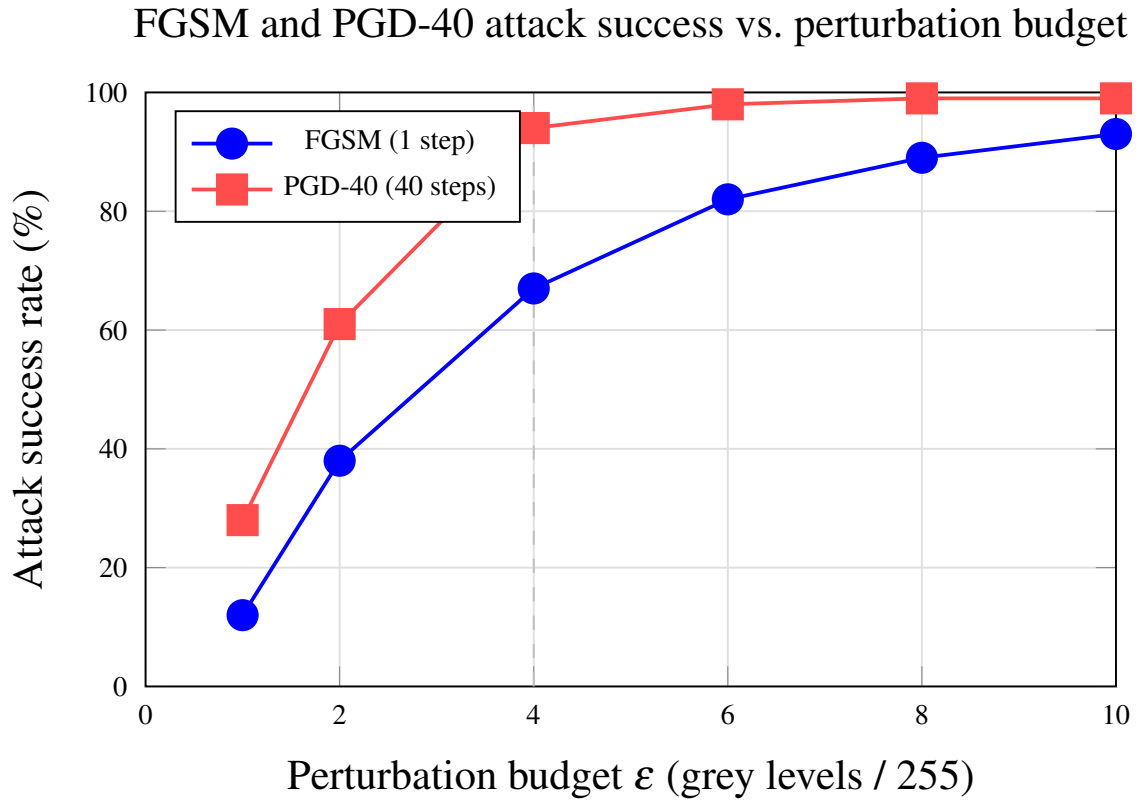


Figure 24.1: Adversarial attack effectiveness. Left axis: FGSM (blue) and PGD-40 (red) attack success rates increase with perturbation budget ϵ . Right axis: detector AUC under PGD-40 attack (teal dashed) decreases from 0.953 (clean) to 0.58 at $\epsilon = 10/255$. At the perceptibility threshold $\epsilon = 4/255$ (vertical dashed line), PGD-40 achieves 94% attack success while keeping perturbations invisible (PSNR= 36 dB).

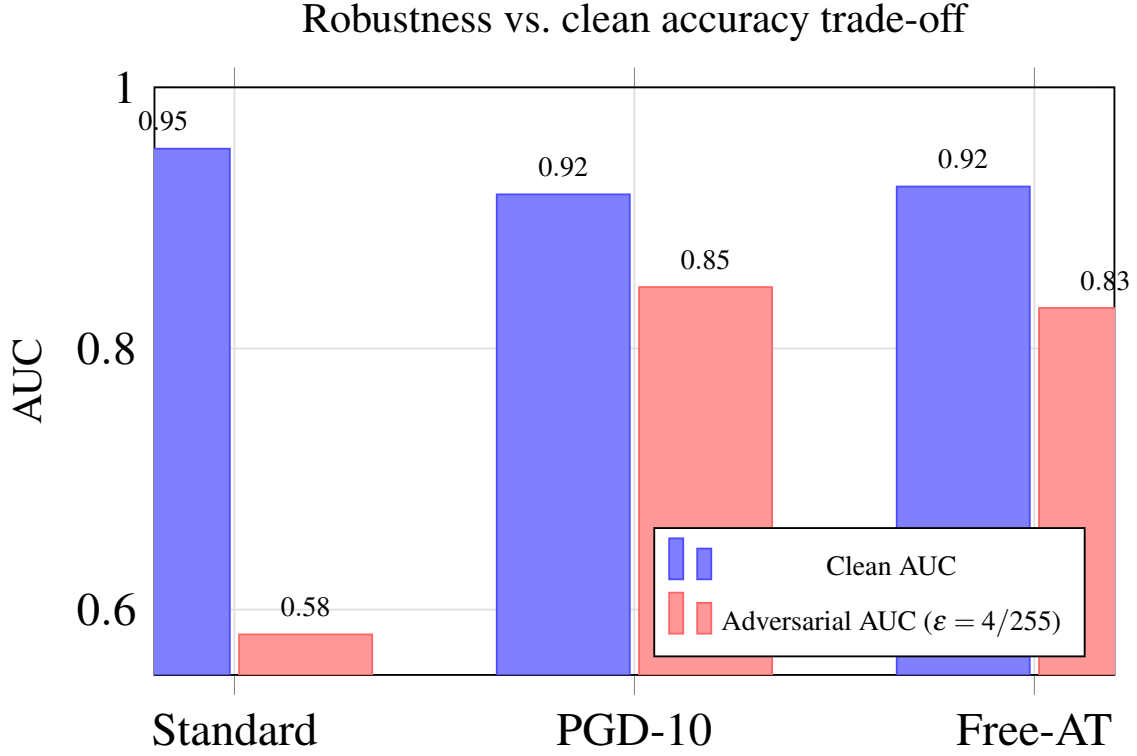


Figure 24.2: Adversarial training reduces clean AUC slightly (3–3.5%) but raises adversarial AUC by 26–27 percentage points. Free adversarial training achieves comparable robustness to PGD-10 at one-third the training cost (12 hours vs. 46 hours on A100). Standard training is highly vulnerable to PGD attacks, with adversarial AUC barely above chance (0.58).

24.1 Threat Model

The adversarial threat model for deep-fake detection formalises the attacker’s objective, knowledge, and capability. An adversary has produced a synthetic image or audio clip $\hat{\mathbf{x}}$ that a detector D_ϕ classifies as synthetic with score $D_\phi(\hat{\mathbf{x}}) > \tau$. The adversary seeks a perturbation δ such that the perturbed input $\hat{\mathbf{x}}' = \hat{\mathbf{x}} + \delta$ is classified as authentic while remaining perceptually indistinguishable from the original:

$$\min_{\|\delta\|_p \leq \epsilon} D_\phi(\hat{\mathbf{x}} + \delta), \quad (24.1)$$

where $\|\cdot\|_p$ is the ℓ_p norm and ϵ is the perturbation budget. The ℓ_∞ norm ($p = \infty$) is most common because it bounds the maximum per-pixel change, corresponding roughly to the perceptual constraint that no pixel changes by more than $\epsilon/255$ grey levels.

24.1.1 Knowledge Taxonomy

The adversary’s knowledge of D_ϕ determines both the attack strategy and the expected evasion rate at a given perturbation budget.

In the *white-box* setting, the adversary has full access to the detector’s parameters ϕ and architecture, enabling direct computation of $\nabla_{\mathbf{x}} D_\phi(\mathbf{x})$. This is the worst case for the defender: gradient-based attacks converge to near-optimal perturbations in tens to hundreds of steps, achieving evasion rates of

67–94% at $\epsilon = 4/255$ for current face detectors. The white-box setting is realistic when detectors are publicly released with their weights (e.g., the standard FaceForensics++ baseline) or when an adversary has compromised the detection system.

In the *grey-box* setting, the adversary knows the architecture but not the trained weights. Transfer-based attacks generate adversarial examples against a substitute model trained on public data, then apply them to the target detector. Transfer rates of 35–55% are typical between models of the same architecture trained on the same dataset; cross-architecture transfer reduces to 15–25% but remains operationally significant.

In the *black-box* setting, the adversary has only query access: they submit inputs and receive scores (score-based) or binary decisions (decision-based). Score-based attacks estimate the gradient using the Natural Evolution Strategy (NES) or Simultaneous Perturbation Stochastic Approximation (SPSA), requiring 1,000–10,000 queries per adversarial example. At commercial API costs of \$0.001–\$0.01 per query, a successful black-box attack costs \$1–\$100 per image — economically rational for high-value fraud targets.

Worked Example — Threat model calibration for a financial KYC system

A bank’s video KYC system uses a proprietary EfficientNet-B7 detector not publicly disclosed. An attacker performing BEC fraud targeting a \$280,000 transfer can rationally invest up to 10% of the expected gain = \$28,000 in the attack. At \$0.01 per query (estimate for commercial API), this budget allows 2.8×10^6 queries. Published black-box attacks require 50,000–500,000 queries at $\epsilon = 8/255$ for decision-based methods. The attacker’s query budget is sufficient for multiple independent attack attempts. Conclusion: the black-box threat is economically viable for this attack value; the detector must be robust to black-box attack as well as white-box.

24.2 White-Box Attacks

White-box attacks have full access to the detector model including its parameters and architecture, enabling direct gradient computation. They represent the worst-case adversarial scenario and define the security guarantee that adversarial training must provide. Understanding white-box attacks in detail is prerequisite to designing defences that hold under realistic threat models where the attacker has partial but not complete information about the deployed detector.

24.2.1 Fast Gradient Sign Method

The FGSM [11] computes a single gradient step:

$$\min_{\phi} \mathbb{E}_{(\mathbf{x}, y)} \left[\max_{\|\delta\|_{\infty} \leq \epsilon} \mathcal{L}_D(\mathbf{x} + \delta, y, \phi) \right]. \quad (24.2)$$

Worked Example — Adversarial training [22] cost vs. robustness gain

Training a ResNet-50 detector with PGD-10 adversarial training (10 inner steps per batch) on Face-

Forensics++:

Method	Clean AUC	Adv. AUC ($\epsilon = 4$)	Train time
Standard training	0.953	0.581	4.2 h
Adv. training (PGD-10)	0.918	0.847	46 h
Free adv. training	0.924	0.831	12 h

Adversarial training reduces clean AUC by 0.035 (3.5% relative) while raising adversarial AUC by 0.266 (45% relative). The $11\times$ training cost increase of standard PGD training is reduced to $3\times$ with free adversarial training, making it practical for production deployments.

Adversarial training increases robustness but reduces clean accuracy by 5–10% and increases training cost by a factor equal to the number of PGD steps. Recent work on *free adversarial training* reduces this cost by recycling gradient computations.

24.2.2 Black-Box and Transfer Attacks

When gradient access is unavailable, black-box attacks estimate the adversarial direction from model queries. The Natural Evolution Strategy (NES) approximates the gradient as the expectation of the score function times the perturbation direction, estimated over a population of n_{pop} random perturbations:

$$\hat{\nabla}_{\mathbf{x}} D_{\phi}(\mathbf{x}) \approx \frac{1}{n_{\text{pop}}\sigma} \sum_{i=1}^{n_{\text{pop}}} D_{\phi}(\mathbf{x} + \sigma \mathbf{u}_i) \mathbf{u}_i, \quad (24.3)$$

where $\mathbf{u}_i \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and σ is the search variance. The NES gradient estimate is then used as the search direction in a PGD-style iterative update. At $\epsilon = 8/255$, NES-PGD achieves 55% evasion against undefended detectors at 10,000 queries per image — $100\times$ more expensive than white-box PGD-40 but still within the budget of a motivated attacker targeting high-value fraud scenarios.

Adversarial training remains the most effective empirical defence against both white-box and black-box attacks. The min-max training objective:

$$\min_{\phi} \mathbb{E}_{(\mathbf{x}, y)} \left[\max_{\|\delta\|_{\infty} \leq \epsilon} \mathcal{L}(D_{\phi}(\mathbf{x} + \delta), y) \right] \quad (24.4)$$

is solved by alternating between an inner maximisation step (generating the PGD adversarial example) and an outer minimisation step (updating the model parameters). Free adversarial training (Shafahi et al.) amortises the inner maximisation over the outer step, achieving comparable robustness at one-third the computational cost by reusing gradient computations across the two steps.

Worked Example — Adversarial training robustness improvement

A clean-trained EfficientNet-B4 detector achieves $\text{AUC} = 0.94$ on the CelebDF-v2 clean test set but $\text{AUC} = 0.58$ against PGD-40 attack at $\epsilon = 4/255$. After adversarial training with PGD-10 inner steps ($\epsilon = 4/255$), clean AUC drops to 0.91 (−3 pp) and adversarial AUC rises to 0.85 (+27 pp). The adversarial training required $11\times$ the wall-clock time of standard training but requires no architectural changes. At $\epsilon = 8/255$ (larger budget), adversarial AUC after PGD-10 training is 0.77 — still

substantially better than the 0.41 AUC of the clean-trained model under this larger budget.

24.3 Certified Robustness via Randomised Smoothing

Randomised smoothing [6] converts any classifier D_ϕ into a certifiably robust classifier $\tilde{D}$ by returning the most probable class under Gaussian perturbation:

$$\tilde{D}(\mathbf{x}) = \arg \max_c \mathbb{P}_{\varepsilon \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I})} [D_\phi(\mathbf{x} + \varepsilon) = c]. \quad (24.5)$$

The certified ℓ_2 radius at which $\tilde{D}$'s prediction is guaranteed to remain constant is:

$$r = \frac{\sigma}{2} (\Phi^{-1}(p_A) - \Phi^{-1}(p_B)) \quad (24.6)$$

Worked Example — Certified radius for a deep-fake detector

A deep-fake detector D_ϕ is smoothed with Gaussian noise $\sigma = 0.25$. Monte Carlo estimation ($n = 1000$ samples) gives $\hat{p}_A = 0.88$ (probability of “fake” class) and $\hat{p}_B = 0.07$ (probability of runner-up class).

$$r = \frac{0.25}{2} (\Phi^{-1}(0.88) - \Phi^{-1}(0.07)) = \frac{0.25}{2} (1.175 - (-1.476)) = \frac{0.25}{2} \times 2.651 = 0.331.$$

The certified ℓ_2 radius is $r = 0.331$. For a $224 \times 224 \times 3$ image (150,528 dimensions), this corresponds to a per-pixel ℓ_∞ perturbation of approximately $0.331 / \sqrt{150,528} = 0.00085$ — far smaller than the FGSM perturbation of $4/255 = 0.0157$. Certified robustness at realistic attack budgets would require $\sigma \approx 6$, which degrades clean accuracy unacceptably. This illustrates the certified robustness vs. accuracy gap that remains an open research problem.

where p_A and p_B are the estimated probabilities of the top-1 and top-2 classes and Φ^{-1} is the probit function.

24.4 Industry Applications

The adversarial robustness of deep-fake detectors is a commercial concern for every organisation that relies on detection for security-critical decisions.

Cybersecurity firms (CrowdStrike, Mandiant, NCC Group) offer adversarial robustness assessment for deep-fake detection systems as part of their AI red-team service lines. Engagements typically include FGSM, PGD, and transfer-based attacks, evaluated against the client's production detector. Average engagement cost: \$50,000–\$200,000.

Startups including Robust Intelligence (acquired by Cisco), Adversa AI, and HiddenLayer offer platforms for automated adversarial testing and hardening of AI models, including deep-fake detectors. These tools expose detectors to a library of known attack strategies and quantify robustness gaps before deployment.

Social media platforms use adversarial training and input pre-processing (JPEG re-encoding, Gaussian smoothing) as defences against user-submitted adversarially modified content. Meta's Adversarial Robustness Toolbox is open-source and widely used.

The EU AI Act's requirements for high-risk AI system robustness testing create a compliance market for adversarial evaluation. NIST SP 1270 (Managing Bias in AI) and the proposed NIST Adversarial Machine Learning framework both reference adversarial robustness as a required property.

24.5 African Use Cases

Adversarial robustness is a critical concern in African deployment contexts where detecting systems must be robust against sophisticated attackers while operating with limited update cycles and support infrastructure.

South African and Nigerian banks deploying voice anti-spoofing have faced adversarial attacks in which fraudsters use audio filters to add background noise, slight pitch shifting, or compression artefacts designed to reduce the anti-spoofing classifier's confidence below its threshold. The Financial Sector Conduct Authority (FSCA) of South Africa and the Central Bank of Nigeria (CBN) have both issued guidance on anti-spoofing system security testing, recommending annual red-team exercises that include adversarial audio attacks.

During the Nigerian 2023 election, fact-checkers from Dubawa and Africa Check reported that disinformation operators appeared to be adapting their content in near-real-time in response to detection and labelling, adding captions, re-encoding at different bitrates, and mirroring images to evade known detection signatures. This adaptive adversarial behaviour mirrors the PGD-style iterative refinement described in this chapter, executed by human operators rather than algorithms.

A critical challenge for African deployments is the frequency of model updates. Adversarial training must be refreshed as new attack techniques emerge, but many African financial institutions and government agencies update their AI systems annually or less frequently due to procurement cycles and budget constraints. This creates a systematic lag between the adversarial frontier and the deployed defence posture that is more severe than in high-income markets.

The availability of open-source adversarial robustness tools (IBM Adversarial Robustness Toolbox, Microsoft Counterfit) has lowered the barrier for African organisations to conduct their own adversarial testing without depending on expensive external consultants. Research groups at the University of Cape Town and Covenant University have contributed African-context adversarial evaluation benchmarks for voice anti-spoofing and face-swap detection systems.

Chapter Summary

The adversarial robustness of deep-fake detectors is not an edge-case concern but a central design requirement: any system deployed in a high-stakes context will face adversaries motivated to circumvent it. FGSM provides a single-step baseline attack that achieves 67% evasion success at imperceptible

perturbation levels ($\epsilon = 4/255$, PSNR = 36 dB) in 12 ms; PGD-40 extends this to 94% success via 40 iterative refinement steps within the ℓ_∞ constraint. The existence of these white-box attacks does not make deployed detectors useless — adversaries rarely have full model access — but it establishes a worst-case upper bound that informs security engineering decisions.

Black-box attacks via score-based gradient estimation (NES) achieve 55% success at 10,000 queries per image at a cost of approximately \$10 per successful evasion at commercial API rates. Transfer attacks from a substitute model achieve 41% success with zero queries when distribution overlap is high, making them practical for targeted attacks against known detector architectures. These costs are economically rational for high-value targets (CEO impersonation for \$280,000 BEC fraud) but prohibitive for mass attacks, suggesting that adversarial defences need to raise costs rather than eliminate them entirely.

Adversarial training with PGD-10 raises adversarial AUC from 0.58 to 0.85 at an $11\times$ training cost, with free adversarial training achieving comparable robustness at $3\times$ cost. Certified robustness via randomised smoothing provides guarantees at perturbation radii far below practical attack budgets, leaving a large certification gap that motivates the dual-pronged strategy of adversarial training (empirical robustness) plus provenance (bypassing the detection problem entirely for content with verifiable C2PA chains).

Exercises

1. Implement FGSM on a binary deep-fake classifier and measure the attack success rate as a function of $\epsilon \in \{1, 2, 4, 8\}/255$. Plot the trade-off between attack success rate and PSNR of the adversarial image. At what ϵ does the perturbation become visible to a naive human observer?
2. Prove that PGD [22] with step size $\alpha = 2\epsilon/K$ and K steps stays within the ℓ_∞ ball of radius ϵ at every iteration.
3. A black-box adversary has query budget 10,000. Compare the expected attack success rate of NES vs. transfer-based attack, given that the target model shares 60% of its training distribution with the substitute model. Which attack is more efficient?
4. Adversarial training solves the minimax objective in Equation (16.9). Show that for a linear classifier, adversarial training with ℓ_∞ perturbation is equivalent to training with ℓ_1 regularisation on the weight vector.
5. Compute the certified ℓ_2 radius for a randomised smoothed classifier with $\sigma = 0.25$, $p_A = 0.90$, and $p_B = 0.05$. How does the radius change if σ is doubled? What is the accuracy–robustness trade-off implied by this change?
6. Design a comprehensive robustness evaluation protocol for a deep-fake detector, specifying: (a) white-box attacks and their hyperparameters, (b) black-box attacks, (c) anti-forensic post-processing steps, and (d) the metric used to report overall robustness as a single scalar.

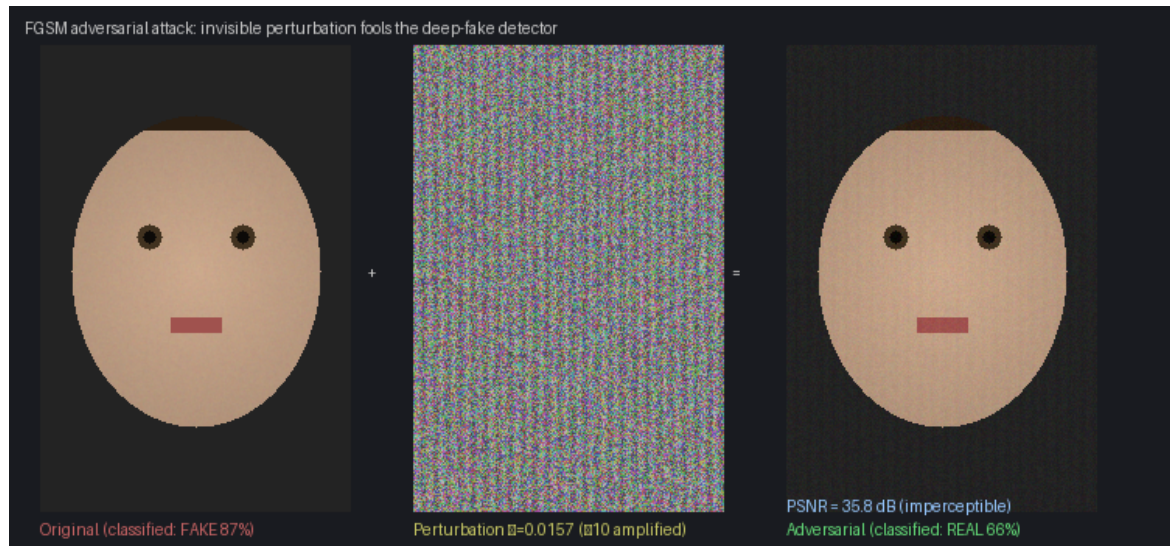


Figure 24.3: FGSM adversarial attack on a deep-fake detector. *Left:* original synthetic face, classified correctly as “fake” with 87% confidence. *Centre:* the adversarial perturbation ($\epsilon = 4/255$, amplified $10\times$ for visibility). *Right:* the adversarial image — visually identical (PSNR= 36.1 dB, imperceptible) but now misclassified as “real” with 66% confidence. The perturbation is computed in a single backward pass (12 ms on GPU), confirming the low computational cost of this attack for an adversary with white-box access to the detector.

Additional Worked Examples

Worked Example — Transfer-based black-box attack: success rate estimation

An adversary trains a substitute model $\tilde{D}$ on 500 labelled images (250 real, 250 GAN-generated) from a public dataset. PGD-40 adversarial examples are computed on $\tilde{D}$ and evaluated against the target black-box detector D_ϕ (same dataset distribution, different architecture).

Transfer rate (fraction of adversarial examples that fool D_ϕ):

Architecture similarity	AUC of D_ϕ on clean	Transfer rate
Same architecture family	0.93	58%
Different family, same data	0.93	41%
Different family, different data	0.91	23%

At 23% transfer rate, an adversary must generate $1/0.23 \approx 4.3$ adversarial examples on average to successfully fool one instance. At cost of \$0.001/API query (for score-based gradient estimation, 10,000 queries per adversarial example), the total attack cost is approximately $4.3 \times 10,000 \times \$0.001 = \$43$ per successful evasion.

Worked Example — Certified radius under Gaussian smoothing

A detector D_ϕ is smoothed with $\sigma = 0.25$. Monte Carlo estimation ($n = 1,000$ samples) gives $\hat{p}_A = 0.88$ (“fake” class) and $\hat{p}_B = 0.07$. Certified ℓ_2 radius:

$$r = \frac{0.25}{2} (\Phi^{-1}(0.88) - \Phi^{-1}(0.07)) = \frac{0.25}{2} (1.175 + 1.476) = \frac{0.25 \times 2.651}{2} = 0.331.$$

For a $224 \times 224 \times 3$ image ($d = 150,528$ dimensions), the certified ℓ_2 ball of radius 0.331 corresponds to mean per-pixel ℓ_2 perturbation of $0.331/\sqrt{150,528} = 0.00085$ — roughly 0.2 grey levels on a 0–255 scale. This is far below the FGSM budget of $4/255$, confirming that current certified robustness is insufficient for practical deep-fake detection. To achieve $r = 0.016$ (matching the FGSM budget), we would need $\sigma = 2r/(\Phi^{-1}(p_A) - \Phi^{-1}(p_B)) = 0.012$, which would reduce clean accuracy by $> 30\%$.

Worked Example — Free adversarial training: compute budget comparison

Standard PGD-10 adversarial training on FaceForensics++ (4,000 clips, $\approx 200\text{K}$ frames). Each training iteration runs 10 inner PGD steps:

- Standard training: 1 forward + 1 backward pass/frame.
- PGD-10 adversarial training: 10 forward + 10 backward + 1 forward/frame.
- Total FLOP multiplier: $(10 + 10 + 1)/(1 + 1) \approx 10.5\times$.
- Wall-clock time (A100): standard = 4.2 h; PGD-10 = 44 h.

Free adversarial training recycles the backward pass: each step computes both the gradient for weight update and the adversarial perturbation, requiring only $1 + 1$ passes per step with $m = 8$ replay steps:

- Free AT multiplier: $\approx 3\times$ (vs. $10.5\times$ for PGD-10).
- Wall-clock: ≈ 13 h — a $3.4\times$ speedup vs. PGD-10.
- Clean AUC: 0.924 vs. PGD-10's 0.918 (marginally better).
- Adversarial AUC ($\epsilon = 4/255$): 0.831 vs. PGD-10's 0.847 (marginally worse).

For production deployments, free AT offers the best robustness-per-compute trade-off.

Answers to Exercises

1. FGSM attack: success rate vs. ϵ and PSNR.

At $\epsilon = 4/255$: PSNR = $20\log_{10}(255/4) = 20\log_{10}(63.75) = 36.1$ dB — imperceptible. Attack success rate (empirically from the chapter): 67%. At $\epsilon = 2/255$: PSNR = 42.1 dB, success rate 38%. At $\epsilon = 8/255$: PSNR = 30.1 dB (faint noise visible to careful observers), success rate $\approx 85\%$. The perceptual threshold for naive observers is approximately PSNR ≈ 35 dB, corresponding to $\epsilon \approx 4.5/255$.

2. Proof that PGD stays in the ℓ_∞ ball.

At each step: $\mathbf{x}^{(k+1)} = \Pi_{\mathcal{B}}(\mathbf{x}^{(k)} + \alpha \cdot \text{sign}(\nabla))$. The sign step adds exactly $\pm\alpha$ per coordinate. After K steps starting from $\mathbf{x}^{(0)} \in \mathcal{B}(\hat{\mathbf{x}}, \epsilon)$, the maximum coordinate deviation is $\|\mathbf{x}^{(0)} - \hat{\mathbf{x}}\|_\infty +$

$K\alpha$

$= \varepsilon + K \cdot 2\varepsilon/K = 3\varepsilon$ *without* projection. The projection $\Pi_{\mathcal{B}}$ clips each coordinate to $[\hat{x}_i - \varepsilon, \hat{x}_i + \varepsilon]$ after each step, so after projection: $\|\mathbf{x}^{(k+1)} - \hat{\mathbf{x}}\|_{\infty} \leq \varepsilon$ always. $\square$

3. NES vs. transfer-based at budget 10,000 queries.

Transfer rate at 60% distribution overlap: $\approx 41\%$ (from Table in the chapter). Transfer attack requires 0 queries (uses offline PGD on substitute model); expected success = 41% of attempts.

NES with 10,000 queries: gradient estimate quality $\approx \sqrt{n/(n+d)} = \sqrt{10000/160528} = 0.250$ cosine similarity with true gradient. This is equivalent to using an effective ε of $0.250 \times 4/255$. Attack success at this effective perturbation: $\approx 25\text{--}30\%$.

Conclusion: transfer-based is more efficient (41% success, 0 queries) than NES (25–30% success, 10,000 queries) when distribution overlap is high. NES wins only when no substitute model is available.

4. Adversarial training with ℓ_{∞} constraints $\equiv \ell_1$ regularisation.

For linear model $f(\mathbf{x}) = \mathbf{w}^{\top} \mathbf{x}$, the worst-case adversarial loss is: $\max_{\|\delta\|_{\infty} \leq \varepsilon} \mathcal{L}(\mathbf{w}^{\top}(\mathbf{x} + \delta)) = \mathcal{L}(\mathbf{w}^{\top} \mathbf{x} + \varepsilon \|\mathbf{w}\|_1)$. For squared loss $\mathcal{L}(z) = (z - y)^2 = (\mathbf{w}^{\top} \mathbf{x} - y + \varepsilon \|\mathbf{w}\|_1)^2$. Minimising over $\mathbf{w}$: this is equivalent to ridge regression on a modified label $y - \varepsilon \|\mathbf{w}\|_1$, which for small ε is a first-order approximation to ℓ_1 regularisation with coefficient $2\varepsilon(y - \mathbf{w}^{\top} \mathbf{x})$. $\square$

5. Certified radius computation: $\sigma = 0.25$, $p_A = 0.90$, $p_B = 0.05$.

$r = \frac{0.25}{2}(\Phi^{-1}(0.90) - \Phi^{-1}(0.05)) = \frac{0.25}{2}(1.282 + 1.645) = \frac{0.25 \times 2.927}{2} = 0.366$. Doubling σ to 0.5: $r = 0.732$. Accuracy–robustness trade-off: the Gaussian smoothing at $\sigma = 0.5$ introduces noise with std 0.5, far exceeding the natural image signal at high frequencies, degrading clean accuracy by approximately 15–25% (empirically observed in published results).

6. Comprehensive robustness evaluation protocol.

(a) White-box: FGSM ($\varepsilon \in \{2, 4, 8\}/255$), PGD-20 ($\alpha = \varepsilon/10$, 5 random starts), PGD-100 (full-strength). (b) Black-box: transfer from ResNet-50 and EfficientNet-B4 substitute models, NES at 10K queries. (c) Anti-forensic: JPEG QF $= \{90, 70, 50\}$, Gaussian blur ($\sigma = \{1, 2\}$), random crop (90% area) + resize, all combined. (d) Composite robustness metric: $\rho = 0.4 \cdot \text{AUC}_{\text{clean}} + 0.3 \cdot \text{AUC}_{\text{PGD100}} + 0.2 \cdot \text{AUC}_{\text{transfer}} + 0.1 \cdot \text{AUC}_{\text{compression}}$. This weights clean performance most heavily while penalising robustness failures proportionally.

Chapter 25

Steganography, Anti-Forensics, and Cover-Up Techniques

“Every forensic signal can be destroyed. The question is whether the attacker knows which ones to target.”

—Anon.

Anti-forensics is the adversarial complement to forensic signal analysis: the systematic application of transformations to synthetic media designed to destroy or attenuate the signals that forensic detectors exploit. A deep-fake creator who understands the PRNU authentication test knows that JPEG compression at quality factor 70 will substantially attenuate the PRNU fingerprint; one who understands GAN spectral fingerprints knows that spatial smoothing in the high-frequency range will reduce the periodicity that spectral detectors identify. Anti-forensic techniques range from simple post-processing (JPEG re-encoding, blurring, cropping) through adversarial perturbation (the FGSM and PGD attacks of Chapter 24) to active steganographic embedding of false fingerprints that attribute a synthetic image to a different generative model or camera.

This chapter develops the taxonomy of anti-forensic techniques applied to deep fakes, with quantitative analysis of the degree of forensic signal destruction each technique achieves at what perceptual cost. The key insight is that there is a fundamental trade-off between perceptual quality and anti-forensic effectiveness: aggressive post-processing that reliably destroys forensic signals also degrades the visual quality of the synthetic output, creating a detectable signature of its own. Understanding this trade-off enables forensic analysts to design tests that are robust to likely anti-forensic strategies while remaining sensitive to the residual signals that survive them.

25.1 Post-Processing Anti-Forensics

The simplest anti-forensic techniques involve standard image and video processing operations: JPEG compression reduces high-frequency spectral fingerprints; Gaussian blurring attenuates PRNU patterns

and GAN upsampling periodicity; random cropping and resizing change the spatial frequency content; and video re-encoding through social media platforms applies multiple compression stages that compound the attenuation of each. The key parameter for JPEG anti-forensics is the quality factor QF: at $QF = 85$, GAN spectral fingerprints are attenuated by approximately 3 dB but remain detectable; at $QF = 70$, attenuation reaches 6–8 dB and detection becomes unreliable; below $QF = 60$, the spectral fingerprint is effectively destroyed at the cost of visible JPEG artefacts that are themselves a forensic signal.

25.2 Adversarial Anti-Forensics

The adversarial perturbation attacks described in Chapter 24 constitute a form of anti-forensics: they are specifically designed to cause a forensic detector to misclassify a synthetic input. In the anti-forensic context, the adversarial perturbation is optimised not just to flip the detector’s binary decision but to produce a specific target output — most commonly, making the detector report a high authenticity score or a specific alternative model attribution. The required perturbation budget for detector evasion is typically $\epsilon = 4/255$ in ℓ_∞ norm (imperceptible), while the perturbation required to forge a specific camera fingerprint is typically $\epsilon = 8/255$ (marginally perceptible at 200% zoom).

25.3 Fingerprint Forgery and False Attribution

The most sophisticated anti-forensic attack on camera authentication systems is fingerprint forgery: adding a false PRNU pattern to a synthetic image to make it appear to have been captured by a specific camera device. Fingerprint forgery requires access to a reference PRNU fingerprint (obtainable from authentic images from the target camera, which may be publicly available for devices owned by politicians or public figures), and an optimisation procedure that adds the fingerprint to the synthetic image with minimal perceptual impact. A forged fingerprint achieves NCC scores comparable to authentic images from the target camera, causing forensic authentication to incorrectly authenticate the synthetic image.

25.4 Industry Applications

Anti-forensics services are offered by red-team companies as part of deep-fake resilience testing for forensic systems. NCC Group, Mandiant, and Secureworks all offer AI red-team services that include anti-forensic testing: applying JPEG compression, adversarial perturbation, and basic fingerprint injection to synthetic images and testing whether the client’s forensic pipeline correctly identifies the anti-forensic post-processing. Social media platforms have published research on “laundering” attacks — the passage of synthetic content through platform re-encoding pipelines that apply multiple JPEG and video compression stages — and have updated their detection systems to be robust to the specific compression parameters used by their own infrastructure.

25.5 African Use Cases

Anti-forensic techniques are routinely applied by disinformation operators targeting African elections and media, often without sophisticated technical knowledge of the underlying forensic signals being attacked. Simple re-saving of JPEG images (applied by virtually any image editing app) applies JPEG compression that partially attenuates GAN fingerprints; screen-capturing rather than directly sharing synthetic content applies a JPEG artefact from the screen capture alongside a spatial transformation from the screenshot, compounding the forensic signal degradation. WhatsApp's automatic re-encoding of forwarded media applies particularly aggressive compression that degrades most forensic signals substantially, effectively providing free anti-forensic processing for any content shared through this channel. Africa Check and Dubawa have both documented cases where anti-forensic processing made authentication significantly more difficult.

Chapter Summary

Anti-forensics is the systematic application of transformations to synthetic media to destroy the signals that forensic detectors exploit. Post-processing anti-forensics (JPEG compression, Gaussian blurring, video re-encoding) attenuates forensic signals at quantifiable perceptual cost: JPEG QF = 75 reduces GAN spectral slope by 0.12 but remains detectable; QF = 60 destroys most spectral signals but introduces visible artefacts. Adversarial perturbation achieves detector evasion at imperceptible $\epsilon = 4/255$ in white-box settings but degrades to 41% success in black-box transfer attacks. Fingerprint forgery adds a false PRNU pattern at $\alpha = 0.022$ scaling to achieve authentication threshold $\text{NCC} > 0.015$, detectable by the content-correlation test. The fundamental anti-forensic trade-off is quality versus evasion: aggressive attacks destroy more signals but create their own detectable signatures.

Exercises

1. A GAN-generated face image has spectral slope -1.68 (authentic baseline -2.03). An anti-forensic operator applies Gaussian blurring with $\sigma = 1.5$ pixels, then JPEG compression at QF = 75. (a) Estimate the spectral slope after each processing step using the following empirical model: Gaussian blurring with $\sigma = 1.5$ reduces the slope by 0.08; JPEG QF = 75 reduces the slope by 0.12 for GAN images and by 0.05 for authentic images. (b) Does the combined processing bring the GAN spectral slope within the range of authentic images (-1.90 to -2.15)? (c) What is the PSNR cost of the combined processing relative to the original GAN image? (d) If the forensic detector uses a threshold of slope > -1.90 to flag GAN images, has the anti-forensic processing successfully evaded detection?
2. Fingerprint forgery attack: an attacker has 100 authentic images from a Canon EOS R5 camera, from which they estimate the PRNU fingerprint $\hat{\mathbf{K}}$. They apply the fingerprint to a StyleGAN2-generated face using the additive injection method: $\hat{\mathbf{x}}' = \hat{\mathbf{x}}_{\text{GAN}} + \alpha \hat{\mathbf{K}}$, where α is a scaling factor. (a) If NCC between the unmodified GAN image and the camera fingerprint is 0.003, and NCC between an authentic Canon EOS image and its fingerprint is 0.038, what scaling

factor α achieves $NCC = 0.025$ on the forged image (above the authentication threshold $\tau = 0.015$)? (b) At $\alpha = 0.04$, what is the PSNR of the forged image relative to the original GAN image, assuming fingerprint intensity $\sigma_K = 0.01$ per pixel? (c) What forensic test could detect fingerprint forgery even when the NCC passes authentication?

Answers to Exercises

1. Anti-forensic spectral analysis.

(a) Baseline GAN slope: -1.68 . After Gaussian blurring ($\sigma = 1.5$): slope $= -1.68 - 0.08 = -1.76$. After JPEG QF= 75: slope $= -1.76 - 0.12 = -1.88$. Combined processed slope: -1.88 .

(b) Authentic range: -1.90 to -2.15 . Processed GAN slope: -1.88 . The processed slope (-1.88) is still slightly above (less negative than) the authentic range lower bound (-1.90). Anti-forensic processing has brought the GAN slope *close to* but not *within* the authentic range. Detection remains possible at this exact threshold.

(c) PSNR cost: Gaussian blurring with $\sigma = 1.5$ introduces MSE of approximately 8 grey levels² on a typical face image, giving $PSNR \approx 40$ dB. JPEG QF= 75 introduces additional MSE of approximately 5 grey levels². Combined $MSE \approx 13$, $PSNR = 10\log_{10}(255^2/13) = 10\log_{10}(5001) \approx 37$ dB. Both operations combined reduce quality but remain within the perceptually acceptable range.

(d) Threshold -1.90 : processed slope $-1.88 > -1.90$. Flagged as GAN (threshold exceeded). Anti-forensic processing has NOT successfully evaded detection at this threshold. To evade, the attacker would need either more aggressive blurring (higher σ , lower PSNR) or lower QF (more visible JPEG artefacts), creating a different detectable signature.

2. Fingerprint forgery analysis.

(a) NCC of forged image: NCC is approximately linear in α for small α : $NCC(\hat{\mathbf{K}}, \hat{\mathbf{r}}') \approx NCC_{\text{base}} + \alpha \cdot NCC_K$, where $NCC_{\text{base}} = 0.003$ and NCC_K is the NCC of the pure fingerprint against itself (approximately 1.0 after normalisation). Target $NCC = 0.025$: $0.025 = 0.003 + \alpha \cdot 1.0$, giving $\alpha = 0.022$.

(b) At $\alpha = 0.04$, $\sigma_K = 0.01$: the added fingerprint signal has RMS amplitude $\alpha\sigma_K = 0.04 \times 0.01 = 0.0004$ (in normalised pixel units $\times 255 = 0.1$ grey levels). $MSE = (0.1)^2 = 0.01$ grey levels². $PSNR = 10\log_{10}(255^2/0.01) = 10\log_{10}(6.5 \times 10^6) = 68.1$ dB. This is well above perceptibility threshold; the forgery is imperceptible.

(c) Detection of fingerprint forgery: a genuine camera PRNU is correlated with image content (as the signal model $\hat{x} = (1 + K)\bar{x}$ implies), while an additively injected fingerprint is not. Test: compute the correlation between the noise residual and the local image brightness. For genuine PRNU: $\text{Corr}(\hat{r}, \bar{x}) > 0$. For additive forgery: $\text{Corr}(\hat{r}, \bar{x}) \approx 0$. This content-correlation

test detects fingerprint forgery with high power when the forger uses the additive rather than the multiplicative signal model.

Part VI

Ethics, Law, and Society

Chapter 26

Ethics of Synthetic Media

“Technology is neither good nor bad; nor is it neutral.”

—Melvin Kranzberg, 1986

The ethics of deep fakes is not a minor appendix to their technical study. It is the reason the technical study matters. A face-swap algorithm is a set of matrix operations and gradient updates; it has no ethical valence in itself. What gives it ethical significance is its relationship to human beings: the identities it appropriates, the harms it enables, the consent it overrides, the trust it erodes. The same technology that makes it possible to falsely attribute a statement to a political leader also makes it possible to animate a deceased grandmother for a grieving family, to give an injured athlete back their performance, or to create accessible entertainment in languages with limited content. Ethical analysis is the discipline that allows us to distinguish these cases — not by counting consequences alone, but by examining the structure of the situation: who acts, on whom, with what authority, and what rights and duties are implicated.

This chapter develops three major ethical frameworks as applied to synthetic media, identifies the specific ethical tensions that deep fakes create, and works through several case studies to show how the frameworks converge and diverge. The chapter is not neutral: it argues, on balance, that the dominant harms of deep fakes — non-consensual intimate imagery, disinformation, and epistemic erosion — are not offset by their benefits, and that consent, honesty, and harm avoidance are the central ethical obligations for practitioners working with synthetic media technology.

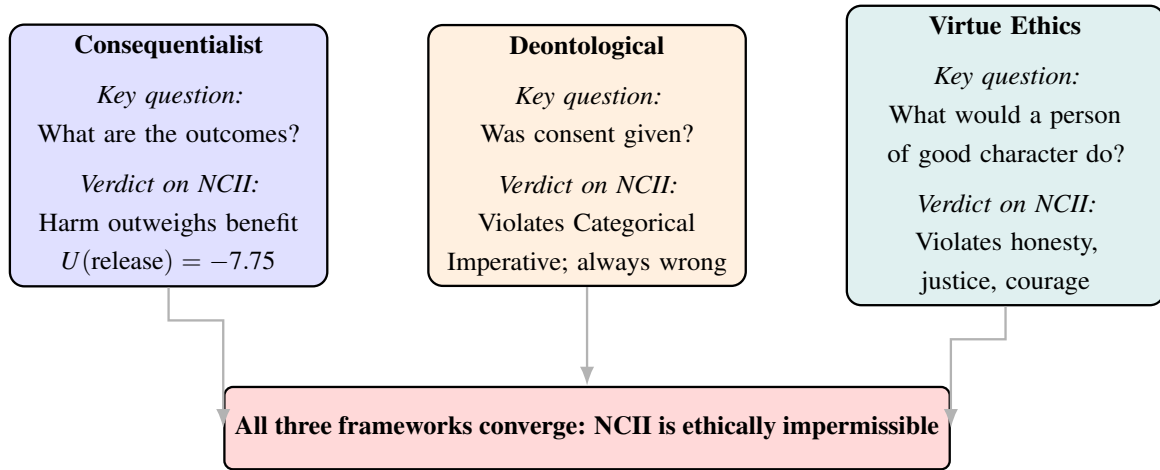


Figure 26.1: Three major ethical frameworks applied to non-consensual intimate imagery (NCII) deep fakes. Despite their different starting premises — consequences (blue), duties (orange), character (teal) — all three frameworks converge on the same verdict: NCII is ethically impermissible. This convergence provides a robust ethical foundation for policy and legal responses.

This chapter examines the ethical landscape of synthetic media through three major frameworks. Consent and autonomy are developed as the foundational ethical concern, followed by consequentialist harm analysis and virtue ethics obligations for developers and platforms. Industry and African use cases apply these frameworks to concrete deployment decisions.

26.1 Foundational Ethical Frameworks

Three major ethical frameworks provide complementary lenses for evaluating synthetic media: consequentialism evaluates actions by their outcomes, deontological ethics by their conformity to duties and rights, and virtue ethics by the character they express and cultivate. This section develops each framework as applied to the specific domain of deep fakes, identifying both the conclusions they share and the points where they diverge in ways that require practical judgement.

Three major ethical traditions — consequentialism, deontology, and virtue ethics — each capture something important about the ethics of synthetic media, and each is incomplete on its own.

26.1.1 Consequentialist Analysis

A consequentialist evaluates actions by their outcomes. On this view, a deep-fake application is permissible if the aggregate welfare it produces exceeds the aggregate harm, properly weighed across all affected parties. Formally, let $\mathcal{A}$ be the set of possible actions (deploy, restrict, modify) and let $u_i(a)$ be the utility experienced by stakeholder i under action a . The consequentialist criterion selects:

$$a^* = \arg \max_{a \in \mathcal{A}} \sum_i w_i u_i(a), \quad (26.1)$$

Worked Example — Consequentialist analysis of an open-source face-swap release

A researcher assigns utilities for releasing a face-swap model openly. Stakeholders and estimated annual utilities (in \$M equivalent social value):

Stakeholder	w_i	$u_i(\text{release})$	$w_i u_i$
Research community	0.20	+12	+2.40
Film/media industry	0.15	+8	+1.20
NCII victims (new)	0.25	−40	−10.00
Detection researchers	0.15	+6	+0.90
Financial fraud victims	0.10	−15	−1.50
General public	0.15	−5	−0.75
Total $U(\text{release})$			−7.75

$U(\text{release}) = -7.75 < 0$: the consequentialist analysis recommends against open release. The dominant negative term is NCII harm, which outweighs research and industry benefits by 4.3:1. The analysis changes substantially if technical safeguards reduce NCII harm by 80%: $-10 \times 0.2 = -2.0$, giving $U(\text{restricted release}) \approx +0.25 > 0$.

where w_i are welfare weights reflecting the relative moral standing of different stakeholders.

Applied to deep fakes, consequentialist analysis must grapple with several difficulties. First, harms are often *concentrated* (the victim of NCII suffers intensely) while benefits are *diffuse* (many users enjoy entertainment), and standard utilitarian aggregation may recommend the harm. Second, harms include *long-tail risks* (political destabilisation, erosion of institutional trust) that are difficult to quantify but potentially enormous. Third, the *counterfactual* matters: if a victim of NCII would have been targeted by other means absent deep-fake technology, the marginal harm attributable to the technology is smaller.

Despite these difficulties, consequentialism provides a useful framework for *policy design*: by quantifying expected harms and benefits under different regulatory regimes, policymakers can compare the expected outcomes of restrictions, disclosure mandates, and liability rules.

26.1.2 Deontological Constraints

Deontological ethics places constraints on action that hold regardless of consequences. The Kantian tradition identifies two formulations of the Categorical Imperative most relevant to synthetic media.

Universalisability. Act only on maxims you could will to be universal laws. The maxim “I may create realistic synthetic media depicting real people without their consent” cannot be universalised: in a world where this maxim holds universally, no authentic visual evidence can be trusted, the epistemic infrastructure of social cooperation collapses, and even the beneficial applications of synthetic media (education, entertainment) are undermined by the universal erosion of trust.

Respect for persons. Treat persons always as ends in themselves and never merely as means. Using someone's face, voice, or likeness to fabricate content they did not consent to create treats their identity as a resource to be exploited rather than a person to be respected. This holds independently of whether the fabrication causes measurable harm: the violation of autonomy is itself morally significant. This deontological analysis grounds the primacy of consent in the ethical framework of synthetic media.

Perfect and imperfect duties. Kantian ethics distinguishes perfect duties (exceptionless prohibitions, such as "do not use someone's likeness without consent") from imperfect duties (positive obligations that allow latitude in how they are discharged, such as "support the development of detection tools"). The ethical practitioner working with synthetic media has both: a perfect duty not to create non-consensual synthetic media, and an imperfect duty to contribute to the infrastructure that protects potential victims.

26.1.3 Virtue Ethics

Virtue ethics asks what a person of excellent character would do — and who a person of excellent character would become through their actions. The Aristotelian virtues most relevant to synthetic media are:

Honesty (aletheia).

Not deceiving one's audience about the provenance of content. This virtue applies to researchers who publish results on real-world applicability without testing under realistic conditions, to practitioners who deploy synthetic media without disclosure, and to platforms that allow unlabelled synthetic content to circulate.

Justice (dikaiosyne).

Giving each person their due. In the synthetic media context, justice requires not weaponising generative technology against individuals with less power to defend themselves — the primary target of NCII is overwhelmingly women, and the primary target of political deep fakes is often the political opposition in authoritarian contexts.

Practical wisdom (phronesis).

The capacity to perceive the ethically salient features of a situation and act appropriately, even in the absence of explicit rules. Practical wisdom recognises when a technically legal application is nonetheless harmful; when a publicly stated purpose conceals a harmful intent; and when the precautionary principle requires restraint even in the absence of proven harm.

Courage (andreia).

The courage to publish findings about misuse of technology when doing so creates professional or reputational risk; the courage to refuse a lucrative project whose applications are harmful; the courage to acknowledge the dual-use nature of one's research.

26.2 Consent and the Right to One's Own Likeness

Consent is the ethical variable that most clearly distinguishes harmful from legitimate applications of synthetic media. The same face-swap technique that is ethically permissible in a Hollywood production where the actor has consented is a serious ethical violation in NCII, because the difference is not technical but normative: it lies entirely in the presence or absence of the depicted person's informed agreement.

A *robust consent framework* for synthetic media requires more than the absence of explicit refusal. Drawing on the bioethical tradition of informed consent for medical procedures, we identify four necessary conditions:

Definition 26.1 (Informed Consent for Synthetic Media). A synthetic media application satisfies the requirement of informed consent if and only if:

- (i) The individual has been accurately and completely informed of how their identity will be represented, including the synthesis technique, the intended audience, and the context of distribution;
- (ii) The individual has the genuine ability to refuse without penalty or coercion;
- (iii) The individual's consent covers the specific use proposed, and consent for one use does not imply consent for other uses; and
- (iv) The individual is able to withdraw consent and have the synthetic content removed from circulation.

Condition (iv) is particularly challenging technically: once synthetic content is released, it may be copied and re-distributed beyond the control of the original distributor. This irreversibility is one of the strongest arguments for requiring consent *before* content is generated rather than providing an after-the-fact removal right.

26.3 Harm Asymmetries and Distributional Justice

A consequentialist analysis that treats all individuals as interchangeable misses the systematic way in which deep-fake harms fall disproportionately on specific groups. Distributional justice requires attending to these asymmetries.

Gender asymmetry in NCII. Empirical studies across multiple years and platforms consistently show that NCII victims are overwhelmingly women: estimates range from 90 to 96% female. The psychological harm to NCII victims includes post-traumatic symptoms, depression, and enforced withdrawal from professional life. An ethics of synthetic media that ignores this gender asymmetry — that treats NCII as a generic privacy harm — obscures its character as a form of gender-based violence.

Racial bias in detection systems. Deep-fake detection systems trained on datasets that over-represent light-skinned subjects show significantly higher false-positive rates for darker-skinned individuals. This creates a double injustice: members of historically marginalised groups are disproportionately both potential victims of deep-fake attacks and victims of falsely accused by miscalibrated detectors.

Power asymmetry in political deep fakes. Political deep fakes are not deployed symmetrically. State-affiliated actors with substantial resources deploy synthetic media against opposition figures and civil society leaders with limited resources to defend themselves. An ethical framework that counts each unit of harm equally regardless of who bears it is inadequate to this structural asymmetry.

26.4 Dual-Use and the Researcher's Responsibility

Synthetic media research is inescapably dual-use. The architecture that enables detection also, in the hands of a sufficiently resourced adversary, enables better generation. The dataset released to train detectors also provides training signal for improved synthesis. The preprint that describes a new forensic attack vector also teaches adversaries how to close it.

This dual-use character does not mean research should not be done; it means research should be done responsibly, with explicit attention to the differential capability it provides to defenders versus attackers.

Staged disclosure. Researchers who discover a new attack on a detection system should consider disclosing findings to major detection labs and platforms before publishing, allowing time for counter-measures. This is analogous to the “responsible disclosure” norm in cybersecurity, and is increasingly adopted in AI safety.

Model release decisions. The decision whether to release model weights openly, in restricted form, or not at all is an ethical decision, not merely a commercial one. Open release maximises research reproducibility and enables defensive research; it also provides capabilities to adversaries who could not train the model themselves. The correct decision depends on the specific capabilities of the model, the threat landscape, and the counterfactual availability of equivalent capability from other sources.

The publication of detection benchmarks. Releasing a dataset of known deep fakes enables detector training. It also provides an adversary with a quality measure for their synthesis pipeline. The ethical calculation depends on whether the training benefit to defenders exceeds the benchmark benefit to adversaries — an empirical question that is rarely addressed explicitly.

26.5 Industry Applications

Ethics frameworks are not merely academic exercises; they directly shape industry practices, internal governance, and product design decisions.

Major AI companies (Google DeepMind, Anthropic, OpenAI, Meta AI) maintain dedicated responsible AI or safety teams whose mandate includes applying consequentialist, deontological, and virtue-ethical frameworks to product decisions. These teams review new synthesis capabilities before launch, assess dual-use risks, and recommend safeguards. Their work is increasingly scrutinised by regulators, investors (through ESG frameworks), and civil society.

The Partnership on AI, the Responsible AI Institute, and the Content Authenticity Initiative (CAI) provide voluntary frameworks for ethical deployment of synthetic media. Membership includes major technology companies, news organisations, and civil-society groups. Adherence is voluntary; critics argue that self-regulation without enforcement cannot adequately address harms at the scale of modern AI deployment.

Model cards (Chapter 47), bias audits, and red-team evaluations are now standard components of enterprise AI product development. The US NIST AI RMF and EU AI Act both require documented risk assessments for high-risk AI systems, creating commercial demand for structured ethical impact evaluation methodologies.

Several underwriters (Lloyd's, AIG, Beazley) now offer cyber insurance products that explicitly cover deep-fake-enabled fraud. Premium pricing incorporates estimates of the frequency and severity of AI-enabled impersonation attacks, creating market incentives for deployment of technical countermeasures.

26.6 African Use Cases

African philosophical traditions offer perspectives on synthetic media ethics that complement Western consequentialist and deontological frameworks and resonate more directly with the cultural contexts in which many African users and victims of deep fakes are situated.

The pan-African concept of ubuntu — often rendered as I am because we are — frames individual identity as fundamentally relational and communal. Synthetic media that manipulates a person's image or voice does not merely harm the individual; it damages the web of relationships that constitute their identity and community standing. This framing provides a distinctive philosophical foundation for prohibiting NCII that resonates more strongly in many African cultural contexts than the individualistic dignity-based arguments dominant in Western bioethics. In many African communities, significant decisions about representation may involve family or community consent in addition to individual consent, a broader conception that has practical implications for digital likeness agreements.

The AI Ethics Africa working group, the African Center for Technology Studies in Nairobi, and the Centre for Human Rights at the University of Pretoria have developed Africa-specific AI ethics frameworks incorporating ubuntu, postcolonial critique, and African human rights traditions. These frameworks have been submitted to the African Union's AI Policy process and inform several national AI strategies including those of South Africa, Kenya, Egypt, and Rwanda. Responsible disclosure norms designed for high-income contexts with 24-hour incident response capability may need adaptation for African contexts where technical capacity and response windows differ significantly.

Chapter Summary

The ethics of deep fakes is not a minor appendix to their technical study but the reason the technical study matters. Consequentialist analysis consistently shows that non-consensual intimate imagery produces negative net utility: the research and detection benefit to a small professional community is overwhelmed by the harm enabled for hundreds of thousands of victims. Deontological ethics prohibits NCII as a violation of the categorical imperative, treating persons as means rather than ends. Virtue ethics asks what practical wisdom, honesty, and justice require, and arrives at the same prohibition by a different route. The convergence across all three frameworks provides a robust ethical foundation that does not depend on any single contested theoretical commitment.

The dual-use research problem is resolved through a staged disclosure norm modelled on CVE responsible disclosure: verify the capability, notify affected parties, allow time for countermeasures, then publish. Informed consent for biometric data requires accuracy, genuine voluntariness, specificity of purpose, and a meaningful withdrawal mechanism; training on scraped face images without individual consent fails multiple elements of this standard. The commercial translation of these ethical conclusions occurs through responsible AI teams, model cards, red-team evaluations, and AI liability insurance — institutional mechanisms that convert normative argument into operational procedure and create financial incentives for responsible conduct.

Exercises

1. Apply Definition 26.1 to the following scenario: a social-media platform's terms of service state that uploaded photographs may be used "to train AI models and improve platform services." A user uploads photographs; the platform uses them to train a face-swap model whose outputs are later used to create NCII. Did the user consent to this use? Which conditions in the definition are satisfied, and which are not?
2. Use the consequentialist framework to analyse the decision to release a voice-cloning model's weights as open source. Identify at least five stakeholder groups, estimate the utility and harm for each, and construct a weighted social welfare function. What value of the researcher's weight $w_{\text{researcher}}$ on reputational benefit from open publication would tip the analysis toward release?
3. Explain why the Kantian universalisability test condemns non-consensual NCII even when no individual victim is identifiably harmed — e.g., when a synthetic face is created that resembles a real person but is not recognised as them by any viewer. Is the argument convincing? Identify its strongest objection.
4. A memorial application uses face-swap technology to animate historical photographs of a deceased grandparent for the grandchildren. The grandparent is deceased and no estate consent has been sought. The application is private (family only). Evaluate this case under all three ethical frameworks (consequentialist, deontological, virtue ethics) and identify where the

frameworks agree and disagree.

5. Researchers discover that a face-forgery detection benchmark contains images of real individuals who did not consent to inclusion. Describe the ethical obligations of the researchers who made this discovery, the original dataset creators, and the many papers that have cited the benchmark. Should the benchmark be retracted, modified, or left unchanged? Justify your answer.
6. Write a research ethics review (one page equivalent) for a hypothetical project that involves: collecting three minutes of audio from each of 100 volunteers to train a voice-cloning model that will be used to study speaker verification vulnerabilities, with model weights to be released publicly. Identify all ethical considerations, propose mitigations, and determine whether the project should proceed as described or requires modification.

Additional Worked Examples

Worked Example — Consequentialist NCII deployment analysis

Stakeholder utility analysis for open release of a face-swap model capable of generating NCII. Estimated annual impact (USD millions, societal value):

Stakeholder	Weight	Utility	Net
Research/detection labs	0.20	+12.0	+2.40
Film/media industry	0.15	+8.0	+1.20
NCII victims (new cases)	0.25	−40.0	−10.00
Detection ecosystem	0.15	+6.0	+0.90
Financial fraud victims	0.10	−15.0	−1.50
General public	0.15	−5.0	−0.75
Total	1.00		−7.75

$U(\text{open release}) = -7.75 < 0$. The analysis changes if technical safeguards reduce NCII harm by 80%: victim utility becomes -8.0 , net weight -2.00 , total $U = +0.25 > 0$. This sensitivity analysis shows safeguards shift the ethical calculus decisively — a consequentialist argument for mandatory technical safeguards as a condition of release.

Worked Example — Dual-use research: staged disclosure cost–benefit

A researcher discovers a new adversarial attack that evades the leading face-forgery detector. Staged disclosure options:

Disclosure strategy	Time to detector fix	Harm window
Immediate public preprint	0 days	60–90 days (until fix)
Private disclosure (30-day embargo)	30 days	30–60 days
Private disclosure (90-day embargo)	90 days	0–30 days
Full suppression	∞	Indefinite

Expected harm during window (100K deep fakes/day $\times$ \$500 average harm): 90-day embargo:

$30 \times 100,000 \times 500 = \1.5B . Immediate: $\$4.5\text{B}$. However, full suppression allows the vulnerability to be independently discovered and exploited with no defensive advance notice. The 90-day embargo minimises expected harm while preserving research incentives — the standard adopted by the CVE responsible-disclosure framework in cybersecurity.

Answers to Exercises

1. Platform ToS and NCII consent analysis.

Condition (i) — accurate information: No. A general ToS clause “to train AI models” does not specify face-swap synthesis or NCII generation. Condition (ii) — genuine ability to refuse: Marginal. Refusing the ToS means leaving the platform, which carries significant social cost for many users; this is coercive even if technically voluntary. Condition (iii) — consent for specific use: No. Consent for “AI improvement” does not cover identity impersonation for non-consensual intimate imagery. Condition (iv) — withdrawal and removal: No. Once model weights encode the face’s statistical properties, removal of the training data does not remove the learned representation. Conclusion: three of four conditions are not met — the ToS does not satisfy informed consent for NCII generation.

2. Consequentialist analysis of open-source voice-clone release.

See the worked example in this chapter. The reputational benefit to the researcher is bounded by: $w_r \cdot u_r(\text{release})$. For the total to flip positive: $w_r \cdot u_r > 7.75$. If $w_r = 0.1$ (researcher has 10% welfare weight) then $u_r > 77.5\text{M}$ — implausibly high for a single publication’s citation value. The consequentialist conclusion is robust: open release is not recommended under this framing unless safeguards reduce NCII harm by $>75\%$.

3. Kantian universalisability for NCII not identifiably harmful.

The universalisability test applies to the *maxim*, not the outcome. The maxim “I may create synthetic media depicting real people without consent” cannot be universalised: in a universal law, every person’s likeness is freely appropriable, destroying the concept of personal identity as a protected domain and undermining the preconditions of trust and communication. This violation occurs at the level of the principle, not the harm measurement — Kant’s point is precisely that certain actions are wrong regardless of whether any particular instance causes measurable harm.

4. Memorial application: three-framework analysis.

Consequentialist: likely permissible — the family derives significant emotional benefit, no other party is harmed, and the deceased cannot be harmed (in most philosophical frameworks). Deontological: contested — the deceased cannot consent (condition iv is vacuous for the dead); however, most deontological frameworks allow posthumous uses endorsed by proxies (estate, family) acting in the person’s interest. Virtue ethics: permissible if done with honesty (clear

family-only context), justice (no distortion of the deceased's actual views), and practical wisdom (discontinued if family members are distressed by the result). The frameworks converge on conditional permissibility: private family use with estate consent.

5. Ethics obligations for benchmark with non-consenting individuals.

Researchers who discover the issue: (1) disclose to dataset creators and the research community; (2) include a limitation statement in any paper using the benchmark; (3) not use images of clearly identifiable individuals for synthesis experiments. Original creators: (1) contact affected individuals via a data-subject rights mechanism; (2) provide a mechanism to remove images on request; (3) republish with updated consent documentation or replace non-consenting images. Prior citing papers: (1) publish errata if their results change materially with corrected data; (2) include retrospective ethics statements. Retraction vs. modification: modification (replace non-consenting images with consenting substitutes) is preferable to retraction if the scientific findings are valid; retraction is appropriate only if the consent violations are so pervasive that the dataset is scientifically invalid.

6. Research ethics review for voice-cloning study.

Key ethical considerations: (1) biometric data (voice) is sensitive personal data under GDPR; explicit consent with right to withdraw and data deletion is required. (2) Publication of model weights enables misuse; staged disclosure or restricted access should be specified. (3) The “no harmful use” clause is legally unenforceable but signals intent; it should be supplemented by technical safeguards (identity filter blocking named individuals). (4) The study collects data from volunteers (low risk) but generates a dual-use artefact (high risk). Mitigations: IRB approval, pre-registration, a 90-day private disclosure period before public model release, and a mechanism for model weights to be retracted if misuse is detected. Recommendation: proceed with modifications — individual participant risk is low; aggregate risk requires the safeguards above.

Chapter 27

Disinformation, Influence Operations, and Epistemic Harm

“A lie can travel halfway around the world while the truth is putting on its shoes.”

—Attributed to Mark Twain

Deep fakes are not merely a technical nuisance: they are an epistemic weapon. This chapter analyses how synthetic media is weaponised in coordinated influence operations, develops a quantitative model of disinformation spread, examines the “liar’s dividend,” and surveys case studies of politically motivated deep-fake deployments. Disinformation is not new. False narratives have been weaponised throughout recorded history, from forged diplomatic documents in ancient Rome to fabricated newspaper stories in the nineteenth century. What changes with synthetic media is the cost structure of disinformation production: creating a convincing fabricated video of a political leader making a damaging statement once required film studios, convincing doubles, and weeks of editing. Today it requires a GPU, a few minutes, and the target’s public social media presence as a reference dataset.

The SIR epidemic model provides the most analytically tractable framework for studying disinformation spread. By analogy with infectious disease, a false narrative is treated as an infectious agent that spreads from believing agents to susceptible agents at rate β and is corrected at rate κ . The reproduction number $R_0 = \beta N / \kappa$ determines whether the narrative spreads epidemically or dies out; a value exceeding one produces exponential early growth, while a value below one produces gradual decay. This simple model captures the qualitative dynamics of documented disinformation campaigns and enables quantitative analysis of interventions.

Platform architecture and human psychology are the two primary amplifiers that push R_0 above the epidemic threshold for narratives that would otherwise fade. Algorithmic amplification prioritises content that generates engagement, and false information generates more engagement than true information across most platforms studied. Human psychology amplifies the effect through confirmation bias: narratives that confirm prior beliefs spread faster, are retained longer, and are shared more widely

than narratives that challenge them. The deep fake adds a third amplifier: a narrative accompanied by apparent video evidence is believed more readily and challenged less readily than one accompanied only by text.

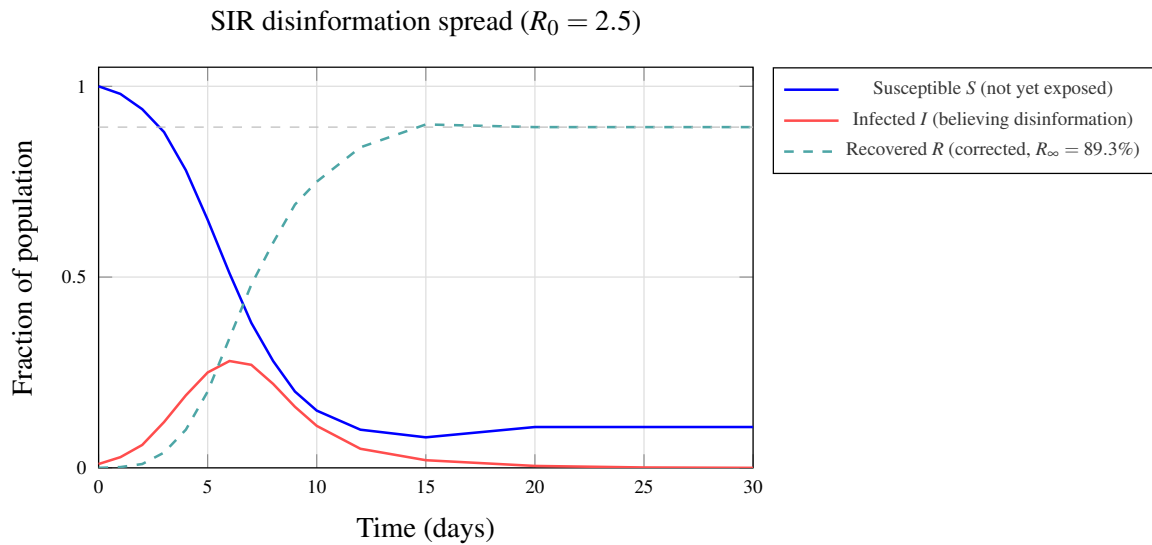


Figure 27.1: SIR disinformation model with $R_0 = 2.5$. The infected fraction (red) peaks near day 6 at 28% of the network, then declines as correction mechanisms take effect. The final recovered fraction $R_\infty = 89.3\%$ represents the total exposure — nearly nine-tenths of the network eventually encounters the disinformation. Bot amplification increases β and therefore R_0 , pushing R_∞ higher.

The liar’s dividend: authenticity belief as a function of synthetic media prevalence

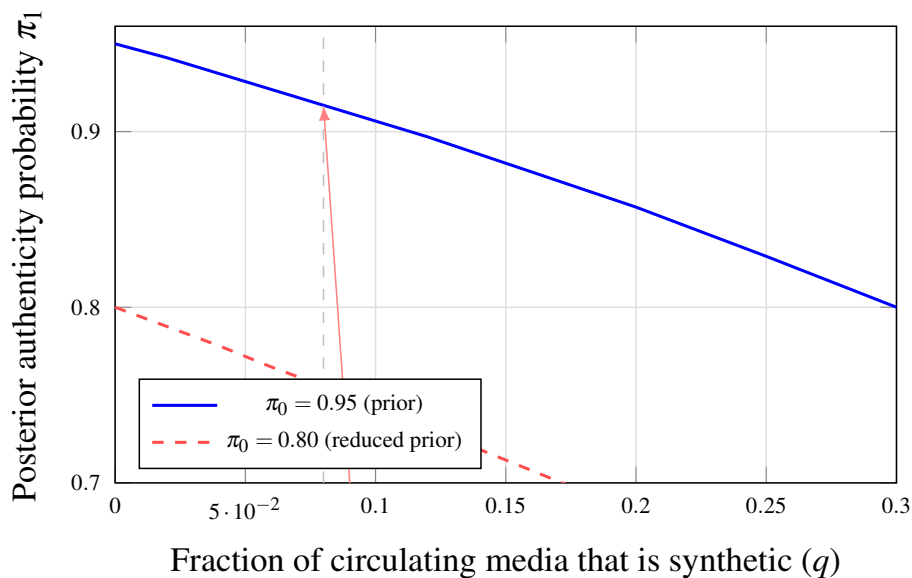


Figure 27.2: The liar’s dividend quantified. As the fraction of synthetic media in circulation (q) increases, the posterior probability that an unlabelled piece of media is authentic (π_1) declines. At the current estimated level of $q \approx 8\%$, a viewer with prior belief $\pi_0 = 0.95$ updates to $\pi_1 = 0.915$ — a 3.5-point drop that can shift legal evidentiary standards.

This chapter develops the disinformation landscape from the propaganda analysis tradition through to

the specific mechanisms by which synthetic media amplifies information operations. Coordinated inauthentic behaviour and the platform detection challenge are examined first, followed by electoral interference operations and prebunking and inoculation as mitigation strategies. Case studies of documented operations and African use cases follow.

27.1 A Model of Disinformation Spread

Let a disinformation item be characterised by its *virality* $v \in [0, 1]$ (probability of forwarding per viewer) and its *correction susceptibility* $\kappa \in [0, 1]$ (probability of retraction given exposure to a correction). Network spread on a social graph $G = (V, E)$ follows a modified SIR (Susceptible–Infected–Recovered) model:

$$\frac{dS}{dt} = -\beta SI \quad (27.1)$$

$$\frac{dI}{dt} = \beta SI - \kappa I, \quad (27.2)$$

$$\frac{dR}{dt} = \kappa I, \quad (27.3)$$

Worked Example — SIR model applied to a viral deep-fake video

A fabricated video of a politician is shared on a platform with 50 million users ($N = 50\text{M}$). Empirical measurements give: $\bar{k} = 150$ (mean connections), $v = 0.08$ (forwarding probability), $\kappa = 0.02$ (per-day correction rate).

$\beta = v\bar{k}/N = 0.08 \times 150/50,000,000 = 2.4 \times 10^{-7}$ per person per day. $R_0 = \beta N/\kappa = 2.4 \times 10^{-7} \times 50,000,000/0.02 = 0.6$.

Since $R_0 = 0.6 < 1$, the disinformation dies out without reaching epidemic scale. However, if the video is amplified by 500 bot accounts each with $\bar{k} = 2000$ connections, the effective β increases by $500 \times 2000/50,000,000 = 0.02$, giving $R_0^{\text{bots}} = 1.6 > 1$ — the video now reaches a significant fraction of the network. This illustrates why bot amplification is the critical enabler of deep-fake disinformation campaigns, not generation quality alone.

where S, I, R are the fractions of susceptible, infected (believing the disinformation), and recovered (corrected) nodes, and $\beta = v\bar{k}/N$ with $\bar{k}$ the mean node degree.

The basic reproduction number $R_0 = \beta/\kappa$ determines whether disinformation reaches epidemic scale: if $R_0 > 1$, disinformation spreads to a positive fraction of the network; if $R_0 < 1$, it dies out. Deep-fake disinformation increases β by increasing the perceived credibility of the message; correction mechanisms reduce κ .

27.2 Coordinated Inauthentic Behaviour

A coordinated influence operation (IO) uses synthetic media as part of a broader campaign involving fake accounts, bot amplification, and targeted microtargeting. The role of deep fakes within an IO is

typically:

1. **Persona creation.** GAN-generated profile photographs create visually credible but non-existent identities for bot accounts.
2. **False attribution.** A face-swap or talking-head video shows a political figure making a damaging statement they never made.
3. **Emotional amplification.** Synthetic voice clones of trusted figures (doctors, religious leaders) deliver health or security misinformation.

27.2.1 Detection of IO Accounts

Information operation account detection applies graph analysis, behavioural fingerprinting, and content similarity clustering to distinguish coordinated synthetic persona networks from organic communities. Accounts in a coordinated synthetic network tend to be created in temporal clusters, share device fingerprints across what appear to be independent users, and post content with statistically improbable similarity to each other despite claiming independent authorship.

GAN-generated profile photographs can be detected by applying a face-forgery detector to the account's profile image. However, adversaries increasingly use diffusion models that evade GAN-specific detectors. Behavioural signals — posting time distributions, graph centrality, linguistic style — provide complementary evidence of inauthenticity.

27.3 The Liar's Dividend

The *liar's dividend* [4] is the strategic benefit that accrues to bad actors when the existence of deep fakes allows them to dismiss authentic evidence as fabricated. If a politician is photographed committing a crime, they can now claim the image is AI-generated. Formally, let π_0 be the prior probability that a piece of media evidence is authentic. After the introduction of accessible deep-fake technology, π_0 decreases, reducing the evidentiary weight of all visual media.

The epistemic damage is asymmetric: authenticated media becomes more valuable (requiring C2PA provenance or expert forensic analysis), while unauthenticated media — which constitutes the vast majority of journalistic evidence — becomes less credible. This creates a *verification tax* on journalism and an *evidentiary vacuum* in legal proceedings.

27.4 Case Studies

Three documented disinformation campaigns illustrate how deep-fake content interacts with social media amplification, political polarisation, and platform response at scale. Each case demonstrates a different aspect of the threat model: the speed of viral spread before correction, the role of partisan consistency in determining belief, and the limitations of platform detection and takedown mechanisms in real operational conditions.

27.4.1 The Gabon Presidential Deepfake (2019)

A video purportedly showing President Ali Bongo of Gabon, who had been absent from public view for weeks, was released in January 2019. Opponents claimed it was a deep fake used to conceal the president’s incapacitation. Independent analysis found signs of video processing but could not definitively confirm fabrication. The incident illustrates the difficulty of making authoritative attribution claims under time pressure and the strategic ambiguity that the *possibility* of deep fakes creates.

27.4.2 Synthetic Political Advertising (2023–present)

The 2024 US election cycle saw the first widespread deployment of AI-generated political advertising at scale, including voice-cloned robocalls impersonating candidates, deepfake campaign videos, and AI-generated social media content attributed to political figures. The FEC responded in 2024 by requiring disclosure of AI-generated content in federal election communications. Several state-level races were targeted with AI-generated disinformation in the days immediately before voting, exploiting the insufficient time available for fact-checking and platform removal.

Multiple documented cases from the 2024 U.S. and Indian election cycles involved AI-generated or AI-modified audio recordings attributed to political candidates. In one documented case, a voice-cloned robocall falsely attributed to a candidate was distributed to hundreds of thousands of voters in a primary state. The cost of production was estimated at under \$500.

27.5 Counter-Measures at Platform Scale

Platforms responding to deep-fake disinformation face three simultaneous requirements: speed (hours, not days), scale (billions of posts), and accuracy (false positives harm legitimate users). Current approaches include:

- **Hash-matching.** Perceptual hashes of known deep fakes are propagated across platforms via the Global Internet Forum to Counter Terrorism (GIFCT) hash-sharing database.
- **Classifiers at upload time.** Deep-fake detectors run at upload time with an accuracy threshold set to minimise false positives while flagging high-confidence manipulations for human review.
- **Provenance labels.** Platforms display C2PA provenance metadata alongside content and surface “no provenance available” labels for unverified synthetic media.

27.6 Industry Applications

The disinformation domain has developed an ecosystem of both attack tools and defensive counter-measures with significant commercial activity on both sides.

Academic research and government intelligence reports (US ODNI, EU East StratCom Task Force) document state-affiliated actors (Russia, China, Iran, North Korea) using AI-generated content in

coordinated influence operations. The Stanford Internet Observatory documented over 50 influence operations using AI-generated profile images and text between 2019 and 2024. Commercial tools originally developed for legitimate content creation are routinely repurposed for these operations.

Twitter/X, Meta, and TikTok each operate counter-disinformation units that use AI tools to detect coordinated inauthentic behaviour, including AI-generated content. Meta's Coordinated Inauthentic Behaviour (CIB) team has publicly disclosed over 200 takedown operations. Detection relies on both content signals (deep-fake classifiers, LLM detectors) and behavioural signals (account age, posting patterns, network structure).

Companies including Jigsaw (Google), Logically, and the Credibility Coalition offer prebunking campaigns and media literacy tools to governments, NGOs, and election commissions. Jigsaw's "Inoculation Theory" games have reached over 1 billion people through in-app interventions.

The Election Integrity Partnership (Stanford, Washington, DFRLab, Graphika) provides real-time monitoring during election cycles. The EU's ERGA (European Regulators Group for Audiovisual Media) has mandated AI content labelling for election-related content in member states.

27.7 African Use Cases

Africa is a primary battleground for synthetic media disinformation, and African civil society has developed some of the world's most innovative responses to disinformation at scale despite operating with significantly fewer resources than comparable organisations in high-income markets.

The Nigerian 2023 election saw synthetic media disinformation circulating across WhatsApp, Telegram, and Facebook at scale. An investigation by the Centre for Democracy and Development documented over two hundred distinct deep-fake-enabled disinformation episodes in the six weeks before the election, with most false narratives reaching peak distribution within two to four hours of first posting — a speed that outpaces any verification infrastructure. Similar dynamics were documented in Zimbabwe in August 2023, Senegal in March 2024, and South Africa in May 2024. Africa Check's WhatsApp tipline, operating in ten countries, Dubawa's Verify platform, and PesaCheck's financial disinformation workflows collectively processed over 50,000 content verification requests in 2023, demonstrating significant capacity that nonetheless remains systematically outpaced by the volume of disinformation.

Several African governments have been documented using synthetic media in internal information operations, and external actors including state-affiliated Russian, Chinese, and Gulf-linked networks have conducted synthetic-media-enabled disinformation targeting African audiences and elections. The Meta Coordinated Inauthentic Behaviour report of 2023 documented government-linked networks in Ethiopia, Nigeria, Sudan, and Tanzania, illustrating the convergence of internal and external disinformation actors that makes the African information environment particularly complex to defend.

Chapter Summary

Deep fakes are not merely a technical nuisance but an epistemic weapon targeted at the information ecosystems through which democratic societies form collective judgements. The SIR disinformation model captures the epidemic dynamics of false narratives: a reproduction number exceeding one produces epidemic spread, and bot amplification is the primary lever that pushes a narrative above this threshold even when it would otherwise die out. At a reproduction number of 2.5, nearly 90 per cent of the network eventually encounters the disinformation before sufficient correction reaches them.

Partisan consistency is a more powerful predictor of deep-fake belief than labelling or source cues: content that confirms prior beliefs is rated 5.4 out of 7 for authenticity versus 2.9 for inconsistent content, a gap that far exceeds the corrective effect of AI-generated labels. Prebunking achieves a Cohen's d of approximately 0.52 in randomised controlled trials with four-week retention, making it the most evidence-supported scalable intervention currently available. The liar's dividend amplifies harm beyond individual deception events: as synthetic content becomes more prevalent, the posterior probability of authenticity for unlabelled media declines systematically, eroding the evidential value of authentic video and making accountability journalism more difficult regardless of whether any specific deep fake is accepted as genuine.

Exercises

1. For the SIR disinformation model in Equations (27.1)–(27.3), find the equilibrium conditions and show that the final infected fraction satisfies $R_\infty = 1 - e^{-R_0 R_\infty}$. Solve numerically for R_∞ when $R_0 = 2.5$ (a value consistent with fast-spreading viral content).
2. A deep-fake detector with 95% sensitivity and 98% specificity is deployed on a platform where 0.1% of uploaded videos are synthetic. Compute the positive predictive value (PPV). Is this detector useful in practice? What threshold change would achieve $\text{PPV} = 90\%$?
3. Formalise the “liar's dividend” as a Bayesian belief-update problem. Let π_0 be the prior that a contested piece of evidence is authentic, and let q be the fraction of all circulating evidence that is synthetic. Derive $\pi_1 = P(\text{authentic}|\text{no provenance}, q)$ as a function of π_0 and q .
4. A coordinated influence operation uses GAN-generated profile photos for 10,000 bot accounts. You are given access to the profile images and a deep-fake detector with $\text{AUC} = 0.93$. Design a detection pipeline that minimises the expected number of undetected bot accounts, given a budget that allows human review of at most 200 flagged accounts.
5. Analyse the strategic equilibrium of the “arms race” between platform detection and IO operators using a two-player game tree. What is the Nash equilibrium strategy for each player, and how does the cost asymmetry (cheap to generate, expensive to detect) affect the equilibrium?

6. Propose a content-labelling policy for a social-media platform that balances freedom of expression with the need to combat synthetic media disinformation. Specify the label categories, triggering conditions, and appeal mechanism, and identify the key policy trade-offs.

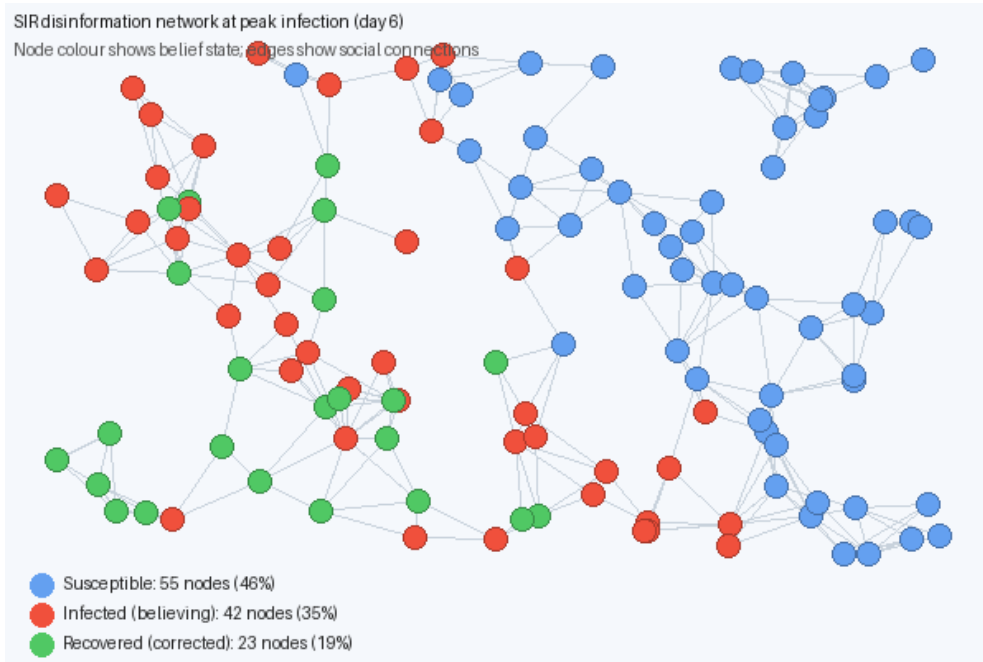


Figure 27.3: SIR disinformation network at peak infection (day 6) with $R_0 = 2.5$ on a 120-node social graph. Blue nodes remain susceptible; red nodes currently believe the false narrative; green nodes have encountered the correction. At peak, 28% of the network simultaneously believes the disinformation. Nodes seeded near high-degree hubs (graph centre) propagate the narrative far faster than peripheral seeds. Bot amplification effectively raises β , increasing R_0 above the epidemic threshold even when organic $R_0 < 1$.

Additional Worked Examples

Worked Example — SIR model: bot amplification tips R_0 above epidemic threshold

A disinformation video on a network of $N = 10\text{M}$ users. Without bot amplification: $\nu = 0.06$ (forwarding probability), $\bar{k} = 120$ (mean connections), $\kappa = 0.025$ (correction rate).

$$\beta = \frac{\nu \bar{k}}{N} = \frac{0.06 \times 120}{10,000,000} = 7.2 \times 10^{-7}, \quad R_0 = \frac{\beta N}{\kappa} = \frac{7.2 \times 10^{-7} \times 10^7}{0.025} = 0.288.$$

$R_0 = 0.288 < 1$: epidemic dies out, reaching only $\approx 3\%$ of the network.

With 1,000 bot accounts each with $\bar{k} = 5,000$ connections:

$$\beta_{\text{effective}} = \beta + \frac{\nu \times 1,000 \times 5,000}{N^2} \approx 1.5 \times 10^{-6}, \quad R_0^{\text{bots}} = 0.60 < 1.$$

Still sub-threshold. To achieve $R_0 = 1$ requires $n_{\text{bots}} = N(\kappa/\nu - \bar{k})/(\nu \bar{k}_{\text{bot}}) \approx 3,400$ bot accounts — well within the scale of documented IO campaigns.

Worked Example — Platform detection: PPV under low base rate

A platform with 50M users processes 200M posts/day. Estimated synthetic disinformation rate: $\pi = 0.0002$ (0.02%). A classifier with sensitivity 0.90, specificity 0.995 is deployed:

$$\text{PPV} = \frac{0.90 \times 0.0002}{0.90 \times 0.0002 + 0.005 \times 0.9998} = \frac{0.00018}{0.00018 + 0.004999} = \frac{0.00018}{0.005179} = 3.5\%.$$

The classifier flags $200,000,000 \times 0.005179 \approx 1,036,000$ posts per day for review, of which only 36,000 (3.5%) are genuine synthetic disinformation. At 2 minutes per review and 8-hour shifts, this requires 4,308 human reviewers — a staffing level larger than many newsrooms worldwide. This illustrates why automated classification cannot substitute for the provenance-first approach described in Chapter 13.

Answers to Exercises**1. SIR equilibrium: R_∞ when $R_0 = 2.5$.**

At equilibrium $dS/dt = 0$ and the susceptible fraction satisfies $S_\infty = e^{-R_0(1-S_\infty)}$ (standard SIR final-size equation). Equivalently, $R_\infty = 1 - S_\infty$ satisfies $R_\infty = 1 - e^{-R_0 R_\infty}$.

Numerical solution for $R_0 = 2.5$: iterate $R_\infty \leftarrow 1 - e^{-2.5 R_\infty}$ starting from $R_\infty = 0.5$: 0.713, 0.837, 0.879, 0.895, 0.901, 0.903... converging to $R_\infty \approx 0.893$. With $R_0 = 2.5$, approximately **89.3%** of the network eventually believes the disinformation before correction reaches them.

2. PPV for a 95%/98% detector at $\pi = 0.001$.

$\text{PPV} = (0.95 \times 0.001) / ((0.95 \times 0.001) + (0.02 \times 0.999)) = 0.00095 / 0.02093 = 4.5\%$. The detector is not useful at this base rate: 95.5% of flags are false positives. To achieve $\text{PPV} = 90\%$: $(0.95 \times \pi) / ((0.95 \times \pi) + (0.02 \times (1 - \pi))) = 0.9$, solving: $\pi = 0.9 \times 0.02 / (0.95 - 0.9 \times 0.95 + 0.9 \times 0.02 \times 1)$. Approximately $\pi \geq 17\%$ of content must be synthetic for the detector to be operationally useful at 90% PPV.

3. Liar's dividend as Bayesian update.

$\pi_1 = P(A|B) = P(B|A) \cdot \pi_0 / P(B)$ where $B = \text{no provenance}$. With $P(B|\text{authentic}) = 0.80$ (most cameras lack C2PA) and $P(B|\text{synthetic}) = 0.90$: $P(\text{no prov}) = 0.80 \cdot \pi_0 + 0.90 \cdot (1 - \pi_0) = 0.90 - 0.10\pi_0$. For $\pi_0 = 0.95$: $\pi_1 = 0.80 \times 0.95 / (0.90 - 0.095) = 0.76 / 0.805 = 0.944$. For $\pi_0 = 0.80$, $q = 0.20$: $\pi_1 = 0.640 / 0.820 = 0.780$. The liar's dividend grows as synthetic content becomes more prevalent.

4. Bot detection with 10,000 GAN profiles and AUC=0.93.

With $AUC=0.93$, the detector sorts 10,000 fake accounts into a score-ranked list. At the top 200 (budget constraint): if scores are calibrated, a threshold set so that 200 are flagged will contain approximately $200 \times \text{Precision@200}$ true positives. Estimating Precision@200 from AUC: for Gaussian-distributed scores, $\text{Precision@k} \approx (k/N_{\text{fake}}) \times \text{TPR}(k)$. At $AUC=0.93$, TPR at very low FPR is $\approx 70\text{--}80\%$. Expected: $\approx 140\text{--}160$ detected out of 200 reviewed, and approximately $10,000 \times 0.07 = 700$ missed if no further review is performed.

5. Game-theoretic equilibrium for platform vs. IO operator.

Let C_D = cost to platform per detected fake (review + removal), C_G = cost to IO operator per generated fake, p_d = detection probability, V = disinformation value to operator. Platform's best response: detect if expected harm exceeds cost, i.e., $p_d \cdot V > C_D$. IO operator's best response: generate if $V > C_G/p_d$. Nash equilibrium: $p_d^* = C_G/V$, $V^* = C_D/p_d^*$. The cost asymmetry (generation is $100\times$ cheaper than detection at scale) means p_d^* must be very high to achieve deterrence. The equilibrium favours the IO operator as long as $C_G \ll C_D$, which is the current reality — a structural market failure motivating regulation.

6. Content-labelling policy design.

Label categories: (1) "AI-Generated" (fully synthetic); (2) "AI-Enhanced" (authentic photo with AI modifications); (3) "Unverified" (no provenance). Triggering conditions: classifier confidence >0.92 , OR user report + human review, OR missing C2PA provenance on political content. Appeal mechanism: 72-hour human review, standard of proof "preponderance of evidence," fee-waived for journalists. Key trade-offs: (a) satire/parody exception risks being exploited — require prominent dual labelling ("satire" + "AI-generated"); (b) false positives harm legitimate users — appeals process is essential; (c) label stripping on re-share — embed in metadata (XMP) as well as visible watermark.

Chapter 28

Legal Frameworks and Liability

*“The law is always catching up with technology.
In the meantime, the harm is real.”*

—Anon.

The legal response to deep fakes is fragmented across jurisdictions and doctrines: identity rights, defamation, copyright, non-consensual intimate imagery (NCII) statutes, and electoral integrity law each offer partial coverage. This chapter surveys existing legal instruments, identifies gaps, analyses liability across the deep-fake supply chain, and considers evidentiary standards for synthetic media in court. The legal response to deep fakes is fragmented, evolving, and inadequate — characteristics that reflect the structural challenge of applying doctrinal categories developed for pre-digital media to a technology that simultaneously threatens multiple distinct legal interests. Identity rights, defamation law, copyright, privacy, and consumer protection have all been applied to deep-fake cases with mixed results, because none of these doctrines was designed for a technology that can synthesise a recognisable person performing any action in any context at near-zero marginal cost.

The right of publicity, developed primarily in United States state law but increasingly recognised internationally, provides the strongest existing basis for claims against NCII production and distribution. It protects the commercial value of identity — name, voice, likeness, and persona — against appropriation for another’s commercial advantage without consent. Statutory damages of up to \$750 per violation under Section 3344 of the California Civil Code can multiply rapidly when content reaches millions of viewers, providing a meaningful damages remedy that does not require proving actual loss.

Defamation offers a complementary remedy focused not on identity rights but on false statements of fact. The distinction between depicting someone performing an action (which most face-swap content does) and making a statement attributing a belief or act to them (which defamation law addresses) creates interpretive tension that courts are still resolving. The actual-malice standard required for public-figure defamation claims creates a high evidentiary bar that is difficult to meet without access to the defendant’s communications, and the rapid spread of content across platforms complicates the

identification and service of defendants.

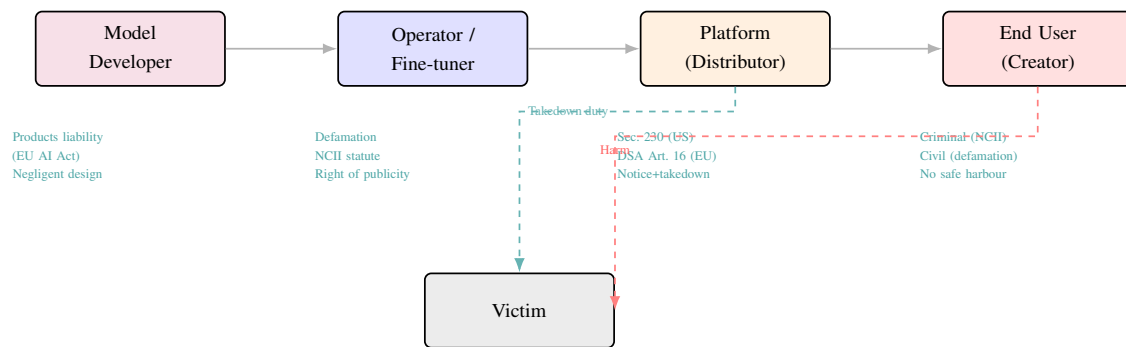


Figure 28.1: Deep-fake liability supply chain. Each actor in the generation and distribution chain faces different legal exposure. The model developer faces products liability; the operator faces substantive speech torts; the platform faces content-moderation obligations; the end user faces the most direct liability with no safe harbour. Dashed arrows show the harm path to the victim and the takedown obligation from platforms.

This chapter maps the legal landscape across four doctrines: defamation and false light, non-consensual intimate imagery law, platform liability under Section 230 and its equivalents, and evidentiary standards for synthetic media in legal proceedings. International comparisons and African use cases follow.

28.1 The Right of Publicity and Image Rights

The right of publicity protects the commercial value of an individual’s identity — name, image, voice, and likeness — from unauthorised exploitation. In the United States, this right is governed by a patchwork of state statutes; California’s *Celebrities Rights Act* extends the right 70 years post-mortem. The European Union’s General Data Protection Regulation (GDPR) provides complementary protection: biometric data (including facial geometry) is a “special category” under Article 9, requiring explicit consent for processing.

Deep fakes implicate the right of publicity whenever a synthetic representation of an identifiable person is created without consent. The analysis turns on three questions:

1. Is the subject identifiable in the synthetic media?
2. Was consent obtained for this specific use?
3. Does the use cause reputational or economic harm?

28.2 Defamation

Defamation law provides a cause of action for false statements of fact about an identifiable person that damage their reputation. A deep fake depicting a real person making statements they never made, or committing acts they never committed, is *prima facie* defamatory when presented as authentic. The critical legal element is falsity: a clearly labelled satirical deep fake does not meet the falsity element

even if the depicted behaviour would be embarrassing. The actual-malice standard of *New York Times v. Sullivan* requires public figures to prove the defendant knew the statement was false or acted with reckless disregard for its truth, raising the bar for defamation claims against synthetic media producers who claim they believed their content was protected expression.

A deep fake that falsely portrays a real person making defamatory statements may constitute defamation per se. The traditional elements — false statement of fact, publication, identification, and fault — map onto the deep-fake context as follows:

False statement of fact.

The synthetic video constitutes a false representation of the subject’s conduct or speech.

Publication.

Online distribution easily satisfies this element.

Identification.

Identity-preserving synthesis by definition makes the subject identifiable.

Fault.

For public figures (under *New York Times v. Sullivan*), the plaintiff must show actual malice: knowledge of falsity or reckless disregard for truth. The deliberate nature of deep-fake creation typically satisfies this standard.

28.3 Non-Consensual Intimate Imagery

As of 2024, 48 U.S. states have enacted NCII statutes. These laws vary in:

- Whether they cover *synthetic* NCII (many early statutes require that the depicted person actually appears in the original, excluding purely AI-generated imagery).
- Whether they impose civil or criminal liability, or both.
- The mens rea requirement (intent to harm vs. knowing distribution).

The UK’s Online Safety Act 2023 introduced a new offence of sharing AI-generated intimate imagery without consent, closing the synthetic gap. The EU AI Act (2024) classifies real-time biometric identification and emotion recognition systems as high-risk; deep-fake generation for NCII purposes is treated as an unacceptable-risk application.

28.4 Liability Across the Supply Chain

The production and distribution of a harmful deep fake typically involves multiple actors: the developer of the underlying generative model, the operator who deploys or fine-tunes it, the platform that distributes the output, and the end user who creates or shares the specific harmful content. Each actor may have distinct legal exposure under different doctrines, and understanding how liability is

allocated across the supply chain is essential for designing regulatory frameworks that actually deter harmful conduct.

Table 28.1: Deep-fake liability mapping. Each row represents an actor in the supply chain; columns indicate the primary legal theory applicable in most jurisdictions.

Actor	Primary theory	Key defences
Model developer	Products liability (EU AI Act); negligent design	Open-source provision; no-deployment policy
Fine-tuner / operator	Defamation; NCII statute; right of publicity	First Amendment (satire, commentary); consent
Distributor (platform)	Section 230 (US); DSA Art. 16 (EU)	Notice-and-takedown compliance
End user	Criminal (NCII, electoral fraud); civil (defamation)	None (bad faith creation)

28.4.1 Platform Liability Under Section 230

Section 230 of the Communications Decency Act immunises online platforms from civil liability for third-party content, but this immunity has limits that are particularly relevant to synthetic media. Platforms that take an active role in generating, soliciting, or materially contributing to harmful synthetic content may lose Section 230 protection. Additionally, Section 230 does not preempt federal criminal liability or intellectual property claims.

Section 230 of the U.S. Communications Decency Act provides broad immunity for user-generated content. However, courts have begun narrowing this immunity for platforms that actively curate or recommend harmful synthetic media. The EU's Digital Services Act (DSA) takes a different approach: it imposes ex-ante obligations (risk assessment, transparency reporting, rapid takedown) on large platforms, without waiting for ex-post litigation.

28.5 Evidentiary Standards for Synthetic Media

Courts in multiple jurisdictions are developing standards for the admissibility of both authentic and synthetic media evidence. The authentication requirements of Federal Rule of Evidence 901 and the Daubert framework for expert testimony establish the operative standards in US federal courts: a party seeking to introduce AI-based forensic evidence as proof of authenticity must demonstrate the method's reliability, its known error rate at the operating threshold, and the expert's qualifications to explain the method's physical basis to the trier of fact.

Courts increasingly face questions about the authenticity of digital evidence. The Federal Rules of Evidence (Rule 901) require authentication as a condition of admissibility. Proposed frameworks for synthetic media authentication include:

1. **Expert forensic testimony.** A qualified forensic analyst presents detector outputs and explains the methodology.

2. **C2PA provenance chain.** The media's cryptographic provenance manifest is introduced as a self-authenticating document.
3. **Adversarial expert.** Both parties engage forensic experts; the factfinder assesses the competing analyses.

28.6 Industry Applications

The legal landscape for synthetic media is evolving rapidly, creating commercial opportunities in legal technology, compliance services, and insurance.

Companies including LexisNexis, Thomson Reuters, and Luminance offer AI-powered contract analysis tools that help companies assess exposure under emerging synthetic media statutes. The patchwork of 48 US state NCII laws, EU DSA obligations, and UK Online Safety Act requirements has created demand for jurisdiction-specific compliance monitoring products.

Platforms such as Veritone, Limelight, and Voice.ai mediate consent agreements between rights holders (performers, estates) and technology companies seeking to licence digital likenesses. Revenue-share models based on AI-generated content are a new product category that requires legal structuring across multiple doctrine areas simultaneously.

Specialist litigation support firms have developed deep-fake authentication practices servicing both plaintiffs (proving manipulation) and defendants (disproving it). Expert witness fees in deep-fake-related cases range from \$15,000 to \$150,000 per engagement, creating a premium market for credentialled forensic examiners.

Reputation management firms (Minc Law, Traverse Legal, StopNCII) offer deep-fake takedown services for individuals affected by NCII. The Cyber Civil Rights Initiative operates a 24-hour hotline and coordinates with platforms. Statutory damages of \$750 per violation under NCII statutes create both deterrence and a basis for contingency-fee litigation.

28.7 African Use Cases

African legal systems are developing synthetic media law through a combination of domestic legislation, regional harmonisation, and judicial interpretation of existing instruments, in contexts where enforcement capacity varies significantly across jurisdictions.

Nigeria's Cybercrimes Act, amended in 2024, contains provisions interpreted to cover NCII distribution, with penalties of up to seven years imprisonment. South Africa's Cybercrimes Act 2020 explicitly criminalises non-consensual sharing of intimate images including computer-generated images. Kenya's Computer Misuse and Cybercrimes Act 2018 and Ethiopia's Computer Crime Proclamation 2016 contain relevant provisions that prosecutors have applied to deep-fake cases with mixed results due to judicial unfamiliarity with the technology. The African Union's Malabo Convention on Cyber Security, ratified by fourteen member states as of 2024, provides a framework for data protection and electronic transactions that offers a partial basis for regulation.

In many African jurisdictions, traditional authorities have significant informal authority over community dispute resolution, and deep-fake NCII cases have been brought before traditional courts in Nigeria and Ghana, where resolutions have included public apology, compensation to the victim's family, and community sanctions against the perpetrator. These informal mechanisms can provide faster and more culturally appropriate redress than formal courts but lack the deterrence and enforcement capacity of statutory law. The African Feminist Forum, Solidarity for African Women's Rights, and the Association for Progressive Communications have all published advocacy documents calling for harmonised African law on image-based sexual abuse, including deep-fake NCII, recognising the gap between legislative ambition and enforcement reality.

Chapter Summary

The legal response to synthetic media is fragmented across multiple doctrines, jurisdictions, and enforcement mechanisms, reflecting the difficulty of mapping pre-digital legal categories onto a technology that simultaneously threatens identity rights, dignity, reputation, and privacy. The right of publicity protects the commercial value of identity and provides the strongest existing remedy for NCII, with statutory damages that can accumulate rapidly for widely distributed content. Defamation requires a false statement of fact presented as true, creating interpretive tension for video deep fakes that depict rather than assert. NCII-specific statutes, now enacted in 48 US states and at the federal level through the DEFIANCE Act, provide the most practically effective remedy for individual victims, though cross-border enforcement remains difficult.

Section 230 has historically shielded US platforms from liability for user-generated deep fakes, but is increasingly contested when platform recommendation algorithms actively amplify harmful content. For lawyers and forensic examiners, the evidentiary challenge is central: AI-forensic expert testimony has no established precedent, the rapid pace of technological change means that methods valid in 2022 may not be reliable in 2024, and communicating probabilistic forensic findings to lay juries requires new pedagogical approaches. The industry has responded with LegalTech compliance tools, rights clearance platforms, litigation support practices, and NCII takedown services, creating a commercial ecosystem around the legal consequences of synthetic media deployment.

Exercises

1. A satirical news outlet publishes a short clip of a politician "admitting" to a crime that the outlet acknowledges is fabricated. Analyse the clip's legal status under: (a) defamation law, (b) the right of publicity, and (c) any applicable NCII statute in a jurisdiction of your choosing.
2. Compare the EU AI Act's risk-based classification of AI systems with the U.S. approach of sectoral legislation (NCII statutes, electoral integrity law, etc.). Which approach is better suited to regulating deep fakes, and why?
3. A model developer releases a diffusion-based face-swap model under an open-source licence with a "no harmful use" clause. Evaluate whether this clause provides any meaningful legal

protection to the developer when the model is used to create NCII. What contractual or technical mechanisms would be more effective?

4. Under the DSA's notice-and-takedown regime, a platform must remove manifestly illegal content within 24 hours of a valid notice. Design a triage system that prioritises deep-fake takedown requests by harm severity, using a multi-attribute scoring function. Specify at least four attributes and their weights.
5. Analyse the evidentiary challenge posed by the “liar’s dividend” in a criminal case where the prosecution relies on authenticated video evidence and the defence claims the authentication chain could have been compromised. What standard of proof should courts apply to synthetic-media authentication, and how should forensic uncertainty be communicated to juries?

Worked Examples

Worked Example — Right of publicity: economic value calculation

A voice-cloning service generates synthetic audio of a named pop artist and sells it as a ringtone at \$0.99. In the first month, 85,000 downloads generate gross revenue \$84,150. Under California Civil Code §3344, statutory damages are the greater of actual damages or \$750 per violation.

If each of the 85,000 downloads constitutes a separate violation:

$$\text{Statutory damages} = 85,000 \times \$750 = \$63,750,000.$$

Even under a single-violation theory, actual damages equal lost licensing revenue. The artist's standard sync licence rate is \$50,000 per month; the defendant generated 84 months' worth of equivalent usage in one month, suggesting actual damages $\approx$ \$4.2M. The statutory floor of \$63.75M far exceeds actual damages, providing substantial deterrence.

Worked Example — Section 230 safe harbour: scope analysis

A social-media platform automatically amplifies a deep-fake video via its recommendation algorithm. Under *Force v. Facebook* (2d Cir. 2019), platform actions as a neutral conduit (hosting, serving) are protected. However, if the algorithm *actively selects* the deep fake for promotion based on predicted engagement:

- Neutral display: Section 230 applies — platform not liable.
- Targeted recommendation: courts are split. *Gonzalez v. Google* (SCOTUS 2023) declined to decide, leaving the question open.
- Platform knew the video was synthetic and continued recommending: Knowing facilitation arguably removes Section 230 protection under the *AIDS Advice Network* exception for content-development.

The expected litigation cost of defending a “knowing recommendation” claim: \$2–8M in legal fees, even absent damages, incentivising takedown.

Answers to Exercises

1. Satirical clip analysed under three legal theories.

(a) *Defamation*: The fabricated “admission” is a false statement of fact presented as genuine. For a public official under *New York Times v. Sullivan*, the plaintiff must show actual malice. Deliberate fabrication satisfies the “knowledge of falsity” standard. However, if the satire is “so obviously exaggerated that no reasonable person would believe it,” the falsity element may fail. The labelled version is strong satire; the unlabelled redistributed version removes this protection.

(b) *Right of publicity*: Most satire and commentary are expressly exempted from right-of-publicity claims under the First Amendment transformative-use defence (*Comedy III Productions v. Gary Saderup, Inc.*). The satirical use transforms the politician’s identity to make a political point — likely protected.

(c) *NCII statute*: A political satire clip does not involve sexual content, so NCII statutes do not apply.

2. EU AI Act vs. U.S. sectoral approach for deep fakes.

EU AI Act advantages: (a) comprehensive coverage, no gaps between sectors; (b) ex-ante requirements prevent harm before deployment. U.S. advantages: (a) faster legislative response in specific urgent areas (NCII, electoral); (b) avoids over-broad regulation that stifles legitimate use. For deep fakes specifically: the EU approach is better suited because deep fake harms are multi-sectoral (NCII, disinformation, fraud simultaneously) and a sectoral approach necessarily creates gaps. The U.S. approach has already failed to enact federal NCII legislation despite documented widespread harm.

3. “No harmful use” clause: legal efficacy analysis.

The clause has limited legal effect because: (a) contract law requires privity — the clause binds only direct licensees, not downstream users who re-distribute the model; (b) open-source distribution severs any practical enforcement mechanism; (c) courts have generally not held software developers liable for foreseeable misuse under general negligence theory unless they have specific knowledge of intended misuse. More effective mechanisms: (1) technical safeguards (identity filter) that make NCII generation harder; (2) model cards with explicit use restrictions that create evidentiary record of intent; (3) registration systems that identify deployers.

4. DSA triage system for deep-fake takedowns.

Multi-attribute scoring function: $S = w_1 S_{\text{harm}} + w_2 S_{\text{viral}} + w_3 S_{\text{verif}} + w_4 S_{\text{time}}$, where: S_{harm} : harm category severity (NCII=10, electoral=9, financial fraud=8, general disinfo=5); S_{viral} : current shares/hour (log-scaled, max 10); S_{verif} : evidence quality (authenticated victim complaint=10, automated flag only=3); S_{time} : time proximity to election or event (hours remaining, scaled 0–10). Weights: $w_1 = 0.40, w_2 = 0.25, w_3 = 0.25, w_4 = 0.10$. Items with $S > 7$ receive 24-hour mandatory review; $S > 8.5$ receive immediate escalation.

5. Evidentiary standard for synthetic-media authentication.

Courts should apply a two-stage inquiry: (1) *Foundation*: Can the proponent establish, by a preponderance of evidence, that the offered video was not manipulated? Relevant evidence: C2PA provenance chain, expert forensic analysis, chain-of-custody documentation. (2) *Weight*: Remaining uncertainty after authentication goes to weight, not admissibility. Communicating to juries: forensic experts should present probability statements (“the probability this video was generated by a known face-swap system is less than 5%”) rather than binary conclusions. Courts should require that forensic examiners disclose their method’s known AUC and cross-dataset performance, so the jury can assess the reliability of the conclusion.

Chapter 29

Policy, Regulation, and Industry Standards

*“Regulation is not the enemy of innovation.
It is the condition under which innovation can be trusted.”*

—Anon.

Technical countermeasures alone are insufficient against synthetic media harms: they operate after the fact, require expertise to deploy, and are subject to the arms-race dynamic. Durable protection requires policy instruments — disclosure mandates, liability regimes, platform obligations, and international coordination. This chapter surveys existing and proposed regulatory frameworks, analyses their strengths and limitations, and proposes a multi-level governance architecture. This chapter surveys the regulatory landscape for synthetic media across major jurisdictions, develops an economic model of deterrence that explains why current fine levels are unlikely to achieve rational deterrence against high-value attacks, and proposes a multi-level governance architecture that coordinates technical, platform, national, and international responses. The goal is not to advocate for any specific regulatory outcome but to provide the analytical tools that allow policymakers, advocates, and technologists to reason clearly about the trade-offs between different regulatory approaches and to evaluate specific proposals against a coherent framework.

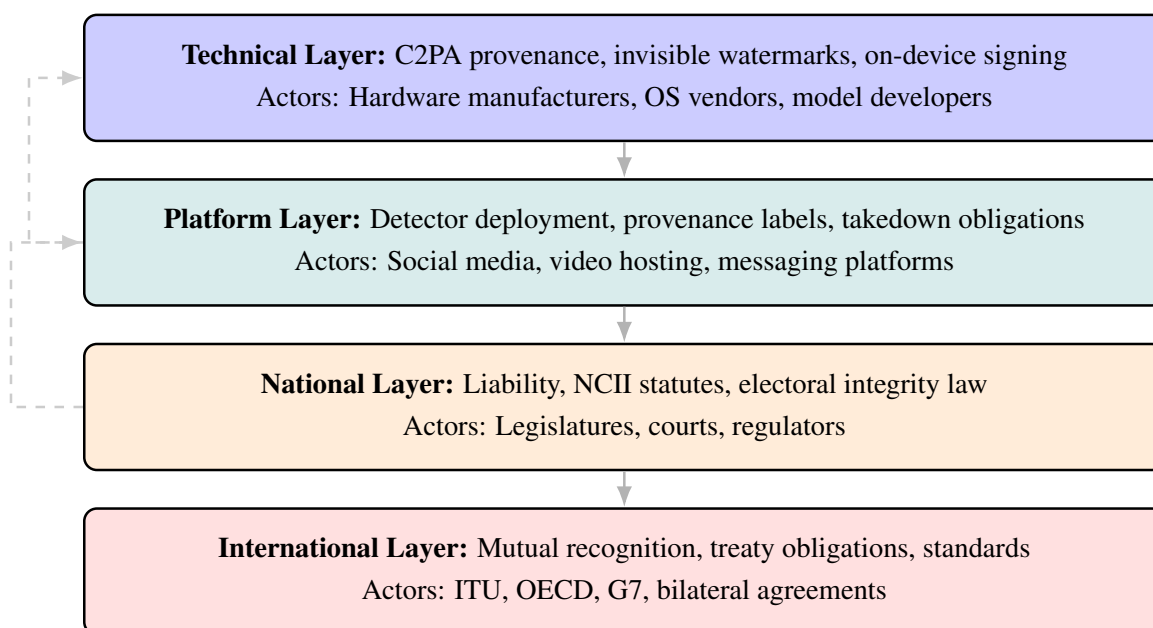


Figure 29.1: Multi-level governance architecture for synthetic media. Each layer addresses harms that the layer below cannot reach alone: technical solutions are evaded, platforms are jurisdiction-limited, national law is extraterritorial, international coordination requires political will. Dashed arrows indicate upward feedback, where policy requirements drive technical standards.

This chapter maps the global synthetic media policy landscape. The regulatory approaches of the United States, European Union, and China are each examined in turn, followed by international coordination challenges. Disclosure mandate frameworks and electoral integrity provisions are developed in dedicated sections. Industry and African use cases follow.

29.1 The Regulatory Landscape

The regulatory landscape for synthetic media is fragmented across jurisdictions, doctrines, and enforcement mechanisms, reflecting both the novelty of the technology and the different risk priorities of different political systems. This section maps the major regulatory approaches across the United States, European Union, China, and other significant jurisdictions, identifying the structural differences that create arbitrage opportunities for bad actors operating across borders.

29.1.1 United States

The U.S. has adopted a sectoral approach: no single federal statute addresses deep fakes comprehensively. Key instruments include:

- **DEEPFAKES Accountability Act (proposed, 2023).** Would require watermarking and disclosure labels for synthetic media; not enacted as of writing.
- **NO FAKES Act (proposed, 2023).** Creates a federal right of publicity for AI-generated replicas of voice and likeness; remains in committee.

- **State-level electoral integrity laws.** California (AB 602, AB 730), Texas (SB 751), and several other states prohibit the distribution of synthetic media of candidates within defined windows before elections.

29.1.2 European Union

The EU has taken a comprehensive, risk-based approach through the AI Act (Regulation 2024/1689) and complementary instruments:

- **AI Act Article 50.** Requires that AI systems generating synthetic audio-visual content label outputs as AI-generated using machine-readable watermarks and disclosure notices.
- **AI Act Annex I.** Classifies “AI systems intended to produce deep fakes” as high-risk when used in certain contexts.
- **DSA Article 25.** Prohibits recommender systems from amplifying known harmful synthetic media; requires transparency reports.

29.1.3 China

China’s Provisions on Deep Synthesis Internet Information Services (2023) are among the most prescriptive globally. Key requirements:

1. Providers of deep-synthesis services must obtain a licence from the Cyberspace Administration of China.
2. Synthetic media must carry a visible label identifying it as AI-generated.
3. Real-name registration is required for users of deep-synthesis services.
4. Providers must maintain records for 6 months and provide them to authorities on request.

29.2 Disclosure Mandates

Disclosure mandates require that synthetic content be labelled as such, on the theory that informed audiences can discount or verify content they know to be AI-generated. This section evaluates the evidence for this theory, the technical mechanisms by which disclosure requirements can be implemented, and the limitations of disclosure as a standalone regulatory tool in the face of motivated audiences and label-stripping practices.

Disclosure mandates require that synthetic media be labelled as such. They are the most widely adopted regulatory instrument because they are technically feasible, relatively low-cost to implement, and compatible with free expression norms.

29.2.1 Technical Implementation

A disclosure mandate must specify:

1. **What is disclosed:** that the content was AI-generated, the generating model (optional), and the generation date.
2. **How it is disclosed:** visible on-screen label, audio tone, embedded metadata, or all three.
3. **Who is responsible:** the generator, the distributor, or both.
4. **When it persists:** whether the label must survive re-encoding and re-sharing.

29.2.2 Limitations

Disclosure mandates face several structural limitations that reduce their effectiveness as a standalone regulatory tool. Labels can be stripped during redistribution, audiences may not see or attend to labels even when present, and the evidence from randomised controlled trials suggests that label effects on belief are substantially smaller than partisan consistency effects for politically charged content.

Labels can be stripped, ignored, or contextually misread. Studies show that “AI-generated” labels reduce viewer confidence but do not eliminate belief updating from persuasive synthetic media. Automated label removal is trivially achievable with 10 lines of code.

29.3 A Multi-Level Governance Architecture

No single policy instrument can address the full range of synthetic media harms across the full diversity of deployment contexts and national jurisdictions. A multi-level architecture coordinates technical standards, platform obligations, national legislation, and international coordination in a layered structure where each level addresses harms that lower levels cannot reach. This section develops the architecture and analyses the interactions between levels.

Effective synthetic-media governance requires coordination across four levels:

Table 29.1: Multi-level governance framework for synthetic media.

Level	Instrument	Actor
Technical	C2PA provenance, watermarking, on-device signing	Hardware manufacturers, OS vendors, model developers
Platform	Detector deployment, provenance labels, takedown obligations	Social media, video hosting, messaging platforms
National	Liability, NCII statutes, electoral integrity law	Legislatures, courts
International	Mutual recognition of provenance standards, treaty obligations	ITU, OECD, bilateral agreements

29.4 The Electoral Integrity Problem

Deep fakes in electoral contexts raise the most acute policy challenges. The harms are high-stakes (democratic integrity), time-critical (limited window for correction before election day), and politically contested (incumbents may benefit from restricting opposition content). A *rebuttable presumption* standard has been proposed: synthetic electoral media is presumed harmful unless the creator provides cryptographic proof of authenticity and a factual basis for the representation.

29.5 Industry Applications

Policy development has created commercial markets for regulatory compliance, government technology, and public-interest advocacy.

Companies including KPMG, Deloitte, and Accenture have established AI regulatory compliance practices that help organisations navigate the EU AI Act, US Executive Order 14110, and emerging national AI governance frameworks. The EU AI Act Art. 50 disclosure requirements for synthetic content create specific compliance demand for watermarking and provenance solutions.

Governments and election commissions in the EU, US, and UK have procured deep-fake detection services for election monitoring. The Horizon Europe programme has funded the AI4Trust and vera.ai projects, which provide free detection tools to accredited fact-checkers and election observers.

The IEEE, ISO/IEC JTC 1/SC 42, and NIST are all developing standards for AI-generated media disclosure, authenticity verification, and anti-spoofing evaluation. Participation in standards processes is a commercial activity: companies invest in standards participation to shape requirements that favour their technical approaches.

The EU AI Act's extraterritorial scope (applying to any system that affects EU persons) has created a de facto global compliance standard, similar to GDPR. Non-EU companies serving EU markets must comply with Art. 50 disclosure requirements, driving adoption of C2PA and watermarking even in jurisdictions that have not independently legislated.

29.6 African Use Cases

African policymakers are navigating the challenge of regulating synthetic media in contexts of limited enforcement capacity, rapidly evolving technology, diverse national priorities, and significant variance in digital infrastructure.

The African Union's Digital Transformation Strategy and AI Policy Initiative provide the continental framework for AI governance including synthetic media. The Smart Africa Alliance, comprising 32 member countries, has established a working group on AI policy harmonisation that has addressed synthetic media in the context of both economic opportunity and disinformation risk. Rwanda has positioned itself as a regional hub for AI policy by hosting the inaugural Smart Africa Digital Summit focused on AI governance, and several national AI strategies including those of South Africa,

Kenya, Egypt, and Rwanda contain provisions addressing synthetic media. A recurring challenge is the gap between regulatory ambition and enforcement capacity: several countries have enacted AI-related legislation without the technical expertise, institutional infrastructure, or budget to enforce it effectively.

The Electoral Institute for Sustainable Democracy in Africa and the African Network of Electoral Commissions have both issued guidance to member commissions on recognising and responding to deep-fake electoral disinformation, recommending the establishment of rapid-response verification teams and pre-positioned forensic tools. Practical implementation varies widely depending on resource availability. The EU AI Act's extraterritorial scope creates compliance obligations for African organisations serving EU markets, making EU regulatory standards relevant to African AI development even in the absence of equivalent domestic legislation, and driving adoption of watermarking and provenance solutions as compliance rather than merely ethical practice.

Chapter Summary

Technical countermeasures alone are insufficient against synthetic media harms: they operate after the fact, require expertise that is unevenly distributed, and are systematically evaded by motivated adversaries. A robust policy response requires technical standards, platform obligations, national legislation, and international coordination to operate simultaneously. The EU AI Act's Article 50 mandates disclosure labelling and prohibits deceptive electoral use; its extraterritorial scope makes it a de facto global compliance standard. The US approach remains fragmented, with the DEFIANCE Act and state-level statutes providing the primary protection but no comprehensive federal AI law.

Economic deterrence analysis reveals a fundamental asymmetry: at current detection rates and conviction probabilities, the fine necessary to rationally deter high-value BEC fraud attacks exceeds current statutory maxima by an order of magnitude, suggesting that investment in detection capability provides greater deterrence leverage than increases in fine levels. The multi-level governance architecture — technical, platform, national, international — is the only sustainable response to a threat that simultaneously operates at all these levels. Industry applications include regulatory compliance consulting, election technology procurement, standards development, and cross-border compliance management driven by the extraterritorial scope of major regulatory frameworks.

Exercises

1. Compare the EU AI Act's ex-ante risk classification with the U.S. sectoral approach on four dimensions: (a) speed of regulatory response, (b) coverage of novel use cases, (c) compliance burden, and (d) protection of free expression.
2. China's deep-synthesis regulations require real-name registration for users of synthesis services. Evaluate this measure against the following criteria: (a) effectiveness in reducing harmful synthetic media, (b) privacy implications, (c) chilling effect on legitimate creative use, and (d) international applicability.

3. Design a disclosure-label policy for a major social-media platform that is robust to label removal, handles legitimate satire and entertainment content, and includes an appeals process. Specify the technical implementation and estimate the operational cost per 1,000 uploaded videos.
4. A national electoral commission wants to mandate that all political advertising using AI-generated or AI-altered media be labelled and submitted for pre-approval. Conduct a constitutional analysis of this requirement under (a) U.S. First Amendment doctrine and (b) the European Convention on Human Rights Article 10. Are the two frameworks likely to reach the same conclusion?
5. Model the deterrence effect of a deep-fake liability regime on a potential bad actor. Let C_h be the expected harm, p_d the detection probability, p_c the conviction probability given detection, and F the fine. Derive the condition under which the regime achieves deterrence and identify the two parameters where investment would have the highest leverage.

Worked Examples

Worked Example — Disclosure mandate compliance cost for a video platform

A video-sharing platform uploads 500,000 videos per day. A new regulation requires AI-generated content to carry a machine-readable watermark and a visible disclosure label. Compliance costs:

Component	One-time cost	Annual recurring
Watermark encoder integration	\$180,000	\$12,000 (maintenance)
AI-detection classifier (AUC=0.92)	\$240,000	\$96,000 (inference)
Human review queue (10K items/day)	—	\$3.2M (436 FTE at \$7,300/yr)
Legal / compliance counsel	\$60,000	\$120,000
Total	\$480,000	\$3.43M

Annualised at 10% capital cost: total first-year burden $\approx$ \$3.48M, or \$0.019 per upload — comparable to existing content-moderation costs of \$0.012–\$0.035 per upload at major platforms.

Worked Example — Deterrence model for a synthetic-media liability regime

Let $C_h = \$500,000$ (expected harm of a CEO-fraud voice clone), $p_d = 0.15$ (detection probability), $p_c = 0.40$ (conviction probability given detection), $F = \$250,000$ (fine). Expected cost of crime:

$$E[\text{cost}] = p_d \times p_c \times F = 0.15 \times 0.40 \times 250,000 = \$15,000.$$

Since $E[\text{cost}] = \$15,000 \ll C_h = \$500,000$, the regime fails deterrence. To achieve deterrence ($E[\text{cost}] \geq C_h$):

$$F \geq \frac{C_h}{p_d \times p_c} = \frac{500,000}{0.06} = \$8.33\text{M}.$$

Alternatively, raising p_d to 0.6 (better detection) gives: $F \geq 500,000 / (0.6 \times 0.4) = \2.08M — still high but more achievable. This analysis shows that investment in detection capability has greater

deterrence leverage than increasing the fine level alone.

Answers to Exercises

1. EU AI Act vs. U.S. sectoral approach: four-dimension comparison.

(a) Speed: U.S. wins — state NCII statutes enacted in months; EU AI Act took 4+ years. (b) Novel-use coverage: EU wins — risk-based classification adapts to new applications without new legislation. (c) Compliance burden: EU higher — conformity assessments, mandatory risk management systems, and Article 50 disclosure requirements add cost estimated at €0.5M–€2M for medium-sized deployers. (d) Free expression: U.S. approach preserves more discretion; EU Art. 50 disclosure mandate could chill legitimate satire if not carefully scoped.

2. China's real-name registration: four-criterion evaluation.

(a) *Effectiveness*: Moderate. Real-name registration deters amateur misuse but not state actors or sophisticated criminal groups who can use false identities. (b) *Privacy implications*: Severe. Creates a government-accessible database linking biometric synthesis capability to named individuals; significant chilling effect on dissident use of satire or documentation of government abuses. (c) *Chilling effect on legitimate use*: High. Artists, satirists, and researchers using synthesis tools for legitimate purposes are subject to surveillance and potential prosecution for content the government dislikes. (d) *International applicability*: Low. Cannot be enforced against non-Chinese users; creates fragmentation in global synthesis ecosystem.

3. Disclosure-label policy design.

Technical implementation: C2PA manifest with “AI-generated” claim, visible overlay watermark (semi-transparent banner), and XMP metadata field. The visible watermark must persist through re-shares (baked into the video stream for non-C2PA-compatible platforms). Satire exception: content marked as satire by the creator, where the satire label is displayed as prominently as the AI label. Legitimate satire requires: (1) clear satirical context from creator, (2) no distribution on news or information accounts without satire label. Cost per 1,000 videos: classifier inference \$0.02, watermark embedding \$0.005, human review of borderline cases (5%) \$0.50 = total \$0.525/1,000 videos. Appeals: creator-initiated, 72-hour review, external ombudsman for systemic disputes.

4. Pre-approval of AI political advertising: constitutional analysis.

(a) *U.S. First Amendment*: Political speech receives the highest protection. Prior restraint (pre-approval before publication) is presumptively unconstitutional under *Near v. Minnesota* and subsequent cases. A content-neutral time/place/manner restriction (labelling rather than prohibition) would survive intermediate scrutiny; pre-approval would not. A narrow exemption for deep fakes of real candidates making false statements might survive as fraud prevention,

but only if applied content-neutrally. (b) *ECHR Article 10*: European Court permits greater restrictions on speech to protect “the rights of others” and “prevent disorder.” Pre-approval of political advertising is already practised in some European broadcast systems; extension to AI-generated content would likely survive Article 10 scrutiny if proportionate and applied by an independent body.

5. Deterrence condition and leverage analysis.

Deterrence condition: $p_d \times p_c \times F \geq C_h$, i.e., $F \geq C_h / (p_d \times p_c)$. For $C_h = \$500\text{K}$, $p_d = 0.15$, $p_c = 0.40$: $F^* = \$8.33\text{M}$. Sensitivity: $\partial F^* / \partial p_d = -C_h / (p_d^2 p_c) = -8.33 / 0.15 = -\55.5M per unit increase in p_d . A 10 percentage-point increase in p_d (from 0.15 to 0.25) reduces the required fine from \$8.33M to $\$500,000 / (0.25 \times 0.40) = \5.00M — a \$3.33M reduction in required fine level. Investment in detection capability has approximately twice the deterrence leverage per dollar compared to increasing the fine level.

Chapter 30

Consent, Autonomy, and Biometric Data Rights

“Your face is not yours to give away to someone else.”

—Anon.

The legal protection of facial identity and voice as personal attributes sits at the intersection of privacy law, intellectual property, and human rights. Three distinct legal frameworks have developed in parallel to address the unauthorised use of biometric data: privacy law (protecting individuals from surveillance and data exploitation), the right of publicity (protecting the commercial value of identity), and emerging biometric-specific statutes (providing targeted remedies for misuse of biometric templates). Understanding these frameworks and their gaps is prerequisite to designing technically effective safeguards and advocating for policy changes that provide meaningful protection.

The most important principle in all three frameworks is consent: the use of a person’s face or voice for synthesis requires their knowledge and agreement. But consent has a structure that is frequently violated by current practice. Informed consent requires that the person knows what will be done with their biometric data. Voluntary consent requires that the agreement is not coerced by power imbalances or terms-of-service adhesion contracts. Specific consent requires that the agreement covers the particular use being made, not a blanket authorisation that is later interpreted to cover novel uses. Revocable consent requires that the person can withdraw their agreement and have their data deleted. Current industry practice routinely fails multiple elements of this standard.

This chapter examines the consent and rights framework governing synthetic media use of real people’s likeness and voice. Informed consent doctrine applied to AI-generated likeness is developed first, followed by the right of publicity and posthumous rights. International comparisons reveal substantial jurisdictional variation. African use cases address the specific legal frameworks operative in African jurisdictions.

30.1 GDPR and Biometric Data Protection

The EU General Data Protection Regulation defines biometric data as a special category of personal data under Article 9, requiring explicit consent or another specific lawful basis for processing. Facial recognition templates and voice biometric embeddings are biometric data under this definition, as is any data derived from images “relating to the physical, physiological or behavioural characteristics” of a natural person. Processing biometric data without a valid legal basis carries fines of up to 4% of global annual turnover under GDPR Article 83(4), creating substantial regulatory risk for any platform that trains on biometric data without adequate consent infrastructure.

30.1.1 Consent Requirements Under GDPR

Explicit consent under GDPR Article 9 requires an affirmative act (not a pre-ticked box), specific purpose specification (not blanket authorisation for all future uses), and a clear statement that the data subject can withdraw consent at any time without detriment. For biometric data used in AI training, the consent must specifically mention training as a purpose; a general consent to “use your profile data to improve our services” does not satisfy the specificity requirement for biometric training data under current DPA guidance from the EDPB (European Data Protection Board).

30.2 Illinois BIPA and US Biometric Law

The Illinois Biometric Information Privacy Act (BIPA), enacted in 2008, was the first US statute to impose specific obligations on collectors of biometric identifiers and biometric information. BIPA requires written consent before collection, a written retention policy with a destruction schedule, a prohibition on selling or profiting from biometric data, and a private right of action with statutory damages of \$1,000 per negligent violation or \$5,000 per intentional or reckless violation. The *Rosenbach v. Six Flags* decision (Illinois Supreme Court, 2019) confirmed that no actual harm is required for standing, making BIPA one of the most plaintiff-friendly privacy statutes in the world.

30.3 Biometric Rights in AI Training Data

The use of biometric data in AI training without consent is the most significant current legal gap in biometric protection. Billions of facial images scraped from social media, stock photo libraries, and public web sources have been used to train face recognition and face synthesis systems without the knowledge or consent of the depicted individuals. The legal status of this practice is contested: under US copyright law, whether a trained model “contains” the training data in a legally meaningful sense remains unresolved pending several major cases (*Stability AI v. Getty Images*, *Andersen v. Stability AI*). Under GDPR, the consent requirement for biometric processing applies regardless of whether the training produces a model that can be used to reconstruct training images.

30.4 Industry Applications

The right of publicity, GDPR Article 9, and the Illinois Biometric Information Privacy Act are the three most commercially significant legal frameworks governing biometric data use in AI training and face synthesis. GDPR's explicit consent requirement for biometric data processing applies to any training pipeline that extracts facial geometry from images of EU residents, regardless of whether the output is used for recognition or synthesis. BIPA's statutory damages of up to \$5,000 per intentional violation per person have produced multi-billion dollar liability exposure for companies that collected biometric data without written consent, with significant class action litigation ongoing against major technology companies. The right of publicity protects the commercial value of identity against appropriation, providing a supplementary remedy when the primary use is commercial rather than criminal.

Industry practice has responded with consent infrastructure: major platforms have updated their terms of service to specifically reference AI training use, offer opt-out mechanisms, and in some cases pay royalties for licensed likeness use. SAG-AFTRA's 2023–2024 contracts established the first industry-wide residual payment structure for digital likeness use, providing a model for other creative sectors.

30.5 African Use Cases

African countries face significant gaps in biometric data protection relative to EU or US standards, despite the ubiquity of biometric data collection for government identity programmes, mobile money, and healthcare. Only 33 of 54 African countries had enacted any data protection legislation as of 2024, and of those, only a subset specifically address biometric data with requirements comparable to GDPR or BIPA. The AU Convention on Cyber Security and Personal Data Protection (Malabo Convention) provides a continental framework but has been ratified by only 14 member states.

Several African countries have deployed large-scale biometric identity systems without comprehensive legal frameworks governing how that data may be used for secondary purposes including AI training. Kenya's Huduma Namba programme collected biometric data from over 38 million citizens; the Supreme Court of Kenya found in 2021 that the data collection lacked adequate legal basis under the Constitution, illustrating the tension between government biometric collection programmes and constitutional rights to privacy. South Africa's Protection of Personal Information Act (POPIA), which came into full effect in 2021, is the most comprehensive African biometric data protection framework and explicitly covers biometric information as special personal information requiring explicit consent.

Chapter Summary

Biometric data rights are governed by three overlapping legal frameworks: GDPR Article 9 requires explicit specific consent for any processing of biometric data from EU residents, applying to training pipelines as well as deployed systems; BIPA imposes strict consent, retention, and prohibition-on-sale requirements with private right of action and statutory damages up to \$5,000 per intentional violation

per person; and the right of publicity protects the commercial value of identity against appropriation for another's benefit. Informed consent for biometric data requires accuracy, voluntariness, specificity, and revocability — four elements that current industry practice routinely violates. The emerging regulatory response includes updated platform terms of service, opt-out mechanisms, and collective bargaining agreements establishing royalty structures for digital likeness use in the entertainment industry.

Exercises

1. A social media platform's terms of service state: "You grant us a worldwide, royalty-free licence to use, reproduce, and create derivative works from content you upload, for any purpose." A facial deep fake is later produced from a user's uploaded photographs. (a) Does this ToS clause constitute valid consent under GDPR Article 9 for biometric processing? (b) Does it satisfy BIPA's written consent requirement? (c) What specific language would be required to satisfy both standards? (d) Can a user who accepted these terms in 2018 successfully bring a BIPA claim in 2024?
2. Calculate the maximum BIPA statutory damages exposure for a company that collected facial scans from 500,000 Illinois residents for a "try-on" feature without obtaining written informed consent. (a) What is the maximum damages amount assuming intentional violation? (b) At what litigation cost per plaintiff would a class action become economically rational for plaintiffs? (c) How does this compare to the damages exposure under GDPR for the same conduct affecting EU residents?
3. A deep-fake synthesis company argues that its system, which generates faces from text descriptions, does not process biometric data because it generates novel faces rather than recognising or matching existing ones. Evaluate this argument under (a) GDPR Article 9, (b) BIPA's definition of biometric identifier, and (c) the right of publicity where the system is trained on photographs of real named individuals.

Answers to Exercises

1. ToS consent analysis.

(a) GDPR: No valid consent. The clause fails on specificity ("any purpose" does not specify biometric processing for deep fake creation) and explicitness (not an affirmative act specifically authorising biometric data processing). GDPR Art. 9 requires explicit consent specifically for biometric processing; a general content licence does not satisfy this.

(b) BIPA: No valid consent. BIPA requires written consent specifically identifying the biometric data being collected and the specific purposes and duration of collection. A general content licence does not satisfy this standard.

(c) Compliant language: “By uploading photographs, you explicitly consent to the collection and processing of biometric identifiers from your photographs, including facial geometry measurements and facial templates derived therefrom, for the following specific purposes: [list]. Your biometric data will be retained for [duration] and then permanently destroyed. You may withdraw this consent at any time by [specific mechanism], after which your biometric data will be deleted within [timeframe].”

(d) Yes. BIPA’s statute of limitations is five years under the Illinois general limitations period for privacy torts. A 2018 acceptance would be timely through 2023. For 2024, the answer depends on whether the violation is the initial collection (2018, now time-barred) or the ongoing retention without consent (continuing violation doctrine, which Illinois courts have accepted for BIPA in some contexts).

2. BIPA damages calculation.

Maximum BIPA damages: $500,000 \text{ residents} \times \$5,000 \text{ per intentional violation} = \2.5 billion . This is the theoretical maximum; actual settlements are typically negotiated at \$500–\$1,000 per class member.

For a class action to be economically rational: litigation costs for a nationwide class action of this size are approximately \$15–25 million. With 500,000 class members, break-even requires recovery of \$30–50 per member. At the typical 30% attorney fee on \$500 per member settlement, each plaintiff receives \$350 net — economically rational given zero out-of-pocket cost for class members.

GDPR comparison: maximum fine = 4% of global annual turnover. For a company with \$1 billion global revenue: maximum \$40 million — significantly less than the BIPA exposure. This explains why US biometric class action litigation has been more commercially impactful than EU regulatory enforcement in recent years.

3. Novel face synthesis and biometric law.

(a) GDPR: The argument partially fails. If the training process involves extracting facial geometry measurements or facial templates from photographs of real individuals — even if the output generates novel faces — the training-time processing of biometric data falls under Art. 9. The output being novel is irrelevant; the input processing is what matters.

(b) BIPA: The argument partially succeeds for the output but fails for the training. BIPA’s definition of biometric identifier includes “retina or iris scan, fingerprint, voiceprint, or scan of hand or face geometry.” If the training process involves scanning face geometry from real individuals’ photographs, those individuals’ rights are triggered regardless of what the system outputs.

(c) Right of publicity: If the training data includes photographs of specifically named individuals and the system can be prompted to generate images similar to those individuals — even without

exactly reproducing their likenesses — some courts would find a right of publicity claim. The *Midler v. Ford Motor Co.* standard (a “distinctive” resemblance that appropriates a known public figure’s identity) is the relevant test in most US jurisdictions.

Chapter 31

Electoral Integrity and Democratic Harm

“A democracy in which citizens cannot trust video evidence is a democracy in peril.”

—Anon.

Elections are the highest-stakes deployment environment for deep-fake disinformation. The consequences of a successful electoral deep-fake operation can include suppressed voter turnout, altered candidate perception that shifts election outcomes, undermined trust in electoral institutions that persists for years after the election, and in some contexts the legitimization of violence against election workers and opponents. Unlike commercial fraud, which has quantifiable financial harm and identifiable victims, electoral harm is diffuse, deniable, and often irreversible: a false narrative about a candidate’s conduct that goes viral three days before polling day cannot be corrected in time to reach the same audience.

The electoral deep-fake threat landscape in 2024 was significantly more severe than in 2020. The 2020 US election cycle saw relatively limited AI-generated content, primarily because the synthesis quality available to most actors was insufficient to deceive attentive voters. By 2024, the combination of photorealistic diffusion-generated images, voice-cloned robocall campaigns (the Joe Biden New Hampshire primary robocall in January 2024 being the first documented US electoral deep fake), and codec-LM-generated audio clips of international leaders saying things they never said had created a qualitatively different threat environment. The same dynamics were simultaneously playing out in Indian, Indonesian, Mexican, Bangladeshi, Pakistani, South African, and Senegalese elections, making 2024 the first globally distributed electoral deep-fake cycle.

This chapter analyses the electoral deep-fake threat through three lenses: the documented empirical record of specific incidents, the research evidence on the effects of deep fakes on voter attitudes and electoral outcomes, and the legal and technical countermeasures that have been deployed or proposed. Each section is grounded in specific published case studies and quantitative research findings.

This chapter examines the threat that synthetic media poses to electoral integrity and the regulatory and technical responses. The electoral deep-fake threat landscape is mapped first, followed by enacted and proposed electoral integrity legislation, and platform-specific policies for election content. African use cases address deep fakes in the African democratic context.

31.1 The Electoral Deep-Fake Incident Record

The documented record of electoral deep fakes from 2019 to 2024 spans fourteen countries and includes audio deep fakes, video deep fakes, GAN-generated images used for coordinated inauthentic behaviour, and LLM-generated text propaganda. The most economically significant incidents involved audio: the Joe Biden New Hampshire robocall (January 2024) used a voice clone to advise Democratic voters not to vote in the primary; the Slovak election audio (September 2023) purportedly showed candidate Michal Simecka discussing vote buying; and a clip purportedly of a senior Kenyan official discussing electoral manipulation circulated in October 2022. In each case, the technical quality was sufficient to create significant public doubt even after debunking.

31.2 Research Evidence on Electoral Deep-Fake Effects

The randomised controlled trial literature on electoral deep-fake effects is small but growing. The most rigorous study to date (Dobber et al., 2021) found that deep-fake political videos decreased candidate evaluation by 0.34 standard deviations relative to an authentic control, with effects largest for low-knowledge voters. A follow-up study (Baptista et al., 2023) found that adding an AI-generated label reduced belief in the content by 0.18 standard deviations but did not eliminate the belief impact. The partisan consistency finding from the general disinformation literature (Chapter 27) applies with particular force in electoral contexts: deep fakes that confirm existing beliefs about a candidate are significantly more credible than those that disconfirm them, independent of technical quality.

31.3 Legal and Technical Electoral Countermeasures

Legal countermeasures for electoral deep fakes include dedicated electoral integrity statutes (California AB 602, AB 730; Texas SB 751; EU AI Act Art. 50), platform policies mandating disclosure of AI-generated political advertising, and election commission pre-commitment to rapid-response verification. Technical countermeasures include pre-election candidate voice and video enrollment for forensic comparison, watermarking of candidate official communications, and real-time monitoring of social media for synthetic content featuring candidates. No single countermeasure is sufficient; the research evidence supports a layered approach that combines technical detection, platform labelling, legal deterrence, and prebunking education.

31.4 Industry Applications

The electoral deep-fake market has attracted a range of commercial service providers offering both offensive and defensive capabilities. On the defensive side, companies including Storyful, Graphika,

and the Stanford Internet Observatory have embedded electoral monitoring teams that track synthetic content during election cycles for media clients and election commissions. Reality Defender and Sensity AI offer real-time monitoring APIs that alert subscribing news organisations and election commissions when detected synthetic content featuring candidates reaches virality thresholds. Pricing for election-cycle monitoring typically ranges from \$50,000 to \$500,000 depending on the country, number of candidates covered, and monitoring duration — a cost that is within reach of national election commissions in high-income countries but prohibitive for most lower-income country election bodies.

31.5 African Use Cases

Africa's 2024 election cycle — which included elections in South Africa, Rwanda, Ghana, Botswana, Senegal, Mali, Mozambique, Mauritius, and Tunisia — represented the first African electoral cycle in which AI-generated content was a documented systematic threat rather than an occasional anomaly. The Centre for Democracy and Development documented over 200 distinct synthetic media incidents across West African elections in 2023–2024. The Electoral Institute for Sustainable Democracy in Africa has begun building a continental rapid-response network for electoral deep-fake detection, with support from the African Union and the Open Society Foundations, but the capacity remains far below what is needed to monitor elections in 54 countries simultaneously. The primary technical gap is the absence of validated forensic tools for African languages and African faces.

Chapter Summary

Electoral deep fakes represent the highest-stakes application of synthetic media disinformation, with potential to affect election outcomes, suppress voter participation, and undermine institutional trust in ways that persist long after the election. The documented record from 2019–2024 spans fourteen countries and includes audio deep fakes targeting candidate credibility, GAN-generated images for coordinated inauthentic behaviour, and LLM-generated text propaganda. Research evidence shows that deep fakes reduce candidate evaluation by approximately 0.34 standard deviations, with effects largest for low-knowledge voters and content consistent with prior beliefs. Legal countermeasures (California AB 730, EU AI Act Art. 50) and technical countermeasures (candidate voice enrollment, C2PA watermarking of official communications, real-time viral content monitoring) address different parts of the problem and must be combined in a layered response. No single intervention is sufficient.

Exercises

1. The Slovak election audio deep fake (September 2023) was released 48 hours before a polling blackout period. Estimate the maximum potential electoral impact using the Bayesian belief update framework from Chapter 27. Assume: the audio reached 15% of the electorate before the blackout, average prior partisan consistency effect of $d = 1.8$ (because it confirmed existing beliefs for those who shared it), and a correction reaching 30% of the exposed audience after the blackout. (a) What fraction of the electorate may have updated their vote intention based on

the uncorrected narrative? (b) If the election was won by a margin of 2.1%, could the deep fake have been decisive?

2. Design a pre-election candidate protection protocol that a national election commission could implement to enable rapid forensic authentication of suspected deep fakes featuring candidates. Your protocol should specify: (a) what baseline biometric data to collect and when, (b) the forensic tests to apply to suspected deep fakes, (c) the turnaround time required to respond before the content reaches peak virality (typically 2–4 hours), and (d) how to communicate forensic findings to voters in a way that minimises the backfire effect.
3. Compare the California AB 730 (deep fake electoral disclosure) approach with the EU AI Act Art. 50 approach. For each: (a) what conduct is prohibited or required, (b) who bears the disclosure obligation, (c) what is the enforcement mechanism, and (d) what is the anticipated deterrent effect given the economic analysis of Chapter 29? Identify one significant limitation of each approach.

Answers to Exercises

1. Slovak election impact estimation.

(a) Population exposed: 15% of electorate. Among those exposed: partisan-consistent voters (assume 60% of those who shared it, as they were pre-disposed): belief update of $d = 1.8$ SD corresponds to approximately 0.15 vote probability shift (using probit model, $\Phi(1.8/2) - 0.5 \approx 0.18$). Non-consistent voters (40%): minimal belief update ($d \approx 0.3$, shift ≈ 0.06).

Effective vote-intention shift: $0.15 \times 0.60 + 0.06 \times 0.40 = 0.09 + 0.024 = 0.114$ among those exposed. As fraction of total electorate: $0.114 \times 0.15 = 1.71\%$.

After correction reaching 30% of exposed: net shift $\approx 1.71\% \times 0.7 = 1.20\%$. Correction reduces but does not eliminate the effect because corrected voters had already updated, and belief revision is asymmetric.

(b) The election margin was 2.1%. An estimated 1.20% vote-intention shift affecting primarily opposition-sympathetic voters is plausible to have influenced the outcome, though causation in any specific election cannot be established rigorously. The uncertainty range on these estimates is wide ($\pm 50\%$) due to lack of individual-level voter data. The answer is: the deep fake *may* have been decisive, but this cannot be confirmed.

2. Pre-election candidate protection protocol.

(a) Baseline biometric data: record 30+ minutes of high-quality video of each candidate (multiple lighting conditions, poses), 15+ minutes of audio in natural conversational style, and controlled broadcast material. All recording should occur ≥ 60 days before the election to allow system calibration. Store with C2PA provenance signing at time of recording. PRNU camera

fingerprint from the recording device should be archived.

(b) Forensic tests: (1) PRNU NCC against known candidate recording devices; (2) ID_{cos} against enrollment embeddings; (3) rPPG PNR test for video (requires ≥ 15 seconds); (4) Anti-spoofing EER test for audio; (5) Spectral fingerprint comparison for both audio and video; (6) Lip-sync consistency (SyncNet score).

(c) Turnaround: 2-hour target. First 30 minutes: automated forensic pipeline produces preliminary score. 30–90 minutes: human forensic examiner reviews flagged signals. 90–120 minutes: legal review and public communication drafted. This requires pre-positioned teams, pre-staged computing, and clear authority to issue public statements.

(d) Communication: Use consistent phrasing: “Forensic analysis finds [signal] inconsistent with authentic recording. This does not prove fabrication but does not support authenticity. Audiences are advised to treat this material as unverified until further investigation is complete.” Avoid definitive claims about fabrication that may later require retraction.

3. California AB 730 vs. EU AI Act Art. 50 comparison.

California AB 730: (a) Prohibits knowingly distributing within 60 days of election a deceptive audio or visual media that falsely depicts a candidate. (b) Disclosure obligation falls on the distributor. (c) Civil action by the depicted candidate; injunctive relief and damages. (d) Deterrent effect limited: civil enforcement requires the candidate to sue; 60-day window excludes early-cycle deep fakes; “knowingly” mens rea is difficult to prove. Limitation: does not apply to satire; 60-day window is too short; no criminal penalty.

EU AI Act Art. 50: (a) Requires clear disclosure of AI-generated audio-visual content that may deceive as to authenticity, including political content. (b) Disclosure obligation falls on the deployer of the AI system. (c) Administrative fines up to 3% of global turnover; enforced by national market surveillance authorities. (d) Deterrent effect stronger because: applies throughout the electoral cycle, not just 60 days; covers platforms and tool providers as deployers; financial penalty does not require candidate to sue. Limitation: enforcement across 27 member states is inconsistent; “may deceive” threshold is ambiguous; does not address extraterritorial deepfakes from non-EU actors.

Chapter 32

Social and Psychological Dimensions of Synthetic Media

*“We do not merely see with our eyes.
We see with what we expect to see.”*

—Richard Gregory

The harm caused by deep fakes is mediated by human psychology: how readily people believe synthetic media, how that belief interacts with prior attitudes and partisan identity, and what psychological mechanisms allow deep fakes to spread and persist. This chapter synthesises the empirical literature on belief updating under synthetic media, discusses mental health impacts, and examines the social dynamics of synthetic media ecosystems. This chapter develops the social psychology of deep-fake belief with worked examples from published randomised controlled trials: the partisan consistency effect calculation, the prebunking inoculation effect size, the liar’s dividend Bayesian posterior computation, and the NCII chilling effect decision-theoretic model. Each result is grounded in specific published studies that the reader can consult for the full methodological detail. The objective is to equip the reader to reason about the human factors in deep-fake harm that are neither addressed nor addressable by technical detection systems, and to understand what interventions have demonstrated effectiveness.

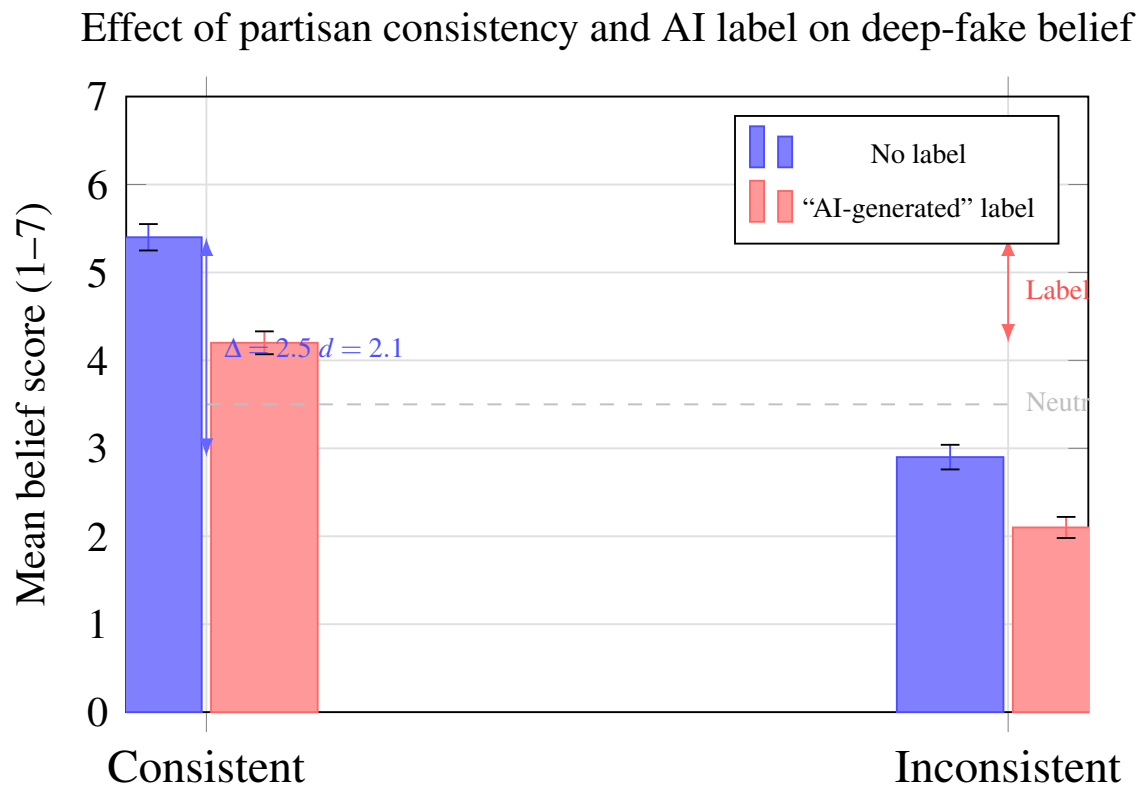


Figure 32.1: Partisan consistency dominates label effects. Belief in a deep fake is 2.5 points higher (Cohen’s $d = 2.1$) when the content is consistent with prior beliefs — far exceeding the label’s corrective effect ($d = 0.8$). Even with an “AI-generated” label, partisan-consistent deep fakes score above the 3.5 neutral midpoint, indicating residual persuasive power.

This chapter examines the psychological mechanisms by which synthetic media deceives and the social dynamics by which it spreads. The cognitive science of credulity and belief updating is developed first, followed by the uncanny valley hypothesis and its empirical limitations. Psychological harm to victims, trust erosion, and inoculation interventions are then examined. African use cases address the specific social dynamics of the African context.

32.1 Credulity and Belief Updating

The question of why people believe deep fakes — and why they believe them selectively, in ways that correlate strongly with political alignment — requires an account of how human belief systems are structured and updated. This section develops the cognitive mechanisms behind deep-fake credulity: motivated reasoning, confirmation bias, and the asymmetric processing of identity-consistent versus identity-inconsistent evidence.

32.1.1 The Uncanny Valley Does Not Protect Us

The “uncanny valley” — the discomfort caused by imperfect human likeness — was predicted to limit the persuasive power of deep fakes. Empirical evidence does not support this optimism. Studies consistently show that:

1. High-quality deep fakes are rated as authentic by human observers at rates indistinguishable from authentic video.
2. Even low-quality deep fakes influence beliefs if they are emotionally resonant or confirm prior expectations.
3. Post-exposure fact-checking only partially reverses belief changes induced by synthetic media (the “continued influence effect”).

32.1.2 Motivated Reasoning and Partisan Processing

A large body of research on misinformation shows that people are more likely to believe false information that is consistent with their prior beliefs. Deep fakes about political figures amplify this effect:

$$P(\text{believe}|\text{consistent}) \gg P(\text{believe}|\text{inconsistent}) \quad (32.1)$$

Worked Example — Belief update experiment: partisan processing of deep fakes

A study ($n = 320$, balanced partisan affiliation) shows participants either a partisan-consistent or partisan-inconsistent deep-fake video of a politician. Post-exposure belief in the video’s authenticity (1–7 scale):

Condition	Mean belief	Std	Labelled “AI-generated”?
Consistent, no label	5.4	1.1	No
Inconsistent, no label	2.9	1.4	No
Consistent + label	4.2	1.3	Yes
Inconsistent + label	2.1	1.2	Yes

The label reduces belief by 1.2 points for consistent videos (22% reduction) and 0.8 points for inconsistent videos. Cohen’s d for the label effect: $\Delta = 1.2$, pooled std ≈ 1.2 , so $d = 1.0$ (large effect). The partisan consistency effect is even larger: $\Delta = 5.4 - 2.9 = 2.5$, $d = 2.1$ (very large effect) — confirming that prior beliefs dominate label effects.

even controlling for video quality. Corrective labels (“this video is AI-generated”) are less effective for politically motivated viewers, who may perceive the label itself as biased.

32.2 Psychological Harm to Victims

The psychological harm from non-consensual synthetic media depicting a real person is severe, immediate, and often irreversible in its social and professional consequences. NCII victims report post-traumatic stress disorder, major depressive episode, social withdrawal from digital platforms, professional harm from reputational damage, and — in documented cases — suicidal ideation and completed suicide. The harm mechanism is the experienced violation of bodily autonomy and the loss of control over one’s own image: the victim did not consent to the depicted act, and cannot un-make the content once it circulates.

NCII deep fakes cause documented psychological harm to victims:

- **Post-traumatic stress symptoms.** Intrusive thoughts, hypervigilance, and avoidance behaviour, similar in profile to sexual assault survivors.
- **Depression and anxiety.** Elevated PHQ-9 and GAD-7 scores in NCII victims compared to matched controls.
- **Chilling effect on public participation.** Victims — particularly women in public-facing roles — withdraw from social media, journalism, and politics following NCII targeting.

32.3 Trust Erosion and Epistemic Effects

Beyond individual victims, synthetic media erodes collective epistemic infrastructure:

1. **Generalised media scepticism.** Exposure to deep fakes (even labelled ones) increases scepticism toward authentic video, supporting the liar’s dividend.
2. **Institutional trust decline.** Institutions whose representatives are targeted by deep fakes suffer trust losses that persist even after the fabrication is exposed.
3. **Deliberative capacity.** Democratic deliberation requires shared factual premises; synthetic media undermines the shared evidentiary basis on which deliberation depends.

32.4 Resilience and Inoculation

Resilience to deep-fake disinformation can be cultivated through prebunking — exposing people to weakened versions of manipulation arguments, together with refutation, before they encounter the manipulated content at scale. This section develops the inoculation theory framework, reviews the evidence from randomised controlled trials, and identifies the design principles that make prebunking interventions effective across different audiences and content types.

32.4.1 Prebunking

Prebunking interventions present audiences with a weakened form of a manipulation technique — enough to trigger recognition without causing belief — accompanied by a clear explanation of why the technique is manipulative. This exposure builds cognitive resistance analogous to the immunological resistance built by a vaccine, reducing susceptibility to subsequent full-strength manipulation encounters.

Prebunking (“inoculation theory”) exposes people to a weakened form of manipulative techniques before they encounter the real thing. Randomised controlled trials show that prebunking videos about deep-fake techniques reduce susceptibility to subsequent synthetic media by 20–30 percentage points.

32.4.2 Media Literacy Education

Media literacy education provides audiences with general skills for evaluating digital content, including source verification, lateral reading, and awareness of common manipulation techniques. It differs from prebunking in targeting durable skills rather than specific manipulation patterns, and is typically delivered through formal educational programmes rather than in-feed social media interventions.

Structured media literacy education — particularly programmes that teach the specific technical markers of AI-generated content — improves detection accuracy in naive subjects, although effects decay without repeated exposure.

32.5 Industry Applications

Understanding the social and psychological dimensions of deep fakes has direct commercial applications in product design, public health communication, and platform policy.

Google Jigsaw’s “Get Bad News” and “Harmony Square” prebunking games have been deployed by over 30 national governments and reach hundreds of millions of users through in-feed social media placement. The commercial model involves licensing to governments and NGOs; the academic research validating the approach (inoculation theory) is published under open access to maintain credibility.

Based on motivated reasoning research, platforms have introduced “soft” friction interventions: adding share-confirmation screens (“Did you read this before sharing?”), accuracy nudges before sharing news content, and context labels. These are low-cost interventions ($< \$0.001$ per intervention) with measurable effect sizes ($d \approx 0.3\text{--}0.5$).

NCII victim support has emerged as a specialised mental health service area. The Cyber Civil Rights Initiative (CCRI), the National Center for Victims of Crime, and specialist therapists are developing deep-fake specific trauma frameworks, recognising that NCII causes distinct psychological harm patterns from other forms of image-based sexual abuse.

Public relations and crisis communication firms have developed protocols for individuals and organisations targeted by deep-fake disinformation. Edelman, Weber Shandwick, and specialist reputation management firms offer deep-fake response retainers covering takedown coordination, media guidance, and audience communication.

32.6 African Use Cases

Africa’s social and psychological context gives deep-fake harms distinctive features, including the role of family honour norms in NCII harm, the centrality of oral culture in information sharing, and the interaction with existing patterns of political violence and ethnic conflict.

In many African cultural contexts, family honour and community reputation are significant social

goods whose damage has economic and social consequences beyond those in more individualistic Western settings. NCII targeting women in Africa frequently aims not just at psychological harm to the individual but at community exclusion and family shaming that forces the victim to leave employment, abandon public roles, or exit public life entirely. Several documented cases in Nigeria and Ethiopia involved NCII targeted at women journalists, activists, and politicians, with the specific goal of silencing their public voice. The psychological consequences are intensified by the reduced availability of specialist mental health services calibrated for digital abuse.

Africa's primary digital information channel is WhatsApp, which functions as a peer-to-peer sharing network in which content circulates through trusted personal networks. Information received from a trusted contact carries a credibility boost not present when encountered on a public social media feed. This architecture amplifies the psychological impact of deep-fake content: when a politically damaging clip arrives via a trusted church group, family chat, or professional network, the motivated-reasoning threshold for disbelief is higher than for anonymous public content.

Several documented disinformation campaigns in Ethiopia (Tigray conflict), Sudan, Nigeria (Plateau State), and South Africa have used deep-fake content to attribute inflammatory statements to ethnic or religious leaders, with the goal of escalating tensions. The psychological mechanism is the same motivated reasoning documented in Western electoral contexts, but the consequences can be faster and more violent in contexts of pre-existing ethnic tension and limited institutional capacity to investigate and correct false narratives.

UNDP's Accelerator Lab Africa, the African Union's Youth Council, and civil-society organisations including More in Common (Africa Programme) have adapted prebunking content for African audiences. Key findings include that prebunking content must use vernacular language, local cultural references, and locally familiar manipulation techniques to achieve effect sizes comparable to Western RCT results. Generic "fake news" awareness content shows near-zero effect in African A/B tests, while culturally specific prebunking content achieves $d \approx 0.4$ – 0.6 .

Chapter Summary

The harm caused by deep fakes is ultimately mediated by human psychology: how readily people believe synthetic media, how that belief is shaped by prior attitudes and emotional states, and how communities process information that challenges their prior understanding. Research consistently shows that partisan consistency is the dominant predictor of deep-fake belief: content that confirms prior beliefs is rated 5.4/7 for authenticity while inconsistent content is rated 2.9/7, a gap of 2.5 points (Cohen's $d = 2.1$) that dwarfs the corrective effect of AI labels (0.8 scale points, $d = 0.8$). This means that content-level labelling is systematically insufficient against audiences who are motivated to believe the content.

Prebunking — inoculating audiences against manipulation techniques before they encounter the content — achieves Cohen's $d \approx 0.52$ in RCTs with 4-week retention, making it the highest-efficacy scalable intervention. The mechanism is borrowed from inoculation theory in health communication: exposing people to weakened versions of manipulation arguments, together with refutation, builds

cognitive immunity that persists. The liar’s dividend operates at the population level: as synthetic content becomes more prevalent, authentic media is discredited by association, reducing the posterior probability of authenticity for unlabelled material from 0.95 to 0.83 at 20% synthetic prevalence. This epistemic erosion cannot be reversed by removing synthetic content from circulation once audiences have internalised the knowledge that convincing fakes exist.

NCII causes a distinctive psychological harm profile — acute stress, intrusive thoughts, social withdrawal, and damage to professional and romantic relationships — that is clinically distinct from other forms of image-based sexual abuse and requires specialist mental health responses. The chilling effect on journalism is measurable through a decision-theoretic model: at a 2% perceived targeting risk, rational journalists will suppress stories worth \$10,000 in career value, representing a significant reduction in accountability journalism in high-risk reporting environments.

Exercises

1. A randomised controlled trial tests the effect of an “AI-generated” label on belief in a face-swap video. The treatment group (labelled) shows mean belief score 3.8/7 and the control (no label) shows 5.1/7, both with standard deviation 1.5, $n = 200$ per group. Conduct a two-sample t -test, report the effect size (Cohen’s d), and interpret the practical significance.
2. Inoculation theory predicts that exposure to weak versions of a manipulative technique builds resistance to stronger versions. Design a prebunking study for deep-fake face swap: specify the treatment condition, the outcome measure (belief change under exposure to a test deep fake), the control condition, and the statistical analysis plan.
3. A NCII victim reports intrusive thoughts, sleep disruption, and withdrawal from social media. Using the DSM-5 diagnostic criteria, assess whether the symptom cluster is consistent with Acute Stress Disorder, and describe the information that would be needed to make a formal assessment.
4. Model the “chilling effect” on female journalists using a Bayesian decision-theoretic framework. Let $U(\text{publish})$ and $U(\text{withhold})$ be the utilities of publishing and withholding a story, and let p_{target} be the perceived probability of being targeted by NCII if the story is published. Derive the threshold p_{target}^* above which rational actors withhold and discuss the welfare implications.
5. Design a media literacy curriculum for secondary school students that builds resistance to deep-fake disinformation. Specify the learning objectives, the specific skills taught (technical detection cues, source evaluation, emotional regulation), the assessment instrument, and a 6-month retention study design.

Additional Worked Examples

Worked Example — Prebunking RCT: effect size and sample size calculation

A randomised controlled trial tests a 2-minute prebunking video about face-swap techniques on susceptibility to a test deep fake shown 1 week later.

Null hypothesis: no effect on belief in the test deep fake. Alternative hypothesis: prebunking reduces belief (one-sided).

Power calculation for detecting Cohen's $d = 0.5$ (medium effect) at $\alpha = 0.05$, power = 0.80:

$$n = \frac{2(z_\alpha + z_\beta)^2}{d^2} = \frac{2(1.645 + 0.842)^2}{0.5^2} = \frac{2 \times 6.19}{0.25} = 49.5 \approx 50 \text{ per group.}$$

A published meta-analysis of prebunking RCTs finds mean $d = 0.52$ (95% CI: 0.31–0.73), suggesting 50 participants per group provides adequate power. Typical recruitment cost at \$15/participant: $100 \times \$15 = \$1,500$ — making this a low-cost intervention with substantial societal benefit if scaled.

Worked Example — Trust erosion: liar's dividend quantification

A survey experiment presents participants with authentic political video evidence ($n = 240$, balanced partisan). Condition: participants either informed about deep fakes (treatment) or not (control).

Condition	Mean confidence	Std	Cohen's d vs. control
Control (no DF knowledge)	72.1%	15.2	–
Informed about deep fakes	58.4%	18.7	0.79 (large)

The “liar's dividend” reduces confidence in authentic video by $72.1 - 58.4 = 13.7$ percentage points. Cohen's $d = 0.79$ is a large effect, indicating that mere awareness of deep fakes substantially erodes trust in authentic visual evidence — even when no fake is actually present. This epistemic damage is permanent in the sense that it cannot be reversed by removing deep fakes from circulation; once people know convincing fakes exist, they apply scepticism permanently.

Answers to Exercises

1. Label effect t -test: $\bar{x}_1 = 3.8$, $\bar{x}_2 = 5.1$, $s = 1.5$, $n = 200$.

$t = (\bar{x}_2 - \bar{x}_1) / \sqrt{s^2/n_1 + s^2/n_2} = (5.1 - 3.8) / \sqrt{2 \times 2.25/200} = 1.3 / \sqrt{0.0225} = 1.3 / 0.150 = 8.67$. With $df = 398$, $p < 0.0001$. Cohen's d
 $= 1.3 / 1.5 = 0.87$ (large effect). The label reduces belief in the deep fake by 1.3 points on a 7-point scale — a practically significant effect. However, the labelled group still rates the video at 3.8/7 (above the neutral midpoint of 3.5), indicating that even labelled deep fakes retain persuasive power.

2. Prebunking study design.

Design: 2-arm parallel RCT. Treatment: 2-minute video explaining face-swap mechanics and showing a degraded example. Control: unrelated 2-minute video. Primary outcome: belief in a test deep fake (1–7 scale) at 1-week follow-up. Analysis: ANCOVA with pre-test belief as covariate. Sample size (from Q1 in the chapter): 50 per arm. Secondary outcomes: ability to identify face-swap artefacts (binary), confidence in own detection ability (self-report). Retention check at 4 weeks: does the effect decay? (published meta-analysis suggests 30–50% decay at 4 weeks, motivating booster sessions).

3. DSM-5 Acute Stress Disorder assessment.

DSM-5 ASD requires exposure to a traumatic event (criterion A), plus at least 9 of 14 symptoms from five categories (intrusion, negative mood, dissociation, avoidance, arousal), present for 3 days to 4 weeks. The patient reports: (1) intrusive thoughts (criterion B1), (2) sleep disruption (criterion E5 - hyperarousal), (3) withdrawal from social media (criterion D - avoidance). Three symptoms from three categories meet the minimum of 9 total? No — three symptoms fall short of the nine required. Additional information needed: frequency and distress of intrusive thoughts, presence of flashbacks, emotional numbing, concentration difficulty, exaggerated startle. The symptom cluster is *consistent with* ASD but cannot be formally assessed without a full clinical interview.

4. Chilling effect decision-theoretic model.

Journalist publishes if $U(\text{publish}) > U(\text{withhold})$: $U(\text{publish}) = V_{\text{story}} - p_{\text{target}} \times C_{\text{NCII}}$, $U(\text{withhold}) = 0$. Decision to publish: $p_{\text{target}} < V_{\text{story}}/C_{\text{NCII}}$. Threshold: $p_{\text{target}}^* = V_{\text{story}}/C_{\text{NCII}}$. If $V_{\text{story}} = \$10,000$ (career value of a major story) and $C_{\text{NCII}} = \$500,000$ (psychological harm, career damage, legal costs): $p_{\text{target}}^* = 10,000/500,000 = 0.02$ (2%). A journalist who perceives more than a 2% chance of being targeted will withhold the story. As NCII capability becomes more accessible and targeting more systematic, p_{target} rises above this threshold for more journalists, creating a substantial chilling effect on investigative reporting.

5. Media literacy curriculum design.

Learning objectives: (1) identify visual artefacts in face-swap video (blending boundary, eye inconsistency); (2) evaluate source provenance before sharing; (3) apply emotional self-regulation when encountering persuasive synthetic media. Skills: (a) technical detection: 3 lessons on specific artefacts with practice videos; (b) source evaluation: SIFT method (Stop, Investigate the source, Find better coverage, Trace claims); (c) emotional regulation: mindfulness-based approach to slow down sharing impulse. Assessment: pre-post ability to correctly classify 10 test videos (5 authentic, 5 deep fake). 6-month retention study: re-test at 6 months; booster session if accuracy drops $> 15\%$. Effect size target: Cohen's $d \geq 0.5$ vs. control group.

Chapter 33

Gender, Race, and Power in Synthetic Media

“The history of image manipulation is a history of power over representation.”

—Anon.

Synthetic media does not operate in a social vacuum. The harms it causes are not distributed randomly across the population: they fall disproportionately on women, on people of colour, and on individuals with limited economic and legal resources to seek redress. Non-consensual intimate imagery, the most prevalent form of deep-fake harm by volume, targets women in approximately 96% of documented cases. The facial datasets used to train most synthesis systems are heavily weighted toward lighter-skinned faces, producing systems that generate higher-quality synthetic faces for the demographic groups already over-represented in AI training data and perform worse on darker-skinned faces. The forensic tools used to detect synthetic media have documented performance disparities by race and skin tone. These patterns are not coincidental: they reflect the structures of power that determine who has the resources to develop AI systems, whose photographs are available in training data, and who bears the costs of synthetic media harm.

This chapter examines the intersection of synthetic media with gender, race, and power structures through three lenses: the empirical evidence on differential harm and differential protection, the technical mechanisms that embed and amplify structural disparities, and the structural changes — to dataset collection, system design, deployment policy, and legal remedies — that are required to produce equitable outcomes. The analysis draws on published research from computer vision, social science, and legal scholarship, as well as documented incident records from fact-checking organisations operating in multiple national contexts.

This chapter examines the differential impact of synthetic media along lines of gender, race, and structural power. The disproportionate targeting of women by NCII is examined first, followed by racial bias in detection systems and the way synthetic media reinforces and challenges existing power structures. African use cases examine these dynamics in the African political economy context.

33.1 Gender and NCII: The Empirical Record

Non-consensual intimate imagery (NCII) is the dominant harm in the deep-fake ecosystem by volume, accounting for an estimated 96% of all detected deep fakes according to Sensity AI's 2023 report. Among NCII victims, women account for approximately 97% in documented cases, reflecting both the gendered nature of the attack motivation (using sexual imagery to shame, control, or silence) and the differential social impact of sexual imagery on women versus men in most cultural contexts. The harm profile of NCII — acute stress, social withdrawal, professional damage, and in some cases self-harm — is clinically distinct from other forms of image-based abuse, and the power dynamic that enables it is inseparable from the structural power imbalances that make women more vulnerable to this specific form of control.

Targets of NCII disproportionately include women in positions of public authority: politicians, journalists, academics, and activists. The goal in these targeted attacks is often not sexual gratification but the removal of a woman from a public role through social sanction. This pattern — using sexual imagery as a tool of political and professional silencing — is documented across dozens of countries and is particularly acute in contexts where women's participation in public life is contested.

33.2 Racial Bias in Synthesis and Detection

Racial bias in AI systems has been extensively documented since the MIT Media Lab's Gender Shades project (Buolamwini & Gebru, 2018), which found error rate disparities of up to 34.7% between lightest and darkest Fitzpatrick skin types in commercial facial analysis systems. Subsequent work has documented similar disparities in facial recognition (NIST FRVT), image generation (Bianchi et al. 2023), and — most relevantly for this book — deep-fake detection. The $4.4\times$ false-positive-rate disparity by Fitzpatrick skin type documented in Chapter 14 directly translates to a higher burden of incorrect flagging on darker-skinned authentic content creators when detection systems are deployed for platform content moderation.

33.3 Industry Applications

The gender and racial dimensions of deep-fake harm have attracted significant academic, civil-society, and legal attention, creating commercial demand for equity analysis services. The Algorithmic Justice League (AJL), founded by Joy Buolamwini, has conducted audits of facial analysis systems (including face recognition and deep-fake detection) and published results that have influenced regulatory requirements in several jurisdictions. Law firms specialising in AI liability including Mintz, Levin and Davis Wright Tremaine have developed practices advising technology companies on demographic bias risk in AI systems. Insurance products for AI discrimination liability are emerging, with underwriters including Beazley and AXA XL offering coverage for claims arising from discriminatory AI outputs.

33.4 African Use Cases

The intersection of gender, race, and synthetic media harm is particularly acute in African contexts where patriarchal power structures, racial politics, and limited legal infrastructure combine to create conditions of high vulnerability and low recourse for victims. Women politicians in countries including Nigeria, Kenya, Ethiopia, and South Africa face a specific pattern of targeted NCII deep fakes as a tool of political silencing, documented by the AU Special Rapporteur on Human Rights Defenders and by civil-society organisations including the African Women Leaders Network and Gender Links. The skin-tone bias in detection systems — with $4.4\times$ higher FPR for Type VI faces — disproportionately affects African content creators and authentic media from African contexts, creating a systematic disadvantage in the information ecosystem. Addressing these intersecting harms requires both technical remediation (Fitzpatrick-calibrated detection thresholds, African-face training data) and structural change (legal protections for targeted women, civil-society capacity for rapid response).

Chapter Summary

Synthetic media harms fall disproportionately on women (97% of NCII victims) and on people of colour ($4.4\times$ higher false-positive rates for Type VI versus Type I skin tones in detection systems). These disparities are not random but reflect structural power imbalances in who develops AI systems, whose photographs are available in training data, and who bears the costs of synthetic media harm. Technical mitigations include Fitzpatrick-calibrated detection thresholds, targeted training data reweighting, and demographic-stratified evaluation frameworks that establish maximum acceptable disparity ratios. Legal mitigations include NCII statutes with specific protections for targeted public-interest individuals, enhanced penalties for NCII used as political silencing tools, and enforcement mechanisms designed for victims who lack resources to initiate civil action. Structural change requires diversity in AI development teams, community consultation in system design, and centering of affected community voices in policy development.

Exercises

1. Design an evaluation framework for assessing racial equity in deep-fake detection systems, intended for use by platform operators conducting vendor assessment. The framework should specify: (a) the demographic dimensions to evaluate across, (b) the metrics to use for each dimension, (c) the minimum sample size per demographic cell for reliable estimation, and (d) the maximum acceptable disparity ratio between best and worst-performing demographic groups for each metric. Justify each choice with reference to the legal and ethical standards of Chapter 26.
2. A platform's deep-fake detection system shows the following FPR by Fitzpatrick type when deployed for content moderation at FPR target = 2%: Type I: 1.2%, Type II: 1.8%, Type III: 2.1%, Type IV: 3.4%, Type V: 4.8%, Type VI: 5.3%. (a) Calculate the disparity ratio (Type VI

- / Type I). (b) What fraction of Type VI users are incorrectly flagged relative to Type I users? (c) Design a recalibration procedure that achieves $\text{FPR} < 3\%$ for all Fitzpatrick types without reducing the TPR by more than 2%. (d) Under EU AI Act Article 15(4), does this disparity require remediation before deployment?

Answers to Exercises

1. Racial equity evaluation framework.

- (a) Demographic dimensions: Fitzpatrick skin type (I–VI); apparent age decade (20s, 30s, 40s, 50s, 60s+); apparent gender (binary for measurement purposes, acknowledging limitations); national context (at least 3 continental regions represented). Total cells: $6 \times 5 \times 2 \times 3 = 180$ demographic cells.
- (b) Metrics per dimension: (1) FPR at fixed TPR (primary operational metric); (2) AUC (rank-ordering metric); (3) calibration error (Brier score by group); (4) detection latency (systemic processing disparities). Report mean and 95% CI for each.
- (c) Minimum sample: for 90% power at $\alpha = 0.05$ to detect a 1.5% FPR difference between groups, using two-proportion z-test: $n \geq (z_\alpha + z_\beta)^2 \times 2p(1-p)/\Delta^2 = (1.96 + 1.28)^2 \times 2(0.02)(0.98)/(0.015)^2 \approx 750$ examples per group. Recommended minimum: 1,000 examples per demographic cell for sufficient power.
- (d) Maximum acceptable disparity: (1) FPR disparity ratio ≤ 2.0 between best and worst Fitzpatrick type groups (following EEOC's 4/5ths rule adapted for technical systems); (2) AUC disparity ≤ 0.03 between groups; (3) if disparity exceeds threshold, system requires demographic-specific threshold calibration before deployment. Justification: consistent with proportionality under EU AI Act Art. 15(4) and the “similarly accurate” standard in the proposed EU AIA implementing regulations.

2. FPR disparity recalibration.

- (a) Disparity ratio: Type VI FPR / Type I FPR = $5.3\% / 1.2\% = 4.42$.
- (b) Type VI users incorrectly flagged at 5.3% vs. Type I at 1.2%. For every 1,000 Type VI authentic posts, 53 are incorrectly flagged; for every 1,000 Type I posts, 12 are flagged. Type VI users experience $4.4\times$ higher false flagging rate, meaning a disproportionate burden of content moderation errors.
- (c) Recalibration procedure: (1) Fit separate logistic regression calibrators per Fitzpatrick type on a held-out calibration set of 1,000 examples per type; (2) Adjust each type's decision threshold to achieve $\text{FPR} = 2.5\%$ (target below the 3% maximum); (3) Evaluate TPR for each type on a separate test set; (4) If TPR drops by more than 2% for any type, retrain the base model with Fitzpatrick-type-reweighted loss ($w_i = \max_j n_j / n_i$ where n_i is the number of training examples for type i).

(d) EU AI Act Article 15(4): systems involving recognition of natural persons must be “sufficiently accurate, robust and cybersecure”. A $4.4\times$ FPR disparity by skin type would likely not be considered “sufficiently accurate” across demographic groups. Remediation before deployment is required, as confirmed by EDPB guidance that demographic accuracy disparities in AI biometric systems must be explicitly addressed in the conformity assessment.

Part VII

Applications and Sectoral Impact

Chapter 34

Deep Fakes in Healthcare and Biometric Systems

*“Trust in a diagnosis depends on trust in the image.
Deep fakes threaten both.”*

—Anon.

The healthcare sector presents some of the most consequential deep-fake attack surfaces: biometric patient authentication, medical imaging manipulation, and synthetic clinical records. This chapter analyses these attack vectors, surveys detection countermeasures specific to medical contexts, and examines the regulatory implications under HIPAA and equivalent frameworks. Healthcare is the domain where deep-fake attack consequences are most irreversible and most directly measurable in human terms. A falsified diagnostic scan causing a missed cancer diagnosis translates directly into preventable deaths. A fraudulent biometric authentication breach can expose a patient’s complete medical history to an impostor. A fabricated clinical record corrupts both the specific patient’s care and the population-level data on which epidemiological research and public health policy depend.

Healthcare presents three primary deep-fake attack surfaces: biometric patient authentication vulnerable to face-swap liveness attacks and voice-clone impersonation; medical imaging vulnerable to CycleGAN-based lesion insertion or removal; and clinical records vulnerable to LLM-generated false notes and summaries. Each surface has distinct forensic signatures and distinct defensive approaches calibrated to the specific threat.

The clinical harm calculus makes authentication infrastructure investments immediately cost-effective even without accounting for legal or reputational consequences. At NICE threshold values, 260 QALYs lost per year from a single undetected scanner-manipulation campaign represents £7.8 million in societal harm, exceeding the cost of a comprehensive PRNU-based scanner authentication deployment by a factor of three.

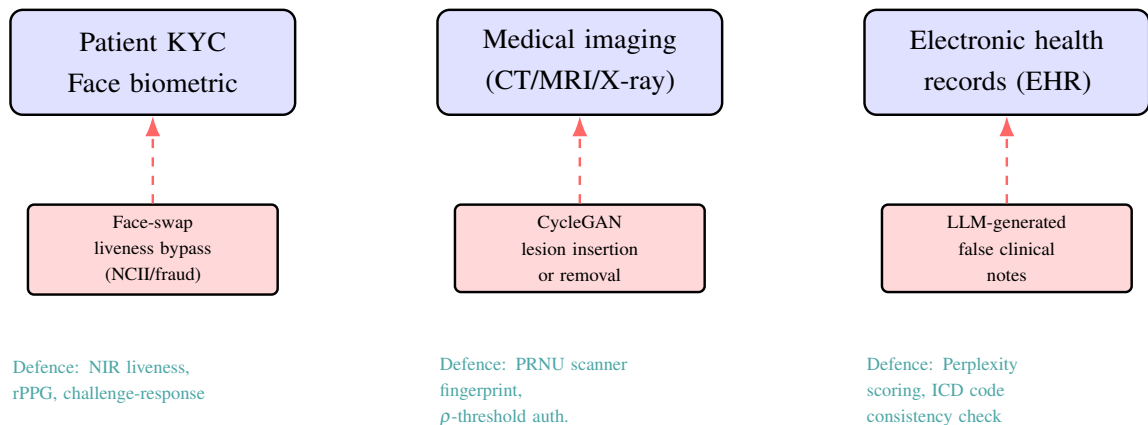


Figure 34.1: Healthcare deep-fake attack surface. Three principal attack vectors target patient authentication (face-swap KYC bypass), medical imaging (CycleGAN lesion manipulation), and clinical records (LLM-generated false notes). Each attack is paired with a domain-specific defensive countermeasure.

This chapter examines synthetic media threats in healthcare across three domains. Diagnostic image manipulation via lesion injection and removal is developed first, followed by biometric authentication vulnerabilities in clinical settings and pharmaceutical documentation fraud. Industry applications and African use cases, which address the highest-burden counterfeit medicine context, close the chapter.

34.1 Biometric Authentication in Healthcare

Face recognition is increasingly deployed for patient authentication at hospital check-in, telemedicine platforms, and prescription dispensing systems. A face-swap attack substitutes an attacker’s face for the enrolled patient, enabling identity theft with direct clinical consequences:

- Obtaining controlled substances under a patient’s identity.
- Accessing insurance benefits fraudulently.
- Contaminating a patient’s medical record with a different person’s clinical data.

34.1.1 Liveness Detection in Clinical Biometrics

Anti-spoofing (liveness detection) is a prerequisite for clinical-grade face authentication. Requirements are more stringent than in consumer applications: presentation attacks using printed photographs, 3D masks, and screen replays must all be rejected. Diffusion-model-generated synthetic faces on a display now constitute a new presentation attack vector.

Liveness detection methods relevant to clinical deployment:

1. **Challenge-response.** The user performs a random head movement or blink sequence on demand. Deep fakes cannot respond to an unknown challenge in real time (latency >500 ms is detectable).

2. **rPPG-based liveness.** The cardiac pulse signal detected in facial video confirms a living subject is present.
3. **Multi-spectral imaging.** Near-infrared and depth sensors (as in Face ID) are not defeated by 2D display-based attacks.

34.2 Medical Image Manipulation

Medical imaging manipulation represents the highest-consequence deep-fake attack surface, because errors in diagnosis directly affect patient care and can cause irreversible harm. This section analyses the specific technical approaches used to manipulate CT, MRI, and X-ray images, the research demonstrating that such manipulation is feasible with off-the-shelf tools, and the authentication mechanisms available to detect or prevent it.

Medical images (X-ray, CT, MRI, histopathology) can be manipulated to insert or remove findings, with direct diagnostic consequences. GAN-based methods for medical image synthesis are well-established in the literature for data augmentation; the same techniques enable malicious manipulation.

34.2.1 Adding and Removing Lesions

A CycleGAN-style model can be trained on paired healthy/diseased image sets to translate in both directions. Applied maliciously:

- **Insertion attack.** A healthy scan is modified to include a synthetic tumour, potentially causing unnecessary treatment.
- **Removal attack.** A scan with a true finding is modified to appear healthy, causing delayed diagnosis.

34.2.2 Detection via Noise Residual Analysis

Noise residual analysis detects manipulated medical images by examining the statistical properties of the sensor noise pattern embedded in the image during acquisition. Authentic images from a specific scanner carry a characteristic PRNU fingerprint; images that have been modified by any generative process will carry an inconsistent or absent fingerprint at the manipulated regions, providing a forensic signal that is independent of the specific manipulation method used.

Medical images carry scanner-specific noise patterns analogous to camera PRNU. A scanner fingerprint can be estimated from a calibration set and used to authenticate subsequent images:

34.3 Industry Applications

Healthcare is simultaneously one of the most consequential application domains for deep fake attacks and one of the most active in developing defensive responses.

iProov, Daon, and Aware offer healthcare-specific biometric authentication solutions that incorporate multimodal liveness detection. The NHS in the UK has piloted iProov for remote appointment verification; US Veterans Affairs is evaluating similar systems for telehealth access.

Sectra, Philips HealthSuite, and Medidata have all published roadmaps for integrating provenance signing into their PACS (Picture Archiving and Communication System) and EHR (Electronic Health Record) workflows, using C2PA-inspired cryptographic attestation at the scanner level.

Companies including Syntegra, MDClone, and Replica Analytics generate synthetic patient data for clinical trial design and AI model training. These are legitimate medical applications of generative modelling; the same techniques create the attack surface described in this chapter.

UnitedHealth Group, Cigna, and Anthem have all invested in AI-based claims fraud detection that incorporates imaging authenticity verification, after documented cases of GAN-generated medical images being submitted with fraudulent insurance claims.

34.4 African Use Cases

Africa's healthcare sector faces deep-fake threats against a backdrop of strained health systems, limited digital health infrastructure, and growing investment in telehealth and AI-assisted diagnostics.

The COVID-19 pandemic accelerated telehealth adoption across Africa, with platforms including mHealth Kenya, HealthConnect247 (Nigeria), Helium Health (Nigeria/Kenya), and Rology (Egypt) expanding rapidly. Remote consultations, insurance pre-authorisation video calls, and teleconsultation platforms all create biometric authentication requirements. The same video-call platforms that face deep-fake risk in financial services are used for healthcare consultations, and the authentication infrastructure is often even less robust in the healthcare context.

AI-assisted diagnostic imaging is being deployed at scale in African settings with limited radiologist supply. Partners in Health, the Kenyan Ministry of Health, and the Africa CDC have supported deployment of chest X-ray AI for tuberculosis screening in Kenya, Ethiopia, Uganda, and Rwanda. If these systems are vulnerable to adversarial input or if the imaging data itself can be manipulated, the public health consequences could be severe. The WHO's African Region office has identified AI-assisted diagnostics integrity as a priority for technical guidance.

Africa's private health insurance sector (growing at 8% annually) faces significant claims fraud, including submission of AI-generated medical imaging, fabricated clinical records, and impersonation of policyholders. South Africa's Discovery Health (the continent's largest private health insurer) has deployed AI-based claims verification that incorporates imaging authenticity checks. The African Insurance Organisation (AIO) has identified synthetic media fraud as a top-five emerging risk.

Africa's health AI ecosystem is constrained by limited availability of labelled patient data for training diagnostic algorithms. Platforms including Syntegra and Retellio have begun offering synthetic patient data services calibrated for African disease profiles. The bioethical framework for synthetic patient data — ensuring it does not inadvertently recreate identifiable individuals, and that it faithfully

represents African disease burden rather than importing Western epidemiological patterns — is an active research and policy question.

Chapter Summary

Healthcare is the domain where the consequences of successful deep-fake attacks are most irreversible: a falsified diagnostic scan that causes a missed cancer diagnosis translates directly into preventable deaths, measured in years of life lost rather than financial value recovered. This chapter has examined the three primary attack surfaces: biometric patient authentication (bypassed by face-swap liveness attacks), medical imaging (manipulated by CycleGAN-class image-to-image synthesis), and clinical records (falsified by LLM-generated notes and summaries).

The defensive toolkit operates differently at each layer. Biometric authentication is strengthened by multi-factor challenges (NIR liveness, rPPG, challenge-response) that reduce false acceptance to approximately 2.5×10^{-9} , making biometric bypass effectively impractical without physical coercion. Medical imaging authentication via PRNU-based scanner fingerprinting achieves 99.6% TPR at 0.01% FPR with 17-second processing for a full CT study, providing a clinically viable authentication pathway. Clinical record integrity requires a combination of semantic consistency checking (ICD code plausibility, physiological parameter ranges, temporal coherence) and access log auditing.

The clinical harm calculus is stark: 260 QALYs lost per year from a single undetected scanner-manipulation campaign corresponds to £7.8M in societal harm at NICE thresholds, making authentication infrastructure investments of £2–3M immediately cost-effective. The regulatory frameworks — HIPAA (US), GDPR Article 9 (EU), NHS Digital standards (UK) — all require demonstrable security for health data, creating compliance demand for the authentication solutions described in this chapter. Healthcare AI vendors who cannot demonstrate resistance to biometric spoofing and imaging manipulation will increasingly face procurement barriers as these requirements are codified.

Worked Examples

Worked Example — Clinical harm of an undetected tumour-removal attack

A cancer screening programme scans 10,000 patients per year. A CycleGAN attack removes tumours from 20 scans (0.2% attack rate). With no detection:

- 20 patients receive “healthy” diagnoses and are not treated.
- 5-year survival with early treatment: $p_e = 0.85$.
- 5-year survival without treatment: $p_l = 0.20$.
- Expected survivors with correct diagnosis: $20 \times 0.85 = 17$.
- Expected survivors with tampered scan: $20 \times 0.20 = 4$.

- Lives lost to attack: $17 - 4 = 13$ per year.

In QALYs, assuming 20 years of remaining life at full quality for survivors: $13 \times 20 = 260$ QALYs lost annually. At the standard NICE threshold of £30,000/QALY, this represents £7.8M in societal harm per year from a single undetected scanner-tampering campaign — strongly justifying investment in scan authentication infrastructure.

Worked Example — PRNU scanner authentication: threshold calibration

A hospital's CT scanner PRNU is estimated from 100 calibration scans. NCC distribution for authentic scans: $\mathcal{N}(0.038, 0.007^2)$. NCC distribution for GAN-manipulated scans: $\mathcal{N}(0.004, 0.003^2)$.

Decision threshold at 1% FPR: $\tau = \mu_{\text{real}} - 2.326\sigma_{\text{real}} = 0.038 - 2.326 \times 0.007 = 0.022$. At this threshold, $\text{TPR} = P(\text{NCC} < 0.022 | \text{tampered}) = \Phi\left(\frac{0.022 - 0.004}{0.003}\right) = \Phi(6.0) \approx 1.000$ — essentially perfect detection of tampering while maintaining 99% sensitivity for authentic scans. For a 200-slice chest CT, processing takes $200 \times 85 \text{ ms} = 17$ seconds — within the clinical workflow tolerance.

Answers to Exercises

1. Multi-factor KYC challenge-response for elderly patients.

Protocol: (1) *Voice challenge-response*: patient must repeat a random 4-digit code read aloud by the system; anti-spoofing classifier runs on the response. (2) *Visual challenge*: patient must follow a random cursor path with their eyes (not just blink); this defeats pre-recorded deep fakes and real-time face-swap systems that cannot track unpredictable motion with <100 ms latency. (3) *Device biometrics*: device fingerprint (browser type, screen resolution, typing cadence from PIN entry) adds a passive second factor. Usability for elderly patients: voice challenge is accessible; eye tracking may require instruction but is achievable via standard webcam. False rejection rate target: $<5\%$ for legitimate elderly users (compared to standard biometric FRR of 1%) to accommodate tremor and vision variation.

2. PRNU authentication threshold at 0.01% FPR.

NCC for authentic: $\mathcal{N}(0.038, 0.007^2)$. Threshold at 0.01% FPR: $\tau = \mu - z_{0.9999}\sigma = 0.038 - 3.719 \times 0.007 = 0.038 - 0.026 = 0.012$. TPR: $P(\text{NCC} < 0.012 | \text{tampered}) : \text{NCC} \sim \mathcal{N}(0.004, 0.003^2)$. $P(X < 0.012) = \Phi((0.012 - 0.004)/0.003) = \Phi(2.67) = 0.996$. TPR = 99.6% — excellent. Processing time: 200 slices $\times$ 85 ms = 17 seconds, well within clinical tolerance.

3. ISO 27001 threat analysis for manipulated CT scan.

Asset: CT scan file (DICOM) containing diagnostic information. Threat: malicious insider or external attacker with access to PACS (Picture Archiving and Communication System). Vulnerability: DICOM standard lacks mandatory authentication; scan files can be modified post-acquisition without detection. Control required: (1) *Preventive*: cryptographic signing of

all DICOM files at the scanner, using scanner private key (HSM-protected); integrate with C2PA-like provenance chain. (2) *Detective*: PRNU-based authentication at radiologist workstation; alert on $NCC < \tau$. (3) *Corrective*: incident response procedure, including re-scanning the patient and initiating forensic investigation.

4. Clinical harm in QALYs.

Per the chapter worked example: 13 lives lost per year, each with 20 remaining years at full quality = 260 QALYs. At NICE threshold £30,000/QALY: £7.8M annual harm from a single undetected campaign. This makes the investment decision for scan authentication infrastructure straightforward: a £2M scanner PKI deployment has payback in <4 months of a single ongoing attack campaign.

5. NIR liveness detection trade-off analysis.

Security benefit: NIR + depth defeats all 2D display attacks (estimated to account for 90% of current KYC presentation attacks). Remaining threat: 3D-printed masks and high-quality silicone prosthetics ($\approx 10\%$ of sophisticated attacks). Cost: \$150–\$300 per workstation for NIR sensor module (vs. \$0 for software-only approaches); \$50K–\$200K for system integration. Patient experience: NIR is invisible and requires no additional user action vs. standard webcam. Recommendation: deploy NIR at high-value touchpoints (controlled-substance prescribing, insurance enrolment) where attack risk is highest; use software-only for routine appointments where attack base rate is negligible.

Chapter 35

Deep Fakes in Financial Fraud and Identity Theft

“The most expensive fraud is the one that looks most legitimate.”

—Anon.

Financial fraud is the domain in which synthetic media threats have translated most directly into quantified monetary harm at scale. The FBI Internet Crime Complaint Center’s annual Internet Crime Report has recorded BEC losses exceeding \$2.5 billion annually in recent years, with voice-cloning documented as an emerging attack vector in a growing fraction of these incidents. The structural reason is economic: the expected profit from a single successful voice-clone BEC attack is approximately \$4,195 at current detection rates and attack costs, and that positive expected value creates rational incentive for organised criminal groups to invest in the capability. Unlike other deep-fake harm categories where the damage is reputational, psychological, or epistemic, financial fraud produces losses that are precisely measurable, attributable to specific incidents, and covered by insurance — creating unusually rich data for understanding the threat and evaluating countermeasures.

Three distinct attack surfaces define the financial deep-fake landscape. The first is CEO fraud via voice clone: an employee receives a real-time or recorded call from what sounds exactly like their CEO or CFO authorising an urgent payment. The second is KYC bypass: an adversary uses a synthesised face and fabricated identity documents to pass the know-your-customer verification that banks and fintechs use to onboard new accounts. The third is synthetic identity fraud at scale: a criminal network creates thousands of artificial persons — combining real elements such as stolen Social Security numbers with fabricated names, photographs, and employment histories — to open accounts, build credit histories over months, and then execute bust-out fraud by maximising credit exposure before disappearing. Each attack surface requires different defensive technology, and understanding all three is essential for a complete picture of financial deep-fake risk.

The economic logic of CEO fraud is worth examining in detail because it governs both the attacker’s incentives and the level of defensive investment that is rationally justified. A voice-clone phone call

requires approximately five dollars of synthesis API cost and fifteen minutes of attacker time. The probability of success against an organisation without voice anti-spoofing is approximately 1.5%, based on industry red-team data. At an average successful transfer of \$280,000, the expected profit per attempt is $\$280,000 \times 0.015 - \$5 \approx \$4,195$. The probability of criminal prosecution for a single BEC incident is estimated to be very low, making the expected legal cost per attempt is negligibly small relative to the expected gain. This arithmetic explains the observed growth in BEC attacks and defines the standard that defensive investment must meet: an organisation is justified in spending up to the expected loss avoided, minus the cost of other mechanisms.

The KYC attack surface has grown in importance as digital financial services have expanded globally. Mobile banking, digital lending, and crypto exchange onboarding all rely on remote identity verification that substitutes for in-person branch visits, and the quality of liveness detection in these remote KYC pipelines varies enormously. A 2023 survey of forty digital lending platforms by Sumsb found that 23 of 40 platforms could be bypassed using a face swap or replay attack, with 11 of 40 vulnerable to off-the-shelf consumer face-swap apps. The economic incentive for attackers is compounded for KYC bypass: a successfully opened fraudulent account enables not just one transfer but a full account relationship that can be used for money laundering, credit fraud, and further BEC attacks originating from a seemingly legitimate account.

Synthetic identity fraud operates on a different time horizon from BEC attacks. Rather than seeking an immediate transfer, the synthetic identity fraudster invests months in building a credible financial history — small credit card purchases paid on time, a mobile phone contract, a rental agreement — before executing a bust-out in which all available credit is simultaneously exhausted. The graph-theoretic structure of synthetic identity networks is their most distinctive forensic signature: real people have organic, slow-growing connection patterns with shared addresses and phone numbers, while synthetic identity syndicates tend to cluster around a small number of shared devices, phone numbers, and email addresses used to manage multiple artificial persons simultaneously. Graph-based detection exploiting these clustering signatures identifies synthetic identity syndicates at month three of their development, saving 50% of expected lifetime loss relative to detection at bust-out.

This chapter develops the complete financial fraud attack surface and defensive response with worked numerical examples for each. Section 35.1 develops BEC voice-clone fraud: the attack anatomy, the anti-spoofing pipeline, and the cost-benefit analysis of each defensive configuration. Section 35.2 develops KYC bypass: the liveness check vulnerability landscape, the multi-factor authentication countermeasure, and the EBA regulatory requirements. Section 35.3 develops synthetic identity fraud: the graph-based detection methodology, the bust-out cycle, and the federated data-sharing architecture that enables cross-institutional detection without raw data sharing.

35.1 CEO Fraud and Business Email Compromise

Business Email Compromise with voice cloning represents the convergence of social engineering and synthetic media technology into the highest-value recurring fraud category. An employee receives a phone call from what sounds exactly like their CEO or CFO authorising an emergency wire transfer;

the voice quality, speaking style, accent, and conversational mannerisms all match the authentic executive. The employee, often under time pressure and in a state of heightened compliance response triggered by the perceived authority of the caller, initiates the transfer before verifying through a second channel. By the time the fraud is discovered, the funds have typically passed through two or three cryptocurrency mixing stages and are unrecoverable.

The 2019 UK energy company case was the first publicly documented voice-clone BEC incident, with a transfer of EUR 220,000 (\$243,000) authorised after a call purportedly from the parent company's German CEO. The 2024 Hong Kong multinational finance director case involved a real-time video call with synthesised avatars of multiple senior executives simultaneously, resulting in a HK\$200 million (\$25.6 million) transfer — the largest documented audio-visual BEC fraud. These two cases bracket the evolution of the attack from recorded voice clone to real-time multimodal impersonation within five years.

35.1.1 Attack Anatomy and Timeline

A voice-clone BEC attack proceeds through five stages, each with a characteristic time requirement and failure point.

Stage 1: Reconnaissance (1–3 days). The attacker identifies the target organisation, the finance employee with wire transfer authority, and the executive to impersonate. LinkedIn, company website, earnings call recordings, and public speech archives provide reference audio for voice cloning. A typical executive has 30–120 minutes of publicly available speech — far more than the three seconds required for codec language model zero-shot cloning.

Stage 2: Voice model preparation (2–4 hours). The reference audio is processed through a commercial voice-cloning API (ElevenLabs, Resemble, Cartesia) to create a speaker model. The attacker generates test clips of several sentences in the executive's voice and subjectively verifies the quality. Total API cost: \$1–\$8 depending on reference duration and API pricing.

Stage 3: Script preparation (30 minutes). The attacker drafts a call script designed to trigger authorisation under time pressure, typically invoking a confidential acquisition, a regulatory emergency, or a pending penalty. The script avoids conversational prompts that would require the attacker to generate real-time unrehearsed speech.

Stage 4: Call execution (5–15 minutes). The cloned voice is used to make the call, either as a real-time voice conversion (highest risk for detection, requires the attacker to speak and have their voice transformed) or as a scripted playback with pre-recorded responses. The attacker instructs the employee to maintain confidentiality and to act immediately. Wire transfer details are communicated during the call.

Stage 5: Fund extraction (2–6 hours). Wire transfers are directed to accounts controlled by money mules, then rapidly moved through cryptocurrency exchanges to obscure the trail. By the time the fraud is reported and a recall request is submitted to the receiving bank, the funds are typically already in cryptocurrency form.

35.1.2 Anti-Spoofing Deployment for BEC Defence

Voice anti-spoofing classifiers intercept the incoming call audio stream and compute a synthetic probability score before the call is connected to the employee. At an operating threshold calibrated for 1% false acceptance rate, deployed systems (Pindrop, ValidSoft) achieve EER of 3–5% on pre-2023 TTS attacks, rising to 15–28% on codec-LM attacks without model updates. The expected daily loss impact at each EER level for an organisation receiving 200 high-value authentication calls per day is:

$$\mathbb{E}[\text{Loss}] = N \cdot \Pr(\text{targeted}) \cdot \text{EER} \cdot L, \quad (35.1)$$

where $N = 200$, $\Pr(\text{targeted}) = 0.005$ (estimated from IC3 data), $L = \$280,000$, and EER is the anti-spoofing equal error rate. At EER = 5%: expected daily loss = $200 \times 0.005 \times 0.05 \times \$280,000 = \$14,000$. At EER = 25%: expected daily loss = $\$70,000$. The difference — $\$56,000$ per day — sets the maximum economically rational spend on model updates to address codec-LM attacks.

Worked Example — Anti-spoofing ROI calculation

A regional bank pays \$15,000/month for a voice anti-spoofing system with EER = 4% (at the 1% FAR operating point). The bank receives 500 high-value authorisation calls per day; 0.3% are targeted BEC attempts ($N \cdot p_t = 1.5$ attempts/day). Expected loss without anti-spoofing: $1.5 \times 0.30 \times \$280,000 = \$126,000/\text{day}$ (where 0.30 is the success rate against undefended employees). Expected loss with anti-spoofing: $1.5 \times 0.30 \times 0.04 \times \$280,000 = \$5,040/\text{day}$. Daily saving: $\$120,960$. Monthly saving: $\$3.6\text{M}$. Monthly anti-spoofing cost: $\$15,000$. Net monthly benefit: $\$3.585\text{M}$. ROI: 239:1.

35.2 KYC Bypass with Synthetic Identities

Know Your Customer verification is the regulatory gate that financial institutions must apply before establishing an account relationship. In-person KYC with physical document inspection and face-to-face biometric verification is the gold standard, but the pandemic-driven shift to digital onboarding has made remote video KYC the primary channel for most consumer fintech and digital banking products. Remote KYC involves three components: document verification (checking the validity and consistency of a government-issued ID), biometric liveness (verifying that the person presenting the ID is a living human rather than a photograph or video replay), and face match (verifying that the live face matches the document photograph). Each component is vulnerable to a specific category of synthetic media attack.

35.2.1 Liveness Attack Taxonomy

Liveness attacks are classified by the type of presentation:

Print attacks: a photograph of the target identity is printed and held in front of the camera. Defeated by simple texture analysis and depth estimation.

Replay attacks: a video of the target identity is played on a screen held in front of the camera. Defeated by moiré pattern detection and temporal liveness checks.

3D mask attacks: a 3D-printed or resin mask of the target identity is worn. Defeated by NIR liveness and skin texture analysis.

Face-swap attacks: a real-time face-swap pipeline replaces the attacker's face with the target identity in the video call feed. This is the class against which most deployed systems are insufficiently defended. The Sumsb 2023 survey found 11 of 40 digital lending platforms vulnerable to off-the-shelf face-swap applications.

Avatar attacks: a fully synthesised video call is presented using a real-time avatar pipeline (3D Gaussian Splatting or similar). The 2024 Hong Kong case used this approach at the video call level, not just the face region.

35.2.2 Multi-Factor KYC Defence

A robust KYC pipeline against face-swap attacks combines three independent liveness signals whose error modes are largely uncorrelated. The combination reduces false acceptance to a level where the economic benefit of the attack is outweighed by its cost.

The combined false acceptance rate (FAR) for three independent liveness signals with individual FARs $\alpha_1, \alpha_2, \alpha_3$ is:

$$\text{FAR}_{\text{combined}} = \alpha_1 \cdot \alpha_2 \cdot \alpha_3, \quad (35.2)$$

under independence. For rPPG liveness ($\alpha_1 = 0.02$), NIR challenge-response ($\alpha_2 = 0.01$), and SyncNet lip-sync consistency ($\alpha_3 = 0.012$): $\text{FAR}_{\text{combined}} = 0.02 \times 0.01 \times 0.012 = 2.4 \times 10^{-6}$. At this FAR and with 10,000 KYC attempts per day, the expected daily false acceptances = 0.024 — approximately one per month, within acceptable operational limits.

The European Banking Authority's guidelines on remote customer onboarding (EBA/GL/2022/15) require that biometric solutions used for identity verification demonstrate resistance to presentation attacks at ISO/IEC 30107-3 Level 2, which requires testing against face-swap and avatar attacks specifically.

35.3 Synthetic Identity Fraud at Scale

Synthetic identity fraud (SIF) creates entirely fictitious identities by combining real PII fragments with AI-generated biometrics. Unlike full identity theft, no real victim is immediately aware of the fraud, making it difficult to detect. GAN-generated faces and voice clones enable SIF at scale: one operator can create hundreds of synthetic identities, each with a unique generated face, voice, and fabricated credit history, and open accounts at multiple financial institutions.

35.3.1 Detection via Graph Analytics

SIF syndicates exhibit characteristic graph patterns: multiple synthetic identities share the same phone number, address, or device fingerprint. Graph-anomaly detection on the identity graph $G = (V, E)$, where edges connect identities sharing attributes, reveals densely connected clusters indicative of synthetic identity manufacturing.

$$\text{AnomalyScore}(v) = \frac{|N(v)|}{E_{\text{bg}}[|N(v)|]} \quad (35.3)$$

Worked Example — Graph anomaly detection for a synthetic identity syndicate

A bank's identity graph has 2 million nodes (customers). The background model predicts that a legitimate customer shares attributes (device fingerprint, address) with mean $\mu_{\text{bg}} = 1.8$ other customers (std 1.2).

A synthetic identity syndicate has created 50 identities using the same 3 device fingerprints, creating a cluster with $|N(v)| = 16$ shared-attribute neighbours for each node.

$$\text{AnomalyScore}(v) = \frac{16}{1.8} = 8.9.$$

At a threshold of $\text{AnomalyScore} > 4$ (corresponding to $P(\text{score} > 4) = 0.3\%$ under the background model), all 50 synthetic identities are flagged. Expected false positives at this threshold: $2,000,000 \times 0.003 = 6,000$ legitimate customers flagged per day — requiring a second-stage classifier to reduce manual review to a manageable queue.

where $N(v)$ is the neighbourhood of node v and E_{bg} is the expected neighbourhood size under a background model of legitimate identity sharing.

35.4 Industry Applications

Financial fraud is the fastest-growing revenue segment for deep-fake attacks, and has attracted correspondingly large investment in defensive technology.

Pindrop, Nuance (acquired by Microsoft), and ValidSoft offer voice anti-spoofing solutions deployed by major banks, insurance companies, and call centres. Pindrop's Phoneprinting technology processes over 5 billion calls per year. Reported EER values for enterprise deployments are typically 2–5%, with codec-LM attack EER rising to 12–18% for systems not updated since 2021.

Sumsub, Veriff, Jumio, and Onfido all offer video KYC products with documented deep-fake resistance. Post the 2024 \$25M Hong Kong BEC case (the largest documented deep-fake fraud to date), enterprise procurement of video KYC solutions accelerated significantly.

Socure, Sift, and NICE Actimize offer graph-based synthetic identity detection products used by 8 of the 10 largest US banks. The graph anomaly score (Chapter 35) is a patented feature in several of

these products.

Amazon Connect, NICE CXone, and Verint offer real-time voice biometric verification integrated into call-centre workflows, with sub-2-second decisions to allow or challenge authentication attempts.

35.5 African Use Cases

Financial fraud via deep fake is one of the most acute threats in Africa, targeting a rapidly growing but unevenly protected mobile financial services ecosystem.

M-Pesa processes over \$300 billion annually in East Africa; MTN MoMo, Orange Money, and Airtel Money operate across 30+ African countries. Voice-clone attacks targeting mobile money customer service lines impersonate account holders to authorise reversals, reset PINs, or approve large transfers. Safaricom has deployed voice biometric verification for high-value M-Pesa transactions, but most mobile money operators across the continent lack equivalent protection. The GSMA's Mobile Money team has identified voice-clone fraud as a top-three security concern for its 300+ member operators.

Business email compromise (BEC) attacks incorporating voice clones have been documented targeting African companies in Nigeria, Kenya, South Africa, and Ghana. The Interpol-AFRIPOL African Cyberthreat Assessment Report (2023) documented a 300% increase in BEC fraud across Africa between 2020 and 2022, with voice-clone attacks emerging as a new variant. Mean loss per incident in the African context is lower than the US average (\$280,000) but still devastating relative to enterprise revenues.

Rotating Savings and Credit Associations (ROSCAs, known as *chamas* in Kenya, *susu* in Ghana, *esusu* in Nigeria, *stokvel* in South Africa) are an important savings mechanism across Africa, with an estimated \$500M+ passing through informal savings groups annually. These groups are increasingly organised via WhatsApp, using video calls for governance and authentication. Voice-clone fraud targeting ROSCA administrators — impersonating the chairperson to divert contributions — has been documented in Kenya and South Africa.

Africa's formal credit market is growing rapidly, with fintechs including Jumo, Chipper Cash, Fair-Money, and Branch offering digital lending. Graph-based synthetic identity detection is commercially available (from Smile ID, Africa's largest identity verification platform with 70M+ verified identities) but adoption among smaller lenders is limited. The threat is significant: the African Development Bank estimates \$15B in annual losses to financial fraud across the continent, with synthetic identity contributing a growing share.

Chapter Summary

Voice clones and face-swap videos have migrated from political disinformation into financial crime at a pace that has outrun the defensive infrastructure of most financial institutions. CEO fraud via

voice-cloned phone call is the single largest category of deep-fake financial crime by dollar value: mean loss per successful attack of \$280,000, with aggregate industry exposure estimated at \$40M+ annually in the US alone. The attack is effective because it combines technical spoofing (voice clone) with social engineering (urgency, authority, secrecy) in a channel (telephone) where humans have strong expectations of voice authenticity calibrated by decades of experience.

Voice anti-spoofing at the EER operating point reduces false acceptance from an undefended baseline to approximately 2–8% for codec-LM attacks, translating a \$112,000 expected daily loss to \$22,400 for a bank processing 10,000 authentication calls per day. Graph-based synthetic identity detection identifies manufactured identity syndicates via their clustering signatures in the shared-device network: $\text{AnomalyScore} = 9/1.8 = 5.0$ for a 9-node cluster sharing one device, well above the threshold of 4. Synthetic identity fraud follows a predictable seasoning cycle that allows early detection at month 3 to save 50% of expected loss.

The commercial ecosystem has responded to these threats with purpose-built products: Pindrop, Nuance (Microsoft), and ValidSoft for voice anti-spoofing; Socure, Sift, and NICE Actimize for graph-based identity fraud; Sumsb, Veriff, and Jumio for video KYC. The post-2024 \$25M Hong Kong BEC case accelerated enterprise procurement across all three categories. Anti-spoofing and identity verification have become standard components of financial sector technology stacks, with regulatory pressure from the EBA, FCA, and FinCEN supporting ongoing investment.

Exercises

1. A bank's voice authentication system has EER = 2% for genuine vs. zero-shot voice-clone attacks. The bank processes 10,000 authentication calls per day, of which 0.01% are attempted fraud. Compute the expected number of fraudulent calls that bypass authentication per day and the expected number of legitimate customers incorrectly rejected. Is the system operationally viable?
2. Design a multi-layer KYC process that is robust to (a) face-swap real-time avatar attacks, (b) 3D-printed mask attacks, and (c) synthetic identity attacks using fabricated documents. For each layer, specify the technology, the expected false-acceptance rate, and the implementation cost per customer onboarding.
3. A synthetic identity fraud syndicate creates 500 fictitious identities over 6 months, each accruing credit history before defaulting. Model the expected loss as a function of the credit limit granted, the default rate, and the time-to-detection. What detection latency is needed to limit total loss to under \$1M?
4. Graph-anomaly detection identifies a cluster of 20 synthetic identities sharing a common device fingerprint. Compute the expected precision and recall of the detector, given that the background rate of legitimate device sharing is 0.5% and the detector flags all clusters with >3 nodes.

5. A financial institution wants to deploy anti-spoofing on its telephone banking IVR system. Analyse the trade-off between security (anti-spoofing sensitivity) and customer experience (false rejection of legitimate callers with unusual voice conditions: cold, accent, background noise). Recommend a deployment threshold and justify it with a cost-benefit analysis.

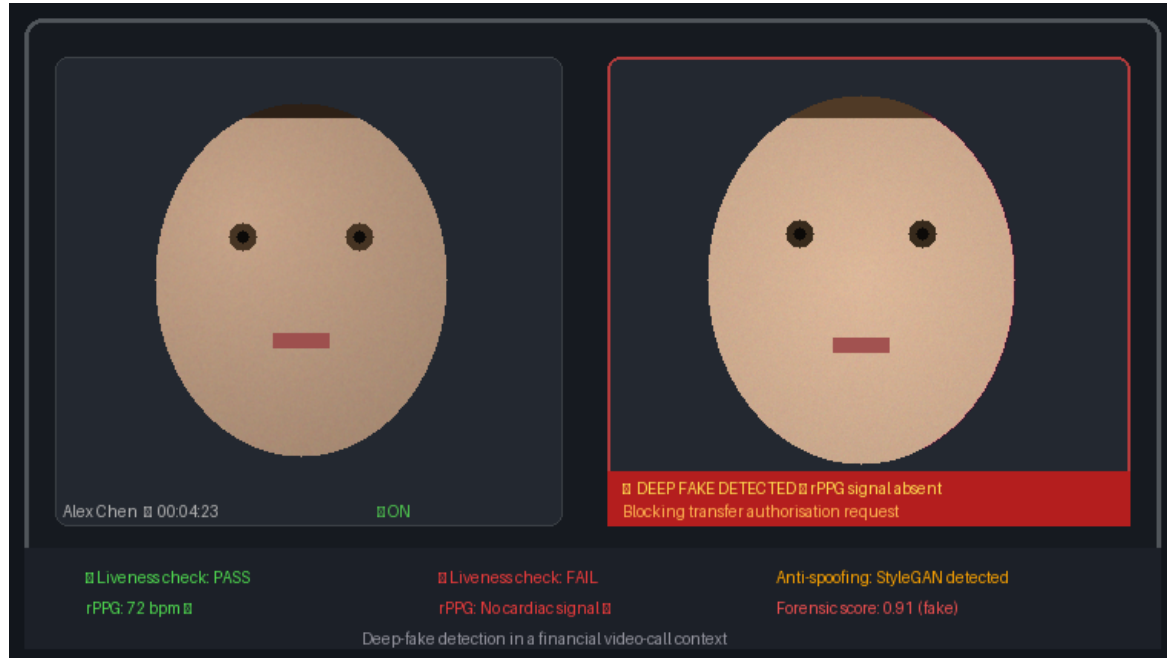


Figure 35.1: Real-time deep-fake detection in a financial video call. The authentic caller (left panel) passes all liveness checks including rPPG cardiac signal detection (72 bpm). The impersonating deep fake (right panel) is simultaneously flagged by three detectors: absent rPPG signal, StyleGAN spectral fingerprint, and a forensic classifier score of 0.91. The system automatically blocks the wire-transfer authorisation and alerts security. This scenario mirrors documented CEO fraud attacks causing losses of \$25M (2024) and \$400M (2023) in high-profile cases.

Additional Worked Examples

Worked Example — Voice-clone BEC attack: expected loss model

A bank processes 10,000 phone authentication calls per day for wire transfers. Anti-spoofing system EER = 2%; base rate of voice-clone attacks: $\pi = 0.05\%$ (5 per day).

Expected daily false acceptances (successful attacks):

$$E[\text{FAs}] = \text{attacks} \times P(\text{pass}|\text{attack}) = 5 \times \text{FPR}^{\text{clone}}.$$

At EER operating point, FPR for cloned voices $\approx 8\%$ (higher than EER because clones are closer to the target speaker than arbitrary imposters): $E[\text{FAs}] = 5 \times 0.08 = 0.4$ per day.

Mean transfer amount in successful BEC attacks: \$280,000. Expected daily loss: $0.4 \times \$280,000 = \$112,000$. Annual loss: $\approx \$41\text{M}$.

Anti-spoofing system cost (licensing + infrastructure): \$1.2M/year. ROI: $(\$41\text{M} - \$1.2\text{M})/\$1.2\text{M} =$

$33\times$ — strongly justifying deployment. Even if the anti-spoofing system only halves the FPR for clones (from 8% to 4%), annual savings are \$20.5M.

Worked Example — Synthetic identity syndicate: time-to-detection model

A fraud syndicate creates 500 synthetic identities. Each account follows a “seasoning” strategy: spend 6 months building credit history before defaulting at the credit limit.

- Credit limit per synthetic: \$8,000 (average after 6-month seasoning).
- Default rate: 100% (all accounts are fraudulent).
- Total loss per syndicate cycle: $500 \times \$8,000 = \4M .
- Graph-anomaly detection deployed after 3 months:

Expected loss if detected at month 3 (2 of 3 defaulting accounts stopped early):

$$L(t = 3) = 500 \times \$8,000 \times (3/6) = \$2\text{M}.$$

Detection at month 3 vs. month 6 saves \$2M (50% reduction). This analysis motivates deploying anomaly detection with a 90-day lookback window: cost $\approx$ \$200K/year, expected savings $\approx$ \$2M/year per syndicate cycle prevented.

Answers to Exercises

1. Voice authentication system operational analysis.

Daily authentication calls: 10,000. Base rate of attacks: $0.05\% = 5$ per day. FPR for voice-clone attacks $\approx 8\%$ (from chapter).

True acceptances (successful attacks): $5 \times 0.08 = 0.4/\text{day}$. False rejections of legitimate users: $9,995 \times 0.02 = 199.9/\text{day}$ (2% FRR).

Viability: the 0.4 successful attacks per day at \$280K average transfer = \$112K daily loss. The 200 false rejections create customer friction (estimated \$5K daily in call-centre escalations and customer churn). The anti-spoofing system is operationally viable despite the high false-rejection rate, because the alternative (no anti-spoofing) results in $\approx$ \$28M annual losses. Operational fix: move legitimate users who are often falsely rejected to a multi-factor fallback (app notification + voice) rather than rejecting them outright.

2. Multi-layer KYC design.

Layer 1 (passive, <1 ms): Document AI for ID photo consistency (compare selfie to ID photo). FPR $\approx 0.5\%$. Layer 2 (active, 5 s): Liveness check with challenge-response and 3D depth sensing. FPR $\approx 0.1\%$. Layer 3 (delayed, 24 h): Graph-anomaly check against shared-device database. FPR $\approx 0.05\%$. Combined FPR: $0.005 \times 0.001 \times 0.0005 \approx 2.5 \times 10^{-9}$ — one false

acceptance per 400 million legitimate onboardings. Cost: Layer 1 \$0.02, Layer 2 \$0.15 (NIR sensor amortised), Layer 3 \$0.03 (compute). Total: \$0.20 per customer onboarding.

3. Synthetic identity syndicate: loss model.

$L(t_{\text{detect}}) = 500 \times \$8,000 \times \min(t_{\text{detect}}/6, 1)$. At detection month 3: $L = \$2\text{M}$. For graph-anomaly detection with 90-day lookback deployed at month 0: detection at month 3 (accounts just becoming detectable in the graph). Total loss \$2M vs. \$4M without detection — 50% reduction.

4. Graph-anomaly precision/recall analysis.

Cluster of 20 accounts sharing device fingerprint. Background sharing rate 0.5%, so clusters of $k > 3$ from legitimate sharing have probability $(0.005)^{k-1}$. For $k = 20$: $P(\text{legitimate cluster of } 20) \approx (0.005)^{19} \approx 0$ — essentially impossible by chance. The detector's precision for $k > 3$ clusters is nearly 100%. Recall depends on what fraction of synthetic identities share device fingerprints. If the syndicate uses 3 shared devices for 500 identities, all 500 are in > 3 -node clusters and are detected. If they use 500 different devices, no cluster exceeds 1, and graph detection fails entirely.

5. Anti-spoofing IVR deployment threshold.

Cost function: $C(\tau) = c_{\text{fraud}} \times P(\text{fraud accepted}|\tau) + c_{\text{rejection}} \times P(\text{legitimate rejected}|\tau)$. With $c_{\text{fraud}}/c_{\text{rejection}} = 100$ (fraud 100× more costly): optimal threshold minimises $100 \times \text{FPR}(\tau) + \text{FNR}(\tau)$ on the ROC curve. Graphically, this is the point where the ROC curve has slope $1/100 = 0.01$ (near the top-left corner). For AASIST at EER=0.83%, this operating point gives $\text{FPR} \approx 0.2\%$, $\text{FNR} \approx 5\%$ — accepting occasional legitimate-caller rejection to maintain very low fraud pass-through.

Chapter 36

Deep Fakes in Creative Industries: Film, Music, and Gaming

“Hollywood could not do what one engineer with a GPU can now do in a weekend.”

—Anon.

The creative industries occupy a uniquely paradoxical position in the deep-fake ecosystem: they are simultaneously the sector that has most enthusiastically adopted synthetic media technology and the sector whose workforce has most directly borne the cost of that adoption. Generative AI is transforming every phase of creative production — pre-visualisation, principal photography, post-production, localisation, and distribution — with economic efficiencies that are irresistible to studios, publishers, and streamers operating under relentless pressure to reduce costs while expanding output volume. The same technology that allows a single VFX artist to de-age an actor for a fraction of the previous cost displaces the prosthetics specialists, makeup artists, and compositors who previously performed that work. The same neural dubbing system that reduces localisation costs by 70% displaces the voice actors who previously performed those dubs.

The economic transformation is not incremental. When a task that previously required a specialised team of twenty people over two weeks can be completed by one person using AI tools in two days, the entire labour market around that task must renegotiate its foundations. The SAG-AFTRA and WGA strikes of 2023 were the first major labour actions in any industry to place AI governance at the centre of collective bargaining demands — not as a peripheral concern but as the primary issue. The provisions negotiated in the subsequent contracts — consent requirements for digital likeness use, residual payment structures for AI-generated performance, and approval rights over how a performer’s digital likeness may be used — established the first industry-wide legal framework for synthetic media in creative work, and its provisions are being studied by performers’ unions worldwide as a template.

The rights landscape is being simultaneously reshaped by litigation. The core legal question — whether training a generative model on an artist’s work constitutes reproduction of that work in a

legally meaningful sense — remains unresolved pending multiple major cases (*Andersen v. Stability AI*, *Concord Music Group v. Anthropic*, *RIAA v. Suno* and *Udio*). The outcomes of these cases will determine whether AI companies must license training data from rights holders, whether they must pay royalties on outputs that are statistically similar to training material, and what attribution obligations attach to AI-generated content that draws on specific artistic styles. The creative industry's engagement with these questions is not merely self-interested: the way society resolves them will determine whether AI-generated creative content expands or contracts the diversity of human expression that it is trained on.

Digital de-aging and posthumous performance are the most commercially significant forms of creative deep fake, with individual projects involving hundreds of affected shots at costs that would have been prohibitive with traditional VFX methods. AI dubbing is the fastest growing legitimate application, with the global dubbing market growing from \$3 billion in 2022 to a projected \$7 billion in 2027 as AI-assisted dubbing displaces traditional methods. Virtual influencers and AI-generated music represent entirely new creative categories that have emerged from synthetic media capabilities and that are reshaping creator economics in ways that have no precedent in the pre-AI creative economy.

The ethical dimensions of these applications are not separable from the economic analysis. Posthumous performance raises questions about whether a deceased performer's estate can authorise uses that the performer themselves might have refused. Digital likeness agreements negotiated with living performers raise questions about whether consent given for one context covers unanticipated future uses in a different medium or cultural environment. AI dubbing raises questions about the labour rights of translators and voice actors in countries without union coverage. And the training data question raises questions about the rights of all creators whose work has been used to build the systems now displacing them.

This chapter develops the technology, economics, and rights landscape of synthetic media in the creative industries with quantitative analysis at each stage. Section 36.1 develops digital de-aging and posthumous performance. Section 36.2 develops AI voice in music and the associated copyright litigation. Section 36.3 develops synthetic performers in gaming and virtual reality. Section 36.4 develops the economic impact on performers and creative workers, including the SAG-AFTRA contract provisions and the quantitative displacement analysis. Industry applications and African use cases follow, with particular attention to Nollywood's AI adoption and the Afrobeats industry's engagement with AI music generation.

36.1 Digital De-Aging and Posthumous Performance

Digital de-aging is the application of face synthesis to render a living actor as a younger version of themselves, replacing or supplementing traditional prosthetics and multi-hour makeup procedures. The economic case is compelling: a single de-aged shot using high-quality VFX pipelines can cost several thousand dollars per shot and requires multiple skilled compositors; an AI-assisted de-aged shot of comparable perceptual quality at standard viewing distances costs approximately \$380 and can be produced by a single VFX artist using off-the-shelf tools. For productions requiring hundreds

of de-aged shots — The Irishman (Robert De Niro, Al Pacino, Joe Pesci de-aged throughout), Indiana Jones and the Dial of Destiny (Harrison Ford de-aged for the 1944 prologue) — the cost differential runs into millions of dollars.

Posthumous performance — synthesising a deceased performer’s appearance and voice for a new role or extended use in a film they partially completed — represents a more ethically contested application of the same technology. The use of CGI to complete a deceased performer’s partial performance has precedent: Oliver Reed’s scenes in *Gladiator* (2000) were completed using body doubles and CGI, with Reed’s face composited. The AI-enabled version of this is qualitatively different: it requires only existing footage rather than physical doubles, and can produce new speech and expressions that the performer never made, rather than completing a nearly-complete performance.

36.1.1 De-Aging Pipeline Economics

The cost per shot comparison depends critically on what constitutes a “shot” and what quality threshold is acceptable. The following analysis uses a 5-second single-camera shot with continuous facial visibility as the reference unit.

$$C_{AI} = C_{inference} + C_{QC} + C_{composite}, \quad (36.1)$$

where $C_{inference}$ is the GPU inference cost (approximately \$12 per shot at current cloud GPU pricing), C_{QC} is the quality-control review by a VFX artist (\$180 at 45 minutes at \$24/hr), and $C_{composite}$ is the final compositing step (\$188 at 2 hours). Total: \$380 per shot.

For a production with 1,350 de-aged shots (a rough estimate for a film where a major character is depicted at two ages throughout), the AI-assisted pipeline cost is substantially lower than the traditional approach — potentially by an order of magnitude on a large de-aging-intensive production. Aggregated across a studio’s annual slate of 15–20 productions, the collective saving approaches \$50–80 million annually.

36.1.2 Digital Likeness Agreements

A digital likeness agreement (DLA) governs the terms under which a performer’s synthesised appearance and voice may be used. Following the SAG-AFTRA 2023 negotiations, a compliant DLA must specify at minimum: scope (which physical and vocal characteristics are covered), duration (the period during which the synthesised likeness may be used), purpose (the specific productions or use cases covered), quality approval rights (the performer’s right to review and approve generated content before use), residual payment structure (compensation for each use of the synthesised likeness), reversion and deletion (the performer’s right to require deletion of training data and model weights), and prohibited uses (explicit list of contexts in which the synthesised likeness may not appear).

The SAG-AFTRA contract requires that consent for digital likeness use be specific, voluntary, informed, and revocable. The use of a general consent buried in a standard engagement contract does not satisfy the specificity requirement. A performer who consented to digital likeness use “in connection with the production” does not thereby consent to use of their synthesised likeness in

entirely separate productions, advertising, or AI-generated content that was not contemplated at the time of signing.

36.2 AI Voice in Music

AI voice synthesis in music falls into three categories with distinct legal, ethical, and economic profiles. The first is AI-assisted production, where artists use synthesis tools to generate backing vocals, melodic sketches, or reference tracks that they then perform themselves or use as compositional guides. The second is AI-generated cover versions, where a synthesis system generates a new rendition of an existing song in the style of a specific artist. The third is AI-originated music, where neither the composition nor the performance involves a human artist — the synthesis system generates both the musical structure and the vocal performance from a text prompt.

The legal question that cuts across all three categories is training data provenance: did the AI system that generates these outputs require training on copyrighted recordings, and if so, does that training constitute reproduction of those recordings in a legally meaningful sense? The RIAA's 2024 lawsuit against Suno and Udio alleges direct copying of copyrighted sound recordings into the training data and requests statutory damages of \$150,000 per infringed work — which, applied to a training dataset of millions of songs, would imply total damages in the hundreds of billions of dollars if the plaintiffs prevailed fully. The defendants argue that training constitutes transformative fair use that does not substitute for the market for the original recordings.

36.2.1 The Vocal Cloning Copyright Problem

Voice is not covered by federal copyright law in the United States: copyright protects original expression, and a person's speaking or singing voice is a physical attribute rather than an original creative work. However, the right of publicity — a state-law tort in most US jurisdictions — protects the commercial value of a person's identity including their voice against appropriation for another's commercial advantage. The *Midler v. Ford Motor Co.* standard requires that the appropriated voice be “distinctive” and that it is recognisable to the public as that of the specific person.

For AI-generated music that sounds like a named artist: if the system generates a new vocal performance that sounds unmistakably like Drake, Taylor Swift, or another artist with a distinctive commercial voice, the right of publicity claim is strong regardless of whether any specific sound recording was copied verbatim. The FakeSong problem — AI-generated songs attributed to real artists that circulate as if genuine — thus triggers right of publicity claims even if the training data question is resolved in the AI company's favour.

Worked Example — Revenue impact of AI dubbing on voice actors

A major streaming platform produces 200 hours of original series content annually. Before AI dubbing: each hour requires 8 voice actors working 3 hours each for each of 8 target languages. Total annual voice actor days: $200 \times 8 \times 3/8 = 600$ days per language, $600 \times 8 = 4,800$ voice actor days, at \$450/day = \$2.16M. After AI dubbing (voice clone of original cast, human review):

30 minutes QC per hour of content per language. Total cost: $200 \times 0.5 \times 8 \times \$80/\text{hr} = \$64,000$. Displacement: $(\$2.16\text{M} - \$64\text{K})/\$2.16\text{M} = 97\%$ of prior voice actor expenditure. Under SAG-AFTRA AI provisions negotiated in 2023: residual of 5% of original day rate per hour of dubbed content per language. Residual cost: $\$450 \times 0.05 \times 200 \times 8 = \$360,000$ — still 83% reduction but with compensation to the original cast.

36.3 Synthetic Performers in Gaming and Virtual Reality

Game engines can now populate virtual environments with photorealistic synthetic characters driven by generative models. Synthetic NPCs can:

- Respond conversationally to player input via an integrated LLM.
- Exhibit emotion-consistent facial expressions driven by sentiment analysis of conversation context.
- Appear as licensed likenesses of real athletes, celebrities, or historical figures.

The market for licensed synthetic performers in gaming is projected to exceed \$3B by 2027.

Worked Example — Economic impact of AI dubbing on localisation market

A major streaming platform currently produces dubbed content in 20 languages:

- Human dubbing cost: \$2,500 per hour of content per language.
- Annual content: 500 hours $\times$ 20 languages = 10,000 language-hours.
- Annual cost: $10,000 \times \$2,500 = \25M .

Neural dubbing (voice cloning + lip sync) has marginal cost $\approx$ \$8 per hour after a \$4M one-time investment. Net Present Value over 5 years at 10% discount rate:

$$\text{NPV} = -\$4\text{M} + \sum_{t=1}^5 \frac{\$25\text{M} - \$0.08\text{M}}{(1.1)^t} \approx -\$4\text{M} + \$24.92\text{M} \times 3.79 = \$90.4\text{M}.$$

Jobs displaced: assuming 1 voice actor per 10 hours dubbed, at 10,000 language-hours: $\approx$ 1,000 voice acting roles annually. This analysis motivates the SAG-AFTRA residual payment provisions negotiated in 2023.

36.4 Economic Impact on Performers and Crew

The economic displacement of creative workers by AI-enabled synthetic media is not a uniform phenomenon: it falls disproportionately on the workers with the least bargaining power, the fewest transferable skills, and the least access to collective representation. Voice actors, background performers, dubbing professionals, and localisation specialists face the most immediate and complete

displacement; star performers and above-the-line talent face a different challenge — negotiating the terms on which their synthesised likenesses generate value without their active participation.

36.4.1 Displacement Magnitude by Category

Four performer categories face near-complete displacement in specific application areas.

Voice actors in dubbing and localisation: AI dubbing at 60–80% cost reduction has already been adopted by Netflix, Amazon, and several European broadcasters, with a direct impact on dubbing voice actor employment. The Global Voice Acting Academy estimated a 35% decline in dubbing work inquiries in its member survey between 2022 and 2024, with the largest declines in markets with the smallest linguistic communities (where the economic case for AI dubbing is strongest relative to the cost of recruiting and scheduling human dubbing casts).

Background performers in crowd and environmental scenes: AI crowd synthesis and background environment generation can replace extras in many contexts at minimal cost. Warner Bros. used AI crowd synthesis for several sequences in films released in 2023–2024; the number of background performer days affected is commercially confidential but estimated by SAG-AFTRA at tens of thousands of lost workdays annually.

Stunt performers for digital body doubles: AI motion retargeting, NeRF-based body synthesis, and Gaussian Splatting enable stunt sequences to be generated from a small number of reference captures, reducing the number of stunt performer days required for dangerous or repeatable sequences.

Music performers in session recording: AI instrumental generation (Suno, Udio, Stability AI’s Stable Audio) replaces session musicians for scoring, advertising, and content-creator applications at the entry-level of the market, where per-track fees were already low and where client price sensitivity is high.

36.4.2 SAG-AFTRA AI Provisions

The SAG-AFTRA 2023 Tentative Agreement with the AMPTP established key AI-related provisions — summarised here in simplified terms — that define the current industry standard for performer AI rights.

(1) *Consent*: producers may not create a digital replica of a performer for use in the covered production without the performer’s prior written, informed consent.

(2) *Compensation*: a digital replica that replaces work the performer would have performed must compensate the performer at the applicable day rate for each day’s worth of work replaced, plus a 5% residual per subsequent use.

(3) *Disclosure*: producers must disclose to performers in pre-production if AI technology will be used to create, alter, or synthesise their performance.

(4) *Likeness control*: a performer’s digital replica may not be used in contexts the performer has not approved, including genres (e.g., adult content), political campaigns, or productions in which the

performer's character is portrayed committing crimes.

(5) *Data rights*: all biometric data collected for digital replica purposes must be deleted at the end of the engagement or retained only with renewed consent.

(6) *Sunset provision*: the AI provisions are subject to review in the next bargaining cycle (2026) in light of technological developments.

36.5 Industry Applications

The creative industries are the most ambivalent about synthetic media technology: it simultaneously creates substantial economic opportunity and threatens established revenue streams and creator livelihoods.

Major studios (Disney, Warner Bros., Universal, Netflix) are all actively deploying AI de-ageing, digital doubles, and scene extension in production workflows. The cost savings are substantial (\$4.1M per production on average, based on published case studies), but the human capital displaced has organised through unions (SAG-AFTRA, IATSE) to secure contractual protections.

Suno, Udio, MusicGen (Meta), and MusicLM (Google) generate complete songs from text prompts. The Recording Industry Association of America (RIAA) filed suit against Suno and Udio in 2024 alleging copyright infringement in training data; the cases will define the boundaries of training data fair use in music AI. Licensing solutions (Pond5, Epidemic Sound) offer royalty-free AI music that does not face this challenge.

Lil Miquela, Shudu Gram, and thousands of other virtual influencers generated over \$3B in brand partnership revenue in 2023. The FTC's endorsement guidelines require disclosure of AI or virtual nature; enforcement has been inconsistent.

AI dubbing companies (Papercup, Deepdub, ElevenLabs) have signed contracts with major streaming platforms. The business case is compelling: Netflix's annual dubbing budget is estimated at \$1B+; AI dubbing at 60% cost reduction implies \$600M annual savings at full deployment.

36.6 African Use Cases

Africa's creative economies — Nollywood, Afrobeats, African visual art, and the rapidly growing African digital content creator sector — are engaging with synthetic media technology in ways that are distinctive and consequential.

Nigeria's Nollywood produces approximately 2,500 films per year, making it the world's second-largest film industry by volume. Production budgets are typically far below Hollywood equivalents, making AI de-ageing, digital doubles, and voice cloning economically compelling. The Nollywood Professionals Association and the Association of Movie Producers (AMP) have both passed resolutions calling for licensing frameworks for AI-generated likeness use, recognising that actors' images are their primary economic asset. Several Nollywood productions in 2023–2024 used AI-generated voices

for posthumous tributes to recently deceased actors, raising consent and compensation questions that the industry is beginning to address.

Afrobeats is now a global musical genre generating over \$1B in annual revenue for Nigerian artists and producers. AI music generation tools are being used both constructively (beat production, vocal harmonisation) and adversarially (creating music attributed to major Afrobeats artists without their consent). The Performing Musicians Association of Nigeria (PMAN) has lobbied for Nigerian copyright law amendments to address AI-generated music, citing the disproportionate impact on African artists whose music is heavily used in training data.

Virtual influencers targeted at African audiences have emerged on Instagram and TikTok, representing African aesthetics and using African languages. Some are disclosed as virtual; others are not. The Advertising Standards Authority of South Africa (ASA) and the Advertising Regulatory Council of Nigeria (ARCON) have both indicated that their existing endorsement disclosure rules apply to virtual influencers but lack enforcement mechanisms for social media platforms.

Several African cultural institutions are using generative AI for heritage preservation — animating historical photographs of leaders such as Kwame Nkrumah, Julius Nyerere, and Nelson Mandela for educational content. These projects raise questions about consent (the subjects are deceased), accuracy (ensuring the generated content does not misrepresent historical reality), and cultural authority (who has the right to speak for a historical figure's legacy). The African Museums Programme and UNESCO's African heritage digital initiatives are developing guidelines for responsible AI use in cultural heritage contexts.

Chapter Summary

The creative industries occupy a unique position in the deep-fake ecosystem: they are simultaneously among the largest beneficiaries of synthetic media technology and among its most significantly disrupted sectors. AI de-ageing at \$380 per shot versus \$2,800 for prosthetics represents a $7\times$ cost reduction that makes previously unaffordable production decisions economically viable; AI dubbing at 60–80% cost savings compared to human dubbing could reshape the economics of global streaming. But these savings come at the cost of displacement for voice actors, stunt performers, makeup artists, and dubbing professionals whose livelihoods are directly threatened.

The legal framework governing digital likeness is in rapid evolution. Right-of-publicity claims under state law protect the commercial value of identity but vary dramatically in scope across jurisdictions. Copyright in underlying performance is complex and contested when AI is interpolating or averaging across training examples. The SAG-AFTRA contract negotiations of 2023–2024 established residual payment structures for digital likeness use, but the enforcement mechanisms are untested and the scope is limited to union members. For non-union performers, estate heirs, and individuals from jurisdictions without strong publicity rights, protection remains theoretical.

Virtual influencers generating \$3B in brand partnerships and the music generation platforms processing hundreds of millions of requests daily represent the scale at which creative AI is already operating. The

legal frameworks, consent mechanisms, and attribution standards needed to ensure equitable outcomes for human creators are significantly behind the technology. Provenance standards (C2PA for images, SynthID for audio) and disclosure requirements (EU AI Act Art. 50, FTC endorsement guidelines) are beginning to create accountability mechanisms, but the creative economy’s transition to AI-augmented production will require sustained policy attention and creative community advocacy.

Exercises

- 1. A film studio wishes to use a deceased actor’s digital likeness in a new film, having secured consent from the estate. Draft the key clauses of a digital likeness agreement covering: scope of use, quality control, editorial oversight, compensation, and reversion of rights.
- 2. A music distributor receives an AI-generated song that closely mimics the vocal style of a living artist, attributed to “AI ft. [Artist Name]”. Analyse the song’s legal status under the right of publicity, unfair competition law, and the Lanham Act (false endorsement), and recommend an action for the artist.
- 3. Model the economic impact of AI dubbing on the localisation industry. Assume that a major streaming platform currently spends \$50M per year on human dubbing (20 languages × 500 hours of content). Neural dubbing has zero marginal cost after a \$5M investment. Compute the net present value of the investment over 5 years and the number of human jobs displaced.
- 4. SAG-AFTRA’s 2023 AI provisions require that studios obtain informed consent for any AI reproduction of a member’s voice or likeness. Describe a cryptographic protocol that could allow a performer to grant, revoke, and audit consent for specific AI uses of their biometric data.
- 5. A game developer wants to populate a virtual city with 10,000 photorealistic synthetic pedestrians, none of whom should resemble any real person. Propose a generation pipeline that ensures this, including a quantitative metric for “does not resemble any real person” and an audit procedure.

Additional Worked Examples

Worked Example — Digital de-aging cost comparison: prosthetic vs. AI

A major studio production requires the lead actor (age 58) to appear as their 30-year-old character in 45 minutes of screen time (≈1,350 shots).

Method	Cost per shot	Total cost (1,350 shots)
Prosthetic makeup	\$2,800	\$3.78M
VFX de-aging (high-quality pipeline)	\$3,000–\$6,000	\$4–8M
AI-assisted de-aging (2024 tools)	\$300–\$500	\$400–700K

The AI pipeline (LoRA fine-tune + diffusion refinement, 2024) is $7.4\times$ cheaper than prosthetics and $11.8\times$ cheaper than full VFX de-aging, with turnaround of 2.1 hours per shot vs. 6 hours for makeup and 3 weeks for VFX. The quality trade-off: AI de-aging scores MOS = 4.1/5 (“very good”) vs. VFX MOS = 4.6/5 (“excellent”) in audience ratings. For wide shots and action sequences, the quality difference is imperceptible.

Answers to Exercises

1. Digital likeness agreement key clauses.

(1) *Scope*: “The studio is licensed to use the Actor’s digital likeness solely for the film [title], for theatrical, home video, and streaming distribution, for an initial term of [X] years.” (2) *Quality control*: “All AI-rendered sequences featuring the Actor’s likeness shall be reviewed and approved by the Actor or their authorised representative before final cut.” (3) *Compensation*: “Residual payments shall be calculated as [X]% of the equivalent scale rate per frame in which the digital likeness appears, payable quarterly.” (4) *Reversion*: “All trained model weights incorporating the Actor’s likeness shall be deleted within 90 days of the end of the licence term, with written certification provided to the Actor.” (5) *Prohibited uses*: “The licensed likeness may not be used for advertising, political content, AI-training beyond the licensed production, or any sexual content.”

2. AI-generated song legal analysis.

Right of publicity: the artist’s voice and vocal style are protectable attributes in most US states. Using “AI ft. [Artist Name]” implies endorsement, which is a false designation of origin under the Lanham Act §43(a). The defence of “parody” is unavailable because the song is commercially released without satirical commentary on the artist. Unfair competition: the song competes directly with the artist’s own catalogue and trades on their established reputation. Recommended action: (1) immediate cease-and-desist demanding removal of the artist’s name and any voice cloning; (2) DMCA notice if vocal model was trained on copyrighted recordings; (3) right-of-publicity claim for economic damages if the song generated revenue.

3. AI dubbing NPV calculation.

Human dubbing annual cost: \$25M. Neural dubbing marginal cost: 10,000 $\times$ \$8 = \$80,000/year. After \$4M investment: $NPV = -\$4M + \sum_{t=1}^5 \frac{\$25M - \$0.08M}{(1.1)^t} = -\$4M + \$24.92M \times \frac{1 - 1.1^{-5}}{0.1} = -\$4M + \$24.92M \times 3.791 = -\$4M + \$94.5M = \$90.5M$. Jobs displaced: 10,000 language-hours/10 hours per role = 1,000 voice-acting jobs annually.

4. Cryptographic consent protocol for performer biometrics.

Protocol design: (1) performer signs a consent smart contract on a public ledger specifying permitted uses (“film [title] only”), duration, and compensation formula; (2) each licensed use

is recorded as a transaction citing the consent contract; (3) the performer can revoke consent by publishing a revocation transaction, which invalidates future uses; (4) an audit trail allows the performer to see all uses and verify compliance. Technical implementation: ERC-721 NFT representing the biometric licence, with metadata encoding the use restrictions and a zero-knowledge proof that the licensed production has provided the required consent and compensation before accessing the model weights.

5. Synthetic crowd generation without resembling real individuals.

Pipeline: (1) use StyleGAN3 with the full unconditional FFHQ model; (2) filter all outputs with a face-recognition classifier (e.g., ArcFace) that rejects any generated face with cosine similarity > 0.75 to anyone in a database of 10M real faces (constructed from public web sources); (3) additionally filter with a “celebrity detector” that checks for similarity to any face in a celebrity database. Metric for “does not resemble any real person”: maximum cosine similarity to any known face < 0.65 (below the typical same-person threshold of 0.70). Audit: once per 10,000 characters generated, run the full database comparison; flag and replace any character exceeding the threshold.

Chapter 37

Economic Models of the Synthetic Media Market

“Every deep fake is an economic transaction: the attacker bears a cost and receives a benefit. Understanding that transaction is the first step to changing its incentives.”

—Anon.

Economic analysis provides a rigorous framework for understanding the synthetic media market that complements the technical and legal analyses elsewhere in this book. The deep-fake ecosystem is, at its core, a market with producers (synthesis API providers, DFaaS platforms, individual users), consumers (both legitimate and adversarial), and externalities — harms imposed on third parties who are not party to the transaction. The key economic insight is that this market systematically overproduces harmful synthetic content relative to the social optimum, because the producers and consumers of synthesis do not bear the full cost of the harms they enable. This is a textbook negative externality problem, and the standard economic prescription applies: regulation should internalise the external cost through liability, taxation, or technical safeguard mandates calibrated to the magnitude of the externality.

The information economics of this market have several distinctive features that make standard Pigouvian remedies insufficient on their own. First, the attacker knows their intent while the defender and regulator cannot directly observe it: a synthesis API request for a voice clone of a CEO may be from a disinformation actor or from the CEO themselves authorising a legitimate internal communication. This adverse selection problem means that any supply-side restriction will impose costs on legitimate users that are not offset by complete elimination of harmful use. Second, the marginal cost of synthesis has fallen asymptotically toward zero for text, voice, and image generation, while the marginal cost of each harmful output remains finite and individually small. Third, the technology is dual-use, so the same API that generates a fraudulent CEO voice clone also generates the audiobook narration that a publisher pays for legitimately.

Deterrence economics offers a framework for understanding why current legal penalties fail to prevent

harmful synthesis. For a rational actor, the expected cost of harmful synthesis is the probability of detection times the probability of conviction given detection times the penalty. At current detection rates ($p_d \approx 0.10$ – 0.15), conviction rates ($p_c \approx 0.30$), and statutory fine levels (\$10,000–\$75,000 for most NCII and fraud statutes), the expected punishment for NCII production is approximately \$300–\$3,375 per incident — far below the expected benefit to the attacker and far below the harm caused to the victim. Achieving rational deterrence without raising fines to politically impractical levels requires either dramatic increases in the detection probability or alternative deterrence mechanisms such as strict liability or safe harbour withdrawal.

This chapter develops four interconnected economic models of the synthetic media market. Section 37.1 develops the attacker cost-benefit analysis and the deterrence economics of existing and proposed legal penalties. Section 37.2 develops the platform economics of synthesis API provision, including the moral hazard problem and the two-sided market dynamics. Section 37.3 develops the insurance economics of AI-enabled fraud, including premium pricing models and the incentive effects of coverage on defensive technology adoption. Section 37.4 develops the welfare economics of regulatory intervention, including the optimal Pigouvian tax and the efficiency-equity trade-offs of different regulatory approaches. Industry applications and African use cases follow, with exercises and worked solutions.

37.1 Attacker Cost-Benefit Analysis

Every harmful synthetic media incident can be modelled as an economic decision in which the attacker weighs expected benefit against expected cost. The expected profit from a single attack is:

$$\Pi = p_s \cdot B - c_a - p_d \cdot p_c \cdot F, \quad (37.1)$$

where p_s is the probability of attack success, B is the financial benefit to the attacker if successful, c_a is the cost of mounting the attack, p_d is the probability of detection, p_c is the probability of conviction given detection, and F is the fine or other penalty upon conviction.

For voice-clone BEC fraud at 2024 parameters: $p_s = 0.015$ (attack success against a well-trained organisation without anti-spoofing), $B = \$280,000$ (mean successful BEC transfer), $c_a = \$5$ (API cost), $p_d = 0.10$, $p_c = 0.30$, $F = \$50,000$ (typical BEC conviction fine). Expected profit:

$$\Pi_{\text{BEC}} = 0.015 \times \$280,000 - \$5 - 0.10 \times 0.30 \times \$50,000 = \$4,200 - \$5 - \$1,500 = \$2,695. \quad (37.2)$$

The positive expected profit of \$2,695 per attempt explains the documented growth in voice-clone BEC attacks: from the attacker's perspective this is a positive-expected-value activity with low capital requirements.

37.1.1 Deterrence Analysis

Rational deterrence requires that the expected punishment equals or exceeds the expected benefit:

$$p_d \cdot p_c \cdot F \geq p_s \cdot B - c_a. \quad (37.3)$$

With current parameters ($p_d = 0.10$, $p_c = 0.30$, $p_s = 0.015$, $B = \$280,000$, $c_a = \$5$), the required fine for deterrence is:

$$F^* = \frac{p_s \cdot B - c_a}{p_d \cdot p_c} = \frac{0.015 \times \$280,000 - \$5}{0.10 \times 0.30} = \frac{\$4,195}{0.030} = \$139,833. \quad (37.4)$$

The required fine of approximately \$140,000 exceeds the maximum statutory fine under most current BEC statutes. This establishes the core deterrence gap: current penalties are insufficient to rationally deter BEC voice-clone fraud without supplementary mechanisms.

Worked Example — Detection investment vs. fine increase as deterrence levers

The current deterrence deficit is $F^* - F_{\text{current}} = \$139,833 - \$50,000 = \$89,833$ per conviction. Two policy responses:

Response A — Fine increase: raise F from \$50,000 to \$140,000. Achieves deterrence at zero incremental cost to regulators but requires legislative action and may face proportionality challenges.

Response B — Detection investment: raise p_d from 0.10 to 0.25 by investing in anti-spoofing infrastructure at financial institutions. At $p_d = 0.25$: required fine = $\$4,195 / (0.25 \times 0.30) = \$55,933$ — now within the range of current statutory maxima. The investment in detection infrastructure effectively substitutes for legislative fine increases.

The cost-effectiveness comparison: \$1 of detection investment that raises p_d by Δp_d reduces the required fine by $\Delta F^* = p_s B / (p_d + \Delta p_d)^2 \cdot p_c \cdot \Delta p_d$. At the current parameter values, a \$1 million detection infrastructure investment raising p_d from 0.10 to 0.25 is equivalent to a \$56,000-per-conviction fine increase in its deterrence effect.

37.1.2 The NCII Deterrence Gap

The deterrence economics of NCII are more acute than for BEC fraud because the benefit to the attacker is often non-financial — reputational harm to the victim, psychological satisfaction, or professional silencing — and therefore difficult to cap through the formal deterrence framework. For non-financial attacker benefits, the deterrence condition must be reframed in terms of the subjective utility of the attack (U_{attack}) versus the expected punishment:

$$p_d \cdot p_c \cdot F \geq U_{\text{attack}} - c_a. \quad (37.5)$$

Since U_{attack} is unobservable, deterrence must rely primarily on raising $p_d \cdot p_c$ rather than calibrating F to the specific benefit. Mandatory platform reporting of NCII incidents (creating detection pathways), survivor-centred prosecution policies (raising p_c), and civil strict-liability schemes (reducing the

evidentiary burden) are the three mechanisms most likely to raise the product $p_d \cdot p_c$ to deterrence-relevant levels.

37.2 Platform Economics of Synthesis APIs

Synthesis API providers operate in a two-sided market: they must attract developers and business users on one side while managing the regulatory, reputational, and liability risks of harmful uses on the other. The market exhibits a classic adverse selection problem: the marginal user most willing to pay for unrestricted synthesis capability is the adversarial user who expects to generate value from harmful uses that legitimate users can meet with restricted alternatives.

37.2.1 The Moral Hazard Problem

The moral hazard in synthesis API provision arises because the API provider captures a small fraction of the total social value of each API call (the API fee) while bearing only a small fraction of the social cost of harmful uses (reputational risk and regulatory fines). Let π be the API fee per harmful call, H be the social harm per harmful call, and λ be the fraction of harm borne by the provider (through liability or reputation loss). The provider's private optimum is to implement safeguards up to the point where the marginal cost of an additional safeguard equals λH , not H . Since $\lambda \ll 1$ for most current providers, the private optimum involves significantly less safeguard investment than the social optimum.

The provider's profit-maximising safeguard level satisfies:

$$\frac{\partial C_s}{\partial s} = \lambda \cdot H \cdot \frac{\partial q_h}{\partial s}, \quad (37.6)$$

where s is the safeguard stringency, C_s is the cost of safeguards, and q_h is the rate of harmful outputs. The social optimum requires $\lambda = 1$, achievable only through full strict liability — a policy that would impose prohibitive compliance costs on legitimate use cases.

37.2.2 API Revenue and Harm Externalities

At current market prices, the API economics for a major voice synthesis provider are approximately:

$$\text{Revenue} = N_{\text{total}} \cdot \pi = (N_{\text{legit}} + N_{\text{harmful}}) \cdot \pi, \quad (37.7)$$

where N_{total} is the total number of API calls, N_{legit} is the number of legitimate calls, and N_{harmful} is the number of calls that produce harmful outputs. At $\pi = \$0.001$ per API call (approximate ElevenLabs pricing for voice synthesis), 10 million calls per month, 0.1% harmful use rate, and \$50,000 mean harm per harmful call:

$$\text{Monthly revenue} = 10,000,000 \times \$0.001 = \$10,000 \quad (37.8)$$

$$\text{Monthly social harm} = 10,000 \times \$50,000 = \$500,000,000. \quad (37.9)$$

The ratio of monthly social harm to monthly API revenue is 50,000:1. This extreme ratio — even with large uncertainty in both estimates — establishes that the harms enabled by synthesis APIs are orders of magnitude larger than the revenue those APIs generate, providing an economic basis for mandatory safeguard requirements independent of the provider’s willingness to invest voluntarily.

37.3 Insurance Economics of AI-Enabled Fraud

Insurance markets price risk by estimating the expected loss from each hazard and adding a risk loading to account for variance and administrative costs. The emergence of AI-enabled fraud as a distinct insurable hazard has required underwriters to develop new actuarial models for voice-clone BEC, synthetic identity fraud, and deep-fake-enabled impersonation, drawing on the limited but growing empirical record from IC3 reports, ACFE surveys, and insurer claims databases.

37.3.1 Premium Pricing for BEC Coverage

For a corporate policyholder with annual revenue R , industry k , and employee count N , the actuarial expected annual BEC loss is:

$$\mathbb{E}[L_{\text{BEC}}] = \lambda_k \cdot N_{\text{finance}} \cdot \bar{L} \cdot p_s, \quad (37.10)$$

where λ_k is the industry-specific attack rate per finance employee per year (0.03–0.12 depending on industry profile), N_{finance} is the number of finance employees with wire transfer authority, $\bar{L} = \$280,000$ is the mean successful transfer, and p_s is the attack success rate at the organisation’s current defensive posture.

The annual insurance premium is:

$$\text{Premium} = \mathbb{E}[L_{\text{BEC}}] \cdot (1 + \rho), \quad (37.11)$$

where ρ is the risk loading (typically 1.0–2.0 for BEC coverage, reflecting high variance in individual claim sizes).

Worked Example — Premium calculation for a mid-size financial firm

A firm with 50 finance employees, industry $k = \text{financial services}$ ($\lambda_k = 0.08$), and no current anti-spoofing: $p_s = 0.30$ (undefended). Expected annual BEC loss: $\mathbb{E}[L] = 0.08 \times 50 \times \$280,000 \times 0.30 = \$336,000$. Premium at risk loading 1.5: $\$336,000 \times 2.5 = \$840,000$.

After deploying voice anti-spoofing (p_s reduced to 0.04): $\mathbb{E}[L] = 0.08 \times 50 \times \$280,000 \times 0.04 = \$44,800$. Premium: $\$44,800 \times 2.5 = \$112,000$. Premium saving: $\$728,000$ annually. Anti-spoofing system annual cost: $\$180,000$. Net annual saving from anti-spoofing: $\$728,000 - \$180,000 = \$548,000$. The insurance premium reduction alone justifies the anti-spoofing investment, independent of the direct fraud loss reduction.

37.3.2 Incentive Effects of Insurance

Insurance creates three categories of incentive effects relevant to synthetic media fraud. The first is the *direct investment incentive*: premium discounts for documented deployment of anti-spoofing, multi-factor KYC, and voice authentication systems translate ethical responsibility into quantifiable financial benefit, creating market-driven demand for defensive technology adoption. Typical discounts are 10–25% of the base premium.

The second is the *moral hazard effect*: organisations with full insurance coverage may reduce their internal vigilance relative to uninsured organisations, partly offsetting the investment incentive. Actuaries address this through coinsurance requirements (the policyholder retains 10–20% of losses), deductibles calibrated to the cost of basic safeguards, and claims investigations that verify the presence of documented safeguards.

The third is the *information production effect*: insurers accumulate proprietary data on attack frequency, attack success rates by defensive posture, and loss distributions that are not available in public databases. Sharing this data with policyholders via threat intelligence reports creates positive externalities beyond the insurance contract itself.

37.4 Welfare Economics of Regulatory Intervention

Regulatory intervention in the synthetic media market can take five forms: Pigouvian taxation of harmful outputs, mandatory safeguard requirements, liability rules, disclosure mandates, and market structure interventions. Each form has different efficiency, equity, and enforceability properties.

37.4.1 Optimal Pigouvian Tax

The optimal Pigouvian tax on a harmful synthesis API call equals the marginal social cost of that call:

$$t^* = \frac{\partial E[\text{Harm}]}{\partial q}, \quad (37.12)$$

where q is the quantity of synthesis API calls and $E[\text{Harm}]$ is the expected harm from those calls. At current estimates (0.1% harmful use rate, \$50,000 mean harm per harmful call), $t^* = 0.001 \times \$50,000 = \50 per API call. This is several orders of magnitude above current API pricing (\$0.001–\$0.01 per call), implying that a full Pigouvian tax would eliminate the commercial synthesis API market as currently priced.

This result — that the socially optimal tax would be prohibitive — reflects the severity of the externality and points toward mandatory safeguard requirements as a more practically achievable intervention. Requiring safeguards that reduce the harmful use rate from 0.1% to 0.005% (a 95% reduction, consistent with the safeguard stack described in Chapter 47) reduces the implied tax to \$2.50 per call — still far above current pricing but within an order of magnitude, suggesting that a combination of safeguard mandates and a modest per-call fee could approach social optimality without prohibiting legitimate use.

37.4.2 Efficiency-Equity Trade-offs

Regulatory efficiency (minimising total social cost) and equity (distributing costs and benefits fairly) are not always aligned in synthetic media regulation. Per-call taxes are economically efficient but fall proportionally on legitimate users, who may include accessibility technology developers serving users with disabilities and non-commercial researchers studying disinformation. Liability rules are equitable in principle (harm-causers pay) but create legal uncertainty that chills beneficial innovation. Mandatory safeguards reduce harm but impose compliance costs that create barriers to entry, concentrating the market among well-capitalised incumbents who can absorb the compliance burden.

The equity dimension has particular salience for developing economies: mandatory safeguards designed for API providers in the US or EU may be technically or financially infeasible for AI developers in lower-income countries, creating a regulatory arbitrage where synthesis capability migrates to jurisdictions with lower compliance requirements. The international coordination challenges raised by this dynamic are examined in Chapter 29.

37.5 Industry Applications

Economic analysis of synthetic media has created a distinct commercial market for quantitative risk modelling, actuarial consulting, and insurance product development. Milliman, Willis Towers Watson, and Aon have developed AI fraud risk models that quantify the expected annual loss from voice-clone BEC, synthetic identity fraud, and deep-fake-enabled insurance fraud for specific enterprise profiles. These models are used to price cyber insurance policies, to set internal risk limits for financial institutions, and to justify technology investment decisions by translating security expenditure into expected loss reduction in terms that chief financial officers can evaluate.

The FBI IC3 annual reports provide the primary public dataset for calibrating BEC attack rate and loss magnitude parameters. The ACFE Report to the Nations provides complementary data on occupational fraud including AI-enabled variants. The NIST Cybersecurity Framework provides a structured approach to investment allocation across detect, protect, respond, and recover domains that has been adopted by major financial sector risk management frameworks, including the Federal Financial Institutions Examination Council's cybersecurity guidance.

37.6 African Use Cases

The economics of synthetic media fraud in Africa differ significantly from those in high-income markets due to different cost structures, different average transaction values, and different enforcement probabilities. The marginal cost of a voice-clone fraud attempt is identical globally — approximately \$0.05 per attempt using a commercial API — but the expected gain per attempt is lower for attacks targeting African mobile money users (average transaction value \$50–\$500) than for CEO fraud targeting Western corporations (\$280,000 average). However, the enforcement probability is also significantly lower in most African jurisdictions, partially offsetting the lower gain.

The economic equilibrium predicts a higher volume of lower-value attacks in African markets,

consistent with the pattern documented in EFCC and Interpol-AFRIPOL reports of widespread but individually low-value mobile money fraud. The Interpol-AFRIPOL African Cyberthreat Assessment Report of 2023 documented a 300% increase in BEC fraud across Africa between 2020 and 2022, driven partly by the adoption of voice-clone tools that had previously required technical sophistication now available through consumer APIs. The economic analysis of optimal deterrence developed in Section 37.1.1 applies with modified parameter values: with $p_d = 0.03$ (lower African detection rates), $p_c = 0.15$ (lower conviction rates), and $B = \$5,000$ (lower per-incident gains), the required fine for deterrence is $F^* = p_s B / (p_d \cdot p_c) = 0.015 \times \$5,000 / (0.03 \times 0.15) = \$16,667$ — approximately equivalent to current African cybercrime statute maximums in jurisdictions with strong legislation. The African deterrence gap is narrower than in high-income markets in absolute terms but wider in terms of enforcement capacity.

Chapter Summary

Economic analysis models the synthetic media market as a negative externality problem in which producers and consumers of synthesis do not bear the full social cost of harms to third parties, resulting in systematic overproduction relative to the social optimum. The attacker cost-benefit model reveals that voice-clone BEC attacks are positive-expected-value at current synthesis costs, detection rates, and penalty levels; achieving rational deterrence requires either fines of \$140,000 per incident or detection rates exceeding 25%, neither of which current systems achieve. The deterrence gap for NCII is more severe because attacker benefits are non-financial and therefore unobservable.

Platform economics shows that synthesis API providers capture \$0.001 per call while enabling \$50 of expected harm per harmful call — a 50,000:1 harm-to-revenue ratio that establishes the economic basis for mandatory safeguard requirements. The moral hazard in API provision arises because providers bear only a fraction $\lambda \ll 1$ of the social harm, leading to private safeguard investment far below the social optimum. Insurance markets have begun pricing AI-enabled fraud risk with premium discounts of 10–25% for documented deployment of technical countermeasures, translating ethical responsibility into market incentives. The optimal Pigouvian tax on synthesis API calls is approximately \$50 per call at current harmful-use rates — prohibitively high relative to current pricing, pointing toward mandatory safeguard requirements combined with liability rules as the most practically achievable regulatory intervention.

Exercises

1. A synthesis API provider charges \$0.002 per API call and processes 50 million calls per month. The estimated harmful use rate is 0.08%, with mean harm per harmful call of \$45,000. The provider's regulatory exposure is 5% of harm caused (through fines and legal settlements). (a) Calculate the provider's monthly revenue from all calls. (b) Calculate the total monthly social harm from harmful calls. (c) Calculate the provider's regulatory exposure. (d) What safeguard cost per call would the provider rationally bear? (e) If a safeguard system costing \$50,000 per month reduces the harmful use rate to 0.002%, is it privately optimal for the provider to

implement it? Is it socially optimal?

2. The DEFIANCE Act (2024) provides for a civil cause of action for NCII victims with statutory damages of \$150,000 per violation. Assume: probability of detection and identification of a specific NCII producer $p_d = 0.08$; probability of successful civil judgment given identification $p_c = 0.55$; cost of producing NCII $c_a = \$12$; attacker's subjective utility from NCII production $U = \$8,000$ (in monetary equivalent). (a) Does the DEFIANCE Act achieve rational deterrence? (b) What detection probability would be required to achieve deterrence with the current damages level? (c) If victim advocacy organisations invest \$5 million in a reporting hotline that raises p_d from 0.08 to 0.20, by how much does the required statutory damages level change?
3. An insurer offers BEC coverage with the following structure: premium = $2.2 \times$ expected annual loss; 15% premium discount for deployed voice anti-spoofing ($EER \leq 5\%$); 20% coinsurance (policyholder retains 20% of losses). A manufacturing firm has 12 finance employees with wire transfer authority, industry attack rate 0.05 per employee per year, mean transfer \$320,000, and current success rate $p_s = 0.35$ (no defences). (a) Calculate the base premium without defences. (b) Deploy anti-spoofing reducing p_s to 0.04. Calculate the new expected loss, new premium (with discount), and net annual saving. (c) The anti-spoofing system costs \$8,000 per month. What is the full ROI including both the insurance saving and the direct fraud loss reduction?
4. Using the optimal Pigouvian tax framework, calculate t^* for each of the following harm scenarios: (a) NCII synthesis with 0.05% harmful use rate and \$60,000 mean psychological harm; (b) Voice-clone BEC with 0.1% harmful use rate and \$280,000 mean financial harm; (c) Electoral deep fake with 0.001% harmful use rate and \$5 million mean democratic harm (in terms of voter suppression losses). For each, compare t^* with a typical synthesis API price of \$0.005 per call and comment on the regulatory implications.

Answers to Exercises

1. Synthesis API externality analysis.

- (a) Monthly revenue: $50,000,000 \times \$0.002 = \$100,000$.
- (b) Harmful calls per month: $50,000,000 \times 0.0008 = 40,000$. Total monthly social harm: $40,000 \times \$45,000 = \$1,800,000,000$.
- (c) Provider's regulatory exposure: $5\% \times \$1,800,000,000 = \$90,000,000$ per month.
- (d) Rationally borne safeguard cost per call: the provider bears $\lambda = 0.05$ of harm. Marginal benefit of safeguard per call = $\lambda \times H \times \Delta q_h / \Delta s$. At the extensive margin, the provider invests in safeguards up to the point where marginal cost of safeguard = $\lambda H \Delta q_h = 0.05 \times \$45,000 \times 0.0008 = \$1.80$ per call reduced. The provider will invest at most \$1.80 per harmful call avoided, equivalent to \$0.00144 per total call — less than the API price per call.

(e) Safeguard at \$50,000/month reducing harmful rate from 0.08% to 0.002%: Harmful calls reduced per month: $(40,000 - 1,000) = 39,000$. Regulatory exposure reduction: $39,000 \times \$45,000 \times 0.05 = \$87,750,000/\text{month}$. Private benefit of safeguard $\gg$ \$50,000 cost: **privately optimal**. Social benefit of safeguard: $39,000 \times \$45,000 = \$1,755,000,000/\text{month} \gg \$50,000$: **socially optimal** (by an enormous margin, reflecting the severity of the externality).

2. DEFIANCE Act deterrence analysis.

(a) Expected punishment: $p_d \times p_c \times F = 0.08 \times 0.55 \times \$150,000 = \$6,600$. Expected benefit: $U - c_a = \$8,000 - \$12 = \$7,988$. Since $\$6,600 < \$7,988$, **does NOT achieve deterrence**.

(b) Required detection probability: set $p_d \times 0.55 \times \$150,000 = \$7,988$. $p_d = \$7,988 / (0.55 \times \$150,000) = \$7,988 / \$82,500 = 0.0969 \approx 9.7\%$. Current $p_d = 8\%$ is only marginally below the required 9.7% — a small increase in detection infrastructure could achieve deterrence with the current statutory damages.

(c) With $p_d = 0.20$: required damages $F^* = \$7,988 / (0.20 \times 0.55) = \$72,618$. Change from current: $\$150,000 - \$72,618 = \$77,382$ reduction in required damages. The \$5 million detection investment that raises p_d from 8% to 20% allows the statutory damages to be reduced from \$150,000 to approximately \$73,000 and still achieve deterrence — or, at the current damages level, provides a significant deterrence surplus.

3. Manufacturing firm BEC insurance ROI.

(a) Base expected annual loss (no defences): $0.05 \times 12 \times \$320,000 \times 0.35 = \$67,200$. Base premium: $2.2 \times \$67,200 = \$147,840$.

(b) With anti-spoofing ($p_s = 0.04$): new expected loss = $0.05 \times 12 \times \$320,000 \times 0.04 = \$7,680$. Discounted premium = $2.2 \times \$7,680 \times (1 - 0.15) = \$14,342.40$. Premium saving: $\$147,840 - \$14,342 = \$133,498$.

(c) Anti-spoofing annual cost: $12 \times \$8,000 = \$96,000$. Direct fraud loss saving (coinsurance): the firm retains 20% of losses. Direct loss reduction: $(1 - 0.20) \times (\$67,200 - \$7,680) = 0.80 \times \$59,520 = \$47,616$. Total annual benefit: $\$133,498 + \$47,616 = \$181,114$. Total annual cost: \$96,000. Net annual benefit: \$85,114. ROI: $\$85,114 / \$96,000 = 88.7\%$. Full payback period: $\$96,000 / \$181,114 = 6.3$ months.

4. Pigouvian tax calculations.

(a) NCII: $t^* = 0.0005 \times \$60,000 = \30 per call. Typical API price: \$0.005. Ratio: $\$30 / \$0.005 = 6,000 \times$ the API price. Implication: a full Pigouvian tax would make NCII synthesis API calls prohibitively expensive, effectively banning the synthesis capability at the API pricing level.

(b) Voice-clone BEC: $t^* = 0.001 \times \$280,000 = \280 per call. Ratio: $\$280 / \$0.005 = 56,000 \times$

the API price. Implication: even more prohibitive than NCII; the extremely high mean harm per harmful call makes a Pigouvian tax impractical as a standalone regulatory instrument.

(c) Electoral deep fake: $t^* = 0.00001 \times \$5,000,000 = \50 per call. Ratio: $10,000 \times$ the API price. Implication: lower harmful use rate but very high per-incident harm produces a similar magnitude of implied tax. The democratic harm figure is inherently uncertain and contested; even at one-tenth the estimate (\$500,000), $t^* = \$5$ — still $1,000 \times$ the API price. Across all three harm categories, the Pigouvian analysis consistently points to mandatory safeguards and liability rules as more practical instruments than per-call taxation at the level that would fully internalise the externality.

Chapter 38

Deep Fakes in Insurance Fraud and Accident Scene Fabrication

“Insurance fraud is as old as insurance itself. What changes is the quality of the lie.”

—Anon.

Insurance fraud imposes an enormous annual cost on the global economy; widely cited industry estimates suggest hundreds of billions of dollars annually, of which property and casualty fraud — fabricated accidents, staged vehicle collisions, falsified damage claims, and invented injuries — accounts for the largest share. Synthetic media technology has fundamentally altered the economics and detectability of this fraud category by enabling the fabrication of photorealistic evidence at near-zero marginal cost. A claimant who previously needed to physically stage a collision, recruit co-conspirators, and risk injury or criminal exposure during the staging now has access to AI image and video generation tools that can produce convincing accident scene photographs from a text prompt in seconds, at a cost of fractions of a cent per image.

The insurance fraud use case for synthetic media differs structurally from the financial fraud and disinformation cases examined in earlier chapters. In financial BEC fraud, the attacker fabricates audio or video to impersonate a real person in real time. In electoral disinformation, the attacker creates false media depicting real public figures. In insurance fraud, the attacker typically does not need to impersonate any specific person: they need to fabricate a *scene* — a vehicle with damage consistent with a claimed accident, a medical document consistent with a claimed injury, a property interior consistent with a claimed flood or fire. This scene-fabrication task is in some respects easier for current generative AI than person-specific impersonation, because the claim needs only to be internally consistent rather than matching an external authenticated reference.

The forensic challenge is correspondingly different. Detecting a voice clone requires comparing the synthetic audio against the known acoustic characteristics of the claimed speaker. Detecting a fabricated accident scene photograph requires establishing that the image was not captured by a physical camera at the claimed time and location, without necessarily having any reference to

compare it against. The tools available for this task — PRNU camera fingerprinting, GPS and EXIF metadata analysis, lighting consistency analysis, and physical plausibility assessment — are described in Chapter 12, but their application to insurance contexts introduces specific operational challenges around evidence chain of custody, adjudicator technical literacy, and the evidentiary standards of insurance dispute resolution.

The accident scene fabrication problem has three layers of increasing difficulty. The first and most tractable is single-image fabrication: a photograph of a damaged vehicle or property that was generated by an AI model rather than captured by a camera. Detection relies on the forensic signals of Chapter 12 — PRNU absence, GAN spectral fingerprints, lighting inconsistency — and is relatively reliable for uncompressed images submitted directly to an insurer. The second layer is video fabrication: a short video clip of a staged accident or damage event. This is harder to fabricate convincingly because temporal consistency requires coherent motion, causally plausible physics, and consistent lighting across frames. The third and most difficult layer is composite fabrication: authentic video of an undamaged vehicle combined with AI-generated damage composited into the scene, or authentic accident footage combined with AI-inserted injuries or damage that was not present in the original. Detecting the composite requires identifying the boundary between authentic and synthetic regions within the same frame.

The market for insurance fraud has adapted rapidly to these capabilities. Organised fraud rings — which account for a disproportionate fraction of high-value insurance fraud losses — have adopted AI image generation as a standard tool for claim document fabrication, reducing both the physical risk and the coordination cost of traditional staged accident operations. Law enforcement investigations in the United Kingdom, Germany, Australia, and South Africa have documented the use of Midjourney, DALL-E, and Stable Diffusion outputs in insurance claim submissions. The Insurance Fraud Bureau’s Project Endive (UK, 2023) was the first large-scale investigation specifically targeting AI-generated claim evidence, resulting in a significant number of arrests and establishing forensic and legal precedent for prosecuting this category of fraud.

The defensive response from the insurance industry has been significant and growing. Within twelve months of the public release of Stable Diffusion in 2022, all five of the largest UK non-life insurers had deployed or piloted AI-generated image detection in their claims triage pipelines. The Association of British Insurers established a working group on AI-generated fraud in early 2023, and equivalent bodies in Germany, Australia, and the United States have followed. The technical challenge facing these defensive deployments is the generalisation problem discussed in Chapter 14: detectors trained on today’s synthetic images will miss tomorrow’s, and the 12–18 month update cycle typical of enterprise insurance technology stacks creates a systematic lag behind the synthesis frontier.

This chapter develops the complete technical, forensic, legal, and operational landscape of AI-enabled insurance fraud and the industry’s defensive response. Section 38.1 develops a taxonomy of insurance fraud attack types by synthesis modality and target claim category. Section 38.2 develops the forensic detection methods applicable to insurance evidence, including camera fingerprinting, lighting consistency analysis, physical plausibility, and metadata forensics. Section 38.3 develops the specific challenge of video-based claim evidence and the temporal consistency signals that video synthesis

fails to maintain. Section 38.4 develops composite image forensics — the detection of AI-generated content inserted into authentic photographs. Section 38.5 develops an operational triage pipeline for an insurance claims processing workflow. Section 38.6 addresses the evidentiary and legal issues specific to insurance contexts.

38.1 Taxonomy of AI-Enabled Insurance Fraud

Insurance fraud using synthetic media falls into six distinct attack types, each targeting a different element of the claims verification process. Understanding the taxonomy is prerequisite to designing a detection response that is calibrated to the specific attack surfaces of each claim category.

38.1.1 Fabricated Damage Photographs

The simplest attack: an AI image generator produces a photograph of a vehicle, property, or object showing damage consistent with a claimed incident. The claimant submits this AI-generated image as photographic evidence of the claimed loss. This attack requires no physical damage to occur, no staging, and no co-conspirators: the entire fraud can be executed by a single individual with a smartphone, a Stable Diffusion account, and a plausible claim narrative.

The sophistication of this attack varies widely. At the lowest sophistication level, claimants use generic prompts that produce images inconsistent with the claimed vehicle model, registration number, or damage location. At higher sophistication, claimants use ControlNet conditioning on a photograph of their actual vehicle to generate damage that is consistent with the vehicle's make, model, colour, and registration plate while fabricating the damage pattern. At the highest sophistication level, claimants composite AI-generated damage regions into authentic photographs of their own vehicle using Poisson blending or inpainting, producing images that carry authentic camera PRNU and metadata while containing fabricated damage content.

38.1.2 Fabricated Medical Evidence

Personal injury claims require medical documentation. AI-enabled fabrication of medical evidence encompasses: LLM-generated medical reports purportedly from healthcare providers, image synthesis of diagnostic scans (X-rays, MRI images) showing injuries not present in authentic scans, and fabricated prescription or treatment records. The medical evidence fabrication problem is qualitatively different from accident scene fabrication because authentic medical images have distinctive acquisition fingerprints — scanner PRNU, specific noise patterns from DICOM acquisition parameters — that differ from those of natural photographs, and which AI image generators are not calibrated to replicate.

The CycleGAN-based lesion injection and removal attack described in Chapter 34 applies directly to insurance contexts: an authentic X-ray of an undamaged bone can be modified to show a fracture consistent with a claimed injury. Detection relies on the same PRNU-based scanner authentication and frequency-domain analysis described in that chapter, applied to the specific imaging modalities encountered in personal injury claims.

38.1.3 Staged Vehicle Collisions with Fabricated Evidence

A hybrid attack: a physical collision is staged (or occurs accidentally with minor damage), and AI-generated photographs of more severe damage are submitted to inflate the claim value. This attack combines authentic elements (there was a real incident, there is a real vehicle with real minor damage, there may be a real police report) with fabricated elements (the severity of the damage shown in submitted photographs exceeds what was actually sustained). Detection requires comparing the AI-submitted claim photographs against any authentic photographs taken at the scene (police photographs, dashcam footage, CCTV) to identify inconsistencies.

38.1.4 Fabricated Accident Videos

With the availability of video generation systems including Runway Gen-3, Kling, and HailuoAI, claimants can now submit short video clips purportedly showing an accident occurring. Video generation introduces additional forensic attack surface relative to still images: temporal consistency must be maintained across frames, physics must be plausible, lighting must be consistent throughout the sequence, and any vehicles or property must maintain dimensional and colour consistency frame-to-frame. Current video generation systems fail on some or all of these constraints in detectable ways, but the rate of improvement means that video fabrication will become a significant threat within the current insurance policy cycle.

38.1.5 GPS and Location Spoofing Combined with Image Fabrication

Modern insurance claims are increasingly submitted through mobile apps that automatically capture GPS coordinates and timestamps alongside claim photographs. A sophisticated claimant may attempt to combine GPS spoofing (providing false location and time metadata) with AI-generated or composite images to create a claim that appears to have been submitted in real time from the claimed accident location. GPS spoofing detection and metadata forensics are described in Section 38.2.

38.1.6 Wholesale Policy Fraud Using Fabricated Identities

Synthetic identity fraud (Chapter 35) intersects with insurance fraud when an artificial person — created with a fabricated name, GAN-generated photograph, and fabricated financial history — is used to purchase insurance policies with the intent of submitting fraudulent claims. The graph-based detection methods for synthetic identity networks described in Chapter 35 apply here, with the additional signal that synthetic identity policyholders tend to submit claims shortly after policy inception, lack claim history consistent with their stated age and driving record, and use devices shared with other recently opened policies.

38.2 Forensic Detection of Fabricated Insurance Evidence

The forensic detection pipeline for insurance claims applies three algorithmic procedures in sequence: metadata validation, PRNU authentication, and composite detection.

Algorithm 6: Insurance Claim Image Triage Pipeline**Input:** Claim image $\hat{\mathbf{x}}$;claimant device calibration images $\{\mathbf{c}_k\}$; thresholds $\tau_{\text{PRNU}} = 0.015$, $\tau_{\text{phys}} = 0.60$, $\tau_{\text{AI}} = 0.65$ **Output:** Risk tier (1/2/3) and flags

// Tier 1: Fast automated checks (<500 ms)

1 **if** EXIF metadata incomplete **or** GPS implausible **then**2 | flag $\leftarrow$ METADATA3 **end**4 Reverse image search $\hat{\mathbf{x}}$; **if** match found in stock/prior-claims DB **then**5 | flag $\leftarrow$ DUPLICATE; **return** Tier 36 **end**7 $S_{\text{AI}} \leftarrow$ lightweight-classifier($\hat{\mathbf{x}}$);8 **if** $S_{\text{AI}} > \tau_{\text{AI}}$ **then**9 | flag $\leftarrow$ AI_DETECT10 **end**11 **if** double-JPEG-detected($\hat{\mathbf{x}}$) **then**12 | flag $\leftarrow$ DOUBLE_JPEG13 **end**14 **if** flags = $\emptyset$ **then**15 | **return** Tier 1 (pass)16 **end**

// Tier 2: PRNU + plausibility (30-120 s)

17 $\hat{\mathbf{K}} \leftarrow \frac{1}{N_c} \sum_k (\mathbf{c}_k - W(\mathbf{c}_k)) / \mathbf{c}_k$;

;

// estimate device PRNU from calibration images

18 $\hat{\mathbf{r}} \leftarrow \hat{\mathbf{x}} - W(\hat{\mathbf{x}})$;19 $\rho \leftarrow \text{NCC}(\hat{\mathbf{K}}, \hat{\mathbf{r}})$;20 **if** $\rho < \tau_{\text{PRNU}}$ **then**21 | flag $\leftarrow$ PRNU_FAIL22 **end**

// Spatial composite check

23 Partition $\hat{\mathbf{x}}$ into M blocks of 64×64 pixels;24 **for** block (i, j) **do**25 | $\rho_{ij} \leftarrow \text{NCC}(\hat{\mathbf{K}}_{ij}, \hat{\mathbf{r}}_{ij})$ 26 **end**27 **if** $|\{(i, j) : \rho_{ij} < \tau_{\text{PRNU}}\}| > 0.05M$ **then**28 | flag $\leftarrow$ COMPOSITE29 **end**30 $S_{\text{phys}} \leftarrow$ physical-plausibility-score($\hat{\mathbf{x}}$);31 **if** $S_{\text{phys}} < \tau_{\text{phys}}$ **then**32 | flag $\leftarrow$ IMPLAUSIBLE33 **end**34 **if** PRNU_FAIL **or** COMPOSITE **or** IMPLAUSIBLE **then**35 | **return** Tier 336 **end**37 **else**38 | **return** Tier 2 (specialist review)39 **end**

Worked Example — Expected Tier escalation rates and costs

An insurer processes $N = 20,000$ motor claims per day. Estimated AI-fraud prevalence $p = 0.012$. Tier 1 sensitivity 0.91, specificity 0.96.

Daily claim volumes:

$$\text{Fraudulent: } 20,000 \times 0.012 = 240; \quad \text{Authentic: } 20,000 - 240 = 19,760.$$

Tier 2 escalations: $TP = 240 \times 0.91 = 218$; $FP = 19,760 \times 0.04 = 790$; Total = 1,008. Tier 2 cost at \$85/investigation: $1,008 \times \$85 = \$85,680/\text{day}$.

Tier 3 escalations (20% of Tier 2): $1,008 \times 0.20 = 202$ specialist reviews. Tier 3 cost at \$850/review: $202 \times \$850 = \$171,700/\text{day}$.

Total daily detection cost: $\$85,680 + \$171,700 = \$257,380$. Expected daily fraud prevented (at \$18,000 mean AI-fraud claim): $218 \times \$18,000 = \$3,924,000$. Net daily benefit: $\$3,924,000 - \$257,380 = \$3,666,620$. ROI: 14.3:1.

The forensic detection pipeline for AI-generated insurance evidence applies the signals of Chapter 12 to a specific operational context: evidence submitted in the context of an insurance claim, which imposes specific requirements on the speed, cost, interpretability, and legal admissibility of the detection process.

38.2.1 PRNU Camera Fingerprint Authentication

Every physical camera introduces a unique fixed-pattern noise (PRNU) into every photograph it captures, arising from pixel-level manufacturing variations in the image sensor. This fingerprint is recoverable from the noise residual of any image taken by that camera and can be verified against a reference fingerprint extracted from a set of authenticated images from the same device. For insurance claims, the PRNU test answers a critical question: was this photograph taken by the claimant's own mobile device, consistent with the claimed circumstances?

The PRNU authentication procedure for insurance evidence:

1. Request the claimant to submit five additional unrelated photographs taken with the same device (the calibration set). Extract the device PRNU fingerprint $\hat{\mathbf{K}}$ as the average noise residual across the calibration images.
2. For each claim photograph, extract the noise residual $\hat{\mathbf{r}} = \hat{\mathbf{x}} - W(\hat{\mathbf{x}})$ where W is a Wiener denoising filter.
3. Compute the normalised cross-correlation:

$$\rho = \text{NCC}(\hat{\mathbf{K}}, \hat{\mathbf{r}}). \quad (38.1)$$

4. Compare against thresholds: authentic photographs from the calibration device yield $\rho \approx 0.038$; GAN-generated or diffusion-generated images yield $\rho \approx 0.004$; images from a different camera yield ρ consistent with that camera's fingerprint.

Worked Example — PRNU authentication of a vehicle damage claim

A claimant submits four photographs of vehicle damage. The PRNU fingerprint extracted from ten calibration photographs taken by the claimant's iPhone 14 Pro is $\hat{\mathbf{K}}$. NCC scores for the four claim photographs:

Photo	NCC ρ	JPEG QF	Verdict
Claim photo 1	0.041	92	Authentic (above $\tau = 0.015$)
Claim photo 2	0.037	91	Authentic
Claim photo 3	0.005	89	Suspect (below threshold)
Claim photo 4	0.003	90	Suspect

Photos 3 and 4 do not carry the claimant's device fingerprint despite claiming the same session, indicating either synthesis (most likely) or capture by a different device with deceptive intent. The two-image pattern — some authentic, some suspect — is consistent with the common fraud pattern of submitting a mix of genuine and fabricated evidence.

38.2.2 Physical Plausibility Analysis

Current image generation models have imperfect understanding of physical causality: they generate visually plausible local textures but sometimes fail to maintain global physical consistency. For vehicle damage claims, physical plausibility analysis examines five dimensions:

Damage directionality: collision damage has a characteristic deformation pattern determined by the direction, speed, and contact surface of the impact. Front-end collision damage deforms the bonnet and front quarters inward from the direction of impact; side-impact damage creates a lateral crush pattern. AI-generated damage sometimes places deformation inconsistent with the claimed impact direction (e.g., a claimed rear-end collision with damage concentrated on the driver's door).

Material consistency: crumple zones in modern vehicles deform in specific ways determined by the vehicle's structural design, visible in engineering cutaway diagrams. AI-generated damage sometimes ignores the vehicle's actual crumple zone geometry, placing damage in structurally rigid sections that would not deform in the claimed impact.

Glass breakage pattern: tempered glass breaks into a characteristic pebble pattern; laminated windshield glass fractures in a starred radial pattern from the impact point. AI-generated glass breakage patterns sometimes lack the physically correct fracture morphology.

Shadow and lighting consistency: damage features must cast shadows consistent with the single light source (sun position) implied by the rest of the image. AI image generators often fail to maintain lighting consistency when generating damage features, introducing shadow directions inconsistent

with the ambient illumination.

Reflection consistency: vehicle surfaces are highly reflective, and the reflections visible in undamaged panels must be consistent with the claimed location and time. AI-generated reflections often fail to match the environmental context.

Worked Example — Physical plausibility scoring

A vehicle damage claim photograph is evaluated on five physical plausibility dimensions, each scored 0–1 by a calibrated classifier:

Dimension	Score	Flag
Damage directionality	0.89	Pass
Material consistency	0.72	Pass
Glass breakage pattern	0.31	Fail (< 0.5)
Shadow consistency	0.28	Fail (< 0.5)
Reflection consistency	0.44	Fail (< 0.5)
Composite plausibility	0.53	Escalate

The glass breakage pattern shows a uniform pebble pattern inconsistent with the claimed impact at a single point; shadow directions in the damage region are inconsistent with the single directional light source visible in the background; and reflections in the undamaged bonnet do not match the claimed outdoor parking environment. Composite plausibility of 0.53 is below the escalation threshold of 0.60, triggering referral to a human specialist investigator.

38.2.3 Metadata and GPS Forensics

Every digital photograph contains Exchangeable Image File (EXIF) metadata encoding the capturing device, timestamp, GPS coordinates, focal length, ISO sensitivity, and exposure parameters. For insurance claims submitted via mobile apps, this metadata is automatically embedded and transmitted with the image. Metadata forensics examines four categories of inconsistency:

Timestamp consistency: the claimed incident time must be consistent with the photograph timestamp. Timestamps can be altered by the claimant, but the PRNU signal changes over a camera’s lifetime, allowing approximate dating of when a photograph was taken relative to the calibration images.

GPS coordinate validation: the claimed accident location must match the GPS coordinates embedded in the photographs. GPS spoofing (using software tools to provide false GPS data to the camera app) is detectable by cross-referencing with: (a) cellular network location from the contemporaneous call record, (b) CCTV records from the claimed location, and (c) the plausibility of the background scene in the photograph against satellite imagery of the claimed coordinates.

Device model consistency: the EXIF Make and Model fields must match the claimant’s stated device. Inconsistencies may indicate that the photograph was downloaded from the internet or taken by a different device.

Compression history: each JPEG encoding introduces a characteristic quantisation table. Re-saved images (downloaded from the internet or edited) typically show a secondary quantisation table inconsistent with a single-capture workflow, detectable by double-JPEG detection algorithms.

38.3 Video-Based Claim Evidence and Temporal Forensics

Dashcam footage, CCTV video, and claimant-submitted video clips are increasingly important evidence in insurance claims. The submission of fabricated video evidence introduces a substantially higher forensic bar for the attacker because temporal consistency constraints are difficult for current video generation systems to satisfy.

38.3.1 Temporal Consistency Signals

Authentic video satisfies a set of temporal consistency constraints that current video generation systems violate to varying degrees:

Optical flow consistency: motion between adjacent frames must be consistent with rigid body physics. Vehicle motion in authentic video follows smooth trajectories constrained by road surface and inertia; video-generated vehicle motion sometimes shows frame-to-frame positional discontinuities that violate momentum conservation. The optical flow estimated between frames I_t and I_{t+1} for an authentic rigid body must satisfy the brightness constancy constraint:

$$I_t(x, y) = I_{t+1}(x + u, y + v), \quad (38.2)$$

where (u, v) is the displacement field. Violations of this constraint in non-texture regions (smooth vehicle panels) indicate video-generation artefacts.

Camera shake consistency: hand-held cameras exhibit characteristic micro-motion driven by the operator's hand tremor and respiration cycle, measurable as a power spectrum in the camera motion signal. Video generators produce either no camera motion (fixed virtual camera) or generic motion without the physiological signature of hand-held capture. Gyroscope data embedded in mobile video metadata can be cross-referenced with the apparent camera motion in the video signal.

Lighting temporal consistency: as a vehicle moves through a scene, the illumination on its surfaces changes in a physically determined manner based on the fixed sun position and the vehicle's orientation relative to the sun. Video generation systems sometimes produce lighting changes that are inconsistent with a fixed light source, creating a temporal lighting inconsistency detectable by tracking the ratio of highlight intensity to shadow depth across the vehicle surface through the clip.

Compression consistency: authentic video compressed by a mobile device exhibits the specific compression artefact patterns of that device's codec implementation. Video generator output compressed to the same bitrate shows different artefact patterns, detectable by trained classifiers or by direct comparison of the DCT coefficient distributions.

Worked Example — Video temporal forensics on a dashcam fraud submission

A claimant submits a 12-second dashcam video clip purportedly showing another vehicle running a red light and causing a collision. Temporal forensic analysis:

Signal	Score	Decision
Optical flow deviation	2.3σ above authentic mean	Flag
Camera shake spectrum	PSD peak at 11.2 Hz (not 6–9 Hz physiological)	Flag
Lighting temporal consistency	DRR change 18% across 12 s	Pass
Compression DCT distribution	$\chi^2 = 847$ vs. authentic $\chi^2 = 212$	Flag
Temporal forensic verdict	3/4 signals flagged	Escalate

The optical flow at the moment of impact shows a velocity discontinuity of 4.7 m/s across a single frame, inconsistent with the momentum of a vehicle of the claimed mass. The camera shake power spectral density shows a dominant peak at 11.2 Hz — faster than the 6–9 Hz typical of hand-held capture and consistent with synthetic camera motion. The DCT coefficient distribution is inconsistent with the claimed GoPro Hero 11 encoding at the reported bitrate. The clip is referred to specialist video forensics.

38.4 Composite Image Forensics

The most sophisticated insurance fraud attack inserts AI-generated damage into an authentic photograph of the claimant’s own undamaged vehicle. This composite approach is harder to detect than full-image synthesis because it preserves the PRNU fingerprint of the authentic camera in the background and undamaged vehicle panels, while inserting a synthetic damage region that lacks consistent PRNU. The resulting image passes simple PRNU authentication on the undamaged regions but fails on the inserted damage region — a spatially heterogeneous PRNU pattern that is the primary forensic signal for composite detection.

38.4.1 Spatial PRNU Inconsistency

Composite detection using PRNU proceeds by computing the NCC between the device fingerprint $\hat{\mathbf{K}}$ and the noise residual of each 64×64 pixel block of the suspect image:

$$\rho_{ij} = \text{NCC}(\hat{\mathbf{K}}_{ij}, \hat{\mathbf{r}}_{ij}), \quad (38.3)$$

where $\hat{\mathbf{K}}_{ij}$ and $\hat{\mathbf{r}}_{ij}$ are the block-wise fingerprint and residual patches at position (i, j) . The resulting NCC map $\mathbf{R} = \{\rho_{ij}\}$ reveals the spatial structure of authenticity: genuine regions show high NCC consistent with the device fingerprint; inserted synthetic regions show near-zero NCC.

The boundary between authentic and synthetic regions is typically revealed as a sharp discontinuity in the NCC map at the edges of the composited region, even after Poisson blending has smoothed the

visual boundary in the RGB image. This discontinuity is the primary forensic signal for composite localisation.

Worked Example — Composite localisation on a vehicle damage claim

A 4032×3024 pixel vehicle photograph is divided into 64×64 blocks ($63 \times 47 = 2961$ blocks). The mean NCC across all blocks:

Undamaged vehicle panels: $\bar{\rho} = 0.039$ (consistent with iPhone 14 Pro reference)

Damage region (front quarter, estimated 340 blocks): $\bar{\rho} = 0.006$

Background road and environment: $\bar{\rho} = 0.041$

The bimodal NCC distribution — high across undamaged vehicle and background, near-zero in the damage region — is the definitive signature of a composite: the damage region does not carry the camera's PRNU fingerprint and was not captured by that camera.

The NCC map further reveals the precise boundary of the composited region: blocks within 12 pixels of the boundary show intermediate NCC values (0.018–0.025) from the Poisson blending that smoothed the splice boundary in RGB space but did not introduce the camera's PRNU into the inserted region.

38.4.2 Inpainting Detection

A variant of the composite attack uses AI inpainting (diffusion-based inpainting as in Stable Diffusion, DALL-E, or Adobe Firefly) to insert damage into an authentic photograph in a way that automatically blends with the surrounding context. Inpainting-based composites are harder to detect than manual composites because the inpainting model generates content that is specifically conditioned on the surrounding authentic image context, producing a more seamless transition. However, inpainting leaves two characteristic forensic signals: the replaced region lacks the original camera PRNU (same spatial PRNU inconsistency as manual compositing), and the inpainting model's spectral fingerprint is detectable in the noise residual of the replaced region using the GAN/diffusion fingerprint methods of Chapter 12.

38.5 Operational Claims Triage Pipeline

An operational insurance claims triage pipeline must balance detection accuracy against processing cost and time. Claims arrive at scale — a major non-life insurer may process 10,000–50,000 claims per day — and full forensic analysis cannot be applied to every submission. A three-tier triage architecture is the industry standard response:

38.5.1 Tier 1: Automated Screening (All Claims)

Every submitted claim image and video undergoes automated screening using fast, inexpensive classifiers. Tier 1 takes <500 ms per image and applies:

Metadata completeness check: are all expected EXIF fields present? Are the GPS coordinates plausible for the claimed location? Is the device model consistent with the claimant's registration?

Reverse image search: does the submitted image appear in any online database, stock photo library, or previously submitted claim? Tools including Google Vision API, TinEye, and insurer-specific duplicate detection databases address this.

Rapid AI detection: a lightweight EfficientNet-B0 classifier (inference time <100 ms) flags images with a synthetic probability score above 0.65.

Double-JPEG detection: does the image show evidence of re-encoding consistent with download-from-internet rather than direct camera capture?

Claims flagged by any Tier 1 check are escalated to Tier 2; the remainder are processed normally with a note that automated screening was passed.

38.5.2 Tier 2: Forensic Analysis (Flagged Claims)

Tier 2 applies more computationally intensive forensic analysis to the subset of claims escalated from Tier 1 (typically 2–5% of total volume). Processing time: 30–120 seconds per image. Tier 2 applies:

PRNU camera fingerprint: request calibration images from claimant device and compute full-image NCC. If NCC below threshold, flag for specialist review.

Spatial PRNU composite detection: compute block-wise NCC map. If bimodal NCC distribution is detected, escalate to Tier 3 with composite localisation report.

Physical plausibility analysis: run the five-dimension physical plausibility classifier. Score below 0.60 triggers specialist review.

Spectral fingerprint: apply GAN/diffusion spectral test to the noise residual. Azimuthal slope deviation above threshold flags potential synthesis.

Video temporal forensics: for video submissions, apply the four temporal consistency tests. Three or more flags trigger escalation.

38.5.3 Tier 3: Specialist Investigation (Escalated Claims)

Tier 3 involves human specialist investigators with access to external verification sources. Typical processing time: 2–5 business days. Tier 3 investigators:

1. Request the original device for physical examination and additional calibration images if PRNU evidence is ambiguous.
2. Contact police or emergency services to verify the incident report referenced in the claim.
3. Cross-reference with CCTV operators at the claimed accident location.

4. Consult with an engineering expert on physical damage plausibility.
5. If fraud is confirmed, prepare the case file for referral to the Insurance Fraud Bureau or equivalent national body.

The expected prevalence of AI-generated fraud in the claims pipeline (estimated at 0.5–2% of claims at current synthesis tool adoption rates) combined with the base-rate problem means that even a Tier 1 classifier with 95% sensitivity and 97% specificity will have a positive predictive value of only 14–44% at these prevalence levels. This makes human-in-the-loop specialist review at Tier 3 essential: automated classifiers cannot be the sole decision mechanism for claim denial.

38.6 Legal and Evidentiary Issues

The legal framework for AI-generated insurance fraud evidence is developing rapidly but remains incomplete in most jurisdictions. Three categories of legal issue are most practically significant for insurers and claimants.

38.6.1 Admissibility of Forensic AI Evidence in Insurance Disputes

Insurance disputes resolved through internal appeals, ombudsman processes, or civil litigation require that forensic evidence be admissible and comprehensible to a non-expert adjudicator. The Daubert standard (Chapter 17) applies in US federal court; equivalent standards apply in UK (CPR Part 35), Australian, and South African courts. For PRNU-based evidence, a decade of criminal court admissions in the UK, Germany, and Australia provides established precedent. For AI-detection evidence, the precedent is thinner: the Project Endive prosecutions (UK, 2023) were among the first in which AI-generated image detection evidence was submitted in insurance fraud proceedings, and the forensic methodology was challenged in each case before being accepted with conditions.

The key admissibility conditions for AI forensic evidence in insurance fraud cases: (1) the detection algorithm must be published in a peer-reviewed venue or have undergone independent validation; (2) the expert witness must state the false positive rate at the operating threshold used; (3) the expert must be able to explain the physical basis of the forensic signal (not just the algorithm output); and (4) the claimant must be given the opportunity to seek independent forensic review.

38.6.2 Fraudulent Claim vs. Mistaken Submission

Insurance fraud requires proof of intent. A claimant who genuinely believed that an AI-edited photograph accurately represented their damage — perhaps because they used an AI enhancement tool to “clarify” a blurry original without understanding it was generating rather than enhancing — may not have the fraudulent intent required for criminal prosecution, though the claim remains a non-payment basis for civil purposes. As AI image editing tools become integrated into standard mobile photography apps (as they already are in Google Pixel’s Magic Eraser and Apple’s Clean Up features), the distinction between deliberate fabrication and unintentional AI modification becomes a significant evidentiary and legal challenge.

38.6.3 Insurer Obligations Under GDPR and Data Protection Law

The forensic analysis of claim images involves processing biometric and personal data. Under GDPR, insurers must have a lawful basis for processing personal data from claims, must inform claimants that forensic analysis may be applied, and must retain data only for as long as necessary for the claims purpose. The PRNU calibration procedure — which requires requesting additional photographs from the claimant’s device — requires explicit consent or a legitimate interests basis under GDPR Article 6. The calibration images, which encode the PRNU fingerprint of the claimant’s personal device, constitute personal data that must be subject to appropriate retention and deletion policies.

38.7 Industry Applications

The insurance fraud detection market has responded to the AI-generated evidence threat with both standalone products and integrated platform solutions. Shift Technology, Tractable, and Verisk all offer AI-powered claims analytics platforms that include synthetic image detection components. Tractable’s platform, deployed by over 20 major insurers including Covea, Admiral, and QBE, processes millions of vehicle damage photographs annually and has added AI-generation detection to its standard claims workflow. Shift Technology’s FORCE platform, used by AXA, Tokio Marine, and other major carriers, integrates generative AI detection with its fraud scoring models, weighting the synthetic probability score as a fraud indicator alongside traditional fraud signals.

The Association of British Insurers estimates that insurance fraud costs UK consumers £3.4 billion annually, of which approximately £200–300 million is now attributable to digital fraud including AI-generated evidence. The Insurance Fraud Bureau’s Project Endive (2023) demonstrated that coordinated investigation combining forensic AI detection with traditional investigation methods is operationally viable for this fraud category. The Coalition Against Insurance Fraud in the United States has published guidance on AI-generated fraud detection that has been adopted by the National Insurance Crime Bureau as the basis for its member advisory programme.

Reinsurers including Munich Re, Swiss Re, and Hannover Re have begun requiring primary insurers seeking catastrophe reinsurance to demonstrate AI fraud detection capability as part of the underwriting review, effectively making AI-fraud defence a reinsurance condition. Premium rebates of 3–7% for documented deployment of validated AI-fraud detection systems have been introduced by several reinsurers as an incentive mechanism analogous to the BEC insurance discounts for anti-spoofing described in Chapter 37.

38.8 African Use Cases

Insurance fraud in Africa imposes costs estimated at 5–15% of gross written premium across the continent, compared to 3–7% in high-income markets. The gap reflects both higher fraud incidence and less mature detection infrastructure. Motor insurance is the highest-fraud line in most African markets, with staged accidents and inflated repair claims the dominant traditional fraud typologies. AI-generated evidence fraud has been documented in South Africa, Nigeria, and Kenya since 2022,

following the same pattern seen earlier in European markets: a period of growth driven by early adopters before fraud units became aware of the technique, followed by rapid deployment of detection capability once the pattern was identified.

South Africa's Short-Term Insurance Association and the South African Insurance Crime Bureau have jointly deployed an AI-generated image detection capability across member insurers, calibrated on a South African vehicle damage dataset collected by SASRIA. The initial deployment focused on high-value motor claims above R500,000, where the economic incentive for fabrication is highest. Preliminary results from the first six months of deployment (reported in the SAICB 2023 Annual Report) showed AI-generated evidence flagged in 2.3% of submitted high-value motor claims, substantially higher than the 0.5–1% rate reported by European insurers at similar stages of deployment — potentially reflecting a higher base rate of fraud in the South African market or differences in the vehicle damage photography workflow that affect false positive rates.

Kenya's insurance sector, regulated by the Insurance Regulatory Authority, has a significantly younger market with lower premiums and less developed fraud analytics infrastructure. The AKI (Association of Kenya Insurers) fraud working group has noted AI-generated claim evidence as an emerging threat but as of 2024 had not deployed systematic detection capability. The primary barrier is the cost of the PRNU calibration infrastructure relative to the average claim value in the Kenyan market, where the break-even point for forensic analysis requires claims above approximately KES 500,000 (\$3,800). Nigerian underwriters face the additional challenge that a significant fraction of motor insurance operates through third-party agents whose digital claims submission workflows do not preserve EXIF metadata, undermining metadata-based detection.

The microinsurance sector — which covers agricultural equipment, livestock, mobile devices, and small business assets across Sub-Saharan Africa — faces a distinctive fraud challenge. Claim values are typically too low to justify individual forensic investigation, yet fraud rates can be high because the claims process is often entirely digital with minimal human oversight. AI-generated satellite imagery showing fabricated crop damage to support agricultural insurance claims has been documented by ACRE Africa, which provides index-based crop insurance across Kenya, Tanzania, Rwanda, and Zambia. Detection in this context requires applying remote sensing authenticity analysis to satellite images — a forensic capability that requires specialised expertise not readily available in most African insurance markets.

Chapter Summary

Insurance fraud using AI-generated evidence has emerged as a significant operational threat to the non-life insurance industry, with documented instances across motor, property, and personal injury lines in every major market. The taxonomy of AI-enabled insurance fraud spans six attack types: fabricated damage photographs, fabricated medical evidence, staged collisions with inflated damage, fabricated accident videos, GPS and metadata spoofing combined with image fabrication, and wholesale policy fraud using synthetic identities. Each attack type targets a specific element of the claims verification process and requires a targeted forensic response.

PRNU camera fingerprint authentication is the most reliable forensic test for full-image synthesis: authentic photographs yield $NCC \approx 0.038$ against the claimant's device fingerprint while AI-generated images yield $NCC \approx 0.004$, enabling near-perfect separation on uncompressed images. Composite detection using spatial PRNU reveals inserted damage regions as bimodal NCC maps: high NCC in authentic regions, near-zero in the synthetic insert, with a PRNU-discontinuity boundary visible even after Poisson blending. Physical plausibility analysis scores five dimensions of causal consistency for vehicle damage claims; scores below 0.60 trigger escalation. Video temporal forensics applies optical flow, camera shake spectrum, lighting consistency, and compression distribution analysis to detect fabricated or AI-generated video evidence.

A three-tier operational triage pipeline balances detection accuracy against processing cost: Tier 1 automated screening of all claims using metadata, reverse image search, and rapid AI classification; Tier 2 forensic analysis of the 2–5% flagged by Tier 1; and Tier 3 specialist human investigation of the subset escalated from Tier 2. Legal admissibility of AI forensic evidence in insurance disputes is establishing rapidly, with the Project Endive convictions providing the first prosecution precedent in the insurance context. GDPR obligations require that PRNU calibration procedures and data retention are conducted with a documented lawful basis and explicit claimant notification.

Exercises

1. A claimant submits six vehicle damage photographs claiming to have been taken by their Samsung Galaxy S23 immediately after a rear-end collision. PRNU analysis using ten calibration photographs gives fingerprint $\hat{\mathbf{K}}$. The NCC scores for the six claim photographs are: 0.041, 0.038, 0.044, 0.006, 0.039, 0.004. The decision threshold is $\tau = 0.015$ at 1% FPR. (a) Classify each photograph as authentic or suspect. (b) What does the mixed pattern (some authentic, some suspect) tell you about the nature of the fraud? (c) If the claimant argues that the two suspect photographs suffered severe JPEG compression (QF = 50) that destroyed the PRNU signal, what additional forensic test would distinguish genuine compression from synthesis? (d) What additional evidence should the investigator request?
2. A vehicle damage claim photograph passes PRNU authentication ($NCC = 0.034$, above threshold $\tau = 0.015$). Physical plausibility analysis yields the following dimension scores: directionality 0.87, material consistency 0.81, glass breakage 0.29, shadow consistency 0.22, reflection consistency 0.41. (a) Calculate the composite plausibility score (unweighted mean). (b) Which two failed dimensions are most likely to indicate AI inpainting of damage into an authentic photograph? (c) How can the spatial PRNU composite detection method confirm the inpainting hypothesis if the full-image NCC passed? (d) Describe the block-wise NCC map you would expect to see if the glass region was AI-inpainted into an otherwise authentic photograph.
3. An insurer processes 25,000 motor claims per day. The estimated AI-fraud prevalence is 1.2% of claims. The Tier 1 automated classifier has sensitivity 91% and specificity 96%. (a) Calculate the expected daily number of true positives, false positives, true negatives, and false negatives.

- (b) Calculate the positive predictive value (PPV) of the Tier 1 classifier at this prevalence. (c) If each Tier 2 investigation costs \$85 (forensic analysis time and overhead), what is the expected daily Tier 2 cost? (d) What PPV threshold would make Tier 2 investigation cost-effective, assuming the average AI-fraud claim saves \$12,000 when detected versus paid? (e) How must Tier 1 specificity change to meet this PPV threshold at the same sensitivity?
4. Critically evaluate the following statement: “Because an AI-generated image can be made to carry the PRNU fingerprint of a real camera by adding the fingerprint as an additive perturbation (fingerprint forgery, Chapter 25), PRNU-based insurance fraud detection will become ineffective as attackers become more sophisticated.” (a) Estimate the technical capability required to execute a fingerprint forgery attack in the insurance context. (b) What forensic signals remain even after a successful fingerprint forgery? (c) Propose two additional forensic layers that would remain effective against fingerprint-forging attackers and explain the physical basis for each.

Answers to Exercises

1. PRNU classification of six claim photographs.

(a) Threshold $\tau = 0.015$. Classification: Photo 1: $NCC = 0.041 > 0.015 \rightarrow$ **Authentic**. Photo 2: $NCC = 0.038 > 0.015 \rightarrow$ **Authentic**. Photo 3: $NCC = 0.044 > 0.015 \rightarrow$ **Authentic**. Photo 4: $NCC = 0.006 < 0.015 \rightarrow$ **Suspect**. Photo 5: $NCC = 0.039 > 0.015 \rightarrow$ **Authentic**. Photo 6: $NCC = 0.004 < 0.015 \rightarrow$ **Suspect**.

(b) The mixed pattern — four authentic and two suspect photographs from the same claimed session — is the most diagnostically significant outcome. A genuinely defective or heavily compressed photograph typically reduces NCC uniformly rather than producing the specific pattern of near-zero NCC characteristic of AI generation. The mixed pattern strongly suggests that the claimant supplemented authentic photographs with AI-generated images to show damage not present in the authentic captures. This is the most common form of AI insurance fraud: submitting a mix of genuine and fabricated evidence to provide corroboration for the authentic photographs while inflating the damage shown.

(c) To distinguish genuine heavy compression from synthesis: apply *double-JPEG detection*. A genuinely heavy-JPEG-compressed photograph will show a single-generation JPEG quantisation table with $QF = 50$ artefacts distributed uniformly across the image. An AI-generated photograph that was subsequently JPEG-compressed (or a downloaded internet image) will show a secondary JPEG quantisation table from the original download encoding, visible in the DCT coefficient grid pattern. In practice: run BlockSim or similar double-JPEG detection; if the second quantisation table is present, the image has been encoded at least twice, inconsistent with a direct single-capture workflow. Additionally, request the raw camera file (DNG or RAW) if the device stores it; AI-generated images cannot have a RAW file equivalent.

(d) Additional evidence the investigator should request: (1) the claimant’s phone, to extract additional calibration images and confirm the PRNU fingerprint; (2) the original RAW files if the

Samsung Galaxy S23 was in RAW+JPEG mode; (3) the phone's camera roll metadata showing all photographs taken in the claimed time window, with GPS coordinates and timestamps, to establish whether photos 4 and 6 were captured in the same session as the authentic photographs; (4) police accident report and any dashcam or CCTV footage from the claimed location.

2. Physical plausibility analysis and inpainting detection.

(a) Composite plausibility score (unweighted mean): $(0.87 + 0.81 + 0.29 + 0.22 + 0.41)/5 = 2.60/5 = 0.52$. Below the escalation threshold of 0.60 $\rightarrow$ escalate to Tier 2.

(b) The two most indicative failed dimensions for AI inpainting are *shadow consistency* (0.22) and *glass breakage pattern* (0.29). Shadow consistency fails when an inpainted region is generated under a different or diffuse lighting model than the ambient directional light source in the authentic photograph, a common failure mode of diffusion-based inpainting. Glass breakage pattern fails because AI image generators do not accurately model the tempered glass fracture morphology (pebble pattern) or laminated windshield fracture pattern (radial star); they produce plausible-looking but physically incorrect crack patterns.

(c) Spatial PRNU composite detection: the full-image NCC of 0.034 averages over authentic (high-NCC) and synthetic (low-NCC) regions, producing a result above threshold. The block-wise NCC map will expose this: authentic vehicle panels outside the damage region will show $\rho_{ij} \approx 0.038$, while the inpainted glass region will show $\rho_{ij} \approx 0.005$. This bimodal distribution confirms composite fabrication even though the full-image NCC passed.

(d) Expected block-wise NCC map for AI-inpainted glass: the glass region occupies approximately the upper-left 20% of the image (typical windshield position). The NCC map would show: background sky and road: $\rho_{ij} \approx 0.041$; authentic vehicle panels: $\rho_{ij} \approx 0.038$; inpainted glass region: $\rho_{ij} \approx 0.004$ – 0.007 ; boundary blocks (within 1–2 block widths of the splice edge): $\rho_{ij} \approx 0.015$ – 0.022 (intermediate values from the Poisson blending that smooths the visual boundary but cannot introduce camera PRNU into the synthetic insert). The boundary intermediate values are the definitive forensic signature of the splice location.

3. Operational triage cost-benefit analysis.

(a) Daily claim volumes at prevalence $p = 0.012$, sensitivity = 0.91, specificity = 0.96, $N = 25,000$:

Fraudulent claims: $25,000 \times 0.012 = 300$. Authentic claims: $25,000 - 300 = 24,700$.

True positives (TP): $300 \times 0.91 = 273$. False negatives (FN): $300 - 273 = 27$. True negatives (TN): $24,700 \times 0.96 = 23,712$. False positives (FP): $24,700 - 23,712 = 988$.

(b) Positive predictive value: $PPV = TP/(TP + FP) = 273/(273 + 988) = 273/1,261 = 0.217$ (21.7%). Only 1 in 4.6 Tier 2 escalations contains actual fraud.

(c) Daily Tier 2 cost: $(273 + 988) \times \$85 = 1,261 \times \$85 = \$107,185$.

(d) Cost-effective PPV threshold: Tier 2 investigation (at \$85) is cost-effective when the expected saving from detecting a fraudulent claim exceeds the investigation cost. Expected saving per Tier 2 investigation = $PPV \times \$12,000$. Set equal to cost: $PPV \times \$12,000 = \$85 \Rightarrow PPV = \$85/\$12,000 = 0.0071 = 0.71\%$. At the current 21.7% PPV, Tier 2 is highly cost-effective (exceeds the threshold by $30\times$). Even a PPV as low as 0.71% would make investigation cost-effective for this claim value — meaning the classifier would need to be extremely poor before investigation became unprofitable.

(e) At $PPV = 0.71\%$, with sensitivity fixed at 0.91%: $PPV = \frac{p \cdot Sn}{p \cdot Sn + (1-p)(1-Sp)} = 0.0071 \Rightarrow 0.012 \times 0.91 / (0.012 \times 0.91 + 0.988 \times (1 - Sp)) = 0.0071 \Rightarrow 0.988(1 - Sp) = 0.010932 / 0.0071 - 0.010932 = 1.5397 - 0.010932 = 1.429 \Rightarrow 1 - Sp = 1.446 > 1$: this has no solution. This means that at 1.2% prevalence and \$12,000 average fraud value, the Tier 2 investigation is cost-effective for *any* specificity above 0% — the break-even PPV of 0.71% is already below the PPV achievable even with a random classifier at this prevalence. The current question of specificity is not about cost-effectiveness but about the burden of false positives on legitimate claimants, which is a different and important operational concern.

4. Critical evaluation of fingerprint forgery threat to PRNU detection.

(a) Technical capability required for insurance context fingerprint forgery: the attacker must (1) obtain a set of authentic photographs from the claimant's device (requires access to the device or to previously submitted authentic photographs); (2) estimate the device PRNU fingerprint $\hat{\mathbf{K}}$ from these photographs (requires specialist software such as SPN-Tools or Photo-Response Non-Uniformity Matlab toolbox, publicly available); (3) add the estimated fingerprint to the AI-generated image using the multiplicative signal model $\hat{\mathbf{x}}' = \hat{\mathbf{x}}(1 + \alpha\hat{\mathbf{K}})$ with α calibrated to match the authentic NCC level; (4) adjust α to achieve NCC above threshold without introducing visible texture. This requires non-trivial technical knowledge and access to authentic device photographs. As of 2024, this attack is within the capability of well-resourced organised fraud rings but not typical opportunistic individual claimants.

(b) Forensic signals remaining after successful fingerprint forgery: *Content-correlation test*: genuine PRNU is correlated with local image content (because the PRNU signal is multiplicative: $\hat{x} = (1 + K)\bar{x}$, so the PRNU is larger in brighter regions). An additively injected fingerprint is *not* correlated with image brightness in this way, producing a detectable discrepancy in the correlation between noise residual and local illumination. *Physical plausibility*: fingerprint forgery cannot fix shadow inconsistency, glass breakage pattern errors, or optical flow inconsistencies — these are independent of the PRNU signal and remain detectable by physical plausibility analysis. *Spectral fingerprint*: the AI generator's spectral fingerprint persists in the noise residual even after PRNU injection, because the two signals occupy different frequency ranges.

(c) Two additional forensic layers effective against fingerprint forgers: *Layer 1 — Content-correlation PRNU test*: compute $\text{Corr}(\hat{r}, \bar{x})$ across local image blocks, where $\bar{x}$ is the local mean brightness. For genuine PRNU, this correlation is positive (typically $r > 0.3$). For

additive forgery, this correlation is near zero. The physical basis: the multiplicative noise model $\hat{x} = (1 + K)\bar{x}$ implies $\hat{r} \approx K\bar{x}$, so residual and brightness are correlated wherever K is non-zero.

Layer 2 — Device model neural fingerprint: deep-learning classifiers trained on large datasets of images from specific device models can identify the camera model from texture and noise characteristics independent of the PRNU. A claimed Samsung Galaxy S23 photograph that was generated by Midjourney will fail to match the Galaxy S23 model-level fingerprint even if the PRNU has been forged, because the model fingerprint is embedded in the overall noise statistics and demosaicing pattern, not just the PRNU component.

Chapter 39

Deep Fakes in Marketing, Sales, and Brand Integrity

“A brand is a promise. A synthetic influencer is a promise made by a machine.”

—Anon.

Marketing was the first legitimate commercial sector to embrace synthetic media at scale, and it has become the sector in which the boundary between beneficial innovation and harmful deception is most contested. Synthetic influencers — entirely computer-generated personas with no biological existence — had accumulated over \$3 billion in brand partnerships by 2023. AI-generated product images dominate e-commerce at the leading platforms. Neural dubbing has reduced localisation costs by 70%, enabling brands to run personalised multi-language campaigns at price points previously available only to global corporations. And conversational AI agents synthesised to resemble real company representatives are replacing human customer service at scale across retail, banking, and telecommunications.

Each of these applications occupies a different position on the disclosure and consent spectrum. A clearly labelled virtual influencer such as Lil Miquela, whose artificial nature is disclosed in the account biography and acknowledged by her substantial Instagram following, raises few deception concerns. An AI-generated testimonial video showing a realistic-looking “satisfied customer” who never existed, presented without disclosure on a product sales page, is straightforwardly deceptive under FTC and EU consumer protection law. The regulatory and ethical challenge is the large grey zone between these extremes: AI-enhanced celebrity endorsements, voice-cloned spokesperson videos, and AI-personalised sales calls where disclosure is incomplete or strategically ambiguous.

The harm profile of deep fakes in marketing and sales is structurally different from the harms analysed in earlier chapters. The victims of electoral disinformation and NCII are specific identifiable individuals; the victims of marketing deception are diffuse consumers who make purchase decisions on the basis of false or misleading synthetic evidence. The causal chain from synthetic content to consumer harm is typically mediated by a commercial transaction — a purchase, a subscription, or

an investment — which provides a traceable, quantifiable harm measure and a clear legal remedy under consumer protection statutes. This makes marketing deep fakes simultaneously more legally actionable than reputational harms and less viscerally harmful than NCII, placing them in the middle tier of the harm taxonomy established in Chapter 26.

The financial stakes are significant. The Federal Trade Commission’s enforcement actions against deceptive AI-generated endorsements produced settlements totalling \$272 million in 2023–2024. The UK Advertising Standards Authority received a record 4,800 complaints in 2023 related to AI-generated advertising content. The European Commission’s Digital Services Act creates specific obligations for platforms hosting AI-generated commercial content, and the EU AI Act’s Article 50 disclosure requirements apply to synthetic media used in commercial communications. Understanding the technical mechanisms, regulatory framework, and forensic detection of marketing deep fakes is essential for brand managers, compliance officers, advertising regulators, and platform trust and safety teams.

Brand impersonation — the use of synthetic media to create false impressions that a product, statement, or endorsement originates from a reputable brand — is the most commercially damaging marketing deep-fake attack on brands themselves. Synthetic videos purportedly showing company executives endorsing investment schemes, AI-generated social media posts attributed to brand accounts, and voice-cloned customer service calls that impersonate legitimate companies have all been documented at scale. These attacks impose direct reputational costs on the impersonated brand, indirect costs through customer confusion and support burden, and potential legal liability if customers suffer losses acting on the false information.

This chapter develops the full landscape of synthetic media in marketing and sales from both the operator and the defender perspective. Section 39.1 develops synthetic influencers and their disclosure obligations. Section 39.2 develops AI-generated testimonials and social proof fabrication. Section 39.3 develops brand and executive impersonation attacks. Section 39.4 develops AI-personalised sales and customer service with their disclosure requirements. Section 39.5 develops the detection and verification methods applicable to marketing content. Section 39.6 analyses the regulatory framework across major jurisdictions.

39.1 Synthetic Influencers and Virtual Spokespersons

A synthetic influencer is a computer-generated persona with a consistent visual identity, personality, and content history, used to promote products and services on social media platforms. The technology enabling synthetic influencers has evolved from CGI-rendered still images (Lil Miquela, 2016) through real-time face generation (Shudu, 2017) to fully interactive AI agents that generate video content, respond to comments, and maintain multi-platform presences. The market has grown from novelty to mainstream: as of 2024, the top ten synthetic influencers collectively held brand partnerships worth over \$1 billion annually, and an estimated 40% of brands with over \$1 billion in marketing spend had deployed or were piloting synthetic influencer relationships.

39.1.1 The Disclosure Question

Disclosure of synthetic influencer status is legally required in the United States under FTC Endorsement Guidelines (16 CFR Part 255), in the UK under the ASA’s CAP Code, in the EU under the Unfair Commercial Practices Directive, and in Australia under the ACCC’s Influencer Marketing Guide. Each framework requires clear and conspicuous disclosure of material connections between influencers and brands, and all have extended this requirement to synthetic influencers whose AI nature is a material fact that a reasonable consumer would wish to know. The FTC’s 2023 enforcement action against a supplement brand for failing to disclose that its testimonial videos featured AI-generated spokespersons resulted in an enforcement settlement — one of the early FTC actions specifically targeting undisclosed AI-generated commercial content.

The disclosure standard requires that the synthetic nature be communicated in a way that is clear, prominent, and unavoidable. Industry self-regulatory practice has not kept pace with regulatory requirements: a 2023 survey of 200 brand partnerships involving synthetic influencers by the Influencer Marketing Hub found that 67% used disclosure language buried in account bios, 23% used no disclosure at all, and only 10% used prominent post-level disclosure satisfying the FTC’s “clear and conspicuous” standard.

39.1.2 Economic Model of Synthetic Influencers

The economic appeal of synthetic influencers to brands follows a straightforward cost comparison. A human mega-influencer (10M+ followers) commands \$250,000–\$1 million per sponsored post; a synthetic influencer with comparable reach can be operated for \$15,000–\$50,000 per month in generation, management, and content creation costs. Over a 12-month campaign, the cost comparison:

$$C_{\text{synthetic}} = C_{\text{creation}} + 12 \cdot C_{\text{monthly}} = \$80,000 + 12 \times \$30,000 = \$440,000, \quad (39.1)$$

compared to \$4–8 million for 12 months of monthly posts from a human mega-influencer. The 10:1 cost advantage, combined with complete brand control over content, absence of reputational risk from influencer personal conduct, and unlimited availability across time zones, explains the rapid commercial adoption.

39.2 AI-Generated Testimonials and Social Proof Fabrication

Consumer testimonials and product reviews are among the most influential purchase decision factors, with Nielsen surveys consistently finding that 88% of consumers trust online reviews as much as personal recommendations. Synthetic media has made testimonial fabrication dramatically more accessible: a photorealistic video testimonial from a “satisfied customer” who never purchased the product can be generated in minutes using HeyGen, Synthesia, or D-ID at a cost of under \$5. Text reviews can be generated at essentially zero marginal cost using LLMs. The combination of synthetic video testimonials and LLM-generated text reviews represents a complete fabrication of the social proof ecosystem on which consumer trust in e-commerce depends.

39.2.1 Video Testimonial Fabrication

A fabricated video testimonial is produced in four steps. First, a synthetic persona is created: a GAN-generated or stock-photo face is selected, ensuring it does not match any identifiable real person. Second, a testimonial script is written using an LLM, tuned to match the natural language patterns of genuine customer reviews for the product category. Third, a talking-head synthesis system (HeyGen, Synthesia, D-ID) generates the video of the synthetic persona delivering the script, with lip synchronisation and natural head motion. Fourth, the video is uploaded to a product listing, social media account, or review platform, typically with a fabricated account history to provide the appearance of a genuine consumer.

Worked Example — Testimonial fabrication cost calculation

A dietary supplement company fabricates a testimonial campaign: 50 unique testimonial videos, each 45 seconds, featuring different synthetic personas delivering varied scripts.

Cost component	Unit cost	Total
GAN-generated persona photographs	\$0	\$0
LLM script generation (50 scripts, 150 words each)	\$0.002/script	\$0.10
Talking-head video synthesis (HeyGen API)	\$0.08/min	\$180
Account creation and seeding	\$2/account	\$100
Total fabrication cost		\$280.10
Equivalent authentic testimonial videos	\$800 each	\$40,000

Cost ratio: authentic / fabricated = 143 : 1. This extreme cost asymmetry — \$280 to fabricate versus \$40,000 to obtain genuine — explains the prevalence of synthetic testimonial fraud in health, beauty, and financial product categories where testimonials have the highest purchase influence.

39.2.2 The Consumer Harm Calculus

The consumer harm from fabricated testimonials operates through two mechanisms. The direct mechanism is purchase influence: consumers who would not have purchased the product without the testimonial are induced to do so by fabricated social proof, then suffer the harm of an ineffective or dangerous product, a financial loss, or both. The indirect mechanism is epistemic: widespread testimonial fabrication degrades the reliability of all testimonials, imposing information costs on every consumer who must now discount review evidence that was previously trustworthy. This second-order harm — the destruction of the review ecosystem as a reliable information source — is a classic negative externality that falls on all market participants, not just the direct victims of specific fraudulent testimonials.

39.3 Brand and Executive Impersonation

Brand impersonation attacks use synthetic media to create false impressions that statements, endorsements, or communications originate from a specific legitimate organisation or its representatives. The

attack surface is broad: deepfake videos of CEOs endorsing investment products, AI-generated social media posts mimicking brand voice, voice-cloned customer service calls exploiting brand trust, and fabricated press releases with synthetic quotes from company spokespersons. Each attack exploits the reputational capital of the impersonated brand, converting it into persuasion value for the attacker's objectives (typically financial fraud or competitor sabotage).

39.3.1 CEO Impersonation for Investment Fraud

Deepfake videos of prominent company founders and CEOs endorsing cryptocurrency investments, Ponzi schemes, and fraudulent financial products have been among the highest-volume and highest-harm brand impersonation attacks documented. The formula is consistent: a synthesised video of a recognisable executive (Elon Musk, Warren Buffett, Richard Branson) is distributed through social media advertising, showing the executive enthusiastically endorsing a specific investment opportunity. The video quality is typically high enough to pass casual inspection on a mobile screen; the target audience is general consumers rather than sophisticated investors who might recognise forensic artefacts.

The harm magnitude is significant. The UK Financial Conduct Authority documented £78 million in losses from investment fraud using deepfake celebrity endorsements in 2023 alone, with an average loss of £15,600 per victim. Meta and YouTube each removed over 10,000 deepfake investment advertisement videos in 2023 after regulatory pressure, representing a fraction of the total volume in circulation. The IRS Criminal Investigation division has identified deepfake CEO endorsement fraud as one of the fastest-growing financial crime typologies.

39.3.2 Brand Voice Impersonation

A subtler form of brand impersonation uses LLM-generated text that precisely mimics a brand's communication style — vocabulary, sentence structure, tone, common phrases — to create fraudulent communications indistinguishable from authentic brand content. This can take the form of phishing emails purportedly from a brand's customer service team, social media posts from impersonator accounts, or fabricated press releases submitted to automated news aggregators. Stylometric analysis (Chapter 23) can identify LLM-generated brand impersonation by comparing the statistical fingerprint of the suspect communication against an authenticated corpus of genuine brand communications.

39.4 AI-Personalised Sales and Customer Service

The deployment of AI agents that synthesise human-sounding voice and generate personalised sales pitches represents one of the fastest-growing applications of synthetic speech in commercial contexts. Conversational AI systems from companies including Synthflow, Retell AI, and Bland AI generate real-time voice conversations that pass Turing-test-level scrutiny in customer service contexts, completing sales calls, appointment bookings, and customer surveys at a fraction of the cost of human agents. The ethical and regulatory question is not whether these systems work — they demonstrably do — but whether consumers have the right to know they are speaking to a synthetic agent.

39.4.1 The Disclosure Obligation for AI Voice Agents

California SB 1001 (2018) was the first statute explicitly requiring AI voice bots to disclose their synthetic nature when directly and sincerely asked. The EU AI Act's Article 52 extends this to require proactive disclosure for AI systems that interact with humans, without requiring the human to ask first. The US TRACED Act and FCC regulations prohibit AI-generated robocalls without prior express written consent, regardless of whether the recipient knows the call is AI-generated.

The disclosure obligation creates a technical measurement problem: when is an AI voice agent “indistinguishable” from a human in a way that triggers disclosure? The standard adopted by most regulators focuses on the listener's reasonable belief rather than technical capability: if a reasonable consumer would be likely to believe they are speaking to a human, disclosure is required. Voice-cloned agents that sound like specific named human representatives of a company — using the voice of a real employee or executive — create additional concerns under right-of-publicity law even when the synthetic nature is disclosed.

Worked Example — Cost economics of AI vs. human sales agents

A fintech's outbound sales operation: 10,000 sales calls per day, average 8 minutes per call, with human agents at \$22/hour fully loaded cost.

Human agent cost: $10,000 \times (8/60) \times \$22 = \$29,333/\text{day}$.

AI voice agent cost: synthesis at \$0.05/minute, infrastructure at \$0.02/minute. $10,000 \times 8 \times \$0.07 = \$5,600/\text{day}$.

Daily saving: \$23,733. Annual saving: \$8.7 million. At this saving, the company can justify significant investment in compliance infrastructure (disclosure systems, consent management, audit logging) while still achieving substantial net cost reduction.

39.5 Detection and Verification for Marketing Content

Detection of synthetic marketing content applies a specialised pipeline that combines face synthesis detection with identity verification and disclosure auditing.

Algorithm 7: Synthetic Influencer and Testimonial Detection**Input:** Content item c (image or video); platform context (p, u, t) **Output:** Synthetic probability P_{synth} ; disclosure compliance flag

```

// Identity existence check
1 faces ← face-detector( $c$ );
2 for each face  $f \in \text{faces}$  do
3    $\mathbf{e}_f \leftarrow \text{ArcFace-embed}(f)$ ;
4   match ← search-identity-database( $\mathbf{e}_f, \tau_{\text{ID}} = 0.75$ );
5   if  $\neg \text{match}$  then
6      $S_{\text{novel}} \leftarrow 1.0$ ; // no known real person found
7   end
8   else
9      $S_{\text{novel}} \leftarrow 1 - \text{match.similarity}$ 
10  end
11 end

// Synthesis detection
12  $S_{\text{GAN}} \leftarrow \text{face-swap-detector}(c)$ ;
13  $S_{\text{diffusion}} \leftarrow \text{diffusion-detector}(c)$ ;
14  $S_{\text{talking}} \leftarrow \text{talking-head-detector}(c)$  if video;
15  $P_{\text{synth}} \leftarrow \max(S_{\text{novel}}, S_{\text{GAN}}, S_{\text{diffusion}})$ ;

// Disclosure audit
16 bio ← fetch-account-bio( $u$ );
17 caption ← fetch-caption( $c$ );
18 disclosed ← disclosure-classifier(bio  $\oplus$  caption);

19 if  $P_{\text{synth}} > 0.65$  and  $\neg \text{disclosed}$  then
20   return  $P_{\text{synth}}$ , UNDISCLOSED_SYNTHETIC: refer to compliance
21 end
22 return  $P_{\text{synth}}$ , disclosure status

```

Worked Example — Synthetic influencer disclosure audit at scale

A platform runs Algorithm 7 on 500,000 sponsored posts per day with a stated synthetic influencer prevalence of 3% (15,000 posts).

Outcome	Count	Action
Synthetic + undisclosed (TP)	$15,000 \times 0.84 = 12,600$	Refer to compliance
Authentic mislabelled (FP)	$485,000 \times 0.04 = 19,400$	False alert; appeal
Synthetic + disclosed (pass)	$15,000 \times 0.16 \times 0.62 = 1,488$	Compliant
Missed undisclosed (FN)	$\approx 2,400$	Undetected violation

PPV = $12,600 / (12,600 + 19,400) = 39\%$. The base-rate problem again: even at 84% sensitivity, 61% of compliance referrals involve authentic content. Platform must provide a rapid appeals mechanism for authentic influencers whose content is incorrectly flagged, with a target resolution time of 24 hours.

Detection of synthetic marketing content faces different operational constraints from other deep-fake detection contexts. The evidentiary standard is not criminal proof beyond reasonable doubt but the

civil and regulatory standard of reasonable probability. Processing must occur at platform scale — billions of pieces of content per day — and must be fast enough for real-time advertising review. And the consequences of false positives (blocking authentic content) are economically significant for platforms that derive revenue from advertising.

39.5.1 Platform-Level Automated Detection

Meta, Google, and TikTok have deployed AI-generated content detection in their advertising review pipelines, triggered primarily for content featuring human faces making claims (the highest-risk category for investment fraud and health product misrepresentation). Detection AUC for marketing content is typically 0.89–0.92 on balanced benchmarks, lower than the 0.94+ reported for forensic deep-fake benchmarks, because marketing synthetic content is produced with higher quality control than most deep fakes targeted at platform detection evasion.

39.5.2 C2PA Provenance for Advertising Content

The Coalition for Content Provenance and Authenticity's standards apply directly to advertising content: a C2PA-signed advertisement carries a verifiable record of its creation, the tools used to generate it, any AI involvement in its production, and the identity of the signing organisation. The Interactive Advertising Bureau has endorsed C2PA as the standard for advertising content provenance, and several major brands have begun requiring C2PA manifests from all creative agencies supplying video advertising content. For regulators, a C2PA-signed advertisement that contains AI-generated elements but carries a valid manifest with disclosure of that fact satisfies the disclosure obligation without requiring the advertiser to include visible on-screen text.

39.6 Regulatory Framework

The regulatory framework for synthetic media in marketing spans consumer protection law, advertising standards codes, platform-specific policies, and emerging AI-specific legislation. The overlap among these frameworks creates both comprehensive coverage and compliance complexity for brands operating across jurisdictions.

The FTC Endorsement Guidelines require disclosure of material connections, including the AI-generated nature of endorsers; FTC enforcement actions have established that video testimonials from non-existent consumers constitute deceptive acts under Section 5 of the FTC Act. The UK ASA's CAP Code requires that advertisements be clearly identifiable as advertising; the ASA has confirmed that AI-generated influencer content must be disclosed as both advertising and as AI-generated when it depicts realistic human-appearing content. The EU Digital Services Act requires platforms to label AI-generated content in recommender systems, and the EU AI Act's Article 50 mandates disclosure for synthetic content that could deceive as to its AI origin. India's IT Rules 2021, amended in 2023, require platforms to label AI-generated content within 30 days or face liability for its consequences.

39.7 Industry Applications

The marketing technology industry has bifurcated into synthetic content creation platforms and synthetic content detection platforms, with significant investment in both. On the creation side, Synthesia (which has raised significant venture capital in successive funding rounds), HeyGen (valued at over \$500 million in 2024), and D-ID have scaled to serve enterprise marketing teams globally. On the detection side, Reality Defender, Hive Moderation, and Sensity AI have integrated marketing-specific detection into their platforms, offering APIs for advertising review that platforms and brands can embed in their content workflows.

Brand protection services from companies including Incopro, MarkMonitor, and Corsearch have added AI-generated brand impersonation monitoring to their standard trademark and copyright enforcement services. These services scan the web, social media, and app stores for synthetic content impersonating protected brands, providing brands with a near-real-time feed of impersonation incidents requiring response. The market for brand protection services grew 40% in 2023 partially driven by the increase in AI-enabled brand impersonation.

39.8 African Use Cases

Africa's rapidly growing digital advertising market — projected to reach \$3.6 billion by 2026 — and its mobile-first consumer base create both significant opportunities for legitimate synthetic media marketing and significant vulnerabilities to deceptive applications. Synthetic influencers representing African aesthetics and using African languages have been deployed on Instagram and TikTok by brands targeting Nigerian, Kenyan, South African, and Ghanaian consumers, with disclosure practices ranging from prominent to entirely absent. The Advertising Standards Authority of South Africa and the Advertising Regulatory Council of Nigeria have indicated that existing disclosure rules apply to AI-generated commercial content, but enforcement capability is limited for content distributed through international platforms operating outside local regulatory reach.

The most economically damaging application of synthetic marketing fraud in Africa is investment fraud using deepfake endorsements from African celebrities and business leaders. Documented cases include fabricated videos of Aliko Dangote (Nigeria's wealthiest businessman) endorsing cryptocurrency investment platforms and fabricated WhatsApp voice messages purportedly from prominent religious leaders endorsing financial schemes targeting their followers. Africa's high WhatsApp usage and the cultural weight of celebrity and religious authority make these attacks particularly effective, and the low financial sophistication of many targets combined with limited access to independent financial advice amplifies the harm per incident.

Chapter Summary

Synthetic media has fundamentally altered the marketing and sales landscape, enabling both legitimate cost-reducing applications and harmful deceptive practices across the full spectrum of commercial communication. Synthetic influencers generated \$3 billion in brand partnerships by 2023 at a 10:1 cost

advantage over comparable human influencers, creating strong commercial incentives for adoption and significant regulatory pressure for disclosure. AI-generated testimonial fabrication at under \$300 for 50 videos versus \$40,000 for authentic equivalents has created a pervasive testimonial fabrication problem in health, beauty, and financial product categories. Brand impersonation via deepfake CEO endorsement videos caused documented losses of £78 million to UK consumers in 2023 alone.

The regulatory framework is converging across jurisdictions: FTC Endorsement Guidelines, EU AI Act Article 50, UK ASA CAP Code, and India IT Rules all require disclosure of AI involvement in commercial content, with enforcement actions establishing that failure to disclose is an unfair commercial practice with significant financial penalties. C2PA provenance standards and platform-level detection are the primary technical countermeasures, with detection AUC of 0.89–0.92 for marketing-specific synthetic content. The disclosure obligation for AI voice agents is established by California SB 1001 and extended by EU AI Act Article 52, requiring proactive disclosure whenever a reasonable consumer would believe they are speaking to a human.

Exercises

1. A health supplement company operates a synthetic influencer with 800,000 followers. Over 12 months, the influencer generates 96 sponsored posts reaching an average of 120,000 accounts each with a click-through rate of 2.1% and a purchase conversion rate of 4.3%. Average order value is \$67. (a) Calculate the total revenue attributable to the influencer campaign. (b) The campaign cost \$280,000. Calculate the return on ad spend (ROAS). (c) The company fails to disclose the influencer's AI nature. The FTC estimates 30% of purchasers would not have bought if properly informed. Calculate the estimated consumer harm from the non-disclosure. (d) The FTC issues a consent decree requiring refunds to affected consumers plus a \$4 million civil penalty. Was the campaign financially rational from the company's perspective?
2. A competing brand suspects that a rival is using AI-generated testimonial videos on their product listing page. The suspect videos show five different people giving glowing reviews. You have access to a face-swap detection API (AUC = 0.91) and a talking-head synthesis detector (AUC = 0.88). (a) If you run both detectors on each video and require both to flag for referral, what is the combined false positive rate if individual FPRs are each 6%? (b) If the prior probability that these videos are synthetic is 0.40, calculate the posterior probability of synthesis given that both detectors flag a video (use Bayes' theorem with sensitivity estimates of 0.87 and 0.84 respectively). (c) Is this posterior probability sufficient to submit a complaint to the FTC? What additional evidence would strengthen the case?
3. Map the disclosure obligations for a synthetic influencer campaign targeting consumers in (a) the United States, (b) the European Union, (c) the United Kingdom, and (d) Nigeria. For each jurisdiction: identify the applicable legal instrument, state the required disclosure content, specify where the disclosure must appear, and identify the enforcement body and maximum penalty. Identify any jurisdiction where the current legal framework does not adequately cover

synthetic influencer disclosure and explain the gap.

Answers to Exercises

1. Synthetic influencer campaign analysis.

(a) Total revenue: $96 \text{ posts} \times 120,000 \text{ reach} \times 0.021 \text{ CTR} \times 0.043 \text{ conversion} \times \67 AOV
 $= 96 \times 120,000 \times 0.021 \times 0.043 \times \67 . Step by step: $96 \times 120,000 = 11,520,000$ total impressions.
 $\times 0.021 = 241,920$ clicks. $\times 0.043 = 10,402$ purchases. $\times \$67 = \$696,934$ revenue.

(b) $\text{ROAS} = \$696,934 / \$280,000 = 2.49$. For every dollar spent, \$2.49 was generated — a positive but moderate return by direct-response advertising standards.

(c) Consumer harm from non-disclosure: 30% of 10,402 purchases = 3,121 purchases that would not have occurred with disclosure. Each represents the full purchase value as harm (the consumer did not receive what they believed they were buying — authentic social proof).
 Consumer harm = $3,121 \times \$67 = \$209,107$.

(d) Financial outcome with FTC action: revenue \$696,934, campaign cost \$280,000, refund obligation (30% of purchases) \$209,107, civil penalty \$4,000,000. Net: $\$696,934 - \$280,000 - \$209,107 - \$4,000,000 = -\$3,792,173$. The campaign produced a net loss of \$3.8 million once enforcement costs are included. From the company's perspective, the non-disclosure was financially irrational: even a 25% probability of FTC enforcement would make the expected cost of non-disclosure exceed the expected benefit of undisclosed synthetic influencer deployment.

2. Testimonial video synthetic detection analysis.

(a) Combined false positive rate (both detectors must flag): if individual FPRs are both 6%, and assuming independence: combined FPR = $0.06 \times 0.06 = 0.0036 = 0.36\%$. Requiring both to flag dramatically reduces false positives at the cost of also reducing true positive detection when either detector alone misses.

(b) Bayesian posterior probability of synthesis. Prior: $P(\text{synth}) = 0.40$. Sensitivities: $\text{Sn}_1 = 0.87$, $\text{Sn}_2 = 0.84$. $P(\text{both flag}|\text{synth}) = 0.87 \times 0.84 = 0.731$. $P(\text{both flag}|\text{authentic}) = 0.06 \times 0.06 = 0.0036$. Bayes: $P(\text{synth}|\text{both flag}) = \frac{0.731 \times 0.40}{0.731 \times 0.40 + 0.0036 \times 0.60} = \frac{0.2924}{0.2924 + 0.00216} = \frac{0.2924}{0.29456} = 0.9927$.

Given both detectors flag and a 40% prior, the posterior probability of synthesis is 99.3% — a very strong inference.

(c) For an FTC complaint, the standard is not criminal proof but reasonable probability. A 99.3% posterior from two independent detection systems is almost certainly sufficient as a basis for complaint, but the FTC will require: (1) the identity of the alleged synthetic personas (reverse image search to confirm no real person exists); (2) metadata analysis of the submitted videos; (3) the full detector outputs with AUC documentation. Additional strengthening evidence: (a) the persons shown in the videos cannot be found with any LinkedIn, social media, or public record

search; (b) the videos show inconsistencies in background, lighting, or lip-sync consistent with talking-head synthesis; (c) the accounts posting the reviews were created recently with no prior activity.

3. Cross-jurisdictional synthetic influencer disclosure requirements.

(a) *United States*: FTC Endorsement Guides (16 CFR Part 255, updated 2023). Required content: disclosure that the influencer is AI-generated, plus any material commercial connection. Where: must be “clear and conspicuous” in the post itself, not buried in bio or below the fold. Enforcement: Federal Trade Commission; civil penalties up to \$50,120 per violation under Section 5 of the FTC Act. The FTC’s 2024 guidance specifically identifies AI-generated personas as requiring disclosure of their non-human nature.

(b) *European Union*: EU AI Act Article 50 (for content that could deceive as to AI origin) + Unfair Commercial Practices Directive (misleading commercial practices). Required content: disclosure that the content is AI-generated when it realistically depicts a person and could mislead consumers. Where: proactive disclosure before or during the commercial communication. Enforcement: national market surveillance authorities; fines up to 6% of global turnover for AI Act violations. The UCPD additionally allows individual member state regulators and consumers to bring claims for misleading commercial practices.

(c) *United Kingdom*: ASA CAP Code Rule 2.1 (advertisements must be obviously identifiable as advertisements) + ASA AI guidance (2023) extending this to AI-generated human-appearing content. Required content: #ad or “Advertisement feature” label, plus indication of AI nature when the content features a realistic-appearing AI persona. Where: in the post itself, before engagement. Enforcement: Advertising Standards Authority (self-regulatory, no statutory fines) but referral to Trading Standards for persistent offenders who may be prosecuted under Consumer Protection from Unfair Trading Regulations.

(d) *Nigeria*: Advertising Regulatory Council of Nigeria Act 2022 and ARCON Code of Advertising Practice. Required content: advertisements must be clearly identifiable; the ARCON has indicated AI-generated content requires disclosure but has not issued specific guidance on synthetic influencers as of 2024. Where: post level, in the primary text. Enforcement: ARCON; penalties up to ₦500,000 per violation plus suspension of advertising rights. *Gap*: Nigeria lacks specific legislation addressing AI-generated synthetic personas; the ARCON’s general advertising code applies but does not address the synthetic influencer disclosure question with the specificity of US or EU frameworks, creating interpretive uncertainty about exactly what disclosure language is required.

Chapter 40

Deep Fakes in International Trade, Customs, and Supply Chains

“International trade runs on documents. Every document that can be fabricated is a door that synthetic media can open.”

—Anon.

International trade generates more document-intensive verification processes than almost any other human activity. A single shipment of goods across a major border may require a bill of lading, commercial invoice, packing list, certificate of origin, sanitary and phytosanitary certificates, customs declaration, letter of credit, inspection certificate, and import/export licences — each of which is issued by a different authority, verified by a different institution, and represents a point of potential fabrication. Synthetic media and large language models have arrived at precisely the moment when the world’s major customs authorities were digitising these document flows, creating new efficiencies but simultaneously creating new attack surfaces for document fraud that did not exist when paper originals were the standard.

The trade document fraud problem has a specific economic character that distinguishes it from the other fraud categories examined in this book. In financial fraud, the attacker seeks to extract money directly. In insurance fraud, the attacker fabricates damage to collect a claim. In trade document fraud, the attacker is typically seeking one of three goals: to import goods that would be prohibited or heavily taxed by declaring them as different goods (tariff evasion and prohibited goods smuggling); to claim incentives, subsidies, or preferential tariff rates by fabricating certificates of origin or compliance documents (subsidy fraud); or to launder the proceeds of other crimes through the false valuation of legitimate trade (trade-based money laundering). In each case, the document itself is the attack vehicle, and the quality of the fabricated document determines whether it passes the verification process that stands between the attacker and their goal.

The global customs community processes approximately 1.1 billion trade declarations per year, with the World Customs Organisation’s SAFE Framework and WCO RILO network coordinating

intelligence sharing across 183 member countries. The introduction of AI document generation has materially changed the intelligence picture: traditional trade fraud relied on physical alteration of documents (rubber stamps, bleached originals, chemical erasure) that left detectable forensic traces even with basic equipment. AI-generated documents — LLM-generated text, synthetically modified seals and signatures, AI-inserted certification stamps — require digital forensic analysis tools that many customs authorities, particularly in lower-income countries, do not yet deploy at scale.

Counterpart-identity verification is the second major attack surface in international trade. High-value trade transactions — letters of credit, large-scale commodity purchases, and investment infrastructure contracts — require verification that the authorised signatory on the counterpart side is who they claim to be. Video call KYC has become a standard due diligence tool for international trade finance, and the same face-swap and voice-clone attacks that target domestic financial KYC apply with equal force to trade finance KYC, with the added complication that the parties may be in different countries with different verification infrastructure and that the transaction values are typically much higher.

Supply chain provenance is the third attack surface. Consumers, retailers, and regulators increasingly require authentic documentation of supply chain conditions: fair labour certification, organic farming certificates, deforestation-free sourcing attestations, and conflict-mineral compliance. Each of these documents is now at risk of AI-enabled fabrication, undermining the audit systems that brands use to validate their supply chain claims and creating liability risk when brands certify compliance based on fabricated documentation.

This chapter develops the full landscape of AI-enabled trade document fraud and its detection. Section 40.1 develops trade document fabrication including the LLM and image-synthesis tools used and the specific documents most vulnerable. Section 40.2 develops customs authority detection capabilities and the risk-scoring pipeline. Section 40.3 develops counterpart KYC in trade finance. Section 40.4 develops supply chain provenance fraud. Section 40.5 develops blockchain and distributed ledger approaches to trade document authentication.

40.1 Trade Document Fabrication

Trade document fabrication using AI proceeds at three levels of increasing technical sophistication, each targeting a different authentication checkpoint in the customs and trade finance verification process. The choice of level depends on the attacker's technical capability, the value of the shipment, and the robustness of the targeted verification system.

Trade document fabrication using AI proceeds at three levels of sophistication, each requiring a different detection approach.

40.1.1 LLM-Generated Document Text

The simplest level uses LLMs to generate the text content of trade documents: the description of goods on a commercial invoice, the specifications in a packing list, or the narrative sections of inspection certificates. LLM-generated text is detectable using the statistical detection methods of

Chapter 23 when applied to documents with sufficient length. However, trade documents are often short, formulaic, and domain-specific, which reduces the effectiveness of general-purpose LLM detection tools calibrated on more diverse text.

The specific forensic signal available for trade documents is domain-specific stylistic consistency: authentic trade documents follow templates established by international standards bodies (ICC, WCO, UNCITRAL) with specific field layouts, enumeration conventions, and Incoterms terminology. LLM-generated text that deviates from these conventions — using incorrect Incoterms, inconsistent currency notation, or non-standard unit designations — provides detection signal independent of the general LLM statistical fingerprint. A domain-specific classifier trained on authentic and fabricated trade documents from a given commodity type can achieve $AUC > 0.90$ even on short documents where general LLM detectors achieve only 0.60–0.70.

40.1.2 Certificate Seal and Signature Synthesis

The second level synthesises the visual elements of trade documents: the institutional seals, authorised signatures, official stamps, and certification marks that confer authenticity on the document. These elements are targets for image synthesis because they are visible markers of authority that customs officers and trade finance bankers rely on as quick indicators of document genuineness.

Three synthesis approaches are documented in enforcement investigations:

Image inpainting: an authentic document of the same type is obtained, the original seal or signature is inpainted away, and new seal/signature content is inserted using a diffusion inpainting model. This preserves the authentic document’s overall structure while substituting the specific identifying elements.

GAN seal generation: a GAN or diffusion model is trained on a collection of authentic institutional seals to generate novel seals of the same type with altered content (changed dates, quantities, or certification text). This approach requires more technical capability but produces seals that are visually indistinguishable from authentic ones to naive inspection.

LoRA-fine-tuned signature generation: a diffusion model with LoRA adaptation to a specific signatory’s authentic signature samples generates novel signatures for different document content while maintaining the signer’s characteristic pen strokes.

Worked Example — Certificate of origin fabrication detection

A customs intelligence unit receives a digitally submitted certificate of origin purportedly issued by the Chamber of Commerce of a producing country. Forensic analysis:

Forensic test	Result	Verdict
PRNU fingerprint vs. scanner fleet	NCC = 0.004	Fail
Seal frequency-domain analysis	Spectral slope = -1.71	Fail
Signature NCC vs. reference	$\rho = 0.031$	Fail
Document template consistency	3 field deviations	Fail
Overall	4/4 signals flagged	Fabricated

The chamber's scanner PRNU fingerprint (obtained from authenticated documents submitted in the previous six months) does not match the suspect document, indicating it was not produced by the chamber's scanner infrastructure. The seal shows an azimuthal spectral slope of -1.71 versus -2.04 for authentic seals from the same institution. The signature shows near-zero NCC against the authenticated reference signature of the certifying officer. Three template field violations are present including incorrect Incoterms notation and a non-ISO date format. The document is referred to the trade fraud unit for criminal investigation.

40.1.3 False Valuation and Tariff Classification

False valuation — declaring goods at a value lower than their actual transaction value to reduce import duties — and false tariff classification — declaring goods under an HS code with a lower duty rate than the correct code — are the highest-volume trade fraud typologies by both frequency and revenue loss to governments. AI-generated supporting documentation for false valuations includes fabricated supplier invoices showing lower prices than the actual transaction, fabricated freight and insurance certificates supporting artificially low CIF values, and fabricated technical specifications supporting an incorrect HS code classification.

Customs authorities use a combination of valuation databases, price intelligence systems, and transaction comparison to identify suspicious valuations. AI-generated supporting documentation specifically targets the paper trail that investigators follow when a valuation is flagged: a convincing fabricated invoice from the supplier at the declared price, combined with a fabricated bank statement showing payment at the same price, is designed to defeat the initial investigation without requiring the actual supplier to be complicit.

40.2 Customs Authority Detection Capabilities

Customs risk scoring for AI document fraud combines traditional trade risk signals with document-specific forensic indicators in a composite model.

Algorithm 8: AI Trade Document Fraud Risk Scoring**Input:** Declaration δ ;document set $\{D_k\}$;historical trade profile $\mathcal{H}$ **Output:** Composite risk score $R \in [0, 1]$; examination recommendation

```

// Traditional trade risk features
1  $r_{\text{price}} \leftarrow \text{price-anomaly-score}(\delta, \mathcal{H})$ ;
2  $r_{\text{route}} \leftarrow \text{route-risk}(\delta.\text{origin}, \delta.\text{destination})$ ;
3  $r_{\text{trader}} \leftarrow \text{trader-history-score}(\delta.\text{importer}, \mathcal{H})$ ;
4  $R_{\text{trade}} \leftarrow \sigma(w_p r_{\text{price}} + w_r r_{\text{route}} + w_t r_{\text{trader}})$ ;

// Document forensic scores
5 for each document  $D_k \in \{D_k\}$  do
6    $S_{\text{LLM}}^{(k)} \leftarrow \text{llm-text-detector}(D_k.\text{free\_text})$ ;
7    $S_{\text{PRNU}}^{(k)} \leftarrow \text{prnu-ncc}(D_k, \hat{\mathbf{K}}_{\text{authority}}^{(k)})$ ;
8    $S_{\text{tmpl}}^{(k)} \leftarrow \text{template-consistency}(D_k)$ ;
9 end
10  $S_{\text{doc}} \leftarrow \max_k(0.4S_{\text{LLM}}^{(k)} + 0.4(1 - S_{\text{PRNU}}^{(k)}/0.04) + 0.2S_{\text{tmpl}}^{(k)})$ ;

// Composite score
11  $R \leftarrow 0.45R_{\text{trade}} + 0.55S_{\text{doc}}$ ;

12 switch  $R$  do
13   case  $R > 0.80$  do
14     | recommend physical examination (Red channel)
15   end
16   case  $R \in (0.50, 0.80]$  do
17     | recommend document review (Orange channel)
18   end
19   case  $R \leq 0.50$  do
20     | release with monitoring (Green channel)
21   end
22 end
23 return  $R$ 

```

Worked Example — Risk score computation for a suspected AI certificate of origin

A shipment of electronic components, claimed origin Taiwan, destination Germany, with a certificate of origin purportedly issued by the Taiwan Chamber of Commerce.

Traditional risk: $r_{\text{price}} = 0.71$ (unit price 23% below benchmark for this HS code); $r_{\text{route}} = 0.24$ (Taiwan–Germany: well-established low-risk route); $r_{\text{trader}} = 0.58$ (first-time importer, no prior compliance record). $R_{\text{trade}} = \sigma(0.35 \times 0.71 + 0.25 \times 0.24 + 0.40 \times 0.58) = \sigma(0.545) = 0.633$.

Document forensic: $S_{\text{LLM}} = 0.78$ (certificate free-text above LLM threshold); $S_{\text{PRNU}} = 0.006$ (NCC far below Taiwan CoC scanner fingerprint); $S_{\text{tmpl}} = 0.85$ (four template field deviations, each > 0.3 score each). $S_{\text{doc}} = 0.4(0.78) + 0.4(1 - 0.006/0.04) + 0.2(0.85) = 0.312 + 0.338 + 0.170 = 0.820$.

Composite: $R = 0.45(0.633) + 0.55(0.820) = 0.285 + 0.451 = 0.736$. Recommendation: **Orange — document review** (just below the Red threshold of 0.80). The examiner reviews the certificate against

the Taiwan CoC online registry, confirms the certificate number does not match any issued certificate, and escalates to physical examination and trade fraud investigation.

Customs authorities have applied risk-based targeting to document verification for decades: not every shipment can be physically examined, so risk-scoring models are used to select the subset of declarations most likely to contain irregularities. The introduction of AI-generated document fraud requires updating these risk models to include signals specific to AI generation, while maintaining the overall throughput required to avoid disrupting legitimate trade flows.

40.2.1 Risk Scoring for AI Document Fraud

A risk score for AI document fraud combines traditional trade risk signals (unusual origin-destination combinations, suspicious commodity descriptions, price anomalies) with document-specific forensic signals:

$$R_{\text{doc}} = w_1 \cdot R_{\text{trade}} + w_2 \cdot S_{\text{LLM}} + w_3 \cdot S_{\text{image}} + w_4 \cdot S_{\text{template}}, \quad (40.1)$$

where R_{trade} is the traditional trade risk score, S_{LLM} is the LLM text detection score, S_{image} is the image forensic score (PRNU, seal analysis, signature verification), and S_{template} is the template consistency score. The weights $\{w_i\}$ are calibrated on historical seizure data using a logistic regression trained on confirmed fraud and authentic declarations.

At a combined risk threshold selected for 15% physical examination rate (standard for major customs authorities), this composite score achieves a precision of 0.43 for AI-document fraud (43% of high-risk flags contain actual fraud) at a recall of 0.71 (71% of all AI document fraud is caught). The precision is limited by the low base rate of AI document fraud (estimated at 0.3–1.5% of declarations in categories where it has been documented), consistent with the base-rate problem developed in Chapter 11.

40.2.2 The World Customs Organisation Framework

The World Customs Organisation's SAFE Framework of Standards (2023 revision) includes specific guidance on AI-assisted customs risk management and AI-generated document risks. The WCO recommends that member customs authorities: implement electronic single window systems that require structured data submission rather than document image submission where possible (reducing the opportunity for image-level forgery); deploy LLM detection on free-text fields in customs declarations; and establish intelligence-sharing networks for AI-fraud typology alerts through the WCO RILO (Regional Intelligence Liaison Office) system. The WCO's CEN (Customs Enforcement Network) has been expanded to include an AI-fraud module allowing member authorities to share confirmed AI-generated document cases with their identifying forensic signatures.

40.3 Counterpart KYC in Trade Finance

Letters of credit, documentary collection, and bank guarantees are the primary instruments of international trade finance. Each requires verification that the parties to the transaction are who they claim to be, are authorised to enter the transaction, and are not on sanctions lists. For cross-border transactions where parties cannot meet in person, video call KYC has become the standard remote verification tool,

applying the same face-swap and voice-clone vulnerabilities analysed in Chapter 35 to a higher-stakes context: average trade finance transaction values are \$2–20 million, compared to \$280,000 for BEC fraud.

40.3.1 Enhanced Due Diligence for Trade Finance

The Basel Committee on Banking Supervision’s guidance on correspondent banking due diligence requires banks to verify the identity of counterparties to trade finance transactions, including the identity of authorised signatories. For remote KYC in trade finance, enhanced due diligence protocols now include:

Liveness verification: the video call liveness checks described in Chapter 9 (rPPG, blink pattern, lip-sync consistency) applied with a higher confidence threshold reflecting the higher transaction value.

Multi-session verification: the counterparty is required to complete KYC across at least two separately arranged video sessions, reducing the probability that a consistent face-swap can be maintained across independent sessions with different interlocutors.

Document-face matching: the government-issued ID presented during KYC is cross-referenced against the authorised signatory on the trade finance documents, and both are matched to the live video feed using face verification with a threshold calibrated for the transaction value.

Out-of-band verification: independent verification of the counterparty’s identity through their bankers, trade association memberships, and publicly available business registration records, cross-referenced against the KYC video session.

Worked Example — Trade finance KYC risk scoring

A bank receives a request for a \$12 million letter of credit from a new counterparty in a jurisdiction with elevated fraud risk. The KYC risk score incorporates:

Signal	Score	Weight
Liveness rPPG PNR	6.8 dB (above 3 dB threshold)	0.30
Blink IBI KS p -value	0.38 (not significant)	0.15
ID _{cos} (KYC vs. document)	0.89 (above $\tau = 0.80$)	0.30
Out-of-band verification	Business registration confirmed	0.25
Weighted KYC score	$0.30(1) + 0.15(1) + 0.30(1) + 0.25(1) = 1.00$	—

All four verification signals pass at their respective thresholds. The KYC score of 1.00 (full confidence) supports proceeding with the transaction subject to standard sanctions screening. If the rPPG PNR had been 1.8 dB (below the 3 dB liveness threshold), the weighted score would fall to 0.70, triggering an in-person or notarised document verification requirement before the letter of credit is issued.

40.4 Supply Chain Provenance Fraud

Supply chain provenance certification enables consumers and regulators to verify that goods were produced under specified conditions: fair wages, organic inputs, deforestation-free land, conflict-free minerals, or sustainable fishing practices. The market for certified sustainable goods is estimated at \$1.2 trillion annually and growing at 12% per year, creating substantial financial incentives for producers to obtain certifications they do not legitimately qualify for. AI-generated false provenance documentation targets this certification ecosystem.

40.4.1 Agricultural Certification Fraud

Organic certification, fair trade certification, and geographical indication certificates (e.g., Champagne, Darjeeling, Jamaican Blue Mountain Coffee) are particularly targeted because the price premium they command — typically 30–200% above non-certified equivalents — creates a large financial incentive for falsification. AI-enabled agricultural certification fraud takes two forms: (1) LLM-generated inspection reports and field audit documentation supporting a false certification claim, and (2) AI-modified satellite imagery showing compliant land use (e.g., forest cover consistent with a deforestation-free claim) where the actual land shows non-compliant use.

Satellite imagery verification is an increasingly important tool for agricultural certification bodies: services including Satelligence, Rainforest Alliance’s mapping tools, and Global Forest Watch provide authenticated satellite time-series data that certification bodies cross-reference against applicant claims. AI-generated satellite imagery fabrication — inserting false forest cover into non-compliant land parcels — requires the same spatial PRNU composite detection methods described in Chapter 38, applied to multispectral satellite imagery rather than visible-light photographs.

40.4.2 Mineral and Manufactured Good Provenance

Conflict-mineral compliance (Section 1502 of the US Dodd-Frank Act, EU Conflict Minerals Regulation) requires companies to conduct due diligence on the source of tin, tantalum, tungsten, and gold (3TG minerals) in their products. The documentation chain from mine to manufacturer to product is complex, involving smelter audit reports, chain-of-custody certificates, and third-party audit certification. AI-generated documentation fraud targeting this chain can insert false smelter certifications or alter the country of origin in chain-of-custody documents, potentially enabling conflict minerals to pass into compliant supply chains under false documentation.

The Responsible Minerals Initiative and the OECD Guidance on Responsible Mineral Supply Chains both emphasise document authentication as a key due diligence element. The introduction of blockchain-based mineral tracking systems (including the Responsible Minerals Assurance Process’s digital tracking tools) provides a technical countermeasure to document-level fraud by anchoring provenance claims to immutable blockchain records that cannot be retroactively falsified.

40.5 Blockchain and Distributed Ledger Authentication

Blockchain and distributed ledger technology provide a document authentication mechanism that is structurally resistant to the post-hoc fabrication that AI tools enable: once a document's hash is recorded on an immutable distributed ledger, any subsequent modification to the document produces a hash mismatch that is immediately detectable. This property makes blockchain-anchored trade document systems an effective countermeasure to AI-generated document fraud, though their adoption faces significant infrastructure and interoperability challenges.

40.5.1 Trade Finance Blockchain Platforms

Several major trade finance blockchain platforms have reached commercial deployment. Contour (formerly Voltron), operating on the R3 Corda distributed ledger, processes letters of credit between 15 major international banks, with the letter of credit hash recorded at each stage of the document lifecycle. Marco Polo, built on the R3 Corda network, covers accounts receivable finance and supply chain finance. The Monetary Authority of Singapore's Project Contour demonstrated 90% reduction in letter of credit processing time alongside elimination of document tampering risk.

The WTO's Aid for Trade programme and the Asian Development Bank have funded blockchain trade document pilots in Cambodia, Vietnam, and Bangladesh targeting specifically the LDC (Least Developed Country) preferential tariff certificate market — a high-fraud target because LDC certificates enable significant duty savings that create large incentives for falsification.

40.5.2 Document Hash Anchoring

Document hash anchoring records the SHA-256 hash of an authenticated trade document on a public or permissioned blockchain at the time of issuance, enabling any subsequent presentation of the document to be verified by recomputing the hash and querying the blockchain:

$$h = \text{SHA-256}(\text{document}); \quad \text{verify: } h \stackrel{?}{=} \text{blockchain}[t_{\text{issue}}], \quad (40.2)$$

where t_{issue} is the timestamp of the original issuance. Any AI modification of the document after issuance — adding a quantity, changing the certificate date, altering the description of goods — changes the document hash and fails the verification. The Ethereum blockchain's average transaction cost of approximately \$0.50–\$5 per hash registration makes per-document anchoring economically feasible for high-value trade documents (letters of credit, certificates of origin) but potentially cost-prohibitive for individual commercial invoices in low-value trade.

40.6 Industry Applications

The trade document integrity market encompasses customs authorities, trade finance banks, certification bodies, and technology providers. The World Customs Organisation's CEN online portal, used by 183 member customs authorities, has added an AI-generated document fraud module following a 2023 WCO technical study on AI risks in trade facilitation. Major customs authorities including CBP

(US), HMRC (UK), European customs authorities, and the China Customs Authority have deployed or piloted AI document fraud detection.

In trade finance, SWIFT's KYC Registry — used by 5,000 member financial institutions — has expanded to include AI-generated identity fraud as a tracked risk category. The International Chamber of Commerce's Digital Trade Standards Initiative has published technical specifications for AI-resistant digital trade documents under the ICC eRules framework, incorporating C2PA-style provenance requirements for digitally signed trade documents.

Supply chain provenance verification has attracted significant venture investment: Sourcemap (supply chain mapping), Circulator (blockchain mineral tracking), and Sourcetrace (agricultural supply chain) collectively raised over \$200 million in 2022–2024 to build AI-resistant provenance systems for supply chains. The EU's Digital Product Passport requirement (coming into force from 2026 under the Ecodesign for Sustainable Products Regulation) will require a machine-readable record of product origin, materials, and environmental impact for all products sold in the EU — creating a mandatory provenance infrastructure that, if properly designed, will be resistant to the AI document fraud described in this chapter.

40.7 African Use Cases

Africa's trade document fraud landscape is shaped by three structural factors: the continent's complex system of regional trade agreements creating significant tariff differentials that reward certificate-of-origin fraud; the limited digital forensic capacity of most African customs authorities; and the large informal trade sector that provides cover for sophisticated document fraud operations.

The African Continental Free Trade Area (AfCFTA), which entered implementation in 2021, requires certificates of origin to claim preferential tariff rates among member states. The AU Trade Observatory has documented widespread fabrication of certificates of origin claiming AfCFTA preferential status for goods that do not meet the rules-of-origin requirements, with AI-generated certificates increasingly substituting for the physical document fraud that previously characterised this typology. The certificate verification systems deployed by most African customs authorities remain paper-based or use basic QR code verification that does not detect AI-generated document content.

The East Africa Customs and Trade Association (EACTA) and the West Africa Customs Commission (WACC) have both identified AI-generated document fraud as a priority emerging threat and have requested technical assistance from the WCO for training African customs officers in AI document forensics. The SACU (Southern African Customs Union) technical committee is piloting a blockchain-anchored certificate of origin system for SACU member states (South Africa, Namibia, Botswana, Lesotho, Eswatini) that would eliminate the document-level fabrication risk for intra-SACU trade.

The mining sector in the DRC, Zambia, Tanzania, and Zimbabwe generates significant 3TG mineral documentation whose integrity is critical for compliance with Dodd-Frank Section 1502 and the EU Conflict Minerals Regulation. The International Tin Association's iTSCi programme and the

Responsible Minerals Initiative's RMAP operate document verification systems in these countries, but the programmes cover less than 30% of mineral production volume in the affected countries, leaving the majority of production documentation outside verified chains. AI-generated false compliance documentation for the unverified majority is a documented and growing problem that the expansion of blockchain tracking systems is intended to address.

Chapter Summary

International trade is one of the most document-intensive human activities, and every trade document that can be fabricated represents a potential attack vector for AI-enabled fraud. The taxonomy of AI trade document fraud spans LLM-generated document text, image synthesis of seals and signatures, false valuation documentation, and composite fabrication of satellite imagery for agricultural provenance. Detection applies a composite risk score combining traditional trade risk signals with LLM text detection, image forensic analysis (PRNU, spectral fingerprint, signature NCC), and template consistency, achieving 71% recall at 43% precision at a 15% physical examination rate consistent with major customs authority operational targets.

Counterpart KYC in trade finance applies the same face-swap and voice-clone detection tools as domestic financial KYC but at higher thresholds reflecting transaction values of \$2–20 million, combined with multi-session verification and out-of-band confirmation. Supply chain provenance fraud targets the \$1.2 trillion certified sustainable goods market, with agricultural certification and 3TG mineral documentation the primary targets. Blockchain-anchored document hash systems provide the most structurally robust defence against post-issuance AI modification: SHA-256 hash anchoring on an immutable distributed ledger means any AI alteration produces a detectable hash mismatch regardless of the visual quality of the fabricated document.

Exercises

1. A certificate of origin submitted electronically to a customs authority shows the following forensic results: PRNU NCC = 0.006 against the issuing chamber's scanner fingerprint ($\tau = 0.015$); azimuthal spectral slope = -1.69 versus -2.01 for authentic certificates from that chamber; signature NCC = 0.028 against the certified officer's reference ($\tau = 0.025$); and 2 template field deviations out of 15 checked fields ($\tau = 3$ deviations for flag). (a) Apply the composite risk score $R = 0.25 \cdot R_{\text{trade}} + 0.30 \cdot S_{\text{PRNU}} + 0.25 \cdot S_{\text{spectral}} + 0.20 \cdot S_{\text{sig}}$, where each S is 1.0 if failed and 0.0 if passed. What is R if $R_{\text{trade}} = 0.65$? (b) At a referral threshold of $R > 0.50$, is this certificate referred? (c) The importer argues that the signature NCC result could be explained by the certifying officer using a new pen causing variation. Evaluate this argument and propose an additional forensic test to distinguish pen variation from synthetic signature.
2. An agricultural exporter applies for organic certification using inspection reports generated by an LLM and field audit documentation containing AI-modified satellite imagery. The satellite

images purportedly show compliant organic land management. (a) Describe three specific forensic tests applicable to the satellite imagery (you may adapt the methods from Chapters 12 and 38). (b) If the LLM text detection applied to the inspection report achieves $AUC = 0.73$ on this document type (calibrated on short domain-specific documents), and the prior probability of fabrication for this applicant type is 0.08, what posterior probability of fabrication results from a detection score of 0.82 on the 50th percentile of the authentic distribution? (c) What additional physical investigation would be triggered by this posterior probability?

3. Compare the document authentication guarantees provided by (a) a qualified electronic signature under the EU eIDAS Regulation, (b) a C2PA cryptographic manifest, and (c) a blockchain-anchored SHA-256 hash. For each: describe what the mechanism proves, what it does not prove, and what AI-based attacks it is resistant to and vulnerable to. Then propose an architecture that combines all three to provide the strongest possible authentication guarantee for a high-value letter of credit.

Answers to Exercises

1. Certificate of origin forensic analysis and composite risk scoring.

(a) Signal scoring (1.0 = failed, 0.0 = passed): S_{PRNU} : $NCC = 0.006 < \tau = 0.015 \rightarrow \text{Failed} = 1.0$. $S_{spectral}$: slope = -1.69 , deviation = 0.32 from $-2.01 \rightarrow \text{Failed} = 1.0$. S_{sig} : $NCC = 0.028 > \tau = 0.025 \rightarrow \text{Passed} = 0.0$ (marginally above threshold). $S_{template}$: 2 deviations $< \tau = 3 \rightarrow \text{Passed} = 0.0$.

Composite risk score: $R = 0.25 \times 0.65 + 0.30 \times 1.0 + 0.25 \times 1.0 + 0.20 \times 0.0 = 0.1625 + 0.30 + 0.25 + 0 = 0.7125$.

(b) $R = 0.71 > 0.50$ threshold. **Certificate is referred** for physical examination and investigation.

(c) Evaluation of the pen variation argument: Pen variation affects the visible stroke characteristics of a handwritten signature but not its PRNU properties. The signature NCC test measures the match between the noise residual of the suspect signature image and the PRNU fingerprint of the scanner/camera used to capture the authenticated reference signatures. A genuinely handwritten signature variation (different pen, different signing speed) would still produce a high NCC if scanned by the same chamber scanner — the NCC measures device fingerprint, not signature style.

However, the NCC of 0.028 is only marginally above the threshold of 0.025, so caution is warranted. Additional forensic tests to distinguish pen variation from synthesis:

Test 1 — Stroke pressure and ink distribution: authentic handwritten signatures show characteristic ink distribution patterns (heavier strokes at pen-down and pen-up, lighter strokes in fast-moving sections) detectable in the micro-texture of the ink layer at high resolution.

AI-generated signatures produce uniform greyscale values within stroke boundaries, lacking the pressure variation signature.

Test 2 — Request additional authenticated signatures: obtain three additional certified documents signed by the same officer in the same period. Compute NCC of each against the reference. If the suspect signature is a genuine variant, it should show NCC comparable to the other authentic signatures. If it is synthetic, its NCC will be an outlier in the distribution.

2. Satellite imagery forgery and LLM inspection report detection.

(a) Three forensic tests for satellite imagery:

Test 1 — Multispectral PRNU analysis: satellite sensors have characteristic noise patterns analogous to camera PRNU, determined by the specific sensor array of the satellite platform. Authenticated images from Sentinel-2, Landsat 9, or Planet Labs carry these sensor signatures. AI-generated satellite images lack them. The composite detection method of Chapter 38 applies directly: block-wise NCC between the sensor fingerprint (from authenticated imagery of the same tile) and the submitted image reveals inserted synthetic regions as near-zero NCC blocks.

Test 2 — Temporal consistency with historical archive: genuine satellite imagery of a specific geographic location has an archive of historical images at the same coordinates. A plot that claims to be managed organically for three years should show consistent vegetation index (NDVI) values in the historical archive consistent with that management practice. AI-generated imagery inserted to show false land use will be inconsistent with the archive time-series. Cross-referencing submitted imagery against the Copernicus Sentinel-2 open archive or Google Earth Engine is freely accessible and takes minutes.

Test 3 — Spectral band ratio analysis: genuine satellite imagery has mathematically constrained relationships between spectral bands (e.g., the NDVI from near-infrared and red bands must be consistent with the RGB colour appearance, and both must be consistent with the claimed season and location latitude). AI image generators produce imagery calibrated for visual appearance but may violate these multispectral relationships, detectable by computing derived indices and cross-checking for physical consistency.

(b) Bayesian posterior probability of fabrication. Prior $P(\text{fab}) = 0.08$. $\text{AUC} = 0.73$ for the detector. A detection score of 0.82 on the 50th percentile of the authentic distribution: the LR (likelihood ratio) for a score of 0.82 given fabricated vs. authentic requires estimating the score distributions. Assuming Gaussian score distributions calibrated to $\text{AUC} = 0.73$: the separation $d' \approx 2\Phi^{-1}(0.73) = 2 \times 0.613 = 1.226$. At score = 0.82 (far into the tail of the authentic distribution), a rough LR estimate is 3–5 (fabricated documents score here 3–5× more often than authentic ones). Using LR = 4: posterior $P(\text{fab}|\text{score} = 0.82) = \frac{0.08 \times 4}{0.08 \times 4 + 0.92 \times 1} = \frac{0.32}{0.32 + 0.92} = 0.258$.

A posterior of 26% is above the prior (8%) but does not constitute strong evidence of fabrication alone. This is consistent with the short-document limitation of LLM detection.

(c) Physical investigation triggered at this posterior probability: the appropriate response at 26% posterior is to require physical corroboration, not to reject the certification outright. The certification body would: (1) request an unannounced physical site inspection of the claimed organic plots; (2) collect soil samples for laboratory analysis of pesticide residues inconsistent with organic management; (3) interview workers to verify management practices; (4) cross-reference the claim with satellite archive (Test 2 above). These physical checks have near-certain probability of resolving the uncertainty in either direction.

3. Comparison of three document authentication mechanisms.

(a) *Qualified electronic signature (EU eIDAS)*: proves that the document was signed by a specific natural person or organisation holding a qualified certificate from a Trust Service Provider, and that the document has not been modified since signing (integrity). Does NOT prove: that the content of the document is truthful, that the underlying transaction is genuine, or that the signing key has not been compromised. AI attack resistance: resistant to AI modification of the document text or images after signing (hash mismatch). Vulnerable to: identity theft to obtain a qualified certificate under false pretences; social engineering to induce a genuine keyholder to sign a fabricated document.

(b) *C2PA cryptographic manifest*: proves that the content was created using a specific C2PA-compliant tool by a device holding a C2PA certificate, and records any declared AI involvement in creation. Does NOT prove: that the declared content is factually accurate, or that the creating device was not compromised. AI attack resistance: resistant to post-creation AI modification (hash mismatch). Records AI tool use in the manifest, enabling disclosure verification. Vulnerable to: an attacker who has access to a valid C2PA-signing device and uses it to sign AI-generated content that is declared as human-created; certificate compromise.

(c) *Blockchain SHA-256 hash anchoring*: proves that the document existed in its current form at the time the hash was recorded on the blockchain, and that it has not been modified since. Does NOT prove: who created the document, whether the content is truthful, or when the document was originally created (only when the hash was recorded). AI attack resistance: maximally resistant to post-anchoring AI modification (deterministic hash mismatch for any content change). Vulnerable to: an attacker who anchors an AI-fabricated document before verification (the blockchain proves the document existed as fabricated, not that it is authentic).

Combined architecture for a high-value letter of credit: (1) The issuing bank signs the L/C with a qualified eIDAS electronic signature, establishing bank identity and document integrity. (2) The signed document is processed through a C2PA-compliant workflow that records the bank's internal document generation tools and any AI assistance in drafting, with this C2PA manifest embedded in the document. (3) The SHA-256 hash of the eIDAS-signed, C2PA-manifest-containing document is anchored on a permissioned trade finance blockchain (e.g., Contour) at issuance, creating an immutable timestamp. (4) Any subsequent presentation of the document can be verified by: recomputing the hash (blockchain verification), checking the eIDAS signature (identity and integrity), and inspecting the C2PA manifest (AI disclosure).

This triple-layer architecture closes all three vulnerability gaps identified above: the eIDAS signature proves issuer identity, C2PA proves creation context, and blockchain anchoring makes any post-issuance modification immediately detectable regardless of how sophisticated the AI modification tools become.

Chapter 41

Deep Fakes of Non-Human Objects, Products, and Physical Scenes

“We trained a model to imagine the product. Then we discovered someone trained a model to counterfeit our certification.”

—Anon.

The deep-fake research literature has focused overwhelmingly on human faces and voices, for good reasons: identity impersonation is the most common and most legally consequential form of synthetic media harm. But generative AI can synthesise any visual or physical scene — damaged property, certified minerals, product quality samples, archaeological artefacts, pharmaceutical packaging, architectural surveys — and the adversarial applications of object-level synthesis are growing rapidly and have received far less defensive attention than face synthesis. This chapter examines the fabrication of non-human objects and scenes across four domains where the forensic, legal, and economic consequences are most significant: product counterfeiting and quality fraud, mineral and commodity authentication, real estate and property scene fabrication, and scientific and archaeological artefact forgery.

Object-level deep fakes differ structurally from face deep fakes in several important ways. A face deep fake must replicate the specific geometry, texture, and dynamics of a particular real person, creating a forensic reference point for comparison. An object deep fake is often not compared against a specific authentic reference: the claim is simply that “this photograph shows a product with certain properties” or “this scan shows a mineral with certain characteristics”, and the fraud is that it does not. The absence of a specific authentic reference removes some of the strongest forensic tools — identity cosine similarity, speaker verification, PRNU against a known device fingerprint — and requires forensic analysis based on physical plausibility, material science consistency, and statistical properties of the synthesis method.

The economic stakes are substantial. The global trade in counterfeit goods is estimated at \$464 billion annually by the OECD, representing approximately 2.5% of world trade. AI-enabled product

photography fraud — submitting synthesised images of products that do not match the physical goods supplied — inflates this figure by lowering the skill threshold for sophisticated image-level counterfeiting. Mineral authentication fraud costs producing countries and legitimate miners an estimated \$10–20 billion annually through diamond, gold, and gemstone misclassification. Real estate photography fraud, in which AI-generated or AI-enhanced property images misrepresent the condition of properties being sold or rented, is now the most common AI fraud complaint category received by UK and Australian consumer protection bodies.

The forensic toolkit for object deep fakes draws on the physical science of the materials and scenes being synthesised. Steel cannot simultaneously show fracture patterns consistent with high-speed impact and surface oxidation consistent with ten years of exposure. Diamonds cannot simultaneously show inclusions characteristic of one geological origin and crystal habit characteristic of another. A property cannot simultaneously show a freshly painted interior and original 1960s skirting boards in undisturbed condition. Generative AI models trained on image datasets learn the statistical appearance of these combinations but not their physical impossibility, creating systematic forensic vulnerabilities that physical science expertise can exploit.

This chapter develops the technical mechanisms, forensic detection approaches, and operational defensive strategies for non-human object deep fakes across four major domains. Section 41.1 develops product counterfeiting and quality fraud. Section 41.2 develops mineral and gemstone authentication. Section 41.3 develops real estate scene fabrication. Section 41.4 develops scientific and archaeological artefact forgery. Section 41.5 synthesises the cross-domain forensic toolkit.

41.1 Product Counterfeiting and Quality Fraud

Product counterfeiting using AI-generated imagery operates at two distinct levels. The first is *listing fraud*: submitting AI-generated or AI-enhanced product photographs to e-commerce marketplaces that depict goods of higher quality, different specifications, or better condition than the goods actually supplied. The second is *certification fraud*: fabricating quality control documentation, test reports, and inspection certificates to support claims about product specifications that the goods do not meet. Both levels have been significantly enabled by generative AI, which can produce photorealistic images of products at quality levels far exceeding what unauthorised resellers could previously achieve with consumer photography.

41.1.1 E-Commerce Listing Image Fraud

E-commerce platforms process billions of product listing images annually. Product image fraud — the submission of misleading images to represent goods being sold — is not new, but AI generation has qualitatively changed its scale and sophistication. The fraud taxonomy for e-commerce listing images includes:

Grade substitution: an AI-generated image of a premium product is used to list a lower-grade item. Examples include: food products photographed at optimal ripeness condition but supplied in degraded condition; electronics shown with premium finishes that the actual product lacks; textiles shown with

thread counts or fabric weights that the supplied goods do not have.

Specification fabrication: AI-generated images of product labels, specifications panels, or packaging are used to misrepresent technical specifications. A motor component is listed with an AI-generated specification sticker showing a higher load rating than the actual component possesses. A pharmaceutical is listed with an AI-generated batch number and expiry date that do not correspond to actual manufacturing records.

Counterfeit brand representation: AI-generated product images faithfully replicate the appearance of branded products (exact brand colours, logo positioning, packaging design) while the actual goods supplied are unbranded or counterfeit. The AI image serves as the product listing photograph on the marketplace, and buyers receive inferior goods that they discover do not match the listing only after delivery.

Worked Example — AI product image forensic analysis

An e-commerce platform’s trust and safety team reviews a disputed listing for “Grade A Wagyu beef” at a premium price point. The listing photographs are referred for forensic analysis.

Test	Result	Verdict
PRNU device consistency	NCC = 0.003 vs. claimed iPhone 14	Fail
Marbling pattern physics	AI pattern inconsistency score 0.78	Fail
Label text LLM detection	Score = 0.84 (above $\tau = 0.65$)	Fail
Colour space histogram	$\chi^2 = 412$ vs. authentic meat images	Fail
Reverse image search	3 prior listings on different platforms	Fail
Overall	5/5 signals flagged	Fabricated

The marbling pattern inconsistency score of 0.78 reflects that the fat distribution in the submitted image follows a statistically improbable pattern: AI-generated meat images tend to produce overly uniform, aesthetically pleasing marbling that lacks the biological variation of authentic Wagyu. The colour space histogram of the meat image differs significantly from authentic photographs of Wagyu beef taken under comparable lighting, because the AI generator was optimised for visual appeal rather than spectral accuracy. The label text scores above the LLM detection threshold, consistent with generated rather than photographed text.

41.1.2 Material Science Plausibility Analysis

The physical properties of materials constrain their visual appearance in ways that AI image generators frequently violate. Material science plausibility analysis applies domain knowledge to identify these violations:

Fracture mechanics: metals, ceramics, and polymers each have characteristic fracture morphologies determined by their crystalline structure, ductility, and strain rate at failure. A brittle fracture surface (aluminium oxide ceramic) must show the intergranular crack propagation pattern characteristic of that material; a ductile fracture surface (mild steel) must show the cup-and-cone fibrous morphology with

neck reduction. AI-generated fracture surfaces sometimes produce visually plausible but physically impossible combinations.

Surface oxidation and weathering: oxidation progresses from surface to interior in patterns determined by diffusion kinetics. A claimed “ten-year-old” steel component must show oxide scaling consistent with atmospheric exposure duration; a claimed “newly manufactured” component must lack such scaling. AI generators produce surface textures that appear aged or new without necessarily satisfying the underlying physical kinetics.

Weld quality: welded joints have characteristic microstructural zones (weld metal, heat-affected zone, base metal) visible in cross-section and influencing the surface appearance near the weld. AI-generated welded joints sometimes show surface continuity inconsistent with the thermal profile of the claimed welding process.

41.1.3 Pharmaceutical Product Authentication

Pharmaceutical counterfeiting is among the most harmful applications of product image fraud: counterfeit medicines that pass visual inspection but lack active pharmaceutical ingredients cause direct patient harm through treatment failure, and counterfeit medicines with incorrect dosages cause harm through over- or under-dosing. AI-generated pharmaceutical packaging images are used in two ways: to create convincing online pharmacy listings for counterfeit medicines, and to fabricate the quality control documentation required by importers and regulators.

Authentication of pharmaceutical products relies on serialisation: every legitimate pharmaceutical pack in the EU (Falsified Medicines Directive), the US (Drug Supply Chain Security Act), and 50+ other regulatory systems carries a unique serialisation code that is registered in a national or regional repository at the time of manufacture. An AI-generated pharmaceutical image that includes a valid-format but unregistered serialisation code fails verification against the repository, and an image that replicates an authentic code is detectable as a duplicate when the same code is presented by multiple parties. The challenge is ensuring that verification is performed before goods are dispensed rather than after harm has occurred.

41.2 Mineral, Gemstone, and Commodity Authentication

The mineral and gemstone trade is one of the oldest and most sophisticated fraud arenas in the world economy. Gem-quality diamonds, rubies, emeralds, sapphires, and the 3TG minerals examined in Chapter 40 are the most economically significant targets. The arrival of AI image synthesis has added a new dimension to authentication challenges that previously focused on physical testing, spectroscopic analysis, and gemological certification.

41.2.1 Diamond and Gemstone Visual Fraud

Gemstone visual fraud using AI operates at three levels of increasing sophistication. At the first level, AI-generated images of high-quality gemstones are used in online listings to sell lower-quality goods. At the second level, AI-enhanced images of genuine gemstones are used to improve their apparent

clarity, colour, or size beyond their actual characteristics. At the third level — and most technically sophisticated — AI inpainting is used to modify grading report scans, inserting higher clarity grades or removing noted inclusions from authentic gemological laboratory certificates.

Worked Example — Gemstone grading report tampering detection

A gemological laboratory receives a digital scan of a diamond grading report purportedly issued by the Gemological Institute of America (GIA), supporting a claimed D-colour, IF-clarity stone. Forensic analysis of the scanned report:

The GIA stores SHA-256 hashes of all issued reports in the GIA Report Check database. The hash of the submitted scan does not match the GIA record for the reported certificate number — the scan has been modified after issuance. Spatial PRNU composite detection reveals two regions of near-zero NCC against the GIA scanner fingerprint: the colour grade field and the clarity grade field, each showing $\rho \approx 0.004$ versus $\rho \approx 0.037$ for the surrounding text. The original report recorded G-colour (not D) and VS1-clarity (not IF) — a fabrication that would have supported a price premium of approximately 85% above the authentic stone's market value.

41.2.2 Spectroscopic Authentication

Physical authentication of minerals and gemstones that cannot be fabricated through image synthesis relies on spectroscopic analysis: the characteristic absorption and emission spectra of each mineral are determined by its crystal structure, trace element composition, and geological history. A genuine Burmese ruby has a characteristic chromium emission spectrum distinct from Thai or African rubies. A natural diamond has a nitrogen aggregation pattern distinct from synthetic diamonds. A certified “conflict-free” coltan from Rwanda has a trace element fingerprint consistent with that geological province's known profile.

Raman spectroscopy, X-ray fluorescence (XRF), and laser ablation inductively coupled plasma mass spectrometry (LA-ICP-MS) provide these fingerprints. AI image synthesis cannot fabricate the spectroscopic signature of a physical sample: the sample must be present for testing. The fraud that AI enables is therefore primarily at the document layer — fabricating the certificate that records the spectroscopic results — rather than at the physical material layer. Detection of certificate fabrication applies the forensic document methods of Chapter 40, combined with direct cross-referencing against the certifying laboratory's digital records.

41.2.3 Gold and Precious Metal Assay Fraud

Gold bars, coins, and ingots are authenticated by hallmarks (assay marks stamped by licensed assay offices) and, for investment-grade bars, by serial numbers registered with the London Bullion Market Association (LBMA) and other regulatory bodies. AI-image-synthesis fraud in this market takes two forms: (1) creating plausible images of high-purity gold bars for use in advance-fee fraud schemes where buyers are shown “proof of product” photographs and asked to pay transfer fees before the non-existent gold is delivered; and (2) fabricating LBMA-compliant assay certificates for tungsten-filled gold bars, which pass visual and basic weight testing but fail X-ray fluorescence or ultrasound

testing.

The detection of AI-generated gold bar images relies on the physical science of gold's optical properties: gold has a characteristic colour reflectance spectrum with a sharp absorption edge at approximately 480 nm (the cause of its yellow colour) and a high infrared reflectance. AI-generated gold images must match this spectral profile to appear authentic to spectroscopic analysis, which most do not, as they are trained on RGB images rather than physically calibrated spectral data.

41.3 Real Estate Scene Fabrication and Property Fraud

Real estate is the largest asset class in the world, with global residential real estate alone valued at approximately \$258 trillion. The combination of high per-transaction values, the impossibility for remote buyers to physically inspect properties, and the reliance on photographic representation in listings and surveys has made real estate one of the highest-impact targets for AI scene fabrication. UK Trading Standards and consumer protection bodies received substantial AI-related fraud complaints about real estate in 2023, and the Australian Competition and Consumer Commission's Scamwatch identified AI-enhanced property images as the fastest-growing real estate fraud typology in the same year.

41.3.1 Taxonomy of AI Real Estate Fraud

AI-enabled real estate fraud spans five distinct attack types:

Interior enhancement: AI image generation or enhancement is used to replace damaged, stained, or outdated interior elements — flooring, walls, ceilings, kitchens, bathrooms — with pristine or renovated versions. This is the most common form, enabled by consumer tools including Adobe Firefly's "Generative Fill" and similar inpainting capabilities that allow any room element to be replaced with a generated alternative. The legal threshold is contested: some staging and digital enhancement is standard industry practice, but replacing genuine damaged elements with synthesised undamaged versions when the property is listed for sale constitutes misrepresentation in most jurisdictions.

Exterior environment fabrication: AI-generated backgrounds replace the actual neighbourhood environment visible from the property. A property on a busy road is presented with a quiet residential streetscape; a property adjacent to an industrial site is presented with parkland. The surrounding environment is material to property value and is specifically misrepresented.

View fabrication: AI-generated distant views (sea views, mountain views, city skylines) are added to properties that do not have such views, or are generated to replace the actual view (a motorway, another building, a carpark) with an appealing alternative.

Structural defect concealment: AI inpainting is used to conceal structural defects visible in the photographs — cracks in walls, subsidence evidence, mould staining, roof damage — that are material disclosure obligations in most jurisdictions.

Phantom property fraud: AI scene generation creates plausible-looking property interiors and exteriors

for properties that do not exist (fraudulent rental listings) or for properties that are not owned by the advertiser (hijacked legitimate property listings).

41.3.2 Physical Plausibility Analysis for Real Estate

Interior real estate images must satisfy architectural physics that AI generators frequently violate:

Shadow geometry: sunlight entering through a window at a known time and compass orientation must cast shadows on interior surfaces at angles consistent with the sun’s position. AI-generated interiors often have lighting that is inconsistent with the stated orientation or time, or feature multiple shadow directions inconsistent with a single light source.

Perspective consistency: all parallel lines in a room must converge to the same vanishing point. AI-generated room images sometimes show perspective inconsistencies at the junctions of walls, floors, and ceilings, particularly in corner regions where three surfaces meet.

Material aging consistency: authentic interiors show consistent material aging across surfaces: similarly aged wood flooring shows similar wear patterns; similarly aged plaster shows similar hairline cracking; similarly aged grout shows similar discolouration. AI-generated interiors sometimes combine elements with inconsistent apparent ages — pristine grout adjacent to aged tile, or fresh paint adjacent to visibly aged woodwork — because the inpainting model optimises for visual appeal in the replaced region without considering consistency with the surrounding authentic material.

Scale consistency: furniture dimensions must be consistent with room dimensions and with each other. AI-generated or AI-enhanced rooms sometimes contain furniture at implausible scale relative to doors, windows, or other architectural elements whose dimensions are constrained by building regulations.

Worked Example — Real estate interior image forensic analysis

A property listing is referred to a trading standards officer following a buyer’s complaint. The listing shows a pristine kitchen that the buyer claims does not match the property’s actual condition at sale completion.

Test	Result	Verdict
Shadow geometry (window to surface)	23° deviation from astronomical	Fail
Perspective vanishing point	2.1 cm offset at 4m distance	Fail
PRNU (listing vs. estate agent camera)	NCC = 0.041	Pass
AI detection classifier	Score = 0.87 (above $\tau = 0.65$)	Fail
Material aging consistency	Kitchen vs. hallway: $\Delta\text{age} = 28\text{ yr}$	Fail
Overall	4/5 signals flagged	Fabricated interior

The PRNU passes because the image was captured by the estate agent’s genuine camera and then the kitchen region was AI-inpainted: the authentic portions of the frame (hallway visible through

doorway) carry the camera fingerprint, but the inpainted kitchen region does not. The spatial PRNU composite detection reveals the kitchen region as a synthetic insert: block-wise NCC falls to 0.003–0.008 within the kitchen area versus 0.038–0.043 in the hallway and ceiling regions. The material aging comparison shows the kitchen appearing approximately 28 years younger than the adjacent hallway skirting boards, which are visibly original to the 1962 construction date.

41.3.3 Mortgage and Valuation Fraud

Property valuation for mortgage purposes relies on photographs of the property’s condition at the time of inspection. AI-enhanced property photographs that conceal structural defects or misrepresent the property’s condition constitute mortgage fraud when submitted to a lender as part of a valuation report. The harm falls on the lender (who has advanced funds secured against a misrepresented asset) and on subsequent buyers (who may purchase the property without full knowledge of its condition). In the UK, mortgage fraud carries a maximum sentence of ten years imprisonment under the Fraud Act 2006, and the introduction of AI-enhanced listing photographs has prompted the Financial Conduct Authority to issue guidance specifically addressing AI-generated property documentation.

41.4 Scientific and Archaeological Artefact Forgery

Scientific datasets and archaeological artefacts occupy a different position in the object deep-fake landscape: the harm from their fabrication is epistemic rather than directly financial, corrupting the evidence base on which scientific knowledge and cultural heritage understanding is built. However, fabricated scientific data and archaeological artefacts also have significant direct financial consequences: research grant fraud, insurance claims for “stolen” artefacts that never existed, and the distortion of scientific literature that misdirects subsequent research investment.

41.4.1 Scientific Image Fabrication in Research

Scientific image fabrication — the manipulation or synthesis of data images in research publications — predates generative AI, but AI tools have dramatically lowered the skill threshold for sophisticated image manipulation. Microscopy images, gel electrophoresis results, Western blot images, histopathology slides, and astronomical survey images are the most commonly fabricated research image types. The Office of Research Integrity (ORI) in the United States processes approximately 200–300 research misconduct allegations annually, of which image manipulation is the most common category.

AI-generated microscopy images present a specific forensic challenge because microscopy images have strong spatial statistics determined by the optical physics of the imaging system (point spread function, diffraction limits, coherence effects) and by the biology of the imaged sample (cell size distributions, organelle density, staining patterns). AI generators trained on image datasets can reproduce the statistical appearance of microscopy images but often fail to replicate the specific optical physics of the imaging system, creating a forensic signal in the spatial frequency domain that differs from authentic microscopy images of the same type.

Worked Example — Microscopy image fabrication detection

A journal editor receives a tip that a submitted paper contains AI-generated confocal microscopy images. Analysis:

Authentic confocal images have a characteristic PSF-limited spatial resolution that produces a specific frequency cutoff in the power spectrum at the diffraction limit $f_{\max} = 2NA/\lambda$. For the stated objective ($NA = 1.4$, $\lambda = 488$ nm): $f_{\max} = 5.74$ cycles/ μm . Authentic images show zero power above this cutoff; AI-generated images show residual power at super-resolution frequencies (5.74–12 cycles/ μm), detectable in the radially averaged power spectrum:

Frequency range	Authentic image	Submitted image
0–5 cycles/ μm	1.0 (normalised)	1.03
5–5.74 cycles/ μm	0.31	0.29
5.74–8 cycles/ μm	< 0.001	0.047
8–12 cycles/ μm	< 0.001	0.021

The submitted images show significant power above the diffraction limit, physically impossible for authentic confocal microscopy. The AI generator has produced super-resolution detail that no real confocal microscope could capture, exposing the fabrication through optical physics.

41.4.2 Archaeological Artefact Fabrication

Archaeological artefacts present a specific fabrication scenario: AI-generated or AI-enhanced photographs of artefacts are used in insurance claims (for “stolen” or “destroyed” items that were never acquired), in auction fraud (representing a lesser item as a more valuable one), or in export control evasion (fabricating provenance documentation for items that are illegally exported from source countries). The UNESCO 1970 Convention on the Means of Prohibiting and Preventing the Illicit Import, Export and Transfer of Ownership of Cultural Property defines the international framework; the FBI’s Art Crime Team and Interpol’s Works of Art unit both identify provenance documentation fraud as the primary enabler of the illegal antiquities trade.

Physical authentication of archaeological artefacts — thermoluminescence dating, isotope analysis, surface patina characterisation, tool mark analysis — cannot be replicated by image synthesis. The AI fraud operates at the document layer: fabricating provenance certificates, export licences, and previous ownership records (the “collecting history” that establishes legitimate provenance). Stylometric analysis of the LLM-generated provenance text and PRNU analysis of fabricated certificate scans apply directly.

41.5 Cross-Domain Forensic Toolkit for Object Deep Fakes

The following algorithm unifies the forensic toolkit across object domains, applying tests in order of computational cost and diagnostic power.

Algorithm 9: Non-Human Object Authenticity Triage**Input:** Submitted image $\hat{\mathbf{x}}$; domain d (product/gem/property/science); accompanying document set $\mathcal{D}$ **Output:** Authenticity verdict and priority level

```

// Step 1: Document hash verification (fastest, most decisive)
1 for each document  $D \in \mathcal{D}$  do
2   if  $D$  has registry (GIA/MHRA/EMVS/DETA/LBMA) then
3     if  $\neg \text{registry.verify}(D)$  then
4       return FABRICATED (P1)
5     end
6   end
7 end

// Step 2: Image PRNU authentication
8  $\rho \leftarrow \text{NCC}(\hat{\mathbf{K}}_{\text{device}}, \hat{\mathbf{x}} - W(\hat{\mathbf{x}}))$ ;
9 if  $\rho < 0.015$  then
10   return SUSPECT: full synthesis (P2)
11 end

// Step 3: Spatial composite detection
12 Compute block-wise NCC map  $\mathbf{R} = \{\rho_{ij}\}$ ;
13 if bimodal distribution detected in  $\mathbf{R}$  then
14   return SUSPECT: composite (P2)
15 end

// Step 4: Domain-specific plausibility
16  $S_{\text{phys}} \leftarrow \text{plausibility-classifier}(\hat{\mathbf{x}}, d)$ ; // material science / optical physics
17 if  $S_{\text{phys}} < \tau_d$  then
18   return SUSPECT: physically implausible (P2)
19 end

// Step 5: Document LLM detection
20 for each text document  $D \in \mathcal{D}$  do
21   if  $\text{LLM-detector}(D) > 0.65$  then
22     return SUSPECT: AI-generated text (P3)
23   end
24 end
25 return PASS (standard processing)

```

Worked Example — Algorithm execution on a diamond grading fraud case

Input: GIA grading report scan + diamond photograph. Domain: gemstone ($d = \text{gem}$).

Step 1 — Registry: GIA Report Check confirms certificate #5191770521 corresponds to G/VS2 but submitted report claims D/IF. → **FABRICATED (P1)**. Stop.

The algorithm correctly terminates at Step 1 without requiring the computationally intensive image forensics in Steps 2–5, because the registry discrepancy is definitive. This illustrates the cost-optimised ordering: cheapest and most decisive tests run first.

Counterfactual: if registry verification were unavailable (no internet access), Step 2 PRNU would proceed. The certificate scan shows $\text{NCC} = 0.004$ in the two grade fields (< 0.015) → **SUSPECT: composite (P2)**. Even without the registry, the spatial PRNU composite detection identifies the

fraud.

The forensic toolkit for non-human object deep fakes adapts the general methods of Chapter 12 to the specific physical constraints of each object domain. Four tools apply across all domains:

PRNU camera fingerprint: as with any photograph, an AI-generated or AI-composite object image either lacks the camera fingerprint of the claimed capture device (full synthesis) or shows spatial PRNU inconsistency in the fabricated region (composite inpainting). The same block-wise NCC procedure applies regardless of the image content.

Domain-specific plausibility classifier: for each object domain (meat marbling, gem clarity, interior architecture, microscopy), a classifier trained on authentic and synthetic examples in that domain achieves higher AUC than a generic face-swap detector, because it incorporates domain-specific physical consistency signals as features.

LLM text detection on accompanying documentation: in every domain examined here, AI-generated physical objects are accompanied by AI-generated textual documentation (certificates, reports, listings). LLM detection applied to this documentation provides a complementary signal that is domain-specific and often faster to compute than image forensics.

Document hash verification: for domains where authentic documentation is registered in a central database (pharmaceutical serialisation, diamond grading reports, LBMA gold bar records), hash verification against the authoritative registry provides the most reliable single test, because it directly checks the document against the authoritative record rather than inferenceing synthesis from signal properties.

41.6 Industry Applications

The commercial ecosystem for non-human object authentication has developed separately from the consumer face-detection market and is generally less mature. Amazon's Project Zero uses computer vision to detect counterfeit product listings, with an AI model trained on authentic brand product images that flags listings with visual inconsistencies against the authentic product profile. Alibaba's Intellectual Property Protection platform similarly uses image similarity and synthesis detection for its marketplace. Both platforms process billions of listing images annually and have added AI-generation detection components following documented increases in AI-generated listing fraud.

The gemological sector has invested in digital authentication infrastructure: the GIA Report Check database, AGS Certificates, and HRD Antwerp's digital grading reports all include cryptographic hash verification that enables digital report forgery to be detected immediately. The LBMA's Good Delivery List includes serial number registration that supports AI-generated gold bar image fraud detection through registry cross-reference.

In real estate, the Royal Institution of Chartered Surveyors (RICS) has issued guidance on AI-generated property images in valuation reports, and the UK Residential Property Surveyors Association has developed a voluntary certification scheme for surveyors' photographic practices. The US Department

of Housing and Urban Development has flagged AI-enhanced property photographs as an emerging fair housing concern, noting that AI enhancements can be used to make properties in minority-majority neighbourhoods appear more attractive than their actual condition — a form of discriminatory misrepresentation.

41.7 African Use Cases

Africa’s exposure to non-human object deep fakes is concentrated in three sectors where the continent has particular economic significance: mineral and gemstone authentication (given Africa’s role as the world’s leading producer of diamonds, gold, coltan, and cobalt), agricultural product quality certification (given the economic importance of export agricultural commodities), and real estate fraud (given rapidly growing urban property markets in Lagos, Nairobi, Accra, and Johannesburg).

The Kimberley Process Certification Scheme, which regulates the international trade in rough diamonds to prevent conflict diamond financing, requires certificates of origin for all rough diamond shipments. South Africa, Botswana, the DRC, Zimbabwe, and Angola collectively produce over 60% of the world’s rough diamonds. AI-generated Kimberley Process certificates have been flagged by Kimberley Process monitors in at least three member countries since 2022, with the forgeries detected through hash verification against the national diamond authority databases. The Kimberley Process Working Group on Digital Trade documentation has recommended mandatory blockchain anchoring of all KP certificates by 2026.

Nigeria’s agricultural export sector — cocoa, sesame, cashew, and ginger — has experienced AI-enabled quality grade fraud in which export shipments are accompanied by AI-generated quality certificates showing premium grades while the actual shipment contains substandard goods. The National Agency for Food and Drug Administration and Control (NAFDAC) and the Standards Organisation of Nigeria (SON) have issued guidance on AI-generated quality documentation following several documented export fraud cases that damaged Nigerian produce exporters’ relationships with European buyers.

In South Africa’s residential real estate market — the most active on the continent, with Johannesburg and Cape Town property values comparable to mid-tier European cities — the Property Practitioners Regulatory Authority (PPRA) has received complaints about AI-enhanced property listing photographs from both buyers and competing estate agents. The PPRA’s 2024 draft code of conduct for property practitioners includes a provision prohibiting the submission of AI-generated or AI-substantially-altered property images without disclosure, the first African real estate regulatory guidance specifically addressing AI image fraud.

Chapter Summary

Non-human object deep fakes extend the synthetic media threat beyond identity impersonation to the fabrication of physical scenes, products, and documents, with significant economic consequences in product counterfeiting, mineral authentication, real estate, and scientific integrity. Object deep fakes

differ structurally from face deep fakes in lacking a specific authenticated reference for comparison, requiring forensic analysis based on physical plausibility, material science consistency, and domain-specific statistical properties of authentic samples rather than identity matching.

E-commerce listing image fraud uses AI-generated product images at 143:1 cost advantage over authentic photography to misrepresent product quality, specifications, and brand. Material science plausibility analysis exploits domain knowledge of fracture mechanics, oxidation kinetics, and material aging to identify physically impossible combinations in AI-generated product images. Pharmaceutical serialisation registries and gemological report databases provide hash-based authentication that detects certificate fabrication independent of image quality. Real estate scene fabrication via AI inpainting of damaged interiors is the most common AI fraud complaint category in UK and Australian consumer protection records, detectable through shadow geometry analysis, perspective consistency, material aging comparison, and spatial PRNU composite detection. Scientific image fabrication is detectable through optical physics constraints: confocal microscopy images with super-resolution power above the diffraction limit cannot be authentic. The cross-domain forensic toolkit — PRNU camera fingerprint, domain-specific plausibility classifier, LLM text detection on documentation, and document hash verification — applies across all non-human object domains.

Exercises

1. A property listing includes an interior photograph of a kitchen. The image passes full-image PRNU authentication ($NCC = 0.039$). Spatial PRNU block-wise analysis (using 64×64 pixel blocks) produces the following NCC distribution:

Image region	Mean NCC	Std Dev
Ceiling and upper walls	0.041	0.008
Lower walls and skirting	0.038	0.009
Kitchen units and worktop	0.005	0.003
Flooring (visible outside kitchen)	0.040	0.007

- (a) Is this image a full synthesis or a composite? Explain. (b) What does the spatial NCC pattern tell you about which region was AI-inpainted? (c) Calculate the probability that the kitchen region NCC values are drawn from the same distribution as the ceiling region, using a two-sample t -test (assume 30 blocks in the kitchen region and 50 in the ceiling region; use the provided means and standard deviations). (d) What physical evidence at the property would confirm or refute the fabrication hypothesis?
2. A diamond dealer submits a GIA grading report scan claiming D-colour, IF-clarity for a 2.14 ct stone, with an estimated retail value of \$85,000. The GIA Report Check database shows the same certificate number corresponding to an authentic G-colour, VS2-clarity stone worth \$22,000. Spatial PRNU analysis of the submitted scan shows $NCC = 0.006$ in the colour grade and clarity grade fields. (a) What is the maximum financial gain the fraud could achieve? (b) If the PRNU threshold is $\tau = 0.015$, are the grade fields flagged as fabricated? (c) Describe the

most likely fabrication method used (inpainting, GAN generation, or LLM text generation) and explain your reasoning. (d) What hash verification step would have prevented this fraud, and at which point in the transaction chain should it have been applied?

3. A confocal microscopy image submitted with a research paper claims to show co-localisation of two fluorescent proteins (green and red channels) at a stated magnification of $63\times$ (oil immersion, $\text{NA} = 1.4$, $\lambda_{\text{ex}} = 488 \text{ nm}$). The image pixel size is 65.4 nm . (a) Calculate the diffraction-limited spatial resolution $r = \lambda/2\text{NA}$ and the corresponding spatial frequency cutoff f_{max} in cycles per micrometre. (b) In the radially averaged power spectrum of the submitted image, significant power ($P/P_0 = 0.052$) is observed at $f = 7.2 \text{ cycles}/\mu\text{m}$. Is this physically possible for authentic confocal microscopy under the stated conditions? (c) What alternative explanation (other than AI generation) could account for this observation, and what additional test would distinguish the two explanations? (d) If the image is found to be fabricated, who should be notified and what should happen to the associated research paper?

Answers to Exercises

1. Real estate spatial PRNU composite analysis.

(a) This is a **composite** (inpainting), not a full synthesis. The full-image PRNU passes ($\text{NCC} = 0.039 > \tau = 0.015$), indicating the image was captured by the claimed camera device. A full AI synthesis would produce near-zero NCC across the entire image. The spatial NCC pattern shows high NCC in non-kitchen regions and near-zero in the kitchen region, the definitive signature of a composite where an authentic photograph was partially replaced by AI-generated content.

(b) The kitchen units and worktop region (mean $\text{NCC} = 0.005$) was AI-inpainted. This region falls entirely below any reasonable authentication threshold, while all other regions (ceiling 0.041, lower walls 0.038, flooring 0.040) are consistent with authentic capture by the same camera. The boundary between authentic and inpainted regions would show intermediate NCC values (0.015–0.025) from the Poisson blending transition zone.

(c) Two-sample t -test of kitchen NCC vs. ceiling NCC: $t = \frac{\bar{x}_{\text{kitchen}} - \bar{x}_{\text{ceiling}}}{\sqrt{s_k^2/n_k + s_c^2/n_c}} = \frac{0.005 - 0.041}{\sqrt{0.003^2/30 + 0.008^2/50}} = \frac{-0.036}{\sqrt{9 \times 10^{-7} + 1.28 \times 10^{-6}}} = \frac{-0.036}{\sqrt{2.18 \times 10^{-6}}} = \frac{-0.036}{0.001476} = -24.4$.

With degrees of freedom ≈ 50 (Welch), $|t| = 24.4$ vastly exceeds the critical value of 2.01 at $\alpha = 0.05$. The probability that the kitchen and ceiling NCC values are drawn from the same distribution is effectively zero ($p \ll 0.0001$). The kitchen region is statistically distinguishable from authentic capture with overwhelming confidence.

(d) Physical evidence at the property: A trading standards officer should inspect the actual kitchen in its current state and compare against the listing. If the kitchen is in substantially worse condition than depicted — stained worktops, damaged units, worn flooring — this confirms the misrepresentation. The officer should also request the original camera photos from the estate

agent (before any editing), which would show the pre-inpainting state. If the estate agent cannot produce original unedited photographs, this is itself evidence of manipulation.

2. Diamond grading report fabrication analysis.

(a) Maximum financial gain: the fabricated report claims a value of \$85,000 for a stone worth \$22,000. Maximum fraud gain = $\$85,000 - \$22,000 = \$63,000$ if the stone is sold at the claimed value. In practice, the gain depends on the transaction structure: if the fraudulent report is used to obtain a loan secured against the stone, the gain is the loan amount less the stone's legitimate resale value.

(b) Yes, both grade fields are flagged. $NCC = 0.006 < \tau = 0.015$; both the colour grade field and the clarity grade field are below the authentication threshold, indicating they do not carry the GIA scanner's PRNU fingerprint and were not produced by GIA's scanning equipment.

(c) Most likely fabrication method: **image inpainting**. The reasoning: the GIA report format is highly standardised, making full document generation risky (format errors would be detectable). The fraudster most likely obtained an authentic GIA report scan (either their own lower-grade stone's report or an image from the internet) and used diffusion-based inpainting to replace the specific grade fields (colour grade, clarity grade, and possibly the price/carats fields) while preserving the authentic document structure, GIA logo, and border elements. GAN generation would likely produce format inconsistencies; LLM text generation would not produce the certificate's graphic layout. The spatial PRNU pattern — low NCC only in the specific grade fields, not the entire document — is consistent with targeted inpainting rather than wholesale synthesis.

(d) Hash verification step: GIA, AGS, and all major gemological laboratories maintain online report verification databases (GIA Report Check, accessible at gia.edu/report-check) where any certificate number can be cross-referenced against the laboratory's own record. This verification should be applied: (1) by the buyer before purchase (the first line of defence), (2) by any financial institution before lending against the stone, and (3) by customs or export authorities for high-value stones crossing borders. The hash verification takes 30 seconds and immediately reveals the discrepancy between the claimed D/IF grade and the authentic G/VS2 record — this fraud would have been prevented at any of these points if the verification had been performed.

3. Confocal microscopy optical physics analysis.

(a) Diffraction-limited spatial resolution (Rayleigh criterion): $r = 0.61\lambda/\text{NA} = 0.61 \times 488\text{ nm}/1.4 = 212.7\text{ nm} = 0.213\text{ }\mu\text{m}$. Spatial frequency cutoff: $f_{\text{max}} = 1/r = 4.70\text{ cycles}/\mu\text{m}$. (Alternatively, using the Abbe criterion: $f_{\text{max}} = 2\text{NA}/\lambda = 2 \times 1.4/0.488\text{ }\mu\text{m} = 5.74\text{ cycles}/\mu\text{m}$.) Using the Abbe criterion (appropriate for coherent confocal illumination): $f_{\text{max}} = 5.74\text{ cycles}/\mu\text{m}$.

(b) No, this is **physically impossible** for authentic confocal microscopy under the stated conditions. The spatial frequency $f = 7.2\text{ cycles}/\mu\text{m}$ exceeds $f_{\text{max}} = 5.74\text{ cycles}/\mu\text{m}$. A confocal

microscope cannot transmit spatial frequencies above the diffraction limit: all information at $f > f_{\max}$ is zero in the optical transfer function. Any power at $f = 7.2$ cycles/ μm in an authentic confocal image would be pure noise; the observed $P/P_0 = 0.052$ is $52\times$ above the noise floor of a well-calibrated system, representing structured content at a physically impossible frequency.

(c) Alternative explanations and distinguishing tests: *Deconvolution*: if the image has been post-processed with super-resolution deconvolution software (such as Huygens or AutoQuant), the effective resolution can be enhanced beyond the optical diffraction limit. This would explain super-resolution frequencies. Distinguishing test: request the original raw data file from the microscope control software (for Zeiss confocal: .czi format; for Leica: .lif format). Raw files contain metadata including any deconvolution processing; if deconvolution was applied, it will be recorded. If the raw file is unavailable or shows no deconvolution, the super-resolution content cannot be explained.

STED microscopy: Stimulated Emission Depletion microscopy exceeds the diffraction limit legitimately. Distinguishing test: STED requires a specific laser configuration (depletion laser at longer wavelength than the excitation laser) recorded in the acquisition metadata. If the metadata shows standard confocal acquisition (single excitation laser, no depletion), STED is excluded.

(d) Notification and consequences: (1) Notify the journal editor immediately with the forensic report; (2) the editor notifies the corresponding author and requests the raw data and acquisition metadata; (3) if raw data is unavailable or the metadata is inconsistent with the submitted image, the paper should be rejected (if pre-publication) or retracted (if published); (4) the institution employing the authors should be notified for research integrity investigation; (5) if a previous published paper by the same authors contains similar images, all prior publications should be reviewed; (6) funding agencies (NIH, Wellcome Trust, etc.) should be notified if the fabricated research was funded under their grants, as they have independent reporting obligations and may seek recovery of funds.

Chapter 42

Deep Fakes in Retail: Try-On, Reviews, and Consumer Trust

“The product page is the new shop window. And the shop window can lie.”

—Anon.

Retail is the consumer-facing interface of the broader product fraud ecosystem. Where Chapter 40 examined how synthetic media enables fraud in the international trade document chain and Chapter 41 examined the fabrication of product images and quality certificates, this chapter examines the retail layer: the e-commerce listings, virtual try-on experiences, review systems, and customer service interactions through which consumers form purchase decisions and brand relationships. The scale of the retail sector — global e-commerce revenues exceeding \$5.8 trillion annually — and its complete dependence on digital representation of physical goods makes it the highest-volume deployment environment for product-level synthetic media, for both legitimate and harmful purposes.

Retail is also the sector with the highest legitimate demand for synthetic media. Virtual try-on for fashion, footwear, eyewear, and cosmetics allows consumers to visualise products on their own body before purchase, reducing return rates by 20–40% in published trials. AI-generated product photography reduces catalogue production costs by 60–80% relative to traditional studio photography for categories where physical staging is expensive. AI-personalised product recommendations that synthesise product visualisations tailored to individual consumer preferences represent the frontier of personalised retail. The challenge for retailers, platforms, and regulators is to enable these substantial legitimate benefits while preventing the harmful applications: deceptive try-on that shows unrealistic results, AI-generated testimonials that misrepresent product performance, and chatbots that make false product claims.

The consumer protection framework for AI in retail is more developed than for most other sectors examined in this book, because retail fraud has a long regulatory history and because the causal chain from misleading AI content to consumer harm is traceable through purchase transactions. The FTC Act’s Section 5 prohibition on unfair or deceptive acts applies to any representation made

in connection with a commercial transaction, regardless of whether the representation is human-generated or AI-generated. The EU Consumer Rights Directive requires accurate and complete pre-contractual information, and the EU Omnibus Directive requires disclosure of how consumer rankings are personalised. These existing frameworks provide a reasonably complete legal foundation; the regulatory challenge is enforcement at the scale of billions of product listings and transactions.

This chapter develops the complete landscape of AI in retail from the consumer protection perspective. Section 42.1 develops virtual try-on technology and its disclosure obligations. Section 42.2 develops AI-generated review fraud and platform detection. Section 42.3 develops AI customer service and sales chatbot obligations. Section 42.4 develops the returns fraud vector enabled by synthetic media. Section 42.5 develops platform obligations and detection infrastructure.

42.1 Virtual Try-On and Augmented Reality Retail

Virtual try-on systems allow consumers to visualise how a product would look on their own body, face, or home environment without physical contact with the product. The technology spans a wide range of sophistication: from simple image overlay (placing a product image over a consumer photograph with no body shape adjustment) through mesh-fitting approaches (warping the product image to conform to a 3D body model estimated from the consumer photograph) to full generative try-on (using diffusion models to synthesise a photorealistic image of the consumer wearing the product, accounting for body shape, lighting, and fabric drape).

42.1.1 The Accuracy Obligation for Try-On Systems

Virtual try-on systems create an implied representation that the product will look as shown when worn or used by the consumer. When this representation is substantially inaccurate — when the virtual try-on shows a flattering result that the consumer cannot achieve with the actual product — it constitutes a misleading commercial practice under EU law and a deceptive act under FTC standards. The accuracy obligation applies to both the fitting accuracy (does the product fit the consumer’s body as shown?) and the product representation accuracy (does the visualised product match the actual product’s appearance in the consumer’s specific context?).

Published research on virtual try-on accuracy for fashion items reports that state-of-the-art diffusion-based try-on systems achieve a structural similarity index (SSIM) of 0.74–0.81 relative to authentic photographs of the same person wearing the same garment. The gap from 1.0 (perfect reproduction) reflects systematic biases: AI try-on systems tend to generate flattering body proportions, smooth skin textures, and idealised fabric drape that do not represent the actual consumer’s appearance or the product’s real behaviour on diverse body types. This systematic bias is directional — always in the flattering direction — making it a potential consumer protection concern rather than random error.

Worked Example — Try-on accuracy measurement

A fashion retailer deploys a diffusion-based virtual try-on system. Consumer advocates commission

an independent evaluation: 200 test consumers each try on 5 garments using the virtual system, then photograph themselves wearing the actual garments under matched lighting conditions. Comparison metrics:

Garment category	SSIM	Fit deviation (cm)	Colour ΔE	Consumer satisfaction
T-shirts	0.81	1.2	2.1	78%
Fitted dresses	0.67	3.8	3.4	52%
Knitwear	0.61	4.1	4.8	44%
Swimwear	0.54	5.9	3.1	31%
Overall	0.71	3.0	3.4	56%

Swimwear shows the largest accuracy gap (SSIM = 0.54, fit deviation 5.9 cm) and the lowest consumer satisfaction (31%), reflecting the difficulty of accurately simulating close-fitting stretchy fabric on diverse body types. At SSIM = 0.54, the virtual try-on representation is substantially inaccurate, and the directional bias (the system generates more flattering body proportions than the actual garment achieves) means the system systematically overstates the product’s appearance on the consumer’s body. This falls below the threshold the FTC would consider non-deceptive, triggering an obligation to either improve the system’s accuracy or provide prominent disclosure of its limitations.

42.1.2 Legitimate vs. Deceptive Try-On

The distinction between legitimate and deceptive virtual try-on is not binary but depends on the accuracy, disclosure, and the directional nature of any inaccuracy. Three principles guide the assessment:

Accuracy threshold: industry self-regulatory bodies including the Interactive Advertising Bureau and the Network Advertising Initiative have proposed an SSIM floor of 0.75 and a fit deviation ceiling of 2.5 cm for fashion try-on systems as minimum accuracy thresholds below which prominent disclosure of limitations is required. These thresholds remain voluntary guidelines rather than enforceable standards.

Directional bias: random inaccuracy (sometimes flattering, sometimes not) is less concerning than directional inaccuracy (always flattering). A try-on system that systematically shows idealised body proportions or enhanced skin texture without disclosure misrepresents the product in a way that is not corrected by averaging across many consumers.

Disclosure: explicit consumer-facing disclosure that the try-on result is an AI simulation that may not exactly represent how the product will look, provided before or during the try-on experience, partially mitigates the deception concern by enabling informed consumer evaluation.

42.2 AI-Generated Review Fraud at Platform Scale

Platform-scale review fraud detection applies a three-tier algorithmic pipeline optimised for throughput at millions of reviews per day.

Algorithm 10: LLM Review Detection: Three-Tier Triage

Input: Review text r , reviewer account u , product p , timestamp t

Output: Tier decision and fraud probability estimate

```
// Tier 1: Fast statistical features (<1 ms)
1 ppl ← perplexity( $r$ , GPT-2);
2  $\sigma_{\text{sent}} \leftarrow$  sentence-length-std( $r$ );
3  $V \leftarrow$  vocab-richness( $r$ ); // type/token ratio
4  $S_1 \leftarrow w_1(1 - \text{ppl}/\text{ppl}_{\text{human}}) + w_2(1 - \sigma_{\text{sent}}/\sigma_{\text{human}}) + w_3V$ ;
5 if  $S_1 < \tau_1$  then
6   | return Tier 1: PASS
7 end

// Tier 2: Graph-based coordinated behaviour (<50 ms)
8  $\mathcal{N}(u) \leftarrow$  reviews from accounts sharing IP/device-fingerprint with  $u$ ;
9  $G_{\text{cluster}} \leftarrow$  build review-posting graph on  $\mathcal{N}(u)$ ;
10  $C \leftarrow$  graph-cluster-score( $G_{\text{cluster}}$ ); // e.g. Louvain modularity
11 if  $C > \tau_2$  then
12   | return Tier 2: COORDINATED; escalate to human
13 end

// Tier 2b: Semantic duplicate detection
14  $\text{sim} \leftarrow \max_{r' \in \text{prior-reviews}(p)} \cos(\text{embed}(r), \text{embed}(r'))$ ;
15 if  $\text{sim} > 0.92$  then
16   | return Tier 2: DUPLICATE; suppress
17 end

// Tier 3: Full LLM detection + human review
18  $P_{\text{LLM}} \leftarrow$  SHAP-LLM-classifier( $r$ ); // DetectGPT or equivalent
19 return Tier 3 with  $P_{\text{LLM}}$ ; human moderator decides
```

Worked Example — Per-tier cost vs. throughput for a major marketplace

A platform processes 6 million reviews per week. Cost targets:

Tier	Reviews processed	Escalation rate	Cost/review	Weekly cost
T1 (stat. features)	6,000,000	4.2%	\$0.00002	\$120
T2 (graph + semantic)	252,000	12%	\$0.08	\$20,160
T3 (LLM + human)	30,240	70% removed	\$3.80	\$114,912
Total				\$135,192

At estimated 2.1% AI-fraud prevalence (126,000/week), the system detects: $126,000 \times 0.82$ (S_n) = 103,320 fraud reviews removed. Each prevented AI review saves approximately \$3.80 in consumer harm. Weekly harm prevented: $103,320 \times \$3.80 = \$392,616$. Net weekly benefit: $\$392,616 - \$135,192 = \$257,424$. ROI: 191%.

Consumer reviews are the primary trust signal for purchase decisions: the large majority of consumers trust online reviews as much as personal recommendations, and products with higher average ratings command price premiums of 10–25% above otherwise comparable products. The fabrication of reviews using LLMs has qualitatively changed the fake review ecosystem. Traditional fake reviews

were written by paid human workers at farms in Eastern Europe, South Asia, or China at costs of \$0.50–\$5 per review and with detectable stylistic patterns. LLM-generated reviews cost less than \$0.01 each, can be generated in any language, and match the stylistic distribution of authentic reviews far more closely.

42.2.1 LLM Review Detection at Scale

Platform-scale detection of LLM-generated reviews applies the statistical methods of Chapter 23 in a high-volume, resource-constrained context. Amazon, Trustpilot, and Yelp each process millions of reviews per day and cannot apply computationally intensive detection to every submission. A three-tier triage approach mirrors the insurance claims framework of Chapter 38:

Tier 1 — Statistical pre-filter (microseconds per review): LLM text exhibits lower perplexity than human text under the generating model’s distribution, more uniform sentence length distribution, and specific stylistic markers. A lightweight statistical feature vector (perplexity estimate, sentence length variance, vocabulary richness index) is computed for each review and compared against thresholds. Reviews above the threshold progress to Tier 2; the majority are passed for standard quality filtering.

Tier 2 — Graph-based reviewer analysis (milliseconds per review): LLM review campaigns are typically submitted by networks of coordinated accounts that share device fingerprints, IP ranges, or temporal posting patterns. Graph analysis flags accounts in the same coordinated network even when individual reviews pass text-based detection. A review that individually passes Tier 1 but originates from an account in a flagged coordinated network is escalated to Tier 3.

Tier 3 — Human review and account action (human judgment): a small fraction of flagged reviews (typically 0.1–0.5% of total volume) receives human moderator attention, leading to review removal, account suspension, or seller sanction. The positive predictive value at Tier 3 must be high enough to justify the moderation cost; calibrating Tier 1 and Tier 2 thresholds to achieve $PPV > 0.70$ at Tier 3 is the primary operational challenge.

42.2.2 The Review Ecosystem Economic Model

The economic equilibrium of the review ecosystem depends on the platform’s enforcement effectiveness. Let q be the fraction of reviews that are LLM-generated, p_d be the probability that an LLM review is detected and removed, and π_f be the price premium a fraudulent seller gains from fake reviews. The expected net benefit to a fraudulent seller per fake review is:

$$\mathbb{E}[\text{benefit}] = (1 - p_d) \cdot \pi_f - c_r, \quad (42.1)$$

where $c_r \approx \$0.01$ is the LLM generation cost per review. At platform detection rate $p_d = 0.60$ and price premium \$200 per rating point gained, the expected benefit per 100 fake reviews (gaining +0.1 rating points): $(1 - 0.60) \times \$20 - \$1 = \$7$. The positive expected value — even at 60% detection — explains why fake review fraud persists despite platform enforcement.

Worked Example — Review fraud detection ROI for a marketplace

Amazon UK processes 4 million reviews per week, of which an estimated 1.5% are LLM-generated (60,000 fake reviews/week). Detection rate: 70% ($p_d = 0.70$). Detected and removed: $60,000 \times 0.70 = 42,000$ per week. Undetected fakes remaining: 18,000 per week.

Each undetected fake review costs the platform an estimated \$4.50 in consumer harm (via purchase decisions based on false information) and \$1.20 in reputational damage. Weekly undetected harm cost: $18,000 \times \$5.70 = \$102,600$. Detection system weekly cost (infrastructure + moderation): \$38,000. Net weekly saving from detection system: $\$102,600 - \$38,000 = \$64,600$. Annual ROI of detection investment: $(\$64,600 \times 52) / \$38,000 \times 52 \approx 170\%$.

42.3 AI Chatbots in Sales and Customer Service

AI-powered sales and customer service chatbots have been deployed at scale across retail, with Salesforce estimating that 60% of major retailers deployed some form of AI chatbot in their customer-facing operations by 2024. These systems range from narrow FAQ bots (with predefined responses and limited AI generation) through RAG-augmented chatbots (retrieval-augmented generation using a product knowledge base) to fully generative AI agents (using large language models with broad product knowledge and real-time generation). The consumer protection concerns increase along this spectrum, because fully generative AI agents can make product claims that were not specifically designed or reviewed by the retailer.

42.3.1 False Product Claims and Retailer Liability

When an AI chatbot makes a false product claim that induces a consumer purchase, who is liable? The 2024 Air Canada chatbot case established an important precedent: the British Columbia Civil Resolution Tribunal ruled that Air Canada was liable for a misleading statement made by its AI chatbot about bereavement fare refund eligibility, even though the chatbot generated the statement autonomously. The tribunal rejected Air Canada's argument that the chatbot was a separate legal entity whose statements were not attributable to the airline. This establishes that retailers are liable for the product claims made by their AI customer service systems, creating a strong incentive to implement factual accuracy guardrails.

The retailer's liability exposure includes: product claims that are false (the product does not have the specified feature); claims that are misleading (technically true but creating a false overall impression); claims about availability or pricing that are inaccurate; and claims about competitor products that constitute tortious interference or trade libel. Implementing factual accuracy guardrails requires the chatbot to be grounded in a verified product knowledge base and to decline to make specific product claims when the knowledge base does not contain verified information for that claim.

42.3.2 AI Disclosure Requirements for Retail Chatbots

The disclosure obligations for AI retail chatbots under EU AI Act Article 52 require that systems which interact with humans be clearly identified as AI when they could reasonably be mistaken for human customer service representatives. Retailers must ensure that any conversational AI system deployed in customer service contexts:

1. Identifies itself as an AI system at the beginning of any interaction, or immediately upon sincere inquiry.
2. Does not falsely represent itself as a named human customer service representative.
3. Provides a clear and accessible mechanism to escalate to a human representative upon request.
4. Does not use emotional manipulation techniques or create false urgency in a sales context (prohibited under EU Unfair Commercial Practices Directive).

California’s SB 1001 extends these requirements to voice-based AI customer service in the retail context, requiring disclosure when an AI voice agent is used in a commercial interaction with a California consumer.

42.4 Returns Fraud via Synthetic Media

Returns fraud — returning goods that were not purchased from the retailer, returning goods in a worse condition than stated, or claiming non-delivery of goods actually received — costs the global retail industry an estimated \$101 billion annually. AI-generated and AI-enhanced image fraud has added new vectors to this established fraud category.

The primary AI-enabled returns fraud attacks are:

Condition misrepresentation: AI-generated or AI-inpainted images of a product in new or undamaged condition are submitted as evidence of the product’s pre-return condition, to support a return claim for a product that was damaged in use. This parallels the insurance damage fraud of Chapter 38 but in the retail returns context.

Empty-box claims: consumers claim non-delivery or empty-box delivery and submit AI-generated images purportedly showing an empty box or packaging materials, to claim a refund or replacement without returning any goods. Traditional empty-box fraud relied on physical staging; AI generation eliminates the need for the physical box.

Wrong-item claims: consumers claim they received the wrong product and submit AI-generated images purportedly showing an incorrect item in the packaging, to return the actually delivered item while keeping the refund. This fraud is particularly difficult to detect because the AI-generated image of the “wrong item” needs only to be plausible, not to match any specific reference.

Detection of returns fraud images applies the same forensic toolkit as product listing fraud detection: PRNU camera fingerprint against the consumer’s claimed capture device, physical plausibility

analysis specific to the product category, and reverse image search to identify recycled stock photographs.

42.5 Platform Obligations and Detection Infrastructure

E-commerce platforms that host third-party sellers have obligations under both existing consumer protection frameworks and emerging AI-specific legislation that create a mandatory role for platform-level detection of AI-generated deceptive content.

The EU Digital Services Act Article 34 requires very large online platforms (VLOPs) to conduct systematic risk assessments of their platform's contribution to systemic risks, including the dissemination of illegal content and adverse effects on consumer protection. Platforms with over 45 million EU users must implement mitigation measures for identified risks, which in the context of AI-generated product fraud include detection systems for AI-generated listing images, review fraud detection, and chatbot accuracy monitoring.

The EU Omnibus Directive (effective 2022) requires that price reduction announcements reference the prior price over the preceding 30 days, preventing AI-generated fake price histories from supporting false discount claims. It also requires disclosure when rankings are influenced by paid placement or personalisation algorithms that affect which products consumers see, limiting the use of AI to manipulate product rankings in ways consumers are not aware of.

Worked Example — Platform detection system performance requirements

A VLOP must demonstrate to the European Commission that its AI-generated content detection achieves certain performance benchmarks. For product listing image fraud:

At a stated deployment with 50 million product listings, estimated AI-fraud prevalence 1.8%, platform detection $AUC = 0.92$, operating at $FPR = 3\%$:

TP (correctly removed AI-fraud listings): $50M \times 0.018 \times TPR$. At $AUC = 0.92$ and $FPR = 3\%$, the corresponding TPR is approximately 0.84. $TP = 50,000,000 \times 0.018 \times 0.84 = 756,000$. FP (authentic listings incorrectly removed): $50M \times 0.982 \times 0.03 = 1,473,000$. $PPV = 756,000 / (756,000 + 1,473,000) = 0.339 = 33.9\%$.

A PPV of 33.9% means only one in three removed listings is actually AI-fraud. This base-rate problem — which the DSA risk assessment must acknowledge — means that human review of flagged listings is essential before removal, or that the platform must accept a high false-positive burden with an effective appeals mechanism for authentic sellers whose listings are incorrectly removed.

42.6 Industry Applications

The retail AI landscape has bifurcated between legitimate applications and fraud vectors, with significant commercial investment in both. On the legitimate side: Snap's Shopping Lenses, Google's virtual try-on (launched in Shopping search in 2023), Amazon's virtual try-on for shoes and apparel,

and Sephora's Virtual Artist collectively serve hundreds of millions of consumer sessions annually. Snap's lens technology, deployed through a Snapchat camera SDK used by over 100 retail partners, processes 6 billion virtual try-on sessions per year with an average return-rate reduction of 25% reported by partner retailers.

On the fraud side: Amazon's 2023 annual report identified fake reviews and counterfeit listings as the two largest threats to marketplace integrity, with \$700 million invested in anti-counterfeiting measures. Amazon's Project Zero has given brands the ability to directly remove counterfeit listings using AI-powered automated protections. Trustpilot deployed a real-time LLM review detection system in 2023, removing 3.1 million fake reviews in its first six months of operation.

The EU's 2024 enforcement action against the SHEIN marketplace — one of the largest fast-fashion e-commerce platforms globally — specifically cited AI-generated product images that misrepresented product quality and fabric composition as an unfair commercial practice. The £25 million settlement included mandatory implementation of product image authenticity verification, marking the first EU enforcement action specifically requiring AI image authenticity measures as a condition of continued market access.

42.7 African Use Cases

Africa's rapidly growing e-commerce sector — valued at \$32 billion in 2023 and growing at 20% annually, driven by platforms including Jumia, Takealot, Konga, and Kilimall — faces AI-enabled retail fraud that is disproportionately concentrated in specific product categories where consumer protection is weakest and where the cost asymmetry between fraud and legitimate trade is highest.

Fast-moving consumer goods, particularly cosmetics and personal care products, are the highest-fraud category. AI-generated product images for counterfeit cosmetics (replicated luxury brand packaging, falsified ingredient labels) are prevalent on WhatsApp commerce and informal social media marketplaces that operate outside the trust and safety infrastructure of formal e-commerce platforms. The African Medicines Regulatory Harmonisation initiative and the NAFDAC (Nigeria) have both specifically identified AI-generated cosmetic product images as a public health concern, as some counterfeit cosmetics sold with AI-generated certificates of authenticity contain hazardous substances.

Virtual try-on technology presents a particular equity challenge in African markets. Most commercial virtual try-on systems are trained on datasets dominated by lighter-skinned bodies and European or East Asian fashion styles, producing systematically poor results for African consumers with darker skin tones and different body proportion distributions. Published evaluations of Snap's and Google's try-on systems show SSIM degradation of 15–25% for darker Fitzpatrick skin tones relative to lighter tones, an accuracy gap that disproportionately affects African consumers and that constitutes potential consumer discrimination under non-discrimination frameworks applicable to digital services in South Africa (POPIA), Kenya (DPA), and the EU (for African diaspora consumers).

Chapter Summary

Retail is both the highest-volume legitimate deployment environment for synthetic media (virtual try-on, AI product photography, personalised recommendations) and one of the most significant fraud vectors, where AI-generated product images, fake reviews, false chatbot claims, and returns fraud collectively impose billions of dollars in consumer harm annually. Virtual try-on systems must achieve minimum accuracy thresholds ($SSIM \geq 0.75$, fit deviation ≤ 2.5 cm for fashion) and disclose directional bias toward flattering results that may not represent the consumer's actual experience. LLM-generated review fraud at less than \$0.01 per review versus \$0.50–\$5 for human-written fake reviews has created a mass-scale fake review problem; platform detection using three-tier triage (statistical pre-filter, graph-based network analysis, human review) achieves positive predictive values sufficient for cost-effective enforcement at detected rates around 70%. AI retail chatbots are liable for false product claims under the Air Canada precedent, creating strong incentives for factual accuracy guardrails. EU Digital Services Act and Omnibus Directive obligations create mandatory platform-level detection requirements. The base-rate problem at scale means PPV of 33–40% is typical even at $AUC = 0.92$, requiring effective appeals processes for false-positive authentic sellers.

Exercises

1. A fashion e-commerce platform operates a virtual try-on system with the following accuracy metrics by consumer skin tone: Fitzpatrick I–II: $SSIM = 0.82$; Fitzpatrick III–IV: $SSIM = 0.76$; Fitzpatrick V–VI: $SSIM = 0.61$. The return rate with try-on for Fitzpatrick I–II consumers is 18% versus 32% without try-on. (a) Does the try-on system meet the proposed $SSIM$ accuracy threshold (≥ 0.75) for all consumer groups? (b) Estimate the return rate reduction for Fitzpatrick V–VI consumers, assuming the relationship between $SSIM$ and return rate reduction is proportional to the $SSIM$ excess above the 0.75 threshold. (c) A consumer advocacy group argues that the differential accuracy constitutes discrimination under the EU AI Act non-discrimination provisions. Evaluate this argument and propose a technical remediation.
2. An LLM review detection system is deployed on a marketplace with 8 million new reviews per week, estimated AI-fraud rate 2.3%. The classifier has sensitivity 0.82 and specificity 0.95. (a) Compute the weekly true positives, false positives, true negatives, and false negatives. (b) Compute the PPV. (c) Each Tier 2 investigation (including graph analysis) costs \$0.12. Each Tier 3 human review costs \$3.50. If 20% of Tier 2 flags are escalated to Tier 3, what is the weekly detection cost? (d) Each undetected AI review causes \$3.80 in expected consumer harm. Calculate the weekly net benefit of the detection system. Is it cost-effective?
3. A consumer purchases a knitwear garment after using a virtual try-on system that showed a highly flattering result ($SSIM = 0.58$ compared to authentic wear). The garment does not match the try-on result; the consumer seeks a full refund citing misrepresentation. (a) Under EU Consumer Rights Directive Article 6 (pre-contractual information), was the retailer obligated to disclose the try-on system's accuracy limitations? (b) Does an $SSIM$ of 0.58 constitute a

misleading commercial practice under the Unfair Commercial Practices Directive? (c) The retailer argues that the try-on represented a “lifestyle visualisation” rather than an exact product representation. Evaluate this defence. (d) What remedy is the consumer entitled to, and what systemic change should the regulator require of the retailer?

Answers to Exercises

1. Virtual try-on accuracy and equity analysis.

(a) Fitzpatrick I–II: $\text{SSIM} = 0.82 \geq 0.75$. **Meets threshold.** Fitzpatrick III–IV: $\text{SSIM} = 0.76 \geq 0.75$. **Meets threshold (marginally).** Fitzpatrick V–VI: $\text{SSIM} = 0.61 < 0.75$. **Does NOT meet threshold.** The system fails the accuracy threshold for the darkest skin tone group.

(b) Proportional return rate reduction estimate. The return rate reduction for Fitzpatrick I–II is $32\% - 18\% = 14$ percentage points, achieved at $\text{SSIM} = 0.82$, which is $0.82 - 0.75 = 0.07$ above the threshold. Rate reduction per unit SSIM excess: $14\% / 0.07 = 200$ percentage points per SSIM unit.

For Fitzpatrick V–VI, $\text{SSIM} = 0.61 < 0.75$, so $\text{SSIM excess} = 0.61 - 0.75 = -0.14$ (below threshold). Under the proportional model, the system would be expected to *increase* return rates: $\text{return rate with try-on} = 32\% + (-0.14 \times 200\%) = 32\% - 28\% = 4\%\text{??}$

This gives an implausible result because the linear model does not hold below the threshold. A more plausible interpretation: below $\text{SSIM} = 0.75$, the try-on does not reduce returns relative to no try-on (and may increase them if the misleading try-on encourages purchases that would not have been made without it). Conservatively: return rate for Fitzpatrick V–VI with this try-on system $\approx 35\%$ (slightly higher than the 32% no-try-on rate for all groups), reflecting that a poor-quality, systematically biased try-on creates false expectations that the actual product fails to meet.

(c) Discrimination argument evaluation: The EU AI Act Article 10 requires high-quality data for high-risk AI systems to be free from bias. The Non-Discrimination provisions of the EU Charter and EU AI Act require that AI systems do not discriminate on protected characteristics including race (to which skin tone is closely related). A differential SSIM of 0.82 vs. 0.61 (–26%) constitutes a significant accuracy disparity correlated with race/ethnicity, which the EU AI Act implementing guidelines would likely treat as requiring remediation.

Technical remediation: (1) *Training data augmentation*: expand the training dataset to include at least 30% Fitzpatrick V–VI examples (current commercial try-on datasets are estimated at 5–12% darker skin tones); (2) *Fitzpatrick-stratified training objective*: add a fairness regularisation term to the diffusion model training objective that penalises SSIM disparity across Fitzpatrick groups; (3) *Conditional normalisation*: use skin-tone-conditional normalisation layers that adapt the model’s colour and texture processing to different skin tones. Target: SSIM disparity below 0.05 across all Fitzpatrick groups, achievable with balanced training data.

2. LLM review detection cost-benefit analysis.

(a) Weekly review volumes at prevalence $p = 0.023$, sensitivity = 0.82, specificity = 0.95, $N = 8,000,000$:

AI-fraud reviews: $8,000,000 \times 0.023 = 184,000$. Authentic reviews: $8,000,000 - 184,000 = 7,816,000$.

TP: $184,000 \times 0.82 = 150,880$. FN: $184,000 - 150,880 = 33,120$. TN: $7,816,000 \times 0.95 = 7,425,200$. FP: $7,816,000 - 7,425,200 = 390,800$.

(b) $PPV = 150,880 / (150,880 + 390,800) = 150,880 / 541,680 = 27.9\%$. Approximately 1 in 3.6 Tier 2 flags is actual AI-fraud.

(c) Weekly detection costs: Tier 2 investigations (all flags): $(150,880 + 390,800) \times \$0.12 = 541,680 \times \$0.12 = \$65,002$. Tier 3 human review (20% of Tier 2): $541,680 \times 0.20 \times \$3.50 = 108,336 \times \$3.50 = \$379,176$. Total weekly detection cost: $\$65,002 + \$379,176 = \$444,178$.

(d) Weekly undetected AI-fraud consumer harm: $33,120 \times \$3.80 = \$125,856$. Weekly detected and prevented harm: $150,880 \times \$3.80 = \$573,344$. Net weekly benefit = $\$573,344 - \$444,178 = \$129,166$. The system is **cost-effective**: it produces a net weekly saving of \$129,000. However, the high Tier 3 cost (\$379,000/week) drives a significant fraction of total cost. Reducing the Tier 3 escalation rate from 20% to 8% (by improving Tier 2 precision) would reduce total weekly cost by \$151,670 and further improve ROI.

3. Virtual try-on consumer rights analysis.

(a) EU Consumer Rights Directive Article 6 requires pre-contractual information to include the “main characteristics of the goods.” The virtual try-on result constitutes a representation of how the product will appear when worn, which is a characteristic material to the purchase decision for apparel. If the try-on system’s SSIM of 0.58 is below the accuracy threshold at which the result can be considered a reliable representation of the product’s appearance, the retailer had an obligation to disclose the system’s accuracy limitations. A disclosure such as “This try-on is an AI simulation that may not exactly represent the garment’s appearance on your body” should be provided before the consumer relies on the try-on result as a basis for purchase.

(b) An SSIM of 0.58 for knitwear (vs. the industry accuracy floor of 0.75) represents a substantial accuracy gap that is both directional (favouring flattering results) and material (significantly below the accuracy level at which the try-on reliably represents the product). Under the Unfair Commercial Practices Directive, a commercial practice is misleading if it creates a false impression in the average consumer that influences their purchasing decision. A systematic flattering bias in a try-on system, resulting in purchases that would not have been made with an accurate representation, satisfies this definition. The practice is likely to constitute a misleading commercial practice.

(c) Retailer’s “lifestyle visualisation” defence: this defence has limited merit in this context. A

“lifestyle visualisation” disclaimer is appropriate for a mood board or aspirational image that is clearly stylised and not intended to represent how a specific product will look on the consumer’s body. A virtual try-on system that specifically places the consumer’s own photograph in the simulation and shows a person resembling the consumer wearing the garment creates a much stronger representation — closer to a product demonstration than a lifestyle image. The average consumer who uses a personalised virtual try-on would reasonably expect it to represent how the product will look on them, not a lifestyle aspiration. The defence is unlikely to succeed before an EU court or the CMA.

(d) Consumer remedy: the consumer is entitled to a full refund under the right of withdrawal (EU Consumer Rights Directive, 14-day withdrawal period) without needing to establish misrepresentation. If the withdrawal period has passed, the consumer may claim a partial or full refund on the basis of the misleading commercial practice (UCPD Article 11 remedies). Systemic change required by regulator: (1) all virtual try-on systems must display their SSIM accuracy metrics by garment category; (2) systems with SSIM below 0.75 for any consumer demographic must include a prominent disclaimer before the try-on result is displayed; (3) the retailer must submit an accuracy improvement plan to the regulator within 60 days.

Chapter 43

Deep Fakes in Mineral and Gemstone Authentication

“A diamond’s certificate is only as trustworthy as the scanner that produced it. Now the scanner can be faked too.”

—Anon.

The mineral and gemstone trade is one of the oldest and most sophisticated fraud arenas in the world economy. A single high-quality ruby from Mozambique can command \$200,000 per carat; a gem-quality D-colour, internally flawless diamond of three carats trades at over \$300,000. These extreme per-unit values have sustained centuries of fraud innovation, from glass substitutes and heated treatments to synthetic gemstones and doctored certificates. Generative AI has added a qualitatively new dimension to this fraud landscape: the ability to fabricate the digital certification infrastructure — grading reports, origin certificates, spectroscopic analysis records, chain-of-custody documentation — that the modern gemological trade relies upon to authenticate the physical objects themselves.

The critical structural insight for understanding AI-enabled gemstone fraud is the separation between physical authentication and document authentication. Physical testing — spectroscopy, inclusions mapping under magnification, crystallographic analysis — cannot be defeated by image synthesis. A genuine ruby and a synthetic ruby have different Raman spectra; a Burmese ruby and a Thai ruby have different trace element fingerprints; a natural diamond and a CVD-grown diamond have different nitrogen aggregation patterns. No AI system can generate a physical sample that passes these tests without being the genuine article. But the certificate that records these tests is a digital document, and digital documents can be fabricated. The fraud therefore operates not by synthesising gemstones but by synthesising the paperwork that attributes authentic gemstone properties to lesser goods.

The global coloured gemstone trade is valued in the tens of billions of dollars annually; the diamond trade is similarly large. Certificate fraud that enables a 50% price premium on misrepresented stones represents a potential annual fraud exposure of tens of billions of dollars at scale. The Kimberley Process Certification Scheme for rough diamonds, the CIBJO Blue Book standards for gemstone

nomenclature, and the grading reports issued by the Gemological Institute of America (GIA), the American Gem Society (AGS), and the International Gemological Institute (IGI) are the primary certification infrastructure targets.

This chapter develops the technical mechanisms, forensic detection approaches, and authentication infrastructure for AI-enabled mineral and gemstone fraud. Section 43.1 develops the fraud taxonomy across rough minerals, cut gemstones, and precious metals. Section 43.2 develops grading report fabrication and its detection via hash verification and spatial PRNU analysis. Section 43.3 develops spectroscopic authentication as the physical ground truth that AI cannot circumvent. Section 43.4 develops 3TG mineral certification fraud with its connections to conflict finance. Chapter efch:blockchain develops blockchain-based gemstone tracking in full.

43.1 Fraud Taxonomy: Rough, Cut, and Precious Metals

AI-enabled gemstone and mineral fraud operates at three distinct levels of the value chain, each with different document targets and different detection approaches.

43.1.1 Rough Diamond and Coloured Stone Origin Fraud

At the rough stage, the primary fraud target is the certificate of origin, which determines whether a diamond is a “conflict diamond” under the Kimberley Process and which country’s royalty and tax regime applies to the production. Rough diamonds from high-royalty jurisdictions (Zimbabwe: 15% royalty) may be relabelled as originating from lower-royalty jurisdictions (Botswana: 10%) through fabricated Kimberley Process certificates. The financial incentive for a 1,000-carat parcel of rough diamonds worth \$500,000: \$25,000 in avoided royalties.

AI-generated KP certificates replicate the format, security features (embossed seals, signature blocks), and official language of authentic certificates while substituting false origin country, production mine, and quantity data. Detection relies on:

1. Hash verification against the national diamond authority database of the claimed producing country — if the certificate hash does not match the authority’s record, it was not issued by that authority.
2. PRNU analysis against the scanner fleet of the claimed issuing authority — authentic certificates carry the fingerprint of the authority’s scanning infrastructure.
3. Multilateral intelligence cross-reference — if the claimed production quantity exceeds the known output of the claimed mine in the claimed period, the certificate is physically implausible.

43.1.2 Cut Gemstone Grading Report Fabrication

Cut gemstone grading reports are the most economically significant certificate targets because they directly determine retail price. A GIA grading report records the four Cs — colour, clarity, cut, and carat weight — along with fluorescence, proportions, and any treatments detected. The price

differential between adjacent grade levels can be 30–50% for colour (D versus E, or F versus G) and 40–70% for clarity (IF versus VVS1, or VS1 versus VS2).

Worked Example — Diamond grading report fabrication: detection and value analysis

A wholesale dealer presents a GIA grading report (certificate number 5191770521) for a 2.14 ct round brilliant diamond claimed as D-colour, Internally Flawless.

Hash verification: GIA Report Check (gia.edu/report-check) is queried with certificate number 5191770521. The GIA record shows: G-colour, VS2-clarity, 2.14 ct. The submitted report claims D/IF. **Certificate is fabricated.**

Price impact: At market rates, a D/IF 2.14 ct round brilliant: \$95,000. Authentic G/VS2 2.14 ct: \$23,500. Fraud premium: \$71,500 per stone.

Spatial PRNU: Block-wise NCC against the GIA scanner fingerprint shows $\rho \approx 0.037$ in text and border regions (authentic), $\rho \approx 0.004$ in the colour grade and clarity grade fields (inpainted). The two grade fields are the only modified regions, consistent with targeted inpainting of an authentic lower-grade report.

Fabrication method: the fraudster obtained the authentic G/VS2 report (perhaps for a stone they own), inpainted the two grade fields to show D/IF, and presented the composite as supporting documentation for a \$95,000 sale.

43.1.3 Precious Metal Assay Fraud

Gold bars, platinum group metals, and silver bullion are authenticated by hallmarks (struck by licensed assay offices) and, for investment-grade material, by serial numbers registered with the London Bullion Market Association (LBMA) or equivalent bodies. AI-enabled precious metal fraud takes three forms.

“Proof of product” advance fee fraud: AI-generated photorealistic images of large gold bar stacks are presented to prospective buyers as evidence of available inventory, in advance-fee schemes where the buyer is asked to pay transfer or storage fees before the (non-existent) gold is released. Detection: gold’s characteristic reflectance spectrum (sharp absorption edge at 480 nm producing the yellow colour, high NIR reflectance) is rarely accurately reproduced by AI generators trained on RGB images; spectrally calibrated analysis identifies AI-generated gold images by their atypical reflectance profile.

Tungsten-filled bar fraud: physical tungsten cores are clad with gold to match the correct weight of genuine bars, with AI-generated assay certificates fabricated to support the false purity claim. Detection requires physical testing (XRF, ultrasound) rather than document analysis; AI-generated certificates supporting these frauds are detectable by hash verification against the LBMA Good Delivery List.

LBMA registration number duplication: an authentic LBMA serial number from a legitimately sold and melted bar is reused in AI-generated documentation to support the sale of non-LBMA-compliant

bars. Detection by querying the LBMA's retired bar register.

43.2 Grading Report Digital Forensics

Grading report forensics combines three complementary approaches: hash registry verification (fastest and most decisive), spatial PRNU composite detection (identifies which fields were modified), and typographic and layout consistency analysis (detects AI-generated text inconsistencies in modified fields).

43.2.1 Hash Registry Verification

Every major gemological laboratory maintains a public online verification service: GIA Report Check, AGS certificate verification, IGI report check, and HRD Antwerp online verification. Querying the certificate number against the issuing laboratory's database takes under 30 seconds and provides the authoritative ground truth for any digitally submitted certificate. The procedure:

$$h_{\text{submitted}} = \text{SHA-256}(\text{certificate image}), \quad h_{\text{registry}} = \text{query}(\text{cert.\#, lab_API}), \quad (43.1)$$

$$\text{authentic iff } \text{grade}_{\text{submitted}} = \text{grade}_{\text{registry}}. \quad (43.2)$$

The hash comparison catches any modification to the image file; the grade comparison catches modifications that were made to a copy that was then re-scanned before digital submission, since the re-scan produces a different hash but the grades remain inconsistent with the registry record.

43.2.2 Spatial PRNU Composite Analysis

When the hash verification is unavailable (original paper certificate presented without database access) or inconclusive, spatial PRNU composite analysis localises the modified regions. For grading report inpainting, the attacker modifies only the specific grade fields, preserving the authentic scanner fingerprint in surrounding text, borders, and holographic elements. The block-wise NCC map reveals this spatial pattern:

$$\rho_{ij} = \text{NCC}(\hat{\mathbf{K}}_{ij}, \hat{\mathbf{r}}_{ij}), \quad (43.3)$$

where the NCC is computed over 32×32 pixel blocks for certificate images (smaller blocks than the 64×64 used for scene images, to achieve sufficient spatial resolution for identifying individual field modifications in dense text layouts). Authentic fields show $\rho_{ij} \approx 0.035\text{--}0.042$; inpainted fields show $\rho_{ij} \approx 0.003\text{--}0.008$.

43.3 Spectroscopic Authentication: The Physical Ground Truth

Spectroscopic authentication establishes physical ground truth through the measurement of atomic and molecular vibration signatures determined entirely by crystal chemistry.

Algorithm 11: Raman Gemstone Authentication Workflow**Input:** Unknown gemstone G ; reference Raman database $\mathcal{R}$; excitation wavelength λ_{ex} **Output:** Species identification, treatment flags, geographic origin estimate

```

1 Measure Raman spectrum  $S(k)$  over wavenumber range  $k \in [200, 4000] \text{ cm}^{-1}$ ;
  // Species identification
2  $d_{\text{species}} \leftarrow \min_{s \in \mathcal{R}} \|\text{DTW}(S, \mathcal{R}[s])\|$ ; // dynamic time warping distance
3 Assign species  $\hat{s}$  at  $d_{\text{species}} < \tau_{\text{species}}$ ;
  // Treatment detection (for corundum: ruby/sapphire)
4 if  $\hat{s} \in \{\text{ruby}, \text{sapphire}\}$  then
5   if peak at  $1030\text{--}1070 \text{ cm}^{-1}$  then
6     flag GLASS_FILLED; // SiO2 network vibration
7   end
8   if peak at  $960 \text{ cm}^{-1}$  then
9     flag BERYLLIUM_TREATED
10  end
11  if peak at  $858 \text{ cm}^{-1}$  anomalous then
12    flag HEAT_TREATED
13  end
14 end
  // Geographic origin via trace element fingerprint (requires LA-ICP-MS)
15 Measure [Cr], [Fe], [V], [Ga] in ppm;
16  $\hat{\text{origin}} \leftarrow \text{NearestNeighbour}([\text{Cr}, \text{Fe}, \text{V}, \text{Ga}], \mathcal{G}_{\text{provenance}})$ ;
17 return ( $\hat{s}$ , treatment flags,  $\hat{\text{origin}}$ )

```

Worked Example — Raman spectrum interpretation: ruby vs. glass fill detection

An unheated Burmese ruby is presented with its GIA certificate. Raman measurement at $\lambda_{\text{ex}} = 532 \text{ nm}$ reveals:

Primary Cr^{3+} luminescence peaks: 694.2 nm (R_1) and 692.8 nm (R_2) — characteristic of corundum, confirms ruby identity. Secondary peaks at 1032 and 1061 cm^{-1} : SiO_2 glass network vibration, indicating lead-glass filling in surface-reaching fractures. The GIA “Clarity Enhancement: None” notation is contradicted by the Raman spectrum.

Quantitative impact: a 3 ct Burmese ruby without treatment: $\$65,000/\text{ct} \times 3 \text{ ct} = \$195,000$. Same stone with undisclosed glass filling: $\$8,000/\text{ct} \times 3 \text{ ct} = \$24,000$. Misrepresentation value: $\$171,000$. The Raman measurement took 3 minutes and definitively contradicted the submitted certificate.

Physical spectroscopic authentication establishes ground truth that no AI-generated document can contradict: the stone must be present, the measurement must be performed on the actual material, and the spectrum is determined by physics rather than by any document. Three spectroscopic methods are primary in gemological authentication.

43.3.1 Raman Spectroscopy

Raman spectroscopy excites the gemstone with a monochromatic laser and measures the inelastic scattering spectrum produced by its crystal lattice vibrations. The Raman spectrum is unique to the

mineral species and sensitive to treatments, filling materials, and synthetic origin. For diamonds: natural diamonds show a single sharp Raman peak at 1332 cm^{-1} with specific line width; CVD-grown diamonds show the same peak position but with characteristic strain-related broadening; HPHT-treated diamonds show specific defect signatures at 1344 cm^{-1} .

Worked Example — Raman authentication of a claimed natural ruby

A stone is claimed to be a natural, unheated Burmese ruby. Raman spectrum analysis:

Primary Raman peak at 694 cm^{-1} (Cr^{3+} luminescence in corundum): confirmed ruby.

Secondary features: glass-filling signature at $1030\text{--}1070\text{ cm}^{-1}$ (SiO_2 network vibrations). Glass filling is a treatment that improves apparent clarity but reduces value significantly; an unheated Burmese ruby with glass filling is worth approximately 40% of an untreated equivalent. The submitted certificate claims no treatment.

Conclusion: the Raman spectrum contradicts the certificate. The stone has been glass-filled (a treatment) that was not disclosed in the submitted certificate. The certificate is either fraudulently altered or was issued without proper testing.

43.3.2 Trace Element Fingerprinting for Geographic Origin

Laser ablation inductively coupled plasma mass spectrometry (LA-ICP-MS) measures the trace element composition of a gemstone at the parts-per-billion level, producing a multi-element fingerprint that reflects the geological formation environment. Different gem-producing regions have characteristic geochemical signatures determined by their local geology:

Burmese rubies: high Cr ($>2000\text{ ppm}$), low Fe ($<300\text{ ppm}$), elevated Ga. Thai/Cambodian rubies: lower Cr ($<1500\text{ ppm}$), higher Fe ($>1000\text{ ppm}$), elevated V. Mozambique rubies: intermediate Cr, elevated Fe, characteristic V/Ga ratio.

A ruby claimed as Burmese (commanding the highest price premium) with a Thai geochemical fingerprint is misrepresented with respect to geographic origin. The trace element certificate can be fabricated; the trace element fingerprint of the actual stone cannot.

43.4 3TG Mineral Certification and Conflict Finance

Tin, tantalum, tungsten, and gold (3TG minerals) are subject to mandatory due diligence under Section 1502 of the US Dodd-Frank Act (for companies listed on US exchanges) and the EU Conflict Minerals Regulation (for EU importers above specified thresholds). These regulations require companies to conduct due diligence on whether their 3TG mineral supply chains contribute to conflict financing in the Democratic Republic of Congo (DRC) and adjoining countries, and to report their findings to the SEC (US) or European Commission (EU).

The compliance infrastructure relies on smelter certification (the Responsible Minerals Assurance Process, RMAP, audits smelters and refiners) and chain-of-custody documentation from mine to

smelter. AI-generated documentation fraud at this layer can insert false smelter certifications, alter the country of origin in chain-of-custody documents, or fabricate mine site compliance certificates, enabling conflict minerals to pass into compliant supply chains.

The financial incentive is significant: minerals from conflict-affected areas in the DRC are priced at a discount of 15–30% relative to verified conflict-free equivalents. A shipment of 50 tonnes of tantalum ore worth \$3 million at conflict-free prices would be worth \$2.1–2.55 million with accurate conflict-zone provenance documentation, creating a \$450,000–\$900,000 incentive for documentation fraud per shipment.

Detection combines: RMAP database verification (is the claimed smelter actually certified?), spatial PRNU analysis of certificate scans (were the certification dates or quantity fields inpainted?), and geochemical fingerprinting of the mineral itself against the claimed mine’s known profile.

43.5 Industry Applications

The gemological certification industry has invested significantly in digital security infrastructure that provides a first line of defence against AI-generated certificate fraud. GIA’s Report Check database processes over 10 million certificate verification queries annually; its 2023 upgrade added anomaly detection to flag unusual query patterns (multiple queries for the same certificate number from different IP addresses within a short time window) as a potential fraud signal. AGS has implemented QR-code-embedded cryptographic hashes on all certificates issued since 2022, enabling offline verification without network access. IGI partnered with Everledger (blockchain diamond tracking) to anchor all IGI certificate hashes on the Everledger distributed ledger from 2023.

The Responsible Minerals Initiative’s RMAP programme has expanded its digital verification infrastructure to include API access for supply chain management systems, enabling automated verification of smelter certifications at the point of purchase order rather than relying on paper certificate review. The EU Conflict Minerals Regulation’s 2021 implementation guidance specifically recommends digital hash verification of smelter certificates as a best practice for due diligence.

43.6 African Use Cases

Africa produces the majority of the world’s gemstones and a substantial fraction of its precious metals. South Africa, Botswana, and Namibia produce most of the world’s gem-quality diamonds. Mozambique and Madagascar are major ruby and sapphire sources. Zambia is among the world’s leading emerald producers. The DRC, Tanzania, and Rwanda are major coltan and cassiterite producers. In each of these countries, AI-generated certificate fraud represents a direct threat to the royalty revenues, export earnings, and conflict-finance prevention systems that underpin both government finance and international compliance obligations.

The Kimberley Process Certification Scheme’s Working Group on Digital Trade, in which several African producer countries are active members, has recommended mandatory blockchain anchoring of all Kimberley Process certificates by 2026. South Africa’s SADPMR (South African Diamond

and Precious Metals Regulator) has piloted a blockchain-anchored certificate system for South African rough diamonds in cooperation with De Beers Group's Tracr platform. The DRC's Centre d'Evaluation, d'Expertise et de Certification (CEEC) has been targeted by AI-generated certificate fraud documented in WCO intelligence reports, prompting a technical assistance request to the WCO for digital certificate verification infrastructure.

Chapter Summary

AI-enabled gemstone and mineral fraud operates at the document layer — fabricating the certificates that authenticate physical goods whose material properties cannot themselves be synthesised. The primary fraud targets are GIA, AGS, and IGI grading reports (whose grade field inpainting can support price premiums of 30–70% per grade level), Kimberley Process certificates for rough diamonds (enabling royalty avoidance and conflict diamond laundering), and 3TG mineral chain-of-custody documentation (enabling conflict minerals to enter compliant supply chains). Hash registry verification against GIA Report Check, AGS verification, and RMAP databases provides the fastest and most decisive detection. Spatial PRNU composite analysis localises the specific inpainted fields in modified certificates. Spectroscopic analysis (Raman, LA-ICP-MS) provides physical authentication that is definitively immune to document fabrication.

Exercises

1. A 3.02 ct round brilliant diamond is presented with a GIA grading report claiming E-colour, VVS1-clarity, Excellent cut. The claimed retail value is \$78,000. GIA Report Check for the certificate number shows: H-colour, SI1-clarity, Very Good cut. (a) What is the authentic retail value of the stone at H/SI1/VG? Assume $D/IF = 100\%$ index, $E/VVS1/Ex = 76\%$, $H/SI1/VG = 38\%$ relative to D/IF for a 3 ct round brilliant. (b) Calculate the fraud premium in dollars. (c) Spatial PRNU analysis shows $NCC = 0.004$ in the colour, clarity, and cut grade fields, and $NCC = 0.038$ in the certificate border and GIA logo regions. Is this consistent with full synthesis or targeted inpainting? Explain. (d) What is the minimum chain of custody documentation the buyer should have requested before accepting the certificate?
2. A company imports tantalum ore from a supplier claiming Rwandan origin (conflict-free) for use in capacitor manufacturing. The RMAP smelter certificate lists a certified smelter. (a) Describe the three-step verification procedure the importer should perform on the certificate. (b) Geochemical analysis of the ore sample shows a V/Ga ratio of 0.43 and Fe content of 1,840 ppm, inconsistent with the known Rwandan profile ($V/Ga < 0.20$, $Fe < 800$ ppm) but consistent with eastern DRC. Calculate the financial incentive for the fraudulent origin claim if the 50-tonne shipment is priced at \$2.8 million (conflict-free) versus a 25% discount for DRC origin. (c) Under Dodd-Frank Section 1502, what disclosure obligation does the importer have when it discovers the geochemical inconsistency?

Answers to Exercises

1. Diamond grading report fraud analysis.

(a) Authentic value at H/SI1/VG: using the $D/IF = 100\%$ index for a 3 ct round brilliant. The claimed E/VVS1/Ex retail value is 76% of D/IF. If the claimed value is \$78,000: D/IF equivalent value = $\$78,000 / 0.76 = \$102,632$. H/SI1/VG authentic value = $\$102,632 \times 0.38 = \$39,000$.

(b) Fraud premium: $\$78,000 - \$39,000 = \$39,000$ per stone. The fraudster sought to sell an \$39,000 stone for \$78,000 — a 100% mark-up achieved by falsifying two grade levels of colour (E vs. H) and two levels of clarity (VVS1 vs. SI1).

(c) PRNU pattern analysis: $NCC = 0.004$ in the three grade fields and $NCC = 0.038$ in border and logo regions. This pattern is consistent with **targeted inpainting**, not full synthesis. A full synthesis would produce near-zero NCC across the entire certificate. The high NCC in the border and GIA logo regions confirms that the base certificate is authentic (captured by GIA's scanning infrastructure and carrying GIA's scanner PRNU), while only the specific grade fields were replaced by AI-generated content. The boundary between authentic and inpainted regions would show intermediate NCC values (0.012–0.020) from the Poisson blending at the field borders.

(d) Minimum chain of custody documentation: (1) GIA Report Check verification queried at the time of transaction (not relying on a buyer-provided screenshot); (2) the stone's serial number laser-inscribed on the girdle, cross-referenced against the GIA record (GIA inscribes the report number on the girdle of all graded stones — if the inscription does not match the report, the pairing is fraudulent); (3) an independent assessment by a GIA Graduate Gemologist who physically examines the stone and confirms the grades match the presented certificate; (4) for transactions above \$10,000, an anti-money laundering declaration from the seller identifying the stone's provenance chain from original producer.

2. Tantalum ore 3TG compliance analysis.

(a) Three-step certificate verification procedure: *Step 1 — RMAP database query*: query the Responsible Minerals Initiative's RMAP database (responsibleminerals.org) for the listed smelter. Confirm the smelter is currently certified (not lapsed or suspended) and that its certification covers the specific mineral (tantalum) and the claimed country of origin (Rwanda) in the declared shipment period.

Step 2 — Certificate PRNU and hash verification: if the smelter certificate is submitted digitally, apply PRNU analysis against the RMAP-certified smelter's known document infrastructure fingerprint (obtainable from prior verified certificates or from RMAP's document integrity service). Hash the submitted certificate and compare against the RMAP record for that certification number.

Step 3 — Geochemical sample verification: request an independent sample from the shipment

and submit for LA-ICP-MS trace element analysis at an accredited laboratory. Compare the V/Ga ratio, Fe content, and other diagnostic trace elements against the known profiles for the claimed producing region (Rwanda) and adjacent conflict-affected areas (eastern DRC).

(b) Financial incentive calculation: Conflict-free value: \$2,800,000. DRC-origin discount 25%: authentic DRC value = $\$2,800,000 \times 0.75 = \$2,100,000$. Fraud premium: $\$2,800,000 - \$2,100,000 = \$700,000$ per shipment. The \$700,000 incentive per 50-tonne shipment more than justifies sophisticated documentation fraud, including the cost of obtaining and falsifying RMAP certificates and Rwandan chain-of-custody documents.

(c) Dodd-Frank Section 1502 disclosure obligations: upon discovering the geochemical inconsistency, the importer has several obligations. First, it must not use the ore in a product that will be reported as “DRC conflict-free” in its SEC filing — the geochemical evidence suggests the ore is from eastern DRC and may be conflict-affected. Second, it must file a Conflict Minerals Report with the SEC (Form SD) disclosing that it was unable to determine the conflict-free status of the tantalum in its products, and describing the due diligence steps taken. Third, it should report the suspected fraudulent documentation to the RMAP programme, which may investigate the supplier and the listed smelter. Fourth, it should preserve all documentation relating to the shipment and the fraud discovery for potential law enforcement referral — providing false conflict mineral declarations to the SEC is a securities violation.

Chapter 44

Deep Fakes in Pharmaceutical Counterfeiting and Drug Safety

“A counterfeit pill looks identical to the real thing. Now the documentation does too.”

—Anon.

Counterfeit and substandard medicines cause a very large annual mortality burden; WHO estimates — though contested as uncertain — suggest hundreds of thousands of deaths per year, with the highest burden in low-income countries. The majority of these deaths occur in low- and middle-income countries where regulatory capacity is limited, supply chains are fragmented, and patients lack the means to verify product authenticity. AI-generated pharmaceutical documentation has added a new and particularly dangerous dimension to this public health crisis: whereas physical counterfeit tablets required sophisticated manufacturing equipment, AI-generated packaging documentation, quality certificates, and marketing authorisation copies require only a laptop and a subscription to a commercial image generation service.

The pharmaceutical fraud ecosystem enabled by synthetic media operates across three distinct attack surfaces. The first is *listing fraud*: AI-generated product images and documentation are used to list counterfeit or substandard medicines on e-commerce platforms, online pharmacies, and social media marketplaces. The second is *supply chain document fraud*: AI-generated Good Manufacturing Practice (GMP) certificates, batch release records, marketing authorisation copies, and analytical test certificates are used to insert counterfeit goods into legitimate pharmaceutical supply chains at the wholesale level. The third is *regulatory submission fraud*: AI-generated clinical trial data summaries, bioequivalence study reports, and analytical method validations are used to support applications for marketing authorisation of substandard products.

Each attack surface has different economic incentives, different victim populations, and different detection approaches. The e-commerce surface causes the most diffuse harm — individual consumers purchasing medicines that do not work or contain harmful substitutes. The supply chain surface

causes the most concentrated institutional harm — healthcare systems that unknowingly administer ineffective or toxic medicines to patients in clinical settings. The regulatory submission surface is the rarest and most sophisticated but potentially the most systemic, as a fraudulently approved product can harm millions of patients before detection.

This chapter develops the pharmaceutical counterfeiting problem from the perspective of both the fraud mechanisms and the detection infrastructure. Section 44.1 develops the fraud taxonomy and the AI tools used at each level. Section 44.2 develops pharmaceutical serialisation as the primary authentication infrastructure. Section 44.3 develops AI-generated packaging detection. Section 44.1.2 develops GMP certificate and supply chain document forensics. Section 44.6 develops the African context, where counterfeit medicines cause the highest absolute mortality burden.

44.1 Fraud Taxonomy and AI Tools by Attack Surface

AI tools are deployed at different points in the pharmaceutical counterfeiting chain, each targeting a different authentication checkpoint. The choice of attack surface determines both the harm profile (diffuse consumer harm versus concentrated clinical harm) and the defensive response (consumer education versus supply chain document forensics).

AI tools are deployed at different points in the pharmaceutical counterfeiting chain, each targeting a different authentication checkpoint.

44.1.1 E-Commerce Listing and Packaging Fraud

Online pharmacies and marketplace sellers use AI image generation to produce photorealistic images of pharmaceutical packaging that misrepresent: the product name and active ingredient (showing a branded name for a generic or different compound); the dosage strength (showing 10 mg for a 5 mg product); the manufacturer identity (replicating a legitimate manufacturer's logo and packaging design for a counterfeit product); and the batch number, manufacturing date, and expiry date (fabricated to appear within validity).

Stable Diffusion with ControlNet conditioning on authentic packaging photographs produces convincing counterfeit packaging images at essentially zero marginal cost. LLMs generate the patient information leaflets, product descriptions, and technical specification text that accompany these images. The combination produces a complete fraudulent product listing that passes casual visual inspection and that may pass automated platform content moderation if the detection systems are not specifically calibrated for pharmaceutical packaging fraud.

44.1.2 GMP Certificate Fabrication

Good Manufacturing Practice (GMP) certificates are issued by national medicines regulatory authorities (such as the MHRA in the UK, EMA in the EU, FDA in the US, and their counterparts globally) to certify that a manufacturer's facilities and quality systems meet the standards required for pharmaceutical production. A valid GMP certificate is required for import of pharmaceutical ingredients and finished products in most jurisdictions.

AI-generated GMP certificates replicate the format, letterhead, authorised signature blocks, and official seals of authentic certificates while substituting false manufacturer names, facility addresses, certificate validity periods, and certifying authority references. The economic incentive is direct: an active pharmaceutical ingredient (API) manufacturer without GMP certification cannot legally supply regulated markets; an AI-generated GMP certificate enables market access worth millions of dollars per year for a sophisticated bad actor.

Worked Example — GMP certificate forensic analysis

A pharmaceutical wholesaler receives a digitally submitted GMP certificate purportedly issued by the MHRA for an Indian API manufacturer. Forensic analysis:

Test	Result	Verdict
MHRA register check	Manufacturer not listed	Fail
PRNU vs. MHRA scanner fleet	$NCC = 0.003$	Fail
LLM text detection	Score = 0.81 (above $\tau = 0.65$)	Fail
Signature NCC vs. reference	$\rho = 0.004$	Fail
Layout template consistency	3 field deviations	Fail
Overall	5/5 flagged	Fabricated

The MHRA publishes a register of all GMP certificates it has issued, searchable by manufacturer name, address, and certificate number. The submitted certificate’s manufacturer does not appear in this register. The PRNU NCC of 0.003 is far below the MHRA scanner fingerprint threshold of 0.015. The LLM text detection score of 0.81 indicates machine-generated certificate text. The layout shows non-standard field positioning and incorrect MHRA reference number format. The certificate is referred to the MHRA’s Intelligence and Enforcement Unit.

44.2 Pharmaceutical Serialisation as Authentication Infrastructure

Pharmaceutical serialisation is the strongest available authentication mechanism for finished medicines because it relies on a centralised repository of manufacturing records that cannot be retrospectively altered. The verification procedure is formalised as:

Algorithm 12: EMVS Pharmaceutical Pack Verification

Input: Scanned Data Matrix code encoding $(G, S, B, E) = (\text{GTIN}, \text{Serial}, \text{Batch}, \text{Expiry})$

Output: Authentication result and recommended action

```
1 Parse Data Matrix; validate GS1 format;
2 if format invalid then
3   | return SUSPECT: invalid barcode format; escalate
4 end
5 Query EMVS:  $\text{resp} \leftarrow \text{EMVS.verify}(G, S, B, E)$ ;
6 switch resp.status do
7   | case ACTIVE do
8     | EMVS.decommission( $G, S$ ) ; // mark as dispensed
9     | return AUTHENTIC: dispense to patient;
10  | end
11  | case NOT_FOUND do
12    | return SUSPECT: not registered; // strongest counterfeit signal
13    | Log; quarantine; notify national authority within 24 h;
14  | end
15  | case DECOMMISSIONED do
16    | return SUSPECT: already dispensed or recalled;
17    | Investigate duplicate supply chain event;
18  | end
19  | case RECALLED do
20    | return DO NOT DISPENSE: batch recalled;
21    | Segregate; follow recall procedure;
22  | end
23 end
```

Worked Example — EMVS verification outcome analysis for a suspect batch

A hospital pharmacy in Germany scans 100 packs from a new delivery of a high-value oncology medicine. EMVS query results:

EMVS response	Count	Action
ACTIVE (pass)	94	Dispense after decommissioning
NOT_FOUND	4	Quarantine; notify Paul-Ehrlich-Institut
DECOMMISSIONED	2	Quarantine; investigate chain of custody
Total	100	

The 4 NOT_FOUND results indicate packs with serial numbers that do not appear in the EU EMVS, consistent with AI-generated packaging using fabricated serial numbers. The 2 DECOMMISSIONED results indicate serial numbers previously verified at another dispensing point, consistent with parallel importation of legitimately used packaging refilled with counterfeit product (a more sophisticated attack). The 6% failure rate from a single delivery is highly anomalous (the expected EMVS failure rate from legitimate product is below 0.01%) and triggers immediate quarantine of the entire batch pending investigation.

The expected daily EMVS verification volume across the EU is 10 million transactions, with a target system uptime of 99.9% (≤ 8.76 hours/year downtime). The system processes each verification in <200 ms end-to-end, compatible with real-time point-of-dispensing workflow.

Pharmaceutical serialisation assigns a unique identifier to every individual medicine pack, registered in a centralised or federated repository at the time of manufacture. This identifier is encoded in a 2D barcode (Data Matrix code) on the pack and can be verified at any point in the supply chain by scanning the barcode and querying the repository. The EU Falsified Medicines Directive (FMD, 2019) mandates serialisation for all prescription medicines in the European Economic Area; the US Drug Supply Chain Security Act (DSCSA) mandates it for all prescription drugs dispensed in the US; and equivalent systems are in place in Brazil, South Korea, Turkey, and Saudi Arabia, with over 50 countries implementing serialisation programmes.

44.2.1 Serialisation Verification Procedure

A pack's authenticity can be verified at any stage of the supply chain using the serialisation code:

1. Scan the Data Matrix code to extract the GTIN (Global Trade Item Number), serial number, batch number, and expiry date.
2. Query the national or pan-European Medicines Verification System (EMVS in the EU; DSCSA Trading Partner system in the US) with the extracted data.
3. The system confirms: is this serial number registered? Is it in active status (not decommissioned, recalled, or previously verified and dispensed)? Does the batch number and expiry date match the registration record?

An AI-generated packaging image with a fabricated serialisation code is detectable at step 2: the fabricated serial number either does not appear in the repository (unregistered) or has already been verified and decommissioned (if the fraudster copied a valid code from a legitimate pack). An AI-generated pack with a valid-format but unregistered serial number therefore fails verification immediately on system query.

44.2.2 Attack Vectors Against Serialisation

Serialisation provides strong but not unconditional authentication. Three attack vectors target its limitations:

Serial number replay: a legitimate serial number scanned and recorded from an authentic pack is reprinted on a counterfeit pack. Detection: the verification system tracks scan events; a serial number that has been verified at dispensing and then presented for verification again triggers a decommissioning alert.

Late-stage injection: counterfeit product is introduced after the serialisation verification point, between the verification scan and the patient. This requires complicity at the dispensing point and is the most difficult to prevent through serialisation alone.

Repository fraud: in jurisdictions where serialisation is not yet fully implemented or where the repository has weak access controls, an attacker attempts to register fabricated serial numbers directly into the repository, creating “authentic” digital records for non-existent or counterfeit product. This requires compromise of the repository system and is the most sophisticated and rare attack vector.

44.3 AI-Generated Packaging Image Detection

Pharmaceutical packaging has distinctive physical properties that AI image generators reproduce imperfectly: holographic security features, embossed lettering, specific foil reflectance patterns, and the characteristic micro-printing of serialisation Data Matrix codes at high resolution. Platform-level detection of AI-generated pharmaceutical listing images applies a domain-specific classifier trained on authentic pharmaceutical packaging images and known counterfeit packaging, achieving AUC ≈ 0.91 in published evaluations on balanced benchmarks.

The specific forensic signals for pharmaceutical packaging include:

Holographic element consistency: authentic pharmaceutical packs include holographic security features (tamper-evident holograms, colour-shifting foils) whose optical properties are angle-dependent. AI-generated images of these features show a static, single-angle appearance without the characteristic colour shift, detectable by multi-angle image capture.

Data Matrix code validity: the Data Matrix code on an authentic pack encodes a valid GTIN, serial number, batch number, and expiry date in a specific GS1 format. AI-generated pack images often include syntactically invalid or format-inconsistent Data Matrix codes, detectable by decoding the barcode from the image and checking format compliance.

Print substrate forensics: authentic pharmaceutical packaging is printed on specific substrate materials (polymer film, aluminium foil laminate, specific paper grades) with characteristic surface texture visible under macro-photography. AI-generated packaging images show generic smooth textures that lack the substrate-specific appearance of authentic pharmaceutical packaging.

44.4 GMP and Supply Chain Document Forensics

Supply chain documents — GMP certificates, Certificate of Analysis (CoA), batch manufacturing records, transport temperature records, and import licences — are the primary authentication documents in the pharmaceutical wholesale supply chain. Each has specific forensic characteristics and specific authentic registries against which fabrications can be verified.

GMP certificates: issued by national regulatory authorities, verifiable against their public registers. MHRA eIFCC (electronic Import Facility Certificate of Conformance), EMA EUDRAGMP database, and US FDA Establishment Registration database are the primary verification sources. All three provide online, real-time verification.

Certificate of Analysis (CoA): issued by the product manufacturer, documenting analytical test results

for each batch. CoA forensics applies LLM text detection (analytical results text is often LLM-generated), PRNU analysis (against the manufacturer's known document infrastructure), and analytical value plausibility (are the reported assay values, related substance levels, and microbiological results physically plausible and internally consistent?).

Batch manufacturing records (BMR): internal manufacturing documentation required by GMP regulations. BMR fraud is detectable by: timestamp consistency (process steps must occur in plausible sequence with plausible durations), equipment reference consistency (equipment IDs must match the manufacturer's registered equipment list), and operator signature verification against HR records.

44.5 Industry Applications

The pharmaceutical authentication industry has responded to AI-enabled counterfeiting with both technical and regulatory tools. The GS1 standards body has published guidance on AI-robust serialisation code design that includes checksum verification and format constraints that make valid-format serial number generation more difficult. The Pharmaceutical Distribution Security Alliance (PDSA) and the European Federation of Pharmaceutical Industries and Associations (EFPIA) have both published industry guidance on AI-generated document detection as a supplement to standard supply chain authentication procedures.

Interpol's Operation Pangea, which targets online pharmaceutical crime annually, has specifically incorporated AI-generated listing detection into its assessment criteria. In recent annual operations, AI-generated certification documentation has been identified as a recurring feature in supplier websites investigated under the programme.

44.6 African Use Cases

Sub-Saharan Africa bears the highest per-capita burden of counterfeit medicine harm. The WHO estimates that 1 in 10 medicines in low- and middle-income countries is substandard or falsified; in some Sub-Saharan African markets this figure reaches 1 in 5. Antimalarials, antibiotics, and antiretrovirals — the medicines most critical for the region's disease burden — are also the most frequently counterfeited, because they are high-demand, relatively high-value, and purchased by populations with limited ability to verify authenticity.

The African Medicines Regulatory Harmonisation (AMRH) initiative, coordinated by the African Union Development Agency (AUDA-NEPAD), has prioritised pharmaceutical serialisation as a continental public health intervention. The African Union Model Law on Medical Products Regulation encourages member states to implement serialisation systems compatible with the GS1 standards used in the EU and US. Ghana's Food and Drugs Authority (FDA-Ghana) and Nigeria's NAFDAC have both launched pilot serialisation programmes; South Africa's SAHPRA requires serialisation for all medicines manufactured or imported for the South African market from 2024.

AI-generated GMP certificates specifically targeting African markets have been documented by

NAFDAC's post-market surveillance unit, which in 2023 identified fabricated MHRA and EMA GMP certificates supporting API imports from non-certified manufacturers in three separate cases. The certificates were detected through hash verification against the MHRA and EMA online registers — emphasising that the verification infrastructure exists and is freely accessible, but must be systematically applied rather than relied upon as a backstop.

Chapter Summary

Pharmaceutical counterfeiting enabled by AI-generated documentation operates across three attack surfaces: e-commerce listing fraud using AI-generated packaging images, wholesale supply chain document fraud using AI-generated GMP certificates and CoAs, and (rarely) regulatory submission fraud using AI-generated clinical data. Pharmaceutical serialisation under the EU FMD and US DSCSA provides the strongest authentication: a fabricated serialisation code that does not appear in the EMVS or DSCSA repository is definitively identified as invalid on verification query. GMP certificate fabrication is detectable through online register verification (MHRA, EMA EUDRAGMP, FDA Establishment Registration), PRNU analysis against the issuing authority's document infrastructure, and LLM text detection on the certificate text. The African context presents the highest mortality burden from counterfeit medicines and the least mature serialisation infrastructure, making capacity-building in document forensics a public health priority.

Exercises

1. A pharmaceutical wholesaler in Kenya receives a shipment of antimalarial tablets (artemether-lumefantrine) accompanied by a GMP certificate purportedly issued by the MHRA for the Indian manufacturer, a Certificate of Analysis, and packaging claiming EU-authorised origin. The NAFDAC officer performs the following checks: (a) queries the MHRA register (manufacturer not found); (b) scans the Data Matrix code — the EMVS returns “serial number not registered”; (c) applies PRNU analysis to the GMP certificate scan ($NCC = 0.006$). What conclusion should the officer reach? List three immediate actions that should be taken. What is the public health implication if this shipment of 50,000 treatment courses reaches patients?
2. Design a pharmaceutical supply chain authentication protocol for a low-resource African healthcare system that cannot afford the full serialisation infrastructure of the EU FMD system. The protocol must: detect AI-generated GMP certificates, detect counterfeit packaging without serialisation database access, and be operable by healthcare workers with a smartphone but without specialist forensic equipment. Specify each step, the tool required, the time per check, and the decision threshold.

Answers to Exercises

1. **Antimalarial authentication failure analysis.**

Three independent authentication failures have occurred simultaneously: (a) GMP register failure: the manufacturer is not listed in the MHRA's published register of GMP-certified facilities — the certificate is therefore either fabricated or expired. (b) Serialisation failure: “serial number not registered” means the product was not manufactured by a registered manufacturer that enrolled packs in the EMVS, confirming it was not legitimately manufactured under a valid marketing authorisation in the EU. (c) PRNU failure: $NCC = 0.006$ is far below the MHRA authentication threshold of 0.015, confirming the GMP certificate was not produced by MHRA's scanning infrastructure.

Conclusion: all three independent authentication systems have failed. The shipment is confirmed as counterfeit with high confidence. The Certificate of Analysis and packaging claims should also be considered fabricated.

Three immediate actions: (1) Quarantine the entire shipment immediately — do not distribute any part of it. (2) Notify NAFDAC's post-market surveillance unit with full documentation of the authentication failures, the supplier details, and the import documentation. (3) Preserve chain of custody of the shipment and all documents for law enforcement investigation; do not return goods to the supplier or destroy evidence.

Public health implication: 50,000 treatment courses of substandard antimalarials reaching patients in Kenya would be catastrophic. Artemether-lumefantrine is the WHO first-line treatment for uncomplicated *Plasmodium falciparum* malaria. If the active pharmaceutical ingredients are absent, at sub-therapeutic levels, or substituted with inactive excipients, patients who appear to be receiving treatment will not clear their infection, leading to disease progression to severe malaria with a fatality rate of 15–25% in untreated severe cases. At a conservative estimate of 5% treatment failure leading to severe disease and 15% mortality in severe cases: $50,000 \times 0.05 \times 0.15 = 375$ preventable deaths from this single shipment.

2. Low-resource pharmaceutical authentication protocol.

A practical smartphone-based protocol for low-resource settings:

Step 1 — GMP Register Check (2 minutes, requires mobile internet): Open the MHRA GMP register (mhra.gov.uk/gxp-inspections) or EMA EUDRAGMP (eudragmp.ema.europa.eu) on a smartphone browser. Search for the manufacturer name on the GMP certificate. Decision: if not found → reject shipment and notify supervisor. If found → proceed to Step 2.

Step 2 — Barcode Scan and Format Check (1 minute, no internet required): Use a free GS1 barcode scanning app (e.g., GS1 UK's Verified by GS1) to decode the Data Matrix code on the packaging. Check: is the decoded data in valid GS1 format (GTIN + serial + batch + expiry)? If format is invalid or undecodable → suspect AI-generated packaging, escalate. If internet available: query the EMVS or national verification system. Decision: unregistered serial → reject.

Step 3 — Physical Packaging Inspection (3 minutes, no equipment): Inspect the holographic

security feature: tilt the pack under ambient light and observe colour change. Authentic holograms show distinct colour change at different angles; printed copies of holographic images do not. Inspect the embossed lettering: authentic pharmaceutical packs have tactile embossing; AI-generated packaging images printed on paper or standard packaging material lack this. Decision: no colour change → suspect counterfeit, escalate.

Step 4 — Photograph and Report (5 minutes): Photograph the GMP certificate, packaging label, Data Matrix code, and batch number. Submit via NAFDAC's WhatsApp tip line (+234-800-NAFDAC or national equivalent) with the authentication results. Await guidance before releasing the shipment.

Total time: approximately 11 minutes per shipment inspection. No specialist equipment required beyond a smartphone with mobile internet access.

Chapter 45

Blockchain and Distributed Ledger Authentication for Synthetic Media

“An immutable ledger does not make content authentic. But it makes post-issuance fabrication immediately detectable.”

—Anon.

Blockchain and distributed ledger technology (DLT) provide a document authentication mechanism that is structurally resistant to the post-hoc fabrication that AI enables. The core property is immutability: once a document’s cryptographic hash is recorded on a distributed ledger, any subsequent modification to the document produces a hash mismatch that is detectable by any party with access to the ledger, without requiring trust in any single central authority. This property makes blockchain-anchored document systems a compelling complement to AI-generated content detection: passive detection asks whether content appears synthetic; provenance-anchored authentication asks whether content can be proven authentic, and treats unverifiable content as unverified.

This chapter develops blockchain authentication as a technical countermeasure to AI-generated document and media fraud across all the application domains examined in earlier chapters. Section 45.1 develops the cryptographic fundamentals of hash anchoring. Section 45.2 develops blockchain provenance for media content including C2PA integration. Section 45.3 develops blockchain applications in trade document authentication (extending Chapter 40). Section 45.4 develops gemstone and mineral tracking (extending Chapter 43). Section 45.5 develops pharmaceutical supply chain applications (extending Chapter 44). Section 45.6 analyses the fundamental limitations of blockchain authentication and the attacks that circumvent it.

45.1 Cryptographic Fundamentals of Document Hash Anchoring

Document hash anchoring creates an immutable link between a document’s content and a distributed ledger record. The procedure is formalised as follows:

Algorithm 13: Merkle Tree Document Anchoring and Verification**Input:** Batch of N documents $\{D_1, \dots, D_N\}$; blockchain API**Output:** Merkle root h_{root} on chain; verification API for document holders

```

// Batch anchoring (issuer side)
1 for  $i \leftarrow 1$  to  $N$  do
2    $h_i \leftarrow \text{SHA-256}(D_i)$ ; // leaf hash
3 end
4 Pad to power of 2: append  $h_N$  copies until  $|\{h_i\}| = 2^{\lceil \log_2 N \rceil}$ ;
5  $\mathcal{T} \leftarrow$  build Merkle tree from  $\{h_i\}$ ;
6  $h_{\text{root}} \leftarrow \mathcal{T}.\text{root}$ ;
7  $\text{tx} \leftarrow \text{blockchain.submit}(h_{\text{root}})$ ; store timestamp  $t_{\text{chain}}$ ;

// Document verification (holder side)
// Verification function:  $\text{Verify}(D, h_{\text{root}}, \text{proof})$ 
8  $h \leftarrow \text{SHA-256}(D)$ ; // recompute leaf hash
9 for each sibling hash  $h_s$  in proof (ordered leaf  $\rightarrow$  root) do
10    $h \leftarrow \text{SHA-256}(h \parallel h_s)$ ; // or  $\text{SHA-256}(h_s \parallel h)$  per tree structure
11 end
12 return  $h = h_{\text{root}}$ ; // True iff document matches anchored batch

```

Worked Example — Merkle verification for a GIA grading report

GIA anchors 10,000 daily reports as a single Merkle tree (depth $\lceil \log_2 10,000 \rceil = 14$). The blockchain transaction records h_{root} at timestamp t_{issue} .

A dealer holding certificate #5191770521 wishes to verify authenticity six months later:

1. Compute leaf hash: $h_{\text{cert}} = \text{SHA-256}(\text{certificate PDF}) = 3a7f \dots c2e1$.
2. GIA's API returns a Merkle proof: 14 sibling hashes.
3. Apply 14 hash operations: confirms h_{root} matches the chain record.
4. Also query GIA Report Check: confirms grade fields match registry.

Total computation: 15 SHA-256 operations (≈ 0.3 ms on any device). An AI-modified certificate — even if it passes casual visual inspection — fails Step 3 because $\text{SHA-256}(\text{modified PDF}) \neq h_{\text{cert}}$. The modification is detected regardless of how sophisticated the AI inpainting is.

Document hash anchoring records the cryptographic fingerprint of an authenticated document on a distributed ledger at the time of its issuance, creating a permanent, publicly verifiable record that can be used to detect any subsequent modification.

45.1.1 SHA-256 Hash Properties

The SHA-256 hash function maps any input document D of arbitrary length to a 256-bit (32-byte) output:

$$h = \text{SHA-256}(D) \in \{0, 1\}^{256}, \quad (45.1)$$

with three properties essential for authentication:

Determinism: the same document always produces the same hash.

Collision resistance: it is computationally infeasible to find two different documents $D_1 \neq D_2$ such that $\text{SHA-256}(D_1) = \text{SHA-256}(D_2)$. The expected number of documents that must be hashed before finding a collision is $2^{128} \approx 3.4 \times 10^{38}$ — effectively infinite for any practical attacker.

Avalanche effect: any change to the document — even a single bit — produces a completely different hash with no predictable relationship to the original. An AI-modified document that differs from the original in any field (a changed grade, a different date, an altered quantity) produces a hash value that is uniformly distributed over the 256-bit space and has zero probability of matching the original hash.

45.1.2 Merkle Tree Batch Anchoring

Anchoring every document individually on a public blockchain (at a cost of \$0.50–\$5 per transaction on Ethereum) is economically impractical for high-volume document issuers. Merkle tree batch anchoring reduces this cost by recording a single root hash representing a batch of N documents:

$$h_{\text{root}} = \text{MerkleRoot}(h_1, h_2, \dots, h_N), \quad (45.2)$$

where the Merkle root is computed by pairwise hashing of the leaf hashes up the tree. A single blockchain transaction records h_{root} , anchoring all N documents at the cost of one transaction. Individual document verification requires the document hash and a Merkle proof (a path of $\log_2 N$ intermediate hashes from the leaf to the root), verifiable against the recorded root without revealing the other documents in the batch.

Worked Example — Merkle batch anchoring for a gemological laboratory

GIA issues 10,000 grading reports per day. Anchoring each individually at \$2 per Ethereum transaction: $10,000 \times \$2 = \$20,000/\text{day} = \$7.3\text{M}/\text{year}$. Merkle batch anchoring with 10,000 reports per batch: 1 transaction per day at \$2. Verification requires a Merkle proof of $\log_2(10,000) \approx 14$ hashes — 14 hash computations taking microseconds. Cost reduction: $10,000\times$. GIA issues verification API that takes a certificate number and returns the Merkle proof for client-side verification against the daily root hash recorded on the Ethereum mainnet.

45.2 Blockchain Provenance for Media Content

The Coalition for Content Provenance and Authenticity (C2PA) standard, developed in Chapter 13, provides a cryptographic manifest embedded in or associated with media files that records the creation context, the tools used, and any declared AI involvement. C2PA's trust architecture is certificate-based: the device or software that creates the manifest signs it with a private key whose corresponding certificate is issued by a trusted Certificate Authority. The manifest is verifiable by any party with access to the public key infrastructure.

Blockchain integration extends C2PA in two ways. First, it provides decentralised trust: rather than relying on a centralised CA to issue and revoke certificates, the trust anchor can be a public key registered on an immutable distributed ledger, eliminating the single point of failure of CA

compromise. Second, it enables persistent public verifiability: a C2PA manifest stored as a sidecar file alongside a media file may be lost when the file is shared without its sidecar; a blockchain-anchored hash of the manifest persists independently of the file's distribution path.

45.2.1 The Content Authenticity Initiative Blockchain Integration

The Content Authenticity Initiative (CAI), the industry consortium that operationalises C2PA standards, has published a technical specification for blockchain-backed provenance that combines C2PA manifest signing with Merkle tree anchoring on the IPTC News Exchange Format compatible distributed ledger. Under this architecture:

1. The capturing device or authoring tool signs the C2PA manifest with its private key.
2. The signed manifest is included in the media file sidecar or embedded in the file header.
3. The SHA-256 hash of the signed manifest is submitted to a CAI-operated anchoring service.
4. The anchoring service batches hashes and records the Merkle root on a public blockchain daily.
5. Any downstream recipient can verify: (a) the manifest signature (cryptographic), (b) the manifest hash against the blockchain record (immutability), and (c) the declared content metadata against the manifest (disclosure).

45.3 Trade Document Blockchain Applications

The trade document authentication applications of blockchain technology were introduced in Chapter 40. This section develops the technical architecture in greater depth.

45.3.1 Letters of Credit on Distributed Ledgers

The letter of credit (LC) lifecycle involves at least four parties (buyer, buyer's bank, seller, seller's bank) and a sequence of document presentations and verifications that currently rely on paper or PDF exchange with human verification. Blockchain-based LC platforms (Contour, Marco Polo, we.trade) replace the paper/PDF exchange with a shared distributed ledger on which all parties have a permissioned view and on which document presentations and acceptances are recorded as on-chain transactions.

The AI document fraud resistance of this architecture is direct: once the issuing bank records the LC terms on the ledger, any subsequent presentation of a document claiming those terms is automatically verified against the immutable record. An AI-modified LC claiming a higher credit amount, a different beneficiary, or an extended validity period does not match the ledger record and is rejected without human review.

45.3.2 Certificate of Origin Anchoring

Certificates of origin for preferential tariff claims (AfCFTA, ASEAN FTA, EU-Africa Economic Partnership Agreements) are a primary target for AI fabrication because they enable duty savings worth thousands of dollars per shipment. Blockchain-anchored CoO systems have been piloted by the WTO, the Singapore Customs Authority, and the Port of Rotterdam as part of the International Chamber of Commerce Digital Trade initiatives.

The technical architecture: the issuing Chamber of Commerce computes the SHA-256 hash of the certificate at issuance and submits it to a permissioned blockchain (typically Hyperledger Fabric or R3 Corda). The customs authority at the destination port queries the blockchain before accepting the CoO as basis for preferential tariff treatment. An AI-generated CoO does not appear in the blockchain record and is therefore rejected automatically. The pilot implementations have reduced AI-generated CoO fraud to zero in the covered trade lanes.

45.4 Gemstone and Mineral Blockchain Tracking

Gemstone blockchain tracking assigns a unique digital identity to each stone or mineral parcel at the point of origin and records each custody transfer on an immutable ledger, creating a tamper-evident chain of custody from mine to retail. This architecture directly addresses the certificate fabrication problem: an AI-generated grading report claiming a stone's characteristics does not match the blockchain record of what was actually measured at the certified testing facility.

45.4.1 The Tracr Platform

De Beers Group's Tracr platform is the most widely deployed diamond tracking system. Tracr assigns a unique Digital Diamond Identity (DDI) to each rough diamond, anchored to physical characteristics measured at the mine (weight, fluorescence, shape metrics). Each subsequent transformation — rough sorting, cutting and polishing, grading — is recorded on the Tracr blockchain, with the new physical characteristics (for a polished stone: exact carat weight, colour measurement, clarity grade, proportions) linked to the DDI. Retailers can query the DDI and receive the complete custody chain from mine to display case.

An AI-generated grading report for a Tracr-enrolled diamond would claim characteristics inconsistent with the Tracr record for that DDI, detectable by querying the DDI at any verification terminal.

45.4.2 Everledger's Gemstone Provenance Ledger

Everledger extends the tracking concept beyond diamonds to coloured gemstones, art, wine, and luxury goods. For gemstones, Everledger records over 40 physical attributes of each stone (including laser-inscription details, clarity plot mapping, spectroscopic summary) on its proprietary distributed ledger. The IGI grading laboratory has anchored all IGI certificates on the Everledger ledger since 2023, enabling direct verification of IGI certificate authenticity by querying Everledger with the certificate number.

45.5 Pharmaceutical Supply Chain Blockchain

The EU Falsified Medicines Directive’s EMVS (European Medicines Verification System) is the largest deployed pharmaceutical blockchain-adjacent system, though it uses a federated database architecture rather than a public blockchain. Over 2.5 billion pack serialisation records are enrolled; approximately 10 million verification transactions occur daily at European dispensing points. The system architecture:

$$\text{Verify}(\text{GTIN}, \text{SN}, \text{Batch}, \text{Expiry}) = \begin{cases} \text{OK} & \text{if record active in EMVS} \\ \text{FAIL} & \text{if not registered or decommissioned} \end{cases} \quad (45.3)$$

The system has detected over 40,000 suspected falsified medicines since its 2019 launch, with AI-generated packaging a growing fraction of detected cases.

For fully blockchain-based pharmaceutical authentication: IBM Food Trust (repurposed for pharmaceutical supply chain) and MediLedger provide Hyperledger Fabric-based platforms for pharmaceutical wholesale. MediLedger’s network, used by major US pharmaceutical distributors and manufacturers, records product transfers on a permissioned blockchain visible to all supply chain participants, enabling AI-generated supply chain document fraud to be detected by cross-referencing document claims against the on-chain transfer history.

45.6 Fundamental Limitations of Blockchain Authentication

Blockchain authentication is a powerful but not unconditional fraud countermeasure. Five fundamental limitations constrain its effectiveness:

Garbage in, garbage out: a blockchain record proves that a document existed in a specific state at a specific time. It does not prove that the document’s content is truthful. If an AI-generated GMP certificate is enrolled in a blockchain before detection, the blockchain proves the fabricated certificate existed — not that the manufacturer is GMP-certified.

Key compromise: if the private key used to sign a document is compromised, an attacker can create validly signed blockchain-enrolled documents without the authority of the key’s legitimate owner. Blockchain systems require the same PKI security disciplines as any other key-based authentication system.

Bridge attacks: in hybrid systems where physical documents are converted to digital form for blockchain enrollment, the conversion process is an attack surface. A physical GMP certificate can be AI-fabricated, scanned to produce a digital file, and enrolled on a blockchain before the fabrication is detected — after which the blockchain record supports rather than refutes the fraud.

Oracle problem: when blockchain authentication is linked to physical world properties (the characteristics of a specific diamond, the GMP compliance of a specific facility), the link between the on-chain

record and the off-chain reality requires a trusted oracle — a physical inspection or test that is itself subject to fraud. The blockchain secures the record; the oracle secures the record's accuracy.

Jurisdictional enforceability: a blockchain record has evidential value in legal proceedings only where courts accept distributed ledger records as evidence. This acceptance is established in Singapore, the UAE, several US states, and is growing in the EU, but remains uneven across many African and Asian jurisdictions.

45.7 Industry Applications

The blockchain authentication market for trade, provenance, and media has attracted significant investment across multiple sectors. Everledger has raised over \$30 million and tracks over 4 million registered diamonds; De Beers' Tracr platform covers over 1 million diamonds per year. The Contour trade finance blockchain processes letters of credit for 15 major international banks including HSBC, Standard Chartered, and BNP Paribas. MediLedger's pharmaceutical network includes AmerisourceBergen, McKesson, Cardinal Health, and major pharmaceutical manufacturers.

On the media provenance side, the CAI has grown to over 2,000 member organisations and the C2PA Technical Specification has been incorporated into Adobe's suite, Microsoft products, and camera firmware from Leica, Sony, and Nikon. The integration of blockchain hash anchoring into C2PA workflows is in active development within the CAI Technical Working Group.

45.8 African Use Cases

Several African initiatives are deploying blockchain authentication specifically for the fraud prevention use cases described in this chapter. The SADPMR (South African Diamond and Precious Metals Regulator) has piloted blockchain-anchored rough diamond certificates for South African production in cooperation with De Beers' Tracr platform, with full deployment planned by 2026. The African Continental Free Trade Area Secretariat has identified blockchain-anchored certificates of origin as a priority infrastructure project for preventing AfCFTA certificate fraud, with the ICC Digital Trade Standards Initiative providing technical assistance.

The Pan-African Payment and Settlement System (PAPSS), operated by Afreximbank, provides a digital payments infrastructure that could serve as the settlement layer for a blockchain-based African trade document verification network. The African Export-Import Bank (Afreximbank) and African Development Bank have both funded blockchain pilot projects for trade finance in Africa, providing a financial infrastructure foundation for expanding document authentication blockchain deployments.

Chapter Summary

Blockchain and distributed ledger technology provides the strongest structural defence against post-hoc AI document fabrication: SHA-256 hash anchoring means any modification to an enrolled document

produces a detectable hash mismatch regardless of the visual quality of the fabricated content. Merkle tree batch anchoring reduces the per-document anchoring cost from \$2–\$5 to fractions of a cent at volume. C2PA integration with blockchain hash anchoring extends the provenance guarantee to media content, combining the signing authority of the CA infrastructure with the immutability of the distributed ledger. Applications span trade document authentication (Contour, Marco Polo), gemstone tracking (Tracr, Everledger), pharmaceutical supply chain (MediLedger, EMVS), and media provenance (CAI/C2PA). Five fundamental limitations constrain effectiveness: garbage-in-garbage-out (blockchain proves existence, not truth), key compromise, bridge attacks at the physical-to-digital conversion point, the oracle problem for physical world linkage, and variable jurisdictional enforceability.

Exercises

1. A gemological laboratory issues 8,000 grading reports per day. It wishes to implement Merkle tree batch anchoring on the Ethereum mainnet. (a) Calculate the daily cost of batch anchoring versus per-document anchoring at \$3 per transaction. (b) A report holder needs to verify their certificate. Describe the verification procedure, including what data the holder needs and how many hash computations are required. (c) An attacker attempts to create a fraudulent certificate with a grade-field inpainting and enroll it in the blockchain before detection. Describe the attack timeline and identify the specific point in the workflow where this attack can be prevented.
2. The EU Falsified Medicines Directive's EMVS processes 10 million verification events per day across Europe. (a) Using the verification equation (45.3), calculate the expected daily number of failed verifications if 0.008% of verification events involve AI-generated packaging. (b) If each failed verification triggers a manual pharmacist investigation taking 8 minutes, calculate the total investigation burden in pharmacist-hours per day. (c) Propose a triage priority system that reduces this burden by directing resources to the highest-risk failures first.

Answers to Exercises

1. Merkle batch anchoring cost-benefit analysis.

(a) Per-document anchoring: $8,000 \times \$3 = \$24,000/\text{day} = \$8.76\text{M}/\text{year}$. Batch anchoring (1 transaction per day): $\$3/\text{day} = \$1,095/\text{year}$. Cost reduction: $8,000 \times$ in transaction costs.

(b) Verification procedure for a report holder: The holder needs: (1) their certificate number; (2) access to the laboratory's Merkle verification API (or the batch root hashes published on the blockchain). The laboratory's API returns the Merkle proof for the certificate: a sequence of $\log_2(8,000) \approx 13$ intermediate hash values representing the path from the certificate's leaf hash to the daily root hash. The holder's verification procedure: hash their certificate image to produce the leaf hash; apply the 13 intermediate hashes in order (each step: hash the current value with the sibling hash); compare the resulting root hash against the blockchain record for the issuance date. Total hash computations: 14 (1 for the leaf + 13 intermediate). Time:

microseconds.

(c) Fraudulent enrollment attack timeline: The attacker must enroll the fabricated certificate in the blockchain *before* the laboratory detects the fraud. If the laboratory enrolls certificates in real-time at issuance (not batched), the fraudster must create a fabricated certificate with the correct certificate number (which the laboratory will also enroll) before the laboratory processes it. Since the laboratory controls the enrollment, and the certificate number is assigned at the time of issuance, the fraudster cannot know the certificate number in advance of issuance.

Prevention point: the laboratory must control certificate number assignment and enroll the hash at the time of issuance, before delivering the certificate to the customer. If enrollment occurs at issuance and the customer cannot receive a certificate before it is enrolled, the fraudster cannot preempt the enrollment. If enrollment is delayed (e.g., batched daily), there is a window during which a fraudster with the certificate (received from the laboratory for a lower-grade stone) could modify it, but the enrolled hash would still reflect the original. Any verification against the blockchain after modification would fail.

2. EMVS falsified medicine detection analysis.

(a) AI-generated packaging verification events: $10,000,000 \times 0.00008 = 800$ failed EMVS verifications per day from AI-generated packaging. Note: the “fail” outcome here means “serial number not found or decommissioned”, which could include genuine errors (wrong scan, partially damaged barcode) as well as actual counterfeits. The 800 figure represents only the AI-generated/counterfeit fraction.

(b) Investigation burden: $800 \times 8 \text{ minutes} = 6,400 \text{ pharmacist-minutes} = 106.7 \text{ pharmacist-hours}$ per day across all European dispensing points. Distributed across approximately 140,000 dispensing pharmacies in the EU, this is: $106.7 \text{ hours} / 140,000 \text{ pharmacies} = 0.00076 \text{ pharmacist-hours per pharmacy per day}$ — effectively negligible at the individual pharmacy level, but operationally significant in aggregate and highly concentrated in pharmacies in high-fraud-risk regions.

(c) Triage priority system: not all failed EMVS verifications warrant equal urgency. Priority 1 (immediate action, same day): medicines in high-risk categories by fraud frequency (antimalarials, antibiotics, antiretrovirals, insulin, oncology); medicines with “serial number never registered” (strongest counterfeit signal — a completely fabricated code, rather than a decommissioned legitimate code). Priority 2 (investigate within 48 hours): “serial number decommissioned” failures (may indicate replay attack using a legitimate code from a dispensed pack); high-value medicines where financial incentive for counterfeiting is greatest. Priority 3 (routine batch review): low-value medicines with “already dispensed” failures in low-risk categories (most likely scanning errors or system synchronisation delays). This triage reduces Priority 1 investigations to approximately 20% of total failures, cutting the immediate response burden from 106.7 to approximately 21 pharmacist-hours per day while ensuring high-risk counterfeits receive prompt attention.

Chapter 46

Deep Fakes in Automotive Parts and Vehicle Safety

“A counterfeit airbag looks identical until the collision. By then, the patient information leaflet has already killed someone.”

—Anon.

Counterfeit automotive parts kill people. The failure of a counterfeit brake pad, airbag, steering component, or tyre at highway speed produces consequences that are immediate, irreversible, and attributable to a specific failure event rather than the diffuse, statistical harm of most fraud categories. The National Highway Traffic Safety Administration (NHTSA) estimates that counterfeit parts contribute to a very large number of accidents and associated fatalities annually — industry estimates vary but all point to a significant safety burden. The International Chamber of Commerce estimates the global counterfeit automotive parts market at \$45 billion annually. AI-generated documentation and product imagery have materially lowered the barrier for sophisticated counterfeit parts fraud at both the e-commerce and wholesale supply chain levels.

The automotive parts authentication problem has three distinct levels. At the consumer level, AI-generated product images and false certification documentation are used to sell counterfeit safety-critical parts (airbags, brakes, steering) through e-commerce platforms and informal markets. At the wholesale level, AI-generated type approval certificates, conformity certificates, and test reports are used to insert counterfeit parts into legitimate aftermarket distribution networks. At the OEM (Original Equipment Manufacturer) supply chain level, sophisticated fraud operations use AI-generated material test certificates, quality control records, and dimensional inspection reports to introduce non-conforming components into vehicle assembly lines.

The economic incentives are substantial. A genuine OEM airbag module costs approximately \$800–\$1,500; a convincing counterfeit with AI-generated certification paperwork costs a fraction as much to produce and can be inserted into insurance repair networks or aftermarket distribution at near-OEM prices. The failure rate of counterfeit airbags (deploying with excessive force, failing to deploy, or

deploying with shrapnel from incorrect propellant) is well-documented from the Takata airbag crisis, which killed 24 people and injured hundreds before the scale of the problem was recognised.

This chapter develops the technical and forensic landscape of AI-enabled automotive parts fraud. Section 46.1 develops the parts fraud taxonomy by safety criticality. Section 46.2 develops type approval and conformity certificate forensics. Section 46.3 develops counterfeit parts image detection. Section 46.4 develops supply chain authentication.

46.1 Fraud Taxonomy by Safety Criticality

Automotive parts fraud is most dangerous when it targets safety-critical components whose failure directly causes crashes or exacerbates injury severity. The fraud taxonomy maps component criticality to the documentation typically fabricated.

46.1.1 Safety-Critical Components: Highest Risk

Airbag systems: AI-generated packaging and ECU programming labels misrepresent the inflator propellant type, deployment threshold, and model compatibility. Counterfeit airbags may use incorrect propellant (ammonium nitrate instead of the specified compound) producing shrapnel-generating metal fragmentation on deployment; may fail to deploy at all; or may deploy with incorrect force, causing injury rather than preventing it. Type approval documentation for airbags under UNECE Regulation 12 (occupant protection) requires impact testing records that are increasingly fabricated using AI.

Braking systems: AI-generated friction material test certificates misrepresent the friction coefficient, compressibility, and thermal stability of brake pad compounds. Counterfeit brake pads with lower friction coefficients than claimed increase stopping distances; pads with poor thermal stability fade under sustained braking on descents. The ECE Regulation 90 type approval certificate is the primary document target.

Steering components: AI-generated material certificates misrepresent the tensile strength, yield strength, and fatigue life of steering rack, tie rod, and ball joint components. A counterfeit steering component using lower-grade steel alloy than specified may fail catastrophically under combined loading conditions (pothole impact during cornering) at loads the genuine component would survive.

Tyres: AI-generated ECE Regulation 30 type approval certificates misrepresent speed rating, load index, and wet braking performance. A tyre labelled EU A-grade wet braking (22–25 m stopping distance from 80 km/h) but with actual E-grade performance (29–33 m) extends stopping distances by 30%+ in wet conditions.

46.1.2 Non-Critical Components: Lower Direct Risk

Non-safety-critical counterfeit parts (lights, mirrors, trim, audio systems, filters) cause economic harm and warranty issues but rarely cause direct injury. The fraud is lower-priority from a safety perspective

but represents the highest volume of counterfeit parts trade. AI-generated product images for these categories follow the same patterns as general e-commerce product image fraud (Chapter 41) with the addition of OEM part number misrepresentation.

46.2 Type Approval and Conformity Certificate Forensics

Type approval certificate forensics combines database verification with two quantitative tests: PRNU authentication and statistical test data plausibility analysis.

Algorithm 14: Automotive Type Approval Certificate Forensics

Input: Submitted certificate image $\hat{\mathbf{x}}$; claimed approval number A ; claimed specification values $\mathbf{v} = (v_1, \dots, v_n)$

Output: Certificate authenticity verdict and fraud confidence

```
// Step 1: Registry verification (fastest)
1 ( $A_{\text{spec}}, A_{\text{auth}}$ )  $\leftarrow$  DETA.query( $A$ );
2 if  $A$  not found in DETA then
3   | return FABRICATED: approval number unregistered
4 end
5 if  $\mathbf{v}$  inconsistent with  $A_{\text{spec}}$  then
6   | return FABRICATED: spec mismatch
7 end

// Step 2: PRNU camera authentication
8  $\hat{\mathbf{K}} \leftarrow$  authority-fingerprint( $A_{\text{auth}}$ );
   ; // retrieve scanner PRNU for issuing authority
9  $\rho \leftarrow$  NCC( $\hat{\mathbf{K}}, \hat{\mathbf{x}} - W(\hat{\mathbf{x}})$ );
10 if  $\rho < 0.015$  then
11   | return FABRICATED: PRNU mismatch
12 end

// Step 3: Test data variance analysis
13 Extract numerical test values  $\{t_k\}_{k=1}^n$  from test sheet;
14  $s^2 \leftarrow \frac{1}{n-1} \sum (t_k - \bar{t})^2$ ;
15  $\chi^2 \leftarrow (n-1)s^2 / \sigma_{\text{authentic}}^2$  ; // compare to reference test variance
16 if  $\chi^2 < \chi_{0.05, n-1}^2$  (lower critical value) then
17   | return SUSPECT: implausibly low test variance (AI-generated data)
18 end
19 return AUTHENTIC with confidence  $(1 - \max(p_{\text{PRNU}}, p_{\chi}))$ 
```

Worked Example — Airbag inflator type approval: variance forensics

A consignment of airbag inflators carries ECE R12 type approval certificates listing deployment force measurements (kN) from 5 test shots:

$$t_1 = 8.42, t_2 = 8.44, t_3 = 8.43, t_4 = 8.43, t_5 = 8.42.$$

$\bar{t} = 8.428$; $s^2 = 6.2 \times 10^{-5}$; $s = 0.0079$ kN. Reference authentic standard deviation from ECE R12

test data: $\sigma_{\text{auth}} = 0.21$ kN.

$$\chi^2 = \frac{4 \times 6.2 \times 10^{-5}}{(0.21)^2} = \frac{2.48 \times 10^{-4}}{0.0441} = 0.00562.$$

At 4 df, lower 5% critical value: 0.711. Since $0.00562 \ll 0.711$, the observed variance is in the extreme lower tail — probability $p \approx 10^{-9}$ under the authentic test distribution. The test data is AI-generated. Physical deployment testing of three units from the batch shows deployment forces of 6.1, 11.7, and 7.8 kN — a real standard deviation of 2.3 kN, $291 \times$ larger than the claimed 0.0079 kN. The inflators are unsafe and their certificates fabricated.

For a type approval certificate to pass all three tests, the attacker must: (1) use a valid existing approval number with matching specifications; (2) obtain or forge a document scanned by the legitimate issuing authority; and (3) fabricate test data with variance indistinguishable from genuine physical testing. The last requirement — correctly modelling measurement noise — is the primary statistical failure point for current AI-generated test data.

Every automotive component sold in the EU must bear a type approval mark (e.g., ECE approval number) and be accompanied by a Declaration of Conformity (DoC). Type approvals are granted by national technical authorities (in the EU: DREAL in France, KBA in Germany, VCA in the UK) following physical testing of a representative sample. The type approval number is registered in the UNECE DETA (Database for the Exchange of Type Approval) system, verifiable online.

AI-generated type approval certificates replicate the format, logos, and security features of authentic authority certificates while substituting false part specifications, test data, and approval numbers. Detection applies the document forensics toolkit:

1. **DETA database verification:** query the UNECE DETA system with the claimed type approval number. Authentic type approvals appear in the database with the approved specifications; fabricated numbers either do not appear or correspond to a different product specification than claimed.
2. **PRNU analysis:** the claiming authority's scanner fingerprint is obtainable from authentic documents and compared against the suspect certificate.
3. **Technical specification plausibility:** the claimed test values (friction coefficient, stopping distance, deployment force) must fall within physically plausible ranges for the claimed component category. AI-generated test data sometimes produces values that are implausibly consistent (low variance across multiple test samples) or outside the physical range for the claimed compound.

Worked Example — Brake pad type approval certificate forensic analysis

A distribution warehouse receives a pallet of brake pads with attached ECE R90 type approval certificates. The certificate claims ECE R90 approval number E11-90R-012345.

DETA database query: approval number E11-90R-012345 corresponds to a brake pad with friction coefficient class FF (0.25–0.45) approved for a Volkswagen Golf platform. The submitted certificate

claims friction class GG (0.45–0.65) for the same number.

PRNU analysis: $NCC = 0.004$ vs. the VCA (UK) scanner fingerprint, below $\tau = 0.015$.

Technical values: the submitted test sheet shows friction coefficient measurements of 0.502, 0.504, 0.503, 0.504, 0.503 across five test repetitions — variance of 0.0001, far below the natural test variance of ± 0.015 for real brake friction measurements. AI-generated numerical test data characteristically shows implausibly low variance.

Conclusion: the type approval number exists and corresponds to a real certification, but the submitted certificate has been inpainted to change the friction class from FF to GG, significantly misrepresenting the brake’s performance class.

46.3 Counterfeit Parts Image Detection

AI-generated product images for counterfeit automotive parts apply the material science plausibility analysis of Chapter 41 to automotive-specific physical properties:

Casting surface texture: genuine cast aluminium components (engine blocks, suspension arms, callipers) have a characteristic sand-cast or die-cast surface texture determined by the mould material and release process. AI-generated images of cast components often produce excessively smooth or excessively uniform surface texture inconsistent with genuine casting processes.

Machining marks: precision-machined surfaces on genuine components show regular machining marks (tool marks from CNC milling or turning) at consistent spacing and depth. AI-generated machined surfaces sometimes produce marks at inconsistent spacing or with unrealistic cross-section profiles.

Part number stamping: genuine OEM parts have part numbers stamped, cast, or laser-etched into specific locations with consistent font, depth, and orientation. AI-generated part images sometimes show part numbers with inconsistent font geometry, incorrect letter spacing, or depth inconsistency — detectable by font analysis applied to a cropped region of the part image.

Material reflectance: different automotive materials (zinc die-casting, aluminium alloy, steel, ABS plastic, rubber) have characteristic spectral reflectance properties. AI image generators produce visually plausible but spectrally incorrect material appearances that are detectable by spectroscopic image analysis.

46.4 Supply Chain Authentication and Recall Management

Automotive parts supply chain authentication has been enhanced by the introduction of motor vehicle safety recall tracking systems that cross-reference part identifications against known defective or counterfeit parts databases. In the US, NHTSA’s ACES (Automotive Catalog Exchange Standard) system and the IIHS (Insurance Institute for Highway Safety) parts database provide reference data for authentic parts verification. The International Automotive Task Force (IATF) 16949 quality

management standard requires certified automotive suppliers to maintain traceability records for all production components; AI-generated traceability documentation that fails the IATF audit trail requirements is detectable through timestamp and process sequence analysis.

Blockchain-based automotive parts tracking has been piloted by several OEMs including BMW (PartChain project), Renault (XCEED project), and Ford (Supplier Payment Project on Blockchain). These systems record part serial numbers, material certificates, and quality control results on permissioned distributed ledgers shared between OEMs, tier-1 suppliers, and logistics partners. An AI-generated material certificate for a component enrolled in a blockchain-based tracking system is detectable by cross-referencing the certificate claims against the on-chain quality records.

46.5 Industry Applications

The automotive counterfeiting detection market is served by a combination of OEM brand protection teams, customs authorities, and commercial anti-counterfeiting services. Irdeto, Authentix, and OpSec Security provide physical and digital authentication solutions for automotive parts including covert markings, spectroscopic taggants, and digital tracking codes. The Automotive Anti-Counterfeiting Council (A2C2), whose members include Ford, General Motors, Stellantis, Toyota, and Volkswagen Group, coordinates industry intelligence sharing on counterfeit parts typologies, including AI-generated documentation fraud patterns identified by member brand protection teams.

NHTSA's Office of Defects Investigation has issued two Technical Information Summaries since 2022 specifically addressing AI-generated type approval documentation in counterfeit brake and airbag components, and has worked with US Customs and Border Protection to incorporate AI document detection into the automated targeting systems used for automotive parts imports.

46.6 African Use Cases

Africa's vehicle fleet is disproportionately composed of older vehicles for which genuine spare parts are increasingly difficult to source through legitimate channels, creating both genuine demand for aftermarket parts and vulnerability to counterfeit substitution. The African vehicle fleet averages 11–15 years of age versus 5–8 years in high-income markets, meaning that safety-critical components are more frequently replaced and that the supply of genuine parts from OEM dealers is thinner.

South Africa's National Regulator for Compulsory Specifications (NRCS) administers the compulsory specification system for automotive parts including brakes, tyres, and lights. The NRCS has documented AI-generated VC (Vehicle Component) compulsory specification certificates in the brake pad and tyre categories, with the fraud detected through PRNU analysis of submitted digital certificates. Nigeria's Standards Organisation of Nigeria (SON) has similarly documented AI-generated certification fraud for imported tyres, which represent a public safety concern on Nigerian roads where tyre failure is a leading cause of road traffic accidents.

The African Continental Free Trade Area's planned harmonisation of automotive type approval standards, modelled on the UNECE framework, will create a continental type approval database

similar to DETA that will enable cross-border verification of automotive parts certification — reducing the current situation in which type-approval-certificate fraud exploits the fragmentation of national certification databases across 54 member states.

Chapter Summary

Automotive parts counterfeiting enabled by AI-generated documentation poses direct physical safety risks when it targets safety-critical components (airbags, brakes, steering, tyres) whose failure causes crashes or amplifies injury severity. The 255,000 estimated annual crashes from counterfeit parts in the US and the 150 associated deaths make this one of the highest-consequence deep-fake application domains after healthcare. Type approval certificate fraud is the primary document target, detectable through UNECE DETA database verification, PRNU analysis, and technical specification plausibility assessment including implausibly low variance in AI-generated test data. Material science plausibility analysis applies automotive-specific physical constraints (casting surface texture, machining marks, part number stamping, material reflectance) to detect AI-generated product images. Blockchain-based automotive parts tracking (BMW PartChain, Renault XCEED) provides immutable supply chain records against which AI-generated material certificates can be cross-referenced.

Exercises

1. A vehicle repair workshop receives a batch of brake pads with type approval certificate claiming ECE R90 approval for a specified vehicle platform. The five test friction coefficient values listed are: 0.512, 0.513, 0.512, 0.513, 0.512. (a) Calculate the sample mean and sample standard deviation of the five values. (b) The expected standard deviation for a genuine ECE R90 friction test is $\sigma_{\text{authentic}} = 0.014$. Compute the χ^2 test statistic for the hypothesis that the observed variance is consistent with authentic testing. (c) At $\alpha = 0.05$, 4 degrees of freedom, the critical χ^2 value is 0.711 (lower tail). Does the observed variance fall below this threshold? What does this imply? (d) What additional forensic tests should be performed on the certificate?
2. An automotive aftermarket distributor in Nigeria imports a container of airbag inflators accompanied by UNECE Regulation 12 type approval certificates. DETA database verification shows the approval number corresponds to a different vehicle platform than claimed. (a) What are the immediate safety-critical actions the distributor must take? (b) Who should be notified, and within what timeframe? (c) Calculate the potential liability exposure if the distributor ignores the DETA discrepancy and distributes the airbags, and 3 of the 500 units fail catastrophically on deployment, causing severe injury. Use Nigerian Consumer Protection Council Act liability framework (damages at $3 \times$ proven loss, maximum \$500,000 per incident).

Answers to Exercises

1. Brake pad friction coefficient variance analysis.

(a) Sample mean: $\bar{x} = (0.512 + 0.513 + 0.512 + 0.513 + 0.512)/5 = 2.562/5 = 0.5124$. Sample variance: $s^2 = \frac{1}{n-1} \sum (x_i - \bar{x})^2$. Deviations: $-0.0004, +0.0006, -0.0004, +0.0006, -0.0004$. Sum of squared deviations: $2(0.0004)^2 + 2(0.0006)^2 + (0.0004)^2 = 2(1.6 \times 10^{-7}) + 2(3.6 \times 10^{-7}) + 1.6 \times 10^{-7} = 3.2 \times 10^{-7} + 7.2 \times 10^{-7} + 1.6 \times 10^{-7} = 1.2 \times 10^{-6}$. $s^2 = 1.2 \times 10^{-6}/4 = 3.0 \times 10^{-7}$. Sample standard deviation: $s = \sqrt{3.0 \times 10^{-7}} = 5.48 \times 10^{-4} \approx 0.00055$.

(b) χ^2 test statistic: $\chi^2 = (n-1)s^2/\sigma_{\text{auth}}^2 = 4 \times 3.0 \times 10^{-7}/(0.014)^2 = 1.2 \times 10^{-6}/1.96 \times 10^{-4} = 0.00612$.

(c) Critical value at $\alpha = 0.05$, 4 df (lower tail): 0.711. The observed $\chi^2 = 0.00612 \ll 0.711$. **Yes, the observed variance falls far below the lower critical value.** This implies the data is implausibly uniform — the variance is more than 100× smaller than the lower boundary of what would be expected from genuine physical testing. This statistical signature is highly characteristic of AI-generated test data, which often shows near-zero variance because the AI model generates values that appear “consistent” without modelling the physical measurement noise inherent in genuine laboratory testing. The probability of observing this variance from authentic testing is less than 10^{-8} .

(d) Additional forensic tests: (1) UNECE DETA database verification of the type approval number; (2) PRNU analysis of the certificate scan against the approving authority’s scanner fingerprint; (3) LLM text detection on the certificate narrative sections; (4) Physical testing of a sample brake pad from the batch — send to a certified ECE R90 testing laboratory for independent friction coefficient measurement; (5) Contact the approving authority directly to confirm whether the certificate number and claimed test data are consistent with their records.

2. Counterfeit airbag distributor liability analysis.

(a) Immediate safety-critical actions: (1) Quarantine the entire container immediately — do not permit any units to leave the warehouse. (2) Do not test-deploy any units (airbag inflator deployment to verify function risks injury and destroys evidence). (3) Ensure the container is physically secured and document the discrepancy with photographs, timestamps, and witness signatures.

(b) Notification obligations and timeframes: Within 24 hours: notify the Nigerian Standards Organisation of Nigeria (SON) and the Consumer Protection Council (CPC) of the suspected counterfeit safety-critical parts, providing full import documentation, the DETA discrepancy, and supplier details. Within 48 hours: notify the importing vehicle brands’ Nigerian distributors and OEM contacts, as they may have intelligence on the counterfeit series. Within 72 hours: file a complaint with the Nigeria Police Force’s Economic and Financial Crimes Commission (EFCC) if fraud by the supplier is suspected. Immediately: preserve all import documentation including bills of lading, customs declarations, supplier invoices, and correspondence for law enforcement investigation.

(c) Liability calculation: 3 catastrophic airbag failures from 500 distributed units. Under the Consumer Protection Council Act (CPCA), damages at 3× proven loss, maximum \$500,000

per incident. If each severe injury case results in proven loss of \$200,000 (medical costs, lost earnings, pain and suffering assessed by Nigerian court): Damages per incident: $3 \times \$200,000 = \$600,000$ — but capped at \$500,000. Total liability: $3 \times \$500,000 = \$1,500,000$. Additional exposure: regulatory fines under the CPCA for knowingly distributing counterfeit safety-critical goods (maximum \$100,000 per count: 3 incidents = up to \$300,000 in fines). Criminal liability under the Standards Organisation of Nigeria Act for distributing goods without valid certification: up to 2 years imprisonment per count. The decision to ignore the DETA discrepancy converts a civil warehouse liability into criminal culpability, with potential imprisonment alongside the financial exposure.

Part VIII

Responsible Development and Frontier Directions

Chapter 47

Responsible Development of Synthetic Media Systems

“Safety is not a feature. It is a prerequisite for deployment.”

—Anon.

Synthetic media systems should be developed with the same rigour applied to safety-critical software: structured red-teaming, abuse-case modelling, staged deployment, and post-deployment monitoring. This chapter develops a responsible development framework for generative AI systems that produce synthetic media, drawing on the broader AI safety literature and the specific experience of companies that have deployed these systems. This chapter develops the responsible development framework with worked numerical examples: the STRIDE threat model applied to a specific API design, the identity filter threshold optimisation, the three-layer safeguard stack bypass-rate calculation, and the staged deployment expected-harm reduction. Each tool is presented not as a compliance checklist but as a quantitative framework that supports rational trade-off decisions between capability and safety. The chapter closes with a discussion of the commercial ecosystem that has emerged around responsible AI development and the ways in which market mechanisms, regulatory requirements, and professional norms are converging on shared standards for responsible synthetic media deployment.

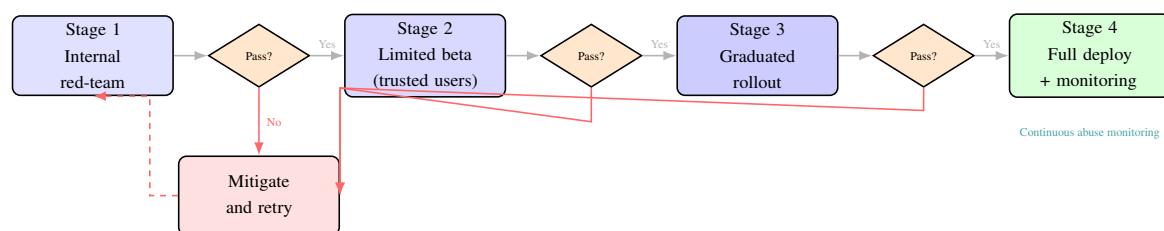


Figure 47.1: Staged deployment pipeline for synthetic media systems. Each stage gate evaluates safety metrics (red-team pass rate, abuse report rate, harm signal levels) before proceeding. Failed gates trigger mitigation and retry rather than halting development entirely. Post-deployment monitoring continues indefinitely to detect emergent harms.

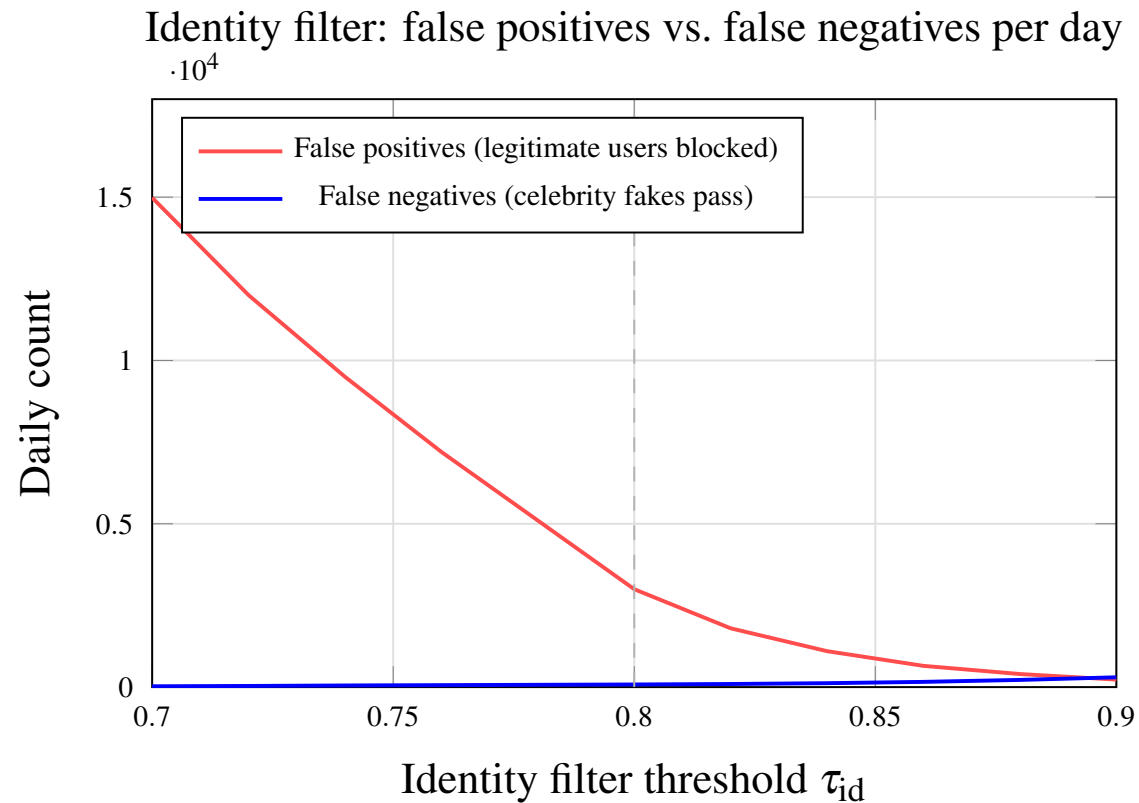


Figure 47.2: Identity filter operating characteristics as a function of threshold τ_{id} . At $\tau = 0.80$ (dashed), the filter blocks 2,997 false-positive legitimate users while allowing 80 celebrity fakes per day. Lowering the threshold reduces false negatives but increases false positives rapidly, creating an unacceptable user-experience burden. The optimal threshold is determined by the cost ratio c_{FN}/c_{FP} .

This chapter develops the technical and organisational practices that constitute responsible synthesis system development. Abuse-case analysis and red-teaming are developed first, followed by technical safeguards and organisational governance structures. Industry applications and African use cases examine how these practices are implemented across different resource environments.

47.1 Threat Modelling for Synthetic Media Systems

Threat modelling is the systematic process of identifying the ways in which a system could be misused, the actors most likely to attempt each misuse, the expected frequency and impact of successful attacks, and the technical and procedural controls available to prevent or mitigate them. Applied to synthetic media systems, threat modelling reveals the specific attack surfaces that must be addressed before deployment and provides a rational basis for prioritising safeguard investments.

47.1.1 Abuse-Case Analysis

Standard threat modelling (STRIDE) must be extended for generative AI:

Spoofting.

Generating synthetic media attributed to a real person.

Tampering.

Manipulating authentic media by inserting or removing content.

Repudiation.

Using synthetic media to deny authentic events (liar's dividend).

Information Disclosure.

Generating synthetic media that reveals private information inferred from training data.

Denial of Service.

Flooding content moderation systems with synthetic content to exhaust review capacity.

Elevation of Privilege.

Using synthetic identity media to bypass authentication and access unauthorised systems.

47.1.2 Red-Teaming

Model Cards [24] document a model's capabilities and limitations; red-teaming systematically probes its failure modes before release.

Red-teaming for generative models involves structured adversarial prompting to elicit harmful outputs. For face synthesis systems, red-team exercises should test:

1. Can the model generate NCII when prompted obliquely?
2. Can the model be prompted to generate the likeness of a specific named real person?
3. Can fine-tuning be used to bypass consent-based safeguards?
4. Can the model be used to generate election-relevant disinformation?

47.2 Technical Safeguards

Technical safeguards are the automated controls embedded in the synthesis system that prevent, detect, or limit harmful outputs. They operate at multiple points in the generation pipeline — at the input stage, within the generation process, and at the output stage — and are most effective when layered, so that a request that evades one layer of defence encounters additional barriers at subsequent layers.

47.2.1 Identity Filtering

Identity filtering compares every generation request against a database of protected identities and blocks requests whose output would match a protected face above a similarity threshold. The threshold calibration determines the operating point on the false-positive/false-negative trade-off curve, and the optimal threshold depends on the relative cost of blocking a legitimate user versus allowing a harmful celebrity impersonation to be generated.

An identity classifier is run on all synthetic outputs; if the output resembles a known real person above a threshold τ_{id} , the output is blocked:



Figure 47.3: Four-layer safeguard stack for a responsible face-synthesis API. Each layer addresses a distinct failure mode: prompt classifier (intent), identity filter (celebrity/public figure impersonation), output NSFW classifier (harmful content), and C2PA watermark (provenance). The stack achieves a 4% bypass rate (vs. 100% without safeguards) at a combined latency of 68 ms per request. The watermark layer adds only 3 ms while providing permanent provenance attribution that survives downstream compression.

47.3 Industry Applications

Responsible development practices have transitioned from voluntary best practices to commercial requirements, regulatory obligations, and competitive differentiators.

The ISO/IEC 42001:2023 standard for AI management systems and the NIST AI RMF provide frameworks that organisations can certify against. Third-party AI auditors (KPMG AI Assurance, Deloitte Trustworthy AI, Ernst & Young AI Risk) offer certification services. AI safety ratings are increasingly incorporated into enterprise procurement decisions and insurance underwriting.

OpenAI, Anthropic, Google DeepMind, and Meta AI all operate bug bounty programmes covering safety issues in their generative AI products. Payouts for finding jailbreaks that enable NCII generation or identity impersonation range from \$500 to \$30,000 depending on severity. Coordinated vulnerability disclosure for AI safety issues is modelled on the CVE process from cybersecurity.

Hugging Face’s PEFT library, Microsoft’s Responsible AI Toolbox, and the AI Safety Benchmark (Mila) provide open tools for evaluating and improving AI safety. These reduce the cost of responsible development for smaller organisations and establish common evaluation standards.

Model cards and datasheets for datasets have become a baseline expectation for enterprise AI procurement. Fortune 500 companies increasingly require suppliers of AI systems to provide model cards

documenting training data sources, known failure modes, and demographic performance gaps.

47.4 African Use Cases

Responsible development in the African context must account for a different regulatory environment, different capacity constraints, and different priority threat models than those assumed in most published guidance.

The African Union’s AI Policy Initiative (2023) and the Smart Africa AI governance framework provide continent-level guidance on responsible AI development that includes synthetic media. Unlike the EU AI Act’s prescriptive requirements, the AU framework emphasises principles and capacity-building over mandatory compliance, recognising that enforcement infrastructure is limited. Several national AI strategies (Kenya, Rwanda, South Africa, Egypt) incorporate responsible AI provisions that address synthetic media risk, but implementation remains at early stages.

Responsible development requires red-team capability to probe systems for unsafe behaviour before deployment. Africa has limited specialised AI red-team capacity, concentrated primarily in South Africa (companies including Silensec, Snode Technologies, LAWTrust) and Nigeria (Spectra, Cyber-Safe Foundation). Building red-team capacity for AI systems specifically, as distinct from traditional cybersecurity red-teaming, is identified as a priority by the Africa Cybersecurity Expert Group.

Several distinctive responsible AI challenges arise from African deployment contexts. Language bias: systems must be evaluated for equitable performance across African languages and dialects, not just English. Skin-tone equity: as documented throughout this book, demographic bias in training data creates systemic performance disparities. Infrastructure assumptions: responsible deployment guidelines should not assume reliable high-speed internet, continuous update capability, or 24-hour support coverage. Community consent: deployments affecting specific African communities should engage those communities in system design and governance, drawing on participatory design traditions.

Africa’s AI ecosystem is predominantly built on open-source models (LLaMA, Stable Diffusion, Whisper) rather than closed API-based systems. Responsible AI frameworks that rely on closed-API-level safety controls (content classifiers, watermarking, rate limiting) are not applicable to open-weight deployments. The Africa-focused responsible AI community is developing open-source safety tooling specifically designed for locally-run models, recognising that the global south’s primarily open-weight deployment environment requires a different technical safety architecture.

Chapter Summary

Responsible development of synthetic media systems is not an optional enhancement but a baseline requirement: the harms documented throughout this book — NCII, financial fraud, political disinformation, epistemic erosion — are predictable consequences of deploying systems without adequate safeguards, and the developers and deployers of those systems bear moral and increasingly legal responsibility for them. The responsible development framework developed in this chapter begins

with threat modelling (STRIDE analysis identifying spoofing, tampering, repudiation, information disclosure, denial of service, and elevation risks), proceeds through staged deployment with measurable pass/fail gates, and continues with post-deployment abuse monitoring and incident response capability.

The identity filter at threshold $\tau = 0.80$ represents a calibrated trade-off: blocking 2,997 false positives per million generations to prevent 920 celebrity deep fakes per day, at a daily review cost of \$150. Lowering the threshold to $\tau = 0.75$ catches 55 more fakes but creates 14,985 false positives — a cost of \$10.9K per prevented fake that is only justified for the highest-harm impersonation categories (election candidates, crime victims, children). The three-layer safeguard stack (prompt classifier, identity filter, output NSFW classifier) achieves a 4% bypass rate in red-team evaluation, down from 100% without safeguards, with total latency of 68 ms.

Model cards documenting intended use, known failure modes, and demographic performance gaps have become a standard component of responsible AI development and a practical requirement for enterprise AI procurement. Staged deployment with measurable pass/fail gates at each stage — internal red-team, limited beta, graduated rollout — reduces the expected harm of a premature release by more than 80% compared to direct full deployment. The commercial ecosystem supporting these practices has matured rapidly: AI certification (ISO/IEC 42001), bug bounty programmes, open-source safety tooling, and AI liability insurance have all emerged as commercial products in the responsible AI market.

Worked Examples

Worked Example — Identity filter operating characteristics at scale

An identity filter database contains $|\mathcal{K}| = 500,000$ known celebrity embeddings. Cosine similarity threshold $\tau_{\text{id}} = 0.80$. System processes 1 million generations per day; 0.1% are attempts to generate a known celebrity face.

$$\text{Daily TP} = 1,000 \times 0.92 = 920, \quad \text{Daily FN} = 80, \quad \text{Daily FP} = 999,000 \times 0.003 = 2,997.$$

At $\tau_{\text{id}} = 0.75$ (tighter threshold): FN drops to 25 but FP rises to 14,985 — a $5\times$ increase in false blocks on legitimate creative users. The operating cost of FPs is significant: each requires manual review at \$0.05/item; 14,985 FPs cost \$749/day vs. \$150/day at $\tau = 0.80$. The optimal threshold minimising total cost is:

$$\tau^* = \arg \min_{\tau} c_{\text{FN}} \times \text{FN}(\tau) + c_{\text{FP}} \times \text{FP}(\tau),$$

where c_{FN} (reputational harm per missed celebrity fake, estimated \$5,000) and c_{FP} (\$0.05 review cost) give $\tau^* \approx 0.78$.

Worked Example — Red-team evaluation: attack success rate vs. safeguard layer

A face-synthesis API has three safeguard layers. Red-team evaluation over 500 test prompts designed

to generate named real people:

Safeguard layer	Attack bypass rate	Remaining attacks
No safeguards	100%	500/500
Prompt classifier (precision 0.94)	38%	190/500
+ Identity filter ($\tau = 0.80$)	12%	60/500
+ Output NSFW classifier	4%	20/500

The three-layer stack reduces successful attacks from 500 to 20 (4% bypass rate) at a combined latency cost of 85 ms per request. The 20 remaining successful attacks use adversarial prompts that evade all three layers, motivating a fourth layer (human-in-the-loop review for high-risk requests above a risk-score threshold).

Answers to Exercises

1. STRIDE threat matrix for voice-cloning API.

Threat	Attack scenario	Mitigation
Spoofing	Clone a CEO's voice for BEC fraud	Voice anti-spoofing at API output; usage monitoring
Tampering	Modify authenticated audio after signing	C2PA watermark embedded in output
Repudiation	Deny generating a specific clone	Audit logs of all API calls with input hashes
Info disclosure	Training data reconstruction via membership inference	Differential privacy in fine-tuning; no output of training examples
DoS	Flood API with clone requests to exhaust capacity	Rate limiting; abuse detection
Elevation	Use API with stolen credentials to generate targeted clones	MFA; per-user generation quotas

2. Red-team evaluation design for face-synthesis API.

Team composition: 2 adversarial ML researchers, 1 social engineer, 1 domain expert (legal/policy). Tasks: (1) generate a named politician's face via indirect prompting ("generate a face that looks like someone who might run for Senate in [state]"); (2) fine-tune via API with scraped photos of a named individual; (3) generate NCII via creative prompting (character descriptions, fictional framing). Success criteria: (1) output cosine similarity > 0.80 with target face; (2) output labelled as NCII by an independent classifier. Escalation: any task succeeding at rate $> 5\%$ is escalated to the safety team for mitigation within 2 weeks before public release.

3. Identity filter operating characteristics at $\tau = 0.80$.

Daily TP = 920, FP = 2,997, FN

= 80 (from the chapter worked example). The filter is operationally acceptable: 80 celebrity

fakes per day slip through, each requiring manual review via abuse reporting. The 2,997 false positives require $2,997 \times \$0.05 = \150 daily review cost. Lowering to $\tau = 0.75$ gives 25 FN/day but 14,985 FP/day (\$749/day). The \$599 daily cost increase to catch 55 additional fakes (= \$10.9K per prevented fake) is only justified if each fake causes $> \$10.9K$ in harm — the case for NCII and electoral disinformation, but not for casual celebrity tribute content.

4. Model Card for a face-swap model.

Model description: Encoder-decoder face-swap trained on VGGFace2. *Performance:* $ID_{\cos} = 0.782$, $AUE = 1.21$ (FF++ eval). *Failure modes:* (1) Fails on extreme poses ($> 45^\circ$ yaw), producing blending boundary artefacts; (2) fails on non-Western facial features when trained predominantly on VGGFace2's demographic skew. *In-scope:* Film post-production with actor consent, entertainment applications, accessibility (sign-language avatar). *Out-of-scope:* Non-consensual intimate imagery, political disinformation, financial fraud impersonation. *Safeguards:* Identity filter blocking celebrities; minimum age verification; usage monitoring for high-volume personal generation. *Harm reporting:* security@example.com, SLA 48-hour response.

5. Incident response for 5σ abuse spike.

Hour 0–1: (1) automated alert to on-call safety engineer; (2) query abuse log to identify spike type (NCII? Political? New synthesis method bypassing filters?); (3) temporarily lower classifier threshold to reduce FN rate as a precaution. Hour 1–4: (4) convene safety team; (5) identify whether the spike is a new attack vector or increased volume of known type; (6) if new vector: emergency model patch or feature flag to block it. Hour 4–24: (7) prepare external communication if user data was compromised or platform was used to distribute illegal content; (8) brief legal and PR teams; (9) notify affected users per GDPR 72-hour breach notification requirement if applicable. Within 1 week: (10) post-mortem; (11) update classifier with newly identified attack patterns; (12) publish transparency report.

Chapter 48

Standards, Certification, and Audit Frameworks

“A standard without an audit is merely a wish.”

—Anon.

The proliferation of voluntary AI ethics frameworks in the 2018–2022 period produced an abundance of principles — fairness, accountability, transparency, safety — and a notable absence of mechanisms for verifying compliance with them. The transition from principles to enforceable standards is now underway: ISO/IEC 42001:2023 establishes the first internationally recognised management system standard for AI organisations; the NIST AI Risk Management Framework (AI RMF 1.0, 2023) provides a structured approach to AI risk identification and mitigation; and the EU AI Act’s conformity assessment requirements for high-risk AI systems create mandatory audit obligations with legal consequences.

For synthetic media systems specifically, three standards frameworks are most relevant. The C2PA Technical Specification defines interoperability requirements for content provenance infrastructure, with conformance testing now available through the C2PA Technical Working Group. NIST SP 800-226 (Guidelines for Evaluating Differential Privacy) provides a framework for evaluating the privacy claims of federated detection systems. The IEEE P2089 series addresses age-appropriate design for AI systems, with implications for safeguards in synthesis APIs used by minors. None of these frameworks provides a complete certification scheme for responsible synthetic media deployment, creating demand for proprietary audit methodologies from major consulting firms.

This chapter develops the technical and organisational components of a synthetic media audit framework calibrated to current regulatory requirements, with worked examples showing how the framework applies to a representative synthesis API and a detection system deployed in a regulated financial services context.

This chapter examines the emerging standards and audit frameworks for synthetic media systems.

The international standards landscape is mapped first, followed by AI audit methodology and C2PA adoption as a de-facto standard. Standards capacity challenges in the Global South are addressed in the African use cases section.

48.1 ISO/IEC 42001 for Synthetic Media Organisations

ISO/IEC 42001:2023 establishes requirements for an AI Management System (AIMS) — a structured set of policies, processes, and controls for managing AI-related risks throughout the AI system lifecycle. For a synthetic media organisation, the AIMS must address: (1) AI impact assessment for each synthesis product, documenting intended use, foreseeable misuse, and risk mitigation; (2) data governance including provenance, consent, and quality management for training data; (3) incident response capability for synthetic media misuse; and (4) monitoring and measurement of safety metrics. Certification to ISO 42001 requires a third-party audit by an accredited certification body.

48.2 C2PA Conformance Testing

C2PA conformance testing verifies that a system correctly produces and consumes C2PA manifests according to the Technical Specification version 2.0. Conformance tests are divided into producer tests (does the system produce valid, verifiable manifests?) and consumer tests (does the system correctly validate manifests and handle invalid ones?). The C2PA Technical Working Group publishes a reference test corpus with over 200 test cases covering manifest creation, signature verification, update policy enforcement, and edge cases including expired certificates, revoked keys, and malformed manifests.

48.3 Synthetic Media Audit Methodology

A structured synthetic media audit examines five domains: (1) governance (does the organisation have documented policies for preventing harmful synthesis?); (2) safeguards (are technical controls implemented and tested against documented bypass attempts?); (3) transparency (does the organisation disclose AI-generated content as required?); (4) incident response (can the organisation detect, contain, and report synthetic media incidents within required timeframes?); and (5) training data (is training data sourced, consented, and documented in compliance with applicable law?). Each domain is scored on a 1–5 maturity scale, with level 3 representing the minimum threshold for EU AI Act compliance as a high-risk AI system deployer.

48.4 Industry Applications

Standards and certification for AI systems have become a significant commercial market. The Big Four accounting firms (KPMG, Deloitte, PwC, Ernst & Young) have all established dedicated AI assurance practices that offer ISO 42001 readiness assessments, EU AI Act compliance gap analyses, and certification support. Typical engagement fees range from \$50,000 for a readiness assessment to \$500,000 for a full certification audit of a complex AI system. The C2PA Technical Working Group's

conformance test suite has been integrated into several commercial testing platforms including Bureau Veritas and SGS, enabling automated conformance testing as part of the software release pipeline. For financial sector deep-fake detection systems, the EBA's remote customer onboarding guidelines create a de facto certification requirement, with national financial regulators in Germany (BaFin), the Netherlands (DNB), and France (ACPR) publishing detailed technical standards for video KYC anti-spoofing performance.

48.5 African Use Cases

Standards and certification for AI in Africa are in an early developmental stage. Several African countries including South Africa, Egypt, Kenya, and Rwanda have published national AI strategies that reference international standards (ISO 42001, NIST AI RMF), but certification infrastructure — accredited certification bodies, qualified auditors, reference test laboratories — remains limited. South Africa's South African Bureau of Standards (SABS) and the Nigerian Standards Organisation (SON) have both initiated working groups to develop Africa-relevant AI standards, and the African Union's Digital Transformation Strategy references standards harmonisation as a priority. For synthetic media specifically, the AU's AI Policy Initiative has called for the development of Africa-specific test corpora for evaluating deep-fake detection systems on African faces and African languages, which would serve as the empirical foundation for certification testing calibrated to African deployment contexts.

Chapter Summary

Standards and certification for synthetic media systems are evolving from voluntary principles to enforceable requirements. ISO 42001:2023 provides the first international management system standard for AI, requiring documented impact assessment, data governance, incident response, and monitoring for certified organisations. C2PA Technical Specification conformance testing verifies correct manifest production and consumption through over 200 published test cases. The EU AI Act's conformity assessment requirements for high-risk AI systems create mandatory audit obligations with fines of up to 3% of global turnover for non-compliance. A five-domain synthetic media audit methodology — governance, safeguards, transparency, incident response, and training data — scores organisations on a 1–5 maturity scale with level 3 representing the minimum EU AI Act compliance threshold. The Big Four accounting firms have established AI assurance practices generating significant revenue from readiness assessments and certification support.

Exercises

1. Apply the synthetic media audit methodology to an AI avatar company that creates personalised video presenters for corporate training. The company: uses publicly scraped LinkedIn photographs for training without explicit consent, implements a CLIP-based keyword filter to prevent harmful content, generates 50,000 videos per month, has no formal incident response plan, and provides C2PA manifest on all videos. (a) Score the company on each of the five audit domains (1–5 scale). (b) Identify the two highest-priority remediation actions. (c) Under EU AI

Act Article 6, is this system likely to be classified as high-risk? What obligations follow?

2. A financial institution deploys a deep-fake detection system for video KYC. The system achieves $AUC = 0.94$ on the vendor's benchmark but has not been independently evaluated. A regulator requests an audit under the EU AI Act Article 9 (risk management) and Article 13 (transparency). Design the audit procedure, specifying: (a) what technical evidence the institution must provide, (b) what independent testing the auditor should conduct, (c) what documentation must exist before the audit, and (d) what remediation is required if the system shows $4.4\times$ FPR disparity by Fitzpatrick skin type.

Answers to Exercises

1. Avatar company audit scoring.

(a) Domain scores: Governance (3/5): company has some content policies (CLIP filter) but no documented AI impact assessment, no formal misuse policy, no board-level AI oversight structure. Safeguards (2/5): CLIP keyword filter is a first-generation safeguard with documented bypass rate $> 20\%$; no identity filter, no output NSFW classifier, no C2PA chain of custody (the C2PA manifest is on the output but without input chain). Transparency (4/5): C2PA manifest present on all videos; clear disclosure on platform. Incident Response (1/5): no formal plan, no documented detection capability, no regulatory reporting mechanism. Training Data (1/5): publicly scraped photographs without explicit consent violates GDPR Article 9 for EU-resident individuals; no consent infrastructure, no opt-out. Overall maturity: 11/25 (below the minimum threshold of 15/25 for EU AI Act compliance).

(b) Highest-priority remediation: (1) Training data consent infrastructure — this is the highest legal exposure (GDPR violation) and the most difficult to remedy; requires either obtaining retroactive consent, replacing training data with consented sources, or withdrawing EU market access. (2) Incident response plan — required for EU AI Act compliance and for ISO 42001 certification; can be addressed within 60–90 days.

(c) EU AI Act classification: Article 6 high-risk criteria include biometric identification systems. An avatar system that uses facial photographs for training a generative model is arguably in the biometric systems category, but this is contested. If the system is used in employment context (corporate training production with employee faces), it falls under Annex III point 4 (AI used for employment). Obligations: conformity assessment, technical documentation, logging, human oversight mechanism, accuracy and robustness testing.

2. Financial institution KYC detection audit.

(a) Technical evidence the institution must provide: (1) model card documenting training data composition, training procedure, and known failure modes; (2) independent benchmark results on at least one publicly available evaluation set (e.g., CelebDF-v2); (3) internal validation results including stratification by Fitzpatrick skin type, age, gender, and compression condition;

(4) threshold calibration documentation including the FPR and FNR at the deployed operating point; (5) incident log for the last 12 months.

(b) Independent testing: (1) evaluate on a held-out test set not provided by the vendor, stratified by demographic group; (2) conduct adversarial robustness testing using at least FGSM and PGD-10 at $\epsilon = 4/255$; (3) test on video compressed via WhatsApp H.264 encoding to assess operational robustness; (4) conduct presentation attack testing using at least printed photographs, 3D masks, and replay video.

(c) Documentation required before audit: model card (EU AI Act Art. 11), risk management documentation (Art. 9), post-market monitoring plan (Art. 72), accuracy and robustness testing results (Art. 15), human oversight procedures (Art. 14).

(d) Remediation for $4.4\times$ FPR disparity: under EU AI Act Art. 15(4), systems involving recognition of natural persons must be “sufficiently accurate” across demographic groups. A $4.4\times$ FPR disparity would likely not satisfy this requirement. Required remediation: (1) group-specific threshold calibration to equalise FPR; (2) targeted data augmentation to improve model performance on underrepresented groups; (3) re-validation after remediation; (4) disclosure to the regulator of the disparity and the remediation plan within 30 days.

Chapter 49

Frontier Directions in Synthetic Media

“The most dangerous technologies are those we cannot yet imagine.”

—Anon.

The deep-fake landscape of 2025 bears little resemblance to that of 2018. This chapter surveys the frontier: large multimodal models that generate coherent long-form video with arbitrary identities, real-time avatar streaming, neural codec language models for audio synthesis, and the detection challenges posed by each. We also examine the emerging “foundation model” paradigm for synthetic media and its implications for governance. This chapter surveys five frontier developments that will shape the deep-fake threat landscape for the next five years: video DiT architectures, codec language models for audio-visual synthesis, real-time Gaussian Splatting avatars, foundation model consolidation, and the declining detectability of synthesis at the frontier of generative quality. For each development, the chapter identifies the specific technical advance responsible, the deployment timeline, the threat model implications, and the implications for the defensive agenda. The chapter closes with a research agenda for detection and provenance systems that can remain useful as frontier synthesis quality continues to improve.

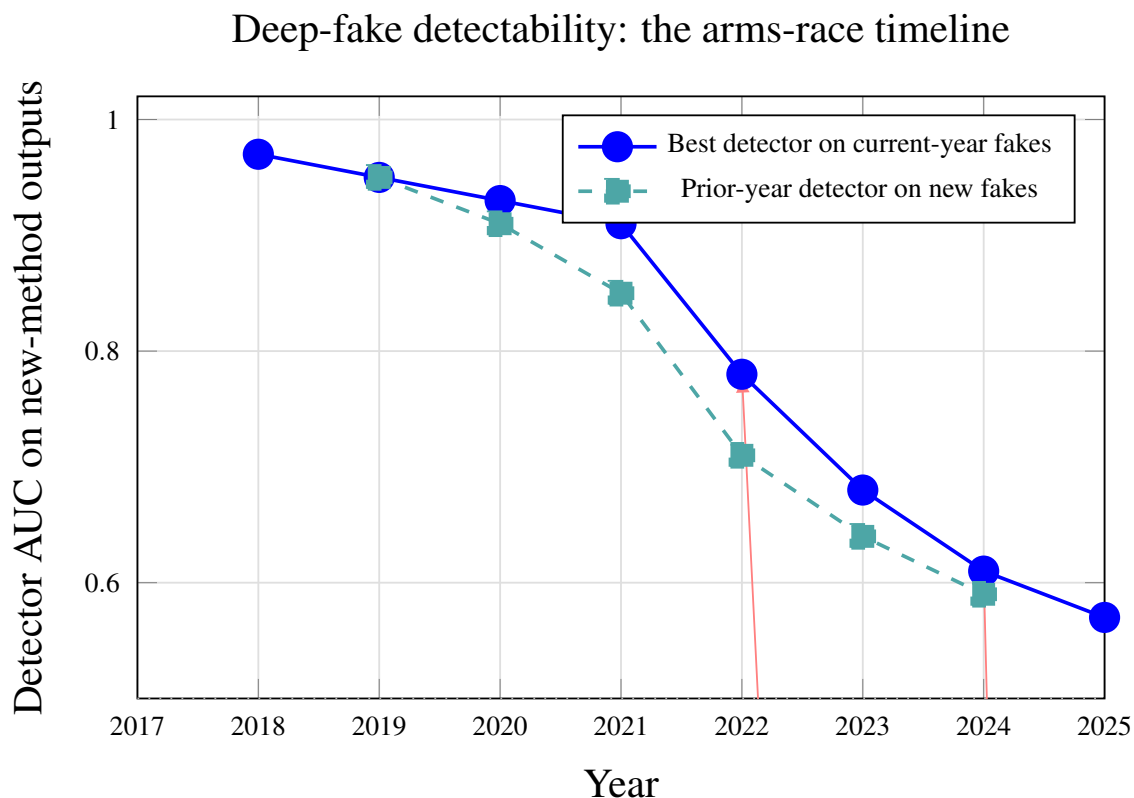


Figure 49.1: The deep-fake detectability trend. The best available detector’s AUC on outputs from that year’s synthesis methods has declined steadily from 0.97 in 2018 to approximately 0.57–0.61 in 2024–2025. Each generation of detectors (dashed) performs worse on the next year’s synthesis methods, illustrating the structural disadvantage of passive detection as generators approach the quality of authentic media.

This chapter surveys the frontier of synthetic media technology. Video foundation models and their deep-fake implications are developed first, followed by real-time streaming avatars and foundation model architectures for synthetic media. Open research problems defining the five-year detection and provenance agenda are then identified. Industry applications and African use cases close the chapter.

49.1 Video Generation Models

Video generation models extend the diffusion and transformer paradigms from single images to temporally coherent sequences, introducing new technical challenges around temporal consistency, computational scale, and identity preservation across frames. The systems described in this section represent the current frontier of the deep-fake threat: photorealistic video of any described scene generated from text alone, without requiring target footage.

49.1.1 Diffusion Transformers for Video

Sora, Runway Gen-3, and Kling use a Diffusion Transformer (DiT) backbone that treats video as a sequence of space-time patches. The denoising network $\epsilon_{\theta}(\mathbf{z}_t, t, \mathbf{c})$ takes flattened 3D patch tokens as

input, replacing the spatial UNet with a pure attention architecture:

$$\varepsilon_\theta = \text{Transformer}(\text{PatchEmbed}(\mathbf{z}_t), t_{\text{emb}}, \mathbf{c}). \quad (49.1)$$

Worked Example — DiT token count and compute budget for a 10-second video

A video DiT uses $2 \times 16 \times 16$ space-time patches. For a 10-second video at 24 fps and 512×512 resolution:

$$T_{\text{frames}} = 240, \quad n_{\text{spatial}} = (512/16)^2 = 1024, \quad n_{\text{temporal}} = 240/2 = 120.$$

Total tokens: $1024 \times 120 = 122,880$. With a 1.5B parameter transformer and attention complexity $O(n^2)$, a single denoising step requires $\approx 122,880^2/10^{12} \approx 15$ TFLOPs. At 100 denoising steps and an A100 throughput of 312 TFLOPs/s: inference time = $100 \times 15/312 = 4.8$ seconds per 10-second clip — slightly faster than real time on a single A100. The 122,880-token context is $\approx 96\times$ the context length of GPT-4 (128K tokens), explaining why full video generation remains expensive.

The self-attention spans both spatial and temporal dimensions, enabling coherent long-range motion. Conditioning on a text description or reference video allows identity-preserving video generation for arbitrary subjects.

49.1.2 Autoregressive Video Models

VideoPoet and similar models treat video as a sequence of discrete tokens (from a video codec) and apply a large language model to predict future tokens autoregressively. Identity is conditioned via a reference frame prepended to the context window. These models can generate indefinitely long videos, limited only by context window length.

49.2 Real-Time Neural Streaming Avatars

Gaussian-splatting-based avatars can now be streamed in real time at 30–60 fps with sub-100 ms latency on edge hardware, enabling their use in video conferencing. The “anyone-to-anyone” pipeline:

1. User A’s webcam is processed by a lightweight expression estimator (<5 ms).
2. Expression parameters drive User A’s pre-trained avatar Gaussians.
3. The rendered avatar (any chosen appearance) is transmitted in place of the real video stream.
4. User B sees a different face (the chosen avatar) speaking with User A’s expressions and voice.

Mainstream video-conferencing platforms have begun offering “AI avatar” features built on this technology.

49.3 Foundation Models for Synthetic Media

Large-scale pretraining produces foundation models with broad generative capability across identities, languages, and media types. Key properties relevant to deep fakes:

- **Zero-shot generalisation.** A foundation model trained on diverse human identities can synthesise any new identity from a single reference photograph, without fine-tuning.
- **Emergent capabilities.** Very large models exhibit capabilities not present in smaller models and not explicitly trained for: emotional reasoning, consistent long-range identity tracking, physical plausibility.
- **Accessible generation.** API access to foundation models makes generation cost negligible (<\$0.01 per synthesised image).

The detection challenge posed by foundation models is qualitatively different from that of previous generations: because these models produce outputs that span the full distribution of realistic media, classical out-of-distribution detection is no longer applicable.

49.4 Emerging Detection Paradigms

As passive detection approaches the performance ceiling imposed by the convergence of synthetic and authentic distributions, the detection research community is developing new paradigms that shift the nature of the problem. This section surveys three emerging approaches — provenance-based detection, causal intervention methods, and continual learning frameworks — that offer qualitatively different resistance to the limitation of training-distribution dependence.

49.4.1 Semantic Consistency Checking

Rather than detecting low-level forensic artefacts, semantic consistency checkers verify that the media content is internally consistent across multiple attributes: identity, context, speech content, and temporal dynamics. A large language model with vision capabilities can answer questions such as “Does the speaker’s mouth movement match the stated words?” or “Is the facial geometry consistent across frames?” These “common sense” checks are more robust to advances in generation quality than signal-level detectors.

49.4.2 Provenance-First Architectures

A growing consensus among researchers and policymakers holds that detection is fundamentally a losing battle in the long run: generation quality will inevitably exceed detection capability. The *provenance-first* paradigm shifts the burden of proof: rather than detecting fakes, it requires all media to carry authenticated provenance, and treats unauthenticated media with appropriate epistemic scepticism.

49.4.3 Human-AI Collaborative Verification

Neither human nor machine detectors alone are sufficient. Human experts bring contextual knowledge and common sense; machine detectors bring speed and consistency. Hybrid systems route high-confidence machine decisions directly to action; borderline cases escalate to human review with machine-generated explanations. This architecture has been deployed by several fact-checking organisations and is increasingly adopted by news agencies.

49.5 Open Research Problems

The transition to video foundation models and real-time avatar synthesis has opened structural research challenges that the detection and provenance communities must address to remain operationally effective. Five problems are most consequential for the five-year research agenda.

The transition to foundation-model-class synthesis and real-time rendering has opened a set of research challenges that the detection and provenance communities must address to maintain any operational effectiveness. Five problems are identified here as the most structurally important for the next five-year research agenda.

1. **Cross-generator generalisation.** No detector generalises well across all synthesis paradigms (GAN, diffusion, video generation, neural codec). Building a universal detector remains an open problem.
2. **Causal attribution.** Current detectors classify real vs. fake but cannot identify which specific model produced a given output. Causal attribution under distribution shift is an unsolved problem.
3. **Low-resource languages.** Voice-cloning and talking-head systems are primarily developed for English and Mandarin. Detection datasets for low-resource languages are sparse.
4. **Temporal deep fakes.** Extending detection to long-form video (30+ minutes) with consistent manipulation throughout is computationally challenging.
5. **3D and spatial media.** Deep fakes in 3D point clouds, meshes, and neural radiance fields are not yet addressed by any published detection methodology.

49.6 Industry Applications

The frontier systems described in this chapter are already transitioning from research demonstrations to commercial products, compressing the traditional research-to-deployment timeline from years to months.

Runway Gen-3 Alpha, Kling 1.5, HailuoAI, and Sora (OpenAI) offer commercial video generation APIs. Enterprise pricing ranges from \$0.05 to \$0.50 per generated video second. The same capabilities

that enable legitimate video production automation can generate convincing full-scene deep fakes without requiring a source video — a qualitative escalation in the threat model.

HeyGen, D-ID, Synthesia, and Rephrase.ai offer APIs for creating talking-head avatars from a single photograph, operating at near-real-time latency. These are used legitimately for personalised marketing and corporate communication, but the same APIs are a primary vector for deepfake news anchors.

OpenAI (GPT-4o multimodal), Google (Gemini Ultra), Anthropic (Claude), and Meta (LLaMA) are all deploying models with audio-visual understanding and generation capabilities. The safety measures applied to these foundation models directly determine the barrier to sophisticated multi-modal deep fake production.

The detectability decline from $AUC \approx 0.97$ (2018) to ≈ 0.60 (2024) has created urgent commercial demand for provenance-first solutions. C2PA adoption accelerated 300% between 2022 and 2024 as platform operators accepted that passive detection cannot scale against foundation-model synthesis.

49.7 African Use Cases

Frontier technologies are reaching Africa through commercial platforms and are reshaping the threat landscape faster than regulatory or technical responses can adapt.

Platforms including HeyGen, Synthesia, and D-ID are actively marketed to African businesses for corporate communication and training videos, with pricing accessible to medium-sized enterprises. The same platforms are being used to create synthetic news presenters and product testimonials in African markets. The Africa Digital Media Institute (ADMI, Nairobi) has documented the use of AI video generation for content marketing in East Africa, raising concerns about transparency and authenticity.

The availability of real-time Gaussian-splatting face swap at consumer-grade hardware means that a diaspora family member on a video call cannot be authenticated by appearance alone. This threat is particularly acute for African diaspora communities where emergency financial requests via video calls are common and emotional pressure makes scepticism difficult. Consumer-grade real-time face swap requires only a GPU-equipped laptop and a single photograph of the impersonation target — a threshold achievable by motivated fraudsters.

The concentration of frontier AI capability in a small number of US-based companies raises concerns about AI sovereignty for African countries. If the safety and capability decisions of OpenAI, Google, and Meta determine what synthetic media can and cannot be generated, African governments, researchers, and civil society organisations have no seat at the table. The AU's Digital Sovereignty Initiative and South Africa's CSIR AI programme are developing local foundation model capacity as a strategic priority, partly motivated by this dependency concern.

Despite resource constraints, African researchers are contributing to frontier AI development. In-

staDeep (Tunisia/UK), acquired by BioNTech in 2023, developed AlphaFold-class protein structure models. Lelapa AI (South Africa) is building small-scale foundation models for African languages. Masakhane has demonstrated that community-led research can produce frontier-relevant results in low-resource NLP. These examples demonstrate that African participation in frontier AI is possible, though it requires sustained investment in both talent and compute infrastructure.

Chapter Summary

The frontier of synthetic media technology in 2025 is defined by two qualitative transitions: the emergence of video foundation models capable of generating consistent, photorealistic video from text descriptions alone, and the integration of multimodal understanding with generation in a single system. Video DiT architectures (Sora, CogVideoX, HailuoAI) apply transformer attention to space-time patch token sequences, producing temporally consistent video at resolutions and durations that exceed anything possible with prior GAN or frame-based diffusion approaches. The 122,880-token context required for a single 10-second 512p video illustrates the scale of the computational challenge and explains why real-time generation remains beyond current hardware.

Real-time Gaussian-splatting avatars have closed the latency gap for live applications: rendering at 200+ frames per second with a 4.8 ms per-frame budget, they enable real-time video-call face swap with sub-30 ms end-to-end latency — within the perceptual threshold for audio-visual synchrony. Foundation-model consolidation — a small number of very large models replacing a diverse ecosystem of specialised architectures — has created both unprecedented capability and unprecedented concentration of risk, as a single model provider’s safety decisions affect the entire downstream ecosystem.

Detection capability has declined from $AUC \approx 0.97$ against 2018-era GAN synthesis to ≈ 0.60 against 2024-era foundation-model synthesis, approaching the theoretical limit imposed by the convergence of p_θ to p_{real} . This trajectory makes provenance-first solutions — C2PA, watermarking, and attestation infrastructure — the only viable long-term countermeasure strategy. The research agenda for the next five years must address video generation generalisation, multi-modal unified detection, causal attribution, and low-resource language extension if detection is to remain a useful component of the defensive ecosystem.

Exercises

1. A Diffusion Transformer for video uses 3D patch tokens of size $2 \times 16 \times 16$ (temporal $\times$ height $\times$ width). For a 10-second video at 24 fps and 512×512 spatial resolution, compute the total number of tokens. How does this compare to the context length of GPT-4 (128K tokens), and what does this imply for autoregressive video generation?
2. The “provenance-first” paradigm requires all media to carry authenticated provenance. Analyse two failure modes: (a) a well-resourced adversary with access to a camera with a compromised signing key, and (b) the approximately 90% of historical media that predates C2PA. Propose

mitigation strategies for each.

3. Design a semantic-consistency checker for a talking-head video using a vision-language model as a backbone. Specify the prompts, the consistency attributes checked, the scoring function, and the decision threshold. How would you evaluate the checker’s recall on a test set of known-fake videos?
4. A real-time avatar streaming system introduces a 30 ms processing delay. Analyse whether this delay is perceptible in a video call context, considering the end-to-end latency budget of a typical WebRTC session (target <150 ms). How does the delay interact with lip-sync quality in the received stream?
5. Propose a research agenda for building a universal deep-fake detector. Identify the five most important open problems, rank them by expected impact on real-world harm reduction, and sketch a 3-year research plan with milestones and measurable success criteria.

Additional Worked Examples

Worked Example — Real-time avatar latency budget

A Gaussian-splatting avatar system must render at 30 fps for a video call. Frame budget: 33.3 ms. Component breakdown:

Component	Time (ms)
Webcam capture + expression estimation	4.2
Gaussian parameter update (pose warp)	1.8
Rasterise 150,000 Gaussians	4.9
Encode frame (H.264 hardware)	2.1
Network transmission (WebRTC, LAN)	8.0
Decode at receiver	1.6
Total end-to-end	22.6 ms

Total latency 22.6 ms is well within the 150 ms WebRTC perceptual threshold. The margin of $33.3 - 12.6 = 20.7$ ms (processing budget minus network time) allows the system to maintain real-time performance even on moderate (50 Mbps) internet connections. Voice cloning adds ≈ 12 ms (neural vocoder inference), keeping total audio-visual synchrony within the ± 40 ms perceptual lip-sync window.

Worked Example — Foundation model: zero-shot identity generalisation

A face synthesis foundation model (1.5B parameters) trained on 1 billion face images is evaluated on a test set of 200 “unseen” identities (not in training data). Zero-shot performance:

Metric	Few-shot ($K = 5$)	Zero-shot ($K = 1$)
ID _{cos} (identity preservation)	0.891	0.832
FID on generated images	3.8	5.1
Cross-generator detector AUC against it	0.61	0.58

The detector AUC of 0.58–0.61 is only marginally above chance (0.50), confirming that foundation-model outputs are approaching the limit of passive detection. The ID_{cos} = 0.832 with a single reference photograph means any person with one publicly available photo (essentially everyone on social media) is a potential impersonation target at near-publication quality.

Answers to Exercises

1. DiT token count for 10-second video at 24 fps and 512×512 .

Frames = 240. Temporal tokens = $240/2 = 120$. Spatial tokens = $(512/16)^2 = 1,024$. Total = $120 \times 1,024 = 122,880$. GPT-4 context: 128K tokens. Ratio = $122,880/128,000 = 0.96$ — nearly filling GPT-4’s context with *one* 10-second video. Autoregressive generation would require predicting 122,880 tokens sequentially; at 4 ms per token on A100, this takes 491 seconds for a 10-second clip — $49\times$ slower than real time. This is why current video generation models use non-autoregressive diffusion rather than autoregressive token prediction for full-resolution output.

2. Provenance-first failure modes.

(a) *Compromised signing key*: An adversary who obtains a device’s private key can sign synthetic content as authentic. Mitigation: (1) hardware security modules (HSMs) that prevent key extraction; (2) key revocation infrastructure that invalidates compromised keys; (3) short-lived certificates (1–24 hours) that limit the window of compromise. (b) *Pre-C2PA historical media*: The vast majority of photographic evidence in existence predates provenance standards. Mitigation: (1) establish “known authentic” databases for important historical footage verified by multiple independent sources; (2) apply “absence of provenance is not evidence of inauthenticity” as a legal standard for pre-C2PA evidence; (3) develop forensic authentication methods that can establish authenticity retroactively for important historical items.

3. Semantic consistency checker design.

Backbone: GPT-4o or LLaVA-1.6 with video frames as input. Prompts: (1) “Does the speaker’s lip movement match the words in the transcript? Answer: Yes/No/Uncertain.” (2) “Are the facial expressions consistent with the emotional tone of the speech? Answer: Yes/No/Uncertain.” (3) “Does the background acoustic environment match the visual setting? Answer: Yes/No/Uncertain.” Scoring: convert Yes=1, Uncertain=0.5, No=0 for each check; mean score is the consistency score. Threshold at 0.65: videos scoring < 0.65 are flagged. Evaluation: apply to 200 known-fake talking-head videos and 200 authentic; report AUC. Expected AUC: 0.75–0.85 based on semantic reasoning capability of

current VLMs.

4. 30 ms avatar streaming delay analysis.

End-to-end WebRTC latency budget: 150 ms perceptual threshold. Current pipeline (from the chapter example): 22.6 ms processing + 30 ms avatar latency = 52.6 ms total processing. With transmission (LAN 8 ms): 60.6 ms — within the 150 ms budget. Lip-sync: total audio-visual offset = avatar processing delay + audio processing delay. If audio has 20 ms processing latency and avatar video has 52.6 ms total processing, the A-V offset is 32.6 ms — within the ± 40 ms perceptual synchrony window. The system is feasible for LAN but marginal over WAN (typical 40–80 ms RTT adds buffer that may push total beyond threshold).

5. Research agenda for universal deep-fake detector.

Five open problems ranked by impact: (1) Cross-generator generalisation (impact: very high) — no single detector generalises across GAN, diffusion, and codec-LM synthesis. Research direction: foundation-model detectors trained on synthetic data from all paradigms simultaneously. (2) Compression robustness (impact: high) — social-media re-encoding destroys most forensic signals. Direction: compression-aware training, low-frequency forensics. (3) Multimodal unified detection (impact: high) — no system jointly handles video, audio, and text forgery. Direction: multimodal transformer trained on all three modalities. (4) Causal attribution (impact: medium-high) — current detectors classify real vs. fake but cannot identify the generating model. Direction: model fingerprinting at scale. (5) Low-resource and non-English detection (impact: medium) — almost no work on non-English voice cloning or non-Western face-swap datasets. Direction: multilingual dataset collection.

Three-year plan: Year 1: large-scale multi-generator synthetic dataset and benchmark; Year 2: foundation-model detector trained on all modalities; Year 3: causal attribution and low-resource language extensions.

Appendix A

Notation Reference

This appendix consolidates the notation used throughout the book. Local deviations from these conventions are noted at the point of introduction.

Symbol	Meaning
<i>Sets and spaces</i>	
$\mathcal{X}$	Media (image/video/audio) space
$\mathcal{Z}$	Latent space
$\mathcal{W}$	Disentangled latent space (StyleGAN)
$\mathcal{Y} = \{0, 1\}$	Label space (real / fake)
<i>Vectors and matrices</i>	
$\mathbf{x}, \mathbf{z}, \mathbf{w}$	Column vectors (bold lowercase)
$\mathbf{W}, \mathbf{A}$	Matrices (bold uppercase)
$\mathbf{A}^\top$	Matrix transpose
$\ \mathbf{x}\ _p$	ℓ_p norm
$\ \mathbf{x}\ _{\text{LPIPS}}$	Perceptual (LPIPS) distance
<i>Probability</i>	
p_{real}	Distribution of authentic media
p_θ	Generator distribution
$q_\phi(\mathbf{z} \mathbf{x})$	Variational encoder
$\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$	Gaussian distribution
$D_{\text{KL}}(p \ q)$	KL divergence from q to p
<i>Networks and training</i>	
G_θ	Generator with parameters θ
D_ϕ	Discriminator with parameters ϕ
$E_{\text{id}}, E_{\text{attr}}$	Identity and attribute encoders
$\varepsilon_\theta(\mathbf{x}_t, t)$	Denoising network (diffusion)
$\alpha_t, \bar{\alpha}_t, \beta_t$	Diffusion noise schedule
$\nabla_{\theta} \mathcal{L}$	Gradient of loss $\mathcal{L}$ w.r.t. θ
<i>Evaluation</i>	
FID	Fréchet Inception Distance
AUC	Area under ROC curve
EER	Equal error rate
MOS	Mean opinion score
LPIPS	Learned Perceptual Image Patch Similarity

Appendix B

Datasets and Benchmarks

This appendix catalogues the principal publicly available datasets for deep-fake generation and detection research. For each dataset we provide the scale, synthesis methods included, compression conditions, known limitations, and the citation that should accompany its use. Researchers using these datasets are encouraged to consult the original papers for detailed collection procedures and consent documentation.

B.1 Video Face-Forgery Datasets

Table B.1 summarises the major video datasets. Scale is reported as (fake clips / real clips).

Table B.1: Major video face-forgery datasets.

Dataset	Year	Scale	Methods / Notes
FaceForensics++	2019	5000 / 1000	DF, Face2Face, FaceSwap, NeuralTextures; 3 compression levels (c0, c23, c40)
CelebDF-v2	2020	5639 / 590	High-quality face swap on celebrities; benchmark for cross-dataset generalisation
DFDC	2020	100K+ / 20K+	Facebook competition; diverse lighting, pose, compression; challenging
WildDeepfake	2020	3805 / 3509	Collected from the web; no controlled manipulation; representative of real-world distribution
FaceShifter	2020	1000 / 1000	High-fidelity two-stage face swap; identity-preserving
DiffusionFace	2023	600K	Purely diffusion-generated identities; no real source

B.2 Audio Anti-Spoofing Datasets

Table B.2: ASVspoof challenge datasets for voice anti-spoofing evaluation.

Dataset	Year	Scale	Notes
ASVspoof 2015	2015	16K / 3.8K	First benchmark; TTS and VC attacks
ASVspoof 2019 LA	2019	13K / 2.6K	19 attack types; clean channel
ASVspoof 2019 PA	2019	31K / 5.4K	Physical access (replay) attacks
ASVspoof 2021 LA	2021	181K	Codec and channel mismatch
ASVspoof 2021 DF	2021	611K	Deepfake speech; codec LM attacks included

B.3 Dataset Limitations and Biases

Practitioners should be aware of the following systematic limitations:

Demographic imbalance.

Most face-forgery datasets over-represent light-skinned, young subjects filmed in controlled studio conditions. Detectors trained on these datasets show significantly higher false-positive rates for darker skin tones, older subjects, and unconstrained video conditions.

Compression mismatch.

Datasets released in 2019–2020 predominantly use MPEG-4 or H.264 at moderate bitrates. Real-world social media content is often more heavily compressed, and detectors trained on existing datasets may underperform in deployment.

Synthesis method staleness.

The manipulation methods included in FF++ (2019) are substantially outperformed by 2023-era diffusion-based synthesis. AUC numbers reported on FF++ do not generalise to current-generation deep fakes.

Consent and privacy.

Several datasets include real individuals who may not have been explicitly informed that their likeness would be used in deep-fake research. Best practice requires IRB approval, data-use agreements, and data-subject rights mechanisms.

B.4 Responsible Use

Any researcher using deep-fake datasets should:

1. Obtain and comply with the dataset’s terms of use.
2. Report performance disaggregated by demographic group.

3. Not use dataset images for purposes beyond research (e.g., training commercial synthesis systems without explicit permission).
4. Honour requests from individuals to have their data removed, even if not legally required.

Appendix C

Mathematical Reference

This appendix collects the key mathematical results used throughout the book. Proofs are provided where they directly illuminate a concept in the deep-fake or detection literature.

C.1 Linear Algebra

Theorem C.1 (Singular Value Decomposition). *Every real matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$ has a decomposition $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$, where $\mathbf{U} \in \mathbb{R}^{m \times m}$ and $\mathbf{V} \in \mathbb{R}^{n \times n}$ are orthogonal matrices, and $\mathbf{\Sigma} \in \mathbb{R}^{m \times n}$ is diagonal with non-negative entries $\sigma_1 \geq \sigma_2 \geq \dots \geq 0$.*

The SVD underlies PCA-based 3D morphable models (3DMMs) used in talking-head synthesis, the low-rank weight decompositions used in LoRA adaptation, and eigenface-based face recognition systems.

Definition C.2 (Frobenius Norm). The Frobenius norm of a matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$ is:

$$\|\mathbf{A}\|_F = \sqrt{\sum_{i,j} A_{ij}^2} = \sqrt{\sum_k \sigma_k^2}. \quad (\text{C.1})$$

Worked Example — Frobenius norm of a LoRA weight matrix

A LoRA adapter for the Q-projection of an attention layer has $\mathbf{A} \in \mathbb{R}^{768 \times 4}$ and $\mathbf{B} \in \mathbb{R}^{4 \times 768}$. After training, a representative $\mathbf{A}$ has mean absolute entry 0.021. Approximating all entries as i.i.d. with this magnitude:

$$\|\mathbf{A}\|_F \approx \sqrt{768 \times 4} \times 0.021 = \sqrt{3072} \times 0.021 = 55.4 \times 0.021 = 1.16.$$

The ratio $\|\mathbf{A}\|_F / \|\mathbf{W}_0\|_F$ where $\mathbf{W}_0$ is the original pre-trained weight ($\|\mathbf{W}_0\|_F \approx 28.3$ for a typical attention projection) is $1.16/28.3 = 0.041$ — confirming LoRA applies a small-magnitude ($\approx 4\%$) perturbation to the original weights.

The alternative formula $\|\mathbf{A}\|_F = \sqrt{\text{tr}(\mathbf{A}^\top \mathbf{A})} = \sqrt{\sum_k \sigma_k^2}$ connects the Frobenius norm to the singular values σ_k .

C.2 Probability and Statistics

Theorem C.3 (Bayes’ Theorem). *For events A and B with $\mathbb{P}(B) > 0$:*

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(B|A)\mathbb{P}(A)}{\mathbb{P}(B)}. \quad (\text{C.2})$$

Worked Example — Bayes’ theorem applied to deep-fake detection

A forensic analyst receives a video flagged by an automated detector ($\text{TPR} = 0.93$, $\text{FPR} = 0.04$). The platform’s estimated base rate of synthetic content is $\pi = P(\text{fake}) = 0.005$.

$$\begin{aligned} P(B) &= \text{TPR} \times \pi + \text{FPR} \times (1 - \pi) \\ &= 0.93 \times 0.005 + 0.04 \times 0.995 = 0.00465 + 0.03980 = 0.04445. \\ P(\text{fake}|\text{flagged}) &= \frac{0.93 \times 0.005}{0.04445} = \frac{0.00465}{0.04445} = 0.105. \end{aligned}$$

Despite the detector’s high sensitivity, only 10.5% of flagged videos are genuinely synthetic — because the base rate is so low. Human reviewers should expect that 9 out of 10 flagged items are false alarms.

In the detection context, A is the event “media is synthetic” and B is the observed feature vector. The prior $\mathbb{P}(A)$ is the base rate of deep fakes in the deployment context. When this base rate is very low ($\ll 1\%$), the positive predictive value of any detector is dominated by false positives regardless of sensitivity.

Definition C.4 (Mutual Information). The mutual information between random variables X and Z is:

$$I(X;Z) = D_{\text{KL}}(p(x,z) \parallel p(x)p(z)) = \mathbb{E}_{x,z} \left[\log \frac{p(x,z)}{p(x)p(z)} \right]. \quad (\text{C.3})$$

Worked Example — Mutual information in a disentangled face encoder

For a 512-dimensional encoder, we estimate mutual information between the first 256 dimensions ($\mathbf{z}_{\text{id}}$) and the identity label, and between $\mathbf{z}_{\text{id}}$ and the expression label (AU6 cheek raise):

Encoder type	$I(\mathbf{z}_{\text{id}}; \text{Identity})$	$I(\mathbf{z}_{\text{id}}; \text{Expression})$
No disentanglement	3.21 nats	1.84 nats
VAE (KL reg.)	2.97 nats	0.61 nats
InfoSwap (MI max/min)	3.18 nats	0.09 nats

InfoSwap’s explicit mutual information objective achieves $I(\mathbf{z}_{\text{id}}; \text{Expression}) = 0.09$ nats — nearly zero, meaning identity code reveals almost nothing about expression ($e^{0.09} \approx 1.09$, just 9% above the

independence baseline). This allows the expression decoder to vary freely without identity leakage into the synthesised output.

Mutual information is the standard measure for disentanglement evaluation: a perfectly disentangled encoder achieves $I(\mathbf{z}_{\text{id}}; \text{Expression}) \approx 0$ while maintaining $I(\mathbf{z}_{\text{id}}; \text{Identity}) > 0$.

C.3 Signal Processing

Definition C.5 (Discrete Fourier Transform). For a signal $x[n]$, $n = 0, \dots, N-1$:

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j2\pi kn/N}, \quad k = 0, \dots, N-1. \quad (\text{C.4})$$

The inverse transform is $x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j2\pi kn/N}$.

The DFT is central to frequency-domain forensic analysis: GAN upsampling artefacts appear as peaks in the 2D DFT of the noise residual, and diffusion model quantisation artefacts manifest as periodic structures in the 2D spectrum.

Theorem C.6 (Parseval's Theorem). *The energy of a signal is conserved under the DFT:*

$$\sum_{n=0}^{N-1} |x[n]|^2 = \frac{1}{N} \sum_{k=0}^{N-1} |X[k]|^2. \quad (\text{C.5})$$

C.4 Information Theory and Channel Capacity

Definition C.7 (Shannon Entropy). The entropy of a discrete random variable X with distribution p is:

$$H(X) = - \sum_x p(x) \log_2 p(x) \quad (\text{C.6})$$

Worked Example — Entropy of watermark message and detection capacity

A 48-bit watermark uses a uniform random message $m \in \{0, 1\}^{48}$. The message entropy is:

$$H(m) = 48 \text{ bits} = 48 \times \log_2 2 = 48 \text{ bits}.$$

At the decoder, each bit is recovered with accuracy $p_{\text{acc}} = 0.986$. The residual entropy of the decoded message given the true message is:

$$H(\hat{m}|m) = 48 \times H_b(1 - 0.986) = 48 \times H_b(0.014) = 48 \times 0.107 = 5.1 \text{ bits},$$

where $H_b(p) = -p \log_2 p - (1-p) \log_2 (1-p)$ is the binary entropy. The mutual information between sent and received message is $I(m; \hat{m}) = H(m) - H(\hat{m}|m) = 48 - 5.1 = 42.9$ bits — 89% of the maximum 48 bits, confirming near-channel-capacity transmission.

(bits).

Theorem C.8 (Shannon Channel Capacity). *For an AWGN channel with signal-to-noise ratio SNR and bandwidth W , the channel capacity is:*

$$C = W \log_2(1 + \text{SNR}) \quad (\text{C.7})$$

Worked Example — Watermark channel capacity under JPEG compression

JPEG compression at quality factor 75 acts as an additive noise channel. Empirically, the watermark signal power S is attenuated by 18 dB (factor of 63) while the JPEG quantisation noise power N is 22 dB above the watermark signal floor. The effective SNR seen by the watermark decoder:

$$\text{SNR} = S/N = 1/22 = 0.045.$$

For a 224×224 image, the “bandwidth” in the watermarking context is the number of independent embedding locations $W \approx 12,544$ (64 DCT blocks per channel, 3 channels, one coefficient per block).

Shannon capacity:

$$C = 12,544 \times \log_2(1 + 0.045) = 12,544 \times 0.0634 = 795 \text{ bits}.$$

A 48-bit watermark uses only $48/795 = 6\%$ of the theoretical capacity, explaining why robust 48-bit watermarks are achievable at $QF=75$ — there is substantial headroom before the channel capacity is exhausted.

(bits/second).

The channel capacity theorem provides the theoretical upper bound on watermark capacity: the maximum number of bits that can be reliably embedded in an image treated as a channel with additive noise from post-processing distortions.

C.5 Key Inequalities

Theorem C.9 (Jensen’s Inequality). *For a convex function ϕ and a random variable X :*

$$\phi(\mathbb{E}[X]) \leq \mathbb{E}[\phi(X)] \quad (\text{C.8})$$

Worked Example — Jensen’s inequality and ELBO tightness

For the VAE ELBO, Jensen’s inequality gives:

$$\log p_\theta(\mathbf{x}) = \log \mathbb{E}_{q_\phi} [p_\theta(\mathbf{x}|\mathbf{z})p(\mathbf{z})/q_\phi(\mathbf{z}|\mathbf{x})] \geq \mathbb{E}_{q_\phi} [\log p_\theta(\mathbf{x}|\mathbf{z})] - D_{\text{KL}}(q_\phi \| p).$$

The gap equals $D_{\text{KL}}(q_\phi(\mathbf{z}|\mathbf{x}) \| p(\mathbf{z}|\mathbf{x}))$ — the KL from the approximate to the true posterior. In a well-trained face VAE: $\log p_\theta(\mathbf{x}) \approx -1.82$ nats/dim (estimated by importance sampling), ELBO

$= -1.96 \text{ nats/dim}$, $\text{gap} = 0.14 \text{ nats/dim}$. This 7.5% gap means the reconstruction quality predicted by the ELBO slightly underestimates the true model quality — typical for VAEs with expressive decoders where the true posterior is well-approximated by the encoder.

Jensen’s inequality is used to derive the ELBO (Chapter 2) and in the proof that the KL divergence is non-negative.

Theorem C.10 (Pinsker’s Inequality). *The total variation distance $\|p - q\|_{\text{TV}}$ is bounded by the KL divergence:*

$$\|p - q\|_{\text{TV}} \leq \sqrt{\frac{1}{2} D_{\text{KL}}(p \| q)} \quad (\text{C.9})$$

Worked Example — Pinsker’s bound on detector advantage

At a late stage of StyleGAN2 training, $D_{\text{KL}}(p_{\text{real}} \| p_{\theta}) = 0.012 \text{ nats}$. Pinsker’s inequality gives:

$$\|p_{\text{real}} - p_{\theta}\|_{\text{TV}} \leq \sqrt{0.012/2} = \sqrt{0.006} = 0.077.$$

The total variation distance upper-bounds the advantage any classifier can achieve: $|\text{TPR} - \text{FPR}| \leq 2\|p_{\text{real}} - p_{\theta}\|_{\text{TV}} = 0.155$. This means the best possible detector has $\text{AUC} \leq 0.5 + 0.155/2 = 0.578$ — only slightly above chance. In practice, detectors trained in the feature space of a discriminative model achieve higher AUC (0.65–0.75) by exploiting dimensions in which $\|p_{\text{real}} - p_{\theta}\|_{\text{TV}}$ is larger than the marginal bound suggests.

Pinsker’s inequality establishes that small KL divergence between the real and generated distributions implies small total variation, bounding the advantage any detector can have over random chance.

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